

Stefano Gentili

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Stefano Gentili

Measure, Integration and a Primer on Probability Theory

Volume 1

Springer

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Preface

"We may forgive a child that is afraid of the darkness. The real tragedy of life is when a man fears the light."

Plato

"Classes and concepts may, however, also be conceived as real objects, namely classes as "pluralities of things" or as structures consisting of a plurality of things and concepts as the properties and relations of things existing independently of our definitions and constructions.
It seems to me that the assumption of such objects is quite as legitimate as the assumption of physical bodies and there is quite as much reason to believe their existence. They are in the same sense necessary to obtain a satisfactory system of mathematics as physical bodies are necessary for a satisfactory theory of our sense perceptions and in both cases it is impossible to interpret the propositions one wants to assert about these entities as propositions about the "data", i.e., in the latter case the actually occurring sense perceptions. (1944)"[1]

Kurt Gödel

This text presents, starting from simple notions, advanced topics in real analysis, measure theory and integration. The various subjects are introduced first in an informal and wordy manner, and then discussed from a technical and scientific point of view. The thorough and complete study thus provides the reader with an extensive and well-founded picture of the more complex, interesting and stimulating concepts of mathematical analysis.

[1]"Russell's mathematical logic", in: A. D. Irvine (ed), Bertrand Russell—Critical assessments, Volume II: logic and mathematics, Routledge, London-New York, 1999, p. 120.

The text's approach, albeit prevailingly theoretical, leads up to the study of applied subjects such as probability, statistics and other areas of applied mathematics (including classical analysis and dynamics of financial markets) plus many more. This aspect will become particularly relevant in Volume Two which discusses, besides Fourier transforms, topics like characteristic functions and the various themes related to convergence, including the laws of large numbers and the central limit theorem. All of this clearly attests to the natural relationship between the notions elaborated in analysis and the possible applications of such theory to the solution of practical problems.

A second feature of the book, which I believe makes it more accessible to novices, are the numerous references to the historical-scientific context in which the topics have developed. These tell about the often misguided and error-laden, yet ingenious, attempts made by scholars that contributed to the birth of novel branches of mathematical analysis, while at the same time leading to a revision and deep reorganisation of the entire subject. Notions that earlier appeared to be rock-solid truths, and part of a system made of attuned laws without odd behaviours, thanks to the critical investigations begun in the 1800s are now called into question. The ensuing virtuous revision of several fields has made mathematics more rigorous as a whole, and has laid the foundations of the contemporary landscape's various facets.

The focus of this book is the Lebesgue integral, which was created and studied by the French namesake at the beginning of the 1900s, in order to overcome the limitations of the Riemann integral which hindered a satisfactory development of Fourier's theory. In due course it became manifest that a reasonably general theory of integration, capable of outdoing Riemann's theory, could only be attained by at the same time strengthening the understanding of the measure of an object. In turn, this prompted the development of a full theory of measure able to interpret new aspects and conceptual differences that would otherwise not be detectable.

Throughout the treatise a number of important contributions are presented, made by the likes of Lejeune Dirichlet, Émile Borel, Ulisse Dini, Giuseppe Peano, Georg Cantor, Camille Jordan, Giuseppe Vitali, Vito Volterra, Constantin Carathéodory, René-Louis Baire, Felix Hausdorff, Heinrich Tietze, Wacław Sierpiński, and Kazimierz Kuratowski, to name a few.

The books is meant for STEM and Economics students with a background in real analysis in one variable. Notwithstanding the rather elementary inception, the text allows to learn the subject's tools and understand its sophisticated techniques. It also introduces the reader to the more elaborate topics of contemporary mathematics as we know them today.

Needless to say, the contents are the result of the work of several scholars which the author has only strived to reproduce, in order to contextualise, logically and historically, a rather profound matter. Due merits are therefore exclusive to all those mathematicians that contributed to the development of the subject treated here.

At last, my heartfelt thanks go to Francesca Bonadei and Francesca Ferrari at Springer Italy, who with tremendous professionalism shepherded the author to the final publication. Special thanks to Simon G. Chiossi (Fluminense Federal University—UFF) who, besides translating the book from Italian with unparalleled expertise and competence, called my attention to a few points that were eventually fleshed out. The author clearly remains the sole responsible for the contents.

Tolentino, Italy Stefano Gentili
April 2019

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Part I
Sets

Chapter 1
Round-Up of Topology



1.1 Topology, Neighbourhoods, Closure, Interior and Frontier

The concept of a topological space took shape in the mid-1800s with the study of $\mathbb{R}$ and more generally Euclidean space, and the properties of continuous maps on such spaces.

Associated to the notion of topological space is the primitive concept of open set, from which there descend neighbourhoods, closed sets, the closure, the interior, the frontier, limit points and isolated points, derived sets etc.

The term "set", in particular, is used in mathematics to indicate a collection or family of objects, and is therefore a primitive notion.

The objects that form a set are called its elements, and for a set to be well defined, i.e. for it to have a meaning in mathematics, we must be able to decide without ambiguity whether an element belongs in it or not.

Definition 1.1.1 (*Topology and topological spaces*) A topology on a non-empty set X is a family $\mathcal{T}$ of subsets of X satisfying the following properties:

1. $\emptyset$ and X belong to $\mathcal{T}$;
2. arbitrary unions of elements of $\mathcal{T}$ belong to $\mathcal{T}$;
3. finite intersections of elements of $\mathcal{T}$ belong to $\mathcal{T}$.

The set X is called space, while the pair $(X, \mathcal{T})$ is called topological space. The sets belonging in $\mathcal{T}$ are called open sets of the topological space $(X, \mathcal{T})$. ◇

Definition 1.1.2 (*Closed sets*) Given a topological space $(X, \mathcal{T})$, a subset $F \subseteq X$ is said to be closed in X in the topology $\mathcal{T}$, or in $(X, \mathcal{T})$, if $(X \setminus F)$ is an open set of the topology $\mathcal{T}$. ◇

The following fact holds:

Proposition 1.1.1 (Properties of closed sets in a topological space) *Given a topological space* $(X, \mathcal{T})$*:*

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1. *$\emptyset$ and X are closed sets;*
2. *arbitrary intersections of closed sets in $(X, \mathcal{T})$ are closed in $(X, \mathcal{T})$;*
3. *finite unions of closed sets in $(X, \mathcal{T})$ are closed in $(X, \mathcal{T})$.*

Proof

1. $\emptyset$ and X are closed sets, for their respective complements are the open sets X and $\emptyset$.
2. Given an arbitrary collection $\{F_\alpha\}_{\alpha \in I}$ of closed sets in $(X, \mathcal{T})$, by De Morgan's laws we have
$$X \setminus \bigcap_{\alpha \in I} F_\alpha = \bigcup_{\alpha \in I} (X \setminus F_\alpha).$$
As the $X \setminus F_\alpha$ are open sets, the expression on the right-hand side is an open set, so $\bigcap_{\alpha \in I} F_\alpha$ is closed.
3. Similarly, if F_i, for $i = 1, \ldots, n$, are closed in $(X, \mathcal{T})$, still by De Morgan's laws
$$X \setminus \bigcup_{i=1}^{n} F_i = \bigcap_{i=1}^{n} (X \setminus F_i)$$
where the term on the right is open, and hence $\bigcup_{i=1}^{n} F_i$ is closed in $\mathcal{T}$.

■

We have seen that X and $\emptyset$ are simultaneously open and closed in every topological space. But there exist topological spaces admitting a larger selection of both open and closed sets.

From now onwards, unless otherwise specified, we shall assume the general topological space $(X, \mathcal{T})$ given, and we shall denote it simply by X.

Definition 1.1.3 (*Neighbourhoods*) Given a set $A \subseteq X$, we call neighbourhood of $x \in A$ any open set $U_x \subseteq A$ such that $x \in U_x$. ◇

From the above definition it follows that if A is an open set, we may take $U_x = A$, so we may also say that A is open in $\mathcal{T}$ if $A = \bigcup_{x \in A} U_x$ for any $x \in A$ (property 2. in the definition of topology).

Definition 1.1.4 (*Closure of a set*) Let $A \subseteq X$ be a subset and consider the collection of all closed sets $\mathcal{F}_A = \{F\}$ containing A. We call closure of A the set $\bar{A} = \bigcap_{F \in \mathcal{F}_A} F$. ◇

In practice the closure of A is the smallest closed set containing A, and a set A is closed if and only if it coincides with its closure. Note furthermore that the collection of all closed sets containing A is not empty, since the whole space X, which is closed, contains $\bar{A}$.

Proposition 1.1.2 (Proprieties of the closure operator) *Given subsets A, B in the space X, the closure operator enjoys the following properties:*

1. $\emptyset = \bar{\emptyset}$ *and* $X = \bar{X}$;
2. $A \subseteq \bar{A}$;
3. $A \subseteq B \Rightarrow \bar{A} \subseteq \bar{B}$;
4. $\overline{A \cup B} = \bar{A} \cup \bar{B}$;
5. $\bar{A} = \overline{(\bar{A})}$;
6. $\overline{A \cap B} \subseteq \bar{A} \cap \bar{B}$;
7. $\bigcap_{\alpha \in I} A_\alpha \subseteq \bigcap_{\alpha \in I} \bar{A}_\alpha$, *where* $\{A_\alpha\}_{\alpha \in I}$ *is an arbitrary family of subsets in* X.

Proof

1. $\emptyset$ is both open and closed, as is the space X;
2. this is a direct consequence of the definition of closure;
3. as $\mathcal{F}_A \subseteq \mathcal{F}_B$, we have $\bar{A} \subseteq \bar{B}$;
4. since $A \subseteq A \cup B$, then $\bar{A} \subseteq \overline{A \cup B}$ and similarly $\bar{B} \subseteq \overline{A \cup B}$, so

$$\bar{A} \cup \bar{B} \subseteq \overline{A \cup B}.$$

   Conversely, since $A \subseteq \bar{A}$ and $B \subseteq \bar{B}$, then $A \cup B \subseteq \bar{A} \cup \bar{B}$. The latter set is closed, being the union of two closed sets, so by definition of closure

$$\overline{A \cup B} \subseteq \bar{A} \cup \bar{B},$$

   whence the equality.
5. The closure of $\bar{A}$ is the smallest closed set containing $\bar{A}$, so $\bar{A} \subseteq \overline{(\bar{A})}$. Moreover $\bar{A}$ is closed and contains $\bar{A}$, so $\overline{(\bar{A})} \subseteq \bar{A}$, and the equality follows.
6. We know $\overline{(A \cap B)}$ is the smallest closed set containing $(A \cap B)$. Since $A \subseteq \bar{A}$ and $B \subseteq \bar{B}$, then $(A \cap B) \subseteq (\bar{A} \cap \bar{B})$. The latter is closed and contains $(A \cap B)$, and differs from the smallest closed set $\overline{(A \cap B)}$ containing $(A \cap B)$. Therefore $\overline{(A \cap B)} \subseteq (\bar{A} \cap \bar{B})$.
7. For any $J \subseteq I$, let $B_J = \bigcap_{\alpha \in J} A_\alpha$. Then by item *3.* we have

$$\begin{aligned} B_J &= \bigcap_{\alpha \in J} A_\alpha \subseteq A_\alpha, \forall \alpha \in J \\ \Leftrightarrow \bar{B}_J &= \overline{\bigcap_{\alpha \in J} A_\alpha} \subseteq \bar{A}_\alpha, \forall \alpha \in J \\ \Leftrightarrow \bar{B}_J &= \overline{\bigcap_{\alpha \in J} A_\alpha} \subseteq \bigcap_{\alpha \in J} \bar{A}_\alpha, \forall J \subseteq I. \end{aligned}$$

   As this holds for every $J \subseteq I$, the claim follows. ■

Definition 1.1.5 (*Interior of a set*) Take a set $A \subseteq X$ and the family of all open sets $\mathcal{G}_A = \{G\}$ contained in A. One calls interior of A the set $\overset{\circ}{A} = \bigcup_{G \in \mathcal{G}_A} G$. ◇

Concretely, the interior of a A is the largest open set contained in A, and a set A is open if and only if it equals its interior. The collection of open sets contained in A is non-empty since $\emptyset$, which is open, is a subset of $\overset{\circ}{A}$.

The following proposition is an immediate consequence of the definition of interior.

Proposition 1.1.3 (Interior points) *Given a set $A \subseteq X$, a point $x \in X$ belongs to the interior $\overset{\circ}{A}$ if and only if there exists a neighbourhood U_x such that $U_x \subseteq A$.* ■

There are remarkable properties relating the operators of closure and interior. The first regards the open set $\overset{\circ}{A}$ and the closed set $(\overline{X \setminus A})$.

Proposition 1.1.4 (Remarkable properties of closure and interior) *For any subset $A \subseteq X$*

1. $\overset{\circ}{A} = X \setminus \overline{(X \setminus A)}$;
2. $\overline{\left((\bar{A})^c\right)} = (\overset{\circ}{A})^c$.

Proof

1. Clearly $(X \setminus A) \subseteq \overline{(X \setminus A)}$, so

$$X \setminus \overline{(X \setminus A)} \subseteq X \setminus \underbrace{(X \setminus A)}_{A^c} = X \setminus A^c = A.$$

   Furthermore, $X \setminus \overline{(X \setminus A)}$ is an open set contained in A, whence

$$X \setminus \overline{(X \setminus A)} \subseteq \overset{\circ}{A}.$$

   To show the opposite inclusion observe that for any open set U contained in A we have $(X \setminus A) \subseteq (X \setminus U) = \overline{(X \setminus U)}$. Moreover[1] $\overline{(X \setminus A)} \subseteq (X \setminus U)$. In addition $X \setminus \underbrace{(X \setminus U)}_{U^c} \subseteq X \setminus \overline{(X \setminus A)} \Leftrightarrow U \subseteq X \setminus \overline{(X \setminus A)}$.

   This is true for every open set U contained in A, so by taking $U = \overset{\circ}{A}$ we obtain

$$\overset{\circ}{A} \subseteq X \setminus \overline{(X \setminus A)}.$$

2. As $\overline{(X \setminus A)} = (\overset{\circ}{A})^c$, the set $(\overline{X \setminus A})$ is the complement of the open set $\overset{\circ}{A}$, and as such is closed.

   $(\bar{A})^c = X \setminus \bar{A} = \overbrace{(X \setminus A)}^{\circ}$ is open since complementary to the closed set $\bar{A}$. Hence its closure is

[1] Note $A \subseteq B \Leftrightarrow \bar{A} \subseteq \bar{B}$.

$$\overline{\left((\bar{A})^c\right)} = \overline{(X \setminus A)} = (\overset{\circ}{A})^c.$$

■

The latter interesting relation, together with Proposition 1.1.2, implies the next property.

Proposition 1.1.5 (Further property of the closure operator) *Given subsets A, B of a space X, the closure operator satisfies the following property*

$$\bar{A} \setminus \bar{B} \subseteq \overline{A \setminus B}.$$

Proof $\bar{A} \setminus \bar{B}$ is not closed, being the intersection of the closed set $\bar{A}$ and the open set $(\bar{B})^c$. The smallest closed set containing $\bar{A} \setminus \bar{B}$ is

$$\overline{(\bar{A} \setminus \bar{B})} \subseteq \overline{(\bar{A})} \cap \left(\overline{(\bar{B})^c}\right) = \bar{A} \cap (\overset{\circ}{B})^c = \bar{A} \setminus \overset{\circ}{B} = \overline{(A \setminus B)}.$$

The claim then follows.

■

Proposition 1.1.6 (Properties of the interior operator) *Given subsets A, B in X, the interior operators satisfies:*

1. $\overset{\circ}{X} = X$ *and* $\overset{\circ}{\emptyset} = \emptyset$;
2. $\overset{\circ}{A} \subseteq A$;
3. $A \subseteq B \Rightarrow \overset{\circ}{A} \subseteq \overset{\circ}{B}$;
4. $\overbrace{(A \cap B)}^{\circ} = \overset{\circ}{A} \cap \overset{\circ}{B}$;
5. $\left(\overset{\circ}{A}\right) = \overset{\circ}{A}$;
6. $\overset{\circ}{A} \cup \overset{\circ}{B} \subseteq \overbrace{A \cup B}^{\circ}$;
7. $\bigcup_{\alpha \in I} \overset{\circ}{A}_\alpha \subseteq \overbrace{\bigcup_{\alpha \in I} A_\alpha}^{\circ}$.

Proof All claims follow directly from Propositions 1.1.2, 1.1.4 and De Morgan's law.

■

Definition 1.1.6 (*Frontier of a set*) We call frontier of $A \subseteq X$ the set

$$\partial A = \bar{A} \cap \overline{(X \setminus A)} = \bar{A} \cap (\overset{\circ}{A})^c.$$

◇

Proposition 1.1.7 (Boundary points) *Given a set $A \subseteq X$, a point $x \in X$ belongs to the frontier ∂A if and only if for every neighbourhood U_x*

$$U_x \cap A \neq \emptyset \ \text{ and } \ U_x \cap A^c \neq \emptyset.$$

Proof For the necessary implication let us assume $x \in \partial A$, so that x is not an interior nor an exterior point of A. Then we claim $x \notin \overset{\circ}{A}$, so no neighbourhood U_x exists that is completely contained in A. This means $U_x \cap A^c \neq \emptyset$. A similar reasoning shows that as x is not exterior to A, x cannot be interior to $(\overset{\circ}{A})^c$, and therefore there is no neighbourhood U_x totally contained in A^c, meaning that $U_x \cap A \neq \emptyset$.

Suppose now $x \in X$ is such that for every neighbourhood U_x we have $U_x \cap A \neq \emptyset$ and $U_x \cap A^c \neq \emptyset$. Take $x \in \bar{A} = \overset{\circ}{A} \cup \partial A$. If $x \in \overset{\circ}{A}$ there exists a U_x contained in A such that $U_x \cap A^c = \emptyset$, contradicting the assumption. Hence necessarily $x \in \partial A$. ■

Proposition 1.1.8 (Properties of the frontier operator) *Let A and B be subsets in X. The frontier operator satisfies the following properties:*

1. $\overset{\circ}{A} = A \setminus \partial A$;
2. $\bar{A} = A \cup \partial A$;
3. $\partial(A \cup B) \subseteq \partial A \cup \partial B$;
4. $\partial(A \cap B) \subseteq (\bar{A} \cap \partial B) \cup (\bar{B} \cap \partial A)$;
5. $\partial(X \setminus A) = \partial A$;
6. $X = \overset{\circ}{A} \cup \partial A \cup \overset{\circ}{(X \setminus A)}$;
7. *A is open if and only if $\partial A = \bar{A} \setminus A$;*
8. *A is closed if and only if $\partial A = A \setminus \overset{\circ}{A}$;*
9. *A is open and closed if and only if $\partial A = \emptyset$.*

Proof

1. By Definition 1.1.6 we obtain

$$A \setminus \partial A = A \setminus \left(\bar{A} \cap (\overset{\circ}{A})^c\right) = A \cap \left(\bar{A}^c \cup \overset{\circ}{A}\right) = \underbrace{(A \cap \bar{A}^c)}_{=\emptyset} \cup \underbrace{(A \cap \overset{\circ}{A})}_{=\overset{\circ}{A}} = \overset{\circ}{A}.$$

2. Still by definition of frontier

$$\partial A \cup A = \bar{A} \cap \underbrace{(\overset{\circ}{A})^c \cup A}_{X} = \bar{A}$$

3.

$$\begin{aligned}\partial(A \cup B) &= \overline{A \cup B} \cap \overline{X \setminus (A \cup B)} \\ &= (\bar{A} \cup \bar{B}) \cap \overline{(X \setminus A) \cap (X \setminus B)} \\ &\subseteq (\bar{A} \cup \bar{B}) \cap \overline{(X \setminus A)} \cap \overline{(X \setminus B)} \\ &\subseteq (\bar{A} \cap \overline{(X \setminus A)}) \cup (\bar{B} \cap \overline{(X \setminus B)}) \\ &= \partial A \cup \partial B.\end{aligned}$$

4.

$$\begin{aligned}\partial(A \cap B) &= \overline{A \cap B} \cap \overline{X \setminus (A \cap B)} \\ &\subseteq (\bar{A} \cap \bar{B}) \cap \overline{X \cap (A^c \cup B^c)} \\ &= (\bar{A} \cap \bar{B}) \cap \overline{(X \setminus A) \cup (X \setminus B)} \\ &= (\bar{A} \cap \bar{B}) \cap \left[\overline{(X \setminus A)} \cup \overline{(X \setminus B)}\right] \\ &= [(\bar{A} \cap \bar{B}) \cap \overline{(X \setminus A)}] \cup [(\bar{A} \cap \bar{B}) \cap \overline{(X \setminus B)}] \\ &= [\bar{B} \cap (\bar{A} \cap \overline{(X \setminus A)})] \cup [\bar{A} \cap (\bar{B} \cap \overline{(X \setminus B)})] \\ &= (\bar{B} \cap \partial A) \cup (\bar{A} \cap \partial B).\end{aligned}$$

5. The frontier of the complement of a set equals

$$\begin{aligned}\partial(X \setminus A) &= (\overline{X \setminus A}) \cap (\overline{X \setminus X \setminus A}) = (\overset{\circ}{A})^c \cap (\overline{X \setminus A^c}) \\ &= (\overset{\circ}{A})^c \cap (\overline{X \cap A}) = (\overset{\circ}{A})^c \cap \bar{A} = \partial A.\end{aligned}$$

6. Item *6.* is self-evident.
7. Suppose A is open, hence $A = \overset{\circ}{A}$, and so

$$\partial A = \bar{A} \cap (\overset{\circ}{A})^c = \bar{A} \cap A^c = \bar{A} \setminus A.$$

   Vice versa let us assume $\partial A = \bar{A} \setminus A$. Then

$$\bar{A} \setminus A = \partial A = \bar{A} \cap (\overset{\circ}{A})^c = \bar{A} \setminus \overset{\circ}{A} \Leftrightarrow A = \overset{\circ}{A}.$$

8. If A is closed then $A = \bar{A}$, so

$$\partial A = \bar{A} \cap (\overline{X \setminus A}) = A \cap (\overline{X \setminus A}) = A \cap (\overset{\circ}{A})^c = A \setminus \overset{\circ}{A}.$$

   Conversely if $\partial A = A \setminus \overset{\circ}{A}$,

$$A \setminus \overset{\circ}{A} = \partial A = \bar{A} \cap (\overset{\circ}{A})^c = \bar{A} \setminus \overset{\circ}{A} \Leftrightarrow A = \bar{A}.$$

9. As A is open and closed $\overset{\circ}{A} = A = \bar{A}$, whence

$$\partial A = \bar{A} \setminus \bar{A} = \overset{\circ}{A} \setminus \overset{\circ}{A} = \emptyset.$$

■

1.2 Limit Points and Isolated Points

Let us now define limit points and isolated points.

Definition 1.2.1 (*Limit points and isolated points*) Given a subset $A \subseteq X$:

1. a point $x \in X$ is a limit point of A whenever

$$x \in \overline{A \setminus \{x\}};$$

2. a point $x \in X$ is an isolated point of A if there exists U_x such that

$$U_x \cap A = \{x\}.$$

◇

Proposition 1.2.1 (Characterisation of limit points) *A point $x \in X$ is a limit point of a given set $A \subseteq X$ if and only if every neighbourhood U_x contains at least one point of A other than x. Formally:*

$$U_x \cap (A \setminus \{x\}) \neq \emptyset.$$

Proof For the necessary part we assume x is a limit point of A, so x is not isolated. Then for every neighbourhood U_x we have $U_x \cap (A \setminus \{x\}) \neq \emptyset$, so U_x contains some point of A different from x.

For the sufficient part, each neighbourhood U_x contains some point of A other than x. If $x \notin \overline{A \setminus \{x\}}$, then $x \in X \setminus \overline{(A \setminus \{x\})}$. As the latter set is open, there exists a neighbourhood U'_x contained in it, but against the hypothesis $U'_x \cap (A \setminus \{x\}) = \emptyset$. The contradiction thus proves the claim. ■

Proposition 1.2.2 (Characterisation of isolated points) *Given $A \subseteq X$ and a point $x \in A$, $\{x\}$ is open in A if and only if x is isolated in A.*

Proof If $\{x\}$ is open in A, there is $U_x \subseteq X$ such that $U_x \cap A = \{x\}$, i.e. there exists a neighbourhood of x in A only containing x, whence x is isolated in A.

Suppose conversely x is an isolated point of A, so by definition there exists a neighbourhood U_x such that $U_x \cap A = \{x\}$. Then $\{x\}$ is open in A. ■

Let us now apply the theory seen thus far to the number set $\mathbb{R}$. By a **linear set of points** we shall mean a **set of points chosen arbitrarily on the real line** $\mathbb{R}$. The simplest instance of linear set is an interval. If a and b are any two points on $\mathbb{R}$ with $a \leq b$, we shall say:

- $[a, b]$ is a closed bounded interval;
- (a, b) is an open bounded interval;
- $(a, b]$ is a left-open, right-closed bounded, interval;
- $[a, b)$ is a right-open, left-closed bounded interval;
- $[a, a]$ is a degenerate interval; the closed interval reduces to the set $\{a\}$, in the other cases we have $\emptyset$;
- $[a, +\infty)$ is a left-closed, right-unbounded interval;
- $(a, +\infty)$ is a left-open, right-unbounded interval;
- $(-\infty, b]$ is a left-unbounded, right-closed interval;
- $(-\infty, b)$ is a left-unbounded, right-open interval;
- $\mathbb{R} = (-\infty, +\infty)$ is the left- and right-unbounded interval.

Regarding the extended real numbers:

- $[a, \infty]$ is a left-closed, right-unbounded interval;
- $(a, \infty]$ is a left-open, right-unbounded interval;
- $[-\infty, b]$ is a left-unbounded, right-closed interval;
- $[-\infty, b)$ is a left-unbounded, right-open interval;
- $\overline{\mathbb{R}} = [-\infty, \infty]$ is the left- and right-unbounded closed interval, representing the set of extended real numbers.

Using the same process of Proposition 1.2.1 the reader should prove the characterisation of limit points in $\mathbb{R}$. We may therefore adopt the following definition of limit point in $\mathbb{R}$.

Definition 1.2.2 (*Limit points in* $\mathbb{R}$) Let A be a linear set of points. A point $x \in \mathbb{R}$ is a limit point of A if every open interval I_x contains at least one point of A distinct from x:

$$I_x \cap (A \setminus \{x\}) \neq \emptyset.$$

$\diamond$

Proposition 1.2.3 (Open intervals with limit points) *Let x be a limit point of a linear set A. Then every open interval (a, b) containing x must contain infinitely many points of A.*

Proof Let us assume by contradiction that the interval (a, b), to which x belongs, only contains a finite number of points of A. Call $x_1, x_2, \ldots, x_n$ the points of $A \cap (a, b)$ different from x. Indicate by δ the smallest among the positive numbers $|x - x_1|, |x - x_2|, \ldots, |x - x_n|, |x - a|, |x - b|$. The open interval $(x - \delta; x + \delta)$ contains none of $a, x_1, x_2, \ldots, x_n, b$, so it cannot contain any point of A distinct from x. This contradicts the fact that x is a limit point of A, whence the claim. ■

Let us now examine the structure of open and closed, bounded linear sets in $\mathbb{R}$.

Theorem 1.2.1 (Georg Cantor[2] —1882—Open subsets of $\mathbb{R}$) *Any open set $G \subseteq \mathbb{R}$ is a countable union (at most) of pairwise disjoint open intervals.*

Proof Let G be an open linear set in $\mathbb{R}$. Given $x \in G$, call $\mathcal{U}_x$ the family of neighbourhoods U_x contained in G. The family is obviously non-empty since G is open.

The union $\bigcup_{U_x \in \mathcal{U}_x} U_x = I_x$ therefore exists and becomes the largest open interval containing x and contained in G. Take $y \in G$ and let $I_y \subseteq G$ be an interval containing y. If $(I_x \cap I_y) \neq \emptyset$, then $(I_x \cup I_y)$ is an open interval containing x and y, and $I_x \subseteq (I_x \cup I_y)$. Since, furthermore, I_x is the largest open interval containing x, clearly $(I_x \cup I_y) \subseteq I_x$. Then $I_x = (I_x \cup I_y)$. So either $I_x = I_y$ or $(I_x \cap I_y) = \emptyset$, meaning the intervals contained in an open subset G of $\mathbb{R}$ either coincide or are pairwise disjoint.

Let now

$$\left\{ I_{q_i} : \bigcup_{U_{q_i} \in \mathcal{U}_{q_i}} U_{q_i} = I_{q_i}, \forall q_i \in (G \cap \mathbb{Q}) \right\}$$

be the collection of intervals contained in G, each one containing a rational number of G. It is clear that this family is at most countable, since $\mathbb{Q}$ is at most countable in G and the intervals I_{q_i} are pairwise disjoint. At last, G open implies that the union of these open intervals must coincide with G.

■

The converse statement holds by definition of open set. Knowing that the complement of an open set is closed, the previous theorem gives the following observation.

Remark 1.2.1 (*Closed subsets of* $\mathbb{R}$) A non-empty bounded subset $F \subseteq \mathbb{R}$ is closed if either it is a closed interval $[a, b]$, or it is obtained from a closed interval $[a, b]$ by removing at most countably many pairwise disjoint open intervals with endpoints in F. ♣

[2] G. Cantor, *Über unendliche, lineare Punktmannigfaltigkeiten*, Math. Ann. 20 (1882), 113–121.

Chapter 2
Types of Sets



2.1 Derived Set, Set of Isolated Points and Boundary

Definition 2.1.1 (*Derived set and set of isolated points*) Given a set $A \subseteq X$ we call:

1. derived set of A the set of limit points of A, denoted by A';
2. $(A \setminus A')$ the set of isolated points of A. ◇

Proposition 2.1.1 (Characterisation of closed sets) *A set $A \subseteq X$ is closed if and only if it contains all of its limit points, that is $A' \subseteq A$.*

Proof Let $A \subseteq X$ be closed and $x \in X$ a limit point of A, so $x \in A'$. If $x \notin A$, since $\overline{A \setminus \{x\}} \subseteq \bar{A} = A$ then $x \notin \overline{A \setminus \{x\}}$, so x is not a limit point of A. The contradiction thus proves the claim.

For the sufficient implication we suppose $A' \subseteq A$. It suffices to show A^c is open. For this let y be an arbitrary point in A^c. Observe that $y \notin A'$ since $y \notin A$, so $y \notin \overline{A \setminus \{x\}}$. Negifying Proposition 1.2.1 gives a neighbourhood U_y that does not contain points of A different from x, i.e. $U_y \cap A \setminus \{x\} = \emptyset$. Moreover, as $y \notin A$,

$$U_y \cap A \setminus \{x\} = U_y \cap A = \emptyset.$$

Therefore $U_y \subseteq A^c$ and y is an internal point of A^c. But $y \in A^c$ is arbitrary, so A^c is open. ■

Proposition 2.1.2 (Derived sets) *All iterated derived sets A', A'', A''', $\ldots$, $A^{(n)}$, $\ldots$ of a given set $A \subseteq X$ are closed and such that*

$$A' \supseteq A'' \supseteq A''' \supseteq \cdots \supseteq A^{(n)} \supseteq \cdots .$$

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S. Gentili, *Measure, Integration and a Primer on Probability Theory*, UNITEXT 125, https://doi.org/10.1007/978-3-030-54940-4_2

Proof We begin from the latter claim and suppose $x \in A^{(n)}$, for $n \geq 2$, is such that $x \notin A'$. A neighbourhood U_x will contain a finite number of elements of A at most. Consequently U_x will not contain points of A', and hence no points of A'', A''', ..., so no point of $A^{(n)}$ for every $n \geq 2$. This goes against the assumption and proves the statement.

Let us prove the first claim. To show A' is closed it is enough to prove $X \setminus A'$ is open. Suppose $x \in X \setminus A'$, so $x \notin A'$, and then there exists a neighbourhood U_x containing at most finitely many points of A. None of these can be a limit point of A, for otherwise U_x would contain infinitely many points of A. Hence $U_x \subseteq X \setminus A'$. As this is true for any $x \in X \setminus A'$, it follows $X \setminus A'$ is open and so A' is closed.

By induction, we suppose $X \setminus A^{(n-1)}$ is open and $A^{(n-1)}$ closed.

Take $x \in X \setminus A^{(n)}$, so $x \notin A^{(n)}$, and then there is a neighbourhood U_x containing a finite number of points of $A^{(n-1)}$ at most. None is a limit point of $A^{(n-1)}$, otherwise U_x would have infinitely many points in common with $A^{(n-1)}$, thus violating the hypothesis. Therefore $U_x \subseteq X \setminus A^{(n)}$. As this holds for any $x \in X \setminus A^{(n)}$, necessarily $X \setminus A^{(n)}$ is open, as requested. ■

From this proposition we infer that the set of isolated points of A is $A \setminus A'$, an open set in X.

A set $A \subseteq X$ with n derived sets, where the nth derived set is clearly finite, is said to be of first species of order n. In particular all sets of first species of order zero contain only a finite number of points; all sets of first species of order 1, have the first derived set which contains only a finite number of points. In general for all $n \in \mathbb{N}$ all sets of first species of order n have the nth derived set which contains only a finite number of points.

A set $A \subseteq X$ is called of second species if it has infinitely many derived sets.[1]

Example 2.1.1 (*Derived sets of sets of first and second species in* $\mathbb{R}$)

1. An example[2] of set of first species and order 1 is

$$A = \left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{3}, \ldots, \frac{1}{n}, \ldots\right\},$$

   whose derived set $A' = \{0\}$ contains the element 0 that does not belong in A.
2. An example[3] of set of first species and order four is:

$$A = \left\{\frac{1}{3^{n_1}} + \frac{1}{5^{n_2}} + \frac{1}{7^{n_3}} + \frac{1}{11^{n_4}} : \forall n_i \geq 1, \quad i = 1, 2, 3, 4\right\}.$$

[1] The notion of derived set was introduced in 1872 by Cantor. In 1874 Du Bois-Reymond created a hierarchy of limit points based on their order, including infinite order.

[2] Cantor, Math. Ann. vol. V (1872).

[3] Ascoli, Ann. of Mat., Serie II. vol. VI, p. 56, 1875.

The first derived set of A is the following set

$$A' = \Big\{0; \frac{1}{3^{n_1}} + \frac{1}{5^{n_2}} + \frac{1}{7^{n_3}}; \frac{1}{3^{n_1}} + \frac{1}{5^{n_2}} + \frac{1}{11^{n_4}}; \frac{1}{3^{n_1}} + \frac{1}{7^{n_3}} + \frac{1}{11^{n_4}};$$
$$\frac{1}{5^{n_2}} + \frac{1}{7^{n_3}} + \frac{1}{11^{n_4}}; \frac{1}{3^{n_1}} + \frac{1}{5^{n_2}}; \frac{1}{3^{n_1}} + \frac{1}{7^{n_3}}; \frac{1}{3^{n_1}} + \frac{1}{11^{n_4}};$$
$$\frac{1}{5^{n_2}} + \frac{1}{7^{n_3}}; \frac{1}{5^{n_2}} + \frac{1}{11^{n_4}}; \frac{1}{7^{n_3}} + \frac{1}{11^{n_4}};$$
$$\frac{1}{3^{n_1}}; \frac{1}{5^{n_2}}; \frac{1}{7^{n_3}}; \frac{1}{11^{n_4}} : \forall n_1, n_2, n_3, n_4 \geq 1\Big\}.$$

The second derived set of A is

$$A'' = \Big\{0; \frac{1}{3^{n_1}} + \frac{1}{5^{n_2}}; \frac{1}{3^{n_1}} + \frac{1}{7^{n_3}}; \frac{1}{3^{n_1}} + \frac{1}{11^{n_4}};$$
$$\frac{1}{5^{n_2}} + \frac{1}{7^{n_3}}; \frac{1}{5^{n_2}} + \frac{1}{11^{n_4}}; \frac{1}{7^{n_3}} + \frac{1}{11^{n_4}};$$
$$\frac{1}{3^{n_1}}; \frac{1}{5^{n_2}}; \frac{1}{7^{n_3}}; \frac{1}{11^{n_4}} : \forall n_1, n_2, n_3 \; and \; n_4 \geq 1\Big\}.$$

The third derived set of A is

$$A''' = \Big\{0; \frac{1}{3^{n_1}}; \frac{1}{5^{n_2}}; \frac{1}{7^{n_3}}; \frac{1}{11^{n_4}} : \forall n_1, n_2, n_3 \; and \; n_4 \geq 1\Big\}.$$

The fourth derived set consists in the singleton $\{0\}$.

3. In the light of the above example we may say that the set[4]

$$A = \Big\{\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n} : \forall a_i \geq 1, \; with \; i = 1, \ldots, n\Big\},$$

is of first species and order n.

4. Also the following set[5]

$$A = \Big\{\frac{1}{2^{m_1}} + \frac{1}{2^{m_1+m_2}} + \cdots + \frac{1}{2^{m_1+m_2+\cdots+m_n}}$$
$$: \forall m_i \geq 0, \; and \; n \in \mathbb{N}\Big\},$$

is of first species and order n.

[4] H. J. S. Smith, Proc. Lond. Math. Soc., vol. VI, p. 145, 1875.

[5] Mittag-Leffler, Acta Math., vol. IV, p. 58, 1884.

5. Here is an example[6] of second species, i.e. with derived sets up to the first infinite ordinal ω:

$$A = \left\{ \frac{1}{2^n} + \frac{1}{2^{n+m_1}} + \frac{1}{2^{n+m_1+m_2}} \cdots + \frac{1}{2^{n+m_1+m_2+\cdots+m_n}} \right.$$
$$\left. : \forall n \geq 0; \forall\, m_1, \ldots, m_n \geq 0 \right\},$$

   where the m_i, for $i = 1, \ldots, n$ have all positive integral values including zero, independently of one another. For every n the derived set $A^{(n)}$ contains the numbers $1/2^n, 1/2^{n+1}, \ldots$, so the derived set $A^{(\omega)}$ exists and reduces to 0 only.
6. Another subset of second species in $\mathbb{R}$ are the rationals contained in the interval $[0, 1]$. The first derived set contains every real number in $[0, 1]$, and each successive derived set coincides with the first. ★

Several statements in the next proposition are immediate from Proposition 1.1.2, but we reckon it is a useful exercise to prove them explicitly.

Proposition 2.1.3 (Properties of derived sets) *Let A, B be subsets of X. Then:*

1. $\bar{A} = A \cup A'$;
2. *if A is a subset of B, every limit point of A is a limit point of B:* $A \subseteq B \Rightarrow A' \subseteq B'$;
3. $(A \cup B)' = A' \cup B'$;
4. $(A \cap B)' \subseteq (A' \cap B')$;
5. $\bigcup_{\alpha \in I} A'_\alpha \subseteq \left(\bigcup_{\alpha \in I} A_\alpha\right)'$.

Proof

1. Regarding the first inclusion, $A \subseteq \bar{A}$ and by Proposition 2.1.1 $A' \subseteq \bar{A}$. Hence $A \cup A' \subseteq \bar{A}$.
   For the opposite inclusion we need to show first $A \cup A'$ is closed, i.e. $(A \cup A')^c$ is open. Since $y \in (A \cup A')^c \Rightarrow y \notin A'$, there exists U_y such that $U_y \cap A \setminus \{x\} = \emptyset$. Moreover $y \notin A$, so $U_y \cap A \setminus \{x\} = U_y \cap A = \emptyset$. But $A' \subseteq A$ implies $U_y \cap A' = \emptyset$. Overall $U_y \cap (A \cup A') = \emptyset$, and so $U_y \subseteq (A \cup A')^c$. As $y \in (A \cup A')^c$ is arbitrary $(A \cup A')^c$ is open, and $A \cup A'$ closed. Furthermore, $A \cup A'$ is closed and contains A. As $\bar{A}$ is the smallest closed set containing A, clearly $\bar{A} \subseteq A \cup A'$.
2. Now $x \in A' \Leftrightarrow U_x \cap A \setminus \{x\} \neq \emptyset$. As $A \subseteq B$, then $U_x \cap A \setminus \{x\} \subseteq U_x \cap B \setminus \{x\} \neq \emptyset$. Therefore $x \in B'$, i.e. $A' \subseteq B'$.
3. Since $A \subseteq A \cup B$ and $B \subseteq A \cup B$, by the previous item $A' \subseteq (A \cup B)'$ and $B' \subseteq (A \cup B)'$, hence $A' \cup B' \subseteq (A \cup B)'$.
   Conversely, suppose $x \in (A \cup B)'$, so every neighbourhood U_x contains infinitely many points of $(A \cup B)$. This means that every neighbourhood U_x contains infinitely many points of A, infinitely many points of B, or both. If x is a limit point of A and/or B, then $x \in A'$ and/or $x \in B'$, whence $x \in (A' \cup B')$. Therefore $(A \cup B)' \subseteq (A' \cup B')$. From the latter two inclusions the assertion follows.

[6] Mittag-Leffler, Acta Math., vol. IV, p. 58, 1884.

4. Now $(A \cap B) \subseteq A$, so item *2.* forces $(A \cap B)' \subseteq A'$. In addition $(A \cap B) \subseteq B$, so still by *2.* we have $(A \cap B)' \subseteq B'$. By these two we have $(A \cap B)' \subseteq (A' \cap B')$.
5. Take $x \in \bigcup_{\alpha \in I} A'_\alpha$, so $x \in A'_\alpha$ for some $\alpha \in I$. Then there is a neighbourhood U_x such that $U_x \cap A_\alpha \setminus \{x\} \neq \emptyset$. *A fortiori*, then,

$$U_x \cap \left(\bigcup_{\alpha \in I} A_\alpha \setminus \{x\} \right) \neq \emptyset \Leftrightarrow x \in \left(\bigcup_{\alpha \in I} A_\alpha \right)',$$

proving the claim. ■

Remark 2.1.1

1. Item *1.* in Proposition 2.1.3 implies

$$\overline{A \setminus \{x\}} = (A \setminus \{x\}) \cup (A \setminus \{x\})' \quad \text{and} \quad (A \setminus \{x\})' \subseteq \overline{A \setminus \{x\}}.$$

2. Regarding the frontier of a set $A \subseteq X$:

   - an isolated point x of A is not a limit point of A, so it does not belong to A'. As internal points are limit points, x cannot belong in the interior of A, and must then belong to the frontier ∂A;
   - an internal point x of A cannot belong to the frontier ∂A;
   therefore ∂A consists of the isolated points and the non-internal limit points of A:

$$\partial A = (A \setminus A') \cup (A' \setminus \overset{\circ}{A}).$$

3. The set of limit points A' of $A \subseteq X$ is made of the internal points of A and the frontier points that are not isolated:

$$A' = \overset{\circ}{A} \cup (\partial A \setminus (A \setminus A')).$$

4. As the closure[7] $\bar{A}$ of $A \subseteq X$ obviously contains the isolated points of A, it can be seen as the union of the derived set A' and the set of isolated points $(A \setminus A')$

$$\bar{A} = A' \cup (A \setminus A').$$

   Likewise, the union of the interior $\overset{\circ}{A}$ and the frontier ∂A is still the closure of A

$$\bar{A} = \overset{\circ}{A} \cup \partial A.$$ ♣

Informally speaking, the cardinality of the set of isolated points is at most countable. In fact if x is an isolated point of A, hence not a limit point, it is possible to choose a

[7] Points in the closure of a set are at times called adherent to the set.

neighbourhood such that $U_x \cap A = \{x\}$. By doing the same with every point of the set and choosing the size of the neighbourhoods U_x so that they are pairwise disjoint, we obtain a family of neighbourhoods that is at most countable. Yet not all countable set is made of isolated points. What we have seen goes to show that a closed subset $F \subseteq X$ differs from its derived set at most by countably many points. In fact $F \backslash F'$ does not contain limit points, and as such it consists of isolated points, whence it must be at most countable.

Let us now examine a number of properties of $\mathbb{R}$ related to these issues.

Theorem 2.1.1 (Bolzano–Weierstraß theorem) *Every infinite bounded subset $A \subseteq \mathbb{R}$ possesses at least one limit point, which may or not belong to A.*

Proof As A is bounded, there exists a closed interval $[a, b]$ containing it. Take the point

$$c = \frac{a+b}{2}.$$

At least one of the closed subintervals $[a, c]$ and $[c, b]$ must contain infinitely many points of A, otherwise A would be finite. Clearly, if both subintervals contain infinitely many points we simply choose one, and rename it $[a_1, b_1]$. Now pick the point

$$c_1 = \frac{a_1 + b_1}{2}$$

and call $[a_2, b_2]$ the subinterval, among $[a_1, c_1]$ and $[c_1, b_1]$, that contains infinitely many points of A. Iterating this process we generate a nested sequence

$$[a, b] \supset [a_1, b_1] \supset [a_2, b_2] \supset \cdots,$$

where each interval contains infinitely many points of A. As

$$b_n - a_n = \frac{b-a}{2^n},$$

the length of $[a_n, b_n]$ tends to zero as $n \to \infty$, and there exists a point x contained in all of the $[a_n, b_n]$, and such that

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n = x.$$

We claim x is a limit point of A. Pick an open interval (α, β) containing x. Clearly for n large enough

$$[a_n, b_n] \subset (\alpha, \beta).$$

The open interval (α, β) contains infinitely many points of A, and the theorem is proved. ■

The condition of I being bounded is essential for Theorem 2.1.1 to hold. To explain the importance of this observation think of $\mathbb{N}$. It is infinite, but it has no limit points since it is unbounded.

Now we shall introduce Cantor's intersection theorem for $\mathbb{R}$.

Theorem 2.1.2 (Cantor's intersection property for $\mathbb{R}$) *Let $\{F_n\}_{n=1}^{\infty}$ be a decreasing sequence of non-empty, closed and bounded subsets in $\mathbb{R}$. Then there exists a unique point $x \in \mathbb{R}$ such that $x \in F_n$ for every $n \in \mathbb{N}$, i.e.*

$$F = \bigcap_{n=1}^{\infty} F_n \neq \emptyset.$$

Therefore F is non-empty, closed and bounded in $\mathbb{R}$.

Proof Let $\{F_n\}_{n=1}^{\infty}$ be a decreasing sequence of non-empty, closed and bounded subsets of $\mathbb{R}$. Clearly

$$F = \bigcap_{n=1}^{\infty} F_n$$

is closed, and since $F \subseteq F_1$, with F_1 bounded, also F is bounded.

There remains to prove F is not empty. For this pick

$$x_1 \in F_1, x_2 \in F_2, \ldots, x_n \in F_n, \ldots.$$

The sequence $\{x_n\}_{n\geq 1}$ may be finite or infinite. The former situation occurs when starting from a certain n', for every $n \geq n'$ the x_n stabilise to a specific value ξ, so $\xi \in F_n$ for every $n \geq n'$. But since the sequence of subsets $\{F_n\}$ is decreasing, necessarily $\xi \in F_n$ for every $n < n'$ as well, so eventually $\xi \in F$. In the second case $A = \{x_1, x_{n_1}, \ldots, x_{n_k}, \ldots\}$, with $1 < n_1, \ldots, n_k, \ldots$. Clearly A is a subset of F_1 so, besides infinite, it must be bounded. The Bolzano–Weierstraß theorem (Theorem 2.1.1) implies A has a limit point, say ξ.

We claim that for every $k \in \mathbb{N}$ the set $(A \setminus \{x_1, \ldots, x_{n_k}\})$ has the same limit points as A. Let in fact y be a limit point of A, and take a neighbourhood U_y, which contains an infinite number of points of A. At most finitely many elements in U_y belong to $\{x_1, \ldots, x_{n_k}\}$. The rest is still made of infinitely many points, all falling in $\left(A \setminus \{x_1, \ldots, x_{n_k}\}\right)$, making y a limit point of $\left(A \setminus \{x_1, \ldots, x_{n_k}\}\right)$. Analogously, one proves that if y is a limit point of $\left(A \setminus \{x_1, \ldots, x_{n_k}\}\right)$, then it is a limit point of A, too. As all this holds for every $k \in \mathbb{N}$,

$$A' = \left(A \setminus \{x_1, \ldots, x_{n_k}\}\right)', \ \forall k \in \mathbb{N}.$$

But ξ for every $k \in \mathbb{N}$ is a limit point of $\left(A \setminus \{x_1, \ldots, x_{n_k}\}\right)$, and

$$\left(A \setminus \left\{x_1, \ldots, x_{n_k}\right\}\right) \subseteq F_{n_k},$$

so ξ is a limit point of F_{n_k}. Now, F_{n_k} closed implies $\xi \in F_{n_k}$, for every $k \in \mathbb{N}$. By assumption the sequence $\{F_n\}$ decreases as n grows, so $\xi \in F_n$ for every n and, ultimately, $\xi \in F$, meaning F is non-empty. ■

Definition 2.1.2 (*Open covers and finite subcovers*) An open cover (or open covering) of a set $A \subseteq \mathbb{R}$ is a collection of open subsets $\mathcal{G} = \{G_\alpha\}_{\alpha \in \mathcal{I}}$ in $\mathbb{R}$ such that for every $x \in A$ there exists a $G_\alpha \in \{G_\alpha\}_{\alpha \in \mathcal{I}}$ such that $x \in G_\alpha$, so

$$A \subseteq \bigcup_{\alpha \in \mathcal{I}} G_\alpha.$$

A finite subcover of an open cover $\mathcal{G}$ of A is a finite subcollection $\{G_{\alpha_1}, G_{\alpha_2}, \ldots, G_{\alpha_n}\}$ of subsets in $\mathcal{G}$, such that for every $x \in A$ there exists an open set G_{α_i}, $1 \leq i \leq n$, with $x \in G_{\alpha_i}$. So again

$$A \subseteq \bigcup_{i=1}^{n} G_{\alpha_i}.$$

◇

We are now ready to prove the important covering theorem of Heine–Pincherle–Borel. The result states that if a closed bounded set $F \subseteq \mathbb{R}$ is covered by a family of open intervals, one can discard many intervals and still cover the set with the rest. As a matter of fact a finite number is sufficient to cover F. The theorem might seem trivial at first sight, but reveals its depth in its multiple and crucial applications.

Theorem 2.1.3 (Heine–Pincherle–Borel theorem for closed bounded subsets of $\mathbb{R}$) *If a closed bounded set $F \subseteq \mathbb{R}$ is covered by a family of open intervals $\mathcal{I}$, there exists a finite family of intervals $I_1, I_2, \ldots, I_n$ of $\mathcal{I}$ that still covers F.*

Proof We suppose by contradiction that there is no finite open subcover of $\mathcal{I}$ for F.

As F is bounded there are finitely many closed intervals $\{J_i\}$, each of length less than 1, covering F. At least one of these contains the portion of F that cannot be covered by a finite number of open intervals of $\mathcal{I}$. Let us call it J_1, and define $F_1 = J_1 \cap F$. As F_1 is bounded, there are finitely many closed intervals, each of length less than $1/2$, that cover F_1. As before, one of these contains the part of F_1 not covered by finitely many open intervals of $\mathcal{I}$. Call it J_2 and let $F_2 = J_2 \cap F_1$. Proceeding thus we generate a decreasing sequence of closed bounded sets

$$F_1 \supseteq F_2 \supseteq \cdots \supseteq F_n \supseteq \cdots$$

such that for every $n \in \mathbb{N}$, the set F_n is covered by a closed interval J_n of length less than $1/n$, but cannot be covered by a finite number of open intervals from $\mathcal{I}$.

By Theorem 2.1.2

$$F = \bigcap_{n=1}^{\infty} F_n$$

and $x \in F$. But $\mathcal{I}$ is an open cover, so there exists $I' = (a, b) \in \mathcal{I}$ such that $x \in I'$. There is a $k \in \mathbb{N}$ for which the length of J_k satisfies

$$\ell(J_k) < \frac{1}{k} < \min\{x - a, b - x\},$$

so $J_k \subseteq I'$. Since $F_k \subseteq J_k$, we have $F_k \subseteq I' \in \mathcal{I}$. But this means F_k admits a finite open cover; in the case at hand the cover consists of a single interval of the open cover $\mathcal{I}$. This contradicts the assumption, so the claim is proved. ■

The subsets in the space $\mathbb{R}$ that enjoy the Heine–Pincherle–Borel property are contained in a finite number of open intervals, for which reason they are called **compact**. Compact sets are way more complicated than open or closed sets, and we want to explain why compactness is a good property for topological spaces, in particular metric spaces (see Appendix A).

Compactness is important because compact spaces share some of the relevant features of finite spaces, despite there being compact spaces that are not finite. Compact objects can be handled and controlled as if they were finite objects. Compact spaces might be called, by an abuse of language, "pseudo-finite", for they resemble finite spaces. They can be studied and managed by similar methods, allowing to extract useful and important properties.

To make an analogy, take a finite set A. Any map $f : A \mapsto \mathbb{R}$ has a maximum and a minimum, and any map $f : A \mapsto \mathbb{R}^n$ is bounded. Similarly, if A is compact, any continuous map $f : A \mapsto \mathbb{R}$ has a maximum and a minimum and any continuous map $f : A \mapsto \mathbb{R}^n$ is bounded, because $f(A)$ inherits its compactness from A because of the continuity of f. Another similarity: just as every sequence in a finite set A possesses a constant subsequence, every sequence in a compact set A possesses a convergent subsequence. Compact sets therefore behave in a similar manner to finite sets which, as we know, are easier to understand and manipulate than uncountable sets. Ultimately, compactness is the closest notion to finiteness. That is why it is often useful to turn a non-compact space into a compact one by adding something small, so not to alter the initial space too much. This procedure is called compactification and consists in building a new compact space $\bar{X}$ containing an open dense[8] subset homeomorphic to the starting space X.

Definition 2.1.3 (*Compact sets in* $\mathbb{R}$) A set $A \subseteq \mathbb{R}$ is compact if any open cover of A admits a finite subcover (Heine–Pincherle–Borel property). ◇

By this definition we immediately see that any closed bounded interval and any closed bounded subset of $\mathbb{R}$ are compact in $\mathbb{R}$.

[8] See Definition 2.2.2 and Remark 2.2.2.

For general topological spaces we have the following.

Proposition 2.1.4 (Compactness of closed-compact intersections) *Let F be closed and K compact in a space X. Then $(F \cap K)$ is compact in X.*

Proof Calling $\mathcal{G}$ an open cover of $F \cap K$, we should prove that the set has the Heine–Pincherle–Borel property.

The family $\mathcal{G} \cup (X \setminus F)$ is patently an open cover of K. As K is compact, there is a finite subcover of $\mathcal{G} \cup (X \setminus F)$ covering K, say $\mathcal{G}_K$. Since $(F \cap K) \subseteq K$ it follows $\mathcal{G}_K$ is a finite cover of $(F \cap K)$, whence the claim. ■

Proposition 2.1.5 (Closed and compact subsets of $\mathbb{R}$) *Every closed linear set is the countable union of compact subsets of $\mathbb{R}$ (any such is also known as σ-compact).*

Proof Let F be a closed subset of $\mathbb{R}$ and $\{K_n\}_{n\in\mathbb{N}}$ the increasing sequence of compact subsets of $\mathbb{R}$ where $K_n = \{x \in \mathbb{R} : |x| \leq n\}$ for every $n \in \mathbb{N}$. Clearly

$$\bigcup_{n=1}^{\infty} K_n = \mathbb{R} \Rightarrow F = F \cap \bigcup_{n=1}^{\infty} K_n = \bigcup_{n=1}^{\infty} (K_n \cap F).$$

By Proposition 2.1.4 $(F \cap K_n)$ is compact for every $n \in \mathbb{N}$. Therefore F is the countable union of compact sets in $\mathbb{R}$. ■

Definition 2.1.4 (*Boundary of a set*) We call boundary of a set $A \subseteq X$ the set of frontier points that actually belong in A:

$$B(A) = A \cap \partial A = A \cap \overline{(X \setminus A)}. \tag{2.1.1}$$

◇

Remark 2.1.2 Note that if $A \cap \partial A$ is the boundary of A, the boundary of the complement of A is the part of frontier that does not belong to A, and hence lies in A^c:

$$B(A^c) = B(X \cap A^c) = A^c \cap \partial A.$$

Consequently $\bar{A} = A \cup B(X \setminus A) = A \cup \partial A$. ♣

Definition 2.1.5 (*Boundary*) A set $A \subseteq X$ is said to be a boundary when every element of A is an element of the boundary of A:

$$A = B(A).$$ ◇

Proposition 2.1.6 (Properties of boundaries) *Given a set $A \subseteq X$:*

1. *A is a boundary if and only if*

$$\overline{(X \setminus A)} = X \quad \textit{and} \quad \overset{\circ}{A} = \emptyset.$$

These further imply

$$\overline{(X \setminus A)} = X = (X \setminus \overset{\circ}{A}).$$

2. *If A is a boundary, any subset of A is a boundary.*
3. *The boundary of $B(A)$ is a boundary:*

$$B(B(A)) = B(A).$$

Proof

1. To prove the first relation observe that if A is a boundary, then $A = B(A)$ and

$$A \subseteq \overline{(X \setminus A)}. \tag{2.1.2}$$

From this, since $X \setminus A \subseteq \overline{(X \setminus A)}$, using (2.1.2) we get

$$X = A \cup (X \setminus A) \subseteq \overline{(X \setminus A)}.$$

For the opposite inclusion note $\overline{(X \setminus A)} \subseteq X$. The latter, together with the previous inclusion produces

$$\overline{(X \setminus A)} = X. \tag{2.1.3}$$

As $A \subseteq X$, moreover, the above implies $A \subseteq \overline{(X \setminus A)} = X$, which is precisely the initial inclusion (2.1.2). Therefore (2.1.2) and (2.1.3) are equivalent, and so

$$A \subseteq \overline{(X \setminus A)} \Leftrightarrow \overline{(X \setminus A)} = X.$$

For the second relation recall that the interior of $A \subseteq X$ is $\overset{\circ}{A} = A \setminus \overline{(X \setminus A)}$. Using (2.1.3), if A is a boundary

$$\overset{\circ}{A} = A \cap X^c = \emptyset.$$

2. Take $B \subseteq A$, with $A, B \subseteq X$, and then $\overset{\circ}{B} \subseteq \overset{\circ}{A}$. If A is a boundary $\overset{\circ}{A} = \emptyset$ and $\overset{\circ}{B} = \emptyset$. Hence B, too, is a boundary.
3. From (2.1.1) we have

$$B(B(A)) = B(A) \cap \overline{(X \setminus B(A))}; \tag{2.1.4}$$

obviously $B(A) = A \cap \overline{(X \setminus A)}$ and

$$\begin{aligned}\overline{(X \setminus B(A))} &= \overline{(X \cap B(A)^c)} \\ &= \overline{(X \cap (A \cap \overline{(X \setminus A)})^c)} \\ &= \overline{(X \cap A^c) \cup (X \cap \overline{(X \setminus A)^c})} \\ &= \overline{(X \setminus A) \cup (\underbrace{X \setminus \overline{(X \setminus A)}}_{=\overset{\circ}{A} \text{ n.1. prop. 1.1.4}})} \\ &= X;\end{aligned}$$

therefore

$$B(B(A)) = A \cap \overline{(X \setminus A)} \cap X = A \cap \overline{(X \setminus A)} = B(A). \qquad \blacksquare$$

2.2 Nowhere Dense, Dense and Perfect Sets

Definition 2.2.1 (*Nowhere dense sets*) A set $A \subseteq X$ is nowhere dense if the closure $\bar{A}$ equals the boundary:

$$\bar{A} = B(A). \qquad \diamond$$

Remark 2.2.1 (*Boundaries*)

1. By Definition 2.2.1 and item *1.* in Proposition 2.1.6 we infer that A is nowhere dense iff

$$\overline{(X \setminus \bar{A})} = X. \tag{2.2.1}$$

Item *1.* in Proposition 2.1.6 also tells that A is nowhere dense iff the interior of the closure is empty, i.e.

$$\overset{\circ}{(\bar{A})} = \emptyset; \tag{2.2.2}$$

This statement may be taken as alternative definition to 2.2.1.

2. A subset in a nowhere dense set is nowhere dense. In other words the property of being nowhere dense is hereditary for sets. To show this, take $A \subseteq B$, so $\bar{A} \subseteq \bar{B}$. If B is nowhere dense then $\bar{B} \subseteq \overline{(X \setminus \bar{B})}$ by (2.1.2), so $\bar{A} \subseteq \bar{B} \subseteq \overline{(X \setminus \bar{B})}$, proving our claim. ♣

By virtue of (2.2.2) we provide below an analogous definition of a nowhere dense set. In it we also define dense subsets and sets that are dense in themselves.

Definition 2.2.2 (*Dense sets, nowhere dense sets, and sets dense in themselves*) Let A, B be subsets of X:

1. A is dense in B if every $x \in B$ is a limit point of A, or a point of A (or both), i.e. $B \subseteq A \cup A'$;
2. A is everywhere dense if it is dense in the entire space X;
3. A is nowhere dense in X if the interior of the closure $\bar{A}$ is empty;
4. A is dense in itself if $A \subseteq A'$, i.e. every point of A is a limit point of A. ◇

Remark 2.2.2 (*Everywhere dense sets*) If A is everywhere dense, item *1.* in Definition 2.2.2 forces $X \subseteq A \cup A'$. And since $A \cup A' = \bar{A}$, we have $X \subseteq \bar{A}$. But X is closed and contains $\bar{A}$, so $\bar{A} \subseteq X$. Therefore to say A is everywhere dense in X (**or more simply "dense" in** X) is the same as saying the closure of A coincides with X, i.e. $\bar{A} = X$. Sometimes we shall use the equivalent formulation: $A \subseteq X$ is dense in X iff $\bar{A} = X$. ♣

Theorem 2.2.1 (Characterisation of dense sets) *A set $A \subseteq X$ is dense in a space X if and only if every non-empty open subset of X meets A, i.e. it contains points of A.*

Proof Let A be a dense set in X and G an open, non-empty subset of X such that $A \cap G = \emptyset$. Then $X \setminus G$ is closed in X and contains A, which is absurd since the smallest closed set containing the dense set A is X.

Vice versa, suppose for every non-empty open set G in X we have $A \cap G \neq \emptyset$. Let F be a closed set in X, so $X \setminus F$ is non-empty and open in X, for which $(X \setminus F) \cap A \neq \emptyset$. This means A does not intersect F. As F is arbitrary, the only closed set that can meet A is X, and so $\bar{A} = X$. ■

Remark 2.2.3 (*Interior of dense sets and density of* $\mathbb{Q}$ *and* $\mathbb{R} \setminus \mathbb{Q}$ *in* $\mathbb{R}$) By Proposition 1.1.3 and Theorem 2.2.1 a dense set has empty interior. In fact let A be dense in X. Any neighbourhood U_x of a point $x \in X$, irrespective of how small, will always contain points of $X \setminus A$. Therefore there is no neighbourhood U_x entirely contained in A, whence the only neighbourhood in A is $\emptyset$. Furthermore $\bar{A} = X$, and since X is open by definition, the interior of $\bar{A}$ is the whole space X. From this it follows that a set A and its closure $\bar{A}$ will have, in general, different interiors.

The sets of rational and irrational numbers are both dense in $\mathbb{R}$. There is no neighbourhood $U_x \subseteq \mathbb{R}$, irrespective of size, that only contains points of $\mathbb{Q}$ or only of $\mathbb{R} \setminus \mathbb{Q}$. This shows that in $\mathbb{R}$ there are two dense disjoint subsets, complementary to each other. These subsets have different cardinalities. While $\mathbb{Q}$ is countable, $\mathbb{R} \setminus \mathbb{Q}$ is uncountable.

Cantor was the mathematician who proved $\mathbb{Q}$ has the same cardinality as the natural numbers. To show this fact directly we can establish without loss of generality a one-to-one correspondence between $\mathbb{N}$ and $\mathbb{Q}^+$ by organising all fractions in a table like the one in Fig. 2.1.

All fractions on the first row have numerator equal to 1, those on the second row have numerator 2 and so forth, while numbers in the first column have 1 as denominator, those in the second column denominator 2 etc. If we follow the path of the

arrows and eliminate equivalent fractions we eventually obtain a 1-1 correspondence between the natural numbers and the rationals. This means the two sets have the same cardinality. It must be noted, though, that the sequence does not respect the ordering of the rationals, because the density of $\mathbb{Q}$ in $\mathbb{R}$ prevents we attain a complete list of rationals ordered in the standard way.

Now we shall prove that there are more real numbers than rationals. This fact, too, was proved by Cantor by means of the diagonal argument he devised. Cantor compared the cardinality of the interval $[0, 1]$ with that of natural numbers and discovered the former are more numerous than the latter.

So suppose the contrary, that the set of real numbers $x \in [0, 1]$ is countable. Let us represent these elements in decimal notation $x_n = 0.d_{n1}d_{n2} \ldots d_{np} \ldots$, where $d_{np} \in \{0, 1, 2, \ldots, 9\}$. Thus we produce the following infinite listing of all real numbers $x \in [0, 1]$

$$
\begin{aligned}
&0.d_{11}d_{12} \ldots d_{1m} \ldots \\
&0.d_{21}d_{22} \ldots d_{2m} \ldots \\
&\ldots\ldots\ldots\ldots\ldots \\
&0.d_{n1}d_{n2} \ldots d_{nm} \ldots \\
&\ldots\ldots\ldots\ldots\ldots
\end{aligned}
$$

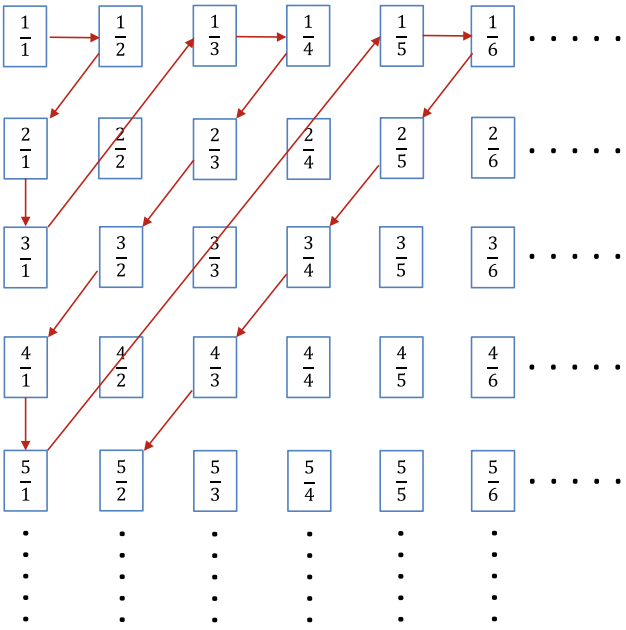


Fig. 2.1 Table of rational numbers

Now consider the real number $y = 0.b_1 b_2 \dots b_n \dots$, where $b_n = 1$ if $d_{nn} \geq 5$, while $b_n = 6$ if $d_{nn} < 5$. Clearly y is greater than 0 and less than 1, and also distinct from all the x_n. Therefore the countable list we built does not contain all reals $x \in [0, 1]$. The contradiction proves that the real numbers in $[0, 1]$ are more than countable, i.e. their cardinality is that of the continuum. The cardinality of $\mathbb{N}$ and any equinumerous set is denoted by $\aleph_0$, whereas the cardinality of any set in 1-1- correspondence with $\mathbb{R}$ is indicated by $\mathbf{c}$ and equals $2^{\aleph_0}$. Consequently $\mathbb{Q}$ is countable and dense in $\mathbb{R}$, while $\mathbb{R} \setminus \mathbb{Q}$, still dense in $\mathbb{R}$, has the cardinality of the continuum.

In conclusion, in a topological space X there might coexist several dense subsets with distinct cardinalities. ♣

Theorem 2.2.2 (Characterisation of nowhere dense sets) *A subset $A \subseteq X$ is nowhere dense in X if and only if, for every non-empty neighbourhood $U \subseteq X$, there exists a non-empty neighbourhood $V \subseteq U$ such that $V \cap A = \emptyset$.*

Proof Let A be nowhere dense, and take any non-empty open set $U \subseteq X$. Since the closure $\bar{A}$ has empty interior there always exists a non-empty open subset $V = U \setminus \bar{A}$ of U such that $V \cap A = \emptyset$.

Vice versa, suppose every non-empty open set $U \subseteq X$ contains a non-empty open set V such that $V \cap A = \emptyset$, so $A \subseteq X \setminus V$. Moreover, as $X \setminus V$ is closed and $\bar{A}$ is the smallest closed set containing A, we have $\bar{A} \subseteq (X \setminus V)$. From this

$$U \setminus \bar{A} \supseteq U \setminus (X \setminus V) = V \neq \emptyset.$$

Given that U is arbitrary, $\bar{A}$ contains no (non-empty) neighbourhoods, i.e. the interior of the closure of A is empty. ■

Since a set is nowhere dense if its closure has empty interior, we have another way of characterising this class of sets.

Theorem 2.2.3 (Second characterisation of nowhere dense sets) *A subset $A \subseteq X$ is nowhere dense if and only if $\bar{A}$ contains no neighbourhood.*

Proof Necessary implication: let A be nowhere dense in X. If $\bar{A}$ contained a neighbourhood then $\bar{A}$ would have internal points, which cannot be since $\overset{\circ}{(\bar{A})} = \emptyset$ by definition.

Sufficient implication: suppose $\bar{A}$ contains no neighbourhoods, and despite this A is not nowhere dense in X. If so, there exists a non-empty neighbourhood $U \subseteq X$ such that $V \cap A \neq \emptyset$ for every non-empty neighbourhood $V \subseteq U$, i.e. $U \cap A$ is dense in U. Take $x \in U$ and suppose $x \notin U \cap A$. By assumption $U \cap A$ is dense in U, so every neighbourhood of x intersects $U \cap A$. By definition, then, x is a limit point of $U \cap A$. But $x \in U$ is arbitrary, so $U \subseteq \overline{(U \cap A)}$. But $\overline{(U \cap A)} \subseteq \bar{A}$, and therefore $U \subseteq \bar{A}$. The contradiction yields the claim. ■

By Definition 2.2.1 and item *1.* in Proposition 2.1.6 (plus item 1. in Remark 2.2.1) we obtain immediately the following fact.

Proposition 2.2.1 (Relationship between dense and nowhere dense sets) *A set $A \subseteq X$ is nowhere dense in X if and only if the complement of its closure $(\bar{A})^c$ is dense in X:*

$$A \quad \text{is nowhere dense in } X \Leftrightarrow \overline{(X \setminus \bar{A})} = X. \qquad \blacksquare$$

Proposition 2.2.2 (Subsets of nowhere dense sets) *In a space X every subset of a nowhere dense set is also nowhere dense in X.*

Proof Let A be nowhere dense in X. If the subset B of A is empty, it is necessarily nowhere dense, for the interior of its closure is empty. If instead $B \neq \emptyset$, its closure is a subset of the closure of A. As the latter's interior is empty, *a fortiori* $\overset{\circ}{(\bar{B})} = \emptyset$.■

Remark 2.2.4 (*Dense sets and nowhere dense sets*) Consider two sets $A, B \subseteq X$. If A is dense in B then by definition every $x \in B$ is a limit point of A, or a point of A (or both). Proposition 1.2.1 guarantees that every neighbourhood U_x contains some point of A other than x.

By Theorem 2.2.2 we may then say:

"A set $A \subseteq X$ is nowhere dense in X if and only if for every non-empty neighbourhood $U \subseteq X$ there exists a non-empty neighbourhood $V \subseteq U$ such that $V \cap A = \emptyset$, i.e. A is not dense in U."

Since this holds for every neighbourhood $U \subseteq X$, we may also rephrase as follows:

"*$A \subseteq X$ is nowhere dense in X if and only if A is not dense in any neighbourhood $U \subseteq X$,*"

or equivalently

"*$A \subseteq X$ is not nowhere dense in X if and only if it is dense in some neighbourhood $U \subseteq X$.*" ♣

Example 2.2.1 (*Nowhere dense sets*) A trivial example of a nowhere dense set in $\mathbb{R}$ is

$$A = \left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots, \frac{1}{n}, \ldots\right\},$$

whose closure is precisely the boundary set $A \cup \{0\}$. As $A' = \{0\}$, A is not closed.

On the other hand, if we consider

$$A = \left\{0, 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots, \frac{1}{n}, \ldots\right\},$$

then $A' = \{0\} \subseteq A$ and A is closed.

Another nowhere dense set in $\mathbb{R}$ is the set of integers $\mathbb{Z}$, whose closure is $\mathbb{Z}$ itself. ★

A special type of closed sets are perfect sets.

Definition 2.2.3 (*Perfect sets*) A set $A \subseteq X$ is perfect if $A = A'$, i.e. when it is closed and dense in itself. $\diamond$

2.3 Types of Sets in Metric Spaces

Metric spaces form an important subclass of topological spaces. By this we mean that a metric space is a topological space but not conversely.

For an example, suppose we were to define open sets in $\mathbb{R}$ in the following fashion: we declare a subset $G \subseteq \mathbb{R}$ to be open in $\mathbb{R}$ iff for every $x \in G$ there exists a neighbourhood

$$U(y, r) = \left\{ y \in \mathbb{R} : |x - y| < r \right\}$$

with $r > 0$, such that $x \in U(y, r) \subseteq G$. Call $\mathcal{G}$ the collection of open sets just defined. Let us check that this defines a topology on $\mathbb{R}$:

- first of all, for every $x \in \emptyset$ there exists $U(\emptyset) \subseteq \emptyset$, so $\emptyset$ is in $\mathcal{G}$; moreover $\mathbb{R} = (-\infty, \infty)$ is the neighbourhood $U(y, \infty)$, whence $\mathbb{R} \in \mathcal{G}$;
- now we show that arbitrary unions belong to $\mathcal{G}$. Let $\{G_i\}_{i \in \mathcal{I}}$ be a family of open sets of $\mathcal{G}$, where $\mathcal{I}$ is a collection of indices. If $x \in \bigcup_i G_i$, then $x \in G_i$ for some i. So, there exists a neighbourhood $U(y, r)$ such that $x \in U(y, r) \subseteq G_i \subseteq \bigcup_i G_i$. Given x is arbitrary, $\bigcup_i G_i$ is in $\mathcal{G}$;
- finally, take $G_1, G_2 \in \mathcal{G}$. If $x \in (G_1 \cap G_2)$ then $x \in G_1$, so there is a neighbourhood $U(y, r_1)$ such that $x \in U(y, r_1) \subseteq G_1$, and similarly for G_2. Set $U(y, r) = U(y, r_1) \cap U(y, r_2)$, from which $x \in U(y, r) \subseteq G_1 \cap G_2$. But again $x \in G_1 \cap G_2$ is arbitrary, and the claim holds.

In a similar way one proves that the following class $\mathcal{G}$ of subsets of $\mathbb{R}^n$ is a topology on $\mathbb{R}^n$: we call $G \in \mathcal{G}$ open iff, for every $x \in G$, there is an open ball

$$B(y, r) = \left\{ x \in \mathbb{R}^n : \|x - y\| < r \right\}$$

where $r > 0$ and $\|x - y\| = \left[\sum_{i=1}^n (x_i - y_i)^2 \right]^{1/2}$, such that $x \in B(y, r) \subseteq G$.

These topologies are called standard or Euclidean topologies of $\mathbb{R}$ and $\mathbb{R}^n$, and they are actually special instances of metric topologies.

Definition 2.3.1 (*Metric spaces*) A metric space is a set X equipped with a real function $d : X \times X \mapsto [0, \infty)$, called distance or metric, satisfying:

1. $d(x, y) \geq 0$ for every $x, y \in X$;
2. $d(x, y) = 0 \Leftrightarrow x = y \in X$;
3. $d(x, y) = d(y, x)$ for every $x, y \in X$;

4. $d(x, z) \leq d(x, y) + d(y, z)$ for every $x, y, z \in X$.

In general one indicates a metric space as a pair (X, d). ◇

By this definition we see that the spaces $\mathbb{R}$ or $\mathbb{R}^n$, whose distance function induces the Euclidean topology, are metric spaces:

- $(\mathbb{R}, |\cdot|)$, i.e. the real numbers with Euclidean distance $d(x, y) = |x - y|$;
- $(\mathbb{R}^n, \|\cdot\|_n)$, i.e. real n-tuples $x = (x_1, \ldots, x_n)$ with distance

$$d(x, y) = \left[\sum_{i=1}^{n}(x_i - y_i)^2\right]^{1/2}.$$

Further instances of metric spaces are:

- $(\mathbb{Q}, |\cdot|)$, the rationals with distance $d(x, y) = |x - y|$;
- $(H, \langle\cdot, \cdot\rangle)$, i.e. real sequences $\{x\} = \{x_1, x_2, \ldots, x_n, \ldots\}$ such that $\sum_{i=1}^{\infty} x_i^2 < \infty$, equipped with metric

$$d(x, y) = \left[\sum_{i=1}^{\infty}(x_i - y_i)^2\right]^{1/2}.$$

  Here one should prove that if $\sum_{i=1}^{\infty} x_i^2 < \infty$ and $\sum_{i=1}^{\infty} y_i^2 < \infty$ then

$$\sum_{i=1}^{\infty}(x_i - y_i)^2 < \infty,$$

  besides proving the axioms of a metric space.

Definition 2.3.2 (*Balls in metric spaces*) Given a metric space (X, d), a point $x \in X$ and a number $r > 0$, we call open ball with centre x and radius r the set

$$B(x, r) = \big\{y \in X : d(x, y) < r\big\},$$

and closed ball with centre x and radius r the set

$$\bar{B}(x, r) = \big\{y \in X : d(x, y) \leq r\big\}.$$ ◇

In general we may define the topology on X induced by the metric d as the collection $\mathcal{T}_d$ of sets $G \subseteq X$ such that, for every $x \in G$, there exists an open ball $B(x, r) = \big\{y \in X : d(y, x) < r\big\}$ of radius $r > 0$ and centre x such that $B(x, r) \subseteq G$. The sets G are called open sets of the topology $\mathcal{T}_d$ induced by d.

Several theorems and results valid in $(\mathbb{R}, \|\cdot\|)$ generalise to arbitrary metric spaces, since metric spaces rely on properties of the real numbers that guarantee a similar behaviour in terms of distance, convergence of sequences, completeness, separability etc. Before to define the notions of limit and convergence in metric spaces, we define the concept of separable metric space.

Definition 2.3.3 (*Separable metric spaces*) A metric space (X, d) is called separable if X contains a countable and dense subset. ◇

For example $\mathbb{R}$ is separable for it contains the rationals, which are countable and dense. Hence the metric space $(\mathbb{R}, |\cdot|)$, where $|\cdot|$ is the Euclidean distance $d(x, y) = |x - y|$, is separable. Just as a real number can be approximated, as well as one wants, by rational numbers, a separable space possesses countable subsets using which one can get as close as one wishes—in the sense of limits—to any element.

Definition 2.3.4 (*Limits of sequences in metric spaces*) Given a metric space (X, d), a sequence $\{x_n\}_{n\geq 1} \subseteq X$ converges to the limit x_0 if for every $\epsilon > 0$ there exists an n' such that for every $n \geq n'$

$$d(x_0, x_n) < \epsilon.$$ ◇

Theorem 2.3.1 (Cauchy condition for convergent sequences in metric spaces) *Let (X, d) be a metric space and $\{x_n\}_{n\geq 1} \subseteq X$ a sequence converging to x_0. Then the sequence satisfies the Cauchy condition: for every $\epsilon > 0$ there exists $n' \in \mathbb{N}$ such that for every $n, m \geq n'$*

$$d(x_n, x_m) < \epsilon.$$

Such a sequence is called a Cauchy sequence.

Proof If $\{x_n\}_{n\geq 1}$ converges to x_0, for every $\epsilon > 0$ there exists n' such that $d(x_0, x_n) < \epsilon/2$ for every $n \geq n'$. Take n, m larger than n'. By the triangle inequality

$$d(x_n, x_m) \leq d(x_0, x_n) + d(x_0, x_m) < \epsilon,$$

and the statement follows. ■

Not every Cauchy sequence converges in a given space X. Consider the metric space (X, d), where $X = (0, 1)$ and $d(x, y) = |x - y|$. The sequence $\{x_n\}_{n>1} = \{1/n\}$ is Cauchy, but has no limit in X. It is a Cauchy sequence because it converges in $(\mathbb{R}, d)$, but not in X, since its limit 0 does not belong to X. The property of a space that every Cauchy sequence converges in the space is called completeness, and is detailed below.

Definition 2.3.5 (*Complete metric spaces*) A metric space (X, d) is complete when every Cauchy sequence converges (has a limit) in X. ◇

All the examples of metric spaces presented earlier are complete metric spaces except for the rational numbers. In $\mathbb{Q}$, in fact, sequences may converge to a limit in $\mathbb{R}$.

Complete separable metric spaces are also known as Polish spaces, because the first mathematicians to investigate their properties, at the beginning of the XX century, belonged to the Polish School. The main results on Polish spaces are linked to the names of Wacław Sierpiński, Kazimierz Kuratowski and Alfred Tarski.

We shall prove a number of metric properties regarding the sets described earlier. The first result is due to Ulisse Dini. Dini's theorem say that sets of first species are always nowhere dense and hence somehow neglectable. This happens when we can enclose points in finitely many neighbourhoods chosen so small that their measure may be neglected without consequence.

Theorem 2.3.2 (Negligibility of sets of first species) *Let (X, d) be a complete and separable metric space. A set $A \subseteq X$ of first species is nowhere dense and negligible.*

Proof Let A be of first species of order n in X, and $U \subseteq X$ an open set. If U contains no points of A and of each derived set, then the claim is proved.

Suppose then the points $x_1, \ldots, x_{m_n}$ of $A^{(n)}$ belong to U. Given any $\epsilon > 0$, let $0 < r_n < \frac{\epsilon}{[m_n \cdot (n+1)]}$ such that $\{B(x_{i_n}, r_n)\}_{i_n=1}^{m_n}$ be a cover of these points, with $\bar{B}(x_{i_n}, r_n) \cap \bar{B}(x_{j_n}, r_n) = \emptyset$ for $i_n \neq j_n$, $i_n, j_n = 1, \ldots, m_n$, and

$$\left[\left(\bar{A} \setminus A^{(n)}\right) \cap U\right] \cap \left(\bigcup_{i_n=1}^{m_n} \bar{B}(x_{i_n}, r_n)\right) = \emptyset,$$

where $B(x_{i_n}, r_n) = \left\{x \in X : d(x, x_{i_n}) < r_n\right\}$, for $i_n = 1, \ldots, m_n$. Then we remove from U the cover, where the total length of the radii is less than $\frac{\epsilon}{(n+1)}$. Having taken care of the points of $A^{(n)}$, suppose there are $x_1, \ldots, x_{m_{n-1}}$ points from $A^{(n-1)}$ in

$$U_{n-1} = U \setminus \bigcup_{i_n=1}^{m_n} \bar{B}(x_{i_n}, r_n).$$

Let $0 < r_{n-1} < \frac{\epsilon}{[m_{n-1} \cdot (n+1)]}$ such that $\{B(x_{i_{n-1}}, r_{n-1})\}_{i_{n-1}=1}^{m_{n-1}}$ be a cover of these points, with $\{\bar{B}(x_{i_{n-1}}, r_{n-1})\}$ pairwise disjoint,

$$\left(\bigcup_{i_{n-1}=1}^{m_{n-1}} \bar{B}(x_{i_{n-1}}, r_{n-1})\right) \subseteq U_{n-1}$$

and

$$\left[\left(\bar{A} \setminus A^{(n-1)}\right) \cap U_{n-1}\right] \cap \left(\bigcup_{i_{n-1}=1}^{m_{n-1}} \bar{B}(x_{i_{n-1}}, r_{n-1})\right) = \emptyset.$$

The total length of the radii of eliminated balls at the second stage is still less than $\frac{\epsilon}{(n+1)}$. If we continue the process and assume in all subsequent iterations U will contain a finite number of points from $A^{(n-2)}, \ldots, A^{(1)}, A^{(0)}$, at the very end the sum of the radii of all removed balls will be less than ϵ. After the elimination process the set

$$V = U \backslash \left(\bigcup_{k=0}^{n} \bigcup_{i_k=1}^{m_k} \bar{B}(x_{i_k}, r_k) \right) \subseteq U$$

does not contain points of A. Hence in $U \subseteq X$ there is at least one neighbourhood $V \subseteq U$ without points of A. This means A is nowhere dense. Moreover as ϵ is arbitrary, all the points of the infinite set A can be put in a finite number of balls, whose radii can be made as small as one wants. We say that A is negligible. ■

The first examples of non-negligible nowhere dense sets, which are also relevant from a historical perspective, were discovered around the end of the 1800s by the mathematicians Vito Volterra and Henry Smith.

Example 2.3.1 (*Vito Volterra's non-negligible nowhere dense set—1881*)[9] To construct Volterra's non-negligible nowhere dense set we proceed in steps.

Step one is to take the interval [0, 1] and construct an infinite sequence

$$0 < \cdots < a_{n+1} < a_n < \cdots < a_2 < a_1 < 1$$

accumulating at 0

$$\lim_{n \to \infty} a_n = 0.$$

We also require that $d(1, a_1) = \frac{1}{2^2}$, and the distances between successive points

$$d(a_1, a_2); \quad d(a_2, a_3); \quad \ldots; \quad d(a_n, a_{n+1}); \quad \ldots \to 0$$

are decreasing and less than $d(1, a_1) = \frac{1}{2^2}$.

In step two we take out the open interval $(a_1, 1)$, and apply to each remaining interval (a_{n+1}, a_n) the above procedure: we construct a sequence

$$a_{n+1} < \cdots < a^1_{n_{k+1}} < a^1_{n_k} < \cdots < a^1_{n_2} < a^1_{n_1} < a_n$$

converging to a_{n+1}

$$\lim_{k \to \infty} a^1_{n_k} = a_{n+1},$$

such that $d(a_n, a^1_{n_1}) = \frac{1}{2^{2 \cdot 2}} \cdot \frac{1}{2^{2 \cdot 1}}$ and the distances of consecutive points

[9] Volterra was interested in constructing a nowhere dense perfect set with positive exterior measure, which would have enabled him to define an everywhere differentiable function with bounded derivative that is not Riemann integrable on any closed bounded interval. In turn, this would have given a counterexample to Hankel's conjecture, whereby every function with dense discontinuity set is Riemann integrable.

The example is taken from: V. Volterra, Opere Matematiche—Memorie and Note, Roma Accademia Nazionale dei Lincei (1954), Volume I—Alcune osservazioni sulle funzioni punteggiate discontinue (1881), p.11 ff.

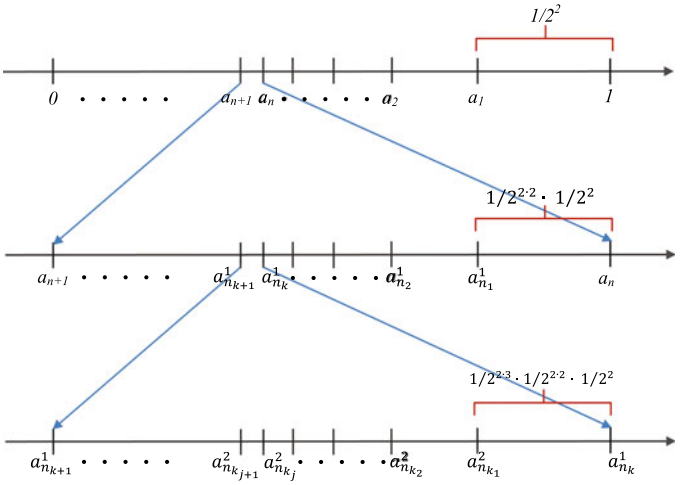


Fig. 2.2 Construction of Volterra's set—the first three iterations

$$d(a^1_{n_1}, a^1_{n_2}); d(a^1_{n_2}, a^1_{n_3}); \ldots; d(a^1_{n_k}, a^1_{n_{k+1}}); \ldots \to a_{n+1}$$

are decreasing and less than $d(a_n, a^1_{n_1}) = \frac{1}{2^{2\cdot 2}} \cdot \frac{1}{2^{2\cdot 1}}$. Observe that whilst in step one there was a unique limit point (zero), now there are infinitely many limit points, namely

$$\{0, \ldots, a_{n+1}, a_n, \ldots, a_2, \}.$$

In step three we take out the intervals $(a^1_{1_1}, a_1), (a^1_{2_1}, a_2), \ldots, (a^1_{n_1}, a_n), \ldots$ and again apply the procedure to each remaining interval $(a^1_{n_{k+1}}, a^1_{n_k})$: build

$$a^1_{n_{k+1}} < \cdots < a^2_{n_{k_{j+1}}} < a^2_{n_{k_j}} < \cdots < a^2_{n_{k_2}} < a^2_{n_{k_1}} < a^1_{n_k}$$

tending to $a^1_{n_{k+1}}$ and satisfying

$$d(a^1_{n_k}, a^2_{n_{k_1}}) = \frac{1}{2^{2\cdot 3}} \cdot \frac{1}{2^{2\cdot 2}} \cdot \frac{1}{2^{2\cdot 1}}$$

and that the distances of consecutive points are decreasing and less than the above quantity (Fig. 2.2).

If we suppose to go on indefinitely, eventually we will produce a linear set $\bar{V}$ containing all limit points and 1. $\bar{V}$ is nowhere dense in $[0, 1]$, since $A = [0, 1] \setminus \bar{V}$ is a disjoint union of open intervals with endpoints in $\bar{V}$. The open intervals of which A is made are precisely the intervals removed during the various steps of the construction of $\bar{V}$:

$$V_0 = (a_1, 1);\ V_{1_1} = (a^1_{1_1}, a_1),\ V_{2_1} = (a^1_{2_1}, a_2), \ldots, V_{n_1} = (a^1_{n_1}, a_n), \ldots$$

In general, then, for every non-empty neighbourhood $U \subseteq [0, 1]$, there exists a non-empty neighbourhood $V_k \subseteq U$ such that $V_k \cap \bar{V} = \emptyset$. The overall length of the intervals constituting A is

$$l(A) < \frac{1}{2^{2\cdot 1}} + \frac{1}{2^{2\cdot 2}} + \frac{1}{2^{2\cdot 3}} + \cdots + \frac{1}{2^{2n}} < \frac{1}{3},$$

so the length of the nowhere dense set $\bar{V}$ is bigger than $2/3$. This is Volterra's non-negligible, nowhere dense set. ★

Example 2.3.2 (*Henry J.S. Smith's non-negligible nowhere dense set—1875*)[10] Divide $[0, 1]$ in $m > 2$ equal parts, and delete the last interval, of length $1/m$, from each successive subdivision.

Now take the $(m - 1)$ remaining intervals and divide them in m^2 parts, deleting the last one produced in each subdivision. At this stage we are taking out $(m - 1)$ pairwise disjoint intervals, all of length $\left(1/m \cdot 1/m^2\right)$.

Divide again the $(m - 1)(m^2 - 1)$ intervals left in m^3 parts, and delete as before the last one produced in each division. Thus we take out $(m - 1)(m^2 - 1)$ pairwise disjoint intervals of length $\left(1/m \cdot 1/m^2 \cdot 1/m^3\right)$. And so on.

At step k the number of pairwise-disjoint excluded intervals is

$$N(k) = 1 + (m - 1) + (m - 1)(m^2 - 1) + \cdots + (m - 1)(m^2 - 1)\cdots(m^{k-1} - 1).$$

Call $I_1, \ldots, I_{N(k-1)}$ such intervals. The total length of these pairwise-disjoint intervals is computed by induction. When $n = 1$ the length equals

$$l(I_1) = \frac{1}{m} = 1 - \left(1 - \frac{1}{m}\right);$$

for $n = 1, 2$

$$\begin{aligned} l(I_1 \cup I_2) &= 1 - \left(1 - \frac{1}{m}\right) + (m - 1) \cdot \frac{1}{m} \cdot \frac{1}{m^2} \\ &= 1 - \left(1 - \frac{1}{m}\right) + \left(1 - \frac{1}{m}\right) \cdot \frac{1}{m^2} \\ &= 1 - \left(1 - \frac{1}{m}\right) \cdot \left(1 - \frac{1}{m^2}\right). \end{aligned} \tag{2.3.1}$$

To use induction we suppose that for $n = 1, 2, \ldots, N(k - 1)$

[10] The collected mathematical papers of Henry John Stephen Smith, edited by J. W. L. Glaisher, Volume II—On the integration of discontinuous functions (1894), p.95 ff.

$$l\left(\bigcup_{n=1}^{N(k-1)} I_n\right) = 1 - \left(1 - \frac{1}{m}\right) \cdot \left(1 - \frac{1}{m^2}\right) \cdots \left(1 - \frac{1}{m^{k-1}}\right).$$

Then for $n = 1, 2, \ldots, N(k)$

$$\begin{aligned} l\left(\bigcup_{n=1}^{N(k)} I_n\right) =& 1 - \left(1 - \frac{1}{m}\right) \cdot \left(1 - \frac{1}{m^2}\right) \cdots \left(1 - \frac{1}{m^{k-1}}\right) \\ &+ (m-1)(m^2-1)\cdots(m^{k-1}-1)\frac{1}{m} \cdot \frac{1}{m^2} \cdots \frac{1}{m^{k-1}} \cdot \frac{1}{m^k} \\ =& 1 - \left(1 - \frac{1}{m}\right) \cdot \left(1 - \frac{1}{m^2}\right) \cdots \left(1 - \frac{1}{m^{k-1}}\right) \\ &+ \left(1 - \frac{1}{m}\right) \cdot \left(1 - \frac{1}{m^2}\right) \cdot \left(1 - \frac{1}{m^{k-1}}\right) \cdot \frac{1}{m^k} \\ =& 1 - \left(1 - \frac{1}{m}\right) \cdot \left(1 - \frac{1}{m^2}\right) \cdots \left(1 - \frac{1}{m^{k-1}}\right) \cdot \left(1 - \frac{1}{m^k}\right). \end{aligned}$$

This sum, as $k \to \infty$, tends to the finite limit

$$1 - E\left(\frac{1}{m}\right),$$

where $E\left(\frac{1}{m}\right)$ is Euler's product $\prod_{k=1}^{\infty}\left(1 - \frac{1}{m^k}\right)$, for which

$$0 < \prod_{k=1}^{\infty}\left(1 - \frac{1}{m^k}\right) < 1.$$

Therefore the length of what remains $A = [0, 1] \setminus \left(\sum_{n=1}^{\infty} I_n\right)$ is

$$0 < \prod_{k=1}^{\infty}\left(1 - \frac{1}{m^k}\right) < 1,$$

which is surely positive. Hence A is non-negligible. Notice that A is also nowhere dense in $[0, 1]$.[11] In fact, for every interval I contained in $[0, 1]$ there always exists an interval $I_n \subseteq I$, deleted in the construction of A, such that $I_n \cap A = \emptyset$. ★

The examples shown together with Theorem 2.3.2 allow us to conclude that

"Sets of first species are negligible and nowhere dense, whereas nowhere dense sets may be non-negligible. Similarly, negligible sets are not necessarily of first

[11] At this point Smith takes a bounded map f that is discontinuous on the entire A, which is closed, nowhere dense and non-negligible in $[0, 1]$, and is made of the endpoints of all intervals plus their limits. He then proves that f is not Riemann integrable.

species. We shall see shortly that the Cantor set C is negligible,[12] *but of second species."*

The construction of Volterra's and Smith's sets abides by the structure prescribed in the following remark, which characterises the general form of compact nowhere dense subsets of $\mathbb{R}$.

Remark 2.3.1 (*Structure of compact nowhere dense subsets of* $\mathbb{R}$)

1. A set $A \subseteq [a, b]$ is nowhere dense and closed iff it consists of the endpoints of a union of pairwise-disjoint open intervals, which is dense in $[a, b]$, together with their accumulation points.
   ($\Rightarrow$) Let $A \subseteq [a, b]$ be closed and nowhere dense. Take any interval (a_1, b_1) of $[a, b]$ and a point $x_1 \in (a_1, b_1)$ not in A. By definition of nowhere dense set, there exists an interval (α_1, β_1) containing x_1 such that $(\alpha_1, \beta_1) \cap A = \emptyset$.
   Suppose (α_1, β_1) is the largest interval containing x_1 and disjoint from A. Clearly α_1 and β_1 belong to A.
   If we continue to construct intervals like the above on $[a, b]$, we end up with an open set $G = \bigcup_{i=1}^{\infty}(\alpha_i, \beta_i)$ representing the real intervals not in A, by Theorem 2.2.3. Hence A will coincide with $\left([a, b] \setminus \bigcup_{i=1}^{\infty}(\alpha_i, \beta_i)\right)$. Its points may be classified as follows:

   (a) common endpoints of adjacent intervals of G, which are therefore isolated in A, like (α_i, β_i) and (β_i, β_j);
   (b) points of A between two given intervals of G, which must appear as endpoints of intervals of G lying between the two given intervals. More specifically, these can be:

   - an endpoint of an interval of G lying between the two given intervals, which must be a one-sided limit of the endpoints of the intervals of G;
   - a two-sided limit of the endpoints of the intervals lying between the two given intervals. This limit point is obviously the endpoint of some interval of G.

   ($\Leftarrow$) Let $\{q_i\}_{i=1}^{\infty}$ denote the sequence of all rational numbers of $[a, b]$. Given $r_1 \in [a, b] \setminus \mathbb{Q}$, consider the interval $I_1 = (q_1 - r_1; q_1 + r_1)$ and let q_{i_2} be the first rational appearing in the remaining sequence $\{q_i\}_{i=2}^{\infty} \setminus I_1$. Clearly $q_{i_2} \neq (q_1 \pm r_1)$, otherwise r_1 would not be irrational. Pick $r_2 \in [a, b] \setminus \mathbb{Q}$, $r_2 \neq r_1$, and define $I_2 = (q_{i_2} - r_2; q_{i_2} + r_2)$ such that $I_1 \cap I_2 = \emptyset$, with the possibility that they have a common endpoint. Let q_{i_3} be the first number in the new remaining sequence $\{q_i\}_{i=3}^{\infty} \setminus (I_1 \cup I_2)$. Here, too, $q_{i_3} \neq (q_{i_2} \pm r_2)$ because r_2 is irrational. Now choose $r_3 \in [a, b] \setminus \mathbb{Q}$, $r_i \neq r_j$ for $i, j = 1, 2, 3$, and call $I_3 = (q_{i_3} - r_3; p_{i_3} + r_3)$ so that $I_i \cap I_j = \emptyset, i, j = 1, 2, 3$, still allowing for the intervals to share an endpoint.

[12] At the time when these concepts were taking shape the idea of a negligible set—which is negligible with respect to some measure, being of zero measure—had not been formalised. That is why the notion remains somehow vague.

Going on like this will generate a sequence $\{I_i\}_{i=1}^{\infty}$ of pairwise-disjoint open neighbourhoods, each centred at a rational number, possibly with common irrational endpoints.
The set $[a, b] \setminus \left(\bigcup_{i=1}^{\infty} I_i\right)$ consists of the irrational endpoints of all the pairwise-disjoint open intervals, among which are the limit points of the endpoints. Such a set is definitely closed, being complementary to an open set, and non-empty since at least the endpoints of the deleted intervals are there.
To show A is nowhere dense, take any non-empty open interval $(\alpha, \beta) \subseteq [a, b]$. Clearly $(\alpha, \beta) \cap \bigcup_{i=1}^{\infty} I_i \neq \emptyset$. As $(\alpha, \beta) \subseteq [a, b]$ is arbitrary, $\bigcup_{i=1}^{\infty} I_i$ is dense in $[a, b]$, and Proposition 2.2.1 allows to conclude.

2. A closed set $A \subseteq \mathbb{R}$ is dense in an interval $[a, b]$ if it coincides with $[a, b]$ itself. A dense set, in fact, is closed when it coincides with the smallest closed set containing it, in this case $[a, b]$.
   Analogously, a closed set $A \subseteq \mathbb{R}$ is dense in a subset of an interval $[a, b]$ when it coincides with an interval $[\alpha, \beta] \subseteq [a, b]$. In fact, take $x \in A$ and call $\mathcal{U}_x$ the family of neighbourhoods such that $(U_x \cap A) \neq \emptyset$. Let

$$\bigcup_{(U_x \cap A) \in \mathcal{U}_x} (U_x \cap A) = (\alpha, \beta) \subseteq [a, b]$$

   be the largest open interval containing x such that $(\alpha, \beta) \cap A \neq \emptyset$. Given $y \in (\alpha, \beta)$, if $y \in A$ there are no problems. If $y \notin A$, instead, since every neighbourhood of y contains points of A, then y is a limit point of A, and as A is closed, $y \in A$. Hence $(\alpha, \beta) \subseteq A$. But α and β are limit points for $(\alpha, \beta) \subseteq A$. As A is closed, in the end $[\alpha, \beta] \subseteq A$.
3. A closed set $A \subseteq \mathbb{R}$ is nowhere dense in an interval $[a, b]$ if for every interval $(\alpha, \beta) \subseteq [a, b]$ we have $(\alpha, \beta) \setminus A \neq \emptyset$.
   To see this, note $[a, b] \setminus A$ is open because by assumption A is closed and nowhere dense in $[a, b]$. Let (α, β) be any subinterval of $[a, b]$. If $(\alpha, \beta) \setminus A = \emptyset$, the open interval is completely contained in A, against the fact that A is nowhere dense in $[a, b]$. Hence $(\alpha, \beta) \setminus A \neq \emptyset$. But $(\alpha, \beta) \subseteq [a, b]$ is arbitrary, whence the claim. ♣

In view of Definition 2.2.3 and item 1. of Remark 2.3.1, a compact and nowhere dense set whose isolated points have been removed is perfect, in particular nowhere dense and perfect. That is because the set equals its own derived set.

Remark 2.3.2 (*Structure of bounded perfect subsets of* $\mathbb{R}$) A non-empty bounded subset P of $\mathbb{R}$ is absolutely perfect if it is a closed interval. It is nowhere dense and perfect if it can be obtained from a closed interval by deleting a dense union of open intervals no two of which have common endpoints (including the initial interval) and such that between any two interval no internal points of P exist. ♣

Put differently, a bounded, nowhere dense and perfect subset P of $\mathbb{R}$ is made of the endpoints of a dense union of open intervals no two of which have common endpoints, together with the limits of said endpoints.

The Volterra set is a nowhere dense, closed subset of a given interval. As we have explained, this set consists of the endpoints of a dense (in [0, 1]) union of disjoint open intervals deleted during the construction, and their accumulation points. An example of a nowhere dense, perfect subset of an interval is the Cantor set.

Example 2.3.3 (*A nowhere dense, perfect set of second species: the Cantor set*) In order to build Cantor's ternary set let us take the closed interval [0, 1] and split it in three equal parts $\left[0, \frac{1}{3}\right]$, $\left(\frac{1}{3}, \frac{2}{3}\right)$, $\left[\frac{2}{3}, 1\right]$. Delete the middle one $\left(\frac{1}{3}, \frac{2}{3}\right)$ and divide the others in three equal subintervals. From the left segment we obtain

$$\left[0, \frac{1}{9}\right], \left(\frac{1}{9}, \frac{2}{9}\right), \left[\frac{2}{9}, \frac{1}{3}\right],$$

while the right one splits into

$$\left[\frac{2}{3}, \frac{7}{9}\right], \left(\frac{7}{9}, \frac{8}{9}\right), \left[\frac{8}{9}, 1\right].$$

Now delete the new middle intervals $\left(\frac{1}{9}, \frac{2}{9}\right)$ and $\left(\frac{7}{9}, \frac{8}{9}\right)$, and then iterate indefinitely the process.

At the first iteration we produce an open interval (deleted) and two closed intervals of length $\frac{1}{3}$; at the second iteration we have two more open intervals (deleted) and four closed intervals, each of length $\frac{1}{3^2}$. At step n we will have deleted

$$1 + 2 + 2^2 + \cdots + 2^n = \frac{1 - 2^n}{1 - 2} = 2^n - 1$$

open intervals, where

$$\begin{aligned}
&\text{step 1,} \quad G_1 = G_{1,1} \\
&\text{step 2,} \quad G_2 = G_{1,1} \cup G_{2,1} \cup G_{2,2} \\
&\text{step 3,} \quad G_3 = G_{1,1} \cup G_{2,1} \cup G_{2,2} \cup G_{3,1} \cup G_{3,2} \cup G_{3,3} \cup G_{3,4} \\
&\cdots\cdots\cdots\cdots\cdots\cdots \\
&\text{step n,} \quad G_n = G_{1,1} \cup G_{2,1} \cup G_{2,2} \cup \cdots \cup G_{i,1} \cup \cdots \cup G_{i,2^{i-1}} \cup \cdots \\
&\qquad\qquad \cup G_{n,1} \cup \cdots \cup G_{n,2^{n-1}}.
\end{aligned}$$

There remain

$$\underbrace{2 \cdot 2 \cdots 2}_{n \text{ times}} = 2^n$$

disjoint closed intervals. At the very end we have deleted from [0, 1] the set

$$G = \left\{ \left(\frac{1}{3}, \frac{2}{3}\right); \left(\frac{1}{9}, \frac{2}{9}\right); \left(\frac{7}{9}, \frac{8}{9}\right); \cdots \right\}$$

made of this countable family of intervals. At each step of the deletion we end up with sets:

$$
\begin{aligned}
C_1 &= \left[0, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right] = C_{1,1} \cup C_{1,2^1} \\
C_2 &= \left[0, \frac{1}{9}\right] \cup \left[\frac{2}{9}, \frac{3}{9}\right] \cup \left[\frac{6}{9}, \frac{7}{9}\right] \cup \left[\frac{8}{9}, 1\right] = C_{2,1} \cup C_{2,2} \cup C_{2,3} \cup C_{2,2^2} \\
C_3 &= \left[0, \frac{1}{27}\right] \cup \left[\frac{2}{27}, \frac{3}{27}\right] \cup \left[\frac{6}{27}, \frac{7}{27}\right] \cup \left[\frac{8}{27}, \frac{9}{27}\right] \cup \left[\frac{18}{27}, \frac{19}{27}\right] \cup \left[\frac{20}{27}, \frac{21}{27}\right] \cup \left[\frac{24}{27}, \frac{25}{27}\right] \\
&\quad \cup \left[\frac{26}{27}, 1\right] = C_{3,1} \cup C_{3,2} \cup C_{3,3} \cup C_{3,4} \cup C_{3,5} \cup C_{3,6} \cup C_{3,7} \cup C_{3,2^3} \\
&\cdots\cdots\cdots\cdots \\
C_n &= C_{n,1} \cup C_{n,2} \cup \cdots \cup C_{n,k} \cup \cdots \cup C_{n,2^n} \\
&\cdots\cdots\cdots\cdots
\end{aligned}
$$

Each C_n is the union of 2^n disjoint closed intervals of length $1/3^n$, hence C_n is closed. The length of the intervals remaining after step n is $m(C_n) = 2^n/3^n$. The sequence $\{C_n\}_{n\geq 1}$ is decreasing, so the deleting process leaves us with the set

$$C = \bigcap_{n=1}^{\infty} C_n = \lim_{n\to\infty} C_n.$$

Both G and C are called Cantor sets.

G is clearly **dense in the interval** $[0, 1]$: given $x \in C$ and $\delta > 0$ arbitrarily small and such that $(x - \delta, x + \delta) \subseteq [0, 1]$, for n large enough there exists at least one deleted interval $G_{i,k}$, $i = 1, \ldots, n$ and $k = 1, 2^1, \ldots, 2^{i-1}$, of length $1/3^n < 2\delta$ and completely contained in $(x - \delta, x + \delta)$.

To identify rapidly the elements of G or C it is useful to write the points of $[0, 1]$ in base three. The base-three representation of a decimal number $x \in [0, 1]$ is found by computing fractions $c_n/3^n$, where $c_n \in \{0, 1, 2\}$ so that

$$x = \sum_{n=1}^{\infty} \frac{c_n}{3^n},$$

for $n = 1, 2, \ldots$ Then x reads, in base three,

$$x_3 = 0.c_1 c_2 \ldots c_n \ldots.$$

For example, regarding the endpoints of the first deleted interval $\left(\frac{1}{3}, \frac{2}{3}\right)$ we have

$$x = \frac{1}{3} = 0.33\bar{3} = \begin{cases} \dfrac{0}{3^0} + \dfrac{1}{3^1} + \dfrac{0}{3^2} + \dfrac{0}{3^3} + \cdots \Leftrightarrow x_3 = 0.1 \\ \dfrac{0}{3^0} + \dfrac{0}{3^1} + \dfrac{2}{3^2} + \dfrac{2}{3^3} + \cdots \Leftrightarrow x_3 = 0.02\bar{2} \end{cases}$$

$$x = \frac{2}{3} = 0.66\bar{6} = \begin{cases} \dfrac{0}{3^0} + \dfrac{1}{3^1} + \dfrac{2}{3^2} + \dfrac{2}{3^3} + \cdots \Leftrightarrow x_3 = 0.122\bar{2} \\ \dfrac{0}{3^0} + \dfrac{2}{3^1} + \dfrac{0}{3^2} + \dfrac{0}{3^3} + \cdots \Leftrightarrow x_3 = 0.2; \end{cases}$$

whilst 4/9 reads, in base three,

$$x = \frac{4}{9} = 0.44\bar{4} = \begin{cases} \dfrac{0}{3^0} + \dfrac{1}{3^1} + \dfrac{1}{3^2} + \dfrac{0}{3^3} + \cdots \Leftrightarrow x_3 = 0.11 \\ \dfrac{0}{3^0} + \dfrac{1}{3^1} + \dfrac{0}{3^2} + \dfrac{2}{3^3} + \cdots \Leftrightarrow x_3 = 0.10\bar{2} \end{cases}$$

From the above expressions we note that rational numbers $x \in [0, 1]$ with finite ternary expansion can be written in a second way. We cannot but observe that only the rational numbers in which the digits 0 and 2 appear (and not 1) belong to the set C. There are rational numbers with finite ternary expansion that admit a second expansion, but both display the digit 1, like 4/9. But this number belongs in the first removed interval, so it does not lie in C.

This leads us to conjecture that the numbers in Cantor's set C have one ternary expansion without the digit 1. The idea is supported in the fact that at each step $n = 1, 2, \ldots$, the coefficient c_n equals 0, 1, or 2 whenever $x \in [0, 1]$, after the trisection at step $n - 1$, belongs in the left, middle, or right subinterval respectively. Therefore to list the elements of C it would suffice to pick all reals in $[0, 1]$ with only the digits 0 and 2 in the ternary expansion. This conjecture is actually true, as we set out to show by induction on $\mathbb{N}$.

Suppose $x \in [0, 1]$ has ternary expansion $0.c_1c_2c_3 \ldots c_n \ldots$. Clearly $x \in C_0 = [0, 1]$. Moreover if $c_1 = 0$, then $x \leq 0.022\bar{2} = 1/3$; if $c_1 = 2$, then $x \geq 0.22\bar{2} = 2/3$. Therefore $x \in [0, 1/3] \cup [2/3, 1] = C_1$.

By induction suppose $x \in C_n$. Then x belongs to one of 2^n closed intervals forming C_n, call it $[a, b]$. By construction every left endpoint in C_n is either a left endpoint in C_{n-1}, or it equals the sum of $2/3^n$ with a left endpoint in C_{n-1}. Trisect $[a, b]$ and delete the middle interval, so to obtain closed, disjoint intervals $[a, a + 1/3^{n+1}]$ and $[b - 1/3^{n+1}, b]$. If $c_{n+1} = 0$

$$x \leq a + 1/3^{n+1} = 0.\underbrace{00\ldots}_{n+1 \text{ zeroes}}22\bar{2},$$

so $x \in [a, a + 1/3^{n+1}]$. If, instead, $c_{n+1} = 2$

$$x \geq a + 2/3^{n+1} = 0.\underbrace{00\ldots}_{n \text{ zeroes}}2,$$

and then $x \in [b - 1/3^{n+1}, b]$. In either case $x \in C_{n+1}$, concluding the proof by induction. Therefore in the ternary expansion of any $x \in C$ the coefficient c_n equals, for every $n \in \mathbb{N}$, either 0 or 2.

With this settled, we note that the Cantor set has a number of other interesting features.

1. ***C* is non-empty**: this is trivial since C contains the endpoints of the intervals of every C_n, but not only. There are other points, like $1/4$ for instance.
   The latter is not an endpoint of an interval in some C_n, since the denominator is not a power of 3. The base-three expansion of $1/4$ is

$$\begin{aligned} x = \frac{1}{4} &= 2 \cdot \frac{1}{9-1} = 2 \cdot \frac{\frac{1}{9}}{\left[1 - \frac{1}{9}\right]} = 2 \sum_{n=1}^{\infty} \left(\frac{1}{9}\right)^n \\ &= \frac{2}{9} + \frac{2}{9^2} + \frac{2}{9^3} + \cdots \\ &= \frac{0}{3^0} + \frac{0}{3^1} + \frac{2}{3^2} + \frac{0}{3^3} + \frac{2}{3^4} + \frac{0}{3^5} + \frac{2}{3^6} + \cdots \\ \Leftrightarrow x_3 &= 0.020202\overline{02}; \end{aligned}$$

   implying $1/4 \in C$, even though $1/4$ is the endpoint of no interval in some C_n. We can also find other values like $1/4$ by exploiting the symmetries of the Cantor set. For every $x \in C$ we have $\frac{1}{3}x \in C$ and $(1 - x) \in C$. Using these, and that $1/4$ is not the endpoint of an interval in some C_n, we deduce that neither $\frac{3}{4}; \frac{1}{12}; \frac{11}{12}; \ldots$ are endpoints.
2. ***C* is compact**: it is closed, as countable intersection of closed sets, and bounded since contained in $[0, 1]$ (Heine-Borel theorem).
3. ***C* has the cardinality of the continuum**. For this we must prove C and $[0, 1]$ are equinumerous. Observe that any $x \in [0, 1]$ has a binary expansion of the sort

$$x = \sum_{n=1}^{\infty} \frac{b_n}{2^n},$$

   where $b_n \in \{0, 1\}$, for every n. To make the representation of periodic (rational) numbers unique, the convention is to take the expansion with tail made only of 1s or 0s.[13] We have also seen that every number y in C has a ternary expression

[13] The binary representation of real numbers in $[0, 1]$ gives us the chance for a digression on probability. Imagine a game of where a coin is tossed indefinitely, and declare tails to be 0 and heads 1. The outcome is then coded by a binary expansion corresponding to a real number in $(0, 1)$ (we are only taking the digits after the binary point; and to make the binary expansion of periodic (rational) numbers unique we need to choose among tails containing only 1s or only 0s. If we go for the latter, for example, $0.0101111\cdots$ (only 1s after the fourth binary digit) must be written as $0.011000\cdots$ (all 0s after the fourth place, 1 in third place). This countable choice of two objects covers all possibilities, which are in 1-1 correspondence with $(0, 1)$. Hence the latter has the cardinality of the continuum.

$$y = \sum_{n=1}^{\infty} \frac{c_n}{3^n},$$

where $c_n \in \{0, 2\}$. Hence any $x \in [0, 1]$ may be put in binary form

$$x = 0.b_1 b_2 b_3 \ldots b_n \ldots,$$

where b_n, for $n = 1, 2, 3, \ldots$, is 0 or 1; whereas every $y \in C$ may be put in ternary form

$$y = 0.c_1 c_2 c_3 \ldots c_n \ldots,$$

where, for every $n = 1, 2, 3, \ldots$, $c_n = 2b_n$.
Given a point $x \in [0, 1]$ with binary form $0.b_1 b_2 b_3 \ldots b_n \ldots$ we can construct a number $y \in C$, with ternary expansion $0.2b_1 2b_2 2b_3 \ldots 2b_n \ldots$, and vice versa. In this way we have defined a bijection from $[0, 1]$ to C

$$f : [0, 1] \mapsto C.$$

In order to show f is indeed bijective—assuming every $x \in [0, 1]$ has a unique binary form—let us take distinct binary points in $[0, 1]$

$$x_1 = 0.b_1^1 b_2^1 b_3^1 \ldots b_n^1 \ldots,$$

$$x_2 = 0.b_1^2 b_2^2 b_3^2 \ldots b_n^2 \ldots.$$

As $x_1 \neq x_2$, in the above expansions there exists a subscript i_0 such that $b_{i_0}^1 \neq b_{i_0}^2$, implying $2b_{i_0}^1 = c_{i_0}^1 \neq c_{i_0}^2 = 2b_{i_0}^2$. Hence

$$\begin{aligned} y_1 &= 0.c_1^1 c_2^1 c_3^1 \ldots c_n^1 \ldots, \\ &= 0.2b_1^1 2b_2^1 2b_3^1 \ldots 2b_n^1 \ldots \end{aligned}$$

$$\begin{aligned} y_2 &= 0.c_1^2 c_2^2 c_3^2 \ldots c_n^2 \ldots, \\ &= 0.2b_1^2 2b_2^2 2b_3^2 \ldots 2b_n^2 \ldots \end{aligned}$$

are the respective images of x_1 and x_2 under f, and necessarily they differ.
In other words the mapping sending x to y is 1-1 and therefore $f : [0, 1] \mapsto C$ is bijective. Then C and the interval $[0, 1]$ have the same cardinality, that of the continuum. But this forces C to contain irrational numbers as well. For exam-

ple $0.20200200020000\ldots$ lives in C: note how the number of zeroes between consecutive 2s grows by one as we move down along the expansion.

4. ***C* is nowhere dense** in $[0, 1]$. For this, take $x_1 < x_2$ in C with ternary expansions $0.c_1^1 c_2^1 c_3^1 \ldots$ and $0.c_1^2 c_2^2 c_3^2 \ldots$.
   Suppose the first digit by which the numbers differ appears in nth place. Then x_1 must have $c_n^1 = 0$ and x_2 must have $c_n^2 = 2$. The number with 1 in that place does not belong to C, and lies between x_1 and x_2. Therefore between any two points in C there is always a number not in C. This is still not enough, for we need to prove that between any two points of C, as close as we want, there exists an entire interval not in C.
   Observe that, irrespective of the distance between x_1 and x_2, both in C, there exists an n such that $1/3^n$ is less than this distance. Recalling that C is built by deleting intervals of length $1/3^n$, we see that one such interval will surely fit between x_1, x_2, even if they are arbitrarily close. Therefore the closure of C, i.e. C itself, has empty interior.
5. ***C* does not have isolated points**,[14] even if any two points in C, as close as we want, will be separated by an interval outside C.
   Take $x \in C$ and an interval $(a, b) \subseteq [0, 1]$ containing x. There exists an interval $C_{n',k}$ that is part of a set $C_{n'}$ such that $x \in C_{n',k} \subseteq (a, b)$. For every $n \geq n'$ we have $C_{n,k} \subseteq (a, b)$, so (a, b) contains the infinite set

$$\bigcap_{n \geq n'} C_n \cap C.$$

   As $x \in C$ is arbitrary, every open interval (a, b) around any element $x \in C$ contains infinitely many other points of C, thus freeing C of isolated points.
6. By *4.* and *5.*, therefore, ***C* is nowhere dense** and **perfect**: indeed, it is closed and dense in itself.
   To prove the last statement take any $x \in C$ and $\delta > 0$. For n large enough there exists a C_n containing a closed interval $C_{n,k}$, $k = 1, \ldots, 2^n$, of length $1/3^n < \delta$, such that $x \in C_{n,k} \subset C_n$. Let a_1 be the endpoint of the interval $C_{n,k}$ distinct from x. Then a_1 belongs to C and

$$|x - a_1| \leq \frac{1}{3^n} < \delta.$$

   As this holds for every $n \in \mathbb{N}$, as n increases the remaining intervals' endpoints, which belong to C, tend to $x \in C$. Hence every $x \in C$ is a limit point of C, and $C \subseteq C'$. ★

Despite Cantor's ternary set is a linear set of points with the cardinality of the continuum, not only it contains no interval, but it is not even dense in any segment of $[0, 1]$, regardless of size. Sets of this sort, albeit of the cardinality of the continuum, do not possess a continuous structure to speak of.

[14] Indeed, it is perfect.

Here is another perfect example not dissimilar to the Cantor set.

Example 2.3.4 (*A perfect set*) Write numbers of [0, 1] in binary form, so that every $x \in [0, 1]$,

$$x = \sum_{n=1}^{\infty} \frac{b_n}{2^n},$$

reads $x = 0.b_1b_2b_3\ldots$ in binary form, with $b_n \in \{0, 1\}$ for every n.

Every rational number in the interval of the form $m/2^n$, $n = 1, \ldots$, has a double binary expansion. These rationals are called dyadic since their denominator is a power of 2. Consider for instance $\frac{1}{4}$, $\frac{1}{2}$ and $\frac{3}{4}$, which read:

$$x = \frac{1}{4} = 0.25 = \begin{cases} \frac{0}{2^0} + \frac{0}{2^1} + \frac{0}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \cdots \Leftrightarrow x_2 = 0.0011\bar{1}; \\ \frac{0}{2^0} + \frac{0}{2^1} + \frac{1}{2^2} + \frac{0}{2^3} + \frac{0}{2^4} + \cdots \Leftrightarrow x_2 = 0.01. \end{cases}$$

$$x = \frac{1}{2} = 0.5 \ = \begin{cases} \frac{0}{2^0} + \frac{0}{2^1} + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \cdots \Leftrightarrow x_2 = 0.011\bar{1}; \\ \frac{0}{2^0} + \frac{1}{2^1} + \frac{0}{2^2} + \frac{0}{2^3} + \frac{0}{2^4} + \cdots \Leftrightarrow x_2 = 0.1; \end{cases}$$

$$x = \frac{3}{4} = 0.75 = \begin{cases} \frac{0}{2^0} + \frac{1}{2^1} + \frac{0}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \cdots \Leftrightarrow x_2 = 0.1011\bar{1}; \\ \frac{0}{2^0} + \frac{1}{2^1} + \frac{1}{2^2} + \frac{0}{2^3} + \frac{0}{2^4} + \cdots \Leftrightarrow x_2 = 0.11. \end{cases}$$

All other real numbers in [0, 1] have a unique binary representation.

By viewing binary expansions as decimal strings we may imagine that every rational number in [0, 1] with double representation defines an open interval, complementary to the points with unique representation. The set of points with unique representation, as complement of the countable union of the aforementioned open intervals, is closed, more than countable, non-empty, perfect and nowhere dense in [0, 1], just like the Cantor set. ★

Lemma 2.3.1 (Baire—countable intersections of open dense sets) *Let (X, d) be a complete metric space and $\{G_n\}_{n\geq 1}$ a sequence of open dense subsets. Then the intersection*

$$\bigcap_{n=1}^{\infty} G_n = G$$

is dense in X.

Proof Take a sequence $\{G_n\}_{n\geq 1}$ of open dense subsets in X. Given $x_0 \in X$ and a neighbourhood U of radius $r_0 > 0$, to prove the statement we should show

$$U \cap G \neq \emptyset.$$

As G_1 is dense in X, $U \cap G_1$ contains a point x_1. But $U \cap G_1$ is open, so it contains a neighbourhood centred at x_1 of radius $r_1 < r_0/2$ such that

$$\bar{B}_1 = \left\{x \in X : d(x, x_1) \leq r_1\right\} \subseteq (G_1 \cap U).$$

Now take a set G_2 dense in X, so $G_2 \cap B_1$ contains a point x_2. As $G_2 \cap B_1$ is open, it contains a neighbourhood centred at x_2 with radius $r_2 < r_1/2$ such that

$$\bar{B}_2 = \left\{x \in X : d(x, x_2) \leq r_2\right\} \subseteq (G_2 \cap B_1),$$

where B_1 denote the open ball centered at x_1 with radius r_1.

At step n, the set G_n is dense in X, so $G_n \cap B_{n-1}$ contains a point x_n, and $G_n \cap B_{n-1}$ being open means it contains a neighbourhood centred at x_n of radius $r_n < r_{n-1}/2$ such that

$$\bar{B}_n = \left\{x \in X : d(x, x_n) \leq r_n\right\} \subseteq (G_n \cap B_{n-1}),$$

where B_{n-1} denote the open ball centered at x_{n-1} with radius r_{n-1}.

Continuing by induction we generate a sequence $\{x_n\}$ in X, which we claim is a Cauchy sequence.[15] Note $\left\{\bar{B}_n\right\}$ is a decreasing sequence of closed sets and $\{r_n\}$ is a positive, real, infinitesimal sequence. Given $\epsilon > 0$ and some $n > 1$ such that $r_n \leq \epsilon$, for $i, j > n$ there exist, by construction,

$$x_i, x_j \in B_n = \left\{x \in X : d(x_n, x) < r_n\right\}$$

such that

$$d(x_i, x_j) \leq d(x_i, x_n) + d(x_j, x_n) < 2r_n \leq 2\epsilon.$$

Hence $\{x_n\}$ is Cauchy, and by virtue of the space's completeness it converges to $x \in X$.

Finally, we need to show $x \in G \cap U$. Take $n > 1$, so for every $m \geq n$

$$x_m \in \bar{B}_n \subseteq \bar{B}_1.$$

As the above two sets are closed,

[15] This step relies on the axiom of dependent choice. If the complete metric space is also assumed to be separable, the lemma is provable in ZF with no additional choice hypotheses.

$$x \in \bar{B}_n \subseteq \bar{B}_1 \Leftrightarrow x \in \bar{B}_n \cap \bar{B}_1.$$

What is more, $\bar{B}_1 \subseteq (G_1 \cap U) \subseteq U$ and $\bar{B}_n \subseteq (G_n \cap B_{n-1}) \subseteq G_n$, so in particular $x \in G_n \cap U$. But $n > 1$ was arbitrary, so

$$x \in G \cap U. \qquad \blacksquare$$

A rather direct consequence of Baire's lemma is:

Corollary 2.3.1 (Countable unions of closed sets with empty interior) *Let (X, d) be a complete metric space and $\{C_n\}_{n\geq 1}$ a sequence of closed sets with empty interior. Then the union*

$$\bigcup_{n=1}^{\infty} C_n = C$$

has empty interior.

Proof By the previous lemma a countable intersection of open dense sets is dense. Since a closed subset $C \subseteq X$ has empty interior iff $A = X \setminus C$ is open and dense,

$$\bigcap_{n=1}^{\infty} A_n = A \text{ is dense } \Leftrightarrow \bigcup_{n=1}^{\infty} A_n^c = A^c \text{ has empty interior.} \qquad \blacksquare$$

Remark 2.3.3 (*A countable intersection of open dense subsets of* $\mathbb{R}$) In complete metric spaces the converse of Lemma 2.3.1 is false, for not every dense subset is a countable intersection of open dense sets.

Take $\mathbb{Q}$, and suppose one could express it as countable intersection of open dense sets U_{q_i}, $q_i \in \mathbb{Q}$:

$$\mathbb{Q} = \bigcap_{q_i \in \mathbb{Q}} U_{q_i}.$$

For every $i = 1, 2, \ldots$, U_{q_i} are open in $\mathbb{R}$, as is $\mathbb{R} \setminus \{q_i\}$. The latter is open since $\{q_i\}$ is closed (the singleton $\{q_i\}$ would be open if q_i were isolated, but $\mathbb{R}$ does not have isolated points). Hence for every $i = 1, 2, \ldots$ the $\big(U_{q_i} \cap \big(\mathbb{R} \setminus \{q_i\}\big)\big)$ are open dense subsets, and

$$\begin{aligned}\bigcap_{q_i \in \mathbb{Q}} \big(U_{q_i} \cap \big(\mathbb{R} \setminus \{q_i\}\big)\big) &= \mathbb{Q} \cap \left(\bigcap_{q_i \in \mathbb{Q}} \big(\mathbb{R} \setminus \{q_i\}\big) \right) \\ &= \mathbb{Q} \cap \left(\mathbb{R} \setminus \bigcup_{q_i \in \mathbb{Q}} \{q_i\} \right) \\ &= \mathbb{Q} \cap \big(\mathbb{R} \setminus \mathbb{Q}\big) = \emptyset.\end{aligned}$$

The countable intersection of these open dense sets is empty. As this contradicts Baire's lemma above, the initial assumption is absurd and the claim follows. ♣

Remark 2.3.4 (*A countable union of nowhere dense subsets of* $\mathbb{R}$) While finite unions of nowhere dense subsets is nowhere dense, for countable unions this might not be the case. Let $q_1, q_2, \ldots, q_n, \ldots$ be a complete enumeration of $\mathbb{Q}$. Any singleton $\{q_i\}$, $i \in \mathbb{N}$, is nowhere dense in $\mathbb{R}$, for $\{q_i\}$ coincides with both its closure and its boundary, and when this happens the closure's interior is empty. On the other hand the countable union of these nowhere dense subsets $\bigcup_{i=1}^{\infty} \{q_i\} = \mathbb{Q}$ is dense in $\mathbb{R}$. ♣

2.4 Meagre and Residual Sets

The fact that a countable union of nowhere dense subsets of a complete and separable metric space is not nowhere dense leads to suspect we might need to define a new type of set. This is exactly what René-Louis Baire did in 1899 in his doctoral thesis *"Sur les fonctions des variables réelles"*.

Baire defined countable unions of nowhere dense sets 'meagre'. The first example of a meagre subset of $\mathbb{R}$, by Remark 2.3.4, is $\mathbb{Q}$. As we shall see, in general a meagre subset of a given closed set $F \subseteq \mathbb{R}$ is an insignificant, or at any rate small, portion of F. What remains in F after we take out a meagre set is called a residual, or comeagre, subset of F. A residual subset, in contrast to a meagre one, forms a substantial part of F.

Definition 2.4.1 (*Meagre sets and sets of second category*) In a space X we say:

1. a set $M \subseteq X$ is meagre, or of first category, in X if it can be written as a countable union of nowhere dense subsets:

$$M = \bigcup_{i=1}^{\infty} A_i;$$

   with $\overset{\circ}{\bar{A}}_i = \emptyset$ for every $i \in \mathbb{N}$;
2. a set $A \subseteq X$ is of second category if it is not meagre. ◇

Definition 2.4.2 (*Comeagre or residual sets*) In a space X a set R is comeagre or residual in X if its complement is meagre in X. ◇

Sets of second category therefore include comeagre/residual sets.

Proposition 2.4.1 (Properties of meagre sets) *In a space* X

1. *the countable union of meagre subsets* $\bigcup_{i=1}^{\infty} M_i$ *is meagre;*
2. *a subset of a meagre set is meagre.*

Proof As for *1.*, it suffices to observe that a countable union of countable unions of nowhere dense subsets is a countable union of nowhere dense subsets.

Regarding *2.*, let B be a subset in the meagre set A, which is a countable union of meagre subsets $\{A_n\}_{n\geq 1}$. Clearly

$$B = B \cap A = B \cap \bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} (B \cap A_n).$$

By *2.* in Proposition 2.2.2 a subset in a nowhere dense set is nowhere dense, so for every $n \in \mathbb{N}$ the set $B \cap A_n \subseteq A_n$ is nowhere dense. Hence $B = \bigcup_{n=1}^{\infty}(B \cap A_n)$ is meagre. ■

Proposition 2.4.2 (Characterisation of residual sets) *Let (X, d) be a complete metric space, a set $R \subseteq X$ is residual, or comeagre, if and only if R contains a countable intersection of open dense subsets of X.*

Proof Suppose R is residual, so $R^c = M$ is meagre. There exists a sequence of nowhere dense subsets $\{R_i\}_{i=1}^{\infty}$ such that $M = \bigcup_{i=1}^{\infty} R_i$, and R_i is nowhere dense in X for every $i \in \mathbb{N}$ iff $(\bar{R}_i)^c = D_i$ is dense and open in X. Therefore

$$M = \bigcup_{i=1}^{\infty} R_i \subseteq \bigcup_{i=1}^{\infty} \bar{R}_i \Leftrightarrow \bigcap_{i=1}^{\infty} (\bar{R}_i)^c = \bigcap_{i=1}^{\infty} D_i \subseteq R.$$

■

Proposition 2.4.3 (Countable intersections of residual sets) *The countable intersection of residual sets $\bigcap_{i=1}^{\infty} R_i$ in a space X is a residual set R.*

Proof The countable intersection of a countable intersection of open dense subsets is a countable intersection of open dense subsets. ■

Theorem 2.4.1 (Baire's category theorem) *Any residual subset R in a complete metric space (X, d) is dense in X.*

Proof Since $A = X \setminus R$ is of first category, then $A = \bigcup_{i=1}^{\infty} A_i$, where each $A_i \subseteq X$ is nowhere dense. Then $A \subseteq \hat{A} = \bigcup_{i=1}^{\infty} \bar{A}_i$, where each $\bar{A}_i$ is closed with empty interior. By corollary 2.3.1 $\hat{A}$ has empty interior, and the inclusion forces A to have empty interior as well. Let us assume that R is not dense. Then there exists a non-empty open set $U \subseteq X$ such that $U \cap R = \emptyset$ and $U \subseteq A$, but A has empty interior. The contradiction proves the claim. ■

Let us illustrate the notions of meagre set and second-category set by applying them to $\mathbb{R}$.

Proposition 2.4.4 (Baire category of intervals) *Intervals in $\mathbb{R}$ are of second category.*

Proof Suppose the interval $I \subseteq \mathbb{R}$ were meagre, i.e. of first category. Then $\mathbb{R} \setminus I$ would be residual and Theorem 2.4.1 would make it dense in $\mathbb{R}$. But $(\mathbb{R} \setminus I) \cap I = \emptyset$, a contradiction, so the statement holds. ■

In general the closure of a meagre set is not meagre. For example $\bar{\mathbb{Q}} = \mathbb{R}$ is of second category, as the next proposition entails.

Proposition 2.4.5 (Baire category of $\mathbb{R}$) $\mathbb{R}$ *is of second category.*

Proof If $\mathbb{R}$ were of first category, we could write

$$\mathbb{R} = \bigcup_{i=1}^{\infty} R_i,$$

with every R_i nowhere dense in $\mathbb{R}$. Let I be an interval of $\mathbb{R}$, so

$$I = \mathbb{R} \cap I = \bigcup_{i=1}^{\infty} R_i \cap I.$$

But now $R_i \cap I$ is nowhere dense for every $i \in \mathbb{N}$, turning I in a countable union of nowhere dense subsets of $\mathbb{R}$. It would follow I is of first category, i.e. meagre. The contradiction proves the claim. ■

Proposition 2.4.6 (Baire category of residual subsets of $\mathbb{R}$) *Residual subsets in* $\mathbb{R}$ *are of second category.*

Proof Suppose a residual set $R \subseteq \mathbb{R}$ were of first category. Then both R and $\mathbb{R} \setminus R$ would be of first category. If so $R \cup R^c = \mathbb{R}$ would be a countable union of nowhere dense subsets, a contradiction. ■

Remark 2.4.1 (*The set* $\mathbb{R} \setminus \mathbb{Q}$) We have seen that the rational numbers, being a countable union of nowhere dense subsets, form a meagre set (Remark 2.3.4). Consequently the set of irrationals $\mathbb{R} \setminus \mathbb{Q}$ is residual and of second category in $\mathbb{R}$, while $\mathbb{R}$ or every interval I in $\mathbb{R}$ is of second category in itself, but they are not residual. Finally we observe that $I, I^c \subseteq \mathbb{R}$ are of second category in themselves too. ♣

A set of second category in a complete metric space (X, d) is always uncountable. As we shall explain in a short while, though, there exist uncountable sets of first category.

Proposition 2.4.7 (Cardinality of residual sets in complete metric spaces) *Let M be a meagre subset in X. Then for any set E of second category in X, the set $E \setminus M$ is uncountable.*

Proof Let $E \subseteq X$ be of second category and M a meagre subset in X, so we may write it as countable union of nowhere dense subsets A_i

$$M = \bigcup_{i=1}^{\infty} A_i.$$

If $(E \setminus M)$ were countable, we would be able to express it as a countable union $\bigcup_{i=1}^{\infty} \{x_i\}$ of singletons $\{x_i\}$, $i = 1, 2, \ldots$.

Then

$$E = \big(E \setminus M\big) \cup (E \cap M) = \left(\bigcup_{i=1}^{\infty} \{x_i\}\right) \cup \left(E \cap \bigcup_{i=1}^{\infty} A_i\right) = \bigcup_{i=1}^{\infty} \{x_i\} \cup \big(E \cap A_i\big).$$

Since $\big(E \cap A_i\big)$ and $\{x_i\}$ are both nowhere dense for every $i \in \mathbb{N}$, E would be of first category, a contradiction. ■

Remark 2.4.2 (*Meagre sets with the cardinality of the continuum*) It was mentioned that there are meagre subsets whose cardinality is the continuum, just like for second-category sets. An example is $\bigcup_{n\in\mathbb{Z}} C_n$, where

$$C_n = C + n = \big\{x + n : x \in C \text{ the Cantor set}\big\}.$$

We proved C is nowhere dense, while $\mathbb{Z}$ is countable. The union $\bigcup_{n\in\mathbb{Z}} C_n$ has the following structure. It resembles a countable replica of the Cantor set obtained by integer translation. To each $n \in \mathbb{Z}$ there corresponds a nowhere dense, closed set $C_n = C + n$ embedded in the real line as a copy of the Cantor set. For $n \neq m$ the sets C_n and C_m have the cardinality of the continuum. So in general

$$\bigcup_{n\in\mathbb{Z}} C_n = \bigcup_{n\in\mathbb{Z}} (C + n),$$

is a countable union of nowhere dense subsets, i.e. a meagre set. Furthermore its cardinality is the continuum, since $2^{\aleph_0} \cdot \aleph_0 = 2^{\aleph_0}$. ♣

Part II
Borel Sets and Baire Functions on $\mathbb{R}$

Chapter 3
Borel sets in $\mathbb{R}$



3.1 Basics on Borel Sets

Open and closed subsets of $\mathbb{R}$ are but special families of sets of a larger class, that of Borel sets. The Borel class consists of all subsets of $\mathbb{R}$ that can be obtained by union and intersection of countably many open, or closed, subsets. This brings to mind the definition of a topological space. For Borel sets, though, only countable operations are allowed, whereas in a topological space we may join and intersect sets arbitrarily. At any rate, starting from a topological space one can always obtain a family of Borel sets generated by the collection of open sets, or closed sets, of the space.

Regarding the specific construction of a Borel family, we call:

- G_0 set: any open subset of $\mathbb{R}$;
- G_1 set: a countable intersection of G_0 sets; a G_1 set is also known as a G_δ set;
- G_2 set: a countable union of G_1 sets; a G_2 set is also known as a $G_{\delta,\sigma}$ set;

$\vdots$

In general, for every countable ordinal $\alpha < \Omega$ (Ω being the first uncountable ordinal) we can define G_α sets.

Once G_β sets have been defined for every $\beta < \alpha$, then G_α sets are the countable intersections or unions of G_β sets, for $\beta < \alpha$, according to whether α is odd or even. If α is a limit ordinal, i.e. with no immediate predecessor, we may assume it even, making odd the successor $\alpha + 1$ to any limit ordinal. The first ordinal 1 is odd, and the successor of an odd/even ordinal is even/odd respectively.

Let A be a G_α set. Then

$$A = A \cup A \cup A \cup \ldots \quad \text{and} \quad A = A \cap A \cap A \cap \ldots;$$

so irrespective of the parity of α, any G_α set A is also a $G_{\alpha+1}$ set. We obtain thus the following sequence of types

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S. Gentili, *Measure, Integration and a Primer on Probability Theory*,
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$$G_0, G_1, \ldots, G_\alpha, \ldots, \quad \alpha < \Omega,$$

where α may be either a limit ordinal or one with immediate predecessor. G_α sets, for $\alpha < \Omega$, are usually called Borel sets, for they arise as unions or intersections of a countable number of open sets. To produce the whole Borel class the process must be repeated for every countable α less than Ω (in other terms, by transfinite induction).

If for instance $\{A_i : i \in \mathbb{N}\}$ is a countable family such that

$$\begin{aligned} &A_1 \text{ is a } G_1 \setminus G_0 \text{ set} \\ &A_2 \text{ is a } G_2 \setminus G_1 \text{ set} \\ &A_3 \text{ is a } G_3 \setminus G_2 \text{ set} \\ &\ldots\ldots\ldots \end{aligned}$$

we have no guarantee $\bigcup_{i=1}^{\infty} A_i$ or $\bigcap_{i=1}^{\infty} A_i$ is one of the G_β sets built previously. Only at iteration Ω we can be sure that those two are G_Ω sets. We may in fact prove that to obtain the entire Borel class over $\mathbb{R}$—which as we shall see has important features, such as stability under countable intersections and unions—it is necessary to repeat the iterations for every countable ordinal α less than Ω. The resulting family has cardinality $2^{\aleph_0}$.

At this point the following recursive definition, for $\alpha \in (0, \Omega)$, seems natural. Call Σ_1^0 the class of open sets in $\mathbb{R}$, i.e. the class of G_0 sets:

$$\Sigma_1^0 = \left\{E^1 \subseteq \mathbb{R} : E^1 \text{ open in } \mathbb{R}\right\}.$$

Call Π_2^0 the class of countable intersections of sets in Σ_1^0:

$$\Pi_2^0 = \left\{H^2 \subseteq \mathbb{R} : H^2 = \bigcap_{i=1}^{\infty} E_i^1, E_i^1 \in \Sigma_1^0, \ \forall i \in \mathbb{N}\right\}.$$

As $E^1 = E^1 \cap E^1 \cap E^1 \cap \ldots$ for every $E^1 \in \Sigma_1^0$, it follows $\Sigma_1^0 \subseteq \Pi_2^0$. If $\beta < \alpha$ is an odd ordinal with immediate predecessor, let Σ_β^0 be the class of countable unions of sets in $\Pi_{\beta-1}^0$:

$$\Sigma_\beta^0 = \left\{E^\beta \subseteq \mathbb{R} : E^\beta = \bigcup_{i=1}^{\infty} H_i^{\beta-1}, H_i^{\beta-1} \in \Pi_{\beta-1}^0, \ \forall i \in \mathbb{N}\right\}.$$

Instead, if $\beta < \alpha$ is an even ordinal with immediate predecessor, let Π_β^0 denote the class of countable intersections of sets in $\Sigma_{\beta-1}^0$:

$$\Pi_\beta^0 = \left\{H^\beta \subseteq \mathbb{R} : H^\beta = \bigcap_{i=1}^{\infty} E_i^{\beta-1}, E_i^{\beta-1} \in \Sigma_{\beta-1}^0, \forall i \in \mathbb{N}\right\}.$$

For the same reasons that $\Sigma^0_1 \subseteq \Pi^0_2$, for every non-limit $\beta < \alpha$ we have $\Sigma^0_\beta \subseteq \Pi^0_{\beta+1}$. Therefore when $\beta \in (0, \alpha)$, with α a limit ordinal,

$$\Sigma^0_1 \subseteq \Pi^0_2 \subseteq \Sigma^0_3 \subseteq \ldots \subseteq \Sigma^0_\beta \subseteq \Pi^0_{\beta+1} \subseteq \ldots. \tag{3.1.1}$$

In general, for every even $\beta < \alpha$ the sets in Π^0_β are countable intersections of sets in $\Sigma^0_{\beta-1}$:

$$\Pi^0_\beta = \left(\Sigma^0_{\beta-1}\right)_\delta.$$

Once we have the classes Σ^0_β and Π^0_β for every $\beta < \alpha$, we can define Π^0_α, i.e. the class of countable intersections of sets in Σ^0_β, $\beta < \alpha$:

$$\Pi^0_\alpha = \left(\bigcup_{\beta<\alpha} \Sigma^0_\beta\right)_\delta.$$

A corresponding family can be produced starting from closed subsets of $\mathbb{R}$. In this situation we call

- F_0 sets: the closed sets of $\mathbb{R}$;
- F_1 sets: the countable unions of F_0 sets, also known as F_σ sets;
- F_2 sets: the countable intersections of F_1 sets, also known as $F_{\sigma,\delta}$ sets;

$\vdots$

An F_α set is an $F_{\alpha+1}$ set as well, so we have another sequence of types

$$F_0, F_1, \ldots, F_\alpha, \ldots, \quad \alpha < \Omega.$$

The picture concerning F_α sets mirrors what was discussed about G_α sets. F_α sets, too, are called Borel sets, since they arise from closed subsets by constructing repeated countable unions and intersections.

As one would expect, we define recursively for $\alpha \in (0, \Omega)$ the following classes. Call Π^0_1 the class of closed subsets of $\mathbb{R}$, i.e. the class of F_0 sets in $\mathbb{R}$:

$$\Pi^0_1 = \left\{H^1 \subseteq \mathbb{R} : H^1 \text{ closed in } \mathbb{R}\right\}.$$

Call Σ^0_2 the class of countable unions of sets in Π^0_1:

$$\Sigma^0_2 = \left\{E^2 \subseteq \mathbb{R} : E^2 = \bigcup_{i=1}^{\infty} H^1_i,\ H^1_i \in \Pi^0_1,\ \forall i \in \mathbb{N}\right\}.$$

Note that $H^1 = H^1 \cup H^1 \cup H^1 \cup \ldots$ for every $H^1 \in \Pi^0_1$, so $\Pi^0_1 \subseteq \Sigma^0_2$. For $\beta < \alpha$ a non-limit odd ordinal let Π^0_β denote the class of countable intersections of sets in $\Sigma^0_{\beta-1}$:

$$\Pi^0_\beta = \left\{ H^\beta \subseteq \mathbb{R} : H^\beta = \bigcap_{i=1}^{\infty} E_i^{\beta-1},\ E_i^{\beta-1} \in \Sigma^0_{\beta-1},\ \forall i \in \mathbb{N} \right\}.$$

For $\beta < \alpha$ a non-limit even ordinal let Σ^0_β be the class of countable unions of sets in $\Pi^0_{\beta-1}$:

$$\Sigma^0_\beta = \left\{ E^\beta \subseteq \mathbb{R} : E^\beta = \bigcup_{i=1}^{\infty} H_i^{\beta-1},\ H_i^{\beta-1} \in \Pi^0_{\beta-1},\ \forall i \in \mathbb{N} \right\}.$$

The argument justifying $\Pi^0_1 \subseteq \Sigma^0_2$ also explains that for every non-limit $\beta < \alpha$ we have $\Pi^0_\beta \subseteq \Sigma^0_{\beta+1}$. Therefore, if $\beta \in (0, \alpha)$ with α limit ordinal,

$$\Pi^0_1 \subseteq \Sigma^0_2 \subseteq \Pi^0_3 \subseteq \ldots \subseteq \Pi^0_\beta \subseteq \Sigma^0_{\beta+1} \subseteq \ldots. \tag{3.1.2}$$

In general, therefore, sets in Σ^0_β, for any even $\beta < \alpha$, are countable unions of sets in $\Pi^0_{\beta-1}$:

$$\Sigma^0_\beta = \left(\Pi^0_{\beta-1}\right)_\sigma.$$

Once we have classes Σ^0_β and Π^0_β for every $\beta < \alpha$, we may introduce Σ^0_α, i.e. the class of countable unions of sets in Π^0_β, $\beta < \alpha$:

$$\Sigma^0_\alpha = \left(\bigcup_{\beta<\alpha} \Pi^0_\beta \right)_\sigma.$$

3.2 Borel F_σ Sets and G_δ Sets in $\mathbb{R}$

Proposition 3.2.1 (Borel type of an interval) *Any interval is both an F_σ set and a G_δ set.*

Proof Let us begin with $[a, b]$. To show $[a, b] \in F_\sigma$ it suffices to observe

$$[a, b] = [a, b] \cup [a, b] \cup \ldots.$$

Moreover $[a, b] \in G_\delta$, since

$$[a, b] = \bigcap_{n=1}^{\infty} \left(a - \frac{1}{n}; b + \frac{1}{n} \right).$$

Similarly for the degenerate interval $[a, a] = \{a\}$ (therefore any singleton $\{x\} = [x, x]$ is clearly closed in $\mathbb{R}$).

As for $[a, b)$,

$$[a, b) = \bigcup_{n=1}^{\infty} \left[a, b - \frac{1}{n}\right] \quad \text{and} \quad [a, b) = \bigcap_{n=1}^{\infty} \left(a - \frac{1}{n}, b\right).$$

In a similar way one proves $(a, b] \in F_\sigma$ and $(a, b] \in G_\delta$.

Regarding $(-\infty, b)$

$$(-\infty, b) = \bigcup_{n=1}^{\infty} \left[b - n; b - \frac{1}{n}\right] \quad \text{and} \quad (-\infty, b) = (-\infty, b) \cap (-\infty, b) \cap \ldots.$$

Not dissimilar is the argument for the intervals (a, ∞), $(-\infty, \infty)$, $(-\infty, b]$, $[a, \infty)$ and (a, b). ■

Proposition 3.2.2 (Borel type of a union of intervals) *A union of intervals is an F_σ set.*

Proof Let $A = \bigcup_{i=1}^{\infty} A_i$, where the A_i may be closed, open, or semi-open intervals in $\mathbb{R}$, and not necessarily pairwise disjoint. In any case A can be written as a union of intervals B_i that are closed, open, or semi-open in $\mathbb{R}$ and pairwise disjoint, each one with rational endpoints belonging to the initial union,

$$A = \bigsqcup_{i=1}^{\infty} B_i.$$

Every B_i, by Proposition 3.2.1, is a countable union of closed sets, so $A = \bigsqcup_{i=1}^{\infty} B_i$ is too. Hence A is of type F_σ. ■

Proposition 3.2.3 (F_σ sets and compact sets in $\mathbb{R}$) *Any linear set of F_σ type is a countable union of compact subsets of $\mathbb{R}$ (also known as a σ-compact set).*

Proof If F is an F_σ set, it is the countable union of closed sets, each of which is a countable union of compact sets by Proposition 2.1.5. Hence F is the countable union of compact sets. ■

Proposition 3.2.4 (Borel type of closed subsets of $\mathbb{R}$) *Any closed linear set has type G_δ.*

Proof Let F be a closed linear set of points. Take on $\mathbb{R}$ the standard metric $d(x, y)$ and define the family $\{G_n\}_{n \geq 1}$ by

$$G_n = \left\{x \in \mathbb{R} : d(x, F) < \frac{1}{n}\right\};$$

where, for every x in G_n,

$$d(x, F) = \inf\{d(x, y) : y \in F\}.$$

We claim the G_n are open sets. To see this, let x_0 be any point in G_n. By definition $d(x_0, F) < 1/n$, so there exists a point $x_1 \in F$ such that $d(x_0, x_1) < 1/n$. Set

$$\frac{1}{n} - d(x_0, x_1) = \delta > 0 \;\Rightarrow\; d(x_0, x_1) = \frac{1}{n} - \delta.$$

We want to prove that the neighbourhood $V = (x_0 - \delta, x_0 + \delta)$ of centre x_0 and radius δ is contained in G_n. So let

$$y \in V = (x_0 - \delta, x_0 + \delta).$$

Then $|y - x_0| < \delta$ and, as $|x_0 - x_1| = 1/n - \delta$,

$$|y - x_1| \leq |y - x_0| + |x_0 - x_1| < \delta + \frac{1}{n} - \delta = \frac{1}{n}.$$

Hence $d(y, x_1) < 1/n$, so $d(y, F) < 1/n$ and $y \in G_n$ by definition of G_n. As $y \in V$ is arbitrary, $(x_0 - \delta, x_0 + \delta) \subseteq G_n$. But also $x_0 \in G_n$ was arbitrary, making G_n open.

Once we show $F = \bigcap_{n=1}^{\infty} G_n$ we are done. If $y \in F$ then $d(y, F) = 0 < 1/n$ for every $n \in \mathbb{N}$, so $y \in \bigcap_{n=1}^{\infty} G_n \Rightarrow F \subseteq \bigcap_{n=1}^{\infty} G_n$. For the converse, let x be a point in $\bigcap_{n=1}^{\infty} G_n$. As $x \in \bigcap_{n=1}^{\infty} G_n$, $x \in G_n$ for every $n \in \mathbb{N}$. Hence there exists some $y \in F$ such that $d(x, y) < 1/n$, let us call it y_n. As $x \in G_n$ for every $n \in \mathbb{N}$, then $x \in G_m$ for every $m \geq n$. Therefore there exists $y_m \in F$ such that $d(x, y_m) < 1/m$, for every $m \in \mathbb{N}$. Eventually we have found a sequence $\{y_m\}_{m \geq n}$ in F accumulating at x. Since F is closed in $\mathbb{R}$, $x \in F$. But x is chosen arbitrarily, so $\bigcap_{n=1}^{\infty} G_n \subseteq F$, and the claim follows. ■

Proposition 3.2.5 (Borel type of open sets in $\mathbb{R}$) *Any open linear set has type F_σ.*

Proof An open set G in $\mathbb{R}$ is the complement of a closed set F. By the previous proposition

$$G = F^c = \left(\bigcap_{i=1}^{\infty} G_i\right)^c = \bigcup_{i=1}^{\infty} (G_i)^c = \bigcup_{i=1}^{\infty} F_i,$$

where F_i is closed for every $i \in \mathbb{N}$. ■

Proposition 3.2.6 (Borel type of difference of closed (open) sets in $\mathbb{R}$) *The difference of closed (open) linear sets is an F_σ set (a G_δ set).*

Proof Let F_1, F_2 be closed linear sets, so

$$F_1 \setminus F_2 = F_1 \setminus \left(\bigcap_{i=1}^{\infty} G_i \right) = \bigcup_{i=1}^{\infty} (F_1 \setminus G_i).$$

As G_i is open for every $i \in \mathbb{N}$, $F_1 \setminus G_i$ is closed for every $i \in \mathbb{N}$. Hence $F_1 \setminus F_2$ is a countable union of closed sets, i.e. an F_σ set.

If G_1 and G_2 are open linear sets,

$$G_1 \setminus G_2 = G_1 \setminus \left(\bigcup_{i=1}^{\infty} F_i \right) = \bigcap_{i=1}^{\infty} (G_1 \setminus F_i).$$

As F_i is closed for every $i \in \mathbb{N}$, $G_1 \setminus F_i$ is open for every $i \in \mathbb{N}$, so $G_1 \setminus G_2$ is a countable intersection of open sets, i.e. a G_δ set. ■

Remark 3.2.1 (*Borel type of* $\mathbb{Q}$) The set $\mathbb{Q}$ is of type F_σ. By contradiction, suppose $\mathbb{Q} = \bigcap_{i=1}^{\infty} G_i$, with G_i open in $\mathbb{R}$ for every $i \in \mathbb{N}$. Then $\mathbb{Q} \subseteq G_i$ for every $i \in \mathbb{N}$. Moreover every G_i is dense in $\mathbb{R}$, since $\mathbb{R} = \bar{\mathbb{Q}} \subseteq \bar{G}_i$, and by assumption $G_i \subseteq \mathbb{R}$. But then Proposition 2.4.2 makes $\mathbb{Q}$ is residual, which is absurd. Hence the claim is proved.

Actually any meagre set in a complete metric space (X, d) is contained in some meagre F_σ set. In fact if M is meagre, then $M = \bigcup_{i=1}^{\infty} A_i$, with A_i nowhere dense in X. Clearly $M \subseteq \bigcup_{i=1}^{\infty} \bar{A}_i$ and the union is a meagre F_σ set, since the $\bar{A}_i$ are nowhere dense and closed.

♣

Remark 3.2.2 (*Borel type of* $[0, 1] \setminus \mathbb{Q}$) The set $[0, 1] \setminus \mathbb{Q}$ is a G_δ set:

$$[0, 1] \setminus \mathbb{Q} = [0, 1] \setminus \bigcup_{n=1}^{\infty} \{q_n\} = \bigcap_{n=1}^{\infty} ([0, 1] \setminus \{q_n\}) = \bigcap_{n=1}^{\infty} \left[\left(0 - \frac{1}{n}, 1 + \frac{1}{n} \right) \setminus \{q_n\} \right].$$

As $\left[\left(0 - \frac{1}{n}, 1 + \frac{1}{n} \right) \setminus \{q_n\} \right]$ is open for every $n \in \mathbb{N}$, the set $[0, 1] \setminus \mathbb{Q}$ is a countable intersection of open sets.

Now suppose $[0, 1] \setminus \mathbb{Q}$ were an F_σ set:

$$[0, 1] \setminus \mathbb{Q} = \bigcup_{i=1}^{\infty} F_i$$

with F_i closed for every $i \in \mathbb{N}$. $[0, 1] \setminus \mathbb{Q}$ is residual in $[0, 1]$ (Remark 2.4.1), so Theorem 2.4.1 implies it is dense in $[0, 1]$, and there exists an F_i and an open interval $I \subseteq [0, 1]$ such that F_i is dense in I. But F_i is also closed, so it contains its limit points, and then F_i contains I. In turn this means F_i contains both rational and irrational numbers, contradicting the irrationality of any number in $[0, 1] \setminus \mathbb{Q}$. This confirms $[0, 1] \setminus \mathbb{Q}$ is not an F_σ set.

♣

Later on we shall see F_σ- and G_δ-type sets are fundamental to describe the sets where real functions are discontinuous or continuous.

In analogy to countable unions of nowhere dense subsets and countable intersections of dense sets, G_δ and F_σ sets satisfy a number of properties.

Proposition 3.2.7 (Baire category of dense G_δ sets) *A dense subset of* $\mathbb{R}$ *of type* G_δ *is residual.*

Proof Let

$$A = \bigcap_{i=1}^{\infty} G_i,$$

where each G_i is open. By assumption A is dense in $\mathbb{R}$, and $A \subseteq G_i$ for every $i = 1, 2, \ldots$, so every G_i is dense in $\mathbb{R}$. By Proposition 2.4.2 $A = \bigcap_{i=1}^{\infty} G_i$ is a residual set. ■

Proposition 3.2.8 (Baire category of F_σ sets) *An* F_σ *set in* $\mathbb{R}$ *is either meagre or it contains an interval.*

Proof Let F be an F_σ set, so $F = \bigcup_{i=1}^{\infty} F_i$ with F_i closed for every $i = 1, 2 \ldots$.

Clearly if the F_i are all nowhere dense in $\mathbb{R}$, F is meagre, i.e. of first category. Assume one of the closed sets F_i is not nowhere dense. It must be dense in some interval $(a, b) \subseteq \mathbb{R}$ (see Remark 2.2.4). As F_i is closed it contains its limit points and therefore $(a, b) \subseteq F_i$. Then F contains an interval. ■

Proposition 3.2.9 (Stability of G_δ and F_σ sets)

1. *The intersection of a countable collection of* G_δ *sets in* $\mathbb{R}$ *is a* G_δ *set.*
2. *The union of a countable collection of* F_σ *sets in* $\mathbb{R}$ *is an* F_σ *set.*

Proof 1. Let $\{A_i\}_{i=1}^{\infty}$ be a countable family of G_δ sets. Then

$$A = \bigcap_{i=1}^{\infty} A_i = \bigcap_{i=1}^{\infty} \left(\bigcap_{j=1}^{\infty} G_{i_j} \right) = \bigcap_{i,j=1}^{\infty} G_{i_j}.$$

Hence A is a countable intersection of open sets of $\mathbb{R}$.

2. The argument is completely analogous. ■

Remark 3.2.3 (*Borel type of the Cantor set and its complement*) The Cantor set $C \subset [0, 1]$ is the countable intersection of closed sets and hence closed, or an F_0 set. As

$$C = C \cup C \cup C \cup \ldots,$$

C is an F_σ set.

But C does not contain intervals, so it is meagre by Proposition 3.2.8. On the other hand the complement $G = [0, 1] \setminus C$ is the countable union of open intervals, hence an open, or G_0, set. Moreover

$$G = G \cap G \cap G \cap \dots$$

implies it is a G_δ set. Being G dense in $[0, 1]$, by Proposition 3.2.7 it is furthermore residual. ♣

3.3 Discontinuity Set of Real Functions over $[a, b]$

At this juncture we are ready to determine the type of the set of discontinuous points of a map $f : [a, b] \mapsto \mathbb{R}$. Before this we introduce the notion of oscillation of f, an important tool for analysing the discontinuity set.

Definition 3.3.1 (*Oscillation of real functions*) Take a map $f : [a, b] \mapsto \mathbb{R}$ and an arbitrary subset E of $[a, b]$:

1. the oscillation of the map f over E is the quantity $\omega(f, E) \in [0, \infty]$ defined by

$$\omega(f, E) = \sup_{x,y \in E} |f(x) - f(y)|.$$

   When f is bounded on E, $\omega(f, E)$ is a positive real number. When f is unbounded, instead, we set $\omega(f, E) = \infty$.
2. Given any $\delta > 0$, a point $x \in [a, b]$ and a neighbourhood $U_\delta(x) = (x - \delta, x + \delta)$, the oscillation of f at the point x is the real number $\omega(f, x) \geq 0$ defined by

$$\omega(f, x) = \lim_{\delta \to 0} \omega\big(f, U_\delta(x) \cap [a, b]\big).$$

◇

Remark 3.3.1 (*Comparing oscillations of real maps*) Since $\omega(f, E)$ is the least upper bound of the image of E under f, for every $E_1 \subseteq E_2 \subseteq [a, b]$ we have $f(E_1) \subseteq f(E_2)$ and

$$\omega(f, E_1) \leq \omega(f, E_2). \tag{3.3.1}$$

♣

Theorem 3.3.1 (Oscillation of real maps at continuous points) *A map $f : [a, b] \mapsto \mathbb{R}$ is continuous at $x_0 \in [a, b]$ if and only if its oscillation at x_0 vanishes:*

$$\lim_{\delta \to 0} \omega\big(f, U_\delta(x_0) \cap [a, b]\big) = 0. \tag{3.3.2}$$

Proof Necessary implication: suppose f continuous at x_0. For any $\epsilon > 0$ there exists $\delta > 0$ such that for every $x \in U_\delta(x_0) \cap [a, b]$

$$|f(x) - f(x_0)| < \frac{\epsilon}{2};$$

i.e.

$$\sup_{x \in U_\delta(x_0) \cap [a,b]} |f(x) - f(x_0)| < \frac{\epsilon}{2}.$$

Hence the oscillation of f on the neighbourhood of x_0 is bounded by ϵ, because

$$\begin{aligned}
\omega\big(f, U_\delta(x_0) \cap [a, b]\big) &= \sup_{x,y \in U_\delta(x_0) \cap [a,b]} |f(x) - f(y)| \\
&= \sup_{x,y \in U_\delta(x_0) \cap [a,b]} |f(x) - f(x_0) + f(x_0) - f(y)| \\
&\leq \sup_{x \in U_\delta(x_0) \cap [a,b]} |f(x) - f(x_0)| + \sup_{y \in U_\delta(x_0) \cap [a,b]} |f(x_0) - f(y)| \\
&< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon. \qquad (3.3.3)
\end{aligned}$$

But

$$\omega(f, x_0) \leq \omega\big(f, U_\delta(x_0) \cap [a, b]\big) < \epsilon$$

and ϵ is arbitrary, whence the claim.

Sufficient implication: suppose (3.3.2) holds. Then for any $\epsilon > 0$ there is a $\delta > 0$ such that

$$\omega\big(f, U_\delta(x_0) \cap [a, b]\big) < \epsilon.$$

But $\omega\big(f, U_\delta(x_0) \cap [a, b]\big) = \sup_{x,y \in U_\delta(x_0) \cap [a,b]} |f(x) - f(y)|$, so

$$\sup_{x,y \in U_\delta(x_0) \cap [a,b]} |f(x) - f(y)| < \epsilon;$$

hence for every $x \in U_\delta(x_0) \cap [a, b]$

$$|f(x) - f(x_0)| < \epsilon.$$

As $\epsilon > 0$ is arbitrary, f is continuous at x_0. ■

Proposition 3.3.1 (Discontinuity set of a real map) *Given a map $f : [a, b] \mapsto \mathbb{R}$ and any $\alpha > 0$, the subset on which the oscillation of f is greater than or equal to α,*

$$D(f) = \big\{x \in [a, b] : \omega(f, x) \geq \alpha\big\},$$

is closed.

Proof Let $\bar{x}$ be a limit point of $D(f)$. By definition there exists a sequence $\{x_n\}_{n\geq 1}$ in $D(f)$ such that $x_n \to \bar{x}$. For every $n \in \mathbb{N}$, $\omega(f, x_n) \geq \alpha$.

Suppose $\bar{x} \notin D(f)$. Then $\bar{x}$ belongs in the continuity set $\big(D(f)\big)^c = C(f)$:

$$\bar{x} \in C(f) = \{x \in [a, b] : \omega(f, x) < \alpha\}.$$

Hence $\omega(f, \bar{x}) < \alpha$. By definition there is a $\delta > 0$ such that for every $x \in U_\delta(\bar{x}) \cap [a, b]$

$$\omega\big(f, U_\delta(\bar{x}) \cap [a, b]\big) = \sup_{x \in U_\delta(\bar{x}) \cap [a,b]} |f(\bar{x}) - f(x)| < \alpha. \tag{3.3.4}$$

But $\bar{x}$ is a limit point of $D(f)$, so there exists some $x_i \in D(f)$ internal to the neighbourhood $U_\delta(\bar{x}) \cap [a, b]$. Therefore we can take a neighbourhood $U_\eta(x_i) \cap [a, b]$ such that

$$U_\eta(x_i) \cap [a, b] \subseteq U_\delta(\bar{x}) \cap [a, b]. \tag{3.3.5}$$

At the same time $x_i \in D(f)$ implies

$$\omega\big(f, U_\eta(x_i) \cap [a, b]\big) = \sup_{x \in U_\eta(x_i) \cap [a,b]} |f(x_i) - f(x)| \geq \omega(f, x_i) \geq \alpha.$$

Remark 3.3.1 tells

$$\omega\big(f, U_\delta(\bar{x}) \cap [a, b]\big) \geq \omega\big(f, U_\eta(x_i) \cap [a, b]\big) \geq \alpha.$$

The contradiction then allows to conclude. ■

Proposition 3.3.2 (G_δ and F_σ subsets of closed real intervals) *A subset $A \subseteq [a, b]$ is a G_δ set if and only if $[a, b] \setminus A$ is an F_σ set.*

Proof Necessary part: let $A \subseteq [a, b]$ be a G_δ set, $A = \bigcap_{i=1}^{\infty} G_i$. Then

$$[a, b] \setminus A = [a, b] \setminus \bigcap_{i=1}^{\infty} G_i = \bigcup_{i=1}^{\infty} \big([a, b] \setminus G_i\big) = \bigcup_{i=1}^{\infty} \big([a, b] \cap G_i^c\big).$$

Clearly $[a, b] \cap G_i^c$, for every $i \in \mathbb{N}$, is closed as intersection of closed sets, so $[a, b] \setminus A$ is an F_σ set, being a countable union of closed sets.

Vice versa, let $[a, b] \setminus A$ be an F_σ set, $[a, b] \setminus A = \bigcup_{i=1}^{\infty} F_i$, with F_i closed for every $i \in \mathbb{N}$. Then

$$A = [a,b] \setminus \bigcup_{i=1}^{\infty} F_i = \bigcap_{i=1}^{\infty} \left([a,b] \cap F_i^c \right).$$

For every $i \in \mathbb{N}$ the set $[a,b] \cap F_i^c$ is open in $[a,b]$, making A a G_δ set. ■

Proposition 3.3.3 (Borel type of countable dense subsets of closed intervals) *A countable set $A \subseteq \mathbb{R}$ that is at most dense in $[a,b]$ cannot be a G_δ set.*

Proof Let A be countable and dense in $[a,b]$, and suppose by contradiction it is of type G_δ, so $A = \bigcap_{i=1}^{\infty} G_i$. As $A \subseteq G_i$ for every $i = 1, 2, \ldots$, every G_i is dense in $[a,b]$. Hence A is the countable intersection of open dense sets in $[a,b]$, and

$$[a,b] \setminus A = [a,b] \setminus \bigcap_{i=1}^{\infty} G_i = \bigcup_{i=1}^{\infty} \left([a,b] \setminus G_i \right)$$

the countable union of nowhere dense subsets in $[a,b]$, i.e. of first category. Therefore A is residual in $[a,b]$ and by virtue of Proposition 2.4.7 it cannot be countable. ■

Remark 3.3.2 (*Discontinuities of a function*)

1. Combining Theorem 3.3.1 with Proposition 3.3.1 we may say a map $f : [a,b] \mapsto \mathbb{R}$ is discontinuous at $x_0 \in [a,b]$ iff its oscillation at that point is $\omega(f, x_0) > 0$.
2. Given $f : [a,b] \mapsto \mathbb{R}$ define for every $n \in \mathbb{N}$

$$D_n(f) = \left\{ x \in [a,b] : \omega(f,x) \geq \frac{1}{n} \right\}.$$

   The discontinuity set of f reads

$$F = \bigcup_{n=1}^{\infty} D_n(f) = \bigcup_{n=1}^{\infty} \left\{ x \in [a,b] : \omega(f,x) \geq \frac{1}{n} \right\},$$

   i.e. a countable union of closed sets, whence an F_σ set.
3. On the other hand, any $\left(D_n(f) \right)^c = C_n(f)$ is open in $[a,b]$ and obviously

$$G = \bigcap_{n=1}^{\infty} C_n(f) = \bigcap_{n=1}^{\infty} \left\{ x \in [a,b] : \omega(f,x) < \frac{1}{n} \right\}$$

   represents the continuity set of f on $[a,b]$, which is patently a G_δ set.
4. Remarks 3.2.1 and 3.2.2 tell us there cannot exist a map $f : \mathbb{R} \mapsto \mathbb{R}$ that is continuous at rational points and discontinuous at irrational points. Put equivalently, there is no map $f : \mathbb{R} \mapsto \mathbb{R}$ whose discontinuity set are the irrational numbers.

♣

Theorem 3.3.2 (Real-valued maps that are continuous on a residual set) *A map $f : [a, b] \mapsto \mathbb{R}$ is continuous on a residual set R of $[a, b]$ if and only if R is dense in $[a, b]$.*

Proof The necessary part is immediate because R is residual by hypothesis, and Baire's theorem (Theorem 2.4.1) implies its density in $[a, b]$.

Sufficient implication: let R denote the continuity set of f on $[a, b]$, and suppose R dense in $[a, b]$. As R is a G_δ set,

$$R = \bigcap_{i=1}^{\infty} G_i,$$

where each G_i is open in $[a, b]$. But $R \subseteq G_i$ for every $i \in \mathbb{N}$, so every G_i is dense in $[a, b]$ and R is the countable intersection of open dense sets. Let us define

$$D = \mathbb{R} \setminus R = \bigcup_{i=1}^{\infty} (\mathbb{R} \setminus G_i),$$

a countable union of nowhere dense subsets. It is of first category and hence R is residual. ■

The discontinuity set of a real function is not necessarily countable, as the next example shows.

Example 3.3.1 (*Discontinuity of the Cantor set's characteristic function*) Using Remark 3.2.3 we may also verify the characteristic (or indicator) function $\varphi_C(x)$ of the Cantor set is discontinuous at every $x \in C$ and continuous at every $x \in G$.

For this, define

$$\varphi_C(x) = \begin{cases} 1 & \text{if } x \in C \\ 0 & \text{otherwise} \end{cases}$$

This map satisfies

$$\lim_{x \to \bar{x}} \varphi_C(x) = 0$$

for every $\bar{x} \notin C$. Hence the limit never exists in C, while it always exists in the complement. Consequently φ_C is continuous on G and discontinuous on C. The discontinuities are essential, in other words rather severe ones.

Pick in fact an $x \in [0, 1]$. As C is closed with empty interior, if $x \in C$ then any neighbourhood U of x has non-empty intersection with $C^c = G$. Therefore for any $\epsilon > 0$ less than $1/2$ there exists $\bar{x} \in U \cap G$ such that

$$|\varphi_C(x) - \varphi_C(\bar{x})| = |1 - 0| = 1 > \epsilon.$$

Hence φ_C is discontinuous at any $x \in C$.

If $x \in G$, we may find a neighbourhood U in the open set G. Given that φ_C is 0 on U, for any $\epsilon > 0$ and every $\bar{x} \in U$

$$|\varphi_C(x) - \varphi_C(\bar{x})| = |0 - 0| = 0 < \epsilon.$$

Hence φ_C is continuous at every point in G.

The characteristic function of the Cantor set is then discontinuous on C, which has the cardinality of the continuum. What we have just proved allows us to say there exists real functions with uncountably many discontinuity points. ★

3.4 Additive and Multiplicative Families of Sets

Until now we have examined the features of Borel subsets of $\mathbb{R}$ of type F_σ and G_δ. As we have seen, they play an important role in classifying the discontinuities of real functions.

Yet to acquire the tools necessary for the Lebesgue-Hausdorff theorem we must also study the properties of the additive class Σ^0_α and the multiplicative class Π^0_α, where α is any ordinal smaller than the first uncountable ordinal Ω. To that end we shall invoke a result about transfinite induction.

The principle of transfinite induction is equivalent to finite induction on positive integers, but only the former generalises to well-ordered sets (like $\mathbb{R}$).

Relatively to the set of ordinals, transfinite induction consists in proving a certain property holds for every ordinal number. Any such proof proceeds in two steps. In the first one proves that a given property $P(x)$ holds for the ordinal 0, i.e. $P(0)$. In the second, inductive, step one proves $P(x)$ for every ordinal $\alpha < \Omega$, namely $P(\alpha)$, where α may either be a limit ordinal or one with immediate predecessor, and Ω is the first uncountable ordinal. The inductive step is typically split into two. In part one assumes the property holds for some successor ordinal β, in which case it is possible to use classical induction. Part two consists in assuming the property for every ordinal β smaller than a limit ordinal α. Here classical induction does not work, for α has no predecessor.

Theorem 3.4.1 (Weak principle of transfinite induction) *Let A be the class of ordinals $\alpha < \Omega$ and call by $P(x)$ a property such that:*

1. *for the first ordinal $0 \in A$ property $P(0)$ holds;*
2. *for every successor ordinal $\beta \in A$, $P(\beta - 1) \Rightarrow P(\beta)$;*
3. *for every limit ordinal $\alpha \in A$, $\big(\forall \beta < \alpha, P(\beta) \Rightarrow P(\alpha)\big)$.*

Then $P(\alpha)$ holds for every ordinal $\alpha < \Omega$.

Proof In such a case $\{P(x)\}$ must correspond to the class of ordinal numbers. To prove the result let us take the above hypotheses and suppose, by contradiction, that there existed an ordinal γ such that $\neg P(\gamma)$ holds. Then

$$B = \{\beta \leq \gamma : \neg P(\beta)\}$$

would not be empty, and therefore it would admit a minimum element, say β'. Clearly $\beta' \neq 0$ since $P(0)$ holds. As β' is the minimum of B, $P(\eta)$ is true for every $\eta < \beta'$. Assumptions *2.* or *3.* would give us $P(\beta')$, which cannot be. Hence the claim. ■

Here is a variant of the above principle.

Theorem 3.4.2 (Strong principle of transfinite induction) *Let A be the class of ordinals $\alpha < \Omega$ and call by $P(x)$ a property such that:*

1. *for the first ordinal $0 \in A$ property $P(0)$ holds;*
2. *for every $\alpha \in A$*

$$\forall \beta < \alpha, \quad P(\beta) \Rightarrow P(\alpha).$$

Then $P(\alpha)$ holds for every ordinal $\alpha < \Omega$.

Proof In analogy to the previous theorem suppose there existed an ordinal γ for which $\neg P(\gamma)$ were true. Then

$$B = \{\beta < \gamma : \neg P(\beta)\}$$

would be non-empty, with minimum $\beta' \in B$. This ordinal cannot be the first element of A, because of $P(0)$. As every ordinal $\eta < \beta'$ is in A, by induction

$$\forall \eta < \beta', \quad P(\eta) \Rightarrow P(\beta')$$

and $P(\beta')$ would hold, against the assumption. ■

Definition 3.4.1 (*Additive and multiplicative classes*) Define in $\mathbb{R}$ the following classes of sets recursively:

1. for $\alpha = 1$ let

$$\Sigma_1^0 = \{E^1 \subseteq \mathbb{R} : E^1 \text{ open}\} \quad \text{and} \quad \Pi_1^0 = \{H^1 \subseteq \mathbb{R} : H^1 \text{ closed}\};$$

2. for $0 < \alpha < \Omega$,
   if α is a successor ordinal

$$\Sigma_\alpha^0 = \left(\Pi_{\alpha-1}^0\right)_\sigma \quad \text{and} \quad \Pi_\alpha^0 = \left(\Sigma_{\alpha-1}^0\right)_\delta;$$

   if α is a limit ordinal, having defined Σ_β^0 and Π_β^0 for every $\beta < \alpha$, we set

$$\Sigma_\alpha^0 = \left(\bigcup_{\beta<\alpha} \Pi_\beta^0\right)_\sigma = \left\{E^\alpha \subseteq \mathbb{R} : E^\alpha = \bigcup_{i=1}^{\infty} H_i^\beta,\ H_i^\beta \in \Pi_\beta^0,\ 0 < \beta < \alpha\right\}$$

$$\Pi^0_\alpha = \left(\bigcup_{\beta<\alpha} \Sigma^0_\beta \right)_\delta = \left\{ H^\alpha \subseteq \mathbb{R} : H^\alpha = \bigcap_{i=1}^{\infty} E_i^\beta,\, E_i^\beta \in \Sigma^0_\beta, 0 < \beta < \alpha \right\}.$$

For every $\alpha \in (0, \Omega)$, the families Σ^0_α and Π^0_α are respectively called additive class and multiplicative class of order α.

Correspondingly, we say $E^\alpha \in \Sigma^0_\alpha$ is a set of class Σ^0_α, $H^\alpha \in \Pi^0_\alpha$ a set of class Π^0_α. ◇

Item *2.* above should be understood as follows:

1. if α is a successor ordinal

$$\Sigma^0_\alpha = \left(\Pi^0_{\alpha-1}\right)_\sigma \Leftrightarrow \Pi^0_{\alpha-1} \subseteq \Sigma^0_\alpha \quad \text{and} \quad \Pi^0_\alpha = \left(\Sigma^0_{\alpha-1}\right)_\delta \Leftrightarrow \Sigma^0_{\alpha-1} \subseteq \Pi^0_\alpha;$$

2. if α is a limit ordinal

$$\Sigma^0_\alpha = \left(\bigcup_{\beta<\alpha} \Pi^0_\beta \right)_\sigma \Leftrightarrow\ \forall \beta < \alpha,\ \Pi^0_\beta \subseteq \bigcup_{\beta<\alpha} \Pi^0_\beta \subseteq \Sigma^0_\alpha$$

$$\Pi^0_\alpha = \left(\bigcup_{\beta<\alpha} \Sigma^0_\beta \right)_\delta \Leftrightarrow\ \forall \beta < \alpha,\ \Sigma^0_\beta \subseteq \bigcup_{\beta<\alpha} \Sigma^0_\beta \subseteq \Pi^0_\alpha.$$

Equivalently, for $\beta \in (0, \alpha)$, every set $H^\beta \in \Pi^0_\beta$ is also a set in Σ^0_α, and every $E^\beta \in \Sigma^0_\beta$ is also in Π^0_α.

Proposition 3.4.1 (Properties of classes Σ^0_α and Π^0_α) *In* $\mathbb{R}$, *for every* $\alpha \in (0, \Omega)$ *the additive and multiplicative classes satisfy the following properties:*

1. *additive and multiplicative families are stable:*
   - *(a)* Σ^0_α *is stable under countable unions and finite intersections;*
   - *(b)* Π^0_α *is stable under countable intersections and finite unions;*
2. *any set of* Σ^0_α *is a set of* $\Sigma^0_{\alpha+1}$, *and any set of* Π^0_α *is a set of* $\Pi^0_{\alpha+1}$*:*

$$\Sigma^0_\alpha \subseteq \Sigma^0_{\alpha+1} \quad \textit{and} \quad \Pi^0_\alpha \subseteq \Pi^0_{\alpha+1};$$

3. *any set* $E \subseteq \mathbb{R}$ *belongs in* Σ^0_α *if and only if* $\mathbb{R} \setminus E = H$ *belongs in* Π^0_α*;*
4. *a countable union of sets in* Σ^0_α *belongs in* $\Pi^0_{\alpha+1}$, *and a countable intersection of sets in* Π^0_α *belongs in* $\Sigma^0_{\alpha+1}$*;*
5. Σ^0_α *and* Π^0_α *are preserved under the pre-image of continuous functions;*
6. *the difference of two sets* E_1, E_2 *in* Σ^0_α, *or* H_1, H_2 *in* Π^0_α, *belongs both in* $\Sigma^0_{\alpha+1}$ *and* $\Pi^0_{\alpha+1}$.

Proof 1. Clearly *(a)* holds for open sets, i.e. for $\alpha = 1$.

Let β be a successor ordinal. Take a sequence $\{E_i^\beta\}_{i\geq 1}$, where by definition $E_i^\beta = \bigcup_{j=1}^\infty H_{i,j}^{\beta-1}$ for every $i \in \mathbb{N}$; hence

$$\bigcup_{i=1}^\infty E_i^\beta = \bigcup_{i=1}^\infty \left(\bigcup_{j=1}^\infty H_{i,j}^{\beta-1} \right) = \bigcup_{i,j} H_{i,j}^{\beta-1} \in \left(\Pi_{\beta-1}^0\right)_\sigma = \Sigma_\beta^0.$$

Let now α be a limit ordinal and suppose *a)* holds for every $\beta < \alpha$. Take a sequence $\{E_i^\alpha\}_{i\geq 1}$, so $E_i = \bigcup_{j=1}^\infty H_{i,j}^\beta$ for every $i \in \mathbb{N}$, where $H_{i,j}^\beta \in \bigcup_{\beta<\alpha} \Pi_\beta^0$. Then

$$\bigcup_{i=1}^\infty E_i = \bigcup_{i=1}^\infty \bigcup_{j=1}^\infty H_{i,j}^\beta = \bigcup_{i,j} H_{i,j}^\beta \in \left(\bigcup_{\beta<\alpha} \Pi_\beta^0 \right)_\sigma = \Sigma_\alpha^0.$$

A similar argument shows that the class Π_α^0 is stable under countable intersections, hence finite ones as well.

As for finite intersections, it suffices to prove the claim for two-fold intersections and then use induction. In case $\alpha = 1$ the assertion is undoubtedly true. Let $\beta < \alpha$ be a successor ordinal and take $E_1^\beta = \bigcup_{i=1}^\infty H_i^{\beta-1}$ and $E_2^\beta = \bigcup_{j=1}^\infty H_j^{\beta-1}$. Recalling $\Pi_{\beta-1}^0$ is stable under finite intersections,

$$\begin{aligned} E_1^\beta \cap E_2^\beta &= \left(\bigcup_{i=1}^\infty H_i^{\beta-1} \right) \cap \left(\bigcup_{j=1}^\infty H_j^{\beta-1} \right) \\ &= \bigcup_{i,j} \underbrace{\left(H_i^{\beta-1} \cap H_j^{\beta-1} \right)}_{\in \Pi_{\beta-1}^0} \in \left(\Pi_{\beta-1}^0\right)_\sigma = \Sigma_\beta^0. \end{aligned}$$

Let now α be a limit ordinal and suppose the claim true for every $\beta < \alpha$. Given $E_1^\alpha = \bigcup_{i=1}^\infty H_i^\beta$ and $E_2^\alpha = \bigcup_{j=1}^\infty H_j^\beta$, where for every $i, j \in \mathbb{N}$ the sets H_i^β and H_j^β belong to some $\Pi_\beta^0 \subseteq \bigcup_{\beta<\alpha} \Pi_\beta^0$, we have

$$\left(H_i^\beta \cap H_j^\beta \right) \in \bigcup_{\beta<\alpha} \Pi_\beta^0,$$

and so

$$\begin{aligned} E_1^\alpha \cap E_2^\alpha &= \left(\bigcup_i H_i^\beta \right) \cap \left(\bigcup_j H_j^\beta \right) \\ &= \bigcup_{i,j} \left(H_i^\beta \cap H_j^\beta \right) \in \left(\bigcup_{\beta<\alpha} \Pi_\beta^0 \right)_\sigma = \Sigma_\alpha^0. \end{aligned}$$

The part of *(b)* about finite unions for class Π_α^0 is completely analogous.

2. When $\alpha = 1$, in $\mathbb{R}$ (with the standard metric) every closed set $H^1 \in \Pi^0_1$ is a countable intersection of sets E^1_i in Σ^0_1 (Proposition 3.2.4):

$$H^1 = \bigcap_{i=1}^{\infty} E^1_i \in \Pi^0_2 \Rightarrow \Pi^0_1 \subseteq \Pi^0_2.$$

Similarly every open set $E^1 \in \Sigma^0_1$ is a countable union of sets H^1_i in Π^0_1 (Proposition 3.2.5):

$$E^1 = \bigcup_{i=1}^{\infty} H^1_i \in \Sigma^0_2 \Rightarrow \Sigma^0_1 \subseteq \Sigma^0_2.$$

Let β be a successor ordinal.
By induction assume

$$\Sigma^0_{\beta-1} \subseteq \Sigma^0_\beta \quad \text{and} \quad \Pi^0_{\beta-1} \subseteq \Pi^0_\beta, \tag{3.4.1}$$

so we need to prove $\Sigma^0_\beta \subseteq \Sigma^0_{\beta+1}$ and $\Pi^0_\beta \subseteq \Pi^0_{\beta+1}$. Start from the first. Given any E^β,

$$E^\beta = \bigcup_{i=1}^{\infty} H_i^{\beta-1},$$

where $H_i^{\beta-1} \in \Pi^0_{\beta-1}$. By hypothesis every $H_i^{\beta-1}$ is in Π^0_β. Therefore every $H_i^{\beta-1}$ is a countable intersection of $\{E_{i,j}^{\beta-1}\}$, whence

$$E^\beta = \bigcup_{i=1}^{\infty} H_i^{\beta-1} = \bigcup_{i=1}^{\infty} \left(\bigcap_{j=1}^{\infty} E_{i,j}^{\beta-1} \right).$$

The set $\left(\bigcap_{j=1}^{\infty} E_{i,j}^{\beta-1}\right)$ is in Π^0_β, so

$$\bigcup_{i=1}^{\infty} \left(\bigcap_{j=1}^{\infty} E_{i,j}^{\beta-1} \right) = E^{\beta+1},$$

and then every set in Σ^0_β is also in $\Sigma^0_{\beta+1}$. The proof of $\Pi^0_\beta \subseteq \Pi^0_{\beta+1}$ is similar.
To prove the second inductive step, by definition

$$\Sigma^0_\beta \subseteq \Pi^0_{\beta+1} \quad \text{and} \quad \Pi^0_\beta \subseteq \Sigma^0_{\beta+1}$$

for every $\beta < \alpha$. Hence for every $\beta < \alpha$

$$\Sigma^0_\beta \subseteq \bigcup_{\beta<\alpha} \Pi^0_\beta \subseteq \left(\bigcup_{\beta<\alpha} \Pi^0_\beta\right)_\sigma = \Sigma^0_\alpha \quad \text{and} \quad \Pi^0_\beta \subseteq \bigcup_{\beta<\alpha} \Sigma^0_\beta \subseteq \left(\bigcup_{\beta<\alpha} \Sigma^0_\beta\right)_\delta = \Pi^0_\alpha. \tag{3.4.2}$$

Induction then guarantees that for every $\beta < \alpha$, α a limit ordinal,

$$\Sigma^0_\beta \subseteq \Sigma^0_{\beta+1} \Rightarrow \Sigma^0_\beta \subseteq \bigcup_{\beta<\alpha} \Sigma^0_\beta \quad \text{and} \quad \Pi^0_\beta \subseteq \Pi^0_{\beta+1} \Rightarrow \Pi^0_\beta \subseteq \bigcup_{\beta<\alpha} \Pi^0_\beta.$$

Take $E^\alpha \in \Sigma^0_\alpha$, so $E^\alpha = \bigcup_{i=1}^\infty H_i^\beta$ for $\beta < \alpha$. By induction every H_i^β belongs in $\bigcup_{\beta<\alpha} \Pi^0_\beta$, and every H_i^β is a countable intersection of sets $E_{i,j}^\beta$ in Σ^0_β, $\beta < \alpha$, so

$$E^\alpha = \bigcup_{i=1}^\infty H_i^\beta = \bigcup_{i=1}^\infty \left(\bigcap_{j=1}^\infty E_{i,j}^\beta\right).$$

The set $\bigcap_{j=1}^\infty E_{i,j}^\beta$, $\beta < \alpha$, is in $\left(\bigcup_{\beta<\alpha} \Sigma^0_\beta\right)_\delta = \Pi^0_\alpha$, so

$$\bigcup_{i=1}^\infty \left(\bigcap_{j=1}^\infty E_{i,j}^\beta\right) = E^{\alpha+1}.$$

Therefore we conclude $\Sigma^0_\alpha \subseteq \Sigma^0_{\alpha+1}$. The inclusion $\Pi^0_\alpha \subseteq \Pi^0_{\alpha+1}$ is similar.

3. When $\alpha = 1$ the proposition is true by definition. Take β a successor ordinal. Suppose for $\beta - 1$

$$E^{\beta-1} = (\mathbb{R} \setminus H^{\beta-1}) \in \Sigma^0_{\beta-1} \Leftrightarrow H^{\beta-1} = (\mathbb{R} \setminus E^{\beta-1}) \in \Pi^0_{\beta-1}.$$

Take $E^\beta = \bigcup_{i=1}^\infty H_i^{\beta-1}$ and $H^\beta = \bigcap_{i=1}^\infty E_i^{\beta-1}$, so immediately

$$E^\beta = \mathbb{R} \setminus H^\beta = \left(\mathbb{R} \setminus \bigcap_{i=1}^\infty E_i^{\beta-1}\right) = \bigcup_{i=1}^\infty \left(\mathbb{R} \setminus E_i^{\beta-1}\right) = \bigcup_{i=1}^\infty H_i^{\beta-1} \in \Sigma^0_\beta$$

$$\Leftrightarrow H^\beta = \mathbb{R} \setminus E^\beta = \left(\mathbb{R} \setminus \bigcup_{i=1}^\infty H_i^{\beta-1}\right) = \bigcap_{i=1}^\infty \left(\mathbb{R} \setminus H_i^{\beta-1}\right) = \bigcap_{i=1}^\infty E_i^{\beta-1} \in \Pi^0_\beta.$$

Let now α be a limit ordinal, and suppose that for every $\beta < \alpha$

$$E^\beta = (\mathbb{R} \setminus H^\beta) \in \Sigma^0_\beta \Leftrightarrow H^\beta = (\mathbb{R} \setminus E^\beta) \in \Pi^0_\beta.$$

Given $E^\alpha = \bigcup_{i=1}^\infty H_i^\beta$ and $H^\alpha = \bigcap_{i=1}^\infty E_i^\beta$, where by assumption, for every $i \in \mathbb{N}$,

$$H_i^\beta \in \Pi^0_\beta \subseteq \left(\bigcup_{\beta<\alpha} \Pi^0_\beta\right) \quad \text{and} \quad E_i^\beta \in \Sigma^0_\beta \subseteq \left(\bigcup_{\beta<\alpha} \Sigma^0_\beta\right).$$

Then

$$E^\alpha = \mathbb{R} \setminus H^\alpha = \left(\mathbb{R} \setminus \bigcap_{i=1}^{\infty} E_i^\beta\right) = \bigcup_{i=1}^{\infty} \left(\mathbb{R} \setminus E_i^\beta\right) = \bigcup_{i=1}^{\infty} H_i^\beta \in \Sigma^0_\alpha$$
$$\Leftrightarrow H^\alpha = \mathbb{R} \setminus E^\alpha = \left(\mathbb{R} \setminus \bigcup_{i=1}^{\infty} H_i^\beta\right) = \bigcap_{i=1}^{\infty} \left(\mathbb{R} \setminus H_i^\beta\right) = \bigcap_{i=1}^{\infty} E_i^\beta \in \Pi^0_\alpha.$$

4. By item *1.*, *(a)* a countable union of sets in Σ^0_α is of the same type. Furthermore, by definition a set of Σ^0_α belongs in $\Pi^0_{\alpha+1}$. The second assertion is proved in a similar way.
5. Suppose $f : X \mapsto Y$ is continuous and let $E \subseteq Y$ be in Σ^0_α. When $\alpha = 1$, clearly $f^{-1}(E) \in \Sigma^0_1$. Take $\alpha < \Omega$ and $E = \bigcup_{i=1}^{\infty} H_i$, where $H_i \in \Pi^0_{\alpha-1}$. Then

$$f^{-1}\left(\bigcup_{i=1}^{\infty} H_i\right) = \bigcup_{i=1}^{\infty} \left(f^{-1}(H_i)\right) \in \Sigma^0_\alpha.$$

   The case of Π^0_α is proved using that $f^{-1}(Y \setminus E) = X \setminus f^{-1}(E)$.
6. Let E_1^α, E_2^α be sets in Σ^0_α, and observe

$$E_1^\alpha \setminus E_2^\alpha = E_1^\alpha \cap (E_2^\alpha)^c = E_1^\alpha \cap H_2^\alpha.$$

   By definition any set of Σ^0_α also lives in $\Pi^0_{\alpha+1}$, and by item *2.* any set in Π^0_α belongs in $\Pi^0_{\alpha+1}$. Then $E_1^\alpha \cap H_2^\alpha$ may be written as

$$H_1^{\alpha+1} \cap H_2^{\alpha+1},$$

   which by item *1.*, *(b)* is in $\Pi^0_{\alpha+1}$. Analogously, since

$$H_1^\alpha \setminus H_2^\alpha = H_1^\alpha \cap E_2^\alpha,$$

   any set in Π^0_α belongs in $\Sigma^0_{\alpha+1}$ by definition, and by *2.* any set in Σ^0_α belongs in $\Sigma^0_{\alpha+1}$. Therefore $H_1^\alpha \cap E_2^\alpha$ may be written as

$$E_1^{\alpha+1} \cap E_2^{\alpha+1}.$$

   By *1.*, *a)* this set belongs in $\Sigma^0_{\alpha+1}$. ■

Now let us examine a new type of set, called ambiguous. The family of ambiguous sets of class α consists of elements of Σ^0_α that also belong in Π^0_α and vice versa. In particular, ambiguous sets of class 1 in $\mathbb{R}$ define the class

$$\Delta_1^0 = \big\{A \subseteq \mathbb{R} : A \text{ is clopen}\big\}.$$

Example 3.4.1 (*Ambiguous sets*)

1. Trivial examples of clopen subsets in $\mathbb{R}$ are $\mathbb{R}$ itself and $\emptyset$.
2. To find less immediate clopen examples in a generic metrisable space X, let us define for every pair $x, y \in X$ the discrete metric

$$d(x, y) = \begin{cases} 1 & \text{if } x \neq y \\ 0 & \text{if } x = y. \end{cases}$$

   We claim any $A \subseteq X$ is open. If A is empty the statement holds. If $A \neq \emptyset$, let $x \in A$ and $B(x, 1/2)$, so if $y \in X$ is such that $d(x, y) < 1/2$ then $x = y \Rightarrow y \in A$. On the other hand $d(x, z) = 1$ for every $z \in X$ such that $z \neq x \Rightarrow z \notin A$. Consequently $y \in A$ for every $y \in X$ such that $d(x, y) < 1/2$, whence A is open. Now, $(X \setminus A) \subseteq X$, but we have just shown that any subset in X with discrete metric is open. Therefore $(X \setminus A)$ is open, so A is closed, and hence clopen.
3. Here is a set both in Σ_2^0 and Π_2^0, but neither in Σ_1^0 nor in Π_1^0: the semi-open interval $[0, 1) \subset \mathbb{R}$. As

$$[0, 1) = \bigcup_{n=1}^{\infty} \left[0, 1 - \frac{1}{n}\right] = \bigcap_{n=1}^{\infty} \left(-\frac{1}{n}, 1\right),$$

   we have $[0, 1) \in \Delta_2^0 = \Sigma_2^0 \cap \Pi_2^0$. ★

In general, for every ordinal $\alpha < \Omega$ we declare $A \in \Delta_\alpha^0$ whenever $A \in \Sigma_\alpha^0$ and $A \in \Pi_\alpha^0$. This leads to the following.

Definition 3.4.2 (*The class of ambiguous sets*) In $\mathbb{R}$, for every $\alpha \in (0, \Omega)$ we define the family of ambiguous sets of class α as

$$\Delta_\alpha^0 = \Sigma_\alpha^0 \cap \Pi_\alpha^0.$$

◇

The above relationship means ambiguous sets of class α are only those common to Σ_α^0 and Π_α^0. For example, an ambiguous set is of class 1 if it is both open and closed, of class 2 if it is both a G_δ set and an F_σ set, etc.

Now let us examine the stability properties enjoyed by class Δ_α^0.

Proposition 3.4.2 (Properties of Δ_α^0) *For every $\alpha \in (0, \Omega)$ the class Δ_α^0 over $\mathbb{R}$ is:*

1. *stable under finite unions, finite intersections and complementation;*
2. *stable under set-difference;*
3. *preserved by pre-images of continuous maps.*

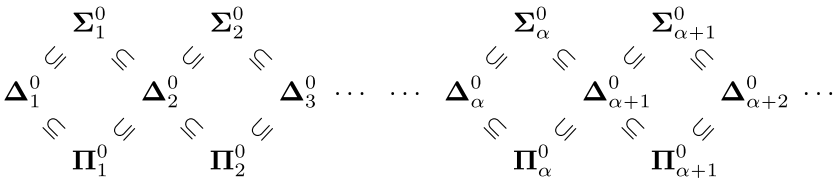


Fig. 3.1 Hierarchy of Borel classes

Proof

1. As $\Delta^0_\alpha = \Pi^0_\alpha \cap \Sigma^0_\alpha$, for every ordinal $\alpha < \Omega$ the class Δ^0_α is made of all sets common to Σ^0_α and Π^0_α. First of all observe that Σ^0_α is stable under finite intersections, and Π^0_α is stable under countable intersections, so when considered simultaneously they are stable under finite intersections only. Similarly, Π^0_α is stable under finite unions, Σ^0_α under countable unions, so they are simultaneously stable under finite unions. Furthermore any set in Σ^0_α is the complement of a set in Π^0_α and vice versa. This property is common to the two classes, so Δ^0_α is stable under complementation.
2. From the stability of ambiguous sets under complementation and finite intersections we deduce that the difference of $A^\alpha_1, A^\alpha_2 \in \Delta^0_\alpha$ stays in Δ^0_α, for in fact $\left(A^\alpha_2\right)^c \in \Delta^0_\alpha$ and $A^\alpha_1 \cap \left(A^\alpha_2\right)^c = A^\alpha_1 \setminus A^\alpha_2 \in \Delta^0_\alpha$.
3. Since $\Delta^0_\alpha = \Sigma^0_\alpha \cap \Pi^0_\alpha$, by Proposition 3.4.1, item *5.*, the class is preserved by the pre-image of a continuous function. ■

Overall, ambiguous sets of class α, for every $\alpha \in (0, \Omega)$, form an algebra of sets. The figure 3.1 depicts the hierarchical structure of the Borel classes.

Proposition 3.4.3 (Prominent relationships between Σ^0_α, Π^0_α, Δ^0_α) *In $\mathbb{R}$ the additive, multiplicative and ambiguous classes fulfil the following identities:*

1. $\Sigma^0_\alpha \cup \Pi^0_\alpha \subseteq \Delta^0_{\alpha+1}$, *for every* $\alpha \in (0, \Omega)$*;*
2. $\Sigma^0_\alpha = \left(\Delta^0_\alpha\right)_\sigma$ *and* $\Pi^0_\alpha = \left(\Delta^0_\alpha\right)_\delta$, *for every* $\alpha \in (1, \Omega)$ *(also* $\alpha = 1$ *for zero-dimensional separable spaces*[1]*);*

Proof

1. The assertion at hand means a set in either Σ^0_α or Π^0_α is ambiguous of class $\alpha + 1$. So let us prove $\Sigma^0_\alpha \subseteq \Delta^0_{\alpha+1}$, for which it is enough to show $\Sigma^0_\alpha \subseteq \Pi^0_{\alpha+1}$ and $\Sigma^0_\alpha \subseteq \Sigma^0_{\alpha+1}$. The former is true by definition. The inclusion $\Sigma^0_\alpha \subseteq \Sigma^0_{\alpha+1}$ holds by *2.* in Proposition 3.4.1. In a completely similar manner $\Pi^0_\alpha \subseteq \Delta^0_{\alpha+1}$. Overall then $\Sigma^0_\alpha \cup \Pi^0_\alpha \subseteq \Delta^0_{\alpha+1}$.
2. Every ambiguous set A^α_i of class α is both a set $E^\alpha_i \in \Sigma^0_\alpha$ and a set $H^\alpha_i \in \Pi^0_\alpha$, so a countable union of A^α_i is still in Σ^0_α, while a countable intersection of A^α_i is in Π^0_α. Therefore

[1] See [19, p.277 ff.].

$$\left(\Delta^0_\alpha\right)_\sigma \subseteq \Sigma^0_\alpha \quad \text{and} \quad \left(\Delta^0_\alpha\right)_\delta \subseteq \Pi^0_\alpha.$$

For the opposite inclusions observe

$$\Pi^0_{\alpha-1} \subseteq \Delta^0_\alpha \Leftrightarrow \left(\Pi^0_{\alpha-1}\right)_\sigma \subseteq \left(\Delta^0_\alpha\right)_\sigma \Leftrightarrow \Sigma^0_\alpha \subseteq \left(\Delta^0_\alpha\right)_\sigma$$
$$\Sigma^0_{\alpha-1} \subseteq \Delta^0_\alpha \Leftrightarrow \left(\Sigma^0_{\alpha-1}\right)_\delta \subseteq \left(\Delta^0_\alpha\right)_\delta \Leftrightarrow \Pi^0_\alpha \subseteq \left(\Delta^0_\alpha\right)_\delta.$$

■

Proposition 3.4.4 (Decomposition of additive sets into ambiguous sets) *For* $\alpha \in (1, \Omega)$ *any set in* Σ^0_α *is a countable union of pairwise-disjoint ambiguous sets of class* α.

Proof Let E be in Σ^0_α, so

$$E = \bigcup_{i=1}^{\infty} H_i,$$

with all H_i in $\Pi^0_{\alpha-1}$. Define

$$A_1 = H_1;\ A_2 = \left(H_2 \setminus H_1\right); \ldots; A_i = H_i \setminus \left(\bigcup_{j=1}^{i-1} H_j\right), \ldots \tag{3.4.3}$$

where $A_i \cap A_j = \emptyset$ for every $i \neq j$. As H_i is in $\Pi^0_{\alpha-1}$, by Proposition 3.4.3, *1.*, it is ambiguous of class α. Then the sets in (3.4.3) are disjoint ambiguous sets of class α such that

$$E = \bigcup_{i=1}^{\infty} A_i. \tag{3.4.4}$$

■

Proposition 3.4.4 implies the following fact.

Proposition 3.4.5 (Decomposition of additive sets into multiplicative sets) *For* $\alpha \in (2, \Omega)$ *any set in* Σ^0_α *is a disjoint countable union of multiplicative sets of class* $< \alpha$.

Proof Take $E \in \Sigma^0_\alpha$. By formulas (3.4.4) and (3.4.3) we have

$$E = \bigcup_{i=1}^{\infty} A_i = H_1 \cup (H_2 \setminus H_1) \cup \ldots \cup H_i \setminus \left(\bigcup_{j=1}^{i-1} H_j\right) \cup \ldots,$$

where each H_i is multiplicative of class $< \alpha$. Of the same class is also $H_1 \cup \ldots \cup H_{i-1}$ for every $i \in \mathbb{N}$. By Proposition 3.4.1, *3.*, for every $i \in \mathbb{N}$ the set $E_i = \mathbb{R} \setminus (H_1 \cup \ldots \cup H_{i-1})$ is additive of class $< \alpha$. Again by Proposition 3.4.4

$$E_i = \bigcup_{n=1}^{\infty} A_n^i,$$

where A_n^i is ambiguous of class $< \alpha$ and $A_n^i \cap A_m^i = \emptyset$ for every $n \neq m$. Note further $H_i \cap E_i = H_i \setminus (H_1 \cup \ldots \cup H_{i-1})$, so

$$E = \bigcup_{i=1}^{\infty} (H_i \cap E_i) = \bigcup_{i=1}^{\infty} H_i \cap \left(\bigcup_{n=1}^{\infty} A_n^i \right) = \bigcup_{i=1}^{\infty} \bigcup_{n=1}^{\infty} (H_i \cap A_n^i); \tag{3.4.5}$$

where each $\left(H_i \cap A_n^i\right)$ is multiplicative of class $< \alpha$. Equation (3.4.5) is then the required decomposition. ■

At this point we shall present the so-called reduction and separation theorems.

Theorem 3.4.3 (Reduction theorem) *For every $\alpha \in (1, \Omega)$ the Borel class Σ_α^0 enjoys the so-called generalised reduction property*[1]*: for every sequence $\{A_i\}_{i\geq 1}$ in additive class α there is a sequence $\{B_i\}_{i\geq 1}$ in the same additive class such that $B_i \subseteq A_i$ for every $i = 1, 2, \ldots$, $B_i \cap B_j = \emptyset$, $i \neq j$ and $\bigcup_i A_i = \bigcup_i B_i$.*

Proof Let $\{E_i^\alpha\}_{i\geq 1}$ be a sequence in Σ_α^0. Clearly $E_i \cap E_j \neq \emptyset$ for $i \neq j$. By Proposition 3.4.4 for every $i \in \mathbb{N}$ we may define $E_i^\alpha = \bigcup_{j=1}^{\infty} A_{i,j}^\alpha$, where $\{A_{i,j}^\alpha\}_{j\geq 1}$ is a sequence of pairwise-disjoint ambiguous sets in Δ_α^0.

We may order the elements of $\left\{A_{i,j}^\alpha : i, j \in \mathbb{N}\right\}$ as a single sequence, by putting $n = \varphi(i, j)$, where φ is a bijection from $\mathbb{N} \times \mathbb{N}$ to $\mathbb{N}$. Thus we produce a sequence $\{D_n\}_{n\geq 1}$, with $D_n = D_{\varphi(i,j)} = A_{i,j}^\alpha$ for every $n \in \mathbb{N}$, and

$$B_n = D_n \setminus \bigcup_{m<n} D_m = A_{i,j}^\alpha \setminus \left(\bigcup_{\varphi(k,l)<\varphi(i,j)} A_{k,l}^\alpha \right) = B_{i,j}^\alpha.$$

Every $B_{i,j}^\alpha$ is ambiguous of class α, for ambiguous classes are stable under finite unions and differences. Hence

$$\bigcup_{i,j} B_{i,j}^\alpha = \bigcup_{i,j} A_{i,j}^\alpha = \bigcup_{i=1}^{\infty} E_i^\alpha.$$

Now let, for every $i = 1, 2, \ldots$, $B_i^\alpha = \bigcup_{j=1}^{\infty} B_{i,j}^\alpha$ be sets in Σ_α^0 (the latter is stable under countable unions). They are pairwise disjoint, so

[1] To fix ideas we might view the reduction property as the procedure that redistributes the common part of two sets proportionally, in such a way that the new union still equals the original union but is now disjoint.

$$\forall i \in \mathbb{N},\ B_i^\alpha \subseteq \bigcup_{j=1}^{\infty} A_{i,j}^\alpha = E_i^\alpha,\quad B_i^\alpha \cap B_k^\alpha = \emptyset,\ \forall i \neq k \quad \text{and} \quad \bigcup_{i=1}^{\infty} B_i^\alpha = \bigcup_{i=1}^{\infty} E_i^\alpha,$$

as required. ■

Theorem 3.4.4 (Separation theorem) *Take $\alpha \in (1, \Omega)$ and a sequence $\{H_i\}_{i\geq 1}$ in Π_α^0 such that $H_1 \cap H_2 \cap \ldots = \emptyset$. There exists an ambiguous sequence $\{A_i\}_{i\geq 1}$ of class α such that*

$$H_i \subseteq A_i, \forall i \quad \textit{and} \quad A_1 \cap A_2 \cap \ldots = \emptyset. \tag{3.4.6}$$

In particular if H, H' are disjoint sets in Π_α^0 there exists an ambiguous set A of class α such that

$$H \subseteq A \quad \textit{and} \quad A \cap H' = \emptyset.$$

Proof Let $\{E_i\}_{i\geq 1}$ be the sequence in Σ_α^0 such that for every $i \in \mathbb{N}$

$$E_i = \mathbb{R} \setminus H_i. \tag{3.4.7}$$

The reduction theorem guarantees the existence of a sequence $\{B_i\}_{i\geq 1}$ of pairwise-disjoint sets in Σ_α^0 such that

$$B_i \subseteq E_i,\ \forall i \in \mathbb{N} \quad \text{and} \quad \bigcup_{i=1}^{\infty} B_i = \bigcup_{i=1}^{\infty} E_i,$$

where the B_i are ambiguous of class α, as well as every

$$A_i = \mathbb{R} \setminus B_i. \tag{3.4.8}$$

By assumption $\bigcap_{i=1}^{\infty} H_i = \emptyset$, so (3.4.7) implies

$$\bigcup_{i=1}^{\infty} E_i = \mathbb{R} \setminus \bigcap_{i=1}^{\infty} H_i = \mathbb{R} = \bigcup_{i=1}^{\infty} B_i. \tag{3.4.9}$$

Furthermore, keeping in account the last equality above and Eq. (3.4.8), we obtain

$$\bigcap_{i=1}^{\infty} A_i = \bigcap_{i=1}^{\infty} (\mathbb{R} \setminus B_i) = \mathbb{R} \setminus \bigcup_{i=1}^{\infty} B_i = \emptyset,$$

i.e. the second equation in (3.4.6). To prove the first equation observe that

$$B_i \subseteq E_i = \mathbb{R} \setminus H_i \quad \text{and} \quad A_i = \mathbb{R} \setminus B_i,$$

for every $i \in \mathbb{N}$. Hence $B_i \subseteq \mathbb{R} \setminus H_i \Leftrightarrow \mathbb{R} \setminus B_i \supseteq H_i \Leftrightarrow A_i \supseteq H_i$ for every $i \in \mathbb{N}$. ■

Using the previous results we have established, for every $H \in \Pi^0_\alpha$ and $E \in \Sigma^0_\alpha$, $\alpha \in (1, \Omega)$, the existence of an ambiguous set A of class α such that $H \subseteq A \subseteq E$. Another proof of this fact, which is worth recalling here, was found by Wacław Sierpiński in 1924. Before that we need a lemma.

Lemma 3.4.1 (Structure of ambiguous sets)[2] *Let $\{E_i\}_{i\geq 1}$ be a decreasing sequence of sets and $\{H_i\}_{i\geq 1}$ an increasing sequence of sets such that $\bigcap_{i=1}^{\infty} E_i \subseteq \bigcup_{i=1}^{\infty} H_i$. Then*

$$\bigcup_{i=1}^{\infty}(H_i \cap E_i) = E_1 \cap \left(\bigcap_{i=1}^{\infty}\left(H_i \cup E_{i+1}\right)\right).$$

Proof Take $\{E_i\}_{i\geq 1}$ and $\{H_i\}_{i\geq 1}$ as in the statement and define

$$A_1 = \bigcup_{i=1}^{\infty}(H_i \cap E_i) \quad \text{and} \quad A_2 = E_1 \cap \left(\bigcap_{i=1}^{\infty}\left(H_i \cup E_{i+1}\right)\right).$$

If x is any element in A_1, there exists an $n \in \mathbb{N}$ such that $x \in (H_n \cap E_n)$. As $\{H_i\}$ is increasing, $x \in H_i$ for any $i \geq n$, whilst $\{E_i\}$ decreasing implies $x \in E_i$ for $i \leq n$. Hence $x \in E_1$ and $x \in (H_i \cup E_{i+1})$ for $i = 1, 2, \ldots$, i.e. $x \in A_2$ and $A_1 \subseteq A_2$.

For the opposite inclusion take $x \in A_2$ and define

$$E = \bigcap_{i=1}^{\infty} E_i \quad \text{and} \quad H = \bigcup_{i=1}^{\infty} H_i.$$

If $x \in E$, since $\bigcap_{i=1}^{\infty} E_i \subseteq \bigcup_{i=1}^{\infty} H_i$ there exists an $n \in \mathbb{N}$ such that $x \in H_n$. As $x \in \bigcap_{i=1}^{\infty} E_i$, then $x \in E_n$ for every n, and so $x \in (H_n \cap E_n)$. Therefore $x \in A_1$. If $x \notin E \Rightarrow x \in H$, there is some $i \in \mathbb{N}$ such that $x \notin E_i$. Let m be the smallest index $i = 1, 2, \ldots$ for which this happens. Clearly $m \neq 1$, since $x \in A_2$, so $x \in E_1$. But then $m = n + 1$, for $n = 1, 2, \ldots$, and so $x \in E_n$. The fact that $x \in A_2 = E_1 \cap \left(\bigcap_{i=1}^{\infty}\left(H_i \cup E_{i+1}\right)\right)$ implies $x \in (H_n \cup E_{n+1})$, so $x \in H_n$. Immediately then $x \in (H_n \cap E_n)$, and hence $x \in A_1$. In this case, too, we conclude $x \in A_1$, so in general $A_2 \subseteq A_1$. Eventually, $A_1 = A_2$. ■

Theorem 3.4.5 (Ambiguous sets of class α)[3] *Let $\alpha \in (1, \Omega)$ and consider two sets $E \in \Sigma^0_\alpha$ and $H \in \Pi^0_\alpha$ such that $H \subseteq E$. Then there exists a set A living in both Σ^0_α and Π^0_α, i.e. $A \in \Delta^0_\alpha$, such that $H \subseteq A \subseteq E$.*

Proof Take $E \in \Sigma^0_\alpha$ and $H \in \Pi^0_\alpha$ such that $H \subseteq E$. Clearly $E = \bigcup_{i=1}^{\infty} H_i^\beta \in \Sigma^0_\alpha$, where $H_i^\beta \in \bigcup_{\beta<\alpha} \Pi^0_\beta$. We can always choose

[2] W.Sierpiński, *Sur une propriété des ensembles ambigus* Fundamenta Mathematicae 6 (1924), 1–5.

[3] W.Sierpiński, *Sur une propriété des ensembles ambigus*, Fundamenta Mathematicae 6 (1924), 1–5.

$$H_1' = H_1 \subseteq H_2' = (H_1 \cup H_2) \subseteq H_3' = (H_1 \cup H_2 \cup H_3) \subseteq \dots,$$

so that $\{H_i'\}_{i\geq 1}$ is an increasing sequence in Π^0_β. At the same time $H = \bigcap_{i=1}^\infty E_i^\gamma \in \Pi^0_\alpha$, where $E_i^\gamma \in \bigcup_{\gamma<\alpha} \Sigma^0_\gamma$. As before we may choose

$$E_1' = E_1 \supseteq E_2' = (E_1 \cap E_2) \supseteq E_3' = (E_1 \cap E_2 \cap E_3) \supseteq \dots,$$

so that $\{E_i'\}_{i\geq 1}$ is a decreasing sequence in $\bigcup_{\gamma<\alpha} \Sigma^0_\gamma$.

We immediately note that the E_i^γ and the H_i^β satisfy the requirements of lemma 3.4.1. Put

$$\bigcup_{i=1}^\infty (H_i^\beta \cap E_i^\gamma) = A = E_1^\gamma \cap \left(\bigcap_{i=1}^\infty \left(H_i^\beta \cup E_{i+1}^\gamma\right)\right).$$

Then

$$A = \bigcup_{i=1}^\infty (H_i^\beta \cap E_i^\gamma) \subseteq \bigcup_{i=1}^\infty H_i^\beta = E^\alpha;$$

$$H^\alpha = \bigcap_{i=1}^\infty E_i^\gamma \subseteq E_1^\gamma \cap \left(\bigcap_{i=1}^\infty \left(H_i^\beta \cup E_{i+1}^\gamma\right)\right) = A;$$

and so $H \subseteq A \subseteq E$. Now to the first claim. By Proposition 3.4.1, *2.*

$$\forall \beta < \alpha,\ \Pi^0_\beta \subseteq \bigcup_{\beta<\alpha} \Pi^0_\beta \subseteq \Sigma^0_\alpha \Rightarrow \forall i \in \mathbb{N},\ H_i^\beta \in \Sigma^0_\alpha;$$

and

$$\forall \gamma < \alpha,\ \Sigma^0_{\gamma+1} \subseteq \Sigma^0_\alpha \Rightarrow \forall i \in \mathbb{N},\ E_i^\gamma \in \Sigma^0_\alpha.$$

What is more, since Σ^0_α is closed under finite intersections,

$$\left(H_i^\beta \cap E_i^\gamma\right) \in \Sigma^0_\alpha.$$

The class Σ^0_α is also stable under countable unions, so

$$A = \bigcup_{i=1}^\infty (H_i^\beta \cap E_i^\gamma) \in \Sigma^0_\alpha.$$

And then, still by item *2.* in the aforementioned proposition,

$$\forall \gamma < \alpha,\ \Sigma^0_\gamma \subseteq \bigcup_{\gamma<\alpha} \Sigma^0_\gamma \subseteq \Pi^0_\alpha \Rightarrow \forall i \in \mathbb{N},\ E^\gamma_i \in \Pi^0_\alpha;$$

and

$$\forall \beta < \alpha,\ \Pi^0_\beta \subseteq \Pi^0_\alpha \Rightarrow \forall i \in \mathbb{N},\ H^\gamma_i \in \Pi^0_\alpha.$$

Given that Π^0_α is stable under finite unions,

$$\left(H^\beta_i \cup E^\gamma_{i+1}\right) \in \Pi^0_\alpha.$$

And since it is stable under countable intersections,

$$A = E^\gamma_1 \cap \left(\bigcap_{i=1}^{\infty} \left(H^\beta_i \cup E^\gamma_{i+1}\right)\right) \in \Pi^0_\alpha.$$

All in all, the set A belongs in both Π^0_α and Σ^0_α. ■

Proposition 3.4.6 (Sequences of ambiguous sets) *Given any $\alpha \in (2, \Omega)$, if a set $A \subseteq \mathbb{R}$ belongs in Δ^0_α, there exists a sequence $\{A_i\}_{i\geq 1}$ of ambiguous sets of class $< \alpha$ such that $A = \lim_{i\to\infty} A_i$.*[4]

Proof Let $A \in \Delta^0_\alpha$ for $\alpha \in (2, \Omega)$. Then A belongs both in Σ^0_α and in Π^0_α, so

$$\bigcup_{i=1}^{\infty} H_i = A = \bigcap_{i=1}^{\infty} E_i,$$

where, for every $i \in \mathbb{N}$, H_i is multiplicative of class $< \alpha$ and E_i additive of class $< \alpha$. Without loss of generality we may assume $H_i \subseteq H_{i+1}$ and $E_{i+1} \subseteq E_i$. Theorem 3.4.5 (or the separation theorem) ensures the existence of an ambiguous set A_i of class $< \alpha$ such that

$$H_i \subseteq A_i \subseteq E_i$$

for every i. If $x \in \limsup_{i\to\infty} A_i$, then $x \in A_i$ for countably many indices i. Hence $x \in E_i$ for every i and

$$\limsup_{i\to\infty} A_i \subseteq A.$$

On the other hand $x \in A$ implies $x \in H_i$ for all but finitely many indices i. Since $H_i \subseteq A_i$ for each $i \in \mathbb{N}$, we obtain

$$A \subseteq \liminf_{i\to\infty} A_i.$$

[4] In case the space is a metrisable zero-dimensional and separable space, the claim holds also when $\alpha = 2$.

The two inclusions allow to conclude. ■

Theorem 3.4.6 (Finite collections of subsets in Π^0_α) *Let $H_1, \ldots, H_k$ be a finite collection of pairwise-disjoint sets in class Π^0_α, with $\alpha \in (1, \Omega)$. There exists a corresponding finite collection of pairwise-disjoint sets $A_1, \ldots, A_k$ in Δ^0_α such that*

$$\mathbb{R} = A_1 \cup \ldots \cup A_k \quad \textit{and} \quad H_i \subseteq A_i, \ \forall i = 1, \ldots, k.$$

Proof The proof is by induction on $i = 1, \ldots, k$. The theorem is clearly true for $k = 1$, so we assume it true for $k - 1$, $k \geq 2$.

Then we have a finite collection $\{A_1, \ldots, A_{k-2}, A^*\}$, made of pairwise-disjoint sets in Δ^0_α such that

$$\mathbb{R} = A_1 \cup A_2 \cup \ldots \cup A_{k-2} \cup A^*, \quad H_1 \subseteq A_1, \ldots, H_{k-2} \subseteq A_{k-2}, \quad (H_{k-1} \cup H_k) \subseteq A^*.$$

As in the proof of the separation theorem, let us apply the reduction theorem to H_{k-1}, H_k and use A^* as ambient space:

$$\begin{aligned} E_{k-1} &= A^* \setminus H_{k-1} \qquad \text{and} \qquad A_{k-1} = A^* \setminus B_{k-1} \\ E_k &= A^* \setminus H_k \qquad \text{and} \qquad A_k = A^* \setminus B_k. \end{aligned} \tag{3.4.10}$$

By assumption $H_{k-1} \cap H_k = \emptyset$, and Eq. (3.4.10) tells

$$A_{k-1} \cap A_k = A^* \setminus (B_{k-1} \cup B_k). \tag{3.4.11}$$

The reduction theorem then gives

$$B_{k-1} \subseteq E_{k-1}, \quad B_k \subseteq E_k \quad B_{k-1} \cup B_k = E_{k-1} \cup E_k \quad \text{and} \quad B_{k-1} \cap B_k = \emptyset, \tag{3.4.12}$$

so

$$E_{k-1} \cup E_k = A^* \setminus (H_{k-1} \cap H_k) = A^* = B_{k-1} \cup B_k. \tag{3.4.13}$$

By assumption A^* is ambiguous of class α, so the sets E_{k-1}, E_k, B_{k-1} and B_k are all of the same type. Comparing the last equality in (3.4.13) with Eq. (3.4.11) implies $A_{k-1} \cap A_k = \emptyset$. Furthermore, comparing (3.4.10) and (3.4.12) gives

$$B_{k-1} \subseteq A^* \setminus H_{k-1} \Leftrightarrow A^* \setminus B_{k-1} \supseteq H_{k-1} \quad \text{and} \quad B_k \subseteq A^* \setminus H_k \Leftrightarrow A^* \setminus B_k \supseteq H_k$$

and

$$A_{k-1} \cup A_k = \big(A^* \setminus B_{k-1}\big) \cup \big(A^* \setminus B_k\big) = A^* \setminus \big(B_{k-1} \cap B_k\big) = A^*,$$

where A_{k-1}, A_k are in class Δ^0_α in A^*, and therefore in $\mathbb{R}$ as well. ■

Chapter 4
Baire Functions on $\mathbb{R}$



4.1 Introduction

The mathematician Peter Gustav Lejeune Dirichlet, while striving to make Fourier theory rigorous and thus prove the conjecture that any map can be represented as a Fourier series, in 1829 gave sufficient conditions for said representation to exist, and demonstrated the following result.

Theorem 4.1.1 (Sufficient conditions for the convergence of Fourier representations) *The Fourier series of a function f converges on the interval $(-\pi, \pi)$ if the function has a finite number of discontinuities, maxima and minima in that interval.*

In such a case the Fourier series converges pointwise to f where f is continuous, and converges to $\frac{[f(x^+)+f(x^-)]}{2}$ where f is discontinuous. ■

In the attempt of weakening the Dirichlet conditions and so extend the reach of the theorem, Dirichlet first of all examined functions with infinitely many discontinuities. In doing that he came up with the celebrated function that nowadays bears his name

$$f(x) = \begin{cases} c & \text{if } x \in \mathbb{Q} \\ d & \text{if } x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

"The function thus defined—says Dirichlet—*assumes given finite values for every value x, and nevertheless it would not be possible to substitute it in the series, because the various integrals appearing in the series would cease to make sense"*. Dirichlet only had the Cauchy integral at his disposal, which is well defined only for maps with at most finitely many discontinuities in a given interval.

The arbitrary nature of the Dirichlet function, although obvious in our eyes, struck at the time as an absolute novelty. Before him other mathematicians—such as Fourier

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S. Gentili, *Measure, Integration and a Primer on Probability Theory*, UNITEXT 125, https://doi.org/10.1007/978-3-030-54940-4_4

and Cauchy—had expressed the need for a more general notion of function, but in practice they only developed their theories using functions with well-behaved curves as graphs. Dirichlet was the first to open Analysis to a broader notion of function, expressed in terms of an arbitrary correspondence between x- and y-values, rather than through an analytical formula.

The Dirichlet function not only seemed to not possess an analytical expression; it could not even be represented graphically, and was among the first examples of a function defined on a proper interval of $\mathbb{R}$. The Dirichlet function opened the gates to a wealth of pathological functions, defined to probe the most disparate concepts of mathematical analysis. As a consequence the geometric point of view completely lost its importance, and the word 'function' acquired a meaning that was clearly independent of any explicit expression. Whilst the old Analysis had erected an extremely tidy and orderly building, that moment witnessed the birth of a major endeavour to find irregularities and exceptions.

Among the greatest scholars seriously contributing to the field was Riemann, who in his 1854 Habilitation thesis extended the concept of Cauchy integral. As the coefficients of a Fourier series are determined by integrals, Riemann enlarged the class of maps admitting a Fourier-series representation. The class of functions to which the new notion of Riemann integral could be applied consisted of bounded maps with infinitely many discontinuities. As we know, Riemann found a discontinuous and bounded function that, like the Dirichlet function, cannot be plotted. In contrast to Dirichlet's example though, Riemann's function was immediately written in its analytical form:

$$R(x) = \frac{(x)}{1^2} + \frac{(2x)}{2^2} + \frac{(3x)}{3^2} + \cdots .$$

A few years later (1872) Weierstraß shocked the mathematical community with the discovery of a continuous, nowhere-differentiable map:

$$f(x) = \sum_{n=1}^{\infty} b^n \cos(a^n \pi x),$$

where a is an odd integer and $b \in (0, 1)$, with $ab > 1 + 3\pi/2$. This example allowed Weierstraß to detach the idea of continuity from that of differentiability, notions that until 1870 had been thought to be deeply intertwined. The most popular analysis textbooks of the time contained proofs that a continuous maps had to be differentiable at all points except for a finite number. Cauchy, too, was convinced of this fact.

Weierstraß demonstrated the great worth of counterexamples for attaining a standard of rigour higher than the typical, plausible, notions of the time. Any truth that had earlier been considered reasonable now needed diligent justifications and proper proofs, also through the pursuit of counterexamples. All in all the mathematical community felt the urge to solidify the foundations of Analysis, and the profound methodological revision that was eventually carried out became known as the 'arithmetisation' of analysis. Weierstraß was one the proponents of this programme, which

by the way inaugurated a novel branch, namely the "theory of functions of real variable". The existence of exceptions to regular patterns slowly became accepted, and the discovery of continuous non-differentiable functions, non-integrable derivatives, non-integrable limits of integrable functions and other pathological occurrences, furnished the tools to uncover the greatest possible number of exceptional phenomena with the aim to find the laws to classify them.

Needless to say not everyone was enthusiastic about the new direction mathematics was heading towards, among which were Hermite and Poincaré. Hermite in 1893 declared his awe and horror for functions that could not be differentiated. In 1899 Poincaré wrote that the exasperated quest of logically unassailable arguments often generated teratological objects.

The times were ripe for the next natural leap forward, which allowed to establish which among the many 'monsters' could be described by effective "analytical expressions". The latter term had been used until then in a very vague, imprecise and informal way. The question of which functions (*à la* Dirichlet) could be represented analytically was first posed by the Italian mathematician Ulisse Dini in 1879, and subsequently addressed by a young René Baire in his 1898 PhD thesis.

For starters, Baire clarified what representing a function analytically should actually mean. He began from the Weierstraß approximation theorem, published in 1885, whereby "any continuous map $f : [a, b] \mapsto \mathbb{R}$ is a uniform limit of polynomials $p : [a, b] \mapsto \mathbb{R}$". He defined functions of class 0 to be the continuous maps $f : [a, b] \mapsto \mathbb{R}$. Then he called class-1 functions those not of class 0 that arise as pointwise limits of sequences of class-0 functions. So if $\{f_n(x)\}_{n\geq 1}$ is a sequence of continuous maps, defined over a common domain $[a, b]$, that converge pointwise on $[a, b]$, the limit

$$f(x) = \lim_{n\to\infty} f_n(x)$$

is of class 1. The importance of class-1 functions resides in the fact that the first derivatives are of class 1. In fact if $f : [a, b] \mapsto \mathbb{R}$ is everywhere differentiable on $[a, b]$, for every $x \in [a, b]$ the derivative

$$f'(x) = \lim_{n\to\infty} \frac{f(x + 1/n) - f(x)}{1/n}$$

exists, and it can be recovered from the sequence

$$\left\{ f_n(x) = \frac{f(x + 1/n) - f(x)}{1/n} \right\}_{n\geq 1}$$

of continuous maps on $[a, b]$ that pointwise converge precisely to $f'(x)$. It is known that the pointwise limit of a sequence of continuous maps is not necessarily continuous, but the limit's continuity is warranted under the additional assumption of uniform convergence. Nonetheless the pointwise limit of a sequence of continuous

maps somehow retains a trace of the sequence's continuity, a point on which Baire focused his attention to formulate important results.

Moving ahead in the construction of the successive classes, Baire called class m the collection of maps that do not belong to any of the preceding classes but can be represented as limits of functions in class $m-1$. The process goes on by transfinite induction to all ordinals α less than the first uncountable ordinal Ω. The class of Baire functions, made of the union of all classes $\alpha < \Omega$, is therefore closed under pointwise limits.

Baire said a function is representable analytically if it belongs to one of the above classes. Put equivalently, a function admits an analytic representation in the sense of Baire if it can be obtained from a variable and constants using a finite or countable number of sums, multiplications and pointwise limits. As one can guess, the class of Baire functions is gigantic. For example, discontinuous maps that admit a Fourier series are of class one. Therefore the functions that can be represented by Fourier series are just a small fraction of Baire's representable functions. This finally answered negatively the question of Fourier on whether any map had a Fourier series. Baire, furthermore, proved that the pathological function of Dirichlet is of class 2, being the pointwise limit of class-one maps.

Lebesgue carried on this study and in 1905 showed that every Baire class is non-empty, and the entire class of Baire functions does not exhaust all functions. There are in fact Lebesgue measurable functions that are not Baire. At the same time Lebesgue proved that to any Borel measurable function f there corresponds a Baire function that differs from f only by a zero-measure set. Lebesgue's examples, which he found using a rather complex method and relying on the axiom of choice, established the existence of functions not captured by Baire's classification, and hence that there are indeed functions without a Baire representation.

Lebesgue's discovery had a deep impact on mathematicians. It revealed that the logical definition of a function is much broader that the purely analytical one, since a logical definition leads to functions that cannot be obtained from polynomials by taking finitely/countably many limits.

The same conclusion can be obtained by a non-constructive argument, i.e. by showing that the class of Baire functions has cardinality $\mathbf{c}$, whereas the class of all real functions has cardinality $2^{\mathbf{c}}$. Consequently there are uncountably many functions that cannot be represented analytically in the sense of Baire.

We should observe, on the other hand, that within analytically representable functions not all maps have an analytic expression in the sense of Baire. There do exist maps with analytic representation that cannot be brought into Baire's form. An example for this situation was given by Nikolai Lusin. For a suitable choice of polynomials $P_{m,n}(x, y)$ the function

$$f(x) = \overline{\lim_{y\to\infty}}\ \overline{\lim_{m\to\infty}}\ \overline{\lim_{n\to\infty}} P_{m,n}(x, y),$$

where the limits on m, n are countable while the limit on y is uncountable, cannot be expressed as Baire function.

To sum up, not every function in Dirichlet's generalised sense, i.e. a completely general correspondence between the dependent and independent variables, possesses an analytical representation in the sense of Baire. That said, finding such an example is extremely complicated.

Many authors, among which Hausdorff, defined Baire classes so to include continuous maps in class 1, continuous maps and class-1 maps in class 2, etc. This definition, as we shall see in Chap. 5, proceeds in parallel to the classification of Borel sets, and makes it immediate to link the two schemes.

4.2 Baire Functions on $\mathbb{R}$

The various classes of Baire functions, as we shall see, are conceptually related to Borel classes.

Before we explain the recursive mechanism defining Baire classes we must explain a little about pointwise and uniform convergence of sequences of $\bar{\mathbb{R}}$-valued maps.

Definition 4.2.1 (*Pointwise convergence of sequences of functions*) A sequence of functions $\{f_n\}_{n\geq 1}$, with $f_n : X \mapsto \bar{\mathbb{R}}$ for every $n \in \mathbb{N}$, converges pointwise to a map $f : X \mapsto \mathbb{R}$ if, given arbitrary $x \in X$ and $\epsilon > 0$, there exists an $\bar{n}$ such that

$$|f_n(x) - f(x)| < \epsilon$$

for every $n \geq \bar{n}$. ◇

Definition 4.2.2 (*Uniform convergence of sequences of functions*) A sequence of functions $\{f_n\}_{n\geq 1}$, with $f_n : X \mapsto \bar{\mathbb{R}}$ for every $n \in \mathbb{N}$, converges uniformly to $f : X \mapsto \mathbb{R}$ if, given any $\epsilon > 0$, there exists an $\bar{n}$ such that

$$|f_n(x) - f(x)| < \epsilon$$

for every $n \geq \bar{n}$ and every $x \in X$. ◇

The classification of Baire functions begins, as we said, from continuous maps, which are in class 0, and which we denote by

$$\mathscr{B}_0 = \left\{ f^0 : \mathbb{R} \mapsto \mathbb{R} : f^0(x) \text{ is continuous on } \mathbb{R} \right\}.$$

Pointwise limits of convergent sequences of continuous maps are in class 1, indicated by

$$\mathscr{B}_1 = \left\{ f^1 : \mathbb{R} \mapsto \mathbb{R} : f^1(x) = \lim_{n\to\infty} f_n^0(x) \text{ and } \forall n,\, f_n^0(x) \in \mathscr{B}_0 \right\}.$$

If β is a successor ordinal, we define Baire functions of order β to form the class

$$\mathscr{B}_\beta = \left\{ f^\beta : \mathbb{R} \mapsto \mathbb{R} : f^\beta(x) = \lim_{n\to\infty} f_n^{\beta-1}(x) \text{ and } \forall n,\, f_n^{\beta-1}(x) \in \mathscr{B}_{\beta-1} \right\}.$$

Suppose we have defined all classes of order $\beta < \alpha$, with α a limit ordinal. Then we call

$$\mathscr{B}_\alpha = \left(\bigcup_{\beta<\alpha} \mathscr{B}_\beta \right)$$

Baire class of order α. By transfinite induction the class of all Baire functions over $\{\alpha : 0 \leq \alpha < \Omega\}$ is the family

$$\mathscr{B}_\Omega = \left(\bigcup_{\alpha<\Omega} \mathscr{B}_\alpha \right).$$

Clearly, any continuous map f can be viewed as pointwise limit of a convergent sequence of functions all equal to f, so $\mathscr{B}_0 \subseteq \mathscr{B}_1$. A similar reasoning holds for every successive Baire class, whence

$$\mathscr{B}_0 \subseteq \mathscr{B}_1 \subseteq \ldots \subseteq \mathscr{B}_\alpha \subseteq \ldots, \text{ for } \alpha \in [0, \Omega).$$

We shall see that the cardinality of Baire functions is $\mathbf{c}$, while the cardinality of Lebesgue integrable functions is $2^{\mathbf{c}}$.

Definition 4.2.3 (*Baire classes of analytically representable functions*) The family of analytically representable functions in the sense of Baire is the smallest family of functions containing:

1. all continuous maps;
2. all limits of pointwise convergent sequences in the family.

Such a collection can be stratified into classes $\mathscr{B}_\alpha$, $\alpha \in [0, \Omega)$, as follows:

(1') class $\mathscr{B}_0$ consists of continuous maps on $\mathbb{R}$;
(2') class $\mathscr{B}_\alpha$, for $\alpha \in (0, \Omega)$, consists of pointwise limits of real functions in class $\mathscr{B}_\beta$, $\beta < \alpha$.

$\diamond$

Proposition 4.2.1 (Operations on Baire functions) *Let $f(x) : \mathbb{R} \mapsto \mathbb{R}$ and $g(x) : \mathbb{R} \mapsto \mathbb{R}$ be functions in Baire class $\mathscr{B}_\alpha$, $\alpha < \Omega$. Then the following functions belong in $\mathscr{B}_\alpha$:*

1. *$k \cdot f(x)$, for every given $k \in \mathbb{R}$;*
2. *$f(x) + g(x)$ and $f(x) \cdot g(x)$;*
3. *$\frac{1}{f(x)}$, provided $f(x) \neq 0$;*

4. $|f(x)|$;
5. $\max(f(x), g(x))$ *and* $\min(f(x), g(x))$.

Proof

1. When $\alpha = 0$ the proposition is true, since $f(x)$ is continuous on $\mathbb{R}$ and so is $k \cdot f(x)$ for every given $k \in \mathbb{R}$.
   As first inductive step assume β is a successor ordinal, and that for every $x \in \mathbb{R}$ and for every given $k \in \mathbb{R}$

$$f^{\beta}(x) \in \mathscr{B}_{\beta} \Rightarrow k \cdot f^{\beta}(x) \in \mathscr{B}_{\beta}.$$

   Given the map

$$f^{\beta+1} = \lim_{n\to\infty} f_n^{\beta},$$

   we must prove $\left(k \cdot f^{\beta+1}\right) \in \mathscr{B}_{\beta+1}$ for every $x \in \mathbb{R}$ and for every given $k \in \mathbb{R}$. Let $\{f_n\}_{n\geq 1}$ be a sequence in Baire class $\mathscr{B}_{\beta}$ with pointwise limit $f^{\beta+1}$. By induction, for every $n \in \mathbb{N}$, $x \in \mathbb{R}$ and $k \in \mathbb{R}$

$$k \cdot f_n^{\beta} \in \mathscr{B}_{\beta} \Rightarrow \lim_{n\to\infty} \left(k \cdot f_n^{\beta}\right) = k \cdot \lim_{n\to\infty} f_n^{\beta} = k \cdot f^{\beta+1}.$$

   For the second inductive step suppose, for every $\beta < \alpha$ with α limit ordinal, every $x \in \mathbb{R}$ and every given $k \in \mathbb{R}$, that

$$f^{\beta}(x) = \lim_{n\to\infty} f_n^{\beta-1}(x) \in \mathscr{B}_{\beta} \Rightarrow k \cdot f^{\beta}(x) = \lim_{n\to\infty} k \cdot f_n^{\beta-1}(x) \in \mathscr{B}_{\beta}.$$

   We claim $f^{\alpha}(x) \in \mathscr{B}_{\alpha}$ forces $k \cdot f^{\alpha}(x) \in \mathscr{B}_{\alpha}$. Since $f^{\alpha}(x) = \lim_{n\to\infty} f_n^{\beta}(x)$, with $\beta < \alpha$ for every $n \in \mathbb{N}$, and by induction the map $k \cdot f_n^{\beta}(x)$ belongs to some $\mathscr{B}_{\beta}$ for every $n \in \mathbb{N}$, with $\beta < \alpha$, then

$$\lim_{n\to\infty} k \cdot f_n^{\beta}(x) = k \cdot \lim_{n\to\infty} f_n^{\beta}(x) = k \cdot f^{\alpha}(x)$$

   for every $x \in \mathbb{R}$. Therefore $k \cdot f^{\alpha}(x) \in \mathscr{B}_{\alpha}$.
   From this if $f(x)$ is a Baire function in class $\mathscr{B}_{\alpha}$, $\alpha < \Omega$, then $-f(x)$ is, too.
2. For $\alpha = 0$ the proposition is true, for if $f(x), g(x)$ are continuous on $\mathbb{R}$, also $f(x) + g(x)$ and $f(x) \cdot g(x)$ are continuous on $\mathbb{R}$.
   For the first inductive step, given f^{β} and g^{β}, β a successor ordinal, let

$$\left(f + g\right)^{\beta} \in \mathscr{B}_{\beta} \quad \text{and} \quad \left(f \cdot g\right)^{\beta} \in \mathscr{B}_{\beta}.$$

   Take sequences $\{f_n^{\beta}\}_{n\geq 1}$ and $\{g_n^{\beta}\}_{n\geq 1}$ pointwise converging to $f^{\beta+1}$ and $g^{\beta+1}$ respectively. Then

$$\lim_{n\to\infty} f_n^\beta + \lim_{n\to\infty} g_n^\beta = \lim_{n\to\infty} \big(f_n + g_n\big)^\beta = \big(f+g\big)^{\beta+1} \in \mathscr{B}_{\beta+1}.$$

The argument for $\big(f \cdot g\big)^{\beta+1}$ is similar.
For the second inductive step suppose, for every $\beta < \alpha$, α a limit ordinal, that the first step holds. Take pointwise convergent sequences

$$\{f_n^\eta\}_{n\geq 1} \quad \text{and} \quad \{g_n^\gamma\}_{n\geq 1},$$

to f^α and g^α respectively. For every $n \in \mathbb{N}$ the function $(f_n^\eta + g_n^\gamma)$ belongs in some class $\mathscr{B}_\beta$, with $\beta = \max(\eta, \gamma) < \alpha$, so

$$\lim_{n\to\infty} f_n^\eta + \lim_{n\to\infty} g_n^\gamma = \lim_{n\to\infty} \big(f_n + g_n\big)^\beta = \big(f+g\big)^{\alpha} \in \mathscr{B}_{\alpha}.$$

The case of $\big(f \cdot g\big)^{\alpha}$ is analogous.
From *1.* and *2.* if $f(x)$, $g(x)$ are Baire functions in $\mathscr{B}_\alpha$, $\alpha < \Omega$, then $f(x) - g(x)$ is in the same class.

3. The proposition is obvious for $\alpha = 0$. We leave the proof of the first inductive step as an exercise.
Now let $\alpha < \Omega$ be a limit ordinal, and suppose by hypothesis that for every $\beta < \alpha$ and every $x \in \mathbb{R}$

$$f(x) \in \mathscr{B}_\beta \Rightarrow \frac{1}{f(x)} \in \mathscr{B}_\beta.$$

Let then $f^\alpha(x) \in \mathscr{B}_\alpha$. Then $f^\alpha(x) = \lim_{n\to\infty} f_n^\beta(x)$, with $\beta < \alpha$, for every $n \in \mathbb{N}$. For every n item *2.* implies $f_n^\beta(x) \cdot f_n^\beta(x) = (f_n^\beta(x))^2$ is again in class $\mathscr{B}_\beta$. Moreover the map $g_n(x) = 1/n$ is continuous for every n and hence in class $\mathscr{B}_0$ for every $n \in \mathbb{N}$. Besides, $g(x) = \lim_{n\to\infty} g_n(x)$, as limit of continuous maps, is in $\mathscr{B}_1$, and so also in $\mathscr{B}_\beta$ for every $n \in \mathbb{N}$. By the previous item *2.*,

$$(f_n^\beta(x))^2 + g_n(x) = (f_n^\beta(x))^2 + \frac{1}{n}$$

is a function in class $\mathscr{B}_\beta$. Since, for every $n \in \mathbb{N}$, $(f_n^\beta(x))^2 + \frac{1}{n} \neq 0$ for every $x \in \mathbb{R}$, we have

$$\frac{1}{(f_n^\beta(x))^2 + \frac{1}{n}} \neq 0, \forall x \in \mathbb{R}.$$

Using *2.* again, for every $n \in \mathbb{N}$

$$\frac{f_n^\beta(x)}{(f_n^\beta(x))^2 + \frac{1}{n}} \in \mathscr{B}_\beta.$$

Therefore for $\beta < \alpha$ and every $n \in \mathbb{N}$

$$\lim_{n\to\infty} \frac{f_n^\beta(x)}{(f_n^\beta(x))^2 + \frac{1}{n}} = \frac{1}{f^\alpha(x)}.$$

But the denominator f^α is never zero, and the limit exists, hence

$$\frac{1}{f^\alpha(x)} \in \mathscr{B}_\alpha.$$

By transfinite induction the proposition holds for every ordinal $\alpha < \Omega$.
What we have proved, together with item *2.*, implies that if $f(x)$, $g(x)$ are in class $\mathscr{B}_\alpha$, and $g(x) \neq 0$ for every $x \in \mathbb{R}$, then $f(x)/g(x)$ belongs in $\mathscr{B}_\alpha$.

4. The proposition is patent for $\alpha = 0$.
First inductive step (β successor): for every $x \in \mathbb{R}$ suppose $f^\beta(x), |f^\beta(x)|$ belong to the same class. Let $\{f_n\}_{n\geq 1}$ be a pointwise convergent sequence in class $\mathscr{B}_\beta$ with limit $f^{\beta+1}$, so by hypothesis

$$\forall n \in \mathbb{N},\ f_n^\beta \Leftrightarrow |f_n^\beta| \Rightarrow \lim_{n\to\infty} |f_n^\beta| = \Big|\lim_{n\to\infty} f_n^\beta\Big| = |f^{\beta+1}|.$$

Second inductive step (the above holds for every $\beta < \alpha$, α limit ordinal): let $\{f_n^\beta\}_{n\geq 1}$ have pointwise limit f^α, where for every $n \in \mathbb{N}$ the map f_n^β belongs to some $\mathscr{B}_\beta$, $\beta < \alpha$. Then

$$\forall n \in \mathbb{N},\ f_n^\beta \Leftrightarrow |f_n^\beta| \Rightarrow \lim_{n\to\infty} |f_n^\beta| = \Big|\lim_{n\to\infty} f_n^\beta\Big| = |f^\alpha|.$$

By transfinite induction the proposition is true for every ordinal $\alpha < \Omega$.

5. Take $f(x)$, $g(x)$ in Baire class $\mathscr{B}_\alpha$, so by *1.*, *2.* and *4.* we have

$$\frac{1}{2}[f(x) + g(x)] \quad , \quad \frac{1}{2}|f(x) - g(x)| \quad \text{and} \quad -\frac{1}{2}|f(x) - g(x)|.$$

Then the functions

$$\max(f(x), g(x)) = \frac{1}{2}[f(x) + g(x)] + \frac{1}{2}|f(x) - g(x)|;$$
$$\min(f(x), g(x)) = \frac{1}{2}[f(x) + g(x)] - \frac{1}{2}|f(x) - g(x)|$$

are in class $\mathscr{B}_\alpha$.

■

Proposition 4.2.2 (Baire class of composites with continuous maps) *If $f : \mathbb{R} \mapsto \mathbb{R}$ is continuous and $g : \mathbb{R} \mapsto \mathbb{R}$ belongs to class $\mathscr{B}_\alpha$, the composite $f\big[g(x)\big]$ is in class $\mathscr{B}_\alpha$.*

Proof The theorem is true for $\alpha = 0$, since in that case f and g are continuous and their composition is continuous.

So let β be a successor ordinal, and by inductive hypothesis suppose $f(g)$ is in class $\mathscr{B}_\beta$. Given a map $g^{\beta+1}$, let $\{g_n\}_{n\geq 1}$ be a sequence in class $\mathscr{B}_\beta$ such that $\lim_{n\to\infty} g_n^\beta = g^{\beta+1}$. Then for every $n \in \mathbb{N}$ the function $f(g_n^\beta)$ is in $\mathscr{B}_\beta$. But as $f(x)$ is continuous on $\mathbb{R}$, then for every $x \in \mathbb{R}$ the limit

$$\lim_{n\to\infty} f\big[g_n^\beta(x)\big] = f\big[g^{\beta+1}(x)\big]$$

exists.

Take now $\alpha < \Omega$ to be a limit ordinal, and suppose the first part holds for every $\beta < \alpha$. Let $g(x)$ be in $\mathscr{B}_\alpha$ and $\{g_n^\beta\}_{n\geq 1}$ a sequence with limit

$$g(x) = \lim_{n\to\infty} g_n^\beta(x),$$

where g_n^β belongs, for every $n \in \mathbb{N}$, to some $\mathscr{B}_\beta$, $\beta < \alpha$. As $f(x)$ is continuous, for every $x \in \mathbb{R}$

$$\lim_{n\to\infty} f\big[g_n^\beta(x)\big] = f\big[g(x)\big]$$

exists. By hypothesis, moreover, for every $n \in \mathbb{N}$ the map $f\big[g_n^\beta(x)\big]$ belongs to some $\mathscr{B}_\beta$, $\beta < \alpha$, so $f\big[g(x)\big]$ is in $\mathscr{B}_\alpha$.

Transfinite induction guarantees that the proposition holds for every ordinal $\alpha < \Omega$. ■

Up to this juncture we know that taking pointwise limits of sequences increases the (order of the) class under exam. In a short while we will see that uniform convergence, on the contrary, is enough to stay in the same class. To fully justify this fact we first need a lemma.

Lemma 4.2.1 (Bounded Baire functions) *Take an ordinal $\alpha < \Omega$ and a Baire function $f : \mathbb{R} \mapsto \mathbb{R}$ in class $\mathscr{B}_\alpha$. Suppose $|f(x)| \leq k$, for $k > 0$, for every $x \in \mathbb{R}$. Then $f(x) = \lim_{i\to\infty} f_i(x)$, where each f_i belongs in some class $\mathscr{B}_\beta$ with $\beta < \alpha$, and $|f_i(x)| \leq k$ for every $x \in \mathbb{R}$.*

Proof Take a sequence $\{g_i^\beta(x)\}_{i\geq 1}$ of functions each of which is in some $\mathscr{B}_\beta$, $\beta < \alpha$, and suppose it converges to the bounded map $f(x)$ pointwise on $\mathbb{R}$.

We may define a sequence $\{f_i^\beta(x)\}_{i\geq 1}$ of bounded maps by setting, for every $i \in \mathbb{N}$ and $x \in \mathbb{R}$,

$$f_i^\beta(x) = \begin{cases} -k & \text{if } g_i^\beta(x) < -k \\ g_i^\beta(x) & \text{if } -k \leq g_i^\beta(x) \leq k \\ k & \text{if } g_i^\beta(x) > k. \end{cases}$$

Every $f_i^{\beta}(x)$ is clearly bounded by the real number k. As $f(x)$ is bounded by k and every $g_i^{\beta}(x)$ has pointwise limit $f(x)$ on $\mathbb{R}$, the sequence $\{f_i^{\beta}(x)\}_{i\geq 1}$ is bounded by k and converges pointwise to $f(x)$ on $\mathbb{R}$. ■

Theorem 4.2.1 (Uniform convergence preserves Baire classes) *If $\{f_j^{\alpha}(x)\}_{j\geq 1}$ is a uniformly convergent sequence of Baire maps of order $\alpha < \Omega$, the limit has order α.*

Proof The statement is true for $\alpha = 0$. Suppose it holds for every $\beta < \alpha$. Given $\alpha < \Omega$, let $\{f_j^{\alpha}(x)\}_{j\geq 1}$ be a sequence in $\mathscr{B}_{\alpha}$ that is uniformly convergent, and hence absolutely convergent, to $f(x)$ on $\mathbb{R}$. Then there exists a convergent series of positive real numbers $\sum_{j=1}^{\infty} k_j$ such that for every j and $x \in \mathbb{R}$,

$$\left|f_{j+1}(x) - f_j(x)\right| \leq k_j.$$

Therefore

$$\left|f_2^{\alpha}(x) - f_1^{\alpha}(x)\right| < k_1; \left|f_3^{\alpha}(x) - f_2^{\alpha}(x)\right| < k_2; \ldots; \left|f_{j+1}^{\alpha}(x) - f_j^{\alpha}(x)\right| < k_j; \ldots;$$

moreover

$$f(x) = \sum_{j=1}^{\infty} \left[f_{j+1}^{\alpha}(x) - f_j^{\alpha}(x)\right] + f_1^{\alpha}(x).$$

By proposition 4.2.1, *2.*, for every $j \in \mathbb{N}$ the map $\left(f_{j+1}^{\alpha}(x) - f_j^{\alpha}(x)\right)$ is in $\mathscr{B}_{\alpha}$. Then lemma 4.2.1 gives a sequence of functions $\{f_{j,i}(x)\}_{i\geq 1}$, in which each $f_{j,i}(x)$ is of some class $\beta < \alpha$, such that for every j and $x \in \mathbb{R}$

$$\lim_{i\to\infty} f_{j,i}(x) = \left(f_{j+1}^{\alpha}(x) - f_j^{\alpha}(x)\right) \quad \text{and} \quad \left|f_{j,i}(x)\right| \leq k_j, \ \forall i \in \mathbb{N}.$$

Consider the sequence $\{h_i\}_{i\geq 1}$, where h_i is defined on $\mathbb{R}$ by

$$h_i(x) = f_{1,i}(x) + f_{2,i}(x) + \ldots + f_{i,i}(x) = \sum_{j=1}^{i} f_{j,i}$$

for every $i \in \mathbb{N}$. Clearly every h_i, being a finite sum of functions of some class $\beta < \alpha$, belongs in some class $\beta < \alpha$.

By the convergence of $\sum_{j=1}^{\infty} k_j$, given any $\epsilon > 0$ there exists a $j' \in \mathbb{N}$ such that, for $j \geq j'$,

$$\sum_{j=j'+1}^{\infty} k_j < \frac{\epsilon}{3},$$

and then for every $x \in \mathbb{R}$ we have

$$\left| f(x) - f_1^{\alpha}(x) - \sum_{j=1}^{j'} \left[f_{j+1}^{\alpha}(x) - f_j^{\alpha}(x) \right] \right| < \frac{\epsilon}{3}.$$

Now for an arbitrary x in $\mathbb{R}$, there exists an integer i' such that for every $j > j'$ and $i > i'$

$$\left| f_{j+1}^{\alpha}(x) - f_j^{\alpha}(x) - f_{j,i}(x) \right| < \frac{\epsilon}{3j'}.$$

Taking $i > \max\{j', i'\}$ then,

$$\begin{aligned}
&\left| \left[f(x) - f_1^{\alpha}(x) \right] - h_i(x) \right| = \\
&= \left| f(x) - f_1^{\alpha}(x) - \sum_{j=1}^{j'} \left[f_{j+1}^{\alpha}(x) - f_j^{\alpha}(x) \right] + \sum_{j=1}^{j'} \left[f_{j+1}^{\alpha}(x) - f_j^{\alpha}(x) \right] - \sum_{j=1}^{i} f_{j,i}(x) \right| \\
&= \left| f(x) - f_1^{\alpha}(x) - \sum_{j=1}^{j'} \left[f_{j+1}^{\alpha}(x) - f_j^{\alpha}(x) \right] + \sum_{j=1}^{j'} \left[f_{j+1}^{\alpha}(x) - f_j^{\alpha}(x) - f_{j,i}(x) \right] - \sum_{j=j'+1}^{i} f_{j,i}(x) \right| \\
&\leq \left| f(x) - f_1^{\alpha}(x) - \sum_{j=1}^{j'} \left[f_{j+1}^{\alpha}(x) - f_j^{\alpha}(x) \right] \right| + \left| \sum_{j=1}^{j'} \left[f_{j+1}^{\alpha}(x) - f_j^{\alpha}(x) - f_{j,i}(x) \right] \right| + \left| \sum_{j=j'+1}^{i} f_{j,i}(x) \right| \\
&< \frac{\epsilon}{3} + j' \frac{\epsilon}{3j'} + \frac{\epsilon}{3} = \epsilon.
\end{aligned}$$

So we obtain $\lim_{i \to \infty} h_i(x) = \left(f(x) - f_1^{\alpha}(x) \right)$. Therefore $\left(f(x) - f_1^{\alpha}(x) \right)$ is of class α, and since

$$f(x) = \left(f(x) - f_1^{\alpha}(x) \right) + f_1^{\alpha}(x)$$

also $f(x)$ is of class α. ■

4.3 Real Functions Over $[a, b]$ in Class $\mathscr{B}_1$

We know the uniform limit of continuous maps is a continuous map, because continuity is preserved by uniform convergence. This is not the case for pointwise convergence. As we shall see in a while, though, the pointwise limit of continuous maps, which is of Baire class one, retains some remnants of the sequence's continuity. The next lemma and theorem prove this fact.

Lemma 4.3.1 (Discontinuity set of $\mathscr{B}_1$ functions defined on $[a, b]$) *Let $f : [a, b] \mapsto \mathbb{R}$ be a function of class one. For every $\alpha > 0$ the set*

$$F_{\alpha} = \left\{ x \in [a, b] : \omega(f, x) \geq \alpha \right\}$$

is closed and nowhere dense.

Proof Proposition 3.3.1 guarantees F_α is closed. To prove the other fact suppose F_α is dense in some interval $I \subseteq [a, b]$.

As F_α is closed, it contains its limit points and so $I \subseteq F_\alpha$. Therefore the oscillation of f on $\overset{\circ}{I}$ is bigger than or equal to α, i.e. $\forall x', x'' \in I$

$$\left|f(x') - f(x'')\right| \geq \alpha. \tag{4.3.1}$$

But f is of class one, so we may write in the form

$$f(x) = \lim_{n\to\infty} f_n(x),$$

with each f_n continuous on $[a, b]$.

By the pointwise convergence of $\{f_n(x)\}_{n\geq 1}$, given any $0 < \epsilon < \alpha/8$ and $x_0 \in \overset{\circ}{I}$, there exists an $n_0 \in \mathbb{N}$ such that

$$\left|f_{n_0}(x_0) - f(x_0)\right| < \epsilon. \tag{4.3.2}$$

The continuity of the f_n gives a closed neighbourhood $[x_0 - \delta_0; x_0 + \delta_0] \cap I = I_0 \subseteq I$ of x_0 such that $\forall x \in I_0$

$$\left|f_{n_0}(x_0) - f_{n_0}(x)\right| < \epsilon. \tag{4.3.3}$$

Inequalities (4.3.2), (4.3.3) imply

$$\begin{aligned}\left|f(x_0) - f_{n_0}(x)\right| &= \left|f(x_0) - f_{n_0}(x) + f_{n_0}(x_0) - f_{n_0}(x_0)\right| \\ &\leq \left|f_{n_0}(x_0) - f_{n_0}(x)\right| + \left|f(x_0) - f_{n_0}(x_0)\right| < 2\epsilon\end{aligned} \tag{4.3.4}$$

for every $x \in I_0$. But $x_0 \in I_0 \subseteq F_\alpha$, so there exists a point $x_1 \in \overset{\circ}{I}_0$ at which the oscillation of f is at least α:

$$\left|f(x_0) - f(x_1)\right| \geq \alpha. \tag{4.3.5}$$

The sequence $\{f_n(x)\}_{n\geq 1}$ converges pointwise, so for $x_1 \in \overset{\circ}{I}_1 \subseteq I_0$ we can find an $n_1 > n_0$ such that

$$\left|f_{n_1}(x_1) - f(x_1)\right| < \epsilon. \tag{4.3.6}$$

By the continuity of f_n there exists a closed neighbourhood $[x_1 - \delta_1; x_1 + \delta_1] \cap I_0 = I_1 \subseteq I_0$ of x_1, with $\ell(I_1) < \ell(I_0)/2$, such that for every $x \in I_1$

$$\left|f_{n_1}(x_1) - f_{n_1}(x)\right| < \epsilon. \tag{4.3.7}$$

Combining (4.3.6) and (4.3.7), for every $x \in I_1$ we have

$$\begin{aligned} \left|f(x_1) - f_{n_1}(x)\right| &= \left|f(x_1) - f_{n_1}(x) + f_{n_1}(x_1) - f_{n_1}(x_1)\right| \\ &\leq \left|f_{n_1}(x_1) - f_{n_1}(x)\right| + \left|f(x_1) - f_{n_1}(x_1)\right| < 2\epsilon. \end{aligned} \tag{4.3.8}$$

Combining instead (4.3.4), (4.3.5) and (4.3.8), for every $x \in I_1$

$$\begin{aligned} \left|f_{n_0}(x) - f_{n_1}(x)\right| &= \left|f_{n_0}(x) - f_{n_1}(x) - f(x_0) + f(x_0) + f(x_1) - f(x_1)\right| \\ &\geq \left|f(x_0) - f(x_1)\right| - \left|f_{n_0}(x) - f(x_0)\right| - \left|f(x_1) - f_{n_1}(x)\right| \\ &> \alpha - 4\epsilon > \alpha - 4 \cdot \frac{\alpha}{8} = \frac{\alpha}{2}. \end{aligned} \tag{4.3.9}$$

Now assume, inductively, that we have (for some index k)

$$\begin{aligned} &I_0 \supseteq I_1 \supseteq \cdots \supseteq I_{k-1} \supseteq I_k \\ &x_0 \in \overset{\circ}{I}_0, x_1 \in \overset{\circ}{I}_1, \ldots, x_{k-1} \in \overset{\circ}{I}_{k-1}, x_k \in \overset{\circ}{I}_k \\ &f_{n_0}, f_{n_1}, \ldots, f_{n_{k-1}}, f_{n_k}, \quad \text{with } n_0 < n_1 < \cdots < n_{k-1} < n_k \end{aligned}$$

such that

$$\left|f(x_{k-1}) - f(x_k)\right| \geq \alpha; \tag{4.3.10}$$

$$\left|f_{n_k}(x_k) - f(x_k)\right| < \epsilon \tag{4.3.11}$$

and for every $x \in I_k$

$$\begin{aligned} \left|f(x_k) - f_{n_k}(x)\right| &= \left|f(x_k) - f_{n_k}(x) + f_{n_k}(x_k) - f_{n_k}(x_k)\right| \\ &\leq \left|f_{n_k}(x_k) - f_{n_k}(x)\right| + \left|f(x_k) - f_{n_k}(x_k)\right| < 2\epsilon. \end{aligned} \tag{4.3.12}$$

As $I_k \subseteq F_\alpha$, there is a $x_{k+1} \in \overset{\circ}{I}_k$ at which the oscillation of f is bigger than or equal to α:

$$\left|f(x_k) - f(x_{k+1})\right| \geq \alpha; \tag{4.3.13}$$

Hence the inductive hypothesis (4.3.10) holds for $k+1$.

The pointwise convergence of $\{f_n(x)\}_{n\geq 1}$ tells that for $0 < \epsilon < \alpha/8$, given $x_{k+1} \in \overset{\circ}{I}_{k+1} \subseteq I_k$ there exists $n_{k+1} > n_k$ such that

$$\left|f_{n_{k+1}}(x_{k+1}) - f(x_{k+1})\right| < \epsilon; \tag{4.3.14}$$

so hypothesis (4.3.11) holds for $k + 1$.

By the continuity of f_n there exists a closed neighbourhood $[x_{k+1} - \delta_{k+1}; x_{k+1} + \delta_{k+1}] \cap I_k = I_{k+1} \subseteq I_k$ of x_{k+1}, with $\ell(I_{k+1}) < \ell(I_k)/2$, such that for every $x \in I_{k+1}$

$$\left| f_{n_{k+1}}(x_{k+1}) - f_{n_{k+1}}(x) \right| < \epsilon. \tag{4.3.15}$$

From (4.3.14) and (4.3.15) we then obtain, for every $x \in I_{k+1}$

$$\left| f(x_{k+1}) - f_{n_{k+1}}(x) \right| < 2\epsilon; \tag{4.3.16}$$

making (4.3.12) true for $k + 1$. Now using (4.3.14), (4.3.15) and (4.3.12), for every $x \in I_{k+1}$ we have

$$\left| f_{n_{k+1}}(x) - f_{n_k}(x) \right| > \alpha - 4\epsilon > \frac{\alpha}{2}.$$

But $\{I_k\}_{k\geq 0}$ are closed nested intervals, $I_k \supseteq I_{k+1}$ for every $k \in \mathbb{N}$, so Cantor's intersection theorem tells $\bigcap_{k=0}^{\infty} I_k$ contains a point x such that

$$\left| f_{n_{k+1}}(x) - f_{n_k}(x) \right| > \frac{\alpha}{2}$$

for every k. This means the sequence $\{f_{n_k}(x)\}_{k\geq 0}$ diverges, i.e. $\{f_n(x)\}_{n\geq 1}$ does not converge for every $x \in [a, b]$. The absurd confirms F_α cannot be dense, whence F_α is nowhere dense. ■

Theorem 4.3.1 (Baire category of the discontinuity set of $\mathscr{B}_1$ maps defined on $[a, b]$.) *The discontinuity set of a map $f : [a, b] \mapsto \mathbb{R}$ of class 1 is at most of first category.*

Proof Let f be in class one, and define

$$F_n = \left\{ x \in [a, b] : \omega(f, x) \geq \frac{1}{n} \right\}, \quad n \in \mathbb{N}.$$

By lemma 4.3.1 this set is nowhere dense. The countable union $D = \bigcup_{n=1}^{\infty} F_n$ describes the discontinuity set of f in $[a, b]$, which is a meagre set, or of first category. ■

Remark 4.3.1 (*Infimum of discontinuity set of $\mathscr{B}_1$ maps defined on* $[a, b]$.) Theorem 4.3.1 implies class-one functions are continuous at least on a residual set (think of functions representable as Fourier series, which are at most in class 1). As continuous maps are of class 0, furthermore, class 1 contains, apart from the latter, also maps with at least finitely or countably many discontinuities.

In order to prove this fact we must build a sequence of continuous maps with pointwise limit f whose discontinuity set is at most countable. Let then $f : [a, b] \mapsto \mathbb{R}$ be continuous except for a countable (at most) set $D = \{x_1, x_2, \ldots, x_n, \ldots\}$. Choose

$$a < x_1 < a_1 < x_2 < a_2 < \ldots < a_n < x_n < a_{n-1} < \ldots < b.$$

Given an arbitrarily small $\delta_1 > 0$, take $(x_1 - \delta_1, x_1 + \delta_1) \subseteq [a, a_1]$ and define $g_1 : [a, a_1] \mapsto \mathbb{R}$ by

$$\begin{cases} g_1(x) = f(x) & \text{if } x \in [a, a_1] \setminus (x_1 - \delta_1, x_1 + \delta_1) \\ f(x_1 - \delta_1) \le g_1(x) \le f(x_1) \text{ linear} & \text{if } x \in (x_1 - \delta_1, x_1] \\ f(x_1) \le g_1(x) \le f(x_1 + \delta_1) \text{ linear} & \text{if } x \in [x_1, x_1 + \delta_1). \end{cases}$$

Given

$$\delta_2 = \min\left\{\frac{\delta_1}{2}; \frac{|x_1 - x_2|}{2}\right\},$$

define $g_2 : [a, a_2] \mapsto \mathbb{R}$ so that

$$\begin{cases} g_2(x) = f(x) & \text{if } x \in [a, a_2] \setminus \bigcup_{i=1}^{2} (x_i - \delta_2, x_i + \delta_2) \\ f(x_i - \delta_2) \le g_2(x) \le f(x_i) \text{ linear} & \text{if } x \in (x_i - \delta_2, x_i] \text{ for } i = 1, 2 \\ f(x_i) \le g_2(x) \le f(x_i + \delta_2) \text{ linear} & \text{if } x \in [x_i, x_i + \delta_2) \text{ for } i = 1, 2. \end{cases}$$

At step n, given

$$\delta_n = \min\left\{\frac{\delta_{n-1}}{2}; \frac{|x_1 - x_n|}{2}; \ldots; \frac{|x_{n-1} - x_n|}{2}\right\},$$

define $g_n : [a, a_n] \mapsto \mathbb{R}$ so that

$$\begin{cases} g_n(x) = f(x) & \text{if } x \in [a, a_n] \setminus \bigcup_{i=1}^{n} (x_i - \delta_n, x_i + \delta_n) \\ f(x_i - \delta_n) \le g_n(x) \le f(x_i) \text{ linear} & \text{if } x \in (x_i - \delta_n, x_i], \text{ for } i = 1, \ldots, n \\ f(x_i) \le g_n(x) \le f(x_i + \delta_n) \text{ linear} & \text{if } x \in [x_i, x_i + \delta_n) \text{ for } i = 1, \ldots, n. \end{cases}$$

Clearly each g_n is continuous, and for every $x \in [a, b]$

$$\lim_{n \to \infty} g_n(x) = f(x).$$

♣

Proposition 4.3.1 (Baire class of the Dirichlet function) *The Dirichlet function belongs to Baire class 2.*

Proof The Dirichlet function is discontinuous at all rational numbers $\mathbb{Q} \cap [0, 1]$. Let us list these as sequence $q_1, q_2, \ldots, q_i, \ldots$. Correspondingly, for every $n \in \mathbb{N}$ and every $x \in [0, 1]$ we define

$$g_n(x) = \begin{cases} 1 & \forall x_i = q_i, \text{ for } i = 1, 2, \ldots, n \\ 0 & \forall x \in [0, 1] \setminus \{q_1, q_2, \ldots, q_n\}. \end{cases}$$

Each g_n has finitely many discontinuities, so it is a function of class one (Remark 4.3.1). Moreover for every $x \in [a, b]$

$$\lim_{n \to \infty} g_n(x) = f(x).$$

Hence $f(x)$ is of class 2. ■

Proposition 4.3.1 implies that the Dirichlet function should have an analytical representation in the sense of Baire, i.e. it must be the limit of limits of continuous maps. This is the content of the next observation.

Remark 4.3.2 (*Analytical representation of the Dirichlet function.*) The generalised Dirichlet function has the following analytical representation

$$f(x) = \begin{cases} c & \text{if } x \in \mathbb{Q} \\ d & \text{if } x \in \mathbb{R} \setminus \mathbb{Q} \end{cases} = (c - d) \lim_{n \to +\infty} \left(\lim_{m \to +\infty} \cos^{2m}(n! \pi x) \right) + d.$$

The familiar function is described, for every $x \in [0, 1]$, by

$$f(x) = \lim_{n \to +\infty} \left(\lim_{m \to +\infty} \cos^{2m}(n! \pi x) \right).$$

In practice if we take $x \in [0, 1] \cap \mathbb{Q}$ the formula returns the value 1, while putting $x \in [0, 1] \setminus \mathbb{Q}$ gives 0.

Let us prove this. Take a rational number x in $[0, 1]$, so $x = \frac{p}{q}$, with p, q integers, $p \geq 0$ and $q > 0$. Given $n \geq q$, one of the factors in $n!$ is q, so $n!$ is a multiple of q and $n!x$ is integer. Hence $\cos(n! x \pi) = \pm 1$, so for every $m \in \mathbb{N}$ we have $\cos^{2m}(n! x \pi) = 1$ and

$$\lim_{m \to +\infty} \cos^{2m}(n! x \pi) = \lim_{m \to +\infty} (\pm 1)^{2m} = 1.$$

This is true for every $n > q$, so

$$\lim_{n \to +\infty} \left(\lim_{m \to +\infty} \cos^{2m}(n! x \pi) \right) = 1.$$

Take now $x \in [0,1]\backslash\mathbb{Q}$ and fix n. Clearly $n!x$ is not an integer, for otherwise $n!x = k$ and $x = k/n!$, making x rational. Therefore $-1 < \cos(n!x\pi) < 1$, and

$$\lim_{m\to+\infty} \cos^{2m}(n!x\pi) = 0.$$

Again, this being true for every n,

$$\lim_{n\to+\infty} \left(\lim_{m\to+\infty} \cos^{2m}(n!x\pi) \right) = 0.$$

The same argument works for the generalised Dirichlet function (in case $x = \frac{p}{q} \in \mathbb{Q}$, we should put $q > 0$ and p of any sign): for $x \in \mathbb{Q}$ we obtain $f(x) = (c - d + d) = c$, for $x \in \mathbb{R}\backslash\mathbb{Q}$ we get $f(x) = d$. ♣

Remark 4.3.3 (*Baire class of monotone functions, functions with bounded variation and first derivatives.*) Monotone functions have at most countable discontinuity sets (see proposition 13.4.1), so they are in Baire class 1 at most. By item *2.*, proposition 4.2.1, the sum, difference and product of monotone functions are at most in class 1, so also functions with bounded variation are at most in Baire class 1, for they are differences of increasing maps (Jordan's theorem 13.3.1).

At last, the first derivative of $f : [a,b] \mapsto \mathbb{R}$, where defined, is at most in Baire class 1. In fact if $f'(x)$ exists for every $x \in [a,b]$, $f(x)$ is continuous (the opposite is not true). Hence for every $n \in \mathbb{N}$ we define

$$f_n(x) = \frac{f\left(x + \frac{1}{n}\right) - f(x)}{\frac{1}{n}},$$

a continuous map as obtained from continuous maps. But for every $x \in [a,b]$

$$f'(x) = \lim_{n\to\infty} f_n(x).$$ ♣

4.4 Cardinality of Classes of Baire Functions

In this section we shall deal with the cardinality of the class of Baire functions. That number will allows us to decide whether Baire functions exhaust all functions covered by Dirichlet's general definition. Put otherwise, we would like to tell whether any function coming from Dirichlet's definition (i.e. any real function) has a Baire analytical representation. To answer the question we must first compute the cardinality of all real functions.

To this end it suffices to observe

$$\left|\left\{f : f : \mathbb{R} \mapsto \mathbb{R}\right\}\right| = |\mathbb{R}|^{|\mathbb{R}|} = \left(2^{\aleph_0}\right)^{2^{\aleph_0}} = 2^{2^{\aleph_0}} = 2^{\mathfrak{c}}.$$

The class of real functions has the same cardinality of the power set of $\mathbb{R}$.

Proposition 4.4.1 (Cardinality of Baire functions) *The collection of Baire functions has cardinality* $\mathbf{c}$.

Proof For starters, observe that continuous maps on $\mathbb{R}$ are of cardinality at least $\mathbf{c}$, because just taking constants $f(x) = k$, $k \in \mathbb{R}$, for every $x \in \mathbb{R}$, already adds up to $\mathbf{c}$. So $|\mathscr{B}_0| \geq \mathbf{c}$.

On the other hand a continuous map is completely determined by the images of the rational points in the domain. Given in fact a continuous map $f(x) : \mathbb{R} \mapsto \mathbb{R}$, the value $f(\xi)$, $\xi \in \mathbb{R}$, is determined by a (countable) sequence $\{q_n\}_{n\geq 1}$ of rational numbers such that $f(q_n) \to f(\xi)$ as $q_n \to \xi$. Hence the number of continuous maps cannot exceed that of maps $f : \mathbb{Q} \mapsto \mathbb{R}$:

$$|\mathscr{B}_0| \leq \mathbf{c}^{\aleph_0} = (2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0} = \mathbf{c}.$$

Overall, $|\mathscr{B}_0| = \mathbf{c}$.

First inductive step: for $\beta < \alpha$ successor ordinal, the cardinality of $\mathscr{B}_\beta$ equals $\mathbf{c}$. As $\mathscr{B}_{\beta+1} \supseteq \mathscr{B}_\beta$, the cardinality of $\mathscr{B}_{\beta+1}$ is not less than $\mathbf{c}$: $\mathscr{B}_{\beta+1} \geq \mathbf{c}$. At the same time any sequence in $\mathscr{B}_\beta$ converges pointwise to some map in $\mathscr{B}_{\beta+1}$, whence the latter's cardinality is at most $\mathbf{c}^{\aleph_0} = (2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0} = \mathbf{c}$, i.e. $|\mathscr{B}_{\beta+1}| \leq \mathbf{c}$. Hence $|\mathscr{B}_{\beta+1}| = \mathbf{c}$.

Second inductive step: for every $\beta < \alpha$, α limit ordinal, the cardinality of $\mathscr{B}_\beta$ equals $\mathbf{c}$. As $\{\beta : \beta < \alpha\}$ has cardinality $\aleph_0$, the cardinality of

$$\bigcup_{\beta<\alpha} \mathscr{B}_\beta$$

equals $\mathbf{c} \cdot \aleph_0 = \mathbf{c}$. Moreover $\mathscr{B}_\alpha \supseteq \mathscr{B}_\beta$ for every $\beta < \alpha$, so

$$|\mathscr{B}_\alpha| \geq |\mathscr{B}_\beta| = \mathbf{c}, \ \forall \beta < \alpha.$$

On the other hand any map in $\mathscr{B}_\alpha$ is a limit of maps in some $\mathscr{B}_\beta$, $\beta < \alpha$. Therefore the cardinality of class $\mathscr{B}_\alpha$ is at most $\mathbf{c}^{\aleph_0} = \mathbf{c}$, and eventually $|\mathscr{B}_\alpha| = \mathbf{c}$.

By transfinite induction the assertion holds for every ordinal $\alpha < \Omega$. ■

To summarise, there exist uncountably many functions (Dirichlet's definition) that cannot be represented analytically (in Baire sense).

Chapter 5
Borel Functions and Baire Functions



5.1 The Lebesgue–Hausdorff Theorem on $\mathbb{R}$

The classification of Baire functions of Sect. 4.2 can be related to the various Borel subsets of $\mathbb{R}$ from Chap. 3, and the Lebesgue–Hausdorff theorem does exactly that. Its proof requires a number of preliminary considerations.

Definition 5.1.1 (*Classification of Borel functions*) A function $f : \mathbb{R} \mapsto \mathbb{R}$ is said to be a Borel function of class $\alpha \in (0, \Omega)$ whenever: either the pre-image

$$f^{-1}(G) = \big\{x \in \mathbb{R} : f(x) \in G\big\}$$

of any open subset $G \subset \mathbb{R}$ is a Borel set of class Σ^0_α, or the pre-image

$$f^{-1}(F) = \big\{x \in \mathbb{R} : f(x) \in F\big\}$$

of any closed subset $F \subset \mathbb{R}$ is a Borel set of class Π^0_α. ◇

Clearly Borel functions of class 1 are those for which the pre-image of an open/closed subset is respectively open/closed, hence they exhaust continuous maps $f : \mathbb{R} \mapsto \mathbb{R}$.

Proposition 5.1.1 (Borel class of a restriction) *Let $f : \mathbb{R} \mapsto \mathbb{R}$ be a Borel function, and $\alpha \in (0, \Omega)$:*

1. *if $\{E_i\}_{i\geq 1}$ is a sequence in class Σ^0_α such that $\mathbb{R} = \bigcup_{i=1}^{\infty} E_i$, and for every $i \in \mathbb{N}$ the restriction $f_{|E_i}$ is in Borel class α, then f belongs in the same Borel class on $\mathbb{R}$;*
2. *if H_1, H_2 in class Π^0_α are such that $\mathbb{R} = H_1 \cup H_2$, and for $i = 1, 2$ the restriction $f_{|H_i}$ is in Borel class α, then f is in the same class on $\mathbb{R}$.*

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Proof

1. Take $\alpha \in (0, \Omega)$ and an open subset G in $\mathbb{R}$, so

$$f^{-1}(G) = f^{-1}_{|E_1}(G) \cup f^{-1}_{|E_1}(G) \cup \ldots,$$

where each $f^{-1}_{|E_i}(G)$, by Definition 5.1.1, is of class Σ^0_α. Since E_i is in Σ^0_α, so is $E_i \cap f^{-1}_{|E_i}(G)$. As $f^{-1}(G)$ is a countable union in class Σ^0_α, it is an additive set in class α, whence f is Borel, of class α.

2. Item *2.* is proved by replacing, in the above argument, G with F and infinite unions with two-term unions.

■

For the time being we shall take simple functions having values only in $\mathbb{R}$. Later, when discussing measurable functions and the theory of integration, we shall suppose simple functions are $\bar{\mathbb{R}}$-valued.

Definition 5.1.2 (*Real-valued simple functions*) Let $A_1, \ldots, A_n$ be subsets in some set X such that $A_i \cap A_j = \emptyset$ and $X = \bigcup_{i=1}^n A_i$. A map $s : X \mapsto \mathbb{R}$ is a simple function when

$$s(x) = \sum_{i=1}^{n} a_i \varphi_{A_i}(x),$$

where $\{a_1, \ldots, a_n\} = \mathscr{A} \subseteq \mathbb{R}$ and φ_{A_i} is the indicator function of A_i, $i = 1, \ldots, n$. ◇

Proposition 5.1.2 (Borel class of real simple functions) *Let $\{H_1, \ldots, H_n\}$ be a finite collection of pairwise-disjoint sets in class Π^0_α, $\alpha \in (1, \Omega)$, and $\mathscr{A}$ a finite set of numbers $\{a_1, \ldots, a_n\} \subseteq \mathbb{R}$. The simple function $s : \mathbb{R} \mapsto \mathscr{A}$, defined for every $x \in \mathbb{R}$ by $s(x) = \sum_{i=1}^n a_i \varphi_{H_i}(x)$, is in Borel class α.*

Proof By Theorem 3.4.6 there is a finite collection of pairwise-disjoint sets $\{A_1, \ldots, A_n\}$ in class Δ^0_α such that

$$\mathbb{R} = A_1 \cup \ldots, \cup A_n \quad \text{and} \quad H_i \subseteq A_i, \forall i = 1, \ldots, n.$$

Now $a_i \varphi_{A_i}(x) = a_i$ for every $x \in A_i$, $i = 1, \ldots, n$. Clearly $\varphi^{-1}(\{a_i\})$, for every $i = 1, \ldots, n$, is in class Σ^0_α as well, so by definition every $a_i \varphi_{A_i}(x)$ is in Borel class α. By Proposition 5.1.1, *1.*, $s(x)$ is in class α on $\mathbb{R}$. ■

Corollary 5.1.1 (Characterisation of Borel-class of simple functions) *A simple function $s : \mathbb{R} \mapsto \mathscr{A}$, with $\mathscr{A} = \{a_1, \ldots, a_n\} \subseteq \mathbb{R}$, is in Borel class $\alpha \in (1, \Omega)$ if and only if the pre-images $s^{-1}(\{a_1\}), \ldots, s^{-1}(\{a_n\})$ are pairwise disjoint and ambiguous of class α.*

Proof Necessary implication: let $s : \mathbb{R} \mapsto \mathscr{A}$ be in Borel class $\alpha \in (1, \Omega)$, with $\mathscr{A} = \{a_1, \ldots, a_n\} \subseteq \mathbb{R}$. As the topological space $\mathscr{A}$ is discrete[1] (in particular, finite), every $\{a_i\}$ is open and closed in $\mathscr{A}$. Take then the open set $\{a_i\}$. Since s is of class α, by definition $s^{-1}(\{a_i\})$ is in Σ^0_α. Similarly $\{a_i\}$ is closed, so by definition $s^{-1}(\{a_i\})$ is in Π^0_α. Therefore $s^{-1}(\{a_i\})$ is ambiguous of class α. Furthermore $s^{-1}(\{a_i\}) \cap s^{-1}(\{a_j\}) = \emptyset$, $i \neq j$, whence the claim.

For the converse, let $\left\{s^{-1}(\{a_1\}), \ldots, s^{-1}(\{a_n\})\right\}$ be a finite collections of pairwise-disjoint ambiguous sets of class α. Then they are in class Π^0_α. Proposition 5.1.2 finishes the proof. ■

Remark 5.1.1 (*Consequence of Corollary* 5.1.1) By the above result if $s : \mathbb{R} \mapsto \mathscr{A}$, $\mathscr{A} = \{a_1, \ldots, a_n\} \subseteq \mathbb{R}$, is of class $\alpha \in (1, \Omega)$, then in general $s^{-1}(A)$ is ambiguous of class α for every subset $A \subseteq \mathscr{A}$. ♣

Lemma 5.1.1 (Pre-image of a pointwise limit) *Let $f(x) : \mathbb{R} \mapsto \mathbb{R}$ be the pointwise limit of a sequence $\{f_n(x)\}_{n\geq 1}$. For any given $k \in \mathbb{R}$ the pre-image of the open set (k, ∞) equals*

$$f^{-1}\big((k,\infty)\big) = \big\{x \in \mathbb{R} : f(x) > k\big\} = \bigcup_{m=1}^{\infty}\bigcup_{i=1}^{\infty}\Big(\bigcap_{n=i}^{\infty}\Big\{x \in \mathbb{R} : f_n(x) \geq k + \frac{1}{m}\Big\}\Big), \tag{5.1.1}$$

while the pre-image of the closed set $[k, \infty)$ is

$$f^{-1}\big([k,\infty)\big) = \big\{x \in \mathbb{R} : f(x) \geq k\big\} = \bigcap_{m=1}^{\infty}\bigcap_{i=1}^{\infty}\Big(\bigcup_{n=i}^{\infty}\Big\{x \in \mathbb{R} : f_n(x) > k - \frac{1}{m}\Big\}\Big). \tag{5.1.2}$$

Proof Take $x_0 \in \mathbb{R}$ with $f(x_0) > k$, i.e.

$$x_0 \in \big\{x \in \mathbb{R} : f(x) > k\big\}.$$

For a suitable $m \in \mathbb{N}$ we have $f(x_0) > k + 1/m$.

As $f(x) = \lim_{n\to\infty} f_n(x)$, there exists an $i \in \mathbb{N}$ such that for every $n \geq i$, $f_n(x_0) \geq k + 1/m$; hence $x_0 \in \big\{x \in \mathbb{R} : f_n(x) \geq k + 1/m\big\}$ and *a fortiori*

$$x_0 \in \bigcap_{n=i}^{\infty}\Big\{x \in \mathbb{R} : f_n(x) \geq k + \frac{1}{m}\Big\}.$$

Then

[1] Every element in a discrete topological space $\mathscr{A}$ is open, i.e. $\{a_i\} = B(a_i, \epsilon)$ for $i = 1, 2, \ldots$ and some $\epsilon > 0$. Moreover any subset $A \subseteq \mathscr{A}$ is open, being a union of open subsets. For every subset $A \subseteq \mathscr{A}$, the set $A^c = \mathscr{A} \setminus A$ is a subset of $\mathscr{A}$ and hence open in $\mathscr{A}$, so A is closed. Consequently any subset $A \subseteq \mathscr{A}$ is ambiguous in $\mathscr{A}$.

$$x_0 \in \bigcup_{m=1}^{\infty}\bigcup_{i=1}^{\infty}\Big[\bigcap_{n=i}^{\infty}\Big\{x \in \mathbb{R} : f_n(x) \geq k + \frac{1}{m}\Big\}\Big],$$

and so

$$f^{-1}\big((k,\infty)\big) \subseteq \bigcup_{m=1}^{\infty}\bigcup_{i=1}^{\infty}\Big(\bigcap_{n=i}^{\infty}\Big\{x \in \mathbb{R} : f_n(x) \geq k + \frac{1}{m}\Big\}\Big).$$

Now the opposite inclusion. For this, take

$$x_0 \in \bigcup_{m=1}^{\infty}\bigcup_{i=1}^{\infty}\Big[\bigcap_{n=i}^{\infty}\Big\{x \in \mathbb{R} : f_n(x) \geq k + \frac{1}{m}\Big\}\Big].$$

Given an integer $m \geq m_0$ there is an i_0 such that for every $n \geq i_0$

$$x_0 \in \bigcap_{n=i_0}^{\infty}\Big\{x \in \mathbb{R} : f_n(x) \geq k + \frac{1}{m}\Big\}.$$

By the pointwise convergence in $\mathbb{R}$ of $\{f_n(x)\}_{n\geq 1}$ to $f(x)$ we have

$$x_0 \in \Big\{x \in \mathbb{R} : f(x) \geq k + \frac{1}{m}\Big\}.$$

But

$$\Big\{x \in \mathbb{R} : f(x) \geq k + \frac{1}{m}\Big\} \subseteq \Big\{x \in \mathbb{R} : f(x) > k\Big\},$$

so $x_0 \in \big\{x \in \mathbb{R} : f(x) > k\big\}$, and so

$$\bigcup_{m=1}^{\infty}\bigcup_{i=1}^{\infty}\Big[\bigcap_{n=i}^{\infty}\Big\{x \in \mathbb{R} : f_n(x) \geq k + \frac{1}{m}\Big\}\Big] \subseteq f^{-1}\big((k,\infty)\big)$$

as required.

The proof of the other assertion is analogous. ■

Before we address the analogous proposition on pre-images of uniform limits we need another lemma.

Lemma 5.1.2 (Uniform continuity of the distance function $d(x, E)$) *In a metric space (X, d), for any $E \subseteq X$ the map $d(x, E) : X \mapsto [0, \infty)$, defined by*

$$d(x, E) = \inf\big\{d(x, y) : y \in E\big\}, \quad x \in X,$$

is uniformly continuous.

Proof Take $E \subseteq X$ and define $d(x, E) : X \mapsto [0, \infty)$ through

$$d(x, E) = \inf\big\{d(x, y) : y \in E\big\}, \quad x \in X.$$

This is the distance function from the set $E \subseteq X$, taking values in $[0, \infty)$.

We claim such function is uniformly continuous. Given $x, y \in X, z \in E$, by the triangle inequality $d(x, z) \leq d(x, y) + d(y, z)$, hence

$$\begin{aligned} & \inf_{z\in E} d(x, z) \leq d(x, y) + \inf_{z\in E} d(y, z) \\ \Leftrightarrow \quad & d(x, E) \leq d(x, y) + d(y, E) \\ \Leftrightarrow \quad & d(x, E) - d(y, E) \leq d(x, y). \end{aligned}$$

Swapping x and y gives $d(y, E) - d(x, E) \leq d(x, y)$, so for every $x, y \in X$

$$|d(y, E) - d(x, E)| \leq d(x, y).$$

Therefore for any $\epsilon > 0$ there exists $\delta_\epsilon \in (0, \epsilon)$ such that

$$|d(y, E) - d(x, E)| \leq d(x, y) < \epsilon$$

for every $x, y \in X$ with $d(x, y) < \delta_\epsilon$.

This means $d(x, E)$ is uniformly continuous on X. ■

Lemma 5.1.3 (Pre-images of uniform limits) *Let $f(x) : \mathbb{R} \mapsto \mathbb{R}$ be the uniform limit of a sequence $\{f_n(x)\}_{n\geq 1}$. Given any $k \in \mathbb{R}$ the pre-image of the closed set $[k, \infty)$ equals*

$$\begin{aligned} f^{-1}\big([k, \infty)\big) &= \bigcap_{m=1}^{\infty} \bigcap_{i=0}^{\infty} \left(\left\{ x \in \mathbb{R} : f_{n_m+i}(x) \geq k - \frac{1}{m} \right\} \right) \\ &= \bigcap_{m=1}^{\infty} \bigcap_{i=0}^{\infty} \left(\big(f_{n_m+i}\big)^{-1}\big([k - 1/m, \infty)\big) \right). \end{aligned} \tag{5.1.3}$$

while the pre-image of the open set (k, ∞) is

$$\begin{aligned} f^{-1}\big((k, \infty)\big) &= \bigcup_{m=1}^{\infty} \bigcup_{i=0}^{\infty} \left(\left\{ x \in \mathbb{R} : f_{n_m+i}(x) > k + \frac{1}{m} \right\} \right) \\ &= \bigcup_{m=1}^{\infty} \bigcup_{i=0}^{\infty} \left(\big(f_{n_m+i}\big)^{-1}\big((k + 1/m, \infty)\big) \right). \end{aligned} \tag{5.1.4}$$

Proof Let $\{f_n\}_{n\geq 1}$ be uniformly convergent to f, so there exists an increasing sequence n_m such that, for every $x \in \mathbb{R}$ and $i \geq 0$,

$$\left|f(x) - f_{n_m+i}(x)\right| < \frac{1}{m}.$$

Based on this we must prove (5.1.3).

($\Rightarrow$) Take $f(x) \in [k, \infty)$. Then $f_{n_m+i}(x) \geq k - 1/m$ for every m and i, since $\left|f(x) - f_{n_m+i}(x)\right| < 1/m$.

($\Leftarrow$) For every $m \in \mathbb{N}$ take $f_{n_m+i}(x) \geq k - 1/m$, whence

$$d\big(f_{n_m+i}(x), [k, \infty)\big) \leq \frac{1}{m}.$$

By Lemma 5.1.2 the distance $d\big(f(x), [k, \infty)\big)$ is a continuous map in $f(x)$, so the condition $f(x) = \lim_{m\to\infty} f_{n_m+i}(x)$ implies

$$d\big(f(x), [k, \infty)\big) = 0.$$

Hence $f(x) \in \big[k, \infty)$, giving (5.1.3). The proof of (5.1.4) is similar. ■

Theorem 5.1.1 (Borel class of pointwise limits) *The limit of a pointwise convergent sequence of Borel functions of class $\alpha \in (0, \Omega)$ is a Borel function of class $\alpha + 1$.*

Proof Let $\alpha \in (0, \Omega)$, and for every $x \in \mathbb{R}$ define $f(x) = \lim_{i\to\infty} f_i(x)$, with every f_i of class α.

Lemma 5.1.1, for arbitrary $k \in \mathbb{R}$, implies

$$\begin{aligned} f^{-1}\big([k, \infty)\big) &= \big\{x \in \mathbb{R} : f(x) \geq k\big\} \\ &= \bigcap_{m=1}^{\infty}\bigcap_{i=1}^{\infty}\left(\bigcup_{n=i}^{\infty}\left\{x \in \mathbb{R} : f_n^{\alpha}(x) > k - \frac{1}{m}\right\}\right) \\ &= \bigcap_{m=1}^{\infty}\bigcap_{i=1}^{\infty}\left(\bigcup_{n=i}^{\infty}\big(f_n^{\alpha}\big)^{-1}\big((k - 1/m, \infty)\big)\right). \end{aligned}$$

The maps f_n are in class α, so by Definition 5.1.1 $\big(f_n^{\alpha}\big)^{-1}\big((k - 1/m, \infty)\big)$ is in class Σ^0_{α}. Then $f^{-1}\big([k, \infty)\big)$ is in class $\Pi^0_{\alpha+1}$. The limit function is therefore Borel, of class $\alpha + 1$.

The other case is proved similarly. ■

We saw in theorem 4.2.1 that a uniform limit stays in the same Baire class as its sequence. The system of Borel functions enjoys a similar property, because the limit function $f : X \mapsto Y$ of a uniformly convergent Borel sequence $\{f_n : X \mapsto Y\}_{n\geq 1}$ of class $\alpha \in (0, \Omega)$ is necessarily a Borel function of class α. The converse is false. To have a sufficient condition of that sort we need $\alpha \in (1, \Omega)$ and the space Y must be separable ($\mathbb{R}$ with the standard metric, for instance).

Theorem 5.1.2 (Necessary condition for the stability of Borel classes under uniform convergence) *The limit of a uniformly convergent sequence $\{f_n\}_{n\geq 1}$ of Borel functions of class $\alpha \in (0, \Omega)$ is a Borel function of class α.*

Proof Let $\alpha \in (0, \Omega)$. The uniform convergence of $\{f_n\}_{n\geq 1}$ in class α gives

$$\begin{aligned} f^{-1}\big([k,\infty)\big) &= \bigcap_{m=1}^{\infty}\bigcap_{i=0}^{\infty}\left(\left\{x\in\mathbb{R} : f^{\alpha}_{n_m+i}(x)\geq k-\frac{1}{m}\right\}\right) \\ &= \bigcap_{m=1}^{\infty}\bigcap_{i=0}^{\infty}\left(\left(f^{\alpha}_{n_m+i}\right)^{-1}\big([k-1/m,\infty)\big)\right). \end{aligned} \tag{5.1.5}$$

It then suffices to observe that $\left(f^{\alpha}_{n_m+i}\right)^{-1}\big([k-1/m,\infty)\big)$ is in class Π^0_α. Therefore $f^{-1}\big([k,\infty)\big)$ is in the same class. ■

Theorem 5.1.3 (Sufficient condition for the stability of Borel classes under uniform convergence) *Any Borel function $f : \mathbb{R} \mapsto \mathbb{R}$ of class $\alpha \in (1, \Omega)$ is the uniform limit of a Borel sequence $\{f_n\}_{n\geq 1}$ of class α, where $f_n(\mathbb{R})$ is discrete for every $n \in \mathbb{N}$.*

Proof Let $f : \mathbb{R} \mapsto \mathbb{R}$ be in class $\alpha \in (1, \Omega)$. For $n \geq 1$ and $k \in \mathbb{Z}$ define

$$H_k = \big\{x\in\mathbb{R} : f(x)\leq k/2^n\big\} \quad \text{and} \quad E_k = \big\{x\in\mathbb{R} : f(x) < (k+1)/2^n\big\}. \tag{5.1.6}$$

As f is in class α, Definition 5.1.1 guarantees H_k and E_k are respectively in class Π^0_α and Σ^0_α and such that $H_k \subseteq E_k$ for every $k \in \mathbb{Z}$. By theorem 3.4.5, for every $k \in \mathbb{Z}$ there exists an ambiguous set A_k in class α such that

$$H_k \subseteq A_k \subseteq E_k. \tag{5.1.7}$$

For every $k \in \mathbb{Z}$ and every $x \in (A_k \setminus A_{k-1})$ put

$$f_n(x) = \frac{k}{2^n}, \forall n = 1, 2, \ldots. \tag{5.1.8}$$

Immediately,

$$f_n(\mathbb{R}) = \{0, \pm 1/2^n, \pm 2/2^n, \ldots\}$$

is discrete for every $n = 1, 2, \ldots$. Spelling out the chain of inclusions

$$(H_k \setminus H_{k-1}) \subseteq (A_k \setminus A_{k-1}) \subseteq (E_k \setminus E_{k-1})$$

gives us

$$\begin{aligned} \big\{x\in\mathbb{R} : (k-1)/2^n < f(x)\leq k/2^n\big\} &\subseteq (A_k \setminus A_{k-1}) \\ &\subseteq \big\{x\in\mathbb{R} : k/2^n \leq f(x) < (k+1)/2^n\big\}. \end{aligned}$$

Hence for every $x \in (A_k \setminus A_{k-1})$ we have $(k-1)/2^n < f(x) < (k+1)/2^n$ and, in general, for every $x \in \mathbb{R}$ and every $n = 1, 2, \ldots$

$$|f(x) - f_n(x)| < \frac{1}{2^n}.$$

Hence $\{f_n\}_{n\geq 1}$ is uniformly convergent to f.

Now fix $\epsilon > 0$, take $M \in \mathbb{R}$, and choose $k, j \in \mathbb{Z}$ so that $k/2^n < k\epsilon < M$ and $j/2^n < j\epsilon \leq M$. From Eq. (5.1.8) we infer

$$\begin{aligned} \{x \in \mathbb{R} : f_n(x) < M\} &= \{x \in \mathbb{R} : f_n(x) \leq k\epsilon\} = A_k \\ \{x \in \mathbb{R} : f_n(x) > M\} &= \{x \in \mathbb{R} : f_n(x) > j\epsilon\} = E_j. \end{aligned}$$

The A_k are ambiguous of class α and the E_j belong in Σ^0_α, so $\{f_n\}_{n\geq 1}$ is a Borel sequence of class α. ■

The next proposition and lemma prepare the ground for the proof of Theorem 5.1.4.

Proposition 5.1.3 (Borel class of simple pointwise limits of simple functions) *Any simple Borel function $s : \mathbb{R} \mapsto \mathscr{A}$ of class $\beta > 2$, where $\mathscr{A} = \{a_1, \ldots, a_k\} \subseteq \mathbb{R}$ and β is a successor ordinal, is the pointwise limit of a sequence $\{s_n\}_{n\geq 1}$ of simple functions of Borel class $\beta - 1$ with values in $\mathscr{A}$.*

If s belongs to class $\alpha + 1$, with α a limit ordinal, then s is the pointwise limit of a sequence $\{s_n\}_{n\geq 1}$ of simple functions of class $\beta < \alpha$.

Proof By hypothesis s is of class β (or, respectively, $\alpha + 1$), so Corollary 5.1.1 implies that any set $A_i = s^{-1}(\{a_i\})$ is ambiguous of class β (or $\alpha + 1$), and moreover $\mathbb{R} = A_1 \cup \ldots \cup A_k$. By proposition 3.4.6,

$$A_i = \lim_{n\to\infty} A_{i_n}, \tag{5.1.9}$$

where every A_{i_n} is ambiguous of class $\beta - 1$ (or $\beta < \alpha$). For every $n \in \mathbb{N}$ define

$$H_{1_n} = A_{1_n}, H_{2_n} = A_{2_n} \setminus A_{1_n}, \ldots, H_{k_n} = A_{k_n} \setminus \left(A_{1_n} \cup \ldots \cup A_{k-1_n}\right).$$

Clearly for every given n these sets are pairwise disjoint and in $\Pi^0_{\beta-1}$ (or in $\bigcup_{\beta<\alpha} \Pi^0_\beta$). By Proposition 5.1.2 there exists a simple function s_n in class $\beta - 1$ (or $\beta < \alpha$), defined on $\mathbb{R}$ and with values in $\{a_1, \ldots, a_k\}$, such that $s_n(x) = a_i$ for every $x \in H_{i_n}$, $i = 1, \ldots, k$.

So we must prove $s(x) = \lim_{n\to\infty} s_n(x)$ for every $x \in \mathbb{R}$. Take any $x' \in \mathbb{R}$ and suppose $x' \in A_i$, so $s(x') = a_i$. As $x' \notin A_j$ for every $j \neq i$, Eq. (5.1.9) implies that for n large enough $x' \in A_{i_n}$ and $x' \notin A_{j_n}$ for every $j_n \neq i_n$. Then

$$x' \in H_{i_n} \quad \text{and so} \quad s_n(x') = a_i = s(x').$$

The arbitrariness of $x' \in \mathbb{R}$ eventually guarantees that $\lim_{n\to\infty} s_n(x) = s(x)$, for every $x \in \mathbb{R}$. ■

Lemma 5.1.4 *Take $\alpha \in (2, \Omega)$ and sequences $\{s_i\}_{i\geq 1}$, $\{t_i\}_{i\geq 1}$ of simple functions in class $< \alpha$ such that*

$$\lim_{i\to\infty} s_i(x) = f(x), \qquad \lim_{i\to\infty} t_i(x) = g(x) \quad \text{and} \quad |f(x) - g(x)| < k$$

for some $k > 0$ and every $x \in \mathbb{R}$. Then there exists a sequence $\{u_i\}_{i\geq 1}$ of simple functions in class $< \alpha$ such that

$$\lim_{i\to\infty} u_i(x) = g(x) \quad \text{and} \quad |u_i(x) - s_i(x)| \leq k \tag{5.1.10}$$

for every $x \in \mathbb{R}$.

Proof Define $d_i(x) = |t_i(x) - s_i(x)|$. The maps t_i, s_i are of class $< \alpha$. Corollary 5.1.1 says $A_{i_k} = t^{-1}(\{a_{i_k}\})$ and $B_{i_j} = s^{-1}(\{b_{i_j}\})$, $k = 1, \dots, n$, $j = 1, \dots m$, are ambiguous of class $< \alpha$. Then the difference $t^{-1}(\{a_{i_k}\}) - s^{-1}(\{b_{i_j}\}) = A_{i_k} \setminus B_{i_j}$ is ambiguous of class $< \alpha$.

The maps d_i are continuous, and by item *3.* in proposition 3.4.2

$$A_i = d_i^{-1}\big([0, k]\big) = \{x \in \mathbb{R} : |t_i(x) - s_i(x)| \leq k\}$$

is ambiguous of class $< \alpha$. Hence d_i is in class $< \alpha$ and its image is finite. Set

$$u_i(x) = \begin{cases} t_i(x) & \text{if } x \in A_i \\ s_i(x) & \text{if } x \in \mathbb{R} \setminus A_i. \end{cases}$$

By Proposition 5.1.1, *2.*, the map u_i is in class $< \alpha$, and for every $x \in \mathbb{R}$

$$|u_i(x) - s_i(x)| \leq k,$$

proving the second relation in (5.1.10).

In order to show the first relation in (5.1.10) take an arbitrary $x' \in \mathbb{R}$. As $\lim_{i\to\infty} s_i(x) = f(x)$ and $\lim_{i\to\infty} t_i(x) = g(x)$, if i is large enough $|t_i(x') - s_i(x')| \leq k$, so $x' \in A_i$ and $u_i(x') = t_i(x')$. Then

$$\lim_{i\to\infty} u_i(x') = \lim_{i\to\infty} t_i(x') = g(x').$$

But this holds for any $x' \in \mathbb{R}$, so

$$\lim_{i\to\infty} u_i(x) = g(x).$$

■

Theorem 5.1.4 (Borel class of sequences of functions and Borel class of pointwise limits) *Any Borel function $f : \mathbb{R} \mapsto \mathbb{R}$ of class $\beta > 2$, where β is a successor ordinal, is the pointwise limit of a sequence $\{f_n\}_{n\geq 1}$ of functions in class $\beta - 1$.*

If f is in Borel class $\alpha + 1$ for some limit ordinal α, then f is the pointwise limit of a sequence $\{f_n\}_{n\geq 1}$ of functions in class $\beta < \alpha$.

Proof Since the interval (c, d), $c < d$, is homeomorphic to $\mathbb{R}$ we may as well take $f : \mathbb{R} \mapsto (c, d)$.

By Theorem 5.1.3, for every $x \in \mathbb{R}$

$$f(x) = \lim_{n\to\infty} f_n(x),$$

where $\{f_n\}_{n\geq 1}$ converges uniformly, and its terms f_n are in class β (respectively $\alpha + 1$) with finite image, so they are simple. By uniform convergence we may assume, without loss of generality,

$$|f_{n+1}(x) - f_n(x)| < \frac{1}{2^n}$$

for every $x \in \mathbb{R}$. If necessary we may replace $\{f_n\}_{n\geq 1}$ by a subsequence. Proposition 5.1.3 says

$$f_n(x) = \lim_{i\to\infty} f_{n,i}(x)$$

for $n \in \mathbb{N}$ and every $x \in \mathbb{R}$, with $f_{n,i}$ in class $\beta - 1$ (or $\beta < \alpha$) and with finite image. Now define $h_{1,i} = f_{1,i}$ for every $i \in \mathbb{N}$. By induction on n define the double sequence $\{h_{n,i}\}_{i\geq 1}$, where $h_{n,i}$ is of class $\beta - 1$ (or $\beta < \alpha$), has finite image, and satisfies

$$\lim_{i\to\infty} h_{n,i}(x) = f_n(x)$$

for every $x \in \mathbb{R}$. The existence of $\{h_{n+1,i}\}_{i\geq 1}$ such that

$$|h_{n+1,i}(x) - h_{n,i}(x)| \leq \frac{1}{2^n}$$

is guaranteed by Lemma 5.1.4 if we take $f = f_n$, $g = f_{n+1}$, $s_i = h_{n,i}$, $u_i = h_{n+1,i}$ and $k = 1/2^n$.

To prove the theorem it then suffices to show that

$$\lim_{i\to\infty} h_{i,i}(x) = f(x)$$

for every $x \in \mathbb{R}$. So, take any $x' \in \mathbb{R}$ and $\epsilon > 0$. Given $n \in \mathbb{N}$ such that $1/2^{n-1} < \epsilon$ and

$$|f_n(x') - f(x')| < \epsilon,$$

call $n_0 > n$ an index such that, for every $i > n_0$,

$$|h_{n,i}(x') - f_n(x')| < \epsilon.$$

For $i > n_0$

$$\begin{aligned}|h_{i,i}(x') - f(x')| =& |h_{i,i}(x') - h_{i-1,i}(x') + \ldots + h_{n+1,i}(x') - h_{n,i}(x') \\ &+ h_{n,i}(x') - f_n(x') + f_n(x') - f(x')| \\ \leq& |h_{i,i}(x') - h_{i-1,i}(x')| + \ldots + |h_{n+1,i}(x') - h_{n,i}(x')| \\ &+ |h_{n,i}(x') - f_n(x')| + |f_n(x') - f(x')| \\ <& \left(\frac{1}{2^{i-1}} + \ldots + \frac{1}{2^n}\right) + \epsilon + \epsilon.\end{aligned}$$

The term in brackets is a partial geometric series, from index n to $(i-1)$, whence

$$\sum_{k=n}^{i-1} \frac{1}{2^k} = \frac{\frac{1}{2^n} - \frac{1}{2^i}}{1 - \frac{1}{2}} = \left(\frac{1}{2^n} - \frac{1}{2^i}\right) \cdot 2 = \underbrace{\frac{1}{2^{n-1}}}_{<\epsilon} - \frac{1}{2^{i-1}} < \epsilon.$$

Therefore $|h_{i,i}(x') - f(x')| < 3\epsilon$. Now recall that $x' \in \mathbb{R}$ and $\epsilon > 0$ were chosen arbitrarily. ■

To show that the previous theorem holds for Borel functions of class $\beta = 2$ we shall rely on a result from functional analysis, Tietze's extension theorem, which is reviewed in Appendix B.

Theorem 5.1.5 (Borel functions of class 2) *Any Borel function $f : \mathbb{R} \mapsto \mathbb{R}$ of class 2 is the pointwise limit of a sequence of continuous maps.*

Proof Without loss of generality f can be thought of as valued in a bounded interval (c, d). To begin with we shall also assume f is simple of class 2, so we rename it $s : \mathbb{R} \mapsto \mathscr{A}$, with $\mathscr{A} = \{y_1, \ldots, y_k\} \subseteq (c, d)$.

For every $i = 1, \ldots, k$ the set $s^{-1}(\{y_i\}) = A_i$ is an F_σ set, i.e.

$$A_i = s^{-1}(\{y_i\}) = \bigcup_{n=1}^{\infty} F_{i,n},$$

where $F_{i,n} \subseteq F_{i,n+1}$, and $F_{i,n} = \bar{F}_{i,n}$. By Tietze's extension (Theorem B.0.2, Appendix B), for every $n \in \mathbb{N}$ there exists a continuous map $f_n : \mathbb{R} \mapsto \{y_1, \ldots, y_k\}$, hence of class 1, such that $f_n(x) = y_i$ for every $x \in F_{i,n}$, $i = 1, \ldots, k$, and whose values lie between the lower and upper bounds of s. As $\mathbb{R} = A_1 \cup \ldots \cup A_k$ and $A_i \cap A_j = \emptyset$ for $i \neq j$, any $x' \in \mathbb{R}$ will belong in A_i, for some $i = 1, \ldots k$. If n is sufficiently large, $x' \in F_{i,n}$, so $f_n(x') = y_i$, i.e. $f_n(x') = s(x')$. But $x' \in \mathbb{R}$ is arbitrary, so for every $x \in \mathbb{R}$

$$s(x) = \lim_{n\to\infty} f_n(x).$$

Now we tackle the general case and take $f : \mathbb{R} \mapsto (c, d)$ of class 2. Just as in Theorem 5.1.3, let $\{f_n\}_{n\geq 1}$ be a sequence of functions in class 2, each with finite image inside (c, d), and uniformly convergent to f. Uniform convergence ensures we may suppose

$$|f_{n+1}(x) - f_n(x)| < \frac{1}{2^n}$$

and $f_0(x) = 0$ for every $x \in \mathbb{R}$, without loss of generality. Clearly for every $x \in \mathbb{R}$ the difference $f_n(x) - f_{n-1}(x)$ is in class 2. From the first step there exists a double sequence of continuous maps $g_{n,i}$, hence of class 1, such that

$$\lim_{i\to\infty} g_{n,i} = f_n(x) - f_{n-1}(x) \quad \text{and} \quad |g_{n,i}(x)| \leq \frac{1}{2^{n-1}}.$$

Define, for every $n \in \mathbb{N}$ and $x \in \mathbb{R}$,

$$h_{n,i}(x) = g_{1,i}(x) + \ldots + g_{n,i}(x);$$

these are still Borel functions of class 1. Hence for every $x \in \mathbb{R}$

$$\lim_{i\to\infty} h_{n,i}(x) = f_n(x) \quad \text{and} \quad |h_{n+1,i}(x) - h_{n,i}(x)| \leq \frac{1}{2^n}.$$

With the same argument used in the final part of Theorem 5.1.4 we obtain

$$\lim_{i\to\infty} h_{i,i}(x) = f(x)$$

for every $x \in \mathbb{R}$. ■

Eventually, here comes the Lebesgue–Hausdorff theorem.

Theorem 5.1.6 (Lebesgue–Hausdorff theorem on $\mathbb{R}$)[2] *For $\alpha \in [0, \Omega)$ the class of Baire functions $\mathscr{B}_\alpha$ coincides with the family of Borel functions of class $\alpha + 1$.*

Proof Assume, without loss of generality, that the range interval of f is contained in the bounded interval (c, d).

It is obvious that $\mathscr{B}_0$ coincides with Borel class 1. Theorems 5.1.5 and 5.1.1 ensure that $\mathscr{B}_1$ coincides with Borel functions of class 2.

First inductive step: the statement holds for some ordinal β with predecessor. By Theorems 5.1.4 and 5.1.1 Borel functions of class $\beta + 1$ are precisely pointwise limits of sequences of Borel functions of class β, that is pointwise limits of sequences

[2]The Lebesgue–Hausdorff theorem was generalised to separable metric spaces by Stefan Banach.

of functions in class $\mathscr{B}_{\beta-1}$. Consequently the Borel class $\beta+1$ coincides with the Baire class $\mathscr{B}_\beta$.

Next, suppose α is a limit ordinal and the theorem is true for every successor ordinal $\beta<\alpha$. Any $f\in\mathscr{B}_\alpha$, for every $x\in\mathbb{R}$, has the form

$$f(x)=\lim_{n\to\infty} f_n(x),$$

where $f_n\in\mathscr{B}_\beta$ and $\beta<\alpha$. By hypothesis, for all $n\in\mathbb{N}$ the map f_n is of some Borel class $\beta+1$. By Theorem 5.1.1

$$\bigcup_{n=i}^{\infty}\left(f_n^{\beta+1}\right)^{-1}\big((k-1/m,\infty)\big)$$

is in Borel class Σ^0_α. Hence $f^{-1}\big([k,\infty)\big)$ is in Borel class $\Pi^0_{\alpha+1}$, and the claim is proved.

Vice versa, let f be Borel in class $\alpha+1$. By Theorem 5.1.4, for every $x\in\mathbb{R}$

$$f(x)=\lim_{n\to\infty} f_n(x),$$

where f_n is of some Borel class $\beta<\alpha$, for every $n\in\mathbb{N}$. Therefore $f_n\in\mathscr{B}_{\beta-1}$ and $f\in\mathscr{B}_\alpha$. ■

We further deduce that the family of Borel sets has cardinality $\mathbf{c}$, whilst we shall see that the family of Lebesgue measurable sets has cardinality $2^{\mathbf{c}}$.

Part III
Families of Sets

Chapter 6
Semi-algebra and Algebra of Sets



The aim of the theory of the Lebesgue measure is to cover an arbitrary linear set of points E with a suitable collection of intervals and extract from their lengths a measure for E. Whereas Peano and Jordan covered the set to be measured with a finite—albeit large—number of partial intervals, Lebesgue decided to employ as covering for E a countable family of Borel sets, whose theory had been in the meantime developed by Emile Borel (1898). This choice gave rise to more elaborate notions, but in return many sets became measurable that were not measurable according to the Peano-Jordan theory.

Remark 6.0.1 According to Lebesgue a good measure theory should satisfy a number of reasonable properties, reflected in the following conditions[1]

1. there exist sets whose measure is non-zero;
2. equal sets have equal measures (translation-invariance of the measure);
3. the measure of a finite or countable number of pairwise-disjoint sets equals the sum of their measures (complete additivity of the measure).

♣

As we will see in the sequel there is no measure satisfying all three conditions simultaneously. The Lebesgue measure, in particular, fulfils 3. only if we restrict it to special collections of sets, called σ-algebras, to which this part is dedicated. By doing so, though, we exclude from the theory all the sets for which $A_i \cap A_j = \emptyset$ for every $i \neq j$, but $m(A_i \cup A_j) \neq m(A_i) + m(A_j)$. This is the so-called *measure problem*.

[1] H. Lebesgue, *Intégrale, Longueur, Aire*, Ann. Mat. Pura Appl., Serie III (1902) 7/4, 231–359.

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S. Gentili, *Measure, Integration and a Primer on Probability Theory*, UNITEXT 125, https://doi.org/10.1007/978-3-030-54940-4_6

6.1 Semi-algebra of Subsets of a Set X

We mentioned that Lebesgue's measure theory begins from suitable covering families of intervals, which serve to determine the measure of linear sets of points.

Hence in this context it becomes very important to study the properties of the class of real intervals, which are the constituents of covers in $\mathbb{R}$. We will show that the collection $\mathcal{I}$ of real intervals is a semi-algebra of sets, and that it generates an algebra of sets. These facts will become useful for understanding how to build the Borel, and Lebesgue, σ-algebras.

Definition 6.1.1 (*Semi-algebra of sets*) Given a non-empty set X, a collection $\mathcal{C}$ of subsets of X is called a semi-algebra of subsets of X if:

1. $\emptyset, X \in \mathcal{C}$;
2. $A \cap B \in \mathcal{C}, \quad \forall A, B \in \mathcal{C}$, i.e. $\mathcal{C}$ is closed under finite intersections;
3. $\forall A \in \mathcal{C},\ A^c \notin \mathcal{C}$, but $A^c = \bigsqcup_{i=1}^{n} C_i, \quad C_i \in \mathcal{C} \quad \text{and} \quad C_i \cap C_j = \emptyset, \quad \forall i \neq j.$

Henceforth we shall refer to such a $\mathcal{C}$ merely as a 'semi-algebra'. ◇

Example 6.1.1 (*The family of real intervals*) Take $X = \mathbb{R}$ and the family of real intervals $\mathcal{I}$. Clearly for every $a \in \mathbb{R}$, $(a, a) = \emptyset \in \mathcal{I}$ and $(-\infty, \infty) \in \mathcal{I}$. Given any two intervals $I_1, I_2 \in \mathcal{I}$, we have $I_1 \cap I_2 \in \mathcal{I}$, for if $I_1 \cap I_2 = \emptyset$ then $\emptyset \in \mathcal{I}$, and if $I_1 \cap I_2 = I_3$ then $I_3 \in \mathcal{I}$ as well. To prove property three take $I = (a, b)$. Then $I^c = \mathbb{R}\backslash(a, b) = (-\infty, a] \cup [b, +\infty)$ is the disjoint union of intervals in $\mathcal{I}$.

The outcome is similar if we take left-open, right-open, or closed intervals.

Eventually the collection $\mathcal{I}$ of intervals in $\mathbb{R}$ is a semi-algebra of subsets of the real line. ★

Example 6.1.2 (*The family of right-closed real intervals*) Let $\tilde{\mathcal{I}}$ denote the class of intervals of $\mathbb{R}$ that are only closed on the right.

Clearly $\forall a \in \mathbb{R}$ the interval $(a, a] = \emptyset$ belongs in $\tilde{\mathcal{I}}$. Similarly, since $\infty \notin \mathbb{R}$, as $a \to \infty$ the interval $(-\infty, a]$ equals $(-\infty, \infty)$, so $\mathbb{R} \in \tilde{\mathcal{I}}$.

Take I_1, I_2 any two intervals in $\tilde{\mathcal{I}}$. We want to see whether $I_1 \cap I_2 \in \tilde{\mathcal{I}}$. There are two possible cases:

- $I_1 \cap I_2 = \emptyset$;
- $I_1 \cap I_2 \neq \emptyset$.

In the first case there is no doubt. In the second case, instead, we need to consider two subcases. If $I_1 \subseteq I_2$ or the other way around, $I_1 \cap I_2 = I_1$ or $I_1 \cap I_2 = I_2$, but anyhow $I_1 \cap I_2 \in \tilde{\mathcal{I}}$. If $I_1 \cap I_2 \neq \emptyset$ and $I_1 \nsubseteq I_2$, then $I_1 \cap I_2$ is right-closed and belongs to $\tilde{\mathcal{I}}$. Hence $I_1 \cap I_2 \in \tilde{\mathcal{I}}$.

Now let us prove the third property. For this, let $I = (a, b]$, so

$$I^c = \mathbb{R}\backslash(a, b] = (-\infty, a] \cup (b, \infty)$$

a union of disjoint members of $\tilde{\mathcal{I}}$.

Overall, the collection $\tilde{\mathcal{I}}$ of right-closed intervals is a semi-algebra of subsets of the real line. This semi-algebra can therefore be described as the family

$$\tilde{\mathcal{I}} = \big\{(a, b] : -\infty < a \leq b < \infty\big\} \cup \big\{(a, \infty) : -\infty < a < \infty\big\}.$$

★

Here is another example of a semi-algebra.

Example 6.1.3 (*The family of singletons of a finite set X*) Given a finite set $X = \big\{x_1, x_2, \ldots, x_n\big\}$, the collection

$$\mathcal{C} = \big\{\{x_i\} : x_i \in X\big\} \cup \big\{\emptyset, X\big\}$$

is a semi-algebra of subsets of X. In fact, by definition $\emptyset, X \in \mathcal{C}$. Given two elements $\{x_i\}, \{x_j\} \subseteq X$, clearly $\{x_i\} \cap \{x_j\} = \emptyset \in \mathcal{C}$ if $i \neq j$ and $\{x_i\} \cap \{x_j\} = \{x_i\} \in \mathcal{C}$ if $i = j$. Finally, $\{x_i\}^c = \bigsqcup_{j \neq i} \{x_j\}$, is a finite disjoint union of sets in $\mathcal{C}$. ★

6.2 Algebra of Sets Generated by a Semi-algebra

Definition 6.2.1 (*Algebra of sets*) Let $\mathcal{A}$ be a collection of subsets in a non-empty set X. We say $\mathcal{A}$ is an algebra of subsets of X if:

1. $\emptyset, X \in \mathcal{A}$;
2. $A \cap B \in \mathcal{A}, \quad \forall A, B \in \mathcal{A}$, i.e. $\mathcal{A}$ is closed under finite intersections;
3. $A^c \in \mathcal{A}, \quad \forall A \in \mathcal{A}$, i.e. $\mathcal{A}$ is closed under taking set-complements.

Henceforth the term algebra of sets will be shortened to 'algebra' for simplicity. ◇

From the last property we deduce that an algebra is a semi-algebra but not the conversely, hence the concept of algebra is stronger than that of semi-algebra.

Combining properties 2 and 3 we obtain

$$A_i \in \mathcal{A} \text{ for } i = 1, \ldots, n \text{ and } A = \bigcap_{i=1}^{n} A_i \in \mathcal{A}, \Leftrightarrow A^c = \bigcup_{i=1}^{n} A_i^c \in \mathcal{A};$$

meaning that **an algebra is also closed under finite unions of non-pairwise-disjoint sets**.

Clearly the largest algebra of subsets of a given set X is the family $\mathcal{P}(X)$. The latter contains all possible subsets of X, so it also contains, together with the empty set and X itself, all finite unions, all finite intersections and every complement set. Clearly $\mathcal{P}(X)$ is a trivial example of an algebra. What we are interested in is whether, given an arbitrary collection $\mathcal{C}$ of subsets in a non-empty set X, there exists a smallest algebra of subsets that contains $\mathcal{C}$. Here is the answer.

Theorem 6.2.1 (Minimal algebra generated by a collection) *Let X be a non-empty set, $\mathcal{C}$ a collection of subsets of X and $\mathscr{F}(\mathcal{C})$ the family of algebras $\mathcal{A}$ of subsets of X containing $\mathcal{C}$. The collection of subsets of X*

$$\mathcal{A}(\mathcal{C}) = \bigcap_{\mathcal{A}\in\mathscr{F}(\mathcal{C})} \mathcal{A},$$

called algebra generated by $\mathcal{C}$, is well defined, and is the smallest algebra of subsets of X containing $\mathcal{C}$.

Proof Define

$$\mathscr{F}(\mathcal{C}) = \{\mathcal{A} : \mathcal{A} \subseteq \mathcal{P}(X), \mathcal{A} \text{ is an algebra}, \mathcal{C} \subseteq \mathcal{A}\}$$

as the family of algebras containing $\mathcal{C}$. Clearly this family is not empty, for $\mathcal{P}(X) \in \mathscr{F}$. Let $\mathcal{A}(\mathcal{C})$ be the intersection of the family $\mathscr{F}(\mathcal{C})$:

$$\mathcal{A}(\mathcal{C}) = \bigcap_{\mathcal{A}\in\mathscr{F}(\mathcal{C})} \mathcal{A}.$$

Clearly $\mathcal{C} \subseteq \mathcal{A}(\mathcal{C})$, since $\mathcal{C}$ is a subset of every algebra in $\mathscr{F}(\mathcal{C})$. Moreover $\mathcal{A}(\mathcal{C})$ exists, since there is at least one algebra containing $\mathcal{C}$, namely the whole power set.

Let us prove $\mathcal{A}(\mathcal{C})$ is an algebra. First, $\emptyset, X \in \mathcal{A}(\mathcal{C})$, since $\mathcal{A}(\mathcal{C})$ is the intersection of all families $\mathcal{A} \in \mathscr{F}(\mathcal{C})$, each of which necessarily contains $\emptyset$ and X, as any algebra does. Let now $A \in \mathcal{A}(\mathcal{C})$, so $A \in \mathcal{A}$, for every $\mathcal{A} \in \mathscr{F}(\mathcal{C})$. Since $\mathcal{A}$ is an algebra, $A^c \in \mathcal{A}$ for every $\mathcal{A} \in \mathscr{F}(\mathcal{C})$, so $A^c \in \mathcal{A}(\mathcal{C})$. Take $A, B \in \mathcal{A}(\mathcal{C})$, so $A, B \in \mathcal{A}$ for every $\mathcal{A} \in \mathscr{F}(\mathcal{C})$. Then $A \cap B \in \mathcal{A}$, for every $\mathcal{A} \in \mathscr{F}(\mathcal{C})$, and hence $A \cap B \in \mathcal{A}(\mathcal{C})$. Therefore $\mathcal{A}(\mathcal{C})$ is an algebra containing $\mathcal{C}$.

Since $\mathcal{A}(\mathcal{C})$ is the intersection of all algebras containing $\mathcal{C}$, it has the obvious property if being the smallest such algebra. We call it the **algebra generated by** $\mathcal{C}$. ■

Proposition 6.2.1 (Algebra generated by a semi-algebra) *If $\mathcal{C}$ is a semi-algebra of subsets in a set $X \neq \emptyset$, then*

$$\mathcal{A}(\mathcal{C}) = \left\{ A \subseteq X : \textit{for some } n \in \mathbb{N},\ A = \bigcup_{i=1}^{n} A_i \textit{ and } A_i \in \mathcal{C} \right\},$$

is the smallest algebra containing $\mathcal{C}$.

Proof Clearly $\mathcal{C} \subseteq \mathcal{A}(\mathcal{C})$, so $\emptyset, X \in \mathcal{C} \Rightarrow \emptyset, X \in \mathcal{A}(\mathcal{C})$. Let $A = \bigcup_{i=1}^{n} A_i \in \mathcal{A}(\mathcal{C})$ with $A_i \in \mathcal{C}$ for every $i = 1, \ldots, n$, and $B = \bigcup_{j=1}^{m} B_j \in \mathcal{A}(\mathcal{C})$ with $B_j \in \mathcal{C}$ for every $j = 1, \ldots, m$. Then

$$A \cap B = \bigcup_{i,j} (A_i \cap B_j),$$

where for every i and j, $(A_i \cap B_j) \in \mathcal{C}$, so $(A \cap B) \in \mathcal{A}(\mathcal{C})$. Let $A = \bigcup_{i=1}^{n} A_i \in \mathcal{A}(\mathcal{C})$ with $A_i \in \mathcal{C}$ for every $i = 1, \ldots, n$. Then

$$A_i^c = \bigsqcup_{j=1}^{n_i} B_j^i$$

where $B_j^i \in \mathcal{C}$ for every $j = 1, \ldots, n_i$. The finite, disjoint union of sets of the semi-algebra $\mathcal{C}$ belongs in $\mathcal{A}(\mathcal{C})$, and we know $\mathcal{A}(\mathcal{C})$ is closed under finite intersections:

$$A^c = \bigcap_{i=1}^{n} A_i^c = \bigcap_{i=1}^{n} \left(\bigsqcup_{j=1}^{n_i} B_j^i \right) \in \mathcal{A}(\mathcal{C}).$$

Hence $\mathcal{A}(\mathcal{C})$ is an algebra.

Now we show $\mathcal{A}(\mathcal{C})$ is the smallest algebra containing $\mathcal{C}$. Suppose $\mathcal{A}$ is another algebra containing $\mathcal{C}$. If $A \in \mathcal{A}(\mathcal{C})$, clearly $A \in \mathcal{A}$. But A is arbitrary, and the claim follows. ■

Remark 6.2.1 Semi-algebra is not closed under set-complementation, since for every $A \in \mathcal{C}$, $A^c \notin \mathcal{C}$. Yet it is interesting to note that the complement of any A in $\mathcal{C}$, albeit not belonging in the family, can be written as a pairwise-disjoint union of elements $C_i \in \mathcal{C}$. Furthermore, since finite unions - even disjoint - do not belong in the semi-algebra, if $\mathcal{C}$ is a semi-algebra of subsets of some X, then

$$\mathcal{A}^+ = \left\{ A \subseteq X : \text{for some } n \in \mathbb{N},\ A = \bigsqcup_{i=1}^{n} A_i \text{ and } A_i \in \mathcal{C} \right\}$$

contains the semi-algebra $\mathcal{C}$.

Imposing on $\mathcal{C}$ that it be closed under finite disjoint unions means demanding closure under complementation. With such an enhancement the semi-algebra $\mathcal{C}$ turns into the algebra $\mathcal{A}(\mathcal{C})$ generated by $\mathcal{C}$. We have, in fact, the following ♣

Proposition 6.2.2 (Algebra generated by pairwise-disjoint sets of a semi-algebra) *If $\mathcal{C}$ is a semi-algebra of subsets in a non-empty set X, the family*

$$\mathcal{A}^+ = \left\{ A \subseteq X : \textit{for some } n \in \mathbb{N},\ A = \bigsqcup_{i=1}^{n} A_i \textit{ and } A_i \in \mathcal{C} \right\}$$

is an algebra, coinciding with $\mathcal{A}(\mathcal{C})$.

Proof Clearly $\mathcal{C} \subseteq \mathcal{A}^+ \subseteq \mathcal{A}(\mathcal{C})$, so there only remains to show $\mathcal{A}^+$ is an algebra.

To begin with, $\emptyset, X \in \mathcal{C} \Rightarrow \emptyset, X \in \mathcal{A}^+$. Let $A = \bigsqcup_{i=1}^{n} A_i \in \mathcal{A}^+$, $A_i \in \mathcal{C}$ for every $i = 1, \ldots, n$, and $B = \bigsqcup_{j=1}^{m} B_m \in \mathcal{A}^+$, $B_j \in \mathcal{C}$ for every $j = 1, \ldots, m$. Then

$$A \cap B = \bigcup_{i,j} (A_i \cap B_j),$$

for every i and j, $(A_i \cap B_j) \in \mathcal{C}$, $(A_i \cap B_j) \subseteq A_i$ and $(A_i \cap B_j) \subseteq B_j$. Moreover, for every $i \neq h$ and $j \neq k$

$$(A_i \cap B_j) \cap (A_h \cap B_k) = \emptyset,$$

so $A \cap B = \bigsqcup_{i,j} (A_i \cap B_j)$, and then $A \cap B \in \mathcal{A}^+$ is closed under finite intersections. Let $A = \bigsqcup_{i=1}^{n} A_i \in \mathcal{A}^+$, $A_i \in \mathcal{C}$ for every $i = 1, \ldots, n$, so

$$A_i^c = \bigsqcup_{j=1}^{n_i} B_j^i$$

with $B_j^i \in \mathcal{C}$ for every $j = 1, \ldots, n_i$. Disjoint finite unions in $\mathcal{C}$ belong to $\mathcal{A}^+$ by construction, and we know the latter is closed under finite intersections:

$$A^c = \bigcap_{i=1}^{n} A_i^c = \bigcap_{i=1}^{n} \left(\bigsqcup_{j=1}^{n_i} B_j^i \right) \in \mathcal{A}^+.$$

Therefore $\mathcal{A}^+$ is an algebra. ■

Remark 6.2.2 Directly from the previous proposition we have an important consequence:

the elements in the algebra $\mathcal{A}(\mathcal{C})$ generated by a semi-algebra $\mathcal{C}$ can always be expressed as disjoint finite unions in $\mathcal{C}$:

$$\begin{aligned} \mathcal{A}(\mathcal{C}) &= \left\{ A \subseteq X : \text{for some } n \in \mathbb{N},\ A = \bigcup_{i=1}^{n} A_i \text{ and } A_i \in \mathcal{C} \right\} \\ &= \left\{ A \subseteq X : \text{for some } n \in \mathbb{N},\ A = \bigsqcup_{i=1}^{n} A_i \text{ and } A_i \in \mathcal{C} \right\} = \mathcal{A}^+. \end{aligned}$$

♣

Proposition 6.2.2 and Remark 6.2.2, in the case $X = \mathbb{R}$, say that the family

$$\mathcal{A} = \left\{ A \subseteq \mathbb{R} : \text{for some } n \in \mathbb{N},\ A = \bigsqcup_{i=1}^{n} I_i,\ I_i \in \mathcal{I} \right\}$$

is the algebra generated by the semi-algebra $\mathcal{I}$.

Should we want to generalise Proposition 6.2.1 to an arbitrary family $\mathcal{C}$ of subsets in X, with no assumption on the family, we would not be able to assume complements of elements of $\mathcal{C}$ to be disjoint finite unions in $\mathcal{C}$. We would have to include them explicitly in the recursive definition in order to generate the algebra $\mathcal{A}(\mathcal{C})$.

Proposition 6.2.3 (Algebra generated by an arbitrary family of sets) *Let X be a non-empty set and $\mathcal{C}$ a family of subsets such that $X \in \mathcal{C}$. Define recursively the families*

$$\mathcal{C}^* = \left\{A \subseteq X : A \text{ or } A^c \underset{m}{\in} \mathcal{C}\right\}$$
$$\mathcal{C}_1 = \left\{A \subseteq X : \text{for some } m \in \mathbb{N},\, A = \bigcup_{i=1}^{m} A_i,\, A_i \in \mathcal{C}^*\right\},$$

for $n \geq 2$

$$\mathcal{C}^*_{n-1} = \left\{A \subseteq X : A \text{ or } A^c \underset{m}{\in} \mathcal{C}_{n-1}\right\}$$
$$\mathcal{C}_n = \left\{A \subseteq X : \text{for some } m \in \mathbb{N},\, A = \bigcup_{i=1}^{m} A_i,\, A_i \in \mathcal{C}^*_{n-1}\right\}.$$

Then

$$\mathcal{A} = \bigcup_{n=1}^{\infty} \mathcal{C}_n$$

is the algebra $\mathcal{A}(\mathcal{C})$ generated by the collection $\mathcal{C}$.

Proof Any $A \in \mathcal{C}$ is the finite union of itself, so for every $A \in \mathcal{C}$ we have $A \in \mathcal{C}_1$, and then $\mathcal{C} \subseteq \mathcal{C}_1$. Similarly, for every $n \in \mathbb{N}$ we have $\mathcal{C}_{n-1} \subseteq \mathcal{C}_n$, hence $\mathcal{C} \subseteq \mathcal{A}$.

By assumption $X \in \mathcal{C}$, so by definition $\emptyset \in \mathcal{C}_1$, and then $\emptyset \in \mathcal{A}$. Let $A \in \mathcal{A}$, then clearly there is some $n \in \mathbb{N}$ such that $A \in \mathcal{C}_n$, so $A^c \in \mathcal{C}_{n+1}$ and then $A^c \in \mathcal{A}$. Take sets $\{A_1, \ldots, A_m\}$ in $\mathcal{A}$. For some $n \in \mathbb{N}$, $A_1, \ldots, A_m \in \mathcal{C}_n$, meaning $\bigcup_{i=1}^{m} A_i \in \mathcal{C}_{n+1} \subseteq \mathcal{A}$.

The family

$$\mathcal{A} = \bigcup_{n=1}^{\infty} \mathcal{C}_n$$

is therefore an algebra. It is, indeed, the algebra generated by $\mathcal{C}$: let $\mathcal{B}$ be another algebra such that $\mathcal{C} \subseteq \mathcal{B}$. Then $\mathcal{C}_n \subseteq \mathcal{B}$ for every n, and so $\mathcal{A} \subseteq \mathcal{B}$. ■

At each step $n \in \mathbb{N}$ we add to $\mathcal{C}_n$ the complements of the elements in the preceding one.[2] This recursive process, together with finite unions, enlarges the initial family $\mathcal{C}$ until $\mathcal{A}(\mathcal{C})$ is achieved.

Example 6.2.1 (*Finite-cofinite algebra*) Let X be a non-empty set and take the set $\mathcal{C} = \{\{x\} : x \in X\}$. We claim the family

$$\mathcal{A}(\mathcal{C}) = \{A \subseteq X : A \text{ or } A^c \text{ is finite}\}$$

is the algebra generated by $\mathcal{C}$.

To check this, note $\emptyset$ is finite by definition, so $\emptyset \in \mathcal{A}(\mathcal{C})$. In order to $X \in \mathcal{A}(\mathcal{C})$, it or its complement, or both, must be finite. Yet, regarding X, we only know it is an arbitrary non empty set. Then, if it is finite, immediately $\mathcal{A}(\mathcal{C}) = \mathcal{P}(X)$, if not, its complement (the empty set) is by definition finite and belongs to $\mathcal{A}(\mathcal{C})$, so $X \in \mathcal{A}(\mathcal{C})$.

Let now $A \in \mathcal{A}(\mathcal{C})$, so $A^c \in \mathcal{A}(\mathcal{C})$ by assumption. Take $A_1, A_2 \in \mathcal{A}(\mathcal{C})$:

1. if both are finite, $A_1 \cup A_2$ is finite and belongs to $\mathcal{A}(\mathcal{C})$;
2. if A_1 is infinite (and similarly for A_2), as $A_1 \in \mathcal{A}(\mathcal{C})$ then A_1^c is finite, and $A_1 \subseteq (A_1 \cup A_2)$, so $(A_1 \cup A_2)^c \subseteq A_1^c$. But A_1^c finite implies $(A_1 \cup A_2)^c$ finite, and by definition $A_1 \cup A_2 \in \mathcal{A}(\mathcal{C})$;
3. if neither of $A_1, A_2 \in \mathcal{A}(\mathcal{C})$ are finite, then A_1^c, A_2^c are finite, so $A_1^c \cap A_2^c = (A_1 \cup A_2)^c$ is finite and $A_1 \cup A_2 \in \mathcal{A}(\mathcal{C})$.

Given that $\mathcal{A}(\mathcal{C})$ is closed under complements, the family is also closed under finite intersections. Hence $\mathcal{A}(\mathcal{C})$ is an algebra of subsets of X.

Now we show it is the smallest algebra containing $\mathcal{C}$. Clearly $\mathcal{C} \subseteq \mathcal{A}(\mathcal{C})$, since every singleton $\{x\} \subseteq X$, being finite, is in $\mathcal{A}(\mathcal{C})$. Take any algebra $\mathcal{A}$ of subsets of X such that $\mathcal{C} \subseteq \mathcal{A}$. Given $A \in \mathcal{A}(\mathcal{C})$, at least one of A, A^c is finite, say $\{x_1, x_2, \ldots, x_n\}$ for $\{x_i\} \in \mathcal{C}$, for every $i = 1, 2, \ldots, n$.

In the first case, $\mathcal{A}(\mathcal{C})$ containing $\mathcal{C}$, we have $A = \bigcup_{i=1}^n \{x_i\}$. As $\mathcal{C} \subseteq \mathcal{A}$, then $\forall i, \{x_i\} \in \mathcal{A}$. By hypothesis $\mathcal{A}$ is an algebra, hence closed under finite unions: $A = \bigcup_{i=1}^n \{x_i\} \in \mathcal{A}$. Hence $\mathcal{A}(\mathcal{C}) \subseteq \mathcal{A}$, and the arbitrary choice of $\mathcal{A}$ allows to conclude.

In the second case it suffices to recall $\mathcal{A}$, as an algebra, is closed under complements, and from there the argument goes as before. ★

Example 6.2.2 (*The family of finite unions of real intervals*) Now take $X = \mathbb{R}$. The collection $\mathcal{A}$ of subsets arising as finite unions of intervals,

$$\mathcal{A} = \left\{ A \subseteq \mathbb{R} : \text{for some } n \in \mathbb{N},\ A = \bigcup_{i=1}^{n} I_i,\ I_i \in \mathcal{I} \right\}$$

is an algebra. That is because the family $\mathcal{I}$ of real intervals, see Example 6.1.1, is a semi-algebra, and $\mathcal{A} = \mathcal{A}(\mathcal{I})$, i.e. $\mathcal{A}$ is the smallest algebra containing $\mathcal{I}$. ★

[2] If $\mathcal{C}$ is a semi-algebra this operation amounts to including in $\mathcal{C}$ finite disjoint unions. See Remark 6.2.1 and Proposition 6.2.2.

Remark 6.2.2 and the above example imply that any element in the algebra generated by $\mathcal{I}$, i.e. any finite union of intervals in $\mathcal{I}$, is representable as finite union of pairwise-disjoint intervals in $\mathcal{I}$.

Chapter 7
Monotone Classes and σ-Algebras



7.1 Monotone Classes and Generation of σ-Algebras

Of pivotal importance in measure theory and integration are the collections of sets known as σ-algebras. Compared to other families, they possess a further, decisive, property: they are closed under countable unions.

Definition 7.1.1 (σ-*algebra of sets*) A family of subsets $\mathcal{S}$ of a non-empty set X is a σ-algebra when:

1. $\emptyset, X \in \mathcal{S}$;
2. $\bigcup_{i=1}^{\infty} A_i \in \mathcal{S}, \quad \forall A_i \in \mathcal{S}, i = 1, 2, \ldots$, i.e. $\mathcal{S}$ is closed under countable unions;
3. $A^c \in \mathcal{S}, \quad \forall A \in \mathcal{S}$.

$\diamond$

Clearly the class of σ-algebras is not empty since $\{\emptyset, X\}$ and $\mathcal{P}(X)$ are σ-algebras, trivially. The former is the smallest, the latter the largest, σ-algebra of subsets of a non-empty set X.

Now we ask, given a family of subsets $\mathcal{C}$ of some non-empty set X, whether there exists a minimal σ-algebra containing $\mathcal{C}$ exists, i.e. the σ-algebra generated by $\mathcal{C}$. The answer is yes, as the next theorem explains.

Theorem 7.1.1 (Minimal σ -algebra) *Let X be a non-empty set, $\mathcal{C}$ a family of subsets of X and $\mathscr{F}(\mathcal{C})$ the collection of all σ-algebras $\mathcal{S}$ of subsets of X containing $\mathcal{C}$. Then the family*

$$\mathcal{S}(\mathcal{C}) = \bigcap_{\mathcal{S} \in \mathscr{F}(\mathcal{C})} \mathcal{S}$$

exists. It is the smallest σ-algebra of subsets of X containing $\mathcal{C}$, and is called σ-algebra on X generated by $\mathcal{C}$.

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Proof Define the collection of σ-algebras containing $\mathcal{C}$:

$$\mathscr{F}(\mathcal{C}) = \left\{\mathcal{S} : \mathcal{S} \subseteq \mathcal{P}(X), \mathcal{S} \text{ is a } \sigma\text{-algebra and } \mathcal{C} \subseteq \mathcal{S}\right\}$$

Clearly, by the earlier argument this is not empty. Let $\mathcal{S}(\mathcal{C})$ denote the intersection of all σ-algebras $\mathcal{S}$ in $\mathscr{F}(\mathcal{C})$:

$$\mathcal{S}(\mathcal{C}) = \bigcap_{\mathcal{S} \in \mathscr{F}(\mathcal{C})} \mathcal{S}.$$

Clearly $\mathcal{C} \subseteq \mathcal{S}(\mathcal{C})$, since $\mathcal{C}$ is a subset of every σ-algebra in $\mathscr{F}(\mathcal{C})$, and $\mathcal{S}(\mathcal{C})$ is well defined because there is at least one σ-algebra containing $\mathcal{C}$, the power set.

Let us show $\mathcal{S}(\mathcal{C})$ is a σ-algebra. First, the empty set and the whole X belong in $\mathcal{S}(\mathcal{C})$, as both belong in every σ-algebra of $\mathscr{F}(\mathcal{C})$, of which $\mathcal{S}(\mathcal{C})$ is the intersection. Take $A \in \mathcal{S}(\mathcal{C})$, so $A \in \mathcal{S}$ for every σ-algebra $\mathcal{S} \in \mathscr{F}(\mathcal{C})$. From $A^c \in \mathcal{S}$ for every $\mathcal{S} \in \mathscr{F}(\mathcal{C})$, we have

$$A^c \in \bigcap_{\mathcal{S} \in \mathscr{F}(\mathcal{C})} \mathcal{S} = \mathcal{S}(\mathcal{C}).$$

Let $A_1, A_2, \ldots, A_n, \ldots$ be a sequence in $\mathcal{S}(\mathcal{C})$, so $A_n \in \mathcal{S}$ for every $\mathcal{S} \in \mathscr{F}(\mathcal{C})$, and $\bigcup_{n=1}^{\infty} A_n \in \mathcal{S}$ for every $\mathcal{S} \in \mathscr{F}(\mathcal{C})$:

$$\bigcup_{n=1}^{\infty} A_n \in \bigcap_{\mathcal{S} \in \mathscr{F}(\mathcal{C})} \mathcal{S} = \mathcal{S}(\mathcal{C}).$$

Moreover $\mathcal{C}$ belongs in $\mathcal{S}(\mathcal{C})$, as an intersection of σ-algebras all containing $\mathcal{C}$.

Because of that, moreover, $\mathcal{S}(\mathcal{C})$ is also the smallest σ-algebra among those containing $\mathcal{C}$, and the proposition is proved. ■

Recall that starting from a semi-algebra $\mathcal{C}$ of subsets of $X \neq \emptyset$ it is always possible to generate the algebra $\mathcal{A}(\mathcal{C})$ by adding finite unions over $\mathcal{C}$. The same trick does not work on an algebra $\mathcal{A}$ to produce the σ-algebra $\mathcal{S}(\mathcal{A})$.

Suppose in fact $\mathcal{A}$ is an algebra of subsets of a non-empty set X, and $\mathcal{E}_1$ is the family of countable unions over $\mathcal{A}$:

$$\mathcal{E}_1 = \left\{ \bigcup_{n=1}^{\infty} A_n : \forall n \in \mathbb{N}, A_n \in \mathcal{A} \right\}.$$

This is not a σ-algebra. Namely, for every $n \in \mathbb{N}$ set

$$B_1 = A_1, \quad B_2 = (A_1 \cup A_2), \ldots, \quad B_n = \bigcup_{i=1}^{n} A_i, \ldots$$

then observe $B_n \in \mathcal{A}$ for every $n \in \mathbb{N}$, and $B_1 \subseteq B_2 \subseteq \ldots \subseteq B_n \subseteq \ldots$, but as $n \to \infty$ generally

$$\lim_{n\to\infty} B_n = \bigcup_{n=1}^{\infty} A_n = A \notin \mathcal{A},$$

and

$$A^c = \left(\bigcup_{n=1}^{\infty} A_n\right)^c = \bigcap_{n=1}^{\infty} A_n^c \notin \mathcal{E}_1.$$

A similar outcome presents itself if we add, instead of countable unions, countable intersections over $\mathcal{A}$. In fact define

$$\mathcal{P}_1 = \left\{\bigcap_{n=1}^{\infty} A_n : \forall n \in \mathbb{N},\, A_n \in \mathcal{A}\right\},$$

and put, as before, for every $n \in \mathbb{N}$

$$B_1 = A_1, \quad B_2 = (A_1 \cap A_2), \ldots, \quad B_n = \bigcap_{i=1}^{n} A_i, \ldots$$

Again $B_n \in \mathcal{A}$ for every $n \in \mathbb{N}$, and $B_1 \supseteq B_2 \supseteq \ldots \supseteq B_n \supseteq \ldots$. Yet as $n \to \infty$ we generally have

$$\lim_{n\to\infty} B_n = \bigcap_{n=1}^{\infty} A_n = A \notin \mathcal{A},$$

and

$$A^c = \left(\bigcap_{n=1}^{\infty} A_n\right)^c = \bigcup_{n=1}^{\infty} A_n^c \notin \mathcal{P}_1.$$

Then call $\mathcal{E}_2$ the family

$$\mathcal{E}_2 = \left\{\bigcup_{n=1}^{\infty} A_n : \forall n \in \mathbb{N},\, A_n \in \mathcal{A}(\mathcal{E}_1)\right\},$$

where $\mathcal{A}(\mathcal{E}_1)$ is a sort of enhanced algebra, for it contains all the elements of $\mathcal{E}_1$, i.e. the countable unions of sequences in $\mathcal{A}$, all finite unions of $\mathcal{E}_1$, and all complements of the above.

Notwithstanding this enlargement, $\mathcal{E}_2$ is still not a σ-algebra for the same reasons mentioned earlier. We must then use transfinite induction. Take $\alpha \in (0, \Omega)$ where Ω is the first uncountable ordinal, and define:

- if $\beta < \alpha$ is a successor ordinal

$$\mathcal{E}_\beta = \left\{ \bigcup_{n=1}^{\infty} A_n : \forall n \in \mathbb{N},\ A_n \in \mathcal{A}(\mathcal{E}_{\beta-1}) \right\};$$

- if $\alpha < \Omega$ is a limit ordinal

$$\mathcal{E}_\alpha = \left\{ \bigcup_{n=1}^{\infty} A_n : \forall n \in \mathbb{N},\ A_n \in \bigcup_{\beta<\alpha} \mathcal{A}(\mathcal{E}_\beta) \right\};$$

- over the set $\{\alpha : \alpha < \Omega\}$

$$\mathcal{S} = \bigcup_{\alpha<\Omega} \mathcal{E}_\alpha.$$

The last family is closed under unions (and intersections) of increasing (decreasing, respectively) sequences in $\mathcal{A}$, and under complements as well.

Taking all this in account, $\mathcal{S}$ should be a σ-algebra. If this was indeed the case, we would have found a means to generate, from an algebra, a σ-algebra that contains it, and in particular the smallest σ-algebra containing $\mathcal{A}$. For the moment though we can only define monotone classes, from which we can try to produce the corresponding σ-algebra.

Definition 7.1.2 (*Monotone class*) We call monotone class a collection $\mathscr{M}$ of subsets of a set $X \neq \emptyset$ enjoying the following monotonicity properties:

1. for every increasing sequence $A_1 \subseteq A_2 \subseteq \ldots \subseteq A_i \subseteq \ldots$ in $\mathscr{M}$,

$$\bigcup_{i=1}^{\infty} A_i \in \mathscr{M},$$

   i.e. $\mathscr{M}$ is closed under countable unions of increasing sequences;
2. for every decreasing sequence $A_1 \supseteq A_2 \supseteq \ldots \supseteq A_i \supseteq \ldots$ in $\mathscr{M}$,

$$\bigcap_{i=1}^{\infty} A_i \in \mathscr{M}$$

   i.e. $\mathscr{M}$ is closed under countable intersections of decreasing sequences. ◇

Not every monotone class is an algebra, as explained in the next example.

Example 7.1.1 (*A monotone class that is no algebra*) Take $X = \{a, b, c, d\}$ and the family $\mathscr{M} = \{\{a\}, \{a, b\}\}$. It is obviously monotone, but not an algebra of sets, for $X \notin \mathscr{M}$.

★

Theorem 7.1.2 (Minimal monotone class) *Let X be a non-empty set, $\mathcal{C}$ a family of subsets of X and $\mathscr{F}(\mathcal{C})$ the collection of monotone classes $\mathscr{M}$ containing $\mathcal{C}$. The family*

$$\mathscr{M}(\mathcal{C}) = \bigcap_{\mathscr{M} \in \mathscr{F}(\mathcal{C})} \mathscr{M}$$

exists and it is the smallest monotone class of subsets of X containing $\mathcal{C}$. It is called monotone class generated by $\mathcal{C}$.

Proof Note first that $\mathscr{F}(\mathcal{C})$ is non-empty. The power set, in fact, contains $\mathcal{C}$ and is closed under countable unions (and intersections) of increasing (decreasing) sequences of subsets of X. We claim $\mathscr{M}(\mathcal{C})$ fulfils the monotonicity properties.

Let $\{A_n\}$ be a decreasing sequence in $\mathscr{M}(\mathcal{C})$. Clearly $A_n \in \mathscr{M}$ for every n, so $\bigcap_{n=1}^{\infty} A_n \in \mathscr{M}$, $\forall \mathscr{M}$, and then $\bigcap_{n=1}^{\infty} A_n \in \mathscr{M}(\mathcal{C})$. Similarly for an increasing sequence $\{A_n\}$. Hence $\mathscr{M}(\mathcal{C})$ is a monotone class.

At last, it is also the smallest monotone class containing $\mathcal{C}$, being the intersection of all monotone classes containing $\mathcal{C}$.

■

From this theorem we deduce immediately that $\mathscr{M}(\mathcal{A})$ is the minimal monotone class containing the algebra $\mathcal{A}$. We will now show that σ-algebras are monotone.

Proposition 7.1.1 (σ-algebra and monotone class) *A σ-algebra is a monotone class.*

Proof A σ-algebra $\mathcal{S}$ on X is closed under arbitrary countable unions, hence under countable unions of increasing sequences. Secondly, given a decreasing sequence $\{A_n\}_{n\geq 1}$ in $\mathcal{S}$, we have $A_n^c \in \mathcal{S}$ for every $n \geq 1$. Hence $\bigcup_{n=1}^{\infty} A_n^c = \left(\bigcap_{n=1}^{\infty} A_n\right)^c \in \mathcal{S}$. By the properties of σ-algebras, $\bigcap_{n=1}^{\infty} A_n \in \mathcal{S}$.

Therefore $\mathcal{S}$ is monotone.

■

Vice versa, not every monotone class is a σ-algebra, as the next example shows.

Example 7.1.2 (*A monotone class that is no σ-algebra*) If X is uncountable, the family

$$\mathscr{M} = \{A \subseteq X : A \text{ is countable}\}$$

is monotone but not a σ-algebra.

Let us show $\mathscr{M}$ is monotone. If $\{A_n\}_{n\geq 1}$ is an increasing sequence in $\mathscr{M}$, each A_n is by definition countable, so

$$A = \bigcup_{n=1}^{\infty} A_n$$

is countable as well, whence it belongs to $\mathscr{M}$. The same argument holds for decreasing sequences in $\mathscr{M}$, so $\mathscr{M}$ is a monotone class.

Yet it is not a σ-algebra: first, it does not contain X, which is uncountable; second, because of this, if A is countable then A^c is not, so $\mathscr{M}$ is not closed under complementation.

★

Proposition 7.1.2 (Algebra, monotone class and σ-algebra) *If $\mathscr{M}$ is a monotone class and an algebra of subsets of X, then $\mathscr{M}$ is a σ-algebra of sets.*

Proof Take $X \neq \emptyset$, and let $\mathscr{M}$ be a monotone class and an algebra on X.

Clearly $\mathscr{M}$ is an algebra, so $\emptyset$, $X \in \mathscr{M}$. For that same reason $A \in \mathscr{M}$ implies $A^c \in \mathscr{M}$. Suppose $A_1, \ldots, A_n \in \mathscr{M}$. Since $\mathscr{M}$ is an algebra then $\bigcup_{i=1}^{n} A_i \in \mathscr{M}$. Furthermore, for every $n \in \mathbb{N}$ the union $\bigcup_{i=1}^{n} A_i$ is increasing, so $\bigcup_{i=1}^{\infty} A_i \in \mathscr{M}$ (the latter is monotone).

■

Theorem 7.1.3 (Halmos's monotone-class theorem) *If $\mathcal{A}$ is an algebra of sets in $X \neq \emptyset$, the monotone class $\mathscr{M}(\mathcal{A})$ is the smallest σ-algebra containing $\mathcal{A}$:*

$$\mathcal{S}(\mathcal{A}) = \mathscr{M}(\mathcal{A}).$$

Proof By Proposition 7.1.1 any σ-algebra is monotone, so $\mathcal{S}(\mathcal{A})$ is a monotone class containing $\mathcal{A}$. Theorem 7.1.2 makes $\mathscr{M}(\mathcal{A})$ the smallest monotone class containing $\mathcal{A}$, so $\mathscr{M}(\mathcal{A}) \subseteq \mathcal{S}(\mathcal{A})$.

To prove the opposite inclusion, that the monotone class $\mathscr{M}(\mathcal{A})$ **is a σ-algebra**, it suffices to show $\mathscr{M}(\mathcal{A})$ **is an algebra** because of Proposition 7.1.2.

Let us first establish $\mathscr{M}(\mathcal{A})$ **is closed under complements**. Given

$$\mathcal{B} = \{A \subseteq X : A^c \in \mathscr{M}(\mathcal{A})\},$$

to prove $\mathscr{M}(\mathcal{A}) \subseteq \mathcal{B}$ means to show for every $A \in \mathscr{M}(\mathcal{A}) \Rightarrow A \in \mathcal{B}$ and, by definition of $\mathcal{B}$, $A^c \in \mathscr{M}(\mathcal{A})$. This argument may be simplified: as $\mathscr{M}(\mathcal{A})$ is the smallest monotone class containing $\mathcal{A}$, it will be enough to show $\mathcal{B}$ is a monotone class containing $\mathcal{A}$.

To see $\mathcal{A}$ is contained in $\mathcal{B}$, take $A \in \mathcal{A}$. As $\mathcal{A}$ is an algebra, $A^c \in \mathcal{A}$ and then $A^c \in \mathscr{M}(\mathcal{A})$, so $A \in \mathcal{B}$. Hence $\mathcal{A} \subseteq \mathcal{B}$. Now let us show $\mathcal{B}$ is monotone. Take an increasing sequence $\{A_n\}_{n\geq 1}$ in $\mathcal{B}$. By definition of $\mathcal{B}$, $A_n^c \in \mathscr{M}(\mathcal{A})$ $\forall n \in \mathbb{N}$, and the sequence is decreasing. As $\mathscr{M}(\mathcal{A})$ is monotone,

$$\bigcap_{n=1}^{\infty} A_n^c \in \mathscr{M}(\mathcal{A}) \Leftrightarrow \left(\bigcup_{n=1}^{\infty} A_n\right)^c \in \mathscr{M}(\mathcal{A}).$$

By definition of $\mathcal{B}$, then, $\bigcup_{n=1}^{\infty} A_n \in \mathcal{B}$, making $\mathcal{B}$ closed under countable unions of increasing sequences. In a similar manner one proves $\mathcal{B}$ is closed under countable intersections of decreasing sequences. Therefore $\mathscr{M}(\mathcal{A}) \subseteq \mathcal{B}$ and $\mathscr{M}(\mathcal{A})$ is complement-closed. Furthermore, $\emptyset \in \mathcal{A}$ implies $\emptyset \in \mathcal{B}$, and by definition of $\mathcal{B}$, $X \in \mathscr{M}(\mathcal{A})$. Similarly, $X \in \mathcal{A}$ implies $X \in \mathcal{B}$ and by definition of $\mathcal{B}$ we have $\emptyset \in \mathscr{M}(\mathcal{A})$.

At last we prove $\mathscr{M}(\mathcal{A})$ **is closed under finite unions**. Fix $A \in \mathcal{A}$ and consider the family

$$\mathscr{M}_A = \left\{B \subseteq X : A \cup B \in \mathscr{M}(\mathcal{A})\right\}.$$

This collection contains the algebra $\mathcal{A}$ and is monotone. In fact, if $B \in \mathcal{A}$, then $A \cup B \in \mathcal{A}$ and by definition $B \in \mathscr{M}_A$, so $\mathcal{A} \subseteq \mathscr{M}_A$. Let $\{B_n\}_{n\geq 1}$ be an increasing sequence in $\mathscr{M}_A$, so $B_n \cup A \in \mathscr{M}(\mathcal{A})$ where $\{B_n \cup A\}_{n\geq 1}$ is an increasing sequence in $\mathscr{M}(\mathcal{A})$. Since $\mathscr{M}(\mathcal{A})$ is monotone,

$$\bigcup_{n=1}^{\infty}(B_n \cup A) \in \mathscr{M}(\mathcal{A}) \Leftrightarrow \left(\bigcup_{n=1}^{\infty} B_n\right) \cup A \in \mathscr{M}(\mathcal{A})$$

By definition of $\mathscr{M}_A$ this means $\bigcup_{n=1}^{\infty} B_n \in \mathscr{M}_A$, i.e. $\mathscr{M}_A$ is closed under countable unions of increasing sequences. In the same fashion we could prove $\mathscr{M}_A$ is stable under decreasing countable intersections, whence $\mathscr{M}_A$ is a monotone class containing $\mathcal{A}$. Being $\mathscr{M}(\mathcal{A})$ the smallest such class,

$$\mathscr{M}(\mathcal{A}) \subseteq \mathscr{M}_A.$$

The above means that

- $A \cup B \in \mathscr{M}(\mathcal{A})$ for every $A \in \mathcal{A}$ and $B \in \mathscr{M}(\mathcal{A})$.

Now fix $B \in \mathcal{A}$ and consider the class

$$\mathscr{M}_B = \left\{A \subseteq X : A \cup B \in \mathscr{M}(\mathcal{A})\right\}.$$

As we did before for $\mathscr{M}_A$, one could prove $\mathscr{M}_B$ is a monotone class such that $\mathscr{M}(\mathcal{A}) \subseteq \mathscr{M}_B$. These inclusions imply $\mathscr{M}(\mathcal{A}) \subseteq \mathscr{M}_A \cap \mathscr{M}_B$, so

$$A \cup B \in \mathscr{M}(\mathcal{A}) \text{ for every } A \text{ and } B \in \mathscr{M}(\mathcal{A}).$$

We have proved $\mathscr{M}(\mathcal{A})$ is an algebra containing $\mathcal{A}$. Being monotone as well, it is a σ-algebra containing $\mathcal{A}$.

But $\mathcal{S}(\mathcal{A})$ is the minimal σ-algebra of subsets of X containing $\mathcal{A}$, so $\mathcal{S}(\mathcal{A}) \subseteq \mathscr{M}(\mathcal{A})$, and finally

$$\mathcal{S}(\mathcal{A}) = \mathscr{M}(\mathcal{A}).$$

■

The above theorem proves that "every σ-algebra is the monotone limit of an algebra". In this light, by induction we may effortlessly generate from an algebra the corresponding σ-algebra, as we set out to prove below.

Theorem 7.1.4 (σ-algebra generated by an algebra) *Let $\mathcal{A}$ be an algebra of sets in $X \neq \emptyset$. Define recursively*

$$\mathcal{E}_1 = \left\{ \bigcup_{n=1}^{\infty} A_n : \forall n \in \mathbb{N},\, A_n \in \mathcal{A} \right\},$$

for $\beta < \alpha$ successor ordinal:

$$\mathcal{E}_\beta = \left\{ \bigcup_{n=1}^{\infty} A_n : \forall n \in \mathbb{N},\, A_n \in \mathcal{A}(\mathcal{E}_{\beta-1}) \right\},$$

and for $\alpha < \Omega$ limit ordinal:

$$\mathcal{E}_\alpha = \left\{ \bigcup_{n=1}^{\infty} A_n : \forall n \in \mathbb{N},\ A_n \in \bigcup_{\beta<\alpha} \mathcal{A}\big(\mathcal{E}_\beta\big) \right\}.$$

Then for any ordinal in $\{\alpha : \alpha < \Omega\}$ the family $\mathcal{S} = \bigcup_{\alpha<\Omega} \mathcal{E}_\alpha$ is the smallest σ-algebra containing $\mathcal{A}$, i.e. $\mathcal{S}(\mathcal{A}) = \bigcup_{\alpha<\Omega} \mathcal{E}_\alpha$.

Proof Clearly $X, \emptyset \in \mathcal{E}_1$, so $X, \emptyset \in \bigcup_{\alpha<\Omega} \mathcal{E}_\alpha$. Let $A \in \bigcup_{\alpha<\Omega} \mathcal{E}_\alpha$, so $A \in \mathcal{E}_\beta$ for some successor ordinal $\beta < \alpha$. Then $A^c \in \mathcal{A}\big(\mathcal{E}_\beta\big)$, and

$$A^c \in \bigcup_{\beta<\alpha} \mathcal{A}\big(\mathcal{E}_\beta\big).$$

This, moreover, holds for every ordinal $\alpha < \Omega$, and so $\bigcup_{\alpha<\Omega} \mathcal{E}_\alpha$ is closed under complementation. Take a sequence $\{A_n\}_{n\geq 1}$ in $\bigcup_{\alpha<\Omega} \mathcal{E}_\alpha$. For every n there is a β smaller than some α such that $A_n \in \mathcal{A}\big(\mathcal{E}_\beta\big)$. Hence there is an $\alpha < \Omega$ such that

$$\bigcup_{n=1}^{\infty} A_n \in \mathcal{E}_\alpha.$$

Even better, for all ordinals $\alpha < \Omega$ we have $\bigcup_{n=1}^{\infty} A_n \in \bigcup_{\alpha<\Omega} \mathcal{E}_\alpha$, whence $\bigcup_{\alpha<\Omega} \mathcal{E}_\alpha$ is closed under countable unions. All in all $\bigcup_{\alpha<\Omega} \mathcal{E}_\alpha$ is a σ-algebra.

There remains to show it is minimal. Let $\mathcal{B}$ be a σ-algebra such that $\mathcal{A} \subseteq \mathcal{B}$. Then clearly $\mathcal{E}_1 \subseteq \mathcal{B}$, for $\mathcal{E}_1$ contains $\mathcal{A}$ and is not closed under complements. The first inductive step is to suppose $\mathcal{E}_\beta \subseteq \mathcal{B}$ with $\beta < \alpha$ a successor. As $\mathcal{A}(\mathcal{E}_\beta)$ is an algebra containing $\mathcal{E}_\beta$ and since $\mathcal{B}$ is a σ-algebra containing $\mathcal{E}_\beta$, then $\mathcal{A}(\mathcal{E}_\beta) \subseteq \mathcal{B}$ and $\mathcal{E}_{\beta+1} \subseteq \mathcal{B}$. As second inductive step, given $\alpha < \Omega$ suppose the above holds for every $\beta < \alpha$, i.e. $\bigcup_{\beta<\alpha} \mathcal{A}(\mathcal{E}_\beta) \subseteq \mathcal{B}$. As $\mathcal{B}$ is a σ-algebra containing $\bigcup_{\beta<\alpha} \mathcal{A}(\mathcal{E}_\beta)$, given any sequence $\{A_n\}_{n\geq 1}$ in $\bigcup_{\beta<\alpha} \mathcal{A}(\mathcal{E}_\beta)$ we have $\bigcup_{n=1}^{\infty} A_n \in \mathcal{B}$, so $\mathcal{E}_\alpha \subseteq \mathcal{B}$.

As $\alpha < \Omega$ is arbitrary, the inclusion $\mathcal{E}_\alpha \subseteq \mathcal{B}$ is true for every $\alpha < \Omega$, and then

$$\mathcal{S} = \bigcup_{\alpha<\Omega} \mathcal{E}_\alpha \subseteq \mathcal{B}.$$

In conclusion $\bigcup_{\alpha<\Omega} \mathcal{E}_\alpha = \mathcal{S}(\mathcal{A})$. ■

Proposition 7.1.3 (σ-algebra generated by pairwise-disjoint subsets of an algebra) *Given a non-empty set X, let $\mathcal{A}$ be an algebra of subsets of X. Define recursively*

$$\mathcal{E}_1^+ = \left\{ \bigsqcup_{n=1}^{\infty} A_n : \forall n \in \mathbb{N},\, A_n \in \mathcal{A} \right\},$$

for $\beta < \alpha$ successor:

$$\mathcal{E}_\beta^+ = \left\{ \bigsqcup_{n=1}^{\infty} A_n : \forall n \in \mathbb{N},\, A_n \in \mathcal{A}(\mathcal{E}_{\beta-1}^+) \right\},$$

for $\alpha < \Omega$ limit ordinal:

$$\mathcal{E}_\alpha^+ = \left\{ \bigsqcup_{n=1}^{\infty} A_n : \forall n \in \mathbb{N},\ A_n \in \bigcup_{\beta<\alpha} \mathcal{A}(\mathcal{E}_\beta^+) \right\}.$$

Then for every ordinal in $\{\alpha : \alpha < \Omega\}$ the class $\mathcal{S}^+ = \bigcup_{\alpha<\Omega} \mathcal{E}_\alpha^+$ is a σ-algebra, and coincides with the σ-algebra $\mathcal{S}(\mathcal{A})$.

Proof Clearly $\mathcal{A} = \mathcal{A}^+ \subseteq \mathcal{S}^+ \subseteq \mathcal{S}(\mathcal{A})$, so we need only prove $\mathcal{S}^+$ is a σ-algebra.

First, $\emptyset, X \in \mathcal{A}^+ \Rightarrow \emptyset, X \in \mathcal{S}^+$. Let $\{A_n\}_{n\geq 1}$ be a sequence of pairwise-disjoint sets of $\mathcal{S}^+$. For every $n \in \mathbb{N}$, $A_n \in \mathcal{E}_\beta^+$ for some $\beta < \alpha$. Hence there is an $\alpha < \Omega$ such that

$$\bigsqcup_{n=1}^{\infty} A_n \in \mathcal{E}_\alpha^+.$$

Consequently for every ordinal $\alpha < \Omega$ we have $\bigsqcup_{n=1}^{\infty} A_n \in \bigcup_{\alpha<\Omega} \mathcal{E}_\alpha^+ = \mathcal{S}^+$. Now given $A \in \mathcal{S}^+$, there exists a successor $\beta < \alpha$ such that $A \in \mathcal{E}_\beta^+$. Therefore $A^c \in \mathcal{A}(\mathcal{E}_\beta^+) \subseteq \mathcal{S}^+$. ■

The above proposition says any countable union of elements in the σ-algebra $\mathcal{S}(\mathcal{A})$ can be written as a disjoint countable union.

As we have seen regarding the algebra of sets of X, the result in Theorem 7.1.4 generalise easily to any family $\mathcal{C}$ of sets in $X \neq \emptyset$. In that case, as observed before Proposition 6.2.3, as $\mathcal{C}$ is a mere family of sets, complements and countable unions must be added to the recursive classes in order to generate the σ-algebra $\mathcal{S}(\mathcal{C})$.

Theorem 7.1.5 (σ-algebra generated by an arbitrary family of sets) *Let $\mathcal{C}$ be any family of sets in $X \neq \emptyset$ such that $X \in \mathcal{C}$. Define the collections:*

$$\mathcal{C}_1 = \left\{ \bigcup_{n=1}^{\infty} A_n : \forall n \in \mathbb{N},\, A_n \text{ or } A_n^c \in \mathcal{C} \right\},$$

for $\beta < \alpha$ successor:

$$\mathcal{C}_\beta = \left\{ \bigcup_{n=1}^{\infty} A_n : \forall n \in \mathbb{N},\, A_n \text{ or } A_n^c \in \mathcal{C}_{\beta-1} \right\},$$

for $\alpha < \Omega$ limit ordinal:

$$\mathcal{C}_\alpha = \left\{ \bigcup_{n=1}^{\infty} A_n : \forall n \in \mathbb{N},\, A_n \text{ or } A_n^c \in \bigcup_{\beta<\alpha} \mathcal{C}_\beta \right\}.$$

For any ordinal in $\{\alpha : \alpha < \Omega\}$, then,

$$\mathcal{S} = \bigcup_{\alpha<\Omega} \mathcal{C}_\alpha$$

is the σ-algebra generated by $\mathcal{C}$, i.e. $\mathcal{S}(\mathcal{C}) = \bigcup_{\alpha<\Omega} \mathcal{C}_\alpha$.

Proof By assumption $X \in \mathcal{C} \subseteq \mathcal{S}$, so $\emptyset \in \mathcal{C}_1 \subseteq \mathcal{S}$. With $A \in \mathcal{S}$, clearly $A \in \mathcal{C}_\beta$ for some $\beta < \alpha$, so $A^c \in \mathcal{C}_{\beta+1}$. This holds for every $\beta < \alpha$, whence

$$A^c \in \bigcup_{\beta<\alpha} \mathcal{C}_\beta \Rightarrow A^c \in \bigcup_{\alpha<\Omega} \mathcal{C}_\alpha = \mathcal{S}.$$

Let $\{A_n\}_{n\geq 1}$ be a sequence in $\mathcal{S}$. Each A_n obviously belongs to some $\mathcal{C}_\beta$ with β less than some α. Then there is an $\alpha < \Omega$ such that

$$\bigcup_{n=1}^{\infty} A_n \in \bigcup_{\beta<\alpha} \mathcal{C}_\beta \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \bigcup_{\alpha<\Omega} \mathcal{C}_\alpha = \mathcal{S}.$$

We claim $\mathcal{S}$ is the smallest σ-algebra containing $\mathcal{C}$. For this, take a σ-algebra of subsets $\mathcal{B}$ containing $\mathcal{C}$. Clearly $\mathcal{C}_1 \subseteq \mathcal{B}$, and for every $\beta < \alpha$

$$\mathcal{C}_\beta \subseteq \mathcal{B} \Rightarrow \bigcup_{\beta<\alpha} \mathcal{C}_\beta \subseteq \mathcal{B}.$$

This holds for every $\alpha < \Omega$, so

$$\mathcal{S} \subseteq \mathcal{B} \Rightarrow \mathcal{S} = \mathcal{S}(\mathcal{C}).$$

■

Proposition 7.1.4 (Equivalence of σ-algebra generating processes) *Let $\mathcal{C}$ be family of subsets in X and $\mathcal{A}(\mathcal{C})$ the algebra generated by it. Then*

$$\mathcal{S}(\mathcal{C}) = \mathcal{S}(\mathcal{A}(\mathcal{C})).$$

Proof Note $\mathcal{C}$ is contained in $\mathcal{A}(\mathcal{C})$, which in turn is contained in $\mathcal{S}(\mathcal{A}(\mathcal{C}))$ by definition. Hence $\mathcal{S}(\mathcal{C}) \subseteq \mathcal{S}(\mathcal{A}(\mathcal{C}))$.

Conversely: $\mathcal{C}$ is contained in $\mathcal{S}(\mathcal{C})$, which is an algebra and hence contains the smallest algebra generated by $\mathcal{C}$, i.e. $\mathcal{A}(\mathcal{C}) \subseteq \mathcal{S}(\mathcal{C})$. Immediately, $\mathcal{S}(\mathcal{A}(\mathcal{C})) \subseteq \mathcal{S}(\mathcal{C})$.

Combining the inclusions leads to the required result. ■

Example 7.1.3 (*Countable/co-countable σ-algebra*) In an uncountable set X a set $A \subseteq X$ is called co-countable if A^c is countable. Let $\mathcal{C}$ indicate the set of singletons of X:

$$\mathcal{C} = \{\{x\} : x \in X\}.$$

The family

$$\mathcal{S} = \{A \subseteq X : A \text{ is countable or co-countable }\}$$

is the σ-algebra generated by $\mathcal{C}$.

We prove first $\mathcal{S}$ is a σ-algebra. As $\emptyset = X^c$ then $X \in \mathcal{S}$. Any $A \in \mathcal{S}$ is either countable or co-countable, so A^c is respectively co-countable or countable, and so $A^c \in \mathcal{S}$. Let $\{A_n\}_{n\geq 1}$ be a sequence in $\mathcal{S}$. If all elements A_n are countable, $\bigcup_{n=1}^{\infty} A_n \in \mathcal{S}$ is countable. If at least one, say A_1, is co-countable then

$$\left(\bigcup_{n=1}^{\infty} A_n\right)^c = \bigcap_{n=1}^{\infty} A_n^c \subseteq A_1^c,$$

which is countable, hence $\bigcup_{n=1}^{\infty} A_n$ is co-countable.

We claim $\mathcal{S}$ is the smallest σ-algebra containing $\mathcal{C}$. Let $\mathcal{F}$ be a σ-algebra containing the class $\mathcal{C}$, and we shall prove $\mathcal{F} \supseteq \mathcal{S}$. Take $A \in \mathcal{S}$, so A or A^c is countable. In the first case

$$A = \{x_1, x_2, \ldots, x_n, \ldots\} = \bigcup_{n=1}^{\infty} \{x_n\}$$

with $\{x_i\}$ in $\mathcal{C}$ and $\mathcal{C} \subseteq \mathcal{F}$. As $\mathcal{F}$ is a σ-algebra, then $A \in \mathcal{F}$. If A is not countable, $A \in \mathcal{S}$ implies A^c countable, and for the same reason $A^c \in \mathcal{F}$. Moreover, as $\mathcal{F}$ is a σ-algebra, $A \in \mathcal{F}$. Eventually $\mathcal{S} \subseteq \mathcal{F} \Rightarrow \mathcal{S} = \mathcal{S}(\mathcal{C})$.

★

7.2 The Borel σ-Algebra and Restricted σ-Algebras

In general the elements of a σ-algebra are not so easy to describe, because the sets it contains can be rather complex and might have a structure that is not representable graphically.

Yet the creation of a σ-algebra begins from rather easy collections of sets, such as open or closed intervals of $\mathbb{R}$, and by transfinite induction arrives at increasingly complex families, in the case at hand called Borel classes. The family of Borel classes in $\mathbb{R}$ is in fact a σ-algebra over $\mathbb{R}$, as explained by the following theorem.

Theorem 7.2.1 (Borel σ-algebra on $\mathbb{R}$) *The family*

$$\mathcal{S} = \bigcup_{\alpha < \Omega} \Sigma^0_\alpha$$

of Borel classes (Definition 3.4.1) is the smallest σ-algebra over $\mathbb{R}$ generated by the class Σ^0_1 of open intervals, i.e. $\bigcup_{\alpha<\Omega} \Sigma^0_\alpha = \mathcal{S}\big(\Sigma^0_1\big)$.

We call $\mathcal{S}\big(\Sigma^0_1\big)$ the Borel σ-algebra, and by definition we set

$$\mathcal{B}(\mathbb{R}) = \mathcal{S}\big(\Sigma^0_1\big).$$

Proof Clearly $\mathbb{R}, \emptyset \in \Sigma^0_1 \Rightarrow \mathbb{R}, \emptyset \in \mathcal{S}$. If $E \in \bigcup_{\alpha<\Omega} \Sigma^0_\alpha$, there exists $\beta < \alpha$ such that $E \in \Sigma^0_\beta$. By item *3.* in Proposition 3.4.1 a set E is in Σ^0_β iff E^c is in Π^0_β. By definition, moreover, $\Pi^0_\beta \subseteq \Sigma^0_{\beta+1}$, so $E^c \in \Pi^0_\beta \subseteq \Sigma^0_{\beta+1}$. Hence $E^c \in \bigcup_{\beta<\alpha} \Sigma^0_\beta$, and by transfinite induction $E^c \in \mathcal{S}$. Let $\{E_i\}_{i\geq 1}$ be a sequence in $\bigcup_{\alpha<\Omega} \Sigma^0_\alpha$. For every $i \in \mathbb{N}$, $E_i \in \Sigma^0_{\beta_i}$, so there is an $\alpha < \Omega$ such that for every $\beta < \alpha$

$$\bigcup_{i=1}^{\infty} E_i \in \bigcup_{\beta<\alpha} \Sigma^0_\beta.$$

By transfinite induction, $\bigcup_{i=1}^{\infty} E_i \in \mathcal{S}$.

This proves $\mathcal{S}$ is a σ-algebra, so now we will show it is the minimal σ-algebra containing the class Σ^0_1 of open intervals. For this, take any σ-algebra of subsets $\mathcal{F}$ containing Σ^0_1. As first inductive step, given $\beta < \alpha$ successor, let $\Sigma^0_\beta \subseteq \mathcal{F}$. We must

show $\Sigma^0_{\beta+1} \subseteq \mathcal{F}$. Given $E \in \Sigma^0_{\beta+1}$, we have $E = \bigcup_{i=1}^{\infty} H_i^{\beta}$, where $H_i^{\beta} \in \Pi^0_{\beta}$. As $\mathcal{F}$ is closed under complements, the family contains, beside Σ^0_{β}, also Π^0_{β}. And $\mathcal{F}$ is closed under countable unions, so

$$E = \bigcup_{i=1}^{\infty} H_i^{\beta} \in \mathcal{F}.$$

As second inductive step suppose the above holds for every $\beta < \alpha$ with α limit ordinal. Let $E \in \bigcup_{\beta<\alpha} \Sigma^0_{\beta}$, so $E = \bigcup_{i=1}^{\infty} H_i^{\beta}$, where $H_i^{\beta} \in \Pi^0_{\beta}$ for some $\beta < \alpha$. Since $\mathcal{F}$, as seen above, contains Π^0_{β}, and is closed under countable unions, necessarily $E \in \mathcal{F}$.

This is true for every ordinal $\alpha < \Omega$, and so

$$\bigcup_{\alpha<\Omega} \Sigma^0_{\alpha} \subseteq \mathcal{F} \Rightarrow \bigcup_{\alpha<\Omega} \Sigma^0_{\alpha} = \mathcal{S}\big(\Sigma^0_1\big).$$

■

Remark 7.2.1 A countable union of sets in the σ-algebra $\mathcal{S}(\Sigma^0_1)$ can also be written as a countable union of pairwise-disjoint sets of the σ-algebra. This is because, by Proposition 6.2.2, $\mathcal{A}^+(\Sigma^0_1) = \mathcal{A}(\Sigma^0_1)$, and Proposition 7.1.3 implies

$$\mathcal{S}^+(\Sigma^0_1) = \mathcal{S}(\Sigma^0_1).$$

♣

The family $\mathcal{B}(\mathbb{R})$ is, apart from the σ-algebra generated by open intervals, also the σ-algebra generated by open sets in $\mathbb{R}$. Since the complement of an open set in X is closed in X, we have the following result.

Theorem 7.2.2 (Borel σ-algebra generated by closed subsets of $\mathbb{R}$) *Let Σ^0_1 be the class of open subsets of $\mathbb{R}$, Π^0_1 that of closed subsets. The σ-algebras generated by Σ^0_1 and Π^0_1 coincide:*

$$\mathcal{S}\big(\Sigma^0_1\big) = \mathcal{S}\big(\Pi^0_1\big).$$

Proof Recall a set is closed iff its complement is open. We shall prove the σ-algebra generated by open sets coincides with the σ-algebra generated by closed subsets. We have

$$\begin{aligned} E \in \Sigma^0_1 &\Rightarrow E \text{ is open} \\ &\Rightarrow E^c \text{ is closed} \\ &\Rightarrow E^c \in \Pi^0_1 \subseteq \mathcal{S}(\Pi^0_1) \\ &\Rightarrow E \in \mathcal{S}(\Pi^0_1); \end{aligned}$$

so $\Sigma_1^0 \subseteq \mathcal{S}(\Pi_1^0)$ and $\mathcal{S}(\Sigma_1^0) \subseteq \mathcal{S}(\Pi_1^0)$.

Conversely

$$\begin{aligned} H \in \Pi_1^0 &\Rightarrow H^c \in \Sigma_1^0 \subseteq \mathcal{S}(\Sigma_1^0) \\ &\Rightarrow H \in \mathcal{S}(\Sigma_1^0) \end{aligned}$$

so $\Pi_1^0 \subseteq \mathcal{S}(\Sigma_1^0)$ and $\mathcal{S}(\Pi_1^0) \subseteq \mathcal{S}(\Sigma_1^0)$. The claim follows. ■

We have shown the Borel σ-algebra over $\mathbb{R}$ is generated by open or closed intervals. Respectively calling $\mathcal{I}$ and $\tilde{\mathcal{I}}$ the collections of all intervals and all right-closed intervals (the same would go for left-closed intervals), we wish to check

$$\mathcal{S}(\mathcal{I}) = \mathcal{S}(\tilde{\mathcal{I}}) = \mathcal{B}(\mathbb{R}).$$

Proposition 7.2.1 (Equivalent ways of generating the Borel σ-algebra on $\mathbb{R}$) *Let $\mathcal{I}$ be the collection of intervals and $\tilde{\mathcal{I}}$ the collection of right-closed intervals. Then*

$$\mathcal{S}(\mathcal{I}) = \mathcal{S}(\tilde{\mathcal{I}}) = \mathcal{B}(\mathbb{R}).$$

Proof By definition $\mathcal{B}(\mathbb{R}) = \mathcal{S}(\Sigma_1^0)$, and the latter contains all subsets of type

$$(a, a) = \emptyset; \quad (a, b); \quad (-\infty, a); \quad (b, \infty); \quad (-\infty, \infty).$$

As $\mathcal{S}(\Sigma_1^0)$ is a σ-algebra, so closed under complements,

$$(-\infty, \infty); \quad (-\infty, b]; \quad [a, \infty); \quad \emptyset$$

belong in $\mathcal{S}(\Sigma_1^0)$. The class is also intersection-closed, so

$$[a, \infty) \cap (-\infty, b] = [a, b]; \quad [a, \infty) \cap (-\infty, a] = [a, a] = \{a\}; \quad [a, b] \cap (a, \infty) = (a, b];$$
$$(-\infty, b) \cap [a, b] = [a, b); \quad \ldots.$$

belong to $\mathcal{S}(\Sigma_1^0)$. Hence $\mathcal{I} \subset \mathcal{S}(\Sigma_1^0)$. One can prove $\mathcal{I} \subset \mathcal{S}(\tilde{\mathcal{I}})$ in the same way, and trivially, $\tilde{\mathcal{I}} \subset \mathcal{S}(\mathcal{I})$. As $\mathcal{S}(\Sigma_1^0)$ is a σ-algebra containing $\mathcal{I}$, then $\mathcal{S}(\mathcal{I}) \subseteq \mathcal{S}(\Sigma_1^0)$. But $\mathcal{S}(\mathcal{I})$ is a σ-algebra containing Σ_1^0, so $\mathcal{S}(\Sigma_1^0) \subseteq \mathcal{S}(\mathcal{I})$ and hence $\mathcal{S}(\Sigma_1^0) = \mathcal{S}(\mathcal{I})$. Analogously, $\mathcal{S}(\tilde{\mathcal{I}})$ is a σ-algebra containing $\mathcal{I}$, so $\mathcal{S}(\mathcal{I}) \subseteq \mathcal{S}(\tilde{\mathcal{I}})$. But $\mathcal{S}(\mathcal{I})$ is a σ-algebra containing $\tilde{\mathcal{I}}$, then $\mathcal{S}(\tilde{\mathcal{I}}) \subseteq \mathcal{S}(\mathcal{I})$ and eventually $\mathcal{S}(\tilde{\mathcal{I}}) = \mathcal{S}(\mathcal{I})$. ■

A σ-algebra generated by some family $\mathcal{C}$ of subsets in a set X may be restricted to a subset Y of X. The restriction takes $\mathcal{C}$ and a suitable subset $Y \subseteq X$, and intersects $\mathcal{C}$ and Y. This gives a new family $\mathcal{C} \cap Y$ of subsets of Y contained in the original family $\mathcal{C}$. From another point of view, the restriction consists in only picking the elements of $\mathcal{C}$ that already are subsets of Y. On the new family we can build the σ-algebra $\mathcal{S}\big(\mathcal{C} \cap Y\big)$.

For a working example we may take the Borel σ-algebra over $\mathbb{R}$ and an interval $[a, b]$. We have seen the Borel σ-algebra can be generated starting from open subsets in $\mathbb{R}$. So we may restrict the σ-algebra $\mathcal{B}(\mathbb{R})$ to $[a, b]$, knowing that such σ-algebra restricted to $[a, b]$ coincides with the σ-algebra generated by open subsets in $[a, b]$.

The question we want to answer is whether the σ-algebra generated by $\mathcal{C}$, then restricted to Y, namely $\mathcal{S}(\mathcal{C}) \cap Y$, is the same as the σ-algebra generated directly by $(\mathcal{C} \cap Y)$.

Proposition 7.2.2 (Restriction of a σ-algebra) *Let $\mathcal{C}$ be a collection of subsets in X, and $Y \subseteq X$ a given subset. Define*

$$\mathcal{C} \cap Y = \{C \cap Y : C \in \mathcal{C}\}.$$

The family $\mathcal{S}(\mathcal{C}) \cap Y = \mathcal{S}(\mathcal{C})_Y$ is a σ-algebra restricted to Y that coincides with the σ-algebra generated by the family $\mathcal{C} \cap Y$ of subsets of Y:

$$\mathcal{S}(\mathcal{C}) \cap Y = \mathcal{S}(\mathcal{C})_Y = \mathcal{S}(\mathcal{C} \cap Y).$$

Proof Let us prove $\mathcal{S}(\mathcal{C}) \cap Y$ is a σ-algebra. Observe $\emptyset \in \mathcal{S}(\mathcal{C})$ so $\emptyset = \emptyset \cap Y \in \mathcal{S}(\mathcal{C}) \cap Y$, and X belongs in $\mathcal{S}(\mathcal{C})$, so $Y = (X \cap Y) \in \mathcal{S}(\mathcal{C}) \cap Y$. Let $E \in \mathcal{S}(\mathcal{C}) \cap Y$, so $E = A \cap Y$ for $A \in \mathcal{S}(\mathcal{C})$. Note $E^c \subseteq Y$, so $E^c = A^c \cap Y$ with $A^c \in \mathcal{S}(\mathcal{C})$. Hence $E^c = A^c \cap Y \in \mathcal{S}(\mathcal{C}) \cap Y$. Let $E_1, E_2, \ldots, E_n, \ldots$ be a sequence in $\mathcal{S}(\mathcal{C}) \cap Y$, so for every E_n there exists $A_n \cap Y$ such that $A_n \in \mathcal{S}(\mathcal{C})$. Therefore

$$\bigcup_{n=1}^{\infty} E_n = \left(\bigcup_{n=1}^{\infty} A_n \right) \cap Y,$$

where $\bigcup_{n=1}^{\infty} A_n \in \mathcal{S}(\mathcal{C})$, and $\left(\bigcup_{n=1}^{\infty} A_n\right) \cap Y = \bigcup_{n=1}^{\infty} E_n \in \mathcal{S}(\mathcal{C}) \cap Y$. All in all, $\mathcal{S}(\mathcal{C}) \cap Y$ is a σ-algebra. Now let us prove $\mathcal{S}(\mathcal{C}) \cap Y$ is the σ-algebra generated by $(\mathcal{C} \cap Y)$. Observe $\mathcal{C} \cap Y \subseteq \mathcal{S}(\mathcal{C}) \cap Y$, so $\mathcal{S}(\mathcal{C} \cap Y) \subseteq \mathcal{S}(\mathcal{C}) \cap Y$. Now we need the opposite inclusion. Consider the family

$$\mathcal{F} = \{E \subseteq X : E \cap Y \in \mathcal{S}(\mathcal{C} \cap Y)\},$$

i.e. the subsets whose intersection with Y belongs in the σ-algebra $\mathcal{S}(\mathcal{C} \cap Y)$.

We claim $\mathcal{F}$ is a σ-algebra containing $\mathcal{C}$. Observe the empty set belongs to $\mathcal{F}$, for $\emptyset = \emptyset \cap Y \in \mathcal{S}(\mathcal{C} \cap Y)$. For the same reason X belongs to $\mathcal{F}$. Let $E \in \mathcal{F}$, so $E \cap Y \in \mathcal{S}(\mathcal{C} \cap Y)$. The latter is a σ-algebra, so closed under complementation relative to Y. Therefore $E^c \cap Y \in \mathcal{S}(\mathcal{C} \cap Y)$ and $E^c \in \mathcal{F}$. Take a sequence $E_1, E_2, \ldots, E_n, \ldots$ in $\mathcal{F}$, so by definition $E_n \cap Y \in \mathcal{S}(\mathcal{C} \cap Y)$, for every n. As $\mathcal{S}(\mathcal{C} \cap Y)$ is a σ-algebra, then $\left(\bigcup_{n=1}^{\infty} E_n\right) \cap Y \in \mathcal{S}(\mathcal{C} \cap Y)$ and so $\bigcup_{n=1}^{\infty} E_n \in \mathcal{F}$.

We have established $\mathcal{F}$ is a σ-algebra. But as $E \cap Y \in \mathcal{S}(\mathcal{C} \cap Y)$ for every $E \in \mathcal{C}$, we have $E \in \mathcal{F}$ and then $\mathcal{C} \subseteq \mathcal{F}$. Moreover $\mathcal{F} \cap Y = \mathcal{S}(\mathcal{C} \cap Y)$, but $\mathcal{S}(\mathcal{C}) \cap Y \subseteq \mathcal{F} \cap Y$ and the statement is proved. ■

Part IV
Measure Theory

Introduction

Measure theory sets off from the concept of length of a segment, which is the core of the real line's construction and represented the first step to transfer relationships between geometrical objects to relationships between real numbers. Segments are special instances of linear sets, i.e. sets of points on the line with a continuous structure. Clearly the line admits other types of sets like finite sets, countable sets or sets with the cardinality of the continuum, which might not carry a continuous structure. Any theory of measure, if it wants to maintain a general nature, cannot simply handle the measures of segments, intervals or pieces of the real line. It must be able to encompass, where possible, the largest number of sets of points or linear subsets of $\mathbb{R}$, of the plane $\mathbb{R}^2$ and so forth. The concept of measure is therefore a generalisation of the length of an interval, the area of a rectangle, the volume of a parallelepiped etc.

We all perceive that the problem of establishing the notion of measure in the above sense has to do with having a satisfactory theory of integration, too. For example, let I be a linear set of points in $[a, b]$ and φ_I its characteristic function, i.e. the map from $[a, b]$ to $\{0, 1\}$ that equals 1 on I and 0 otherwise. The measure of I will be

$$\int_a^b \varphi_I(x)dx = m(I).$$

A similar reasoning works for sets A contained in a rectangle R in the plane, a parallelepiped P in three-space and so on.

The necessity of formalising a notion of measure as in the above broad sense was felt during the 1800s, and in particular with the inception of the theory of Fourier transforms. Among other things the latter required, for instance to define Fourier coefficients, was the possibility of integrating functions with any number of discontinuities, and passing from infinite sums of integrals to integrals of infinite sums. The need to address these issues rigorously was perceived as pressing by many mathematicians, like Augustin-Louis Cauchy (1789–1857) and later Bernhard Riemann (1826–1866).

At the end of the 1700s integration was still seen as the inverse operation of differentiation. Starting from the following century, with Cauchy and Riemann, integrals are defined through limits of sums.

The first to raise the issue of integrating functions with finite discontinuity set was Cauchy, who introduced the notion of improper, or generalised, integral in his 1823 paper *"Résumé des leçons données à l'École Royale Polytechnique sur le calcul infinitésimal"*. In it he gave the first modern definition of integral, which he used for proving the integrability of continuous maps. But Cauchy's construction of the integral worked for maps that were deemed continuous *a priori*. The trend in analysis focusing on the exception rather than the rule, and on the pathologies of weird maps, had still not kicked off, because it was believed that nature was mostly governed by harmonious laws displaying a reassuring and steady behaviour. Cauchy's method was not suited to handle functions with infinitely many discontinuities.

In order to convince mathematicians that studying maps with infinitely many discontinuities was useful and far-sighted, it became necessary to find the analytical expressions generating such functions. Scholars recognised that trigonometric series, in particular, allowed to construct functions with infinite discontinuity sets in an easy manner. As we said in the introduction to Chap. 4, the forefather of this promising movement was Peter Gustav Lejeune Dirichlet (1805–1859). In the celebrated 1829 memoire entitled *"Sur la convergence des séries trigonométriques qui servent à représenter une fonction arbitraire entre des limites données"*, Dirichlet seeked to weaken the result he had proved on Fourier representations (Theorem 4.1.1, introduction to Chap. 4). He **conjectured** that a map $\varphi(x)$ with infinitely many discontinuities could be represented by a convergent trigonometric series, provided it satisfied the following condition:

"It is necessary that the map $\varphi(x)$ be such that, if one denotes by a and b two arbitrary quantities between $-\pi$ and π [the endpoints of integration], *one can place between a and b further sufficiently close quantities r and s so that the map stays continuous on the interval from r to s*."

The integrability condition for maps with infinitely many discontinuities suggests Dirichlet might have been thinking of a nowhere dense subset of an interval. This kind of set's closure has no interior. In this sense for Dirichlet a function is integrable if its discontinuity set E is nowhere dense, hence with no points in the interior of its closure. He then added that if the condition is not met, the integral no longer makes sense. The example he offered is the renowned function that nowadays bears his name: as it does not satisfy the aforementioned condition, it is not (Cauchy-)integrable. In this respect he assumed:

"*$\varphi(x)$ equals a given constant c when the variable x takes a rational value, and equals another constant when this variable is irrational. The function thus defined has finite values for all choice of x, and nevertheless it cannot be substituted in the series, because in that case the various integrals appearing the series would lose their meaning.*"

Even though the integrability condition in Dirichlet's memoire is neither necessary nor sufficient, as we shall see, it certainly is deserving, for it established a relationship

between integrability and the discontinuity set. At the very end he mentioned the intention to return to the matter at a later stage, but eventually never did.

Others pursued the subject inaugurated by Dirichlet. Among them was Lipschitz (1832–1903), who in a 1864 paper considered the derived set E of the set of discontinuities. He stated that improper integrals may be used if E is finite. By the way, this result extends to when the nth derived set of the discontinuity set is finite. Clearly if a set has finite n^{th} derived set, i.e. it is of first species, then it is nowhere dense. Lipschitz was convinced the opposite held, i.e. any nowhere dense set must be of first species, and because of this he believed to have proved the sufficient part of Dirichlet's integrability condition.

Enter Bernhard Riemann who, in the Habilitation thesis submitted at Göttingen University in 1854, proposed a novel theory of integration aimed at encompassing at least discontinuous functions that admitted a Fourier series. The dissertation remained virtually unknown until Dedekind agreed to publish it in 1867.[1] Riemann wanted to know how to define the integral of an arbitrary (not necessarily continuous) bounded map $f(x)$ defined at all points x of an interval $[a, b]$. He began by maps $f(x)$ defined on $[a, b]$, which he assumed only bounded rather than continuous, as Cauchy had, more restrictively. He then applied the familiar procedure that also Cauchy had followed, namely he divided $[a, b]$ in n arbitrary intervals to obtain a finite partition $P = \{a = x_0, x_1, \ldots, x_n = b\}$. To this he associated the two sums

$$\underline{\int_a^b} f(x)dx = s(f, P) = \sum_{i=1}^{n} \left[\inf_{x \in I_i} f(x) \cdot m(I_i) \right]$$

$$\overline{\int_a^b} f(x)dx = S(f, P) = \sum_{i=1}^{n} \left[\sup_{x \in I_i} f(x) \cdot m(I_i) \right].$$

It is easy to show that if $f(x)$ is bounded, as we refine P the lower sum $s(f, P)$ converges to its least upper bound (supremum), called lower integral. Under the same conditions the upper sum $S(f, P)$ tends to the greatest lower bound (infimum), called upper integral (Fig. 1).

Riemann then stated the so-called **first criterion for Riemann integrability**:

"A bounded function $f : [a, b] \mapsto \mathbb{R}$ is integrable if and only if, given any $\epsilon > 0$, there exists a partition $P = \{a = x_0, x_1, \ldots, x_n = b\}$ of $[a, b]$ such that the sum of the products of the width of each of the partition's subintervals times the map's oscillation over that open subinterval is less than ϵ.".

[1] Dedekind is the mathematician to whom we owe the general definition of function as a correspondence between the values of some variable x and well-defined values of a variable y, irrespective of whether the function is representable analytically, or continuous.

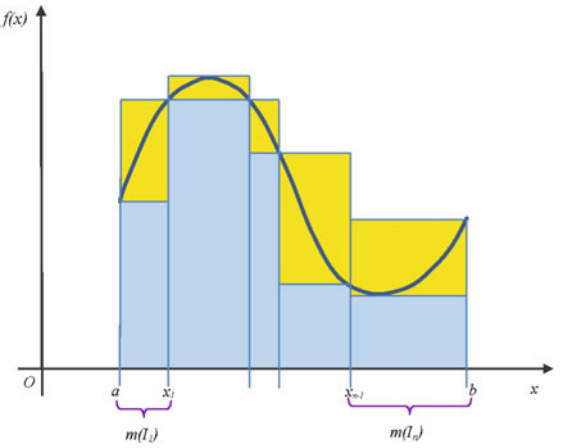


Fig. 1 Construction of the Riemann integral of a function $f(x)$

Paraphrasing,

$$S(f,P)-s(f,P)=\sum_{i=1}^{n}\left[\left(\sup_{x\in I_i}f(x)-\inf_{x\in I_i}f(x)\right)\cdot m(I_i)\right]$$
$$=\sum_{i=1}^{n}\left[\omega(f,I_i)\cdot m(I_i)\right]<\epsilon.$$

The sums of the rectangles (upper and lower) are separated and contiguous sets. The separating element is the integral of f over $[a,b]$, i.e. the area of the region underneath f over $[a,b]$.

Riemann, after introducing the concept of integral, tackled the problem of characterising the class of bounded integrable maps. His solution is now called **second criterion for Riemann integrability**:

"A necessary and sufficient condition for a bounded map $f(x)$ to be integrable is that for every σ and δ positive, there exists a finite partition of $[a,b]$ such that the sum of the lengths of the subintervals on the interior of which the oscillation of f exceeds σ, is less than δ."

To show the reach of his results, and hence that the class of bounded Riemann-integrable maps is larger than that of continuous maps, Riemann constructed a function with a dense discontinuity set. In this way he also proved Dirichlet's necessary condition was wrong. More specifically, he defined the function (x) as the (positive or negative) fractional excess of x from the closest integer, and for x half-way between consecutive integers, (x) is set to 0 (the arithmetic mean of the two differences $\pm\frac{1}{2}$). In formulas, for every $x\in\mathbb{R}$ we define

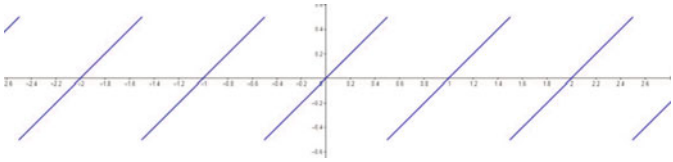


Fig. 2 Graph of $y = (x)$

$$(x) = \begin{cases} x - m(x) & \text{if } x \neq p + \dfrac{1}{2}, \forall p \in \mathbb{Z} \\ 0 & \text{if } x = p + \dfrac{1}{2}, \forall p \in \mathbb{Z}. \end{cases}$$

where $m(x)$ denotes the integer closest to x. The function $y = (x)$ is therefore called **excess of** x (Fig. 2).

To draw its graph we first plot the graph of $f(x) = x$ for $-\frac{1}{2} < f(x) < \frac{1}{2}$; then we copy this segment with period 1, i.e. on all intervals $\left(p - \frac{1}{2}; p + \frac{1}{2}\right)$, $p \in \mathbb{Z}$. Overall, the zeroes are the integers. The discontinuity points of (x) are located at the endpoints of the intervals

$$\left(\frac{2p-1}{2}; \frac{2p+1}{2}\right)$$

whose numerator is always odd while the denominator is always even. The discontinuities are therefore rational with coprime numerator and denominator. In general we can then say (x) is discontinuous at

$$\frac{2p+1}{2}, \text{ where } p \in \mathbb{Z}, \ \gcd(2p+1, 2) = 1.$$

We can then generalise and set, for any $n \in \mathbb{N}$,

$$(nx) = \begin{cases} nx - m(nx) & \text{if } x \neq \dfrac{p}{n} + \dfrac{1}{2n}, \ \forall p \in \mathbb{Z} \\ 0 & \text{if } x = \dfrac{p}{n} + \dfrac{1}{2n}, \ \forall p \in \mathbb{Z}. \end{cases}$$

The graph is similarly made of straight segments over $\left(\frac{p}{n} - \frac{1}{2n}; \frac{p}{n} + \frac{1}{2n}\right)$, $p \in \mathbb{Z}$, each of which is a copy of the graph of $f(x) = nx$, for $-\frac{1}{2} < f(x) < \frac{1}{2}$, so the period is $1/n$. Again, the map has its zeroes at $x = \frac{p}{n}$, for every $p \in \mathbb{Z}$. In general, for any $n \in \mathbb{N}$ the map (nx) is discontinuous at all points x of the type

$$\frac{2p+1}{2n}, \text{ where } p \in \mathbb{Z}, \ \gcd(2p+1, 2n) = 1.$$

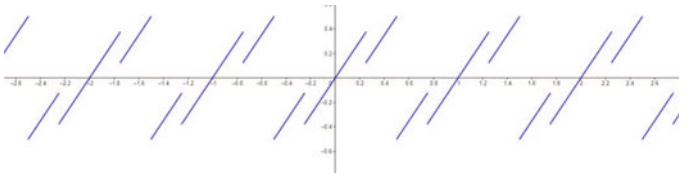


Fig. 3 Graph of Riemann's function $R(x)$ for $n = 2$

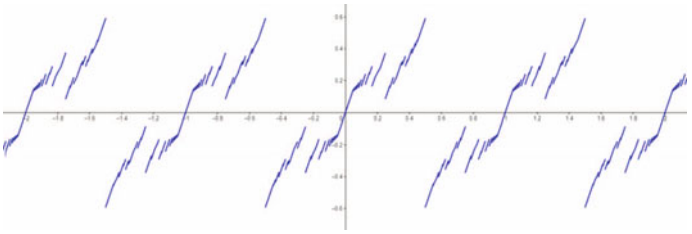


Fig. 4 Graph of Riemann's function $R(x)$ for $n = 10$

At this point Riemann, to obtain a dense discontinuity subset of $\mathbb{R}$, invokes Hermann Hankel's principle of condensation of singularities. The latter was proved rigorously by Ulisse Dini in the 1878 book "*Fondamenti per la teorica delle funzioni di variabili reali*", considered a fundamental work by Henri Lebesgue. The principle consists in summing, for every $n \in \mathbb{N}$, the ratios between (nx) and the corresponding n squared, obtaining (Figs. 3 and 4 represent the map $R(x)$ for $n = 2$ and $n = 10$ respectively)

$$R(x) = \sum_{n=1}^{\infty} \frac{(nx)}{n^2} = \frac{(x)}{1^2} + \frac{(2x)}{2^2} + \frac{(3x)}{3^2} + \cdots + \frac{(nx)}{n^2} + \cdots .$$

It is easy to see the series is totally convergent, and hence also uniformly convergent on $\mathbb{R}$. In fact, remembering that $|(x)| \leq 1/2$, one has

$$\left| \frac{(nx)}{n^2} \right| \leq \frac{1/2}{n^2} < \frac{1}{n^2}.$$

It is well known that the numerical series with general term $1/n^2$ converges, and

$$|R(x)| = \left| \sum_{n=1}^{+\infty} \frac{(nx)}{n^2} \right| \leq \sum_{n=1}^{+\infty} \left| \frac{(nx)}{n^2} \right| \leq \frac{1}{2} \sum_{n=1}^{+\infty} \frac{1}{n^2}.$$

Recalling $\sum_{n=1}^{+\infty} \frac{1}{n^2} = \pi^2/6$ is the Fourier expansion[2] of $f(x) = x^2$ over $[-\pi, \pi]$, if we take $x = \pi$ the above reads

$$|R(x)| \leq \frac{\pi^2}{12} < \infty,$$

proving R is bounded. Given $m \in \mathbb{Z}$ and $n \in \mathbb{N}$, for $x = 2m + 1/2n$ we obtain

$$R\left(\frac{2m+1}{2n} \pm 0\right) = R\left(\frac{2m+1}{2n}\right) + \left[\frac{(x)|_{\left(x=\frac{2m+1}{2} \pm 0\right)}}{1^2 n^2} + \frac{(2x)|_{\left(x=\frac{2m+1}{2\cdot 2} \pm 0\right)}}{2^2 \cdot n^2} + \right.$$
$$+ \frac{(3x)|_{\left(x=\frac{2m+1}{2\cdot 3} \pm 0\right)}}{3^2 \cdot n^2} + \frac{(4x)|_{\left(x=\frac{2m+1}{2\cdot 4} \pm 0\right)}}{4^2 \cdot n^2}$$
$$\left. + \frac{(5x)|_{\left(x=\frac{2m+1}{2\cdot 5} \pm 0\right)}}{5^2 \cdot n^2} + \cdots + \frac{(nx)|_{\left(x=\frac{2m+1}{2\cdot n} \pm 0\right)}}{n^2 \cdot n^2} + \cdots \right].$$

Since $2p+1$ is odd, in the right-hand side the even terms $(2x), (4x), (6x), \ldots$, make the numerator of $\frac{2m+1}{2n}$ an even number, while we saw the discontinuities of $R(x)$

---

[2]The Fourier series of a generic function is:

$$f(x) = \frac{a_0}{2} + \sum_{h=1}^{\infty} (a_h \cos(hx) + b_h \sin(hx)),$$

where: $a_h = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(hx) dx$, $b_h = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(hx) dx$. If $f(x)$ is even, e.g. $f(x) = x^2$,

$$a_h = \frac{2}{\pi} \int_0^{\pi} f(x) \cos(hx) dx; \quad b_h = 0.$$

So in our case: $a_h = \frac{2}{\pi} \int_0^{\pi} x^2 \cos(hx) dx$. Integrating by parts, the above becomes

$$\frac{2}{\pi} \frac{(2\pi)}{h^2} \cos(hx) = (-1)^h \frac{4}{h^2}.$$

Moreover

$$\frac{a_0}{2} = \frac{1}{\pi} \int_0^{\pi} x^2 dx = \frac{1}{\pi} \frac{\pi^3}{3} = \frac{\pi^2}{3}$$

and, in conclusion, for $x \in [-\pi, \pi]$ we have:

$$x^2 = \frac{\pi^2}{3} + \sum_{h=1}^{\infty} (-1)^h \frac{4}{h^2} \cos(hx).$$

If we evaluate the series at $x = \pi$, since $\cos(hx) = (-1)^h$, we get $\pi^2 = \frac{\pi^2}{3} + \sum_{h=1}^{\infty} \frac{4}{h^2}$ whence

$$\frac{\pi^2}{6} = \sum_{h=1}^{\infty} \frac{1}{h^2}.$$

are those with odd numerator. Therefore to compute R at a given discontinuity point we must exclude the values that are not discontinuities for f. The above expression then reduces to this:

$$R\left(\frac{2m+1}{2n}\pm 0\right) = R\left(\frac{2m+1}{2n}\right) + \left[\frac{(x)|_{\left(x=\frac{2m+1}{2}\pm 0\right)}}{1^2n^2} + \frac{(3x)|_{\left(x=\frac{2m+1}{2\cdot 3}\pm 0\right)}}{3^2\cdot n^2} + \right.$$
$$+\frac{(5x)|_{\left(x=\frac{2m+1}{2\cdot 5}\pm 0\right)}}{5^2\cdot n^2} + \frac{(7x)|_{\left(x=\frac{2m+1}{2\cdot 7}\pm 0\right)}}{7^2\cdot n^2} +$$
$$\left. +\frac{(9x)|_{\left(x=\frac{2m+1}{2\cdot 9}\pm 0\right)}}{9^2\cdot n^2} + \cdots\cdots\cdots\cdots\right].$$

Clearly, for $n = 1, 3, 5, 7, \ldots$ the function (nx) vanishes at $x = \frac{2m+1}{2n}$, whilst the right and left limits at said point, i.e. for $x = \frac{2m+1}{2n} \pm 0$, are equal to $\mp\frac{1}{2}$:

$$\frac{(nx)|_{\left(x=\frac{2m+1}{2\cdot n}\pm 0\right)}}{n^2\cdot n^2} = \mp\frac{1}{2}.$$

Substituting these in the formula gives

$$R\left(\frac{2m+1}{2n}\pm 0\right) = R\left(\frac{2m+1}{2n}\right) \mp \left(\frac{\frac{1}{2}}{1^2n^2} + \frac{\frac{1}{2}}{3^2\cdot n^2} + \frac{\frac{1}{2}}{5^2\cdot n^2} + \frac{\frac{1}{2}}{7^2\cdot n^2} + \cdots\right)$$
$$= R\left(\frac{2m+1}{2n}\right) \mp \frac{1}{2n^2}\left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \frac{1}{9^2} + \cdots\cdots\right).$$

The sum on the right is nothing but the Fourier series of $f(x) = |x|$ at $x = 0$, which is

$$\left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots\right) = \sum_{n=1}^{+\infty}\frac{1}{(2n-1)^2} = \frac{\pi^2}{8}.$$

Substituting again,

$$R\left(\frac{2m+1}{2n}\pm 0\right) = R\left(\frac{2m+1}{2n}\right) \mp \frac{1}{2n^2}\cdot\frac{\pi^2}{8} = R\left(\frac{2m+1}{2n}\right) \mp \frac{\pi^2}{16n^2}.$$

Hence the jump of $R(x)$ at $x = \frac{2m+1}{2n}$ equals

$$\left|R\left(\frac{2m+1}{2n}+0\right) - R\left(\frac{2m+1}{2n}-0\right)\right| = \left|-\frac{\pi^2}{16n^2} - \frac{\pi^2}{16n^2}\right| = \frac{\pi^2}{8n^2}.$$

Let us indicate by $\omega(R, x) = \frac{\pi^2}{8n^2}$ this quantity. The discontinuity set of $R(x)$

$$\mathbb{Q}^* = \left\{ x = \frac{2m+1}{2n} : m \in \mathbb{Z},\ 2n \in \mathbb{N},\ \gcd(2m+1, 2n) = 1 \right\}$$

is dense in $\mathbb{R}$.

Now, given its periodic nature, it suffices to show R is integrable on a generic interval $[a, b]$. So given any $\sigma > 0$, let us indicate by

$$\begin{aligned} D_\delta(R) &= \{x \in [a, b] : \omega(R, x_n^m) \geq \sigma\} \\ &= \left\{ \frac{2m+1}{2n} \in [a, b] :\ \gcd(2m+1, 2n) = 1,\ \frac{\pi^2}{8n^2} \geq \sigma \right\} \end{aligned}$$

the points in $[a, b]$ where the oscillation of R is at least σ. Observe $D_\delta(R)$ is at most finite and so negligible, i.e. smaller than any $\delta > 0$. By Riemann's second integrability criterion $R(x)$ is integrable.
Recalling in fact $|R| \leq \pi^2/12$, given any $\epsilon > 0$ there is a partition P of $[a, b]$ such that

$$\delta = \frac{\epsilon}{(4\pi^2/12)}$$

is larger than the total length of the (finitely many) intervals over the interior of which the oscillation of R is at least

$$\sigma = \frac{\epsilon}{2(b-a)}.$$

Call these intervals $I_1, \ldots, I_n$, and let $J_1, \ldots, J_k$ denote the remaining ones in P, on whose interiors R varies less than $\frac{\epsilon}{2(b-a)}$. Then

$$\begin{aligned} S(R, P) - s(R, P) &= \sum_{i=1}^{n} \left[\Big(\sup_{x \in I_i} R(x) - \inf_{x \in I_i} R(x) \Big) m(I_i) \right] + \\ &\quad \sum_{j=1}^{k} \left[\Big(\sup_{x \in J_j} R(x) - \inf_{x \in J_j} R(x) \Big) m(J_j) \right] \\ &= \sum_{i=1}^{n} \big[\omega(R, I_i) m(I_i) \big] + \sum_{j=1}^{k} \big[\omega(R, J_j) m(J_j) \big] \\ &< 2 \cdot \frac{\pi^2}{12} \cdot \frac{\epsilon}{4\pi^2/12} + \frac{\epsilon}{2(b-a)} \cdot (b-a) = \epsilon, \end{aligned}$$

and $R(x)$ is integrable.

Bounded maps with finitely many discontinuities are clearly integrable on the basis of Riemann's criterion, so Riemann integrability implies Cauchy integrability.

The uniform convergence of the series $\sum_{n=1}^{\infty}(nx)/n^2$ to the map R is, as we shall see, too strong a condition for checking integrability in more general contexts.

After Dedekind published Riemann's memoire (1867), the contributions of several scholars followed. Hermann Hankel's work, in particular, had a lot of resonance. In an 1870 paper Hankel believed he had proved any bounded map is integrable iff the discontinuity set of $f(x)$ is nowhere dense. As a matter of fact the condition is not even sufficient. Many mathematicians studied Riemann's work and discovered that according to his criteria a bounded map is not integrable if the discontinuities at which the oscillation is larger than an arbitrary $\sigma > 0$, do not form a negligible set, albeit nowhere dense. Hankel's investigation, if nothing else, had the merit of pinpointing the object to consider, i.e. understand the features of the discontinuity set of the function whose integrability is at issue.

A theorem similar to Hankel's was proved in 1875 by Giulio Ascoli. This time around the result was correct, though, in that Ascoli replaced nowhere-density with negligibility. The notion of negligible set had not yet been related to measures and zero-measure sets, so there was still much uncertainty on the matter, both theoretical and on the terminology.

Also in 1875, in the study of Riemann's theory entitled *"On the integration of discontinuous functions"*, the Irish mathematician Henry John Stephen Smith came up with a non-negligible nowhere dense set (Example 2.3.2), i.e. of positive measure, thus dispelling the myth that a nowhere dense sets had to be negligible. Alas, neither Ascoli's nor Smith's papers got the attention they deserved and, on the contrary, remained for many years almost unknown among mathematicians.

We have already mentioned the confusion lurking among experts concerning the topological features of subsets of the real line, in particular the conceptual distinction of sets of first species, nowhere dense sets and negligible sets. This should not surprise us, for the corresponding theory was still in its infancy, and anyway the mistakes and successive corrections have contributed to clarify many relevant aspects and issues.

The Italian Vito Volterra, unaware of Smith's paper, in the 1881 article *"Alcune osservazioni sulle funzioni punteggiate discontinue"* [3] gave another example of non-negligible nowhere dense set (Example 2.3.1).

In 1878's *Fondamenti* Ulisse Dini, who had been Volterra's teacher at Scuola Normale Superiore in Pisa, was very careful to discriminate among sets of first species, nowhere dense sets and negligible sets. He showed that being of first species implies nowhere dense and negligible (Theorem 2.3.2) and uncovered the fallacy of Hankel's proof. Dini, moreover, introduced in *Fondamenti* four types of derivative at a point x in an interval $[a, b]$, called Dini derivatives (see the introduction to Part VI - Fundamental theorems of integral calculus). These new tools allowed him to state the fundamental theorem of calculus for the Riemann integral: the integral function $F(x) = \int_a^x f(t)dt$ has four derivatives that differ among them, and from $f(x)$,

[3] The article appeared in Giornale di Matematiche, vol. XIX (1881), 76–87.

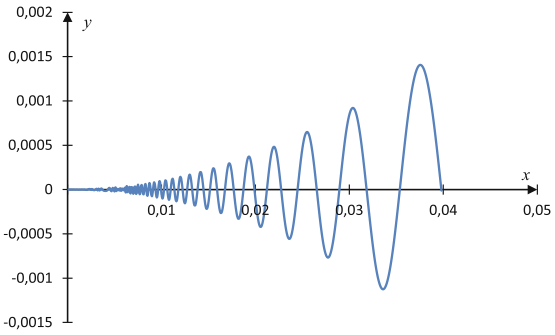


Fig. 5 Graph of $f(x) = x^2 \sin\left(\frac{1}{x}\right)$, as $x \to 0^+$

by functions with zero integral. Dini furthermore showed that if the primitive F is continuous on $[a, b]$ and **one of the four derivatives DF is Riemann integrable on** $[a, b]$, then

$$F(b) - F(a) = \int_a^b DF(x)dx.$$

He stressed the necessity of assuming DF integrable, and conjectured as highly likely the existence of continuous non-constant maps that are constant on some subinterval of any interval. That would mean DF was not Riemann integrable and its discontinuities could not be enclosed in a finite number of arbitrarily small intervals. In turn, that fact would violate the fundamental theorem of integral calculus.

Volterra proved the conjecture in 1881 by constructing an explicit map with bounded derivative on an interval I that is not Riemann integrable on I.[4] The idea was to associate to every interval $[a, b]$ a function with properties similar to (Fig. 5)

$$f(x) = x^2 \sin\left(\frac{1}{x}\right). \tag{IV.1}$$

Let $f_{a,b} : [a, b] \mapsto \mathbb{R}$ be defined by

[4]Published in Giornale di Matematiche, vol. XIX (1881), 333–372.

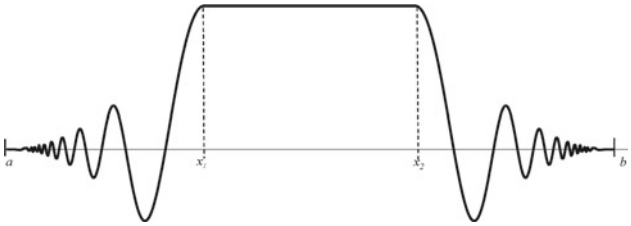


Fig. 6 Graph of $f_{a,b}(x)$, part of Volterra's function

$$f_{a,b}(x) = \begin{cases} 0 & \text{if } x = a \text{ or } x = b \\ (x-a)^2 \sin\left(\dfrac{1}{x-a}\right) & \text{if } a < x \le x_1, \text{ where } x_1 \text{ is the largest} \\ & \text{number} \le \dfrac{a+b}{2} \text{ s.t. } (x-a)^2 \sin\left(\dfrac{1}{x-a}\right) \\ & \text{has a maximum, i.e. derivative 0} \\ (x_1-a)^2 \sin\left(\dfrac{1}{x_1-a}\right) = (b-x_2)^2 \sin\left(\dfrac{1}{b-x_2}\right) & \text{if } x_1 \le x \le x_2 \\ (b-x)^2 \sin\left(\dfrac{1}{b-x}\right) & \text{if } x_2 \le x < b, \text{ where } x_2 \text{ is the smallest} \\ & \text{number} \ge \dfrac{a+b}{2} \text{ s.t. } (b-x)^2 \sin\left(\dfrac{1}{b-x}\right) \\ & \text{has a maximum, i.e. derivative 0.} \end{cases}$$

The function $f_{a,b}(x)$, as $x < x_1$ tends to a, behaves like $f(x) = x^2 \sin\left(\frac{1}{x}\right)$ when $x \to 0^+$: as x approaches a the oscillation width is scaled by the factor $(x-a)^2$ and tends to zero, while the frequency of the oscillations grows indefinitely. For x between x_1 and x_2 the map equals a positive constant. When $x > x_2$ tends to b, $f_{a,b}(x)$ mirrors the behaviour for $x < x_1$: as x tends to b the oscillations become more frequent but their amplitude decreases to zero, as in Fig. 6.
The function $f_{a,b}(x)$ is obviously bounded, since for every $c \in [a, b]$

$$|f_{a,b}(x)| < \begin{cases} (c-a)^2 \\ (b-c)^2 \end{cases} . \tag{IV.2}$$

Clearly $f'_{a,b}(x)$ behaves, around a or b, as the derivative of (IV.1). In fact on a neighbourhood of 0 the map

$$f'(x) = 2x \sin\left(\frac{1}{x}\right) - x^2\left(\frac{\cos(1/x)}{x^2}\right) = 2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right)$$

takes values ± 1:

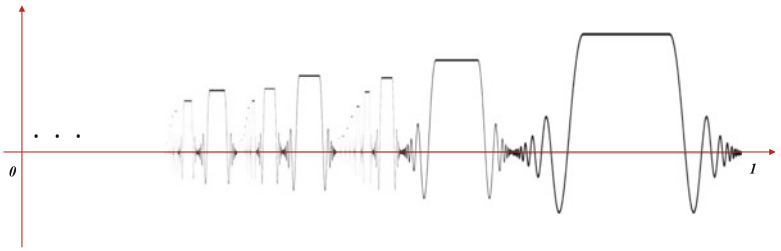


Fig. 7 Graph of the Volterra function

$$f'(0^{\pm}) = 2(0^{\pm}) \sin\left(\frac{1}{0^{\pm}}\right) - \cos\left(\frac{1}{0^{\pm}}\right) = -\cos\left(\frac{1}{0^{\pm}}\right) = \pm 1,$$

and $f'_{a,b}(x)$ oscillates between -1 and 1 as x tends to a or b. Moreover, the derivative $f'_{a,b}(x)$ is bounded:

$$|f'_{a,b}(x)| < 2(b-a) + 1. \tag{IV.3}$$

Now Volterra took $F : [0, 1] \mapsto \mathbb{R}$ defined by

$$F(x) = \begin{cases} f_{a_i,b_i}(x) & \text{if } x \in (a_i, b_i) \text{ for some } (a_i, b_i) \subseteq [0, 1] \setminus \bar{V} = \bigcup_{i=1}^{\infty} (a_i, b_i) \\ 0 & \text{if } x \in \bar{V} \end{cases}$$

where $\bar{V}$ is the Volterra set (Example 2.3.1). The plot of $F(x) : [0, 1] \mapsto \mathbb{R}$ is in Fig. 7. It is clear that

- $F(x)$ is continuous on $[0, 1]$;
- F' is bounded on $[0, 1]$, since (IV.3) implies $|F'(x)| < 3$ for every $x \in [0, 1]$;
- $F'(x)$ is well defined for every $x \in (a_i, b_i) \subseteq [0, 1] \setminus \bar{V}$, and $F'(x)$ exists for every $x \in \bar{V}$. Let $x \in \bar{V}$, for any $\epsilon > 0$, take $|x - t| < \epsilon$. If $t \in \bar{V}$ then

$$\frac{f(t) - f(x)}{t - x} = 0;$$

otherwise $t \notin \bar{V}$, and $t \in (a_i, b_i) \subseteq [0, 1] \setminus \bar{V}$ for some $i \in \mathbb{N}$. Suppose a_i is the endpoint of (a_i, b_i) closest to x. By (IV.2) we have

$$\left|\frac{f(t) - f(x)}{t - x}\right| = \left|\frac{f(t)}{t - x}\right| \leq \left|\frac{f(t)}{t - a_i}\right| \leq \frac{|t - a_i|^2}{|t - a_i|} = |t - a_i| < \epsilon.$$

To show F' is not Riemann integrable we must prove it is discontinuous at every $x \in \bar{V}$. Take $x \in \bar{V}$. For any $\epsilon > 0$, since $\bar{V}$ is nowhere dense there is an $y \in (a_i, b_i) \subseteq [0,1] \setminus \bar{V}$, for some $i \in \mathbb{N}$, such that $|x - y| < \epsilon$. As $F'(y) = f'_{a_i,b_i}(y)$ oscillates between ± 1 as y tends to a_i or b_i, whilst $F'(x) = 0$ when $x \in \bar{V}$, then F' is discontinuous at x. But $x \in \bar{V}$ is arbitrary, so F' is discontinuous everywhere on $\bar{V}$. Note that the measure of the discontinuity set of F' equals the measure of $\bar{V}$, which is bigger than $2/3$. Furthermore, at the discontinuities F' has constant oscillation equal to 2 in absolute value, so the discontinuities of F' that exceed an arbitrarily small $\sigma > 0$ cannot be contained in finitely many intervals of arbitrarily small size $\delta > 0$. This infringes the second integrability criterion.

In the introduction to *"Sui principi del calcolo integrale"* (see footnote 4), Volterra took on one of professor Dini's perplexities,[5] and on page 334 writes: "[…] *in some cases* […] *the usual definition of integral* [may] *not be subsumed by that of Riemann, i.e.* [there are] *maps whose derivates are not apt to be integrated."*. This emphasis should not be understood as Volterra's critique to the definition of Riemann integral. In another section he added that Riemann's definition, in that it did not require the existence of a primitive for integrability, was much superior to previous definitions, which on the contrary made that assumption. Volterra finally highlighted that if G denotes the characteristic function of $[0,1] \setminus \bar{V}$, the upper and lower integrals

$$\overline{G(x)} = \overline{\int_0^x} f\, dt \quad \text{and} \quad \underline{G(x)} = \underline{\int_0^x} f\, dt$$

are continuous, and constant on a dense collection of intervals where the Dini derivatives of $\overline{G(x)}$ and $\underline{G(x)}$ are bounded but not integrable. Exactly as Dini had speculated, for these two functions, as well as for the Volterra function, the fundamental theorem of calculus is meaningless.

1885 was a turning point, for the times were ripe for scientific research to undertake the right path for sorting out the integrability of everywhere-discontinuous, bounded maps in general. Many scholars, including Otto Stolz, Axel Harnack and Georg Cantor, linked integration to the notion of measure of sets, and were asking when an arbitrary set of points, for instance in $\mathbb{R}$, could be called measurable. It is not difficult to grasp the final answer on the issue of integrating real functions is related to understanding the measurability of linear sets. Segments and intervals are easy examples of linear sets. But as we have seen, the real line admits a profusion of set types: some are more interesting even if more complicated to study, like the discontinuity sets of the Dirichlet function (Remark 4.3.2), the Vitali-Cantor map (Example 10.5.1), the Volterra function or other very singular maps. It is exactly on these kinds of sets that measure theory focuses its attention. From a simple method to measure segments, surfaces or volumes, measure theory turns into a tool extending measures to the largest possible number of sets of points, be these on the real line, on the plane, in space etc.

[5] See Dini's *"Fondamenti per la teorica delle funzioni di variabili reali"*, p.276.

The first theories of measure essentially boil down to defining an outer measure for subsets of $\mathbb{R}$. They define the content of a set $A \subset \mathbb{R}$ as the greatest lower bound of (finite) sums of lengths of intervals containing one point of A at least. Albeit permitting the measurement of many sets, this definition produces inadequate measures, and is not additive. Suppose A is a disjoint union of two sets, say the rationals and irrationals between 0 and 1. The content of irrationals equals 1, the content of rationals is also 1, so the two measures add up to give 2, while the union [0, 1] has content 1.

To overcome this predicament in 1887 Giuseppe Peano defined, for a linear set I:

• the inner measure of I as the least upper bound of the measures of pluri-intervals contained in I;

• the outer measure of I as the greatest lower bound of the measures of pluri-intervals containing at least a point of I.

In 1893 Camille Jordan extended the theory to $\mathbb{R}^n$. Thereafter, a set is called Peano-Jordan measurable if the two measures coincide. The Peano-Jordan measure is clearly a big step forward compared to the notions proposed by Stolz, Harnack and Cantor, for it is additive. Yet it does not permit the measurement of extremely complex sets because it is not countably additive. So-called 'pluri-intervals' are made of a finite number of intervals whose possible pairwise intersections have no interior points. The measure of pluri-intervals contained in I defines the inner measure of I, while the measure of those containing at least a point of I defines the outer measure of I.

The reason for not covering linear sets with pluri-intervals made of countably many intervals is that Harnack, back in 1885, had written that the measure $b - a$ of an interval $(a, b) \subset \mathbb{R}$ might in general differ from the sum of the measures of a countable partition of (a, b). Already in 1882 Cantor had proved that any open set in $\mathbb{R}$, hence an open interval too, is the union of a countable family of pairwise-disjoint open intervals. A measure theory contemplating countable additivity was still wanting.

We have to wait until 1898, when Emile Borel in his *"Leçons sur la théorie des fonctions"* (p.41 ff.) proved what Harnack had thought to be false. He did so by emphasising the relevance of a statement made by Heine concerning open covers of compact sets (nowadays known as Heine-Borel theorem): any open cover of a closed, bounded interval admits a finite subcover.

Since the measure of a bounded interval I is unchanged if we exclude one or both endpoints, a bounded interval can be replaced by the corresponding compact interval. By the Heine-Borel theorem the latter, when covered by any (also infinite) number of open intervals, will have length equal to the finite sum of the lengths of the subcover's intervals.

By means of this theorem all intervals of $\mathbb{R}$ become measurable, and from them we can measure any bounded open subset in $\mathbb{R}$. If a set is the countable union of pairwise-disjoint intervals of *"total length s, we shall say the set has measure s. Whenever two sets have no common points, and their measures are s and s', the set obtained by joining them, i.e. their sum, has measure $s + s'$"* (*Leçons*, p.46). In general, if we have a countable family of disjoint sets each of measure s_i, the

union will measure $\sum_{i=1}^{\infty} s_i$. And if $A \subseteq B$ respectively measure s and s', then the difference $B \setminus A$ (points of B that are not in A) has measure $s' - s$. Therefore any open bounded subset of $\mathbb{R}$ is measurable. Starting here and iterating endlessly the above operations, Borel obtained a large class of measurable sets, the so-called Borel sets.

He adds (*Leçons*, p.48) *"The procedure we have employed reduced in reality to this: we have recognised that a definition of measure cannot be useful unless it has certain fundamental properties: we have given these properties* a priori *and it is these that have served us to define the class of sets we consider to be measurable"*.

After, he lists the 'fundamental properties' of a measure:

- the measure of a countable union of (disjoint) measurable sets equals the sum of their measures;
- the measure of the difference of two sets equals the difference of the measures;
- any measure is non-negative;
- a set with non-zero measure must be uncountable.

The theorem on countable additivity, and the corresponding axiomatic definition, were used by Borel merely as technical tools to investigate topics he went on to develop later in the work. They had nothing to do with the possible application to integration. Nonetheless, Borel's measure theory turned out to be high-yielding as soon as it was implemented within the theory of integration. The ramifications of Borel's statements were fully understood by Henri Lebesgue, who in 1902 included Borel's results among his own definitions of measure and integral.

To define the measure of bounded linear sets Lebesgue gave these axioms:

- there exist sets with non-zero measure;
- equal sets have equal measures (translation-invariance of the measure);
- the measure of a finite or countable sum of pairwise-disjoint sets equals the sum of the sets' measures (complete additivity of the measure).

From here Lebesgue started explaining his theory of measure, which can be summarised as follows.

Given a linear set of points E, let $\{I_i\}_{i\geq 1}$ denote a sequence of intervals that contain at least one point of E. The measure $m(E) = m\big(\bigcup_{i=1}^{\infty} I_i\big)$ of E does not exceed the measure $\sum_{i=1}^{\infty} m(I_i)$:

$$m(E) \leq \sum_{i=1}^{\infty} m(I_i).$$

The infimum of the sum is an upper bound for $m(E)$, which we call outer measure of E and indicate by $m_e(E)$:

$$m(E) \leq \inf \left\{ \sum_{i=1}^{\infty} m(I_i) : E \subseteq \bigcup_{i=1}^{\infty} I_i \right\} = m_e(E).$$

Now suppose E is contained in the interval (a, b). As the measure of the complement $(a, b) \setminus E$ is at most $m_e\big((a, b) \setminus E\big)$,

$$m\big((a, b) \setminus E\big) \leq m_e\big((a, b) \setminus E\big) \Leftrightarrow m((a, b)) - m(E) \leq m_e\big((a, b) \setminus E\big),$$

then

$$m(E) \geq m((a, b)) - m_e\big((a, b) \setminus E\big).$$

The measure $\big[m((a, b)) - m_e\big((a, b) \setminus E\big)\big]$ is called inner measure of E, and is denoted $m_i(E)$. Equal sets have the same outer and inner measures. Moreover, since

$$m_e(E) + m_e\big((a, b) \setminus E\big) \geq m((a, b)) \Leftrightarrow m_e(E) \geq m((a, b)) - m_e\big((a, b) \setminus E\big) = m_i(E),$$

the outer measure is never less than the inner measure. Call measurable those sets whose inner and outer measures coincide, and the common value is then called the measure $m(E)$ of the set. Lebesgue, instead of conforming to Peano, used the outer measure of the complement of E to define the inner measure of E, so that the inner measure is given in terms of the outer measure rather than being defined independently, *à la* Peano.

To complete the account of Riemann's theory of integration let us briefly mention another criterion for Riemann integrability. It was proved in 1907 independently by Vitali and Lebesgue. The criterion puts the spotlight on the hypothesis delineated by Ascoli in 1875, that a bounded map is Riemann integrable iff the discontinuity set is negligible. We have also seen that the idea of negligible set was not supported by a suitable measure theory, but after the theoretical work of Peano, Jordan and Lebesgue, the time had come to finally clarify what had remained a well-grounded, yet mere, speculation.

More specifically, to get to define negligible sets (or better, "enclosable" sets) and therefore Peano-Jordan measurable sets, we must first introduce the following important notions.

Fix $A \subseteq \mathbb{R}$. We say a family $\mathscr{I}$ of intervals of $\mathbb{R}$ is an i-cover of A if every point of A is **internal** to one interval of $\mathscr{I}$ at least. An open i-cover is a covering. Moreover if $\mathscr{I}$ is an i-cover of A, the family $\{I \cap A : I \in \mathscr{I}\}$ covers A, while if $\{I \cap A : I \in \mathscr{I}\}$ is a cover of A, we cannot conclude $\mathscr{I}$ is an i-cover of A. Therefore given an i-cover $\mathscr{I}$ of a closed, bounded A, there exists a finite subcover of $\mathscr{I}$ that is still an i-cover of A. In fact, since A is compact the family $\{\overset{\circ}{I} \cap A : I \in \mathscr{I}\}$ is an open cover of A, and the Heine-Borel theorem allows to extract a finite subcover.

Now we are ready to introduce the notion of enclosable sets. A subset A of $\mathbb{R}$ is an enclosable set, or a set of zero measure, if for any $\epsilon > 0$ there exists an i-cover $\mathscr{I}$ of A, made of a finite or countable number of pairwise-disjoint intervals (a_i, b_i), such that $\sum_i (b_i - a_i) < \epsilon$. Examples include:

- sets made of a finite number p of points in $\mathbb{R}$. In fact, for every $\epsilon > 0$ it suffices to take an interval I_i centred at each point of length less than ϵ/p, so that $\sum_{i=1}^{p} m(I_i) < p \cdot \frac{\epsilon}{p} = \epsilon$;
- countable sets $\{x_n\}$. In fact, for every $\epsilon > 0$ it is enough to take an interval I_i, centred at each point of the set, of length less than $\epsilon/2^i$, whence $\sum_{i=1}^{\infty} m(I_i) < \sum_{i=1}^{\infty} (\epsilon/2^i) = \epsilon$;
- the Cantor set C, which is uncountable, is enclosable. Given an arbitrary $\epsilon > 0$, consider C_n: it contains every C_{n+i} and therefore also C, and is made of 2^n pairwise-disjoint compact intervals, so it has measure $2^n/3^n < \epsilon$. Finite unions of compact sets are compact, so there exists a finite number k of disjoint open sets each of measure $m(a_{n,i}, b_{n,i}) < \epsilon/k$, forming an i-cover of C_n, and hence of C, such that

$$C \subseteq C_n \subseteq \bigsqcup_{i=1}^{k} (a_{n,i}, b_{n,i}) \quad \text{and} \quad m(C) \leq m(C_n) \leq \sum_{i=1}^{k} (b_{n,i} - a_{n,i}) < \epsilon.$$

The previous argument cannot be reversed, since a zero-measure set might not be enclosable. Instances include Lebesgue measurable sets that are not Peano-Jordan measurable, in particular the Dirichlet set. In this case, although i-covers exist, an open i-cover would be empty (so strictly speaking it would not be an i-cover), because it would consist of a countable union of intervals $(q_i, q_i) = \emptyset$, with $q_i \in \mathbb{Q}$ for every $i \in \mathbb{N}$.

Based on the above notions, we can at last state the **third criterion for Riemann integrability**, sometimes known as Vitali-Lebesgue integrability criterion:

"A bounded function $f : [a, b] \mapsto \mathbb{R}$ is Riemann integrable if and only if the set of its discontinuities is enclosable."

Chapter 8
Set Functions and Measure



8.1 Set Functions and Finitely Additive Measures

Before we undertake measure theory on completely additive families it is useful to examine the theory on finitely additive families. This approach, apart from respecting the historical evolution of the topic, allows to understand which theoretical developments facilitated the passage from the Peano-Jordan measure theory to the more refined theory of the Lebesgue measure.

Definition 8.1.1 (*Set functions*) Given a collection $\mathcal{C}$ of subsets in a non-empty set X, we call set function any map m defined on $\mathcal{C}$ that assumes non-negative values in the extended real line, i.e.

$$m : \mathcal{C} \mapsto [0, \infty] \quad \text{and} \quad m(A) \geq 0, \ \forall A \in \mathcal{C}.$$

$\diamond$

A general set function might satisfy certain properties, and we summarise below the main ones.

Definition 8.1.2 (*Additional features of set functions*) A set function $m : \mathcal{C} \mapsto [0, \infty]$ is called:

1. monotone if $m(A) \leq m(B)$ for every $A, B \in \mathcal{C}$ such that $A \subseteq B$;
2. finitely additive if, for every pairwise-disjoint finite collection $A_1, \ldots, A_n \in \mathcal{C}$ such that $A = \bigcup_{i=1}^{n} A_i \in \mathcal{C}$,

$$m(A) = m\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} m(A_i);$$

3. completely additive (generalising the above) if, for every pairwise-disjoint collection $\{A_i\}_{i\geq 1}$ in $\mathcal{C}$ such that $A = \bigcup_{i=1}^{\infty} A_i \in \mathcal{C}$,

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$$m(A) = m\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} m(A_i);$$

4. completely subadditive, if for any collection $\{A_i\}_{i\geq 1}$ in $\mathcal{C}$ (not necessarily pairwise disjoint) such that $A = \bigcup_{i=1}^{\infty} A_i \in \mathcal{C}$,

$$m(A) = m\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} m(A_i);$$

5. totally finite if $m(A) < \infty$ for every $A \in \mathcal{C}$;
6. σ-finite if there exists a sequence of pairwise-disjoint sets $\{A_i\}_{i\geq 1}$ in $\mathcal{C}$ such that $m(A_i) < \infty$ for every i and $\bigcup_{i=1}^{\infty} A_i = X$.

◇

In general a set function $m : \mathcal{C} \mapsto [0, \infty]$ is well suited to represent a measure on the collection $\mathcal{C}$ if is possesses some of the previous properties.

Definition 8.1.3 (*Finitely additive measures*) A set function $m : \mathcal{C} \mapsto [0, +\infty]$ is said to be a finitely additive measure on the family of subsets $\mathcal{C}$ of a non-empty set X when:

1. $\emptyset \in \mathcal{C}$ with $m(\emptyset) = 0$ (the empty set is null);
2. given pairwise-disjoint sets $A_1, \ldots, A_n \in \mathcal{C}$ and $A = \bigsqcup_{i=1}^{n} A_i \in \mathcal{C}$,

$$m(A) = \sum_{i=1}^{n} m(A_i).$$

◇

A measure, being a set function, is always non-negative by definition, i.e. $m(A) \geq 0$ for every $A \in \mathcal{C}$ (non-negativity of a measure).

Remark 8.1.1 We may assume the family $\mathcal{C}$ of subsets of X on which the measure m is defined (Definition 8.1.3) is for instance a semi-algebra, for the latter contains all the sets on which a finitely additive set function can be defined easily (consider for instance the length function on the collection $\mathcal{I}$ of Example 8.1.1). Whenever it becomes necessary to measure more general sets, we may pass to the algebra generated by such collections with the help of the properties seen in Sect. 6.2. ♣

Let us examine a special set function of crucial importance in the study of the theories of measure and integration, namely the length function. For the time being we shall prove only the properties that play a role in the measure of Peano-Jordan (PJ, henceforth).

Example 8.1.1 (*Core properties of the length function on the semi-algebra $\mathcal{I}$*) The length function, defined on the family $\mathcal{I}$ of intervals of $\mathbb{R}$ (which we know is a semi-algebra), is the set function $m : \mathcal{I} \mapsto [0, \infty]$ defined by

$$m(I_a^b) = \begin{cases} |b-a| & \text{if } a, b \in \mathbb{R}, \\ +\infty & \text{if } a = -\infty \text{ and/or } b = \infty \end{cases}$$

for every interval $I_a^b \in \mathcal{I}$. We wish to establish whether it enjoys the following important properties: non-negativity, monotonicity, finite additivity, subtractivity, subadditivity and invariance under translations.

1. For $(a, a) = \emptyset$ we have $m((a, a)) = |a - a| = 0$.
2. Every singleton $\{x\}$ can be written as an interval $[x, x]$, so $m([x, x]) = m(\{x\}) = |x - x| = 0$.
3. Let us verify monotonicity. Take $I, J \in \mathcal{I}$ and $I \subseteq J$. Two things can happen:
   - I is bounded or unbounded: given any M we have $I_M = (-M, a]$ and $J = (-\infty, c)$, for $a < c$, whence $m(I_M) < m(J)$. As $M > 0$ is arbitrary, $m(I) \leq m(J)$;
   - both I and J are bounded. Suppose $I = (a, b)$, $a \leq b$. As I is a subset of J, the endpoints of $J = (c, d)$ satisfy $c \leq a \leq b \leq d$, so

   $$(b - a) \leq (d - c) \Leftrightarrow m(I) \leq m(J).$$

4. Now finite additivity. Suppose $I = \bigcup_{i=1}^n I_i$ with pairwise-disjoint sets I_i. If I is unbounded, $m(I_j) = \infty$ for some $j = 1, \ldots, n$, and so

   $$m(I) = +\infty = \sum_{i=1}^n m(I_i).$$

   If $I = [a, b]$ is bounded, take a partition

   $$a = a_1 \leq b_1 = a_2 \leq b_2 \ldots \leq b_{n-1} = a_n \leq b_n = b$$

   of I. Since the endpoints of the subintervals and of the larger interval are irrelevant to the measure of each, without loss of generality we may assume them all closed. Hence

   $$m(I) = (b - a) = b_n - a_1 = \sum_{i=1}^n (b_i - a_i) = \sum_{i=1}^n m(I_i').$$

5. Subtractivity: take intervals $I_2 \subseteq I_1$ with $m(I_2) < \infty$. In case $m(I_1) = \infty$, clearly $I_2 \cap (I_1 \setminus I_2) = \emptyset$, so the length's finite additivity forces

$$m(I_1) = m(I_2) + m(I_1 \setminus I_2) = \infty \Leftrightarrow m(I_1 \setminus I_2) = m(I_1) - m(I_2) = \infty.$$

In case $m(I_1) < \infty$, $I_2 \cap (I_1 \setminus I_2) = \emptyset$, and by monotonicity and finite additivity

$$m(I_2) \leq m(I_1) < \infty \quad \text{and} \quad m(I_2) + m(I_1 \setminus I_2) < \infty.$$

Therefore $m(I_1)$, $m(I_2)$ and $m(I_1 \setminus I_2)$ are finite, non-negative real numbers and

$$m(I_1) = m(I_2) + m(I_1 \setminus I_1) \Leftrightarrow m(I_1) - m(I_2) = m(I_1 \setminus I_2).$$

6. Finite subadditivity: take $I \in \mathcal{I}$ such that

$$I = \bigcup_{i=1}^{n} I_i, \text{ with } I_i \cap I_j \neq \emptyset.$$

Define $J_1 = I_1$ and $J_i = I_i \setminus (I_1 \cup \ldots \cup I_{i-1})$, so that $\{J_i\}_{i=1}^n$ is a finite pairwise-disjoint collection. Then

$$I = \bigsqcup_{i=1}^{n} J_i = \bigcup_{i=1}^{n} I_i.$$

By finite additivity and monotonicity,

$$m\left(\bigcup_{i=1}^{n} I_i\right) = m\left(\bigsqcup_{i=1}^{n} J_i\right) = \sum_{i=1}^{n} m(J_i) \leq \sum_{i=1}^{n} m(I_i).$$

7. The last (for the time being) property of the length is the invariance under translations. Given $I = [a, b]$ and $x \in \mathbb{R}$, we call

$$I + x = \{y + x : y \in I\}$$

interval shifted, or translated, by x. Translation-invariance then reads as follows: shifting any $I \in \mathcal{I}$ by any $x \in \mathbb{R}$ defines an interval in $\mathcal{I}$, and $m(I + x) = m(I)$. In fact, let $I = [a, b]$, so $I + x = [a + x, b + x] \in \mathcal{I}$ and

$$m(I + x) = (b + x) - (a + x) = b - a = m(I).$$

In case $I = [a, +\infty)$, then $I + x = [a + x, \infty) \in \mathcal{I}$ and

$$m(I + x) = \infty = m(I).$$

★

Now we define extensions of set functions from a class $\mathcal{C}$ to a larger class $\mathcal{A}$. In this sense set functions generalise the idea of length, area and volume to arbitrary sets on a line, a surface, in a three-dimensional space etc.

Definition 8.1.4 (*Extension of a set function*) Given collections of subsets $\mathcal{C} \subseteq \mathcal{A}$ in a set X, define set functions

$$m_1 : \mathcal{C} \mapsto [0, \infty] \quad \text{and} \quad m_2 : \mathcal{A} \mapsto [0, \infty]$$

such that $m_1(E) = m_2(E)$ for every $E \in \mathcal{C}$. Then m_2 is called an extension of m_1 to $\mathcal{A}$. ◇

Now we are ready to present the Peano-Jordan measure theory. We will show that this set function is only finitely additive.

The theory's founding idea is to cover a given linear set A with collections of suitably chosen intervals and recover from their lengths the measure of A. In particular, to obtain the Peano-Jordan measure of a bounded set $A \subseteq \mathbb{R}$ one considers two finite collections of subintervals whose possible pairwise intersections have empty interiors: the one containing at least one point of A on one hand, the one contained entirely in A on the other. Then one computes:

- the sum of the lengths m of those subintervals containing at least a point of A;
- the sum of the lengths m of those subintervals completely contained in A.

The greatest lower bound $m_e(A)$ of the sums of the first type is called *outer measure* of A, while the least upper bound $m_i(A)$ of the sums of the second type is the *inner measure* of A.

Remark 8.1.2 Recall that for every non-empty set $A \subseteq [0, \infty]$

- $\sup(A) = \sup(A \cap [0, \infty))$ if $(A \cap [0, \infty)^c) = \emptyset$, i.e. if A is upper bounded, otherwise $\sup(A) = \infty$;
- $\inf(A) = \inf(A \cap [0, \infty))$ if $(A \cap [0, \infty)) \neq \emptyset$, i.e. if A is lower bounded, otherwise $\inf(A) = \infty$.

Consequently $\sup(A)$ and $\inf(A)$ are well defined for every non-empty subset $A \subseteq [0, \infty]$. ♣

In the light of the remark we propose the following general definition.

Definition 8.1.5 (*Inner and outer Peano-Jordan measure of a subset of* $\mathbb{R}$) Let $\mathcal{I}$ be the family of intervals of $\mathbb{R}$ and $m : \mathcal{I} \mapsto [0, \infty]$ the length measure. Define:

1. outer measure induced by m the function $m_e : \mathcal{P}(\mathbb{R}) \mapsto [0, \infty]$ defined, for every $A \subseteq \mathbb{R}$, by

$$m_e(A) = \inf \left\{ \sum_{i=1}^{n} m(I_i) : A \subseteq \left(\bigcup_{i=1}^{n} I_i \right), I_i \in \mathcal{I}, \forall i \text{ and } \overset{\circ}{I}_i \cap \overset{\circ}{I}_j = \emptyset, \ \forall i \neq j \right\} \leq \infty, \tag{8.1.1}$$

2. inner measure induced by m the function $m_i : \mathcal{P}(\mathbb{R}) \mapsto [0, \infty]$ defined, for every $A \subseteq \mathbb{R}$, by

$$m_i(A) = \sup \left\{ \sum_{k=1}^{m} m(J_k) : \left(\bigcup_{k=1}^{m} J_k \right) \subseteq A, J_k \in \mathcal{I}, \forall k \text{ and } \overset{\circ}{J}_k \cap \overset{\circ}{J}_j = \emptyset, \ \forall k \neq j \right\} \leq \infty. \tag{8.1.2}$$

◇

In (8.1.1) we can drop the hypothesis that the I_i intersect with empty interior, because the inf subsumes this requirement.

We call outer pluri-interval and inner pluri-interval of a linear set A the respective sets

$$P^e = \left(\bigcup_{i=1}^{n} I_i \right) \supseteq A, \quad \text{with } I_i \in \mathcal{I}, \forall i \text{ and } \overset{\circ}{I}_i \cap \overset{\circ}{I}_j = \emptyset, \ \forall i \neq j;$$

$$P^i = \left(\bigcup_{k=1}^{m} J_k \right) \subseteq A, \quad \text{with } J_k \in \mathcal{I}, \forall k \text{ and } \overset{\circ}{J}_k \cap \overset{\circ}{J}_j = \emptyset, \ \forall k \neq j.$$

Indicate by $\{P\}$ the collection of pluri-intervals of $\mathbb{R}$.

Clearly for every $A \subseteq \mathbb{R}$ we have $m_i(A) \leq m_e(A)$, which suggests the next definition.

Definition 8.1.6 (*Peano-Jordan measurable sets and Peano-Jordan measure*) A set $A \subseteq \mathbb{R}$ is Peano-Jordan measurable if

$$m_i(A) = m_e(A).$$

Let $\mathcal{A}_{pj}(\mathcal{I})$ be the collection of PJ measurable sets $A \subseteq \mathbb{R}$.

Call Peano-Jordan measure the set function $m_{pj} : \mathcal{A}_{pj}(\mathcal{I}) \mapsto [0, \infty]$ defined by

$$m_{pj}(A) = m_e(A) = m_i(A)$$

for any $A \in \mathcal{A}_{pj}(\mathcal{I})$. ◇

We will see this collection is an algebra generated by $\mathcal{I}$. Such an algebra obviously contains all pluri-intervals $\{P\}$ of $\mathbb{R}$.

Now we prove a number of properties of the PJ measure that will come handy to establish important results on the subject.

Proposition 8.1.1 (Properties of the Peano-Jordan measure) *The PJ measure* $m_{pj} : \mathcal{A}_{pj}(\mathcal{I}) \mapsto [0, \infty]$ *satisfies the following properties:*

1. *for every pluri-interval* $P \in \mathcal{A}_{pj}(\mathcal{I})$ *the measure* $m_{pj}(P)$ *coincides with the length measure* $m(P)$*;*

2. *m_{pj} is monotone, i.e. if for any $A_1, A_2 \in \mathcal{A}_{pj}(\mathcal{I})$ such that $A_1 \subseteq A_2$ then*

$$m_{pj}(A_1) \leq m_{pj}(A_2);$$

3. *m_{pj} is finitely additive, i.e. if $A_1, \ldots, A_n$ are in in $\mathcal{A}_{pj}(\mathcal{I})$ and $A = \bigsqcup_{i=1}^{n} A_i$, then*

$$m_{pj}(A) = \sum_{i=1}^{n} m_{pj}(A_i);$$

4. *m_{pj} is subtractive, i.e. if $A, B \in \mathcal{A}_{pj}$ with $m_{pj}(A) < \infty$ and $A \subseteq B$, then*

$$m_{pj}(B \setminus A) = m_{pj}(B) - m_{pj}(A);$$

5. *m_{pj} is finitely subadditive, i.e. if $A_1, \ldots, A_n$ are non-pairwise-disjoint sets in $\mathcal{A}_{pj}(\mathcal{I})$ and $A = \bigcup_{i=1}^{n} A_i$, then*

$$m_{pj}(A) \leq \sum_{i=1}^{n} m_{pj}(A_i).$$

Proof 1. Let $P \in \{P\}$ be arbitrary, so $P = \bigcup_{i=1}^{n} I_i$, $I_i \in \mathcal{I}$ and $\overset{\circ}{I}_i \cap \overset{\circ}{I}_k = \emptyset$ for every $i \neq k$. Then

$$m_{pj}(P) = \sum_{i=1}^{n} m(I_i) = m(P);$$

since P is arbitrary the claim follows, so the PJ measure m_{pj} extends the length function m to $\mathcal{A}_{pj}(\mathcal{I})$, moreover $m_{pj}(\emptyset) = m(\emptyset) = 0$ and $m_{pj}(\{x\}) = m(\{x\}) = m([x, x]) = 0$. As $m_{pj}(\emptyset) = 0$ we obtain the PJ measure's non-negativity on $\mathcal{A}_{pj}(\mathcal{I})$.

2. Take any two pluri-intervals P_1^i, $P_2^i \in \{P\}$ such that

$$P_1^i = \left(\bigcup_{i=1}^{n} J_i \right) \subseteq A_1 \quad \text{and} \quad P_2^i = \left(\bigcup_{k=1}^{m} J_k \right) \subseteq A_2,$$

where $P_1^i \subseteq P_2^i$. For every $i = 1, \ldots, n$ J_i is at most as long as some $J_k \in P_2^i$. By the subadditivity of m on $(\mathcal{I})$

$$m(P_1^i) = \sum_{i=1}^{n} m(J_i) \leq \sum_{k=1}^{m} m(J_k) = m(P_2^i).$$

As the inequality holds for any pluri-interval contained in A_1 and A_2, it holds for the respective least upper bounds, so

$$m_i(A_1) \leq m_i(A_2).$$

Incidentally we have proved **the inner measure's monotonicity**.
Furthermore, by assumption A_1 and A_2 are PJ measurable, i.e. $m_i(A_1) = m_e(A_1)$ and $m_i(A_2) = m_e(A_2)$, whence **the outer measure's monotonicity** follows, i.e. $m_e(A_1) \leq m_e(A_2)$, and then

$$m_{pj}(A_1) \leq m_{pj}(A_2).$$

3. By assumption $A = \bigsqcup_{i=1}^{n} A_i$ and A_i are in $\mathcal{A}_{pj}(\mathcal{I})$, $i = 1, \ldots, n$, so given arbitrary pluri-intervals

$$P_i^i = \bigcup_{k_i=1}^{n_i} J_{k_i} \subseteq A_i$$

the disjoint union

$$\bigsqcup_{i=1}^{n} P_i^i = \bigsqcup_{i=1}^{n} \left(\bigcup_{k_i=1}^{n_i} J_{k_i} \right) \subseteq \bigsqcup_{i=1}^{n} A_i = A$$

is a pluri-interval P^i contained in A. As $P^i \in \{P\}$,

$$P^i = \bigcup_{j=1}^{m} J_j \subseteq A.$$

For every j

$$J_j = \bigsqcup_{i=1}^{n} \bigcup_{k_i=1}^{n_i} (J_{k_i} \cap J_j),$$

and for every i

$$P_i^i = \bigcup_{j=1}^{m} \bigcup_{k_i=1}^{n_i} (J_{k_i} \cap J_j) \subseteq A_i.$$

By finite additivity of m on $\mathcal{I}$ we have

$$\sum_{j=1}^{m} m(J_j) = \sum_{j=1}^{m} m\left(\bigsqcup_{i=1}^{n} \bigcup_{k_i=1}^{n_i} (J_{k_i} \cap J_j) \right) = \sum_{i=1}^{n} \left(\sum_{j=1}^{m} \sum_{k_i=1}^{n_i} m(J_{k_i} \cap J_j) \right).$$

This holds for every $P_i^i \subseteq A_i$ and $P^i \subseteq A$, so also for the supremum, and then

$$m_i(A) = \sup\left\{\sum_{j=1}^{m} m(J_j)\right\}$$
$$= \sum_{i=1}^{n}\left(\sup\left\{\sum_{j=1}^{m}\sum_{k_i=1}^{n_i} m(J_{k_i}\cap J_j)\right\}\right) = \sum_{i=1}^{n} m_i(A_i).$$

Similarly $m_e(A) = \sum_{i=1}^{n} m_e(A_i)$. As every A_i is measurable, $m_i(A_i) = m_e(A_i)$, $i = 1, \ldots, n$. Then the previous relation becomes

$$m_{pj}(A) = \sum_{i=1}^{n} m_{pj}(A_i),$$

as required.

4. Take P_B^i, $P_A^i \in \{P\}$ such that

$$P_B^i = \left(\bigcup_{i=1}^{n} J_i\right) \subseteq B \quad \text{and} \quad P_A^i = \left(\bigcup_{k=1}^{m} J_k\right) \subseteq A,$$

where clearly $P_A^i \subseteq P_B^i$. By length subtractivity and finite additivity we have

$$m\big(P_B^i \setminus P_A^i\big) = m\left(\bigcup_{i=1}^{n} J_i \setminus \bigcup_{k=1}^{m} J_k\right) = \sum_{i=1}^{n} m(J_i) - \sum_{k=1}^{m} m(J_k)$$
$$= m\big(P_B^i\big) - m\big(P_A^i\big).$$

This is true for every $P_B^i \subseteq B$ and $P_A^i \subseteq A$, so it holds for the sup:

$$m_i(B \setminus A) = m_i(B) - m_i(A).$$

Incidentally we have proved the **the inner measure's subtractivity**. By assumption, moreover, A, B are PJ measurable, so $m_i(A) = m_e(A)$, $m_i(B) = m_e(B)$ and similarly $m_e(B \setminus A) = m_e(B) - m_e(A)$ (**outer measure's subtractivity**). Overall

$$m_{pj}(B \setminus A) = m_{pj}(B) - m_{pj}(A).$$

5. To prove item *5.* let $A = \bigcup_{i=1}^{n} A_i$ with $A_i \cap A_k \neq \emptyset$. Define

$$B_1 = A_1, \ldots, B_i = A_i \setminus \bigcup_{k=1}^{i-1} A_k.$$

Clearly the finite sequence $\{B_i\}_{i=1}^n$ consists of pairwise-disjoint sets that are PJ measurable by subtractivity. Furthermore, $A, A_i \in \mathcal{A}_{pj}(\mathcal{I})$, $i = 1, \ldots, n$, i.e. all are PJ measurable and

$$A = \bigsqcup_{i=1}^{n} B_i \subseteq \bigcup_{i=1}^{n} A_i.$$

Given, for $i = 1, \ldots, n$, arbitrary pluri-intervals

$$P_i^i = \bigcup_{j_i=1}^{n_i} J_{j_i} \subseteq B_i$$
$$Q_i^i = \bigcup_{q_i=1}^{m_i} J_{q_i} \subseteq A_i,$$

we have

$$P^i = \bigsqcup_{i=1}^{n} \left(\bigcup_{j_i=1}^{n_i} J_{j_i} \right) \subseteq A \quad \text{and} \quad J_{j_i} \subseteq J_{q_i} \quad \forall i = 1, \ldots, n.$$

As $P^i \in \{P\}$,

$$P^i = \bigcup_{k=1}^{m} J_k \subseteq A,$$

for every k

$$J_k = \bigsqcup_{i=1}^{n} \bigcup_{j_i=1}^{n_i} (J_{j_i} \cap J_k),$$

and for every i

$$P_i^i = \bigcup_{k=1}^{m} \bigcup_{j_i=1}^{n_i} (J_{j_i} \cap J_k) \subseteq B_i$$
$$Q_i^i = \bigcup_{k=1}^{m} \bigcup_{q_i=1}^{m_i} (J_{q_i} \cap J_k) \subseteq A_i.$$

Applying the finite additivity and monotonicity of m we obtain

$$\sum_{k=1}^{m} m(J_k) = \sum_{k=1}^{m} m\Big(\bigsqcup_{i=1}^{n}\bigcup_{j_i=1}^{n_i}\big(J_{j_i}\cap J_k\big)\Big) = \sum_{i=1}^{n}\Big(\sum_{k=1}^{m}\sum_{j_i=1}^{n_i}\underbrace{m\big(J_{j_i}\cap J_k\big)}_{\leq m(J_{q_i}\cap J_k)}\Big)$$

$$\leq \sum_{i=1}^{n}\Big(\sum_{k=1}^{m}\sum_{q_i=1}^{m_i} m\big(J_{q_i}\cap J_k\big)\Big).$$

As always, the above holds for arbitrary pluri-intervals in A, B_i, A_i for every $i = 1, \ldots, n$, so the inequality is preserved by least upper bounds

$$\sup\Big\{\sum_{k=1}^{m} m(J_k)\Big\} \leq \sum_{i=1}^{n}\Big(\sup\Big\{\sum_{k=1}^{m}\sum_{q_i=1}^{m_i} m\big(J_{q_i}\cap J_k\big)\Big\}\Big)$$

$$\Leftrightarrow m_i(A) \leq \sum_{i=1}^{n} m_i(A_i),$$

which also demonstrates the **inner measure's finite subadditivity**. In a similar way one proves $m_e(A) \leq \sum_{i=1}^{n} m_e(A_i)$, namely the **outer measure's finite subadditivity**. Also every A_i is measurable, so $m_i(A_i) = m_e(A_i)$, $i = 1, \ldots, n$, giving:

$$m_{pj}(A) \leq \sum_{i=1}^{n} m_{pj}(A_i).$$

■

Before we prove an important theorem characterising the PJ measurability of bounded sets in $\mathcal{A}_{pj}(\mathcal{I})$ we need a useful preliminary fact.

Lemma 8.1.1 (Inner measure of the interior and outer measure of the closure of bounded subsets of $\mathbb{R}$) . *The inner measure of a bounded set $A \subseteq \mathbb{R}$ equals the inner measure of the interior of A:*

$$m_i(A) = m_i(\overset{\circ}{A}).$$

The outer measure of a bounded set $A \subseteq \mathbb{R}$ equals the outer measure of the closure of A:

$$m_e(A) = m_e(\bar{A}).$$

Proof Given any $\epsilon > 0$, the quantity $m_i(A) - \epsilon$ is no longer the least upper bound, so we can find a pluri-interval

$$P_\epsilon^i = \bigcup_{i=1}^{n} J_i \subseteq A$$

such that

$$m_i(A) - \epsilon < m_i(P_\epsilon^i) = m_i\left(\bigcup_{i=1}^{n} J_i\right)$$

As, for every $J \in \mathcal{I}$, $m(J) = m(\overset{\circ}{J})$, and since

$$\overset{\circ}{P}{}_\epsilon^i = \bigcup_{i=1}^{n} \overset{\circ}{J}_i \subseteq \overset{\circ}{A},$$

we may replace P_ϵ^i with $\overset{\circ}{P}{}_\epsilon^i$, so by inner measure monotonicity

$$m_i(A) - \epsilon < m_i\left(\bigcup_{i=1}^{n} \overset{\circ}{J}_i\right) = \sum_{i=1}^{n} m(\overset{\circ}{J}_i) \leq m_i(\overset{\circ}{A}).$$

The arbitrary choice of $\epsilon > 0$ guarantees

$$m_i(A) \leq m_i(\overset{\circ}{A}).$$

Moreover $\overset{\circ}{A} \subseteq A$, and again by monotonicity we obtain the opposite inequality, whence $m_i(\overset{\circ}{A}) = m_i(A)$.

The second statement is similar, and left as an exercise. ■

Theorem 8.1.1 (Characterisation of PJ measurability of bounded subsets of $\mathbb{R}$) *For any bounded set $A \subseteq \mathbb{R}$*

$$m_e(\bar{A}) - m_i(\overset{\circ}{A}) = m_e(\partial A).$$

In particular, a bounded set A is PJ measurable if and only if $m_e(\partial A) = 0$.

Proof Take two pluri-intervals $\bar{P}^e(A) = \bigcup_{i=1}^{n} \bar{I}_i$ and $\overset{\circ}{P}{}^i(A) = \bigcup_{k=1}^{m} \overset{\circ}{J}_k$, the former external and the latter internal to A. For $i = 1, \ldots, n$ denote

$$\overset{\circ}{J}_{k_i} = \left(\bar{I}_i \cap \bigcup_{k=1}^{m} \overset{\circ}{J}_k\right),$$

so $\bigcup_{i=1}^{n} \left(\bar{I}_i \setminus \overset{\circ}{J}_{k_i}\right)$ is the pluri-interval containing the frontier of A, hence external, and call it $P^e\left(\partial A\right)$. Then

$$m_e\big(P^e(\partial A)\big) = m_e\Big(\bigcup_{i=1}^{n} \bar{I}_i \setminus \overset{\circ}{J}_{k_i}\Big) = m_e\Big(\bigcup_{i=1}^{n} \bar{I}_i\Big) - m_i\Big(\bigcup_{k=1}^{m} \overset{\circ}{J}_k\Big)$$
$$\geq m_e(\bar{A}) - m_i(\overset{\circ}{A}).$$

Note the inequality holds for any pluri-interval containing ∂A, so

$$m_e(\partial A) \geq m_e(\bar{A}) - m_i(\overset{\circ}{A}).$$

For the converse, take $\epsilon > 0$ and

$$\bar{P}^e(A) = \bigcup_{i=1}^{n} \bar{I}_i \supseteq \bar{A} \quad \text{and} \quad \overset{\circ}{P}^i(A) = \bigcup_{k=1}^{m} \overset{\circ}{J}_k \subseteq \overset{\circ}{A},$$

so that

$$m_e(\bar{A}) > m_e(\bar{P}^e(A)) - \epsilon/2 \quad \text{and} \quad m_i(\overset{\circ}{A}) < m_i(\overset{\circ}{P}^i(A)) + \epsilon/2.$$

As before, for $i = 1, \ldots, n$ we take

$$\overset{\circ}{J}_{k_i} = \Big(\bar{I}_i \cap \bigcup_{k=1}^{m} \overset{\circ}{J}_k\Big),$$

so $\big(\bigcup_{i=1}^{n} \bar{I}_i \setminus \overset{\circ}{J}_{k_i}\big)$ is the pluri-interval $P^e\big(\partial A\big)$ containing the frontier of A. Then

$$m_e(\partial A) \leq m_e(\partial P^e(A)) = m_e(\bar{P}^e(A)) - m_i(\overset{\circ}{P}^i(A)) < m_e(\bar{A}) - m_i(\overset{\circ}{A}) + \epsilon.$$

Varying $\epsilon > 0$ shows

$$m_e(\partial A) \leq m_e(\bar{A}) - m_i(\overset{\circ}{A}).$$

The two inequalities give

$$m_e(\partial A) = m_e(\bar{A}) - m_i(\overset{\circ}{A}).$$

Hence A is PJ measurable iff $m_e(\partial A) = 0$. ■

From Theorem 8.1.1 we deduce that a bounded linear set A is PJ measurable iff the inner measure of $\overset{\circ}{A}$ and the outer measure of $\bar{A}$ coincide, i.e. the frontier $\partial A = (\bar{A} \setminus \overset{\circ}{A})$ has zero outer measure. This happens precisely when the frontier has empty interior. Consequently a set A has zero PJ measure iff $m_e(\bar{A}) = 0$, and A has positive

measure, i.e. $m_i(A) > 0$, iff $\overset{\circ}{A} \neq \emptyset$. Equivalently put, a PJ measurable set has positive measure iff it has interior points.

A careful inspection reveals that Peano-Jordan measure theory determines the measurability of a bounded set $A \subseteq \mathbb{R}$ by approximating the interior of A with open pluri-intervals, and the exterior with compact pluri-intervals. Now it becomes clear the frontier of A will always be involved in the computation of the outer measure and never in that of the inner measure. On the contrary, if we approximate from inside with compact sets and from outside with open sets, we might take points of ∂A that belong to A and exclude others. As we will see, this is the criterion used by the Lebesgue measure theory, which has the further advantage of approximating the measure of $A \subseteq \mathbb{R}$ using the Borel σ-algebra as family. This discussion implies the Lebesgue measure is subtler and more flexible than the Peano-Jordan measure, insofar as any bounded PJ measurable set is Lebesgue measurable. The converse is false, as explained by Proposition 8.1.2.

Remark 8.1.3 An unbounded set $A \subseteq \mathbb{R}$ is PJ measurable if, for every $n \in \mathbb{N}$,

$$A \cap [-n, n]$$

is PJ measurable. Then the criterion whereby a bounded set $A \subseteq \mathbb{R}$ is PJ measurable iff $m_e(\partial A) = 0$ can be extended to any subset A of $\mathbb{R}$.

In fact, for every $A \subseteq \mathbb{R}$

$$A \in \mathcal{A}_{pj}(\mathcal{I}) \Leftrightarrow m_e\big(\partial(A \cap [-n, n])\big) = 0 \ \ \forall n \in \mathbb{N}.$$

Note the family $\mathcal{A}_{pj}(\mathcal{I})$ of PJ measurable sets $A \subseteq \mathbb{R}$ is an algebra of subsets of $\mathbb{R}$:

- $m_e(\partial\mathbb{R}) = 0$ and $m_e(\partial\emptyset) = 0$ imply $\mathbb{R}, \emptyset \in \mathcal{A}_{pj}(\mathcal{I})$;
- $A \in \mathcal{A}_{pj}(\mathcal{I})$ implies $m_e(\partial A) = 0$, and since $\partial(A)^c = \partial A$, then $A^c \in \mathcal{A}_{pj}(\mathcal{I})$;
- for $A, B \in \mathcal{A}_{pj}(\mathcal{I})$ we have $m_e(\partial A) = 0$, $m_e(\partial B) = 0$ and

$$\begin{aligned} & \partial(A \cup B) \subseteq \partial A \cup \partial B \\ \Leftrightarrow \quad & m_e\big(\partial(A \cup B)\big) \leq m_e(\partial A) + m_e(\partial B) = 0 \\ \Leftrightarrow \quad & A \cup B \in \mathcal{A}_{pj}(\mathcal{I}). \end{aligned}$$

♣

Proposition 8.1.2 *The Peano-Jordan measure* $m_{pj} : \mathcal{A}_{pj}(\mathcal{I}) \mapsto [0, \infty]$ *is not completely additive.*

Proof It suffices to find a countable union of sets in $\mathcal{A}_{pj}(\mathcal{I})$ that is not PJ measurable.

We choose $A = [0, 1] \cap \mathbb{Q}$ (which we will show is Lebesgue measurable) and prove it is a countable union of sets with zero PJ measure. Call $\{q_i\}_{i \geq 1}$ the rationals in $[0, 1]$, so each $\{q_i\} = [q_i, q_i]$ is the frontier and closure of itself and hence has empty interior. Then

$$m_e(\partial\{q_i\}) = m_e([q_i, q_i]) - m_i(\emptyset) = 0.$$

Regarding the inner measure, A as a whole does not contain non-degenerate open intervals. In fact, by the density of $(\mathbb{R} \setminus \mathbb{Q})$ in $\mathbb{R}$, in every non-degenerate open interval (q_i, q_j), $q_i, q_j \in \mathbb{Q}$, for $i, j \in \mathbb{N}$ and $i \neq j$, there will always be points of A^c and points of A. Hence the only internal pluri-interval is the empty set. We conclude A has empty interior and

$$m_i(A) = m_i(\overset{\circ}{A}) = m_i(\emptyset) = 0.$$

On the other hand

$$m_e(A) = m_e(\bar{A}) = m_e(\overline{[0, 1] \cap \mathbb{Q}}) = m_e([0, 1]) = 1,$$

so $A = [0, 1] \cap \mathbb{Q}$ is not PJ measurable.

Therefore the PJ measure fails to measure $[0, 1] \cap \mathbb{Q}$: on one hand because the outer measure requires the closure, which does not belong to the set. On the other hand the PJ measure is not completely additive. ■

Now we shall prove a result relating the theory of integration (here, Riemann's) with measure theory (of Peano-Jordan). It clarifies the tight relationship between integration and measure theory for sets, two aspects that seem different at first sight, but are indeed opposite sides of the same coin. In the next theorem, in particular, we prove the equivalence between Riemann's second integrability criterion and the PJ measurability of sets with zero outer measure.

Theorem 8.1.2 (Riemann integrability and PJ measurability) *A bounded map $f : [a, b] \mapsto \mathbb{R}$ is Riemann integrable on $[a, b]$ if and only if, for every $\epsilon > 0$, the discontinuity set $D_\delta(f) = \left\{x \in [a, b] : \omega(f, x) \geq \sigma\right\}$ has zero PJ outer measure (i.e. zero PJ measure).*

Proof Suppose, first, f is bounded and $D_\delta(f)$ has outer measure zero.

Boundedness means there is an $M > 0$ such that $|f(x)| \leq M$ for every $x \in [a, b]$. Zero outer measure entails that, given $\epsilon > 0$, there exists a partition P of $[a, b]$ such that $\delta = \epsilon/(4M)$ is larger than the total length of the intervals in P, on whose interior the oscillation of f is at least

$$\sigma = \frac{\epsilon}{2(b - a)}.$$

Denote by

$$D_\delta(f) = \left\{x \in [a, b] : \omega(f, x) \geq \sigma = \frac{\epsilon}{2(b - a)}\right\}$$

the set of $x \in [a, b]$ at which the oscillation of f is bigger than or equal to σ. The intervals where that happens are $I_1, \ldots, I_n$. On the interior of the remaining ones $J_1, \ldots, J_k$ the map f varies less than $\sigma = \frac{\epsilon}{2(b-a)}$. Hence

$$\begin{aligned} S(f, P) - s(f, P) &= \sum_{i=1}^{n} \left[\Big(\sup_{x \in I_i} f(x) - \inf_{x \in I_i} f(x) \Big) m(I_i) \right] + \\ &\quad \sum_{j=1}^{k} \left[\Big(\sup_{x \in J_j} f(x) - \inf_{x \in J_j} f(x) \Big) m(J_j) \right] \\ &= \sum_{i=1}^{n} \big[\omega(f, I_i) m(I_i) \big] + \sum_{j=1}^{k} \big[\omega(f, J_j) m(J_j) \big] \\ &< 2M \cdot \frac{\epsilon}{4M} + \frac{\epsilon}{2(b-a)} (b-a) = \epsilon \end{aligned}$$

and f is integrable.

Vice versa, take f Riemann integrable. Given $\sigma > 0$ such that

$$m_e(D_\delta(f)) = m_e\left(\left\{ x \in [a, b] : \omega(f, x) \geq \sigma \right\} \right) = K > 0,$$

call $P = \{x_0, x_1, \ldots, x_{i-1}, x_i, \ldots, x_n\}$ a partition of $[a, b]$ associated to $\sigma > 0$, $\mathscr{I}$ the set of indices i such that $[x_{i-1}, x_i] \cap D_\delta(f) \neq \emptyset$ and $\mathscr{I}' \subseteq \mathscr{I}$ the indices i such that $(x_{i-1}, x_i) \cap D_\delta(f) \neq \emptyset$. For $i \in \mathscr{I}'$ we have $\omega(f, [x_{i-1}, x_i]) \geq \sigma$. As f is Riemann integrable, for the given $\sigma > 0$ we can find $\delta > 0$ greater than the total length of intervals, on whose interiors the oscillation of f is at least σ. Take $i \in \mathscr{I} \setminus \mathscr{I}'$, so at least one endpoint of $[x_{i-1}, x_i]$ meets $D_\delta(f)$. Refine P by adding to (x_{i-1}, x_i) two points $y_i, y_i' \in (x_{i-1}, x_i)$ such that $y_i < y_i'$,

$$|x_{i-1} - y_i| < \frac{\delta}{2n} \quad \text{and} \quad |y_i' - x_i| < \frac{\delta}{2n}.$$

For every $i \in \mathscr{I} \setminus \mathscr{I}'$ rename the intervals $[x_{i-1}, y_i]$ and $[y_i', x_i]$ by $J_1, \ldots, J_m$, where $m \leq 2n$. Note $[y_i, y_i'] \cap D_\delta(f) = \emptyset$, so

$$K \leq \sum_{i \in \mathscr{I}'} (x_i - x_{i-1}) + \sum_{j=1}^{m} m(J_j) \leq \sum_{i \in \mathscr{I}'} (x_i - x_{i-1}) + 2n \cdot \frac{\delta}{2n} = \sum_{i \in \mathscr{I}'} (x_i - x_{i-1}) + \delta.$$

Because of integrability, δ can be chosen as small as we want, so $K \leq \sum_{i \in \mathscr{I}'} (x_i - x_{i-1})$ and

$$S(f,P)-s(f,P)=\sum_{i=1}^{n}\omega\big(f,[x_{i-1},x_i]\big)(x_i-x_{i-1})$$
$$\geq\sum_{i\in\mathscr{I}'}\omega\big(f,[x_{i-1},x_i]\big)(x_i-x_{i-1})\geq K\sigma.$$

As this is true for any partition, we reach the contradiction that f is not Riemann integrable. Hence the claim. ■

Proposition 8.1.2 and Theorem 8.1.2 tell that the Dirichlet function is not Riemann integrable. On the other hand, the Cantor set is PJ measurable as shown below.

Example 8.1.2 (*PJ measure of the Cantor set*) Every C_n has measure $m(C_n)=(\frac{2}{3})^n$, $n=1,2,\ldots$, and is a finite union of bounded closed intervals, hence PJ measurable.

Given in fact $\epsilon>0$, there exist a pluri-interval P_ϵ^e and an $n'\in\mathbb{N}$ such that, for every $n\geq n'$, $C_n\subseteq P_\epsilon^e$. Then

$$m_e(C_n)\leq m_e(P_\epsilon^e)<\epsilon,$$

and therefore $m_{pj}(C)=0$. ★

Remark 8.1.4 (*Cardinality of $\mathcal{A}_{pj}(\mathcal{I})$, of Borel sets, and Lebesgue measure's generality*) Regarding the previous example we should observe that each Cantor subset is PJ measurable. In fact any subset of C has content 0 and hence PJ measure 0. As C has cardinality $\mathbf{c}$, the power set has cardinality $2^{\mathbf{c}}$. But as we shall see, the cardinality of Borel measurable sets is $\mathbf{c}$, so the PJ measure applies to a larger class of sets than Borel's measure. Put equivalently, there are more PJ measurable sets than Borel measurable sets.

Still, many Borel measurable sets are not PJ measurable, so the class of Borel measurable sets, albeit smaller, is not a subclass of PJ measurable sets. To fill the cardinality gap it suffices to add to Borel sets all the subsets of zero Borel measure sets, as we did for the subsets of zero measure in the Cantor set.

Borel was aware that it made sense to assign measure 0 to subsets of Cantor's ternary set. In fact he even observed that given three Borel sets $A\subseteq C\subseteq B$, with A, B having the same measure, then C must have that measure. He did not go beyond this simple remark nor explored the deep implications involved. It fell onto a young student, Henri Lebesgue, to grasp the consequences of that plain observation and the weight it would have in setting up a theory of measure and integration that worked for a broader class of sets.

Lebesgue's theory of integration is laid out in *"Leçons sur l'intégration et la recherche des fonctions primitives"*. In chapter III - Geometric definition of the integral, the author showed that the Riemann integral of non-negative bounded maps is just the Peano-Jordan measure of the region underneath the graph of f. But Lebesgue understood that he could not replace the PJ measure with Borel's, also because he would thus have reduced the number of measurable sets. He tried to define a measure that rendered measurable both the elements of the algebra $\mathcal{A}_{pj}(\mathcal{I})$ and those in the

Borel σ-algebra, and even more. There exists in fact Lebesgue measurable sets that are neither PJ- nor Borel measurable (in 1914 the Russian mathematician Suslin found examples).

In particular, Lebesgue decided his measure should satisfy three requirements:

1. there exist sets with non-zero measure;
2. equal sets have equal measure (translation-invariance);
3. the measure of a finite or countable sum of pairwise-disjoint sets equals the sum of the sets' measures (complete additivity of the measure).

Lebesgue was after the largest-possible class of sets for which his measure made sense.[1] The Lebesgue measure, like Borel's, is founded on the concept of **countable cover**. This bestows on the measure a very general character, also courtesy of the Heine-Borel theorem. Yet Lebesgue, in contrast to Borel, Peano and Jordan, in defining the measure dropped the critical invariance under set-difference, which the others had more or less explicitly included.[2]

For the Borel measure the property is part of the definition, while for Peano and Jordan it emerges from the following facts: if $A \subseteq [a, b]$ is PJ measurable, any pluri-interval contained in A

$$\overset{\circ}{P}{}^{i} = \bigcup_{k=1}^{m} \overset{\circ}{J}_k = \bigsqcup_{k=1}^{m} \overset{\circ}{J}_k$$

is a pairwise-disjoint finite union of intervals, whose complement $[a, b] \setminus A$ is another pairwise-disjoint finite union of intervals (admitting degenerate intervals). Hence

$$\begin{aligned} m_i\big(A \cap [a,b]\big) &= \sup\left\{ \sum_{k=1}^{m} m(J_k) : \left(\bigsqcup_{k=1}^{m} \overset{\circ}{J}_k \right) \subseteq (A \cap [a,b]),\ \overset{\circ}{J}_k \cap \overset{\circ}{J}_j = \emptyset, \forall k \neq j \right\} \\ &= \sup\left\{ (b-a) - \sum_{i=1}^{n} m(\bar{I}_i) : ([a,b] \setminus A) \subseteq \left(\bigcup_{i=1}^{n} \bar{I}_i \right),\ \overset{\circ}{I}_i \cap \overset{\circ}{I}_j = \emptyset, \forall i \neq j \right\} \\ &= (b-a) - \inf\left\{ \sum_{i=1}^{n} m(\bar{I}_i) : ([a,b] \setminus A) \subseteq \left(\bigcup_{i=1}^{n} \bar{I}_i \right),\ \overset{\circ}{I}_i \cap \overset{\circ}{I}_j = \emptyset, \forall i \neq j \right\} \\ &= (b-a) - m_e\big([a,b] \setminus A\big). \end{aligned} \tag{8.1.3}$$

To have $m_i\big(A \cap [a,b]\big) = m_e\big(A \cap [a,b]\big)$ we must impose $m_e\big([a,b] \setminus A\big) = (b-a) - m_e\big(A \cap [a,b]\big)$, namely complement-invariance.

---

[1] If one additionally imposes that the Lebesgue measure is defined for every subset A of $\mathbb{R}$, using the continuum hypothesis (Ulam's theorem) it can be proved it is not possible to satisfy the other three properties simultaneously. If we wish to keep requirements 1.–3. we will not be able to measure every subset of $\mathbb{R}$; and if we insist on measuring all subsets of $\mathbb{R}$, then we need to weaken one other property, for instance replacing countable additivity with countable subadditivity. But 1., 2., 3. cannot be renounced by trivial geometric considerations, so the class of Lebesgue measurable sets will have to be a proper subclass of $\mathcal{P}(\mathbb{R})$.

[2] If A, B are PJ (or Borel) measurable and $A \subseteq B$, then $B \setminus A$ is PJ measurable (Borel measurable). The difference's measure equals the measure of B minus the measure of $B \cap A$.

The Lebesgue measure is more complicated, because the complement of a pairwise-disjoint countable union of intervals is not necessarily a pairwise-disjoint countable union of intervals, take e.g. Cantor's ternary set. For this reason Lebesgue defined the inner measure through the complement's outer measure, thus providing a definition of inner content that is more general than the inner measure of PJ theory. ♣

The remarkable Examples 2.3.1, 2.3.2 give us nowhere dense, non-PJ measurable sets with non-zero measure.

8.2 Completely Additive Set Functions

Up to now we have supposed the length function was only finitely additive. We have thus ignored the Heine–Pincherle–Borel theorem's full force for compact sets, which ensures that we may replace (even countable) open covers by finite ones, and with it the many advantages of passing from the infinite to the finite. Since the measure of a generic bounded interval I (open, half-open, or closed) is always the difference of its endpoints, we may suppose I closed and exploit the above theorem to prove that the length function, fundamental in measure and integration theory, is completely additive as well.

To generalise, we may then consider completely additive functions defined on some class $\mathcal{C}$ of subsets in $X \neq \emptyset$.

Definition 8.2.1 (*Completely additive measure*) A set function $m : \mathcal{C} \mapsto [0, +\infty]$ is a completely additive measure on a collection $\mathcal{C}$ of subsets of a non-empty set X when:

1. $\emptyset \in \mathcal{C}$ with $m(\emptyset) = 0$ (the empty set is null);
2. given pairwise-disjoint $A_1, A_2, \ldots \in \mathcal{C}$ and $A = \bigsqcup_{i=1}^{\infty} A_i \in \mathcal{C}$,

$$m(A) = \sum_{i=1}^{\infty} m(A_i).$$

◇

Such a measure is a set function, so it is non-negative, i.e. $m(A) \geq 0$ for every $A \in \mathcal{C}$ (measure's non-negativity).
From this moment onwards a measure will implicitly be completely additive, unless we say otherwise.

Remark 8.2.1 Suppose $\mathcal{C}$ is a collection of subsets of a set X and the set function $m : \mathcal{C} \mapsto [0, \infty]$ is a measure on $\mathcal{C}$. Then m, besides the properties of a measure, enjoys finite additivity only. Additional features of m might come from certain hypotheses made on $\mathcal{C}$, as in the next example.

At any rate the class $\mathcal{C}$ over which m is given (Definition 8.1.3) could just be a semi-algebra, on which a countably additive set function is easily defined (see the next example). Whenever more general sets need to be measured one can always pass to the σ-algebra generated by the semi-algebra (see Proposition 7.1.4). ♣

Definition 8.2.2 (*Countable covers*) A countable cover of set $A \subseteq X$ is a countable family $\{A_i\}_{i\geq 1}$ of subsets of X whose union contains A

$$A \subseteq \bigcup_{i=1}^{\infty} A_i .$$

The A_i are chosen so that any point of A lies in the interior of one subset at least. ◇

Example 8.2.1 (*Length's complete additivity on the semi-algebra $\mathcal{I}$*) Going back to the length function in Example 8.1.1, now we wish to prove, with the help of the Heine–Pincherle–Borel theorem, that the length is completely additive and subadditive.

To verify the other relevant properties we must first illustrate the properties of intervals with finite or infinite measure:

(a) in the first case ($m(I) < \infty$), for every $J \in \big(I \cap \mathcal{I}\big)$ clearly $m(J) < \infty$, so the length function restricted to $\big(I \cap \mathcal{I}\big)$ is totally finite;
(b) in the second case observe $m : (I \cap \mathcal{I}) \mapsto [0, \infty]$ is a σ-finite function, for $\mathbb{R}$ can be expressed as a countable disjoint union of intervals $\mathbb{R} = \bigsqcup_{n\in\mathbb{Z}}(n, n+1]$, each of finite length. So for an unbounded I, say (a, ∞), we have

$$I = I \cap \mathbb{R} = \bigcup_{n\in\mathbb{Z}} \big(I \cap (n, n+1]\big)$$
$$\Leftrightarrow m(I) = m\bigg(\bigcup_{n\in\mathbb{Z}} \big(I \cap (n, n+1]\big)\bigg) = \sum_{n\in\mathbb{Z}} m\big(I \cap (n, n+1]\big).$$

In this light the length of a generic interval always equals the sum of its disjoint subintervals' lengths, each of which is finite.

1. Concerning countable subadditivity, in the case of a bounded interval we may consider $I = [a, b]$ without loss of generality. Let $\{I_i\}_{i\geq 1}$ be a family in $\mathcal{I}$ that covers I:

$$I = \bigcup_{i=1}^{\infty} I_i .$$

Obviously each I_i is bounded. Given $\epsilon > 0$, let J_i be an open interval such that, for every $i \in \mathbb{N}$, $I_i \subseteq J_i$ with $m(J_i) = m(I_i) + \epsilon/2^i$. The sequence $\{J_i\}_{i\geq 1}$ is an open cover of I, and

$$I = \bigcup_{i=1}^{\infty} I_i \subseteq \bigcup_{i=1}^{\infty} J_i.$$

I being compact, by the Heine-Borel theorem there is a finite subcover

$$I = \bigcup_{i=1}^{n} J_i.$$

Finite subadditivity implies

$$\begin{aligned} m(I) = b - a &\leq \sum_{i=1}^{n} m(J_i) = \sum_{i=1}^{n} m(I_i) + \sum_{i=1}^{n} \frac{\epsilon}{2^i} \\ &\leq \sum_{i=1}^{\infty} m(I_i) + \sum_{i=1}^{\infty} \frac{\epsilon}{2^i} = \sum_{i=1}^{\infty} m(I_i) + \epsilon; \end{aligned}$$

and the claim follows, since ϵ is arbitrary.

2. To prove countable additivity when $I = [a, b]$ is bounded, it suffices to show $m(I) \geq \sum_{i=1}^{\infty} m(I_i)$. Take a pairwise-disjoint sequence $\{I_i\}_{i \geq 1}$ such that $I = \bigcup_{i=1}^{\infty} I_i$. Since $\bigcup_{i=1}^{n} I_i \subseteq I$ for n finite, we can apply finite additivity and monotonicity to get

$$\sum_{i=1}^{n} m(I_i) = m\left(\bigcup_{i=1}^{n} I_i \right) \leq m(I)$$

for each n. Letting $n \to \infty$ we obtain

$$\sum_{i=1}^{\infty} m(I_i) \leq m(I).$$

3. For the countable additivity when I is unbounded, suppose $I = \bigcup_{i=1}^{\infty} I_i$ with the I_i pairwise disjoint. As the length on $\mathbb{R}$ is σ-finite,

$$m(I) = \sum_{n \in \mathbb{Z}} m\big(I \cap (n, n+1]\big). \tag{8.2.1}$$

As $\big[I \cap (n, n+1]\big]$ is bounded,

$$I \cap (n, n+1] = \left(\bigcup_{i=1}^{\infty} I_i\right) \cap (n, n+1]$$
$$= \bigcup_{i=1}^{\infty} \big(I_i \cap (n, n+1]\big)$$
$$\Leftrightarrow m\big(I \cap (n, n+1]\big) = \underbrace{\sum_{i=1}^{\infty} m\big(I_i \cap (n, n+1]\big)}_{\text{property 2.}}. \tag{8.2.2}$$

Substituting the latter in (8.2.1), and keeping in account sums can be rearranged since the series have non-negative terms, we obtain

$$m(I) = \sum_{n\in\mathbb{Z}} \left[\sum_{i=1}^{\infty} m\big(I_i \cap (n, n+1]\big)\right] = \sum_{i=1}^{\infty} \left[\sum_{n\in\mathbb{Z}} m\big(I_i \cap (n, n+1]\big)\right]$$
$$= \sum_{i=1}^{\infty} \left[m\left(I_i \cap \bigcup_{n\in\mathbb{Z}} (n, n+1]\right)\right] = \sum_{i=1}^{\infty} \Big[m\big(I_i \cap \mathbb{R}\big)\Big] = \sum_{i=1}^{\infty} m(I_i),$$

whence the claim.

4. When covers are not disjoint, countable additivity becomes countable subadditivity. Take $I = \bigcup_{i=1}^{\infty} I_i$, with the I_i not pairwise disjoint:

$$m(I) = \sum_{n\in\mathbb{Z}} m\big(I \cap (n, n+1]\big)$$
$$\leq \sum_{n\in\mathbb{Z}} \left[\sum_{i=1}^{\infty} m\big(I_i \cap (n, n+1]\big)\right]$$
$$= \sum_{i=1}^{\infty} \left[\underbrace{\sum_{n\in\mathbb{Z}} m\big(I_i \cap (n, n+1]\big)}_{=m(I_i)}\right] = \sum_{i=1}^{\infty} m(I_i),$$

and we are done.

Note countable additivity implies finite additivity if $m(\emptyset) = 0$. In fact for $I = \bigcup_{i=1}^{\infty} I_i$ with $I_i \cap I_j = \emptyset$ and $I_i = \emptyset$ for every $i \geq n$, we have

$$m(I) = \sum_{i=1}^{\infty} m(I_i) = \sum_{i=1}^{n} m(I_i).$$

A similar argument goes for countable subadditivity. ★

So, the length function on the class $\mathcal{I}$ of real intervals is a good candidate to be a completely additive measure on $\mathcal{I}$. Yet the sets one often deals with are not mere intervals, but sometimes very complicated sets of points A on the number line.

A first attempt to gauge the size of a set $A \subseteq \mathbb{R}$ is to cover it with a suitable union of intervals and obtain $m(A)$ by measuring the covering intervals by the length. In this way we have a new set function on the class $\mathcal{F}(\mathcal{I})$.

Suppose for starters we wish to measure finite unions $A \subseteq \mathbb{R}$ of pairwise-disjoint intervals. The set function is defined on the algebra

$$\mathcal{A}^+(\mathcal{I}) = \left\{ A \subseteq \mathbb{R} : A = \bigsqcup_{i=1}^{n} I_i, \text{ and } I_i \in \mathcal{I} \right\}.$$

This extends the length to $\mathcal{A}^+(\mathcal{I})$, and we call this set function

$$m' : \mathcal{A}^+(\mathcal{I}) \mapsto [0, \infty].$$

Clearly $\mathcal{I} \subseteq \mathcal{A}^+(\mathcal{I})$, and for every $A \in \mathcal{I}$

$$m'(A) = m(A).$$

Inspired by this argument let us examine the extended length over $\mathcal{A}^+(\mathcal{I})$.

Example 8.2.2 (*Length's properties over the algebra* $\mathcal{A}^+(\mathcal{I})$) The set function $m' : \mathcal{A}^+(\mathcal{I}) \mapsto [0, \infty]$

$$m'(A) = \sum_{i=1}^{n} m(I_i),$$

where $A \in \mathcal{A}^+(\mathcal{I})$ and m is the length of Examples 8.1.1, 8.2.1, has the following properties.

1. $m'(I) = m(I), \forall I \in \mathcal{I}$, meaning m' is an extension of m, and trivially $m'(\emptyset) = m(\emptyset) = 0$ and $m'(\{x\}) = m(\{x\}) = m([x, x]) = 0$.
2. m' is finitely additive. Let $A \in \mathcal{A}^+(\mathcal{I})$ be such that $A = \bigsqcup_{i=1}^{n} A_i$, with $A_i \in \mathcal{A}^+(\mathcal{I})$. Clearly $A_i = \bigsqcup_{j_i=1}^{n_i} I_{j_i}$ for every $i = 1, \ldots, n$, with $I_{j_i} \in \mathcal{I}$, and $A = \bigsqcup_{k=1}^{p} J_k$ for $J_k \in \mathcal{I}$ for every $k = 1, \ldots, p$. Hence for every k

$$J_k = \bigsqcup_{i=1}^{n} \bigsqcup_{j_i=1}^{n_i} (I_{j_i} \cap J_k)$$

and for every i

$$A_i = \bigsqcup_{k=1}^{p} \bigsqcup_{j_i=1}^{n_i} (I_{j_i} \cap J_k).$$

The finite additivity of m and the definition of m' give

$$
\begin{aligned}
m'(A) &= \sum_{k=1}^{p} m(J_k) = \sum_{k=1}^{p} m\Big(\bigsqcup_{i=1}^{n}\bigsqcup_{j_i=1}^{n_i}(I_{j_i}\cap J_k)\Big)\\
&= \sum_{k=1}^{p}\sum_{i=1}^{n}\sum_{j_i=1}^{n_i} m\big(I_{j_i}\cap J_k\big) = \sum_{i=1}^{n}\Big(\sum_{k=1}^{p}\sum_{j_i=1}^{n_i} m\big(I_{j_i}\cap J_k\big)\Big)\\
&= \sum_{i=1}^{n} m'\Big(\bigsqcup_{k=1}^{p}\bigsqcup_{j_i=1}^{n_i}\big(I_{j_i}\cap J_k\big)\Big) = \sum_{i=1}^{n} m'(A_i).
\end{aligned}
$$

3. We claim m' is monotone. Take $A_1, A_2 \in \mathcal{A}^+(\mathcal{I})$ such that $A_1 \subseteq A_2$, so $A_1 = \bigsqcup_{i_1=1}^{n_1} I_{i_1}$ and $A_2 = \bigsqcup_{i_2=1}^{n_2} I_{i_2}$. Using the monotonicity of m and the definition of m',

$$
m'(A_1) = \sum_{i_1=1}^{n_1} m(I_{i_1}) \le \sum_{i_2=1}^{n_2} m(I_{i_2}) = m'(A_2).
$$

4. Let us show m' is completely additive. Take $A \in \mathcal{A}^+(\mathcal{I})$ such that $A = \bigsqcup_{i=1}^{\infty} A_i$, with $A_i \in \mathcal{A}^+(\mathcal{I})$ for every $i \in \mathbb{N}$. Clearly $A_i = \bigsqcup_{j_i=1}^{n_i} I_{j_i}$ for every $i \in \mathbb{N}$, with $I_{j_i} \in \mathcal{I}$, and $A = \bigsqcup_{k=1}^{p} J_k$, where $J_k \in \mathcal{I}$ for every $k = 1, \ldots, p$. So for every k

$$
J_k = \bigsqcup_{i=1}^{\infty}\bigsqcup_{j_i=1}^{n_i}(I_{j_i}\cap J_k)
$$

and for every i

$$
A_i = \bigsqcup_{k=1}^{p}\bigsqcup_{j_i=1}^{n_i}(I_{j_i}\cap J_k).
$$

The countable additivity of m and the definition of m' force

$$
\begin{aligned}
m'(A) &= \sum_{k=1}^{p} m(J_k) = \sum_{k=1}^{p} m\Big(\bigsqcup_{i=1}^{\infty}\bigsqcup_{j_i=1}^{n_i}(I_{j_i}\cap J_k)\Big)\\
&= \sum_{k=1}^{p}\sum_{i=1}^{\infty}\sum_{j_i=1}^{n_i} m\big(I_{j_i}\cap J_k\big) = \sum_{i=1}^{\infty}\Big(\sum_{k=1}^{p}\sum_{j_i=1}^{n_i} m\big(I_{j_i}\cap J_k\big)\Big)\\
&= \sum_{i=1}^{\infty} m'\Big(\bigsqcup_{k=1}^{p}\bigsqcup_{j_i=1}^{n_i}\big(I_{j_i}\cap J_k\big)\Big) = \sum_{i=1}^{\infty} m'(A_i).
\end{aligned}
$$

5. Complete subadditivity: choose $A \in \mathcal{A}^+(\mathcal{I})$ such that $A \subseteq \bigsqcup_{i=1}^{\infty} A_i$, with $A_i \in \mathcal{A}^+(\mathcal{I})$. Define

$$B_1 = A_1 \cap A,\, B_2 = A_2 \cap A, \ldots, B_i = A_i \cap A, \ldots,$$

so $B_i \cap B_j = \emptyset$ when $i \neq j$, $B_i \in \mathcal{A}^+(\mathcal{I})$ and

$$A = \bigcup_{i=1}^{\infty} B_i \subseteq \bigcup_{i=1}^{\infty} A_i.$$

As before, $B_i = \bigsqcup_{j_i=1}^{n_i} (I_{j_i} \cap A)$ for every $i \in \mathbb{N}$, where $I_{j_i} \in \mathcal{I}$, and $A = \bigsqcup_{k=1}^{p} J_k$, where $J_k \in \mathcal{I}$ for every $k = 1, \ldots, p$. Therefore for each k

$$J_k = \bigsqcup_{i=1}^{\infty} \bigsqcup_{j_i=1}^{n_i} \big((I_{j_i} \cap A) \cap J_k\big)$$

and each i

$$B_i = \bigsqcup_{k=1}^{p} \bigsqcup_{j_i=1}^{n_i} \big((I_{j_i} \cap A) \cap J_k\big) \text{ and } A_i = \bigsqcup_{k=1}^{p} \bigsqcup_{j_i=1}^{n_i} \big(I_{j_i} \cap J_k\big).$$

At last, the definition of m', the complete additivity, and monotonicity of m tell us

$$\begin{aligned}
m'(A) &= \sum_{k=1}^{p} m(J_k) = \sum_{k=1}^{p} m\Big(\bigsqcup_{i=1}^{\infty} \bigsqcup_{j_i=1}^{n_i} (I_{j_i} \cap J_k \cap A)\Big) \\
&= \sum_{k=1}^{p} \sum_{i=1}^{\infty} \sum_{j_i=1}^{n_i} m(I_{j_i} \cap J_k \cap A) = \sum_{i=1}^{\infty} \Big(\sum_{k=1}^{p} \sum_{j_i=1}^{n_i} \underbrace{m\big(I_{j_i} \cap J_k \cap A\big)}_{\leq m(I_{j_i} \cap J_k)} \Big) \\
&\leq \sum_{i=1}^{\infty} \Big(\sum_{k=1}^{p} \sum_{j_i=1}^{n_i} m\big(I_{j_i} \cap J_k\big) \Big) = \sum_{i=1}^{\infty} m'\Big(\bigsqcup_{k=1}^{p} \bigsqcup_{j_i=1}^{n_i} \big(I_{j_i} \cap J_k\big) \Big) \\
&= \sum_{i=1}^{\infty} m'(A_i).
\end{aligned}$$

★

We know m' is completely additive and $m'(\emptyset) = 0$. Hence m' is a measure on $\mathcal{A}(\mathcal{I})$. As $\forall I \in \mathcal{I}, m(I) = m'(I)$, moreover, the measure $m' : \mathcal{A}(\mathcal{I}) \mapsto [0, \infty]$ extends $m : \mathcal{I} \mapsto [0, \infty]$. The length measure m, initially defined on the semi-algebra $\mathcal{I}$ of real intervals, can therefore be extended to a measure m', on the algebra $\mathcal{A}(\mathcal{I})$ generated by $\mathcal{I}$, having the same properties as m.

The following theorem proves that the extension of a measure on a generic semi-algebra $\mathcal{C}$ to the algebra $\mathcal{A}(\mathcal{C})$ is unique.

Theorem 8.2.1 (Extending a measure from a semi-algebra to an algebra) *Let $\mathcal{C}$ be a semi-algebra of subsets in X and m_1 a measure on $\mathcal{C}$. Then the set function $m_2 : \mathcal{A}(\mathcal{C}) \mapsto [0, \infty]$ such that $m_2(A) = m_1(A)$ for every $A \in \mathcal{C}$ is a measure on $\mathcal{A}(\mathcal{C})$ and it is unique.*

Proof By definition m_1 is a measure, so $m_2(\emptyset) = m_1(\emptyset) = 0$. Let us show m_2 is completely additive. Take $A \in \mathcal{A}(\mathcal{C})$ such that $A = \bigsqcup_{i=1}^{\infty} A_i$, with $A_i \in \mathcal{A}(\mathcal{C})$ for every $i \in \mathbb{N}$, so $A_i = \bigsqcup_{j=1}^{n_i} C_j^i$ with $C_j^i \in \mathcal{C}$. As A belongs to the algebra generated by $\mathcal{C}$, it can be written as pairwise-disjoint finite union over $\mathcal{C}$, i.e. $A = \bigsqcup_{k=1}^{m} D_k$, so for every k

$$D_k = D_k \cap \left(\bigsqcup_{i=1}^{\infty} A_i \right) = D_k \cap \left(\bigsqcup_{i=1}^{\infty} \bigsqcup_{j=1}^{n_i} C_j^i \right) = \bigsqcup_{i=1}^{\infty} \bigsqcup_{j=1}^{n_i} (D_k \cap C_j^i),$$

and

$$A_i = A \cap A_i = \left(\bigsqcup_{k=1}^{m} D_k \right) \cap \left(\bigsqcup_{j=1}^{n_i} C_j^i \right) = \bigsqcup_{k=1}^{m} \bigsqcup_{j=1}^{n_i} (D_k \cap C_j^i).$$

Using the countable additivity of m_1 and the definition of m_2,

$$\begin{aligned} m_2(A) &= m_2\left(\bigsqcup_{k=1}^{m} D_k \right) = \sum_{k=1}^{m} m_1(D_k) = \sum_{k=1}^{m} m_1\left(\bigsqcup_{i=1}^{\infty} \bigsqcup_{j=1}^{n_i} (D_k \cap C_j^i) \right) \\ &= \sum_{k=1}^{m} \sum_{i=1}^{\infty} \sum_{j=1}^{n_i} m_1(D_k \cap C_j^i) = \sum_{i=1}^{\infty} \left(\sum_{k=1}^{m} \sum_{j=1}^{n_i} m_1(D_k \cap C_j^i) \right) \\ &= \sum_{i=1}^{\infty} m_2\left(\bigsqcup_{k=1}^{m} \bigsqcup_{j=1}^{n_i} (D_k \cap C_j^i) \right) = \sum_{i=1}^{\infty} m_2(A_i). \end{aligned}$$

Hence m_2 is a measure on $\mathcal{A}(\mathcal{C})$.

Now we establish its uniqueness. Take another measure m_3 on $\mathcal{A}(\mathcal{C})$ such that for every $A \in \mathcal{C}$, $m_3(A) = m_1(A)$. The claim is $m_3(A) = m_2(A)$ for every $A \in \mathcal{A}(\mathcal{C})$. Let $A \in \mathcal{A}(\mathcal{C})$, so A is representable as finite disjoint union of elements C_i in $\mathcal{C}$, i.e. $A = \bigsqcup_{i=1}^{n} C_i$. Then $m_3(A) = m_3\big(\bigsqcup_{i=1}^{n} C_i \big)$. But m_3 is a measure so finitely additive

$$m_3(A) = m_3\Big(\bigsqcup_{i=1}^{n} C_i \Big) = \sum_{i=1}^{n} m_3(C_i) = \sum_{i=1}^{n} m_1(C_i).$$

As $m_1(C_i) = m_2(C_i)$ for every $C_i \in \mathcal{C}$,

$$m_3(A) = \sum_{i=1}^{n} m_3(C_i) = \sum_{i=1}^{n} m_1(C_i) = m_2(A),$$

and the proof ends because $A \in \mathcal{A}(\mathcal{C})$ is arbitrary. ■

Next, we look at properties important to characterise complete additivity for set functions on an algebra $\mathcal{A}(\mathcal{C})$.

Proposition 8.2.1 (Properties of set functions over an algebra) *Let $\mathcal{A}$ be an algebra of subsets in X, $m : \mathcal{A} \mapsto [0, +\infty]$ a finitely additive set function. Then*

1. *m is monotone;*
2. *$m(B \setminus A) = m(B) - m(A)$ for any $B \in \mathcal{A}$, $m(B) < \infty$, and every $A \in (\mathcal{A} \cap B)$ (subtractivity), and $m(\emptyset) = 0$.*

Proof 1. As $B = A \cup (B \backslash A)$ and since m is finitely additive,

$$m(B) = m(A) + m(B \backslash A) \Rightarrow m(B) \geq m(A).$$

2. From the above and $m(B) < \infty$ we have

$$m(A) \leq m(B) < \infty \quad \text{and} \quad m(A) + m(B \backslash A) < \infty.$$

Hence $m(B)$, $m(A)$ and $m(B \backslash A)$ are finite and non-negative, so

$$m(B) = m(A) + m(B \backslash A) \Leftrightarrow m(B) - m(A) = m(B \backslash A).$$

If, further, $A = B$, then $m(\emptyset) = m(B \setminus A) = 0$.

■

Theorem 8.2.2 (1st characterisation of complete additivity for set functions over an algebra) *Let $\mathcal{A}$ be an algebra of subsets of X and $m : \mathcal{A} \mapsto [0, +\infty]$ a set function such that $m(\emptyset) = 0$. Then m is countably additive if and only if m is finitely additive and countably subadditive.*

Proof ($\Rightarrow$) Let m be countably additive and take $A = \bigsqcup_{i=1}^{n} A_i$, $A_i \in \mathcal{A}$ for every $i = 1, \ldots, n$. Define $\forall i > n$, $A_i = \emptyset$, so $A = \bigsqcup_{i=1}^{\infty} A_i$. Since $m(\emptyset) = 0$, countable additivity implies

$$m(A) = \sum_{i=1}^{\infty} m(A_i) = \sum_{i=1}^{n} m(A_i).$$

By finite additivity we obtain monotonicity (item *1.*, Proposition 8.2.1).

Take now $A \subseteq \bigcup_{i=1}^{\infty} A_i$, where A and each $A_i \in \mathcal{A}$, with $A_i \cap A_j \neq \emptyset$, for every $i \neq j$. Put $B_1 = A_1$, $B_2 = A_2 \backslash A_1, \ldots, B_i = A_i \backslash \big(\bigcup_{j=1}^{i-1} A_j \big), \ldots$, with $B_i \cap B_j = \emptyset$ and $\forall i$, $B_i \in \mathcal{A}$ and $B_i \subseteq A_i$. By monotonicity

$$m\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} m(B_i) = \sum_{i=1}^{\infty} m\Bigg(\underbrace{A_i \cap \left(\bigcup_{j=1}^{i-1} A_j\right)^c}_{\leq A_i}\Bigg) \leq \sum_{i=1}^{\infty} m(A_i),$$

and

$$A = \bigcup_{i=1}^{\infty}(A \cap A_i) \subseteq \bigcup_{i=1}^{\infty} A_i$$
$$\Leftrightarrow m(A) = m\left(\bigcup_{i=1}^{\infty}(A \cap A_i)\right) \leq m\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} m(A_i).$$

($\Leftarrow$) Take $A \in \mathcal{A}$, $A = \bigsqcup_{i=1}^{\infty} A_i$, $A_i \in \mathcal{A}$. As m is countably subadditive, $m(A) \leq \sum_{i=1}^{\infty} m(A_i)$. So it is enough to show $m(A) \geq \sum_{i=1}^{n} m(A_i)$ for every n. Observe $A = \bigsqcup_{i=1}^{\infty} A_i \Rightarrow \bigsqcup_{i=1}^{n} A_i \subseteq A$ and $\bigsqcup_{i=1}^{n} A_i \in \mathcal{A}, \forall n \in \mathbb{N}$. But m is finitely additive hence monotone, so for every $n \in \mathbb{N}$

$$m\left(\bigsqcup_{i=1}^{n} A_i\right) \leq m(A) \Leftrightarrow \sum_{i=1}^{n} m(A_i) \leq m(A)$$

and eventually $\sum_{i=1}^{\infty} m(A_i) \leq m(A)$. ■

Theorem 8.2.3 (2nd characterisation of complete additivity for set functions over an algebra) *Let $\mathcal{A}$ be an algebra of subsets of X and $m : \mathcal{A} \mapsto [0, +\infty]$ a set function such that $m(\emptyset) = 0$. Then m is countably additive if and only if m is finitely additive and*

$$m(A) = \lim_{n\to\infty} m(A_n)$$

for every $A = \bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$ where $\{A_n\}_{n\geq 1}$ is an increasing sequence in $\mathcal{A}$.

Proof ($\Rightarrow$) Suppose m countably additive. First, m is also finitely additive on $\mathcal{A}$. For the second claim take $A = \bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$ where $\{A_n\}_{n\geq 1}$ is an increasing sequence in $\mathcal{A}$. Define $B_1 = A_1$, $B_2 = A_2 \backslash A_1, \ldots, B_n = A_n \backslash \left(\bigcup_{i=1}^{n-1} A_i\right), \ldots$, with $B_i \cap B_j = \emptyset$ for every $i \neq j$, and for every $n \in \mathbb{N}$ $B_n \in \mathcal{A}$ and $B_n \subseteq A_n$. Moreover

$$A = \bigcup_{n=1}^{\infty} A_n = \bigsqcup_{n=1}^{\infty} B_n.$$

By countable additivity

$$m(A) = m\left(\bigsqcup_{n=1}^{\infty} B_n\right) = \lim_{k\to\infty} \sum_{n=1}^{k} m(B_n) = \lim_{k\to\infty} \left(m\left(\bigsqcup_{n=1}^{k} B_n\right)\right)$$
$$= \lim_{k\to\infty} \left(m\left(\bigcup_{n=1}^{k} A_n\right)\right) = \lim_{k\to\infty} m(A_k).$$

Conversely, suppose m is finitely additive and, for every $A \in \mathcal{A}$ such that $A = \bigsqcup_{n=1}^{\infty} A_n$, $\{A_n\}_{n\geq 1}$ increasing in $\mathcal{A}$, we have $m(A) = \lim_{n\to\infty} m(A_n)$. Take

$$A = \bigsqcup_{n=1}^{\infty} A_n = \bigcup_{k=1}^{\infty} \left(\bigsqcup_{n=1}^{k} A_n\right),$$

where $\bigsqcup_{n=1}^{k} A_n = B_k \in \mathcal{A}$ and $\{B_k\}_{k\geq 1}$ is increasing. Then

$$m(A) = \lim_{k\to\infty} m(B_k) = \lim_{k\to\infty} m\left(\bigsqcup_{n=1}^{k} A_n\right) = \lim_{k\to\infty} \left(\sum_{n=1}^{k} m(A_n)\right) = \sum_{n=1}^{\infty} m(A_n).$$

■

Theorem 8.2.4 (3rd characterisation of complete additivity for set functions over an algebra) *Let $\mathcal{A}$ be an algebra of subsets of X and $m : \mathcal{A} \mapsto [0, +\infty]$ a set function such that $m(\emptyset) = 0$, and $m(X) < \infty$. Then m is countably additive if and only if m is finitely additive and*

$$m(A) = \lim_{n\to\infty} m(A_n)$$

for every $A = \bigcap_{n=1}^{\infty} A_n \in \mathcal{A}$ where $\{A_n\}_{n\geq 1}$ is a decreasing sequence in $\mathcal{A}$.

Proof ($\Rightarrow$) Suppose m countably additive, so finitely additive. Let $A = \bigcap_{n=1}^{\infty} A_n \in \mathcal{A}$ be given by a decreasing sequence $\{A_n\}_{n\geq 1}$ in $\mathcal{A}$.

We can turn the decreasing sequence $\{A_n\}_{n\geq 1}$ in an increasing sequence as follows: $B_1 = X\backslash A_1$, $B_2 = X\backslash A_2, \ldots, B_n = X\backslash A_n, \ldots$, so $\forall n$, $B_n \in \mathcal{A}$ and $\{B_n\}_{n\geq 1}$ is increasing. By Theorem 8.2.3 and item *2* of Proposition 8.2.1,

$$m(X\backslash A) = \lim_{n\to\infty} m(B_n) = \lim_{n\to\infty} m(X\backslash A_n)$$
$$\Leftrightarrow \underbrace{m(X)}_{<\infty} - m(A) = \lim_{n\to\infty} [\underbrace{m(X)}_{<\infty} - m(A_n)] \Leftrightarrow m(A) = \lim_{n\to\infty} m(A_n).$$

($\Leftarrow$) Take m finitely additive, $A \in \mathcal{A}$ and $\{A_i\}_{i\geq 1}$ increasing and pairwise disjoint in $\mathcal{A}$ such that $A = \bigsqcup_{i=1}^{\infty} A_i$.

Define

$$A = \bigsqcup_{i=1}^{\infty} A_i = \bigcup_{n=1}^{\infty} \left(\bigsqcup_{i=1}^{n} A_i\right),$$

and write $\bigsqcup_{i=1}^{n} A_i = B_n \in \mathcal{A}$ for every $n \in \mathbb{N}$. Then $\{X \setminus B_n\}_{n\geq 1}$ is decreasing, and

$$X \backslash A = \bigcap_{n=1}^{\infty} X \setminus B_n.$$

All hypotheses holding,

$$m(X \setminus A) = \lim_{n\to\infty} m(X \setminus B_n).$$

By Proposition 8.2.1, *2.*, and since $m(X) < \infty$, we obtain

$$\begin{aligned} m(X\backslash A) &= \lim_{n\to\infty} m(X \setminus B_n) = \lim_{n\to\infty} m\left(X \backslash \bigsqcup_{i=1}^{n} A_i\right) \\ \Leftrightarrow \underbrace{m(X)}_{<\infty} - m(A) &= \underbrace{m(X)}_{<\infty} - \lim_{n\to\infty} m\left(\bigsqcup_{i=1}^{n} A_i\right) \\ \Leftrightarrow m(A) &= \lim_{n\to\infty} m\left(\bigsqcup_{i=1}^{n} A_i\right) = \lim_{n\to\infty} \sum_{i=1}^{n} m(A_i) = \sum_{i=1}^{\infty} m(A_i), \end{aligned}$$

proving m is completely additive.

■

One says the set function m is **continuous from below** on $\mathcal{A}$ if it fulfils Theorem 8.2.3, and **continuous from above** on $\mathcal{A}$ if it satisfies Theorem 8.2.4.

Chapter 9
The Lebesgue Measure



9.1 The Outer Measure

It was mentioned that to set up a theory of measure capable of measuring the greatest number of subsets in a given space, Lebesgue started with their outer measure, dropped the property of invariance under complements and as counterpart defined inner measures in terms of outer measures of complements.

Following in his footsteps we begin to lay out Lebesgue's theory from the definition of outer measure. Since it is immaterial for the measure m whether we use a semi-algebra or an algebra, we suppose m is a measure on an algebra $\mathcal{A}$ of sets in some X, and from there we introduce outer measures on X.

In particular, to define the outer measure of a set $A \subseteq X$ we shall use the familiar tool of countable covers. Here is the definition.

Definition 9.1.1 (*Outer measure of a set*) Let $m : \mathcal{A} \mapsto [0, \infty]$ be a measure[1] on an algebra $\mathcal{A}$ of subsets in a set X. We call outer measure induced by m the set function $m^* : \mathcal{P}(X) \mapsto [0, \infty]$ defined by

$$m^*(A) = \inf \left\{ \sum_{i=1}^{\infty} m(A_i) : A \subseteq \bigcup_{i=1}^{\infty} A_i, A_i \in \mathcal{A}, \forall i, A_i \cap A_j \neq \emptyset, \forall i \neq j \right\} \leq \infty$$

for every $A \subseteq X$. ◇

Remark 9.1.1 The set function m^* is well defined on the power set of X:

- the greatest lower bound in Definition 9.1.1 exists, for m is non-negative and hence bounded from below by 0;
- the collection $m^*(\cdot)$ is not empty because $X \in \mathcal{A} \subseteq \mathcal{P}(X)$, and any $A \subseteq X$ is covered by X. Therefore at least the value $m(X)$ exists in the collection $m^*(A)$,

[1] Hence, completely additive.

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S. Gentili, *Measure, Integration and a Primer on Probability Theory*,
UNITEXT 125, https://doi.org/10.1007/978-3-030-54940-4_9

from which one starts to find the inf of the measures of A: this equals ∞ if $m(A) = \infty$, and is finite if $m(A) < \infty$. ♣

The outer measure of $A \subseteq X$ is independent of the fact the cover consists of pairwise-disjoint or overlapping sets. This also follows from Proposition 6.2.2, and can be proved as follows.

Proposition 9.1.1 (Outer measure and cover) *The outer measure $m^*(A)$ of a set $A \subseteq X$ equals the greatest lower bound of all measures of disjoint covers of A:*

$$m^*(A) = \inf \left\{ \sum_{n=1}^{\infty} m(A'_n) : A \subseteq \bigsqcup_{n=1}^{\infty} A'_n,\ A'_n \in \mathcal{A}, \forall n \right\}.$$

Proof Let $\{A_i\}_{i\geq 1}$ be a cover of A such that $A_i \in \mathcal{A}$ for every i, $A_i \cap A_j \neq \emptyset$, $i \neq j$ and $A \subseteq \bigcup_{i=1}^{\infty} A_i \subseteq X$. There exists a corresponding sequence $\{A'_n\}_{n\geq 1}$ of pairwise-disjoint sets, where each $A'_n \in \mathcal{A}$ and

$$A \subseteq \bigsqcup_{n=1}^{\infty} A'_n = \bigcup_{i=1}^{\infty} A_i \subseteq X.$$

As m is a measure on $\mathcal{A}$, it is completely additive hence finitely additive, so monotone by Proposition 8.2.1. Then for every n there is an $i \in \mathbb{N}$ such that $m(A'_n) \leq m(A_i)$. By complete subadditivity and by the arbitrary choice of $\{A_i\}_{\geq 1}$ we deduce

$$m^*(A) \leq \sum_{n=1}^{\infty} m(A'_n) \leq \sum_{i=1}^{\infty} m(A_i) \leq \infty$$

for every cover of A, and so

$$\begin{aligned} m^*(A) &\leq \inf \left\{ \sum_{n=1}^{\infty} m(A'_n) : A \subseteq \bigsqcup_{n=1}^{\infty} A'_n,\ A'_n \in \mathcal{A}, \forall n \right\} \\ &\leq \inf \left\{ \sum_{i=1}^{\infty} m(A_i) : A \subseteq \bigcup_{i=1}^{\infty} A_i,\ A_i \in \mathcal{A}, \forall i \right\} = m^*(A). \end{aligned}$$

■

Proposition 9.1.2 (Outer measure's properties) *The set function $m^* : \mathcal{P}(X) \mapsto [0, +\infty]$, induced by measure $m : \mathcal{A} \mapsto [0, \infty]$, satisfies the following:*

1. *m^* extends $m : \mathcal{A} \mapsto [0, \infty]$:*

$$m^*(A) = m(A) \quad \forall A \in \mathcal{A};$$

2. *$m^*(\emptyset) = 0$ and $m^*(A) \geq 0, \forall A \subseteq X$;*
3. *m^* is monotone: $B \subseteq A \subseteq X \Rightarrow m^*(B) \leq m^*(A)$;*

4. *m^* is completely subadditive: if $A = \bigcup_{i=1}^{\infty} A_i \subseteq X$ with $A_i \subseteq X$ for every i and $A_i \cap A_j \neq \emptyset$, $i \neq j$, then $m^*(A) \leq \sum_{i=1}^{\infty} m^*(A_i)$.*

Proof 1. Any $A \in \mathcal{A}$ covers itself, and $m^*(A)$ is the greatest lower bound of the measures of countable covers of A, so $m^*(A) \leq m(A)$.
To see m^* extends m on $\mathcal{A}$ it suffices to prove the opposite inequality.
If $m^*(A) = \infty$ we are done. Let $m^*(A) < \infty$, so for $\epsilon > 0$ the quantity $m^*(A) + \epsilon$ is not the inf, and there exists a cover $\{A_i\}_{i\geq 1}$ such that

$$m^*(A) + \epsilon > \sum_{i=1}^{\infty} m(A_i). \tag{9.1.1}$$

Note $\left\{\bigcup_{i=1}^{n}(A_i \cap A)\right\}_{n\geq 1}$ is an increasing sequence with limit $A \in \mathcal{A}$, and each set belongs in $\mathcal{A}$. The function m is a measure, so completely additive, and $m(\emptyset) = 0$. By Proposition 8.2.4

$$\begin{aligned}\sum_{i=1}^{\infty} m(A_i) &= \lim_{n\to\infty} \sum_{i=1}^{n} m(A_i) = \lim_{n\to\infty} m\left(\bigcup_{i=1}^{n} A_i\right) \\ &= \lim_{n\to\infty} m\left(\bigcup_{i=1}^{n}(A_i \cap A)\right) = m\left(\bigcup_{i=1}^{\infty}(A_i \cap A)\right) = m(A). \end{aligned} \tag{9.1.2}$$

Combining (9.1.1) with (9.1.2),

$$m^*(A) + \epsilon > m(A).$$

As $\epsilon > 0$ is arbitrary, $m^*(A) \geq m(A)$.

2. From $\mathcal{A} \subseteq \mathcal{P}(X)$ and $m^*(A) = m(A)$ for every $A \in \mathcal{A}$, we infer $m^*(\emptyset) = m(\emptyset) = 0$. But m^* is a set function on $\mathcal{P}(X)$, as well, so non-negative by definition, i.e. $m^*(A) \geq 0$ for every $A \subseteq X$.
3. As for the monotonicity of m^*, take $B \subseteq A$. Let $\{A_i\}_{i\geq 1}, \{B_j\}_{j\geq 1}$ be covers of A, B, with $A_i, B_j \in \mathcal{A}$ for every i and j. For every $B_m \in \{B_j\}_{j\geq 1}$ there is an $A_n \in \{A_i\}_{i\geq 1}$ such that $B_m \subseteq A_n$. By monotonicity and complete additivity of m on $\mathcal{A}$,

$$\sum_{j=1}^{\infty} m(B_j) \leq \sum_{i=1}^{\infty} m(A_i).$$

The covers are arbitrary, so the above holds for the greatest lower bound too, and so

$$m^*(B) \leq m^*(A).$$

4. Take $A = \bigcup_{i=1}^{\infty} A_i$ with $A_i \in \mathcal{A}$ for every i. If $m^*(A_i) = \infty$ for some i, then $m^*(A) \leq \sum_{i=1}^{\infty} m^*(A_i) = \infty$. Suppose then $m^*(A_i) < \infty, \forall i$. Given $\epsilon > 0$ for every i there is a cover $\{A_j^i\}_{j \geq 1} \subseteq \mathcal{A}$ of A_i such that

$$\sum_{j=1}^{\infty} m(A_j^i) < m^*(A_i) + \frac{\epsilon}{2^i}.$$

But

$$A = \bigcup_{i=1}^{\infty} A_i \subseteq \bigcup_{i=1}^{\infty} \bigcup_{j=1}^{\infty} A_j^i,$$

so

$$m^*(A) \leq \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} m(A_j^i) < \sum_{i=1}^{\infty} m^*(A_i) + \sum_{i=1}^{\infty} \frac{\epsilon}{2^i} = \sum_{i=1}^{\infty} m^*(A_i) + \epsilon.$$

Once again by the arbitrary choice of $\epsilon > 0$, $m^*(A) \leq \sum_{i=1}^{\infty} m^*(A_i)$. ■

If we compare m with the outer measure m^* it induces, we immediately note that only the former is not defined on all subsets of X, but it is completely additive and subadditive. By Definition 8.2.1 and its consequences m^* is actually not a measure, for it lacks countable additivity on pairwise-disjoint sets. Let in fact $m : \mathcal{P}(\{0, 1\}) \mapsto \{0, 1\}$ be defined by

$$m(A) = \begin{cases} 0 \text{ if } A = \emptyset \\ 1 \text{ if } A \neq \emptyset. \end{cases}$$

It is easy to show this set function is an outer measure. Yet it is not countably additive, not even finitely additive. Take $A_1 = \{0\}$, $A_2 = \{1\}$, so

$$m(A_1 \cup A_2) = 1 \neq 2 = m(A_1) + m(A_2).$$

A set function m defined on $\mathcal{P}(X)$ satisfying properties *2., 3., 4.* of Proposition 9.1.2 is just an outer measure. Lebesgue's outer measure, apart from possessing the aforementioned properties, is further characterised by the fact it determines the outer extension of a set when we take the greatest lower bound of all countable covers.

With the next proposition we show that any other outer measure on $\mathcal{P}(X)$, extending m from $\mathcal{A}$, is always less than or equal to the outer measure m^* induced by m. In this sense the latter is the most efficient in detecting and measuring element external to sets of $\mathcal{P}(X)$. In other words m^* is the largest outer measure extending m from $\mathcal{A}$.

Proposition 9.1.3 (Comparing outer measures) *Let $m^* : \mathcal{P}(X) \mapsto [0, \infty]$ be the outer measure induced by $m : \mathcal{A} \mapsto [0, \infty]$. If μ^* is another outer measure on $\mathcal{P}(X)$ extending m, then*

$$m^*(A) \geq \mu^*(A)$$

for every $A \in \mathcal{A}$.

Proof Let $\mu^* : \mathcal{P}(X) \mapsto [0, \infty]$ be an outer measure on X such that $\mu^*(A) = m(A)$ for every $A \in \mathcal{A}$.

Given $A \subseteq X$, let $\{A_i\}_{i \geq 1} \subseteq \mathcal{A}$ be a cover of A, so by complete subadditivity

$$\mu^*(A) \leq \sum_{i=1}^{\infty} \mu^*(A_i) = \sum_{i=1}^{\infty} m(A_i).$$

As the inequality holds for any cover, it holds when taking the inf over all covers. But $A \subseteq X$ is arbitrary, so by definition of m^* we have $\mu^*(A) \leq m^*(A)$ for every $A \subseteq X$. ■

9.2 Carathéodory's Criterion of Measurability

Remark 8.1.4 mentioned that Lebesgue employed the outer measure of the complement of $A \subseteq [a, b]$ to define the inner measure of the bounded linear set A.

Specifically, Lebesgue defined the outer measure of $A \subseteq [a, b]$ by taking the infimum over all countable covers of A extracted from $\mathcal{A}([a, b] \cap \mathcal{I})$:

$$m^*\big(A \cap [a, b]\big) = \inf \left\{ \sum_{i=1}^{\infty} m(A_i) : A \subseteq \bigcup_{i=1}^{\infty} A_i,\ A_i \in \mathcal{A}\big([a, b] \cap \mathcal{I}\big), \forall i,\ A_i \cap A_j \neq \emptyset, \forall i \neq j \right\}.$$

From this he defined the inner measure of A as

$$m_*\big(A \cap [a, b]\big) = m\big([a, b]\big) - m^*\big(A^c \cap [a, b]\big).$$

Finally he declared a bounded linear set A measurable whenever

$$\begin{aligned} & m^*\big(A \cap [a, b]\big) = m_*\big(A \cap [a, b]\big) \\ \Leftrightarrow & m^*\big(A \cap [a, b]\big) = m\big([a, b]\big) - m^*\big(A^c \cap [a, b]\big) \\ \Leftrightarrow & m^*\big([a, b]\big) = m^*\big(A \cap [a, b]\big) + m^*\big(A^c \cap [a, b]\big). \end{aligned} \tag{9.2.1}$$

It is not hard to see that the measurability of $A \subseteq [a, b]$ is tested against the measure of a closed interval $[a, b]$ that contains it. Any such definition of measurability has the advantage of being intuitive and immediate, but does not lend itself to generalisations and more abstract contexts.

At this juncture Constantin Carathéodory[2] (1873–1950) stepped on the stage. He had the brilliant idea of testing the measurability of $A \subseteq \mathbb{R}$ using the outer measures of any subset of $\mathbb{R}$. Since any subset $Y \subset \mathbb{R}$ admits an outer measure, Carathéodory replaced the measure of $[a, b]$ with the outer measure of an arbitrary test set $Y \subseteq \mathbb{R}$, so condition (9.2.1) reads: $\forall Y \subseteq \mathbb{R}$

$$m^*(Y) = m^*\big(A \cap Y\big) + m^*\big(A^c \cap Y\big).$$

Carathéodory's definition, albeit more concise and versatile for proofs, is not as intuitive as Lebesgue's. That is why it is not so easy to explain how Carathéodory's definition captures the notion of measurability in its generality. A possible explanation is that certain sets are difficult to measure for their frontier is very irregular, and Carathéodory's approach permits to discard them because his criterion warrants a set's measurability if we can use it to divide any other set in two without adding further irregularities. Every portion of a measurable set's frontier must therefore be reasonably regular, so that the outer measure of the test set being split is equal to the sum of the outer measures of the two pieces. At any rate, any doubt disappears once we recall the outer measure of $Y \subseteq \mathbb{R}$ is completely determined by the measures of the cover's **intervals**. Without further ado let us present Carathéodory's proposal.

Definition 9.2.1 (*Carathéodory's measurability*) A set $A \subseteq X$ is Lebesgue measurable if, for every $Y \subseteq X$,

$$m^*(Y) = m^*(A \cap Y) + m^*(A^c \cap Y). \tag{9.2.2}$$

If A is Lebesgue measurable we call it L-measurable for short, and we indicate with

$$\mathscr{L} = \big\{A \subseteq X : \forall Y \subseteq X, m^*(Y) = m^*(A \cap Y) + m^*(A^c \cap Y)\big\}$$

the class of L-measurable subsets of X. ◇

The outer measure m^*, defined on $\mathcal{P}(X)$, is completely subadditive, so for every $Y \in \mathcal{P}(X)$, $A \subseteq X$

$$m^*(Y) \leq m^*(Y \cap A) + m^*(Y \cap A^c).$$

For $A \subseteq X$ to be L-measurable it suffices to check the opposite inequality

$$m^*(Y) \geq m^*(Y \cap A) + m^*(Y \cap A^c).$$

Theorem 9.2.1 (Carathéodory's measurability—equivalent definitions) *A set $E \subseteq X$ is L-measurable if and only if one of the following equivalent facts holds:*

[2]Constantin Carathéodory came from a privileged Constantinople family (today's Istanbul). His father served as ambassador in Brussels from 1875 to 1900. Constantin grew up in Belgium, where he read engineering. In 1914 he proposed an alternative definition of measurability that allowed to generalise the concept to any subset of $\mathbb{R}$.

1. *for every* $Y \subseteq X$
$$m^*(Y) \geq m^*(E \cap Y) + m^*(E^c \cap Y);$$

2. *for any* $Y \subseteq X$ *with* $m^*(Y) < \infty$
$$m^*(Y) \geq m^*(E \cap Y) + m^*(E^c \cap Y);$$

3. *for every* $A \in \mathcal{A}$
$$m(A) \geq m^*(E \cap A) + m^*(E^c \cap A).$$

Proof 1. Item *1.* implies *2.* because the inequality
$$m^*(Y) \geq m^*(E \cap Y) + m^*(E^c \cap Y)$$
is trivial if $m^*(Y) = \infty$.
2. Item *2.* implies *3.* in a trivial way too: just notice $\mathcal{A} \subseteq \mathcal{P}(X)$, and $m^*(A) \leq m(A)$ for every $A \in \mathcal{A}$, so $m(A) \geq m^*(E \cap A) + m^*(E^c \cap A)$ is obvious.
3. Let us prove *3.*$\Rightarrow$*1.*. Take $Y = A \subseteq X$, so if $m^*(Y) = \infty$ then
$$m^*(Y) \geq m^*(E \cap Y) + m^*(E^c \cap Y).$$
If $m^*(Y) < \infty$, given $\epsilon > 0$ there exists $\{A_i\}_{i \geq 1}$ in $\mathcal{A}$, with $Y \subseteq \bigcup_{i=1}^{\infty} A_i$, such that
$$m^*(Y) + \epsilon > \sum_{i=1}^{\infty} m(A_i).$$
For every $i \in \mathbb{N}$
$$m(A_i) \geq m^*(A_i \cap E) + m^*(A_i \cap E^c),$$
and so
$$\begin{aligned} m^*(Y) + \epsilon > \sum_{i=1}^{\infty} m(A_i) &\geq \sum_{i=1}^{\infty} m^*(A_i \cap E) + \sum_{i=1}^{\infty} m^*(A_i \cap E^c) \\ &\geq m^*\left(E \cap \bigcup_{i=1}^{\infty} A_i\right) + m^*\left(E^c \cap \bigcup_{i=1}^{\infty} A_i\right) \\ &\geq m^*(E \cap Y) + m^*(E^c \cap Y). \end{aligned}$$
The arbitrary choice of $\epsilon > 0$ implies
$$m^*(Y) \geq m^*(E \cap Y) + m^*(E^c \cap Y).$$
■

Remark 9.2.1 Statement *3.* means the algebra $\mathcal{A}$ is contained in the family $\mathscr{L}$ of L-measurable sets, so $\mathcal{A} \subseteq \mathscr{L}$ and $\mathscr{L}$ is not empty.

Recalling the set function $m^* : \mathcal{A} \mapsto [0, \infty]$ coincides with the measure m on $\mathcal{A}$, than it is finitely additive and completely subadditive on $\mathcal{A}$, by Theorem 8.2.2 it is completely additive as well. This suggests there exists a σ-algebra, containing $\mathcal{A}$, to which m^* can be extended. We set out to prove this fact in the following theorem. ♣

Theorem 9.2.2 (Carathéodory's extension theorem) *The family*

$$\mathscr{L} = \left\{A \subseteq X : \forall Y \subseteq X, m^*(Y) = m^*(A \cap Y) + m^*(A^c \cap Y)\right\}$$

is

1. *an algebra of sets, on which m^* is finitely additive;*
2. *a σ-algebra that contains all subsets with zero outer measure*

$$\mathcal{N} = \left\{N \subseteq X : m^*(N) = 0\right\}.$$

The outer measure m^* is completely additive on $\mathscr{L}$ and extends m from $\mathcal{A}$ to $\mathscr{L}$.

Proof 1. By the symmetry of formula (9.2.2), $A \in \mathscr{L} \Leftrightarrow A^c \in \mathscr{L}$. Moreover, for every $Y \subseteq X$,

$$m^*(Y) = \underbrace{m^*(Y \cap \emptyset)}_{m^*(\emptyset)=0} + \underbrace{m^*(Y \cap \emptyset^c)}_{m^*(Y)}.$$

Hence $\emptyset \in \mathscr{L}$, and $X \in \mathscr{L}$ because the family of measurable sets is complement-closed. There remains to prove $A_1, A_2 \in \mathscr{L} \Rightarrow (A_1 \cap A_2) \in \mathscr{L}$. As A_1 is measurable, for every $Y \subseteq X$,

$$m^*(Y) = m^*(Y \cap A_1) + m^*(Y \cap A_1^c). \tag{9.2.3}$$

But A_2 is measurable, too, so replacing Y with $Y \cap A_1 \subseteq X$ gives

$$m^*(Y \cap A_1) = m^*((Y \cap A_1) \cap A_2) + m^*((Y \cap A_1) \cap A_2^c).$$

Substituting this in (9.2.3) we obtain

$$m^*(Y) = m^*((Y \cap A_1) \cap A_2) + m^*(Y \cap A_1^c) + m^*((Y \cap A_1) \cap A_2^c). \tag{9.2.4}$$

Observe

$$\begin{aligned}(Y \cap A_1^c) \cup (Y \cap A_1 \cap A_2^c) &= (Y \cap A_1^c \cup Y) \cap (Y \cap A_1^c \cup A_1) \cap (Y \cap A_1^c \cup A_2^c)\\ &= (Y \cap A_1^c \cup A_1) \cap (Y \cap A_1^c \cup A_2^c)\\ &= Y \cap [Y \cap (A_1^c \cup A_2^c)]\\ &= Y \cap (A_1^c \cup A_2^c),\end{aligned}$$

so finite subadditivity gives

$$m^*(Y \cap (A_1 \cap A_2)^c) \leq m^*(Y \cap A_1^c) + m^*(Y \cap A_1 \cap A_2^c).$$

Inserting the above in (9.2.4) produces

$$m^*(Y) \geq m^*(Y \cap (A_1 \cap A_2)) + m^*(Y \cap (A_1 \cap A_2)^c),$$

whence $A_1 \cap A_2 \in \mathscr{L}$.

Now we show m^* is finitely additive on $\mathscr{L}$. In (9.2.2) put $Y = A_1 \cup A_2$ and $A = A_1$, with $A_1 \cap A_2 = \emptyset$:

$$\begin{aligned} m^*(A_1 \cup A_2) &= m^*(A_1 \cap (A_1 \cup A_2)) + m^*(A_1^c \cap (A_1 \cup A_2)) \\ &= m^*(A_1) + m^*(A_2). \end{aligned}$$

2. To show $\mathscr{L}$ is a σ-algebra it is enough to prove it is closed under countable unions. Take $\{A_i\}_{i\geq 1}$ in $\mathscr{L}$ such that $A = \bigcup_{i=1}^{\infty} A_i$.
As $\mathscr{L}$ is an algebra, A can be written as union of pairwise-disjoint sets in $\mathscr{L}$, i.e. we may assume $A_i \cap A_j = \emptyset$, $i \neq j$. Moreover $\bigcup_{i=1}^{n} A_i \in \mathscr{L}$ for every $n \in \mathbb{N}$, so for every $Y \subseteq X$

$$\begin{aligned} m^*(Y) &= m^*\left(Y \cap \bigcup_{i=1}^{n} A_i\right) + m^*\left(Y \cap \left(\bigcup_{i=1}^{n} A_i\right)^c\right) \\ &\geq \sum_{i=1}^{n} m^*(Y \cap A_i) + m^*\left(Y \cap \left(\bigcup_{i=1}^{\infty} A_i\right)^c\right). \end{aligned} \tag{9.2.5}$$

In the last passage, besides the finite additivity of m^*, we have used

$$\left(\bigcup_{i=1}^{\infty} A_i\right)^c \subseteq \left(\bigcup_{i=1}^{n} A_i\right)^c.$$

As inequality (9.2.5) holds for every $n \in \mathbb{N}$,

$$m^*(Y) \geq \sum_{i=1}^{\infty} m^*(Y \cap A_i) + m^*\left(Y \cap \left(\bigcup_{i=1}^{\infty} A_i\right)^c\right).$$

By the complete subadditivity of m^* we have

$$\sum_{i=1}^{\infty} m^*(Y \cap A_i) \geq m^*\left(Y \cap \bigcup_{i=1}^{\infty} A_i\right),$$

and so

$$m^*(Y) \geq m^*\left(Y \cap \bigcup_{i=1}^{\infty} A_i\right) + m^*\left(Y \cap \left(\bigcup_{i=1}^{\infty} A_i\right)^c\right).$$

This holds for every $Y \subseteq X$, hence for $Y = \bigcup_{i=1}^{\infty} A_i$:

$$m^*\left(\bigcup_{i=1}^{\infty} A_i\right) \geq \sum_{i=1}^{\infty} m^*(A_i).$$

The opposite inequality holds because of countable subadditivity, so m^* is countably additive on $\mathscr{L}$.
That m^* extends m from $\mathcal{A}$ to $\mathscr{L}$ descends directly from Proposition 9.1.2, *1.*, and what we have just proved.
So there remains $\mathcal{N} \subseteq \mathscr{L}$. Take $N \subseteq X$ such that $m^*(N) = 0$. As $Y \cap N \subseteq N$ and m^* is monotone, $m^*(Y \cap N) \leq m^*(N) = 0$ and therefore $m^*(Y \cap N) = 0$, $\forall Y \subseteq X$. Moreover $Y \supseteq Y \cap N^c$, and by monotonicity

$$m^*(Y) \geq m^*(Y \cap N^c) = m^*(Y \cap N^c) + m^*(Y \cap N).$$

Hence $N \in \mathscr{L}$, and by arbitrariness of $N \in \mathcal{N}$, $\mathcal{N} \subseteq \mathscr{L}$. Consequently, **any subset in $\mathcal{N}$, or any subset in a zero-measure set, is Lebesgue measurable**. ■

Remark 9.2.2 It is easy to check that a subset in a zero-measure set is Lebesgue measurable. Take $T \subseteq N$ and $N \in \mathcal{N}$, arbitrarily. Clearly for every $Y \subseteq X$ we have $Y \cap T \subseteq Y \cap N \subseteq N$, so the monotonicity of m^* implies $m^*(T \cap N) = 0$. But $Y \supseteq Y \cap T^c \supseteq Y \cap N^c$, so monotonicity gives

$$m^*(Y) \geq m^*(Y \cap T^c) = m^*(Y \cap T) + m^*(Y \cap T^c).$$

Hence $T \in \mathscr{L}$. But T is arbitrary, so **a subset of a zero-measure set $N \in \mathcal{N}$ is Lebesgue measurable**.

Property *3.* in Theorem 9.2.1 says the algebra $\mathcal{A}$ is contained in the σ-algebra $\mathscr{L}$ of L-measurable subsets. This σ-algebra, called **Lebesgue σ-algebra**, contains the collection $\mathcal{N}$ of subsets with zero outer measure. We have seen it also contains all subsets of zero-measure sets (called negligible). Since $\mathcal{S}(\mathcal{A})$ is the smallest σ-algebra containing $\mathcal{A}$, it follows that

$$\mathcal{S}(\mathcal{A}) \subseteq \mathscr{L}.$$

Clearly $\mathscr{L}$ **is the smallest σ-algebra containing $\mathcal{S}(\mathcal{A})$, the family $\mathcal{N}$ of zero-outer-measure sets, and all subsets of the latter.**

At last notice the outer measure m^* is subtractive, too. The measure is finitely additive on $\mathscr{L}$ (Theorem 9.2.2) and monotone on $\mathcal{P}(X)$ (Proposition 9.1.2, *3.*), so $m^*(B \setminus A) = m^*(B) - m^*(A)$ for every $A, B \in \mathscr{L}$ such that $A \subseteq B, m^*(B) < \infty$. ♣

9.3 Uniqueness of Extension m^* and Measure-Space Completion

In Theorem 9.2.2 we saw m^* is an extension of m from $\mathcal{A}$ to the σ-algebra $\mathscr{L}$. We still do not know if such an extension is unique.

In general it is useful to know whether, given a measure μ that agrees on an algebra $\mathcal{A}$ with m, the extensions of μ and m to the σ-algebra $\mathcal{S}(\mathcal{A})$ still agree, i.e. if $m(A) = \mu(A)$ for every $A \in \mathcal{S}(\mathcal{A})$.

Alas, the extension's uniqueness does not hold if the measure is not σ-finite.

Example 9.3.1 (*Plurality of extensions of non-σ-finite measures*) Let $X = (0, 1] \cap \mathbb{Q}$ be the set of rationals in $(0, 1]$, and $\mathcal{A}$ the algebra generated by real intervals of the form $(a, b]$.

We claim the σ-algebra $(X \cap \mathcal{S}(\mathcal{A}))$ is the same as the power set $\mathcal{P}(X)$. As $(X \cap \mathcal{S}(\mathcal{A})) \subseteq \mathcal{P}(X)$, the other inequality is enough. For every $q \in X$, $\{q\} \in (X \cap \mathcal{A})$, so $\{q\}$ belongs to $(X \cap \mathcal{S}(\mathcal{A}))$. But X is countable, so any $A \in \mathcal{P}(X)$ is a union (at most countable) of singletons. Then $A \in (X \cap \mathcal{S}(\mathcal{A}))$ and $\mathcal{P}(X) \subseteq (X \cap \mathcal{S}(\mathcal{A}))$.

For every non-empty interval $(a, b] \in \mathcal{A}$ the intersection $(a, b] \cap X$ is either empty or countable.

Define, for every $A \in (X \cap \mathcal{A})$, the measure $m_\# : (X \cap \mathcal{A}) \mapsto [0, \infty]$

$$m_\# = \begin{cases} \text{card}(A) = \infty \text{ if } A \neq \emptyset \\ 0 \text{ if } A = \emptyset \end{cases}.$$

This measure, called **counting measure**, is not σ-finite on $(X \cap \mathcal{A})$, otherwise there would exist a sequence $\{A_i\}_{i \geq 1} \subseteq (X \cap \mathcal{A})$ such that $m_\#(A_i) < \infty$ for every $i \in \mathbb{N}$ and

$$\bigcup_{i=1}^{\infty} A_i = X,$$

whence the absurd conclusion: $\text{card}(X) = 0$. However $m_\#$ is σ-finite when extended to $\mathcal{P}(X)$: since X is countable there exists $\{q_i\}_{i \geq 1} \subseteq \mathcal{P}(X)$ such that $m_\#(\{q_i\}) = 1$ and

$$X = \bigcup_{i=1}^{\infty} \{q_i\}.$$

Given $a \in (0, \infty)$, let $m_a : \mathcal{P}(X) \mapsto [0, \infty]$ be the set function $m_a(A) = a \cdot m_\#(A)$, $A \in (X \cap \mathcal{S}(\mathcal{A}))$.

This is a measure on $\mathcal{P}(X)$. In fact, $\emptyset \in \mathcal{P}(X)$ and $m_a(\emptyset) = a \cdot m_\#(\emptyset) = 0$. Take pairwise-disjoint sets $\{A_i\}_{i \geq 1}$ such that $A = \bigsqcup_{i=1}^{\infty} A_i \in \mathcal{P}(X)$, so

$$m_a(A) = m_a\left(\bigsqcup_{i=1}^{\infty} A_i\right) = a \cdot m_{\#}\left(\bigsqcup_{i=1}^{\infty} A_i\right)$$
$$= \sum_{i=1}^{\infty} a \cdot m_{\#}(A_i) = \sum_{i=1}^{\infty} m_a(A_i).$$

The measure $m_a : \mathcal{P}(X) \mapsto [0, \infty]$ is then an extension of $m_{\#}$ to $(X \cap \mathcal{S}(\mathcal{A})) = \mathcal{P}(X)$. But m_a is not unique, since any positive multiple of $m_{\#}$ extends the counting measure to $\mathcal{P}(X)$. ★

Considering only σ-finite measures is not a big constraint. As we shall see along the way, the most interesting and useful cases, both theoretically and in practice, always give rise to σ-finite measures. The instances of measures that are not σ-finite are relegated to pathological situations of little interest.

We have σ-finiteness for the measure m on $\mathcal{I}$ and $\mathcal{A}(\mathcal{I})$ extended to the Borel σ-algebra $\mathcal{S}(\mathcal{A}(\mathcal{I}))$. This measure, although it assigns value $+\infty$ to unbounded intervals, has the property whereby $\mathbb{R} = \bigsqcup_{n=-\infty}^{\infty}(n, n+1]$ with $m\big((n, n+1]\big) = 1$ for every n. It allows to decompose $\mathbb{R}$, and consequently any set in $\mathcal{S}(\mathcal{A}(\mathcal{I}))$, in a countable union of pairwise-disjoint sets of finite length. There is another fundamental example of σ-finite (since totally finite) measure on the probability σ-algebra, where the probability measure satisfies $P(X) = 1$.

In presence of a measure m, furthermore, for every subset $Y \in \mathcal{S}(\mathcal{C})$ we have the restricted σ-algebra $\mathcal{S}(\mathcal{C} \cap Y)$, on which we can put the restriction of m to Y. More precisely,

Definition 9.3.1 (*Restriction of a measure*) Let m be a measure on the σ-algebra $\mathcal{S}(\mathcal{C})$ generated by a family $\mathcal{C}$ over X. The restriction of m to $Y \in \mathcal{S}(\mathcal{C})$, denoted $m_{|Y}(\cdot)$, is the set function

$$m_{|Y} : \mathcal{S}(\mathcal{C} \cap Y) \mapsto [0, \infty]$$

defined as $m_{|Y}(A) = m(A \cap Y)$ for every $A \in \mathcal{S}(\mathcal{C})$. ◇

Such a function is a measure on $\mathcal{S}(\mathcal{C})$. In fact $m_{|Y}(\emptyset) = m(\emptyset \cap Y) = 0$, and

$$m_{|Y}\left(\bigsqcup_{i=1}^{\infty} A_i\right) = m\left(\bigsqcup_{i=1}^{\infty} A_i \cap Y\right) = \sum_{i=1}^{\infty} m(A_i \cap Y)$$

for any sequence $\{A_i\}_{i \geq 1} \subseteq \mathcal{S}(\mathcal{C})$ such that $A = \bigsqcup_{i=1}^{\infty} A_i$.

In the light of Example 9.3.1 we can establish a general uniqueness result for the extension of σ-finite measures from an algebra to the finest σ-algebra generated.

Theorem 9.3.1 (Hahn-Kolmogorov extension theorem—uniqueness of extension of a σ-finite measure) *Let $\mathcal{A}$ be an algebra of subsets of a non-empty set X and $m : \mathcal{A} \mapsto [0, \infty]$ a σ-finite measure on $\mathcal{A}$. There exists a unique extension of m to a measure m' defined on the σ-algebra $\mathcal{S}(\mathcal{A})$.*

Proof Suppose m_1, m_2 are σ-finite measures extending m to $\mathcal{S}(\mathcal{A})$, i.e. $m_1(E) = m(E) = m_2(E)$ for every $E \in \mathcal{A}$. We need to show $m_1(E) = m_2(E)$ for every $E \in \mathcal{S}(\mathcal{A})$. First, $m_1(\emptyset) = m(\emptyset) = m_2(\emptyset) = 0$.

Without loss of generality assume, to begin with, m_1, m_2 totally finite over $\mathcal{S}(\mathcal{A})$, i.e. $m_1(X), m_2(X) < \infty$. Set

$$\mathcal{M} = \{E \in \mathcal{S}(\mathcal{A}) : m_1(E) = m_2(E)\}.$$

In $\mathcal{M}$ belong only elements of $\mathcal{S}(\mathcal{A})$ satisfying $m_1(E) = m_2(E)$, so $\mathcal{M} \subseteq \mathcal{S}(\mathcal{A})$.

So we just need the opposite inequality. By assumption m, m_1, m_2 agree on any $E \in \mathcal{A}$, so $\mathcal{M}$ is an algebra. By Proposition 7.1.2 we just need that $\mathcal{M}$ is monotone. Then take an increasing sequence $\{E_i\}_{i\geq 1}$ in $\mathcal{M}$ with $E = \bigcup_{i=1}^{\infty} E_i$. Now, m_1, m_2 are measures on $\mathcal{S}(\mathcal{A})$ and hence they are completely additive on $\mathcal{M}$. By Theorem 8.2.3 they are continuous from below as well, so

$$m_1(E) = \lim_{i\to\infty} m_1(E_i) = \lim_{i\to\infty} m_2(E_i) = m_2(E).$$

Therefore $E = \left(\bigcup_{i=1}^{\infty} E_i\right) \in \mathcal{M}$. Let now $\{E_i\}_{i\geq 1}$ be a decreasing sequence in $\mathcal{M}$, with $E = \left(\bigcap_{i=1}^{\infty} E_i\right)$. As before, m_1, m_2 are completely additive on $\mathcal{M}$ and totally finite by assumption. Theorem 8.2.4 ensures their continuity from above,

$$m_1(E) = \lim_{i\to\infty} m_1(E_i) = \lim_{i\to\infty} m_2(E_i) = m_2(E).$$

Hence $E = \left(\bigcap_{i=1}^{\infty} E_i\right) \in \mathcal{M}$. All in all $\mathcal{M}$ is a monotone collection containing $\mathcal{A}$. Further, $\mathcal{M}(\mathcal{A})$ is the smallest monotone family containing $\mathcal{A}$, and Theorem 7.1.3 says $\mathcal{S}(\mathcal{A}) = \mathcal{M}(\mathcal{A}) \subseteq \mathcal{M}$. This proves the opposite inclusion for totally finite measures.

For the general case, take a sequence $\{A_i\}_{i\geq 1}$ in $\mathcal{A}$ such that $m_1(A_i) < \infty$, $m_2(A_i) < \infty$ for every $i \in \mathbb{N}$ and $X = \bigcup_{i=1}^{\infty} A_i$. Given any $E \in \mathcal{S}(\mathcal{A})$, call

$$m_{1|A_i}(E) = m_1(E \cap A_i), m_{2|A_i}(E) = m_2(E \cap A_i)$$

the restrictions of m_1, m_2 to A_i, for every i. Both $m_{1|A_i}, m_{2|A_i}$ are finite on $\mathcal{S}(\mathcal{A})$, for $m_{1|A_i}(X) = m_1(A_i) < \infty$ and $m_{2|A_i}(X) = m_2(A_i) < \infty$. Clearly $m_1(A) = m_2(A)$ for every $A \in \mathcal{A}$. The above argument guarantees they agree on $\mathcal{S}(\mathcal{A}) \cap A_i = \mathcal{S}(\mathcal{A} \cap A_i)$ for every $i \in \mathbb{N}$. But $(E \cap A_i) \uparrow E$, from the continuity from below we deduce

$$\begin{aligned} m_1(E) = \lim_{i\to\infty} m_1(E \cap A_i) &= \lim_{i\to\infty} m_{1|A_i}(E) \\ &= \lim_{i\to\infty} m_{2|A_i}(E) = \lim_{i\to\infty} m_2(E \cap A_i) = m_2(E). \end{aligned}$$

Eventually m_1, m_2 agree on $\mathcal{S}(\mathcal{A})$. ■

Now we would like to understand the role played by the Lebesgue σ-algebra $\mathscr{L}$ in relation to the minimal σ-algebra $\mathcal{S}(\mathcal{A})$. In the section's remaining part we shall explain that $\mathscr{L}$ is the completion of $\mathcal{S}(\mathcal{A})$. By this we mean the former contains the collection $\mathcal{N}$ of zero-outer-measure and their subsets, making it a finer σ-algebra than the Borel σ-algebra $\mathcal{S}(\mathcal{A})$.

In the rest of the section, unless specified otherwise, we shall assume m is a σ-finite measure.

Proposition 9.3.1 (Equivalent descriptions of the outer measure) *For every $E \subseteq X$*

$$\begin{aligned} m^*(E) &= \inf\{m^*(A) : A \in \mathcal{S}(\mathcal{A}), E \subseteq A\} \\ &= \inf\{m^*(A) : A \in \mathscr{L}, E \subseteq A\}. \end{aligned}$$

Proof By definition

$$m^*(E) = \inf\left\{ \sum_{i=1}^{\infty} m(A_i) : E \subseteq \bigcup_{i=1}^{\infty} A_i, A_i \in \mathcal{A} \right\}.$$

Take $A = \bigcup_{i=1}^{\infty} A_i \supseteq E$, then there is a sequence $\{B_i\}_{i\geq 1}$ of pairwise-disjoint sets in $\mathcal{A}$ with $A = \bigcup_{i=1}^{\infty} B_i$ and $B_i \subseteq A_i$ for every $i \in \mathbb{N}$. By the monotonicity and complete additivity of m on $\mathcal{A}$, plus the complete subadditivity of m^* on $\mathcal{P}(X)$,

$$\sum_{i=1}^{\infty} m(A_i) \geq \sum_{i=1}^{\infty} m(B_i) = \sum_{i=1}^{\infty} m^*(B_i) \geq m^*(A).$$

Remembering $A \in \mathcal{S}(\mathcal{A})$, we have

$$m^*(E) = \inf\left\{ \sum_{i=1}^{\infty} m(A_i) : E \subseteq \bigcup_{i=1}^{\infty} A_i, A_i \in \mathcal{A} \right\} \geq \inf\left\{m^*(A) : A \in \mathcal{S}(\mathcal{A}), E \subseteq A\right\}.$$

As $\mathcal{S}(\mathcal{A}) \subseteq \mathscr{L}$,

$$\begin{aligned} m^*(E) &\geq \inf\left\{m^*(A) : A \in \mathcal{S}(\mathcal{A}), E \subseteq A\right\} \\ &\geq \inf\left\{m^*(A) : A \in \mathscr{L}, E \subseteq A\right\}. \end{aligned}$$

Moreover, m^* is monotone on $\mathcal{P}(X)$ and $E \subseteq A$, so

$$\begin{aligned} m^*(E) &\leq \inf\left\{m^*(A) : A \in \mathcal{S}(\mathcal{A}), E \subseteq A\right\} \\ &\leq \inf\left\{m^*(A) : A \in \mathscr{L}, E \subseteq A\right\}. \end{aligned}$$

■

A remarkable fact is that any $E \subseteq X$, not necessarily those in $\mathscr{L}$, can always be approximated from the inside and the outside by elements of $\mathcal{S}(\mathcal{A})$. These elements and E have the same outer measure, because the corresponding differences have zero measure.

In particular, $E \subseteq X$ can be approximated 'from outside' by some containing set $G \supseteq E$ in $\mathcal{S}(\mathcal{A})$, so that $m^*(E) = m^*(G)$ and $m^*(G \backslash E) = 0$. For obvious reasons G is called a measurable cover of E.

Similarly, $E \subseteq X$ can be approximated 'from inside' by some contained set $F \subseteq E$ in $\mathcal{S}(\mathcal{A})$ in such a way that $m^*(E) = m^*(F)$ and $m^*(E \setminus F) = 0$. We call such an F a measurable kernel of E.

Definition 9.3.2 (*Measurable cover and kernel of a set*) For every subset $E \subseteq X$:

1. a set $G \in \mathcal{S}(\mathcal{A})$ is called a measurable cover of E if $E \subseteq G$, $m^*(E) = m^*(G)$ and $m^*(A) = 0$ for all $A \subseteq (G \setminus E)$;
2. a set $F \in \mathcal{S}(\mathcal{A})$ is called a measurable kernel of E if $F \subseteq E$, $m^*(E) = m^*(F)$ and $m^*(A) = 0$ for every $A \subseteq (E \setminus F)$.

$\diamond$

Note $A \in \mathcal{S}(\mathcal{A})$ is both a measurable cover and a measurable kernel for itself.

In summary, any $E \subseteq X$ admits a measurable cover and a smaller set inside of it, called measurable kernel, whose relative differences to E have zero outer measure. Said formally,

Proposition 9.3.2 (Measurable cover and kernel of $E \subseteq X$) *If $m : \mathcal{A} \mapsto [0, \infty]$ is a σ-finite measure on the algebra of subsets $\mathcal{A}$ of a non-empty set X, every $E \subseteq X$ admits in $\mathcal{S}(\mathcal{A})$ a measurable cover G and a measurable kernel F.*

Proof Let us begin by the first fact. Take $E \subseteq X$ arbitrary. As m is σ-finite, so is m^*, hence there is a sequence $\{E_i\}_{i \geq 1} \subseteq \mathcal{P}(X)$ such that $E = \bigcup_{i=1}^{\infty} E_i$, with $E_i \cap E_j = \emptyset$, $i \neq j$, and $m^*(E_i) < \infty$ for every i.

Given $E_j \in \{E_i\}_{i \geq 1}$, Proposition 9.3.1 says

$$m^*(E_j) = \inf \left\{ m^*(G_j) : G_j \in \mathcal{S}(\mathcal{A}), E_j \subseteq G_j \right\}.$$

Take $\epsilon = 1/n$, so $m^*(E_j) + 1/n$ is not the greatest lower bound and there exists $G_n \in \mathcal{S}(\mathcal{A})$ such that $m^*(E_j) + 1/n > m^*(G_n)$, for every $n \in \mathbb{N}$. Set $G_k = \bigcap_{n=1}^{k} G_n$ for every $k \in \mathbb{N}$, so $\left\{ \bigcap_{n=1}^{k} G_n \right\}_{k \geq 1}$ is a decreasing sequence in $\mathcal{A}$ and $G_j = \bigcap_{n=1}^{\infty} G_n$ is a cover of E_j. Since $m^*(G_1) < m^*(E_j) + 1 < \infty$, moreover, by the properties of m^* and Theorem 8.2.4

$$m^*(G_j) = \lim_{k \to \infty} m^*(G_k).$$

Clearly

$$m^*(E_j) \le m^*(G_j) = \lim_{n\to\infty} m^*(G_n) \le \lim_{n\to\infty} m^*(E_j) + \frac{1}{n} = m^*(E_j).$$

whence $m^*(E_j) = m^*(G_j)$. Take any $A_j \in \mathcal{S}(\mathcal{A})$ such that $A_j \subseteq (G_j \setminus E_j)$, so $E_j \subseteq (G_j \setminus A_j)$. Since $m^*(G_j) < \infty$ and because of subtractivity,

$$\begin{aligned} & m^*(E_j) \le m^*(G_j \setminus A_j) = m^*(G_j) - m^*(A_j) \\ \Leftrightarrow & m^*(A_j) \le m^*(G_j) - m^*(E_j) = 0. \end{aligned}$$

As A_j is arbitrary, G_j is a measurable cover of E_j, for any $E_j \in \{E_i\}_{i\ge 1}$.

We claim $G = \bigcup_{i=1}^{\infty} G_i$ is a measurable cover of E. By Proposition 9.3.1 we have $m^*(G_i \setminus E_i) = 0$ for every i. Furthermore

$$(G \setminus E) = \left(\bigcup_{i=1}^{\infty} G_i\right) \setminus \left(\bigcup_{i=1}^{\infty} E_i\right) = \bigcup_{i=1}^{\infty} \left(G_i \cap \left(\bigcap_{i=1}^{\infty} E_i^c\right)\right) \subseteq \bigcup_{i=1}^{\infty} (G_i \setminus E_i),$$

so $m^*(G \setminus E) \le \sum_{i=1}^{\infty} m^*(G_i \setminus E_i) = 0$. Since

$$m^*(E) \le m^*(G) \le m^*(E) + m^*(G \setminus E) = m^*(E),$$

it follows that $m^*(E) = m^*(G)$. Now take a set $A \subseteq (G \setminus E)$, so for every $i \in \mathbb{N}$

$$A \cap G_i \subseteq (G \setminus E) \cap G_i \subseteq G_i \setminus E = G_i \setminus E_i.$$

As G_i is a measurable cover of E_i for every $i \in \mathbb{N}$, then for every $i \in \mathbb{N}$

$$m^*(A \cap G_i) = 0.$$

This and $A \subseteq G$ imply

$$m^*(A) = m^*(A \cap G) \le \sum_{i=1}^{\infty} m^*(A \cap G_i) = 0,$$

and the claim is finally settled.

To show the existence of a measurable kernel, take measurable covers G, N of E and $(G\setminus E)$, and define $F = (G\setminus N)$, so

$$F = (G\setminus N) \subseteq (G\setminus(G\setminus E)) = E.$$

Pick $A \subseteq (E\setminus F)$. As $E \setminus F = E \setminus (G \setminus N)$,

$$A \subseteq (E\backslash(G\backslash N)) = \underbrace{(E \cap G^c)}_{=\emptyset} \cup (E \cap N) \subseteq \underbrace{(N \cap G^c)}_{\neq\emptyset} \cup (E \cap N) = (N\backslash(G\backslash E)).$$

But N is a measurable cover of $(G\backslash E)$, so $m^*(A) = 0$. Given that $A \subseteq X$ is arbitrary, F is a measurable kernel of E. ■

Definition 9.3.3 (*Measurable space and measure space*) A measurable space is a pair $(X, \mathcal{S})$ formed by a non-empty set X and a σ-algebra $\mathcal{S}$ of subsets of X.

A measure space is a triple $(X, \mathcal{S}, m)$ with $X, \mathcal{S}$ as above and m a measure on $\mathcal{S}$. Elements in $\mathcal{S}$ are called measurable sets. ◇

If $(X, \mathcal{S}, m)$ is a measure space with finite measure normalised by $m(X) = 1$, we call it a probability space and m a probability. Such a space plays a fundamental part in axiomatic probability theory and represents a mathematical model to develop and analyse statistical experiments. Specifically, X is the set of possible outcomes of some experiment, the σ-algebra $\mathcal{S}$ is the collection of experimentally relevant events E, and the non-negative number $m(E)$ is the probability of occurrence of E.

To understand the differences between the σ-algebras $\mathscr{L}$ and $\mathcal{S}(\mathcal{A})$ we must study the σ-algebra structure of $\mathscr{L}$. We know $\mathscr{L}$ contains $\mathcal{S}(\mathcal{A}), \mathcal{N} = \{N \subseteq X : m^*(N) = 0\}$, and every subset of zero-measure sets, which in turn are Lebesgue measurable.

Some terminology is in order.

Definition 9.3.4 (*Zero-measure sets*) Let $(X, \mathcal{S}, m)$ be a measure space. A set $N \subseteq X$ has zero measure (or is a zero-measure set) if

$$N \in \mathcal{S} \text{ and } m(N) = 0.$$

We indicate by $\mathcal{N}_m$ the family of zero-measure sets (with respect to m):

$$\mathcal{N}_m = \big\{N \in \mathcal{S} : m(N) = 0\big\}.$$

◇

Definition 9.3.5 (*Negligible sets*) Let $(X, \mathcal{S}, m)$ be a measure space. A set $T \subseteq X$ is negligible if there exists an $N \in \mathcal{N}_m$ such that $T \subseteq N$.

We denote $\mathcal{T}_m$ the family of negligible sets (with respect to m):

$$\mathcal{T}_m = \big\{T \subseteq X : \exists N \text{ with } T \subseteq N \text{ and } N \in \mathcal{N}_m\big\}.$$

◇

Definition 9.3.6 (*Complete measure spaces*) A measure space $(X, \mathcal{S}, m)$ is complete if every negligible set is measurable:

$$\mathcal{T}_m \subseteq \mathcal{S}.$$

In that case m is called a complete measure, and $\mathcal{S}$ is a complete σ-algebra (with respect to m). ◇

Remark 9.3.1 The power set $\mathcal{P}(N)$ of any $N \in \mathcal{N}_m$ consists of negligible subsets, so $\mathcal{N}_m \subseteq \mathcal{T}_m$. Moreover $\emptyset \in \mathcal{S}$ and $m(\emptyset) = 0$, so $\emptyset \in \mathcal{N}_m$, then $\emptyset \in \mathcal{T}_m$.

We claim $\mathcal{N}_m$ and $\mathcal{T}_m$ are closed under countable unions. For the first family, given a complete measure space $(X, \mathcal{S}, m)$, let $\{N_i\}_{i\geq 1}$ be in $\mathcal{N}_m$. Since m is completely additive, so Theorem 8.2.2 guarantees m is completely subadditive and finitely additive on $\mathcal{S}$. Finite additivity implies monotonicity (Proposition 8.2.1) which gives

$$m\left(\bigcup_{i=1}^{\infty} N_i\right) \leq \sum_{i=1}^{\infty} m(N_i) = 0,$$

and therefore $\left(\bigcup_{i=1}^{\infty} N_i\right) \in \mathcal{N}_m$. If $\{T_i\}_{i\geq 1}$ is in $\mathcal{T}_m$, for every T_i there exists N_i such that $T_i \subseteq N_i$, so

$$\left(\bigcup_{i=1}^{\infty} T_i\right) \subseteq \left(\bigcup_{i=1}^{\infty} N_i\right)$$

and eventually $\left(\bigcup_{i=1}^{\infty} T_i\right) \in \mathcal{T}_m$. ♣

Proposition 9.3.3 (Characterisation of complete measure spaces) *A measure space $(X, \mathcal{S}, m)$ is complete if and only if*

$$\mathcal{N}_m = \mathcal{T}_m.$$

Proof Suppose $(X, \mathcal{S}, m)$ complete, since $\mathcal{N}_m \subseteq \mathcal{T}_m$, then we only need the other inclusion. If $E \in \mathcal{T}_m$ then $E \in \mathcal{S}$ by completeness. Take $N \in \mathcal{N}_m$ such that $E \subseteq N$. By the measure's monotonicity $m(E) = 0$, so $E \in \mathcal{N}_m$.

Vice versa, suppose $\mathcal{N}_m = \mathcal{T}_m$. If $E \in \mathcal{T}_m$ then $E \in \mathcal{N}_m$, and so $E \in \mathcal{S}$. ■

The next proposition shows the Lebesgue σ-algebra $\mathscr{L}$ is precisely $\mathcal{S}(\mathcal{A}) \cup \mathcal{T}$.

Proposition 9.3.4 (Structure of the σ-algebra $\mathscr{L}$) *Given a measure space $(X, \mathcal{S}(\mathcal{A}), m^*)$, let*

$$\mathcal{T} = \left\{T \subseteq X : \exists N \in \mathcal{S}(\mathcal{A}) \text{ with } T \subseteq N \text{ and } m^*(N) = 0\right\}$$

and

$$\mathscr{L} = \left\{E = F \cup T \subseteq X : F \in \mathcal{S}(\mathcal{A}), T \in \mathcal{T}\right\};$$

then:

1. *$\mathscr{L}$ is a σ-algebra on X containing $\mathcal{S}(\mathcal{A})$;*

2. *m defined on $\mathcal{S}(\mathcal{A})$ by*

$$m(F \cup T) = m^*(F), \forall F \in \mathcal{S}(\mathcal{A}), T \in \mathcal{N}$$

is a complete measure on $(X, \mathscr{L})$ with $m(F) = m^*(F)$ for every $F \in \mathcal{S}(\mathcal{A})$;

3. *$(X, \mathscr{L}, m)$ is the smallest complete measure space containing $(X, \mathcal{S}(\mathcal{A}), m^*)$, in the sense that if $(X, \hat{\mathcal{S}}(\mathcal{A}), \hat{m})$ is a complete measure space with $\mathcal{S}(\mathcal{A}) \subseteq \hat{\mathcal{S}}(\mathcal{A})$ and $\hat{m}(F) = m^*(F)$ for every $F \in \mathcal{S}(\mathcal{A})$, then $\mathscr{L} \subseteq \hat{\mathcal{S}}(\mathcal{A})$.*

Proof First we observe that $X, \emptyset \in \mathcal{S}(\mathcal{A})$ and $\emptyset \in \mathcal{T}$, so $X, \emptyset \in \mathscr{L}$.

Suppose $F \in \mathcal{S}(\mathcal{A})$ and $T \in \mathcal{T}$. Note that $(F \cup T)^c = F^c \cap T^c$. Let $N \in \mathcal{S}(\mathcal{A})$ with $T \subseteq N$ and $m^*(N) = 0$. Then $T^c \supseteq N^c$, so that $T^c = N^c \cup (T^c \setminus N^c) = N^c \cup (T^c \cap N)$. Thus

$$(F \cup T)^c = F^c \cap T^c = F^c \cap \left[N^c \cup (T^c \cap N)\right] = \left[F^c \cap N^c\right] \cup \left[F^c \cap (T^c \cap N)\right].$$

Note that $F^c \cap N^c \in \mathcal{S}(\mathcal{A})$ and $F^c \cap (T^c \cap N) \subseteq N$ with $m^*(N) = 0$. Hence $(F \cup T)^c \in \mathscr{L}$.

Now if $\{F_i\}_{i\geq 1} \subseteq \mathcal{S}(\mathcal{A})$ and $\{T_i\}_{i\geq 1} \subseteq \mathcal{T}$, we have to show that $\bigcup_{i=1}^{\infty} \left(F_i \cup T_i\right) \in \mathscr{L}$. Note

$$\bigcup_{i=1}^{\infty} (F_i \cup T_i) = \left(\bigcup_{i=1}^{\infty} F_i\right) \cup \left(\bigcup_{i=1}^{\infty} T_i\right).$$

Since $T_i \subseteq N_i$ for some $N_i \in \mathcal{S}(\mathcal{A})$, with $m^*(N_i) = 0$, we have

$$\bigcup_{i=1}^{\infty} T_i \subseteq \bigcup_{i=1}^{\infty} N_i, \text{ with } \bigcup_{i=1}^{\infty} N_i \in \mathcal{S}(\mathcal{A}) \text{ and } m^*\left(\bigcup_{i=1}^{\infty} T_i\right) \leq \sum_{i=1}^{\infty} m^*(N_i) = 0.$$

Therefore $\bigcup_{i=1}^{\infty} T_i \in \mathcal{T}$ so that $\bigcup_{i=1}^{\infty} \left(F_i \cup T_i\right) \in \mathscr{L}$.

As to number 2., since $\emptyset \in \mathcal{S}(\mathcal{A})$, $m(\emptyset) = m^*(\emptyset) = 0$. Next, let $\{F_i\}_{i\geq 1} \subseteq \mathcal{S}(\mathcal{A})$ and $\{T_i\}_{i\geq 1} \subseteq \mathcal{T}$ be such that $(F_i \cup T_i) \cap (F_j \cup T_j) = \emptyset$, for $i \neq j$. Then $\{F_i\}$ and $\{T_i\}$ are disjoint families and hence

$$\begin{aligned} m\left(\bigsqcup_{i=1}^{\infty}(F_i \cup T_i)\right) &= m\left(\bigsqcup_{i=1}^{\infty}(F_i) \cup \bigsqcup_{i=1}^{\infty}(T_i)\right) = m^*\left(\bigsqcup_{i=1}^{\infty}(F_i)\right) = \sum_{i=1}^{\infty} m^*(F_i) \\ &= \sum_{i=1}^{\infty} m(F_i \cup T_i). \end{aligned}$$

Thus m is a measure on $\mathscr{L}$.

For showing the completeness of m, let $E \subseteq F \cup T$ for some $F \in \mathcal{S}(\mathcal{A})$ and $T \in \mathcal{T}$ with $m(F \cup T) = 0$. We have to show that $E \in \mathscr{L}$. Note that

$$E = E \cap (F \cup T) = (E \cap F) \cup (E \cap T),$$

where

$$E \cap F \subseteq F \text{ with } m^*(F) = m(F \cup T) = 0$$

so that $E \cap F \in \mathcal{T}$. Also $E \cap T \subseteq T$ with $T \in \mathcal{T}$ so that $E \cap T \in \mathcal{T}$. Since $\mathcal{T} \subseteq \mathscr{L}$ and $\mathscr{L}$ is a σ-algebra, $E = (E \cap F) \cup (E \cap T) \in \mathscr{L}$.
Number *3.*: suppose that $(X, \hat{\mathcal{S}}(\mathcal{A}), \hat{m})$ is a complete measure space with $\mathcal{S}(\mathcal{A}) \subseteq \hat{\mathcal{S}}(\mathcal{A})$ and $\hat{m}(E) = m^*(E)$ for every $E \in \mathcal{S}(\mathcal{A})$. Since $\mathcal{S}(\mathcal{A}) \subseteq \hat{\mathcal{S}}(\mathcal{A})$ and $\hat{m}$ is a complete measure, we have $\mathcal{T} \subseteq \hat{\mathcal{S}}(\mathcal{A})$. Thus, using the fact that $\hat{\mathcal{S}}(\mathcal{A})$ is a σ-algebra, we also obtain $\mathscr{L} \subseteq \hat{\mathcal{S}}(\mathcal{A})$.

Therefore $\mathscr{L}$ is the smallest σ-algebra containing $\mathcal{S}(\mathcal{A})$, zero-outer-measure sets and their subsets. ■

In the end $(X, \mathscr{L}, m)$ is the completion of the measure space $(X, \mathcal{S}(\mathcal{A}), m^*)$.

The above theorem implies any measure space can be completed, and any completion is unique – being the smallest space containing the measure space it completes. When necessary, then, we shall suppose without loss of generality that a given measure is complete. As a matter of fact most measures are already complete. There are exceptions, a typical instance popping up in the proof of Fubini's theorem.

It is therefore convenient to suppose the measure spaces under consideration are complete, i.e. that any zero-measure subset is measurable.

Proposition 9.3.5 (Approximating measurable sets) *Let $(X, \mathscr{L}, m^*)$ denote a complete measure space and m^* the outer measure induced by a measure m on the algebra of subsets $\mathcal{A}$ of X. Given any $E \in \mathscr{L}$ such that $m^*(E) < \infty$, for each $\epsilon > 0$ there exist finitely many sets $F_1, \ldots, F_{n'}$ in $\mathcal{A}$, pairwise disjoint with $F' = \bigsqcup_{n=1}^{n'} F_n$, for which*

$$m^*(E \Delta F') < \epsilon.$$

Proof By assumption $m^*(E) < \infty$. Given an arbitrary $\epsilon > 0$, by definition of outer measure $m^*(E) + \epsilon/2$ is no longer the least upper bound, so there exists a sequence of pairwise-disjoint sets $\{F_n\}_{n \geq 1} \subseteq \mathcal{A}$ for which $E \subseteq \bigsqcup_{n=1}^{\infty} F_n$ and

$$m^*(E) \leq \sum_{n=1}^{\infty} m^*(F_n) < m^*(E) + \frac{\epsilon}{2} < \infty.$$

Hence there is an n' such that $\sum_{n=n'+1}^{\infty} m^*(F_n) < \epsilon/2$. Let us call $F' = \bigsqcup_{n=1}^{n'} F_n$, so

$$E \setminus F' = E \setminus \left(\bigsqcup_{n=1}^{n'} F_n \right) \subseteq \left(\bigsqcup_{n=1}^{\infty} F_n \right) \setminus \left(\bigsqcup_{n=1}^{n'} F_n \right) = \bigsqcup_{n=n'+1}^{\infty} F_n$$

$$\Rightarrow m^*(E \setminus F') \leq \sum_{n=n'+1}^{\infty} m^*(F_n) < \frac{\epsilon}{2}. \tag{9.3.1}$$

Similarly, we have

$$F' \setminus E \subseteq \left(\bigsqcup_{n=1}^{\infty} F_n \right) \setminus E$$

$$\Rightarrow m^*(F' \setminus E) \leq \sum_{n=1}^{\infty} m^*(F_n) - m^*(E) < \frac{\epsilon}{2}.$$

Altogether, then,

$$m^*(E \Delta F') = m^*(E \setminus F') + m^*(F' \setminus E) < \epsilon,$$

whence the claim. ■

The notion of set of zero outer measure just introduced is very important in measure theory and integration theory alike. Zero-measure sets may be ignored completely in certain results. If a property $\mathcal{P}$ holds for every element x in some X except for those in a set N of zero measure, we are entitled to say $\mathcal{P}$ holds almost everywhere on X.

The meaning of "almost everywhere" (a.e.) depends crucially on the measure considered. If the measure for which the property holds a.e. on X is called m, we will use the shorthand m-a.e. on X, unless it is clear from the context which measure we are talking about.

The idea of zero-measure sets allows us to make some observations.

Remark 9.3.2 Given a complete measure space[3] $(X, \mathcal{S}, m)$ we say:

1. a property P is true m-almost everywhere (m-a.e.) on X if

$$\big\{x \in X : P(x)\big\}^c = \big\{x \in X : \neg P(x)\big\} \in \mathcal{N}_m;$$

2. two maps $f, g : X \mapsto \bar{\mathbb{R}}$ are equal m-a.e. on X if

$$\big\{x \in X : f(x) = g(x)\big\}^c = \big\{x \in X : f(x) \neq g(x)\big\} \in \mathcal{N}_m;$$

3. two maps $f, g : X \mapsto \bar{\mathbb{R}}$ satisfy $f > g$ m-a.e. on X if

$$\big\{x \in X : f(x) > g(x)\big\}^c = \big\{x \in X : f(x) \leq g(x)\big\} \in \mathcal{N}_m;$$

[3] In this context imposing **completeness** is essential, because complete spaces contain all zero-measure sets and all negligible subsets, which are measurable too.

4. a map $f : X \mapsto \bar{\mathbb{R}}$ is bounded m-a.e. on X if

$$\{x \in X : |f(x)| < \infty\}^c = \{x \in X : |f(x)| = \infty\} \in \mathcal{N}_m.$$

♣

Equality m-a.e. among maps $f_i : X \mapsto \bar{\mathbb{R}}$ determines a partition of $\bar{\mathbb{R}}^X$ into equivalence classes, where the representatives of one coset might be thought of as one map defined a.e. on X. In particular,

Proposition 9.3.6 (A.e. equality of functions and equivalence relation) *Let $(X, \mathcal{S}, m)$ be a complete measure space. Then m-a.e. equality on $\bar{\mathbb{R}}^X$ is an equivalence relation.*

Proof Reflexivity and symmetry are evident. Let us show transitivity. Take $f, g, h : X \mapsto \bar{\mathbb{R}}$ and

$$N_1 = \{x \in X : f(x) \neq g(x)\} \text{ and } N_2 = \{x \in X : g(x) \neq h(x)\}.$$

Call $N = N_1 \cup N_2$, so $N \in \mathcal{N}_m$ and for every $x \in N^c = N_1^c \cap N_2^c$

$$f(x) = g(x) = h(x).$$

Then

$$N^c \subseteq \{x \in X : f(x) = h(x)\} \Leftrightarrow \{x \in X : f(x) \neq h(x)\} \subseteq N.$$

As $(X, \mathcal{S}, m)$ is complete,

$$\{x \in X : f(x) \neq h(x)\} \in \mathcal{T}_m,$$

whence $f = h$ a.e. on X. ■

9.4 The Lebesgue Measure on $\mathbb{R}$

Now we specialise to $X = \mathbb{R}$, and let $\mathcal{A}$ indicate the algebra $\mathcal{A}(\mathcal{I})$ generated by collection $\mathcal{I}$ of real intervals. We indicate by ℓ the length function of Sects. 8.1–8.2. The outer measure ℓ^* induced on subsets of $\mathbb{R}$ is called Lebesgue outer measure, and in analogy to Definition 9.1.1 and the formula of Proposition 9.1.1 it is defined as follows: for every $E \subseteq \mathbb{R}$

$$\ell^*(E) = \inf \left\{ \sum_{i=1}^{\infty} \ell(I_i) : E \subseteq \bigsqcup_{i=1}^{\infty} I_i, I_i \in \mathcal{I}, \forall i \text{ and } I_i \cap I_j = \emptyset, i \neq j \right\}.$$

The σ-algebra of ℓ^*-measurable sets, obtained in Sect. 9.2, for $X = \mathbb{R}$ is called σ-algebra of Lebesgue measurable sets and denoted $\mathscr{L}(\mathbb{R})$ or simply $\mathscr{L}$.

Clearly the σ-algebra $\mathcal{S}(\mathcal{I})$ generated by real intervals is the Borel σ-algebra on $\mathbb{R}$, which we indicate with $\mathscr{B}(\mathbb{R})$.

The measure ℓ is σ-finite on $\mathcal{I}$, so Theorem 9.3.1 ensures its extension to $\mathscr{B}(\mathbb{R})$ is unique. Moreover, Borel sets in $\mathbb{R}$ are Lebesgue measurable.

Theorem 9.4.1 (Measurability of Borel sets in $\mathbb{R}$) *Any Borel subset of* $\mathbb{R}$ *is Lebesgue measurable.*

Proof We begin by proving the measurability of (a, ∞). Given $A \subseteq \mathbb{R}$ and

$$A_1 = A \cap (a, \infty) \text{ and } A_2 = A \cap (-\infty, a],$$

we must show

$$\ell^*(A) \geq \ell^*(A_1) + \ell^*(A_2).$$

This is obvious if $\ell^*(A) = \infty$, so suppose $\ell^*(A) < \infty$. Given $\epsilon > 0$, there is a family $\{I_i\}_{i\geq 1}$ of intervals covering A such that $\ell^*(A) + \epsilon$ is not the greatest lower bound, so we have $\sum_{i=1}^{\infty} \ell(I_i) < \ell^*(A) + \epsilon$. Define disjoint intervals (possibly empty)

$$I_i^1 = I_i \cap (a, \infty) \text{ and } I_i^2 = I_i \cap (-\infty, a],$$

so

$$\ell(I_i) = \ell(I_i^1) + \ell(I_i^2) = \ell^*(I_i^1) + \ell^*(I_i^2).$$

As $A_1 \subseteq \bigcup_{i=1}^{\infty} I_i^1$, $A_2 \subseteq \bigcup_{i=1}^{\infty} I_i^2$,

$$\ell^*(A_1) \leq \ell^*\left(\bigcup_{i=1}^{\infty} I_i^1\right) \leq \sum_{i=1}^{\infty} \ell^*(I_i^1)$$
$$\ell^*(A_2) \leq \ell^*\left(\bigcup_{i=1}^{\infty} I_i^2\right) \leq \sum_{i=1}^{\infty} \ell^*(I_i^2)$$

and so

$$\ell^*(A_1) + \ell^*(A_2) \leq \sum_{i=1}^{\infty} \left(\ell^*(I_i^1) + \ell^*(I_i^2)\right) \leq \sum_{i=1}^{\infty} \ell(I_i) < \ell^*(A) + \epsilon,$$

whence the claim.

Now, (a, ∞) measurable implies its complement $(-\infty, a]$ is measurable. Observe $(-\infty, b) = \bigcup_{n=1}^{\infty}(-\infty, b - 1/n]$, so $(-\infty, b)$ is measurable. As $(a, b) = (-\infty, b) \cap (a, \infty)$, any open interval (a, b) is measurable. Consequently any open set in $\mathbb{R}$ is

measurable as a union (finite or countable) of open sets. Taking complements, any closed set is measurable, and so on. ■

Now we know that $\mathscr{B}(\mathbb{R}) \subseteq \mathscr{L}$, so

$$\mathscr{L} = \mathcal{S}(\mathcal{I}) \cup \mathcal{T} = \mathscr{B} \cup \mathcal{T}.$$

Given $E \subseteq \mathbb{R}$ such that $\ell^*(E) = 0$, since zero-measure sets are measurable, E is Lebesgue measurable and has zero measure. Then, for every $T \subseteq E$, we have $\ell^*(T) \leq \ell^*(E)$, so $\ell^*(T) = 0$ and T is also Lebesgue measurable of zero measure. But then the Lebesgue measure ℓ^* on $\mathbb{R}$ and the collection $\mathscr{L}$ of Lebesgue measurable sets are complete, so $(\mathbb{R}, \mathscr{L}, \ell^*)$ is the completion of $(\mathbb{R}, \mathscr{B}, \ell^*)$.

We have said in earlier sections that any $E \subseteq X$ can be approximated by a smaller set F and a larger set $G \in \mathcal{S}(\mathcal{A})$. Now we would like to know if there is a similar situation for sets of the class $\mathscr{L}(\mathbb{R})$.

Theorem 9.4.2 (Approximation in $\mathscr{L}(\mathbb{R})$ by open and closed subsets of $\mathbb{R}$) *A set $E \subseteq \mathbb{R}$ is Lebesgue measurable ($\Rightarrow E \in \mathscr{L}$) if and only if, for any $\epsilon > 0$, there exists an open set $G \supseteq E$, or a closed set $F \subseteq E$, in $\mathbb{R}$ such that*

$$\ell^*(G \setminus E) < \epsilon, \text{ or } \ell^*(E \setminus F) < \epsilon.$$

Proof Let us prove the first one (necessary part). Let E be Lebesgue measurable and suppose for a moment $\ell^*(E) < \infty$, so $\forall Y \subseteq \mathbb{R}$

$$\ell^*(Y) = \ell^*(Y \cap E) + \ell^*(Y \cap E^c).$$

Given $\epsilon > 0$ there certainly exists an open set $G \subseteq \mathbb{R}$ containing E, with $\ell^*(G \cap E^c) < \epsilon$, and such that

$$\ell^*(G) = \ell^*(G \cap E) + \ell^*(G \cap E^c).$$

In case $\ell^*(E) = \infty$, we can set $E = \bigcup_{i=1}^{\infty} E_i$, with $E_i \cap E_j = \emptyset, i \neq j$ and $\ell^*(E_i) < \infty$ for every $i \in \mathbb{N}$. As before, since all E_i are measurable, given $\epsilon > 0$, for every $i \in \mathbb{N}$ there is an open set $G_i \subseteq \mathbb{R}$ containing E_i with $\ell^*(G_i \cap E_i^c) < \epsilon/2^i$ and such that

$$\ell^*\big(G_i\big) = \ell^*\big(G_i \cap E_i\big) + \ell^*\big(G_i \cap E_i^c\big)$$

for every $i \in \mathbb{N}$. In general, setting $G = \bigcup_{i=1}^{\infty} G_i$ with $E \subseteq G$,

$$\big(G \setminus E\big) \subseteq \bigcup_{i=1}^{\infty} \big(G_i \setminus E_i\big) \text{ and } \ell^*\big(G \setminus E\big) \leq \sum_{i=1}^{\infty} \ell^*\big(G_i \cap E_i^c\big) < \epsilon.$$

Vice versa, for any $\epsilon > 0$ there exists an open set G containing E, with $\ell^*(E) < \infty$, such that $\ell^*(G \setminus E) < \epsilon$. Given $Y \subseteq \mathbb{R}$,

$$Y \cap E^c = (Y \cap G^c) \cup (Y \cap (G \backslash E)).$$

Subadditivity and monotonicity of ℓ^*, together with the measurability of G, imply

$$\begin{aligned}\ell^*(Y \cap E) + \ell^*(Y \cap E^c) &\leq \ell^*(\underbrace{Y \cap E}_{\subseteq (Y \cap G)}) + \ell^*(Y \cap G^c) + \ell^*(\underbrace{Y \cap (G \backslash E)}_{\subseteq (G \backslash E)}) \\ &\leq \ell^*(Y \cap G) + \ell^*(Y \cap G^c) + \ell^*(G \setminus E) \\ &< \ell^*(Y) + \epsilon.\end{aligned}$$

Hence

$$\ell^*(Y) \geq \ell^*(Y \cap E) + \ell^*(Y \cap E^c),$$

i.e. E is Lebesgue measurable. The case $\ell^*(E) = \infty$ is left as an exercise.

Now we come to the second one (necessary part). The linear set E is in $\mathscr{L}(\mathbb{R})$, so $E^c \in \mathscr{L}(\mathbb{R})$. Using the previous part there exists an open set G such that $E^c \subseteq G$ and $\ell^*(G \setminus E^c) < \epsilon$. Put $F = G^c$, then $F \subseteq E$ is a closed set, and since $G \setminus E^c = E \setminus F$, we have $\ell^*(E \setminus F) < \epsilon$, i.e. the claim.

Conversely for any $\epsilon > 0$ there exists a closed set F contained in E, with $\ell^*(E) < \infty$, such that $\ell^*(E \backslash F) < \epsilon$. Given $Y \subseteq \mathbb{R}$,

$$Y \cap E = (Y \cap F) \cup (Y \cap (E \backslash F)).$$

But ℓ^* is subadditive and monotone, and F measurable, so

$$\begin{aligned}\ell^*(Y \cap E) + \ell^*(Y \cap E^c) &\leq \ell^*(Y \cap F) + \ell^*(\underbrace{Y \cap (E \backslash F)}_{\subseteq (E \backslash F)}) + \ell^*(\underbrace{Y \cap E^c}_{\subseteq (Y \cap F^c)}) \\ &\leq \ell^*(Y \cap F) + \ell^*(Y \cap F^c) + \ell^*(E \setminus F) \\ &< \ell^*(Y) + \epsilon.\end{aligned}$$

Hence

$$\ell^*(Y) \geq \ell^*(Y \cap E) + \ell^*(Y \cap E^c),$$

i.e. E is Lebesgue measurable. The case $\ell^*(E) = \infty$ is left as an exercise. ■

We have seen any $E \in \mathscr{L}(\mathbb{R})$ can be approximated by open and closed sets. This includes elements of $\mathscr{L}(\mathbb{R})$ with finite measure. In this case the closed sets F approximating $E \in \mathscr{L}(\mathbb{R})$, $\ell^*(E) < \infty$, are compact.

Proposition 9.4.1 (Approximation of sets in $\mathscr{L}(\mathbb{R})$ by compact sets) *A set $E \subseteq \mathbb{R}$, with $\ell^*(E) < \infty$, is Lebesgue measurable ($\Rightarrow E \in \mathscr{L}$) if and only if, for any $\epsilon > 0$, there exists a compact set $K \subseteq E$ such that*

$$\ell^*(E \setminus K) < \epsilon.$$

Proof Immediate corollary of the previous theorem. ■

Now it is clear we may approximate Lebesgue measurable subset of $\mathbb{R}$ by open sets, closed sets or compact sets up to sets with arbitrarily small measure. In the next theorem, using countable intersections of open sets or unions of closed sets, we approximate Lebesgue measurable subsets of $\mathbb{R}$ up to zero-measure sets.

Theorem 9.4.3 (Approximation for sets in $\mathscr{L}(\mathbb{R})$ *by* G_δ *and* F_σ *sets*) *A set* $E \subseteq \mathbb{R}$ *is Lebesgue measurable if and only if there exists a* $G \in G_\delta$ $(F \in F_\sigma)$ *such that* $E \subseteq G$ $(F \subseteq E)$ *and*

$$\ell^*(G \setminus E) = 0 (\ell^*(E \setminus F) = 0). \tag{9.4.1}$$

Proof Start with E Lebesgue measurable. By Theorem 9.4.2, for every $n \in \mathbb{N}$ there exist a closed set F_n and an open set G_n such that $F_n \subseteq E \subseteq G_n$ plus

$$\ell^*(G_n \setminus E) < \frac{1}{n} \text{ and } \ell^*(E \setminus F_n) < \frac{1}{n}.$$

Then

$$G = \bigcap_{n=1}^{\infty} G_n \text{ and } F = \bigcup_{n=1}^{\infty} F_n$$

are, respectively, a G_δ set containing E and an F_σ set contained in E. Moreover, for every $n \in \mathbb{N}$

$$\ell^*(G \setminus E) \leq \ell^*(G_n \setminus E) < 1/n \text{ and } \ell^*(E \setminus F) \leq \ell^*(E \setminus F_n) < 1/n,$$

so as $n \to \infty$ we obtain $\ell^*(G \setminus E) = 0$ and $\ell^*(E \setminus F) = 0$.

The sufficient implication is immediate, since the assumptions imply

$$\ell^*(G \setminus E) < \epsilon \text{ and } \ell^*(E \setminus F) < \epsilon$$

and by Theorem 9.4.2 $E \subseteq \mathbb{R}$ is Lebesgue measurable. ■

The properties of ℓ^* on $\mathscr{L}(\mathbb{R})$ considered in Theorems 9.4.2, 9.4.3 can be rephrased to relate L-measurable sets with familiar topological notions, namely open and closed subsets. This is the content of the following proposition.

In the sequel, for simplicity, we shall indicate the restriction of ℓ^* to $\mathscr{L}(\mathbb{R})$ or $\mathscr{B}(\mathbb{R})$ by ℓ. Thus $(\mathbb{R}, \mathscr{L}, \ell)$ will be the Lebesgue measure space, $(\mathbb{R}, \mathscr{B}, \ell)$ the Borel measure space, and ℓ will denote the Lebesgue measure on both $\mathscr{L}(\mathbb{R})$ and $\mathscr{B}(\mathbb{R})$.

Proposition 9.4.2 (Outer regularity of ℓ *on* $\mathscr{L}(\mathbb{R})$) *Let* $E \in \mathscr{L}(\mathbb{R})$. *For every* $\epsilon > 0$ *there exists an open set* $G \subseteq \mathbb{R}$, *with* $E \subseteq G$, *such that* $\ell(G \setminus E) < \epsilon$ *if and only if*

$$\ell(E) = \inf\big\{\ell(G) : G \textit{ is open in } \mathbb{R} \textit{ and } E \subseteq G\big\}.$$

Proof Necessary implication: For any $\epsilon > 0$, we have a $G' \subseteq \mathbb{R}$, $E \subseteq G'$ such that $\ell(G' \setminus E) < \epsilon$. As $E \subseteq G' = E \cup (G' \setminus E)$,

$$\ell(E) \leq \ell(G') = \ell(E) + \ell(G' \setminus E) < \ell(E) + \epsilon.$$

This is for any open G containing E such that $\ell(G \setminus E) < \epsilon$, so the inequality holds when taking the infimum $\inf\{\ell(G)\}$. Also $\epsilon > 0$ is arbitrary, so we obtain the weak inequality

$$\ell(E) \leq \inf\big\{\ell(G) : G \text{ is open in } \mathbb{R} \text{ and } E \subseteq G\big\} \leq \ell(E).$$

Sufficient implication: take $E \in \mathscr{L}(\mathbb{R})$ such that $\ell(E) < \infty$ and

$$\ell(E) = \inf\big\{\ell(G) : G \text{ is open in } \mathbb{R} \text{ and } E \subseteq G\big\}. \tag{9.4.2}$$

Pick $\epsilon > 0$, so $\ell(E) + \epsilon$ is not the greatest lower bound and there exists $G \subseteq \mathbb{R}$ open containing E such that $\ell(E) + \epsilon > \ell(G)$. As $G = E \cup (G \setminus E)$,

$$\ell(G) = \ell(E) + \ell(G \setminus E) < \ell(E) + \epsilon,$$

and then $\ell(G \setminus E) < \epsilon$. In case $\ell(E) = \infty$, the Eq. 9.4.2 is true and we can set $E = \bigsqcup_{i=1}^{\infty} E_i$ where $\ell(E_i) < \infty$ for all i, so $\ell(G_i \setminus E_i) < \epsilon/2^i$ and

$$\ell(G \setminus E) \leq \sum_{i=1}^{\infty} \ell(G_i \setminus E_i) < \epsilon.$$

■

Remark 9.4.1 The dual statement, so to speak, to Proposition 9.4.2 goes like this:

"Let $E \in \mathscr{L}(\mathbb{R})$. For every $\epsilon > 0$ there exists a closed set $F \subseteq \mathbb{R}$, with $F \subseteq E$, such that $\ell(E \setminus F) < \epsilon$ if and only if

$$\ell(E) = \sup\big\{\ell(F) : F \text{ is closed in } \mathbb{R} \text{ and } F \subseteq E\big\}."$$

In the case of sets $E \in \mathscr{L}(\mathbb{R})$, with $\ell(E) < 0$, the statement reads:

"Let $E \in \mathscr{L}(\mathbb{R})$ satisfy $0 < \ell(E) < \infty$. Then for every $\epsilon > 0$ there exists a compact subset $K \subseteq E$ such that $\ell(E \setminus K) < \epsilon$ if and only if

$$\ell(E) = \sup\big\{\ell(K) : K \text{ is compact in } \mathbb{R} \text{ and } K \subseteq E\big\}."$$

♣

The properties of ℓ on $\mathscr{L}(\mathbb{R})$ considered in Proposition 9.4.2 and Remark 9.4.1 highlight the relationship between L-measurable sets and elementary topological families, namely open, closed and compact subsets of $\mathbb{R}$ (provided $\ell(E) < \infty$ for the latter).

These considerations permit to generalize properties of ℓ on $\mathbb{R}$ to a measure m on a generic metric space (X, d).

Definition 9.4.1 (*Outer and inner regular measures*) Given a metric space (X, d), a measure m on the collection of Borel sets $\mathscr{B}(X)$ is called a Borel measure. We further say it is:

1. **outer regular** if, for every $E \in \mathscr{B}(X)$,

$$\begin{aligned} m(E) &= \inf\big\{m(G) : G \text{ is open and } E \subseteq G\big\} \\ &= \sup\big\{m(F) : F \text{ is closed and } F \subseteq E\big\} \end{aligned}$$

2. **inner regular** if, for every $E \in \mathscr{B}(X)$, $0 < m(E) < \infty$, one has

$$m(E) = \sup\big\{m(K) : K \text{ is compact and } K \subseteq E\big\}.$$

$\diamond$

Proposition 9.4.2 and Remark 9.4.1 now tell us that the Lebesgue measure on $\mathbb{R}$ is both inner and outer regular, i.e. just regular.

The next theorem explains that on complete and separable metric spaces (X, d), a finite measure m on $\mathscr{B}(X)$ is (outer and inner) regular.

Theorem 9.4.4 (Regularity conditions for finite Borel measures) *On a metric space (X, d), a* **finite** *Borel measure on $\mathscr{B}(X)$ is* **outer regular**.

If (X, d) is additionally complete and separable, a **finite** *Borel measure on $\mathscr{B}(X)$ is also* **inner regular**, *and hence* **regular**.

Proof For the first claim we must prove that, for every $E \in \mathscr{B}(X)$,

$$\begin{aligned} m(E) &= \inf\big\{m(G) : G \text{ is open and } E \subseteq G\big\} \\ &= \sup\big\{m(F) : F \text{ is closed and } F \subseteq E\big\}. \end{aligned} \tag{9.4.3}$$

By virtue of Proposition 9.4.2 and Remark 9.4.1 proving (9.4.3) amounts to showing that for every $\epsilon > 0$ there exists an open set G_ϵ and a closed set F_ϵ such that[4]

$$F_\epsilon \subseteq E \subseteq G_\epsilon \text{ and } m(G_\epsilon \setminus F_\epsilon) < \epsilon.$$

[4]Note

$$(G \setminus F) = \big[E \cup (G \setminus E)\big] \setminus F = (E \setminus F) \cup (G \setminus E \setminus F) = (E \setminus F) \cup (G \setminus E).$$

.

Define

$$\mathcal{S} = \big\{E \in \mathscr{B}(X) : F_\epsilon \text{ closed}, G_\epsilon \text{ open}, F_\epsilon \subseteq E \subseteq G_\epsilon \text{ and } m(G_\epsilon \setminus F_\epsilon) < \epsilon\big\}.$$

It suffices to prove $\mathcal{S}$ is a σ-algebra containing all closed subsets, so that $\mathcal{S} = \mathscr{B}(X)$.

For this, if $E \in \mathcal{S}$ then given $\epsilon > 0$ there exist F, G such that $F \subseteq E \subseteq G$, and correspondingly $G^c \subseteq E^c \subseteq F^c$ and $m(F^c \setminus G^c) < \eta$. Let $\{E_i\}_{i\geq 1}$ be a sequence in $\mathcal{S}$. Given $\epsilon > 0$, by assumption for every i we can choose an open set G_i and a closed set F_i such that

$$F_i \subseteq E_i \subseteq G_i \text{ and } m(G_i \setminus F_i) < \frac{\epsilon}{2^{i+1}}.$$

Take

$$G = \bigcup_{i=1}^{\infty} G_i \text{ and } C_i = \bigcup_{k=1}^{i} F_k, i \geq 1.$$

Then G is open and C_i closed for every $i \geq 1$, and moreover $\{G \setminus C_i\}_{i\geq 1}$ decreases to $G \setminus \bigcup_{i=1}^{\infty} F_i$. Since m is finite on $\mathscr{B}(X)$, the limit $\lim_{i\to\infty} m(G \setminus C_i)$ equals $m\big(G \setminus \bigcup_{i=1}^{\infty} F_i\big)$, so

$$\begin{aligned}\lim_{i\to\infty} m(G \setminus C_i) &= m\Big(G \setminus \bigcup_{i=1}^{\infty} F_i\Big) = m\Big(\bigcup_{i=1}^{\infty} G_i \setminus \bigcup_{i=1}^{\infty} F_i\Big) \\ &\leq m\Big(\bigcup_{i=1}^{\infty} (G_i \setminus F_i)\Big) \leq \sum_{i=1}^{\infty} m\big(G_i \setminus F_i\big) < \epsilon.\end{aligned}$$

Hence there is an i such that $m(G \setminus C_i) < \epsilon$. Given that $C_i \subseteq \bigcup_{i=1}^{\infty} E_i \subseteq G$, we have $\bigcup_{i=1}^{\infty} E_i \in \mathcal{S}$. To finish the first part there remains to show $\emptyset, X \in \mathcal{S}$ and closed sets $\mathcal{C}$ belong in $\mathcal{S}$. Clearly $\mathcal{C} \subseteq \mathcal{S}$ is enough, for $\emptyset, X \in \mathcal{C}$. So take a closed set F. Given the uniform continuity of the function $d(x, F)$ (see Lemma 5.1.2) we can put, for every $n \in \mathbb{N}$,

$$G_n = \{x \in X : d(x, F) < 1/n\},$$

where every G_n is open and $F = \bigcap_{n=1}^{\infty} G_n$. As m is finite,

$$\lim_{n\to\infty} m(G_n) = m(F).$$

Hence for $\epsilon > 0$ we can choose $n \in \mathbb{N}$ such that $m(G_n) - m(F) < \epsilon$. Clearly $F \subseteq G_n$ and $m(G_n \setminus F) < \epsilon$, i.e. $F \in \mathcal{S}$. But $F \in \mathcal{C}$ is arbitrary, so $\mathcal{S}$ is a σ-algebra containing all closed subsets, i.e. $\mathcal{S} = \mathscr{B}(X)$.

Now we come to the second statement and prove, for every $E \in \mathscr{B}(X)$,

$$m(E) = \sup\big\{m(K) : K \text{ is compact and } K \subseteq E\big\}.$$

This is equivalent to showing that for every $\epsilon > 0$ there exists a compact set K such that $m(E \setminus K) < \epsilon$. Take $\epsilon > 0$ and choose a closed set $F \subseteq E$ such that $m(E \setminus F) < \epsilon/2$. As X is separable, it contains a countable dense set $D = \{x_1, \ldots, x_k, \ldots\}$. Then for every $k \in \mathbb{N}$ we can choose a sequence $\{x_{k_i}\}_{i\geq 1} \subseteq D$ such that the family of balls $\mathcal{B} = \{B(x_{k_i}, 1/k)\}_{i\geq 1}$, where

$$B(x_{k_i}, 1/k) = \big\{y \in X : d(x_{k_i}, y) < 1/k \text{ and } x_{k_i} \in D\big\},$$

is an open cover of F:

$$F \subseteq \bigcup_{i=1}^{\infty} B(x_{k_i}, 1/k).$$

Define

$$\bar{B}(x_{k_i}, 1/k) = \big\{y \in X : d(x_{k_i}, y) \leq 1/k \text{ and } x_{k_i} \in D\big\}.$$

As m is finite on $\mathscr{B}(X)$, for every $k \geq 1$ there exists n_k such that

$$m\left(F \setminus \bigcup_{i=1}^{n_k} \bar{B}(x_{k_i}, 1/k)\right) < \frac{\epsilon}{2^{k+1}}.$$

Now consider

$$K = \bigcap_{k=1}^{\infty} \bigcup_{i=1}^{n_k} \bar{B}(x_{k_i}, 1/k);$$

this is closed and totally bounded in X and, by completeness, Proposition A.0.5 guarantees K is compact. Furthermore, $E \setminus K = (E \setminus F) \cup (F \setminus K)$, so

$$\begin{aligned} m(E \setminus K) &\leq m(E \setminus F) + m(F \setminus K) < \frac{\epsilon}{2} + m\left(F \setminus \left(\bigcap_{k=1}^{\infty} \bigcup_{i=1}^{n_k} \bar{B}(x_{k_i}, 1/k)\right)\right) \\ &= \frac{\epsilon}{2} + m\left(\bigcup_{k=1}^{\infty} F \setminus \left(\bigcup_{i=1}^{n_k} \bar{B}(x_{k_i}, 1/k)\right)\right) \\ &\leq \frac{\epsilon}{2} + \sum_{k=1}^{\infty} m\left(F \setminus \left(\bigcup_{i=1}^{n_k} \bar{B}(x_{k_i}, 1/k)\right)\right) < \epsilon, \end{aligned}$$

proving the claim. ∎

We close the section with a few results on the translation-invariance of the Lebesgue measure. We want to know whether, given a set $E \subseteq \mathbb{R}$ and $x \in \mathbb{R}$, the shift $E + x = \{y + x : y \in E\}$ has measure $\ell(E)$.

Recall the Lebesgue measure ℓ is nothing but the restriction of the outer measure ℓ^* to Lebesgue measurable sets. Given $E \subseteq \mathbb{R}$,

$$\ell(E) = \inf \left\{ \sum_{i=1}^{\infty} \ell(I_i) : E \subseteq \bigcup_{i=1}^{\infty} I_i, \, I_i \in \mathcal{I} \right\}.$$

We want to prove $\ell(E) = \ell(E + x)$. If E can be covered by intervals I_i then $(E + x)$ is covered by intervals $(I_i + x)$. Equivalently, if $(E + x)$ is covered by $(I_i + x)$ then E is covered by $(I_i + x - x) = I_i$:

$$E \subseteq \bigcup_{i=1}^{\infty} I_i \Leftrightarrow (E + x) \subseteq \bigcup_{i=1}^{\infty} (I_i + x).$$

Hence $\ell(E) = \ell(E + x)$, i.e. the Lebesgue measure of a set $E \subseteq \mathbb{R}$ is invariant under translations.

That said, it is important to know whether every translated-set of the family $\mathscr{L}(\mathbb{R})$ is measurable.

Proposition 9.4.3 (L-measurability of $\mathscr{L}(\mathbb{R})$-translated sets) *If $E \in \mathscr{L}(\mathbb{R})$, for every $x_0 \in \mathbb{R}$ the shifted set*

$$E + x_0 = \left\{ (x + x_0) \in \mathbb{R} : x \in E \right\}$$

is L-measurable.

Proof E is Lebesgue measurable, so for every $Y \subseteq \mathbb{R}$

$$\ell(Y) = \ell(Y \cap E) + \ell(Y \cap E^c).$$

But ℓ is translation-invariant, so $\ell(Y \cap E) = \ell((Y \cap E) + x_0) \; \forall Y \subseteq \mathbb{R}$. In the right hand we can swap the order of intersection and translation, to the effect that $\ell(Y \cap E) = \ell((Y + x_0) \cap (E + x_0))$. Similarly

$$\ell(Y \cap E^c) = \ell((Y \cap E^c) + x_0) = \ell((Y + x_0) \cap (E^c + x_0)).$$

As $(E^c + x_0) = (E + x_0)^c$, then,

$$\ell(Y \cap E^c) = \ell((Y \cap E^c) + x_0) = \ell((Y + x_0) \cap (E + x_0)^c).$$

Hence

$$\ell(Y) = \ell((Y + x_0) \cap (E + x_0)) + \ell((Y + x_0) \cap (E + x_0)^c).$$

Since $\ell(Y) = \ell(Y - x_0)$ for any $x_0 \in \mathbb{R}$, the previous relation reads

$$\begin{aligned}\ell(Y) &= \ell((Y + x_0 - x_0) \cap (E + x_0)) + \ell((Y + x_0 - x_0) \cap (E + x_0)^c) \\ &= \ell(Y \cap (E + x_0)) + \ell(Y \cap (E + x_0)^c),\end{aligned}$$

and $E + x_0$ is L-measurable. ■

Overall, the Lebesgue measure on $\mathbb{R}$ is invariant under translations, σ-finite and regular.

9.5 Cardinality of $\mathscr{L}(\mathbb{R})$ and $\mathscr{B}(\mathbb{R})$

We saw the Lebesgue σ-algebra $\mathscr{L}(\mathbb{R})$ is the completion of the Borel σ-algebra $\mathscr{B}(\mathbb{R})$. In this sense $\mathscr{L}(\mathbb{R}) = \mathscr{B}(\mathbb{R}) \cup \mathcal{T}$, so $\mathscr{B}(\mathbb{R}) \subseteq \mathscr{L}(\mathbb{R}) \subseteq \mathcal{P}(\mathbb{R})$.

The aim of this section is to check whether $\mathscr{B}(\mathbb{R})$ is a proper subfamily of $\mathscr{L}(\mathbb{R})$, i.e. if there exist Lebesgue measurable sets that are not Borel sets. There are two ways to address the matter. The first is to build a Lebesgue measurable sets that is not Borel. The explicit construction is, alas, not that easy.[5] The second consists in showing the cardinality of $\mathscr{B}(\mathbb{R})$ is less than the cardinality $\mathscr{L}(\mathbb{R})$.

The first proposal was undertaken by the Russian mathematician Suslin (1894–1919), who found the first example of Lebesgue measurable set that is not Borel. Suslin discovered a very important large family, the so-called analytical sets, which can be described as continuous images of Borel sets, each of which is Lebesgue measurable (assuming bounded). The family contains Borel sets, but is much bigger. It would be interesting to establish if there existed a bounded, non-Lebesgue measurable set. But it is impossible to answer this question with arguments that only rely on cardinality.

What we will do is compare the cardinalities of $\mathscr{B}(\mathbb{R})$ and $\mathscr{L}(\mathbb{R})$. Let us begin with the latter, and take the Cantor set for guidance. Here

$$\ell(G) = \frac{1}{3} + \frac{2}{3^2} + \frac{2^2}{3^3} + \cdots + \frac{2^n}{3^{n+1}} + \cdots = \frac{1}{3}\sum_{n=0}^{\infty}\left(\frac{2}{3}\right)^n = 1.$$

Every C_n is measurable, hence also $C = \bigcap_{n=1}^{\infty} C_n$ is measurable. More specifically, $\ell(C_1) = 2/3$; $\ell(C_2) = (2/3)^2$; $\ldots$, $\ell(C_n) = (2/3)^n$, so the measure's continuity from above (Theorem 8.2.4) gives:

$$\ell(C) = \ell\left(\bigcap_{n=1}^{\infty} C_n\right) = \lim_{n\to\infty} \ell(C_n) = \lim_{n\to\infty}\left(\frac{2}{3}\right)^n = 0.$$

[5] We shall exhibit a non-constructive example of a set that is not Borel when we discuss the Vitali-Cantor map (Sect. 10.5).

C has the cardinality of the continuum (Example 2.3.3), and now we know it has measure zero, so $C \in \mathcal{N}$. As the Lebesgue σ-algebra is complete, $C \in \mathscr{L}(\mathbb{R})$, any $T \subseteq C$ has $\ell(T) = 0$, hence $T \in \mathscr{L}(\mathbb{R})$. Therefore $\mathscr{P}(C) \subseteq \mathscr{L}(\mathbb{R})$, and $|\mathscr{L}(\mathbb{R})| \geq 2^{\mathbf{c}}$. But $\mathscr{L}(\mathbb{R}) \subseteq \mathcal{P}(\mathbb{R})$, so finally $|\mathscr{L}(\mathbb{R})| = 2^{\mathbf{c}}$.

That the cardinalities of $\mathscr{L}(\mathbb{R})$ and $\mathcal{P}(\mathbb{R})$ are the same does not say anything about the possible existence of non-Lebesgue measurable subsets. Ulam's theorem[6] sheds light on the matter: "*assuming the continuum hypothesis, the Lebesgue measure cannot be extended to the collection of all subsets of* $\mathbb{R}$". Hence not all subsets of $\mathbb{R}$ are Lebesgue measurable. Prior to this, in 1905, Giuseppe Vitali[7] had clarified the issue by constructing, with the axiom of choice (AC), a non-Lebesgue measurable subset of the real numbers.

Without the AC or the continuum hypothesis we cannot know if there exist subsets of $\mathbb{R}$ that are not Lebesgue measurable. If we replace the AC with the statement "*every subsets of* $\mathbb{R}$ *is Lebesgue measurable*" in ZF set theory, the new system stays consistent (as shown by R. Solovay in 1970). To the day, the question whether Lebesgue measurable sets exhaust all subsets of $\mathbb{R}$ remains open.

Shifting the focus onto Borel sets, by the Lebesgue-Hausdorff theorem we know $\mathscr{B}(\mathbb{R})$ has cardinality $\mathbf{c}$. In the next proposition we prove it directly.

Proposition 9.5.1 (Cardinality of the Borel σ-algebra) *The Borel* σ*-algebra* $\mathscr{B}(\mathbb{R})$ *has the cardinality of the continuum.*

Proof Recall Σ_1^0 is the family of open sets. Since the Borel σ-algebra can be generated by Σ_1^0, we put $\mathcal{S}(\Sigma_1^0) = \mathscr{B}(\mathbb{R})$, and by Theorem 7.2.1

$$\mathcal{S}(\Sigma_1^0) = \left\{ \bigcup_{\alpha<\Omega} \Sigma_\alpha^0 \right\}.$$

We first focus on Σ_1^0. Observe $\mathbb{R}$ has a countable basis, hence of cardinality $\aleph_0$, made by open rational intervals (p, q), and any open set G is a countable (at most) union of such, so

$$\left|\Sigma_1^0\right| \leq 2^{\aleph_0}.$$

Moreover any closed set $\{x\} = [x, x] \subseteq \mathbb{R}$, whose total cardinality equals the continuum, is Borel and in Π_1^0, so $\left|\Pi_1^0\right| \geq 2^{\aleph_0}$. As the complement of $\{x\}$ is a pair of open sets, then analogously $\left|\Sigma_1^0\right| \geq 2^{\aleph_0}$. Overall

$$\left|\Sigma_1^0\right| = 2^{\aleph_0}.$$

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[6] Stanislaw Ulam (L'viv, Ukraine, 1909 - Santa Fe, US, 1984). The result appeared in Fund. Math. 16 (1930), 140–150.

[7] G.Vitali, *Sul problema della misura dei gruppi di punti di una retta*, Tipografia Gamberini e Parmeggiani (1905), 231–235.

First inductive step: suppose, for β successor ordinal, the number of sets in $\mathcal{S}_\beta(\Sigma_1^0) = \Sigma_\beta^0$ is $2^{\aleph_0}$. We have to determine the cardinality of the next family $\mathcal{S}_{\beta+1}(\Sigma_1^0) = \Sigma_{\beta+1}^0$. By construction we know

$$\Sigma_{\beta+1}^0 = \Sigma_\beta^0 \cup \left\{A^c : A \in \Sigma_\beta^0\right\} \cup \left\{\bigcup_{i=1}^\infty A_i : \{A_i\}_{i\geq 1} \subseteq \Sigma_\beta^0\right\}.$$

To find a set A in $\Sigma_{\beta+1}^0$, knowing that $|\Sigma_\beta^0| = \mathbf{c}$, two things can happen:

- first, we have an obvious bijection from $\left\{A^c : A \in \Sigma_\beta^0\right\}$ to Σ_β^0, whence
$$\left|\left\{A^c : A \in \Sigma_\beta^0\right\}\right| = \left|\Sigma_\beta^0\right| = \mathbf{c};$$
- second, $\left\{\bigcup_{i=1}^\infty A_i : \{A_i\}_{i\geq 1} \subseteq \Sigma_\beta^0\right\}$ is generated by $\left\{f : f : \mathbb{N} \mapsto \Sigma_\beta^0\right\}$, whose cardinality is less than or equal to all sequences with values in Σ_β^0:
$$\left|\left\{\bigcup_{i=1}^\infty A_i : \{A_i\}_{i\geq 1} \subseteq \Sigma_\beta^0\right\}\right| \leq \left|\Sigma_\beta^0\right|^{|\mathbb{N}|} = \left(2^{\aleph_0}\right)^{\aleph_0} = 2^{\aleph_0} = \mathbf{c}.$$

Therefore

$$\begin{aligned}
|\Sigma_{\beta+1}^0| &= \left|\Sigma_\beta^0 \cup \left\{A^c : A \in \Sigma_\beta^0\right\} \cup \left\{\bigcup_{i=1}^\infty A_i : \{A_i\}_{i\geq 1} \subseteq \Sigma_\beta^0\right\}\right| \\
&\leq |\Sigma_\beta^0| + \left|\left\{A^c : A \in \Sigma_\beta^0\right\}\right| + \left|\left\{\bigcup_{i=1}^\infty A_i : \{A_i\}_{i\geq 1} \subseteq \Sigma_\beta^0\right\}\right| \\
&= 2^{\aleph_0} + 2^{\aleph_0} + 2^{\aleph_0} = 2^{\aleph_0} = \mathbf{c}.
\end{aligned}$$

As $|\Sigma_{\beta+1}^0| \geq |\Sigma_\beta^0| = 2^{\aleph_0} = \mathbf{c}$, then $|\Sigma_{\beta+1}^0| = 2^{\aleph_0} = \mathbf{c}$.

As second inductive step suppose the cardinality of Σ_β^0 equals $2^{\aleph_0} = \mathbf{c}$, for every $\beta < \alpha$, α limit ordinal. As $\{\beta : \beta < \alpha\}$ has cardinality $\aleph_0$ at most,

$$\mathcal{S}_\alpha(\Sigma_1^0) = \left\{\bigcup_{\beta<\alpha} \Sigma_\beta^0\right\}$$

has cardinality

$$|\mathcal{S}_\alpha(\Sigma_1^0)| = \left|\bigcup_{\beta<\alpha} \Sigma_\beta^0\right| \leq \sum_{\beta<\alpha} \left|\Sigma_\beta^0\right| = \sum_{\beta<\alpha} \mathbf{c} = \aleph_0 \cdot \mathbf{c} = \mathbf{c}.$$

Moreover, $|\bigcup_{\beta<\alpha} \Sigma_\beta^0| \geq |\Sigma_\beta^0| = \mathbf{c}$, for every $\beta < \alpha$; hence $|\mathcal{S}_\alpha(\Sigma_1^0)| = \mathbf{c}$.

By transfinite induction $|\mathcal{S}_\alpha(\Sigma_1^0)| = \mathbf{c}$ for every ordinal $\alpha < \Omega$. Since $\{\alpha : \alpha < \Omega\}$ has cardinality $2^{\aleph_0} = \mathbf{c}$, a similar argument leads to

$$|\mathscr{B}(\mathbb{R})| = |\mathcal{S}_\Omega(\Sigma_1^0)| = \mathbf{c}.$$

■

We have shown $\mathscr{L}(\mathbb{R})$ has cardinality $2^{\mathbf{c}}$, so there exist Lebesgue measurable sets that are not Borel. From this point of view $\mathscr{B}(\mathbb{R})$ is a sub-σ-algebra of Lebesgue measurable sets.

Furthermore, regardless the fact the cardinality of Lebesgue measurable sets equals the cardinality of the power set of $\mathbb{R}$, to the cardinal number $|\mathcal{P}(\mathbb{R})| = 2^{\mathbf{c}}$ there correspond infinitely many ordinals, e.g.. $\Omega, \Omega + 1, \ldots, 2\Omega, \ldots, \Omega^2, \ldots$. By this type of reasoning we cannot ascertain whether there are subsets of $\mathbb{R}$ that are not Lebesgue measurable. To answer that question we must manufacture a linear set that is not Lebesgue measurable.

In 1930 Ulam, using the continuum hypothesis, proved it is not possible to extend the Lebesgue measure to all subsets of $\mathbb{R}$, thus establishing $\mathscr{L}(\mathbb{R}) \neq \mathcal{P}(\mathbb{R})$.

Already in 1905 Vitali had showed the existence of such sets, as did other authors (F. Bernstein (1908), H. Rademacher (1916)). All these constructions rely crucially on the axiom of choice, so they are admissible only within the boundaries of ZFC mathematics.

It was later discovered that these sets lead to apparently paradoxical results compared to everyday's experience. For instance, in 1914 Hausdorff proved the 2-dimensional sphere can be decomposed in finitely many pieces recombining which one can reconstruct two spheres with the same surface area of the initial one. And in 1924 Stefan Banach and Alfred Tarski proved something along these lines for the volume of a solid ball. Needless to say, the sphere cannot be sliced in the obvious way, but requires instead the AC to isolate a number of points on the sphere so that they cannot be suitably connected, and one ends up with pieces that are not measurable. Still, the result is highly counter-intuitive, and sometimes it is presented to the lay person by saying a pea can be chopped up and reassembled into the sun, and hence referred to as the 'pea and sun paradox'. If we think about it, doubling a sphere or turning a pea into the sun is no less surprising than partitioning an infinite set into subsets of the same cardinality. When we subdivide an infinite set, though, the elements are chosen, and then rearranged. In the Hausdorff and Banach–Tarski theorems, on the contrary, the points on the sphere or ball remain where they were before the decomposition. The various pieces are recombined in exactly the same way they were cut.

To keep the AC's unpleasant consequences at bay, set theorists have come up with a host of alternatives. Among the proposals the most interesting appears to be the *axiom of determinacy*, elaborated by J. Mycielski and H. Steinhaus and published in the paper *A mathematical axiom contradicting the axiom of choice*, Bull. Polish Acad. Sci. 10 (1962), 1–3. It posits the existence of winning strategies for a large class of mathematical games. Apart from offering a rich and coherent description of the continuum, the axiom of determinacy allows to prove any set is measurable, and therefore dissolves the Banach–Tarski paradox. Yet we should be asking ourselves

if the new axiom is actually capable of recovering all of the AC's consequences currently in use in mathematics, like Zermelo's well-ordering theorem, Zorn's lemma (any partially ordered set with upper bounded chains has a maximal element), or Tychonov's theorem in topology (the product of any collection of compact spaces is compact).

A noteworthy contribution to the topic came from the Italian mathematician Leonida Tonelli, who wanted to free the theories of measure and integration from the well-ordering postulate. He succeeded by introducing special kinds of sets, called pseudo-intervals, and functions, called almost-continuous, that form a subclass of Lebesgue measurable sets and maps. To the day no one has managed to construct, without Zermelo's axiom, a concrete linear set or a function of a different kind.

Here is Vitali's example.

Theorem 9.5.1 (Subsets of $\mathbb{R}$ that are not Lebesgue measurable) *There exist in* $\mathbb{R}$ *subsets that are not Lebesgue measurable.*

Proof Define on $[0, 1/2]$ the relation $\sim$ by setting

$$E_x = \{y \in [0, 1/2] : y - x \in \mathbb{Q}\}$$

for any $x \in [0, 1/2]$, i.e. the points of $[0, 1/2]$ differing from x by a rational number.

Immediately, for any $x_1, x_2 \in [0, 1/2]$ we have $E_{x_1} = E_{x_2}$ iff $x_1 - x_2 \in \mathbb{Q}$, whilst $E_{x_1} \cap E_{x_2} = \emptyset$ when $x_1 - x_2 \notin \mathbb{Q}$. As cosets in $\{E_x : x \in [0, 1/2]\}$ are either disjoint or they coincide, $\sim$ is an equivalence relation on $[0, 1/2]$.

There are uncountably many equivalence classes E_x, but each is countable and dense in $\mathbb{R}$. Let $E \subset [0, 1/2]$ denote the set containing exactly one element for every equivalence class, chosen with the AC.

The difference of any two points in E is irrational. For every $q \in \mathbb{Q}$ write $E + q$ to denote the set E translated by q, and note that translated sets corresponding to distinct rational numbers are disjoint. In fact any point common to $E + q_1$, $E + q_2$ would come from distinct points $x_1, x_2 \in E$, but then $x_1 + q_1 = x_2 + q_2$ would imply $x_1 - x_2 = q_1 - q_2 \in \mathbb{Q}$, contradicting the irrationality of the difference.

Suppose E is measurable. Then the sequence

$$E + \frac{1}{2}; E + \frac{1}{3}; \ldots; E + \frac{1}{n}; \ldots$$

is made by pairwise-disjoint measurable sets with the same measure of E, since the Lebesgue measure is translation-invariant. As $\bigcup_{n=2}^{\infty} \left(E + \frac{1}{n}\right) \subset [0, 1]$, the non-negativity, monotonicity and continuity of the Lebesgue measure imply

$$0 \leq \ell\left[\bigcup_{n=2}^{\infty}\left(E + \frac{1}{n}\right)\right] = \sum_{n=2}^{\infty} \ell\left(E + \frac{1}{n}\right) = \lim_{n\to\infty} n \cdot \ell(E) \leq 1.$$

The inequality holds only when $\ell(E) = 0$, so if we assume E measurable then its measure is zero. Take the sequence of all rationals $\{q_i\}$ and consider

$$E + q_1; E + q_2; \ldots; E + q_n; \ldots.$$

This sequence partitions $\mathbb{R}$: if $x' \in \mathbb{R}$, given $q_j \in \mathbb{Q}$ such that $x' + q_j \in [0, 1/2]$, there exists an $y \in E$ such that $x' + q_j \in E_y$. Then $x' + q_j - y = q_k$, $q_k \in \mathbb{Q}$, and so $x' = y + q_k - q_j$. Calling $q_r = q_k - q_j$, then $x' \in E + q_r$. Hence

$$\mathbb{R} = \bigcup_{n=1}^{\infty}(E + q_n),$$

and the $(E + q_n)$ are pairwise disjoint and of zero measure. This would imply the absurd fact

$$\ell(\mathbb{R}) = 0.$$

■

Taking in account the definition of E and that a translation superimposes E onto any $E + q_n$ so to speak, Vitali's result can be stated as follows: *"The number line can be decomposed in countably many non-measurable disjoint sets that can be superimposed on one another by translations."*.

From the example we extract two important observations.

Remark 9.5.1 (*Measurable subsets of the Vitali set*) Any measurable subset A of the Vitali set E has zero measure. As $A + (1/n) \subset E + (1/n) \subset [0, 1]$, for $n \geq 2$ the sequence

$$A + \frac{1}{2}; A + \frac{1}{3}; \ldots; A + \frac{1}{n}; \ldots$$

consists of pairwise-disjoint subsets of $[0, 1]$. All these are measurable and have the same measure of A, so

$$0 \leq \ell\left[\bigcup_{n=2}^{\infty}\left(A + \frac{1}{n}\right)\right] = \sum_{n=2}^{\infty}\ell\left(A + \frac{1}{n}\right) = \lim_{n\to\infty} n \cdot \ell(A) \leq 1.$$

The last inequality is true only if $\ell(A) = 0$. ♣

Remark 9.5.2 (*Non-measurable subsets in sets with positive outer measure*) Every $A \subset \mathbb{R}$ such that $\ell^*(A) > 0$ contains a non-Lebesgue measurable subset.

We argue as follows. Since $\bigcup_{n=1}^{\infty}(E + q_n) = \mathbb{R}$, with the $(E + q_n)$ pairwise disjoint, at least one of the sets $A_n = A \cap (E + q_n)$ will not be measurable, otherwise we would have (see previous remark) $\ell(A_n) = 0$ for any $n \in \mathbb{N}$, and then $A = \bigcup_{n=1}^{\infty} A_n$ would end up having $\ell^*(A) = 0$. ♣

In Theorem 9.5.1 Vitali used the three properties that Lebesgue believed could not be waived:

1. there exist sets with non-zero measure;
2. equal sets have equal measures (translation-invariance);
3. the measure of a finite or countable sum of pairwise-disjoint sets equals the sum of the sets' measures (complete additivity).

As we have seen, though, the Lebesgue measure cannot satisfy all three for every subset of $\mathbb{R}$, irrespective of how reasonable the properties may seem. In fact the Vitali set satisfies 3 only when $\ell^*(E) = 0$, which means that, contrary to what we saw, E is measurable. The contradiction shows that number 3 is only valid for **measurable** disjoint sets.

We conclude there exists no measure defined on all subsets of $\mathbb{R}$ at the same time satisfying the previous elementary properties.

9.6 The Lebesgue-Stieltjes Measure

Should we want to measure, instead of the lengths of segments $I \in \mathcal{I}$, the increments of an increasing map[8] $F(x)$ on real intervals $(a, b] \in \tilde{\mathcal{I}}$, we would have to replace the length $m : \mathcal{I} \mapsto [0, \infty]$,

$$m((a, b)) = id(b) - id(a) = b - a,$$

by the set function $m_F : \tilde{\mathcal{I}} \mapsto [0, \infty]$, defined by

$$m_F((a, b]) = F(b) - F(a), \quad (a, b] \in \tilde{\mathcal{I}}$$

where $F : \mathbb{R} \mapsto \mathbb{R}$ is increasing.

We would also like to know whether m_F is finitely and completely additive, regarding which we have the next proposition.

Proposition 9.6.1 (Finite and complete additivity of m_F) *If $F : \mathbb{R} \mapsto \mathbb{R}$ is an increasing map, the set function $m_F : \tilde{\mathcal{I}} \mapsto [0, \infty]$ induced by F,*

$$\begin{aligned} m_F((a, b]) &= F(b) - F(a) \\ m_F((-\infty, b]) &= \lim_{x \to \infty} [F(b) - F(-x)] \\ m_F((a, \infty)) &= \lim_{x \to \infty} [F(x) - F(a)] \\ m_F((-\infty, \infty)) &= \lim_{x \to \infty} [F(x) - F(-x)], \end{aligned}$$

is well defined and finitely additive on the collection of right-closed intervals $\tilde{\mathcal{I}}$.

If F is additionally right-continuous,[9] moreover, m_F is completely additive.

---

[8] A map $F : \mathbb{R} \mapsto \mathbb{R}$ is increasing if $x_1 < x_2 \Rightarrow F(x_1) \leq F(x_2)$ for all $x_1, x_2 \in \mathbb{R}$.

[9] This means for every $x \in \mathbb{R}$ the limits

Proof First of all m_F is well defined, for it always assumes non-negative real values: by the monotonicity of F, for every $a, b \in \bar{\mathbb{R}}$, $a < b$, we have $F(a) \leq F(b)$ and $F(b) - F(a) \in [0, \infty]$.

As for finite additivity, take $\tilde{I}$ in $\tilde{\mathcal{I}}$ such that $\tilde{I} = \bigcup_{i=1}^{n} \tilde{I}_i$, where $\tilde{I}_i \cap \tilde{I}_j = \emptyset$ for every $i \neq j$. We must prove

$$m_F(\tilde{I}) = \sum_{i=1}^{n} m_F(\tilde{I}_i).$$

As F is increasing the order is preserved, so assuming the I_i ordered we will have

$$\begin{aligned} m_F(\tilde{I}_1) + \cdots + m_F(\tilde{I}_n) =& m_F((a_1, b_1]) + m_F((a_2, b_2]) + \cdots + m_F((a_i, b_i]) + \cdots + m_F((a_n, b_n]) \\ =& F(b_1) - F(a_1) + F(b_2) - F(a_2) + \cdots + F(b_i) - F(a_i) + \cdots + \\ & F(b_n) - F(a_n), \end{aligned}$$

so

$$m_F(\tilde{I}) = F(b_n) - F(a_1) = m_F(\tilde{I}_1) + \cdots + m_F(\tilde{I}_n).$$

Suppose then $F(b_i) - F(a_i) = \infty$ for some i, so

$$m_F(\tilde{I}) = +\infty = \sum_{i=1}^{n} m_F(\tilde{I}_i).$$

Instead if $F(b_i) - F(a_i) < \infty$ for every i, by monotonicity of F

$$\begin{aligned} F(b) - F(a) &= F(b_n) - F(a_1) \\ &= \sum_{i=1}^{n} [F(b_i) - F(a_i)] \\ m_F((a, b]) &= \sum_{i=1}^{n} m_F((a_i, b_i]). \end{aligned}$$

In any case, m_F is finitely additive on $\tilde{\mathcal{I}}$.

$$\lim_{\epsilon \to 0} F(x + \epsilon) = F(x^+)$$

$$\lim_{\epsilon \to 0} F(x - \epsilon) = F(x^-),$$

exist, and $\forall x \in \mathbb{R}$, $F(x^+) = F(x)$, $F(x^-) \leq F(x)$. There are at most countably many points at which inequality $F(x^-) < F(x)$ is strict, and there F jumps by $F(x) - F(x^-) > 0$.

Right- and left-continuity are indispensable notions when we need to examine the behaviour of a map at points we cannot study otherwise. For instance the frontier points a, b of the closed bounded domain $[a, b]$.

Suppose now F is increasing and right-continuous. Given any bounded interval $\tilde{I} = (a, b]$ let $\{\tilde{I}_i\}_{i\geq 1} = \{(a_i, b_i]\}_{i\geq 1}$ be a countable cover of $\tilde{I}$:

$$\tilde{I} = \bigcup_{i=1}^{\infty} \tilde{I}_i.$$

We may assume the union is disjoint. As m_F is finitely additive, it is monotone on the algebra $\mathcal{A}(\tilde{\mathcal{I}})$ (Proposition 8.2.1), so for every $n \in \mathbb{N}$

$$\bigcup_{i=1}^{n} \tilde{I}_i \subseteq \tilde{I} \Leftrightarrow \sum_{i=1}^{n} m_F(\tilde{I}_i) \leq m_F(\tilde{I}),$$

and then

$$\sum_{i=1}^{\infty} m_F(\tilde{I}_i) \leq m_F(\tilde{I}). \tag{9.6.1}$$

Conversely, given $\epsilon > 0$, right-continuity on $(a, b]$ implies there is a $\delta > 0$ such that $F(a + \delta) - F(a) < \epsilon$, and for every $(a_i, b_i]$ there is a δ_i such that $F(b_i + \delta_i) - F(b_i) < \epsilon/2^i$. Clearly the open intervals $(a_i, b_i + \delta_i)$ cover the compact set $[a + \delta, b]$, so we can extract a finite subcover $\{(a_j, b_j + \delta_j)\}_{j=1}^{n}$ with

$$\begin{aligned} a < a + \delta = a_1 < b_1 = a_2 < b_1 + \delta_1 < b_2 = a_3 < b_2 + \delta_2 < \ldots < b_{n-2} \\ = a_{n-1} < b_{n-2} + \delta_{n-2} < b_{n-1} = a_n < b_{n-1} + \delta_{n-1} < b_n = b < b_n + \delta_n. \end{aligned}$$

Hence

$$\begin{aligned} m_F(\tilde{I}) &\leq F(b) - F(a + \delta) + \epsilon \leq F(b_n + \delta_n) - F(a_1) + \epsilon \\ &= F(b_n + \delta_n) - F(a_n) + \sum_{j=1}^{n-1} \big(F(a_{j+1}) - F(a_j)\big) + \epsilon \\ &\leq F(b_n + \delta_n) - F(a_n) + \sum_{j=1}^{n-1} \big(F(b_j + \delta_j) - F(a_j)\big) + \epsilon \\ &= \sum_{j=1}^{n} m_F\big((a_j, b_j + \delta_j]\big) + \epsilon \leq \sum_{i=1}^{\infty} m_F\big((a_i, b_i]\big) + 2\epsilon \\ &= \sum_{i=1}^{\infty} m_F(\tilde{I}_i) + 2\epsilon. \end{aligned}$$

Because $\epsilon > 0$ is arbitrary, and by inequality (9.6.1), we obtain

$$m_F(\tilde{I}) = \sum_{i=1}^{\infty} m_F(\tilde{I}_i).$$

Observe m_F is also σ-finite on the algebra $\mathcal{A}(\tilde{\mathcal{I}})$, so what we proved above carries through for every $n \in \mathbb{Z}$ and every set in $\mathcal{A}(\tilde{\mathcal{I}}) \cap (n, n+1]$. So when $a = -\infty$, the same argument gives

$$m_F(\tilde{I}_n) = m_F\big((-n, b]\big) = F(b) - F(-n) \leq \sum_{i=1}^{\infty} m_F(\tilde{I}_i) + \frac{\epsilon}{2^n}$$

for every $n \in \mathbb{N}$. And choosing $b = \infty$,

$$m_F(\tilde{I}^n) = m_F\big((a, n]\big) = F(n) - F(a) \leq \sum_{i=1}^{\infty} m_F(\tilde{I}_i) + \frac{\epsilon}{2^n}.$$

Taking the limit as $n \to \infty$ concludes the proof. ■

The Hahn-Kolmogorov extension theorem (Theorem 9.3.1) tells $m_F : \mathcal{A}(\tilde{\mathcal{I}}) \mapsto [0, \infty]$, being σ-finite, admits a unique extension to the σ-algebra $\mathcal{S}(\tilde{\mathcal{I}})$. Finally, since $\mathcal{S}(\tilde{\mathcal{I}}) = \mathscr{B}(\mathbb{R})$, the measure space $\big(\mathbb{R}, \mathscr{L}(\mathbb{R}), m_F\big)$ is the completion of $\big(\mathbb{R}, \mathscr{B}(\mathbb{R}), m_F\big)$. The measure m_F is called **Lebesgue-Stieltjes measure** generated by F, and $\big(\mathbb{R}, \mathscr{L}(\mathbb{R}), m_F\big)$ is the **Lebesgue-Stieltjes measure space**. Whenever necessary we shall indicate with $\mathscr{L}_F$ the Lebesgue σ-algebra with measure m_F.

Remark 9.6.1 (*Lebesgue measure as special case of* m_F) If we choose $F(x) = x$, then $m_F((a, b]) = b - a$ and the extension of m_F is the classical Lebesgue measure on $\mathbb{R}$. The identity map $F(x) = x$, in fact, is increasing and right-continuous for each right-closed, left-open interval. ♣

Clearly we can find various increasing and right-continuous maps $F_i, F_j : \mathbb{R} \mapsto \mathbb{R}$, $i \neq j = 1, \ldots, n$ such that $m_{F_i} = m_{F_j}$. This happens iff the difference $F_i(x) - F_j(x)$ is constant for every $x \in \mathbb{R}$, in fact if F_j is increasing and right-continuous, by the previous theorem it induces a measure m_{F_j}, so for $a, b \in \mathbb{R}$

$$m_{F_i}\big((a, b]\big) - m_{F_j}\big((a, b]\big) = F_i(b) - F_i(a) - F_j(b) + F_j(a),$$

and then $m_{F_i} - m_{F_j} = 0$ iff $F_i(a) - F_j(a) = k = F_i(b) - F_j(b)$.

If $\tilde{m}$ is a finitely additive set function on $\mathscr{B}(\mathbb{R})$ that is further finite on bounded Borel sets, the monotonicity of $\tilde{m}$ forces the existence of some increasing map $F : \mathbb{R} \mapsto \mathbb{R}$ such that $\tilde{m}\big((a, b]\big) = F(b) - F(a)$ and $\tilde{m} = m_F$ on any $(a, b]$. If $\tilde{m}$ is completely additive as well, by the continuity of $\tilde{m}$ from above and below F is right-continuous. In practice, the converse to Proposition 9.6.1 holds:

Proposition 9.6.2 (Function induced by $\tilde{m} : \tilde{\mathcal{I}} \mapsto [0, \infty]$) *If $\tilde{m} : \tilde{\mathcal{I}} \mapsto [0, \infty]$ is a finitely additive set function such that, for every $a, b \in \mathbb{R}$, $\tilde{m}\big((a, b]\big) < \infty$, then there*

exists an increasing map $F : \mathbb{R} \mapsto \mathbb{R}$ *satisfying* $\tilde{m}\big((a,b]\big) = F(b) - F(a)$ *for every* $a, b \in \mathbb{R}$.

Furthermore if $\tilde{m}$ *is completely additive,* F *is right-continuous:* $\lim_{y \to x^+} F(y) = F(x^+) = F(x),\ \forall x \in \mathbb{R}$.

Proof Note $\tilde{m}((a,b])$ is given by $F(b) - F(a)$, which suggests how to define F. For instance suppose we fix a point a, then we compute F at any $b \in \mathbb{R}$ by $F(b) = \tilde{m}((a,b]) + F(a)$. A convenient choice for a is the origin, in which case we put

$$F(x) = \begin{cases} \tilde{m}((0,x]) \text{ if } x > 0, \\ 0 \text{ if } x = 0, \\ -\tilde{m}((x,0]) \text{ if } x < 0. \end{cases}$$

We claim this choice satisfies

$$\tilde{m}((a,b]) = F(b) - F(a)$$

for every $(a,b]$. To see this, just compute

$$\begin{aligned} F(b) - F(a) &= \tilde{m}((0,b]) - \tilde{m}((0,a]) \\ &= F(b) - F(0) - F(a) + F(0) \\ &= \tilde{m}((a,b]). \end{aligned}$$

Now let us show F is increasing. Take $x, y \in \mathbb{R}$ such that $x < y$ and suppose they are positive:

$$F(y) = \tilde{m}((0,y]) = \tilde{m}((0,x] \cup (x,y]) = \tilde{m}((0,x]) + \tilde{m}((x,y]) \geq F(x).$$

If both are negative, with $y < x$,

$$\begin{aligned} F(y) = -\tilde{m}((y,0]) &= -\tilde{m}\big((y,x] \cup (x,0]\big) = -\tilde{m}((y,x]) - \tilde{m}((x,0]) \\ &\leq -\tilde{m}((y,x]) + F(x) \leq F(x). \end{aligned}$$

Finally if $y < 0 < x$,

$$0 \leq \tilde{m}((y,x]) = F(x) - F(y) \Leftrightarrow F(y) \leq F(x)$$

and F is indeed increasing.

Now we prove F is right-continuous provided $\tilde{m}$ is completely additive. Take $x \in \mathbb{R}$ and a sequence $\{x_n\}_{n \geq 1}$ with $x_n \downarrow x$. It suffices to show $F(x_n) \to F(x)$.

In case $x \geq 0$, $0 \leq x < \cdots \leq x_n \leq \cdots \leq x_2 \leq x_1$,

$$(0, x_1] = (0,x] \cup \underbrace{(x, x_1]}_{\bigcup_{n=2}^{\infty} (x_n, x_{n-1}]}.$$

By countable additivity,

$$\begin{aligned}
\tilde{m}((0,x_1]) &= \tilde{m}((0,x]) + \tilde{m}((x,x_1]) \\
F(x_1) &= F(x) + \tilde{m}\left(\bigcup_{n=2}^{\infty}(x_n,x_{n-1}]\right) = F(x) + \sum_{n=2}^{\infty}\tilde{m}((x_n,x_{n-1}]) \\
&= F(x) + \lim_{k\to\infty}\left[\sum_{n=2}^{k}\tilde{m}((x_n,x_{n-1}])\right] \\
&= F(x) + \lim_{k\to\infty}\left[\sum_{n=2}^{k}[F(x_{n-1}) - F(x_n)]\right] \\
&= F(x) + \lim_{k\to\infty}\left[F(x_1) - F(x_2) + F(x_2) - \ldots + F(x_{k-1}) - F(x_k)\right] \\
&= F(x) + F(x_1) - \lim_{k\to\infty} F(x_k),
\end{aligned}$$

and hence $\lim_{k\to\infty} F(x_k) = F(x)$ for every $x \geq 0$.

In case $x < 0$, $x \leq \cdots \leq x_n \leq \cdots \leq x_2 \leq x_1 < 0$, and

$$(x,0] = \underbrace{(x,x_1]}_{\bigcup_{n=2}^{\infty}(x_n,x_{n-1}]} \cup (x_1,0].$$

Countable additivity gives

$$\begin{aligned}
\tilde{m}((x,0]) &= \tilde{m}((x,x_1]) + \tilde{m}((x_1,0]) \\
-F(x) &= \tilde{m}\left(\bigcup_{n=2}^{\infty}(x_n,x_{n-1}]\right) - F(x_1) = \sum_{n=2}^{\infty}\tilde{m}((x_n,x_{n-1}]) - F(x_1) \\
&= \lim_{k\to\infty}\left[\sum_{n=2}^{k}\tilde{m}((x_n,x_{n-1}])\right] - F(x_1) \\
&= \lim_{k\to\infty}\left[\sum_{n=2}^{k}[F(x_{n-1}) - F(x_n)]\right] - F(x_1) \\
&= \lim_{k\to\infty}\left[F(x_1) - F(x_2) + F(x_2) - \ldots + F(x_{k-1}) - F(x_k)\right] - F(x_1) \\
&= F(x_1) - \lim_{k\to\infty} F(x_k) - F(x_1),
\end{aligned}$$

and similarly, $\lim_{k\to\infty} F(x_k) = F(x)$ for every $x < 0$. ■

We have shown one can construct completely additive set functions on right-closed intervals provided they arise through right-continuous, increasing maps. Moreover,

the measure $\tilde{m}$ on $\mathscr{B}(\mathbb{R})$ such that $\tilde{m}\big((a,b]\big) < \infty, \forall a, b \in \mathbb{R}, a < b$, can be viewed as the Lebesgue-Stieltjes measure for some increasing, right-continuous F.

The measure $\tilde{m}$ is totally finite iff

$$-\infty < F(-\infty) < F(\infty) < \infty.$$

In that case, in Proposition 9.6.2 the map $F : \mathbb{R} \mapsto \mathbb{R}$ will be given by

$$F(x) = \tilde{m}\big((-\infty, x]\big).$$

Right-continuous increasing maps $F : \mathbb{R} \mapsto \mathbb{R}$, as characterised by Propositions 9.6.1, 9.6.2, are called cumulative distribution functions of $\tilde{m}$ on $\mathbb{R}$. In case $\tilde{m}(\mathbb{R}) = 1$, $\tilde{m}$ is called probability measure and the corresponding F is the cumulative distribution function where

$$\lim_{x\to\infty} \big[F(x) - F(-x)\big] = \tilde{m}(\mathbb{R}) = 1.$$

Chapter 10
Measurable Functions



10.1 Introduction

For a function f to be measurable the pre-image of any measurable set in the codomain must be measurable. We met a similar situation when discussing the Lebesgue–Hausdorff theorem, and more specifically when we defined the class of Borel functions (Definition 5.1.1). On the basis of that definition and the subsequent discussion about measurable sets, we can say Borel functions represent a special instance of family of measurable functions. Now the time has come to address the matter from a broader angle.

Definition 10.1.1 (*Measurable functions*[1]) Let $(X, \mathcal{S}(\mathcal{A}))$ and $(X', \mathcal{S}'(\mathcal{A}))$ be measurable spaces. We shall say that a function $f : (X, \mathcal{S}(\mathcal{A})) \mapsto (X', \mathcal{S}'(\mathcal{A}))$ is $(\mathcal{S}(\mathcal{A}), \mathcal{S}'(\mathcal{A}))$-measurable, or just measurable, if $f^{-1}(A') \in \mathcal{S}(\mathcal{A})$ for every $A' \in \mathcal{S}'(\mathcal{A})$. ◇

Theorem 10.1.1 (Characterisation of measurable functions) *Given measurable spaces* $(X, \mathcal{S}(\mathcal{A}))$, $(X', \mathcal{S}(\mathcal{A}'))$, *a map* $f : X \mapsto X'$ *is* $(\mathcal{S}(\mathcal{A}), \mathcal{S}(\mathcal{A}'))$*-measurable if and only if, for every* $A' \in \mathcal{A}'$,

$$f^{-1}(A') \in \mathcal{S}(\mathcal{A}).$$

Proof ($\Rightarrow$): trivial.
($\Leftarrow$): suppose $f^{-1}(A') \in \mathcal{S}(\mathcal{A})$ for every $A' \in \mathcal{A}'$.
Take

$$\mathcal{F} = \left\{A' \in \mathcal{A}' : f^{-1}(A') \in \mathcal{S}(\mathcal{A})\right\}.$$

[1] Measurable functions are called random variables in probability theory.

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S. Gentili, *Measure, Integration and a Primer on Probability Theory*,
UNITEXT 125, https://doi.org/10.1007/978-3-030-54940-4_10

Clearly $\mathcal{A}' \subseteq \mathcal{F}$. As $f^{-1}(\emptyset) = \emptyset \in \mathcal{S}(\mathcal{A})$, then $\emptyset \in \mathcal{F}$. Let $A' \in \mathcal{F}$, so $f^{-1}(A') \in \mathcal{S}(\mathcal{A})$, and then $\left(f^{-1}(A')\right)^c = f^{-1}\left((A')^c\right) \in \mathcal{S}(\mathcal{A})$, i.e. $(A')^c \in \mathcal{F}$. Hence $X' \in \mathcal{F}$. Let $\{A'_i\}_{i\geq 1}$ be a sequence in $\mathcal{F}$, then by definition $\left\{f^{-1}(A'_i)\right\}_{i\geq 1} \subseteq \mathcal{S}(\mathcal{A})$, and so

$$f^{-1}\left(\bigcup_{i=1}^{\infty} A'_i\right) = \bigcup_{i=1}^{\infty} f^{-1}(A'_i) \in \mathcal{S}(\mathcal{A}).$$

In other words $\mathcal{F}$ is closed under countable unions, whence a σ-algebra containing $\mathcal{A}'$, but $\mathcal{S}(\mathcal{A}')$ is the σ-algebra generated by $\mathcal{A}'$, then $\mathcal{S}(\mathcal{A}') \subseteq \mathcal{F}$, and therefore $f^{-1}(A') \in \mathcal{S}(\mathcal{A})$ for every $A' \in \mathcal{S}(\mathcal{A}')$. ■

Observe that f is measurable if to each set in $\mathcal{S}(\mathcal{A}')$ there corresponds under f^{-1} at least one set in $\mathcal{S}(\mathcal{A})$. Therefore a measurable function f, from the measurable space $(X, \mathcal{S}(\mathcal{A}))$ to the measurable space $(X', \mathcal{S}(\mathcal{A}'))$, is such that the σ-algebra $f^{-1}(\mathcal{S}(\mathcal{A}'))$ is contained in the σ-algebra $\mathcal{S}(\mathcal{A})$. Consequently the source σ-algebra contains as many (measurable) sets as the target σ-algebra.

Definition 10.1.2 (*Borel and Lebesgue measurable functions*) Let $(X, \mathcal{T})$, $(X', \mathcal{T}')$ be topological spaces with respective Borel σ-algebras $\mathscr{B}(\mathcal{T})$, $\mathscr{B}(\mathcal{T}')$, and let $\mathscr{L}(\mathcal{T})$ denote the completion of $\mathscr{B}(\mathcal{T})$. A map $f : X \mapsto X'$ is called Borel measurable if it is $(\mathscr{B}(\mathcal{T}), \mathscr{B}(\mathcal{T}'))$-measurable, and we call it Lebesgue measurable if it is $(\mathscr{L}(\mathcal{T}), \mathscr{B}(\mathcal{T}'))$-measurable. ◇

It is straightforward that a Borel measurable map $f : X \mapsto X'$ is Lebesgue measurable, since $\mathscr{B}(\mathcal{T}) \subseteq \mathscr{L}(\mathcal{T})$.

Proposition 10.1.1 (Borel measurability of continuous maps) *Given topological spaces $(X, \mathcal{T})$, $(X', \mathcal{T}')$, a continuous map $f : X \mapsto X'$ is Borel measurable.*

Proof By continuity, $f^{-1}(A') \in \mathcal{T}$ for every $A' \in \mathcal{T}'$. As open sets in $\mathcal{T}$, $\mathcal{T}'$ generate the Borel σ-algebras $\mathscr{B}(\mathcal{T})$ and $\mathscr{B}(\mathcal{T}')$, by Theorem 10.1.1 f is $(\mathscr{B}(\mathcal{T}), \mathscr{B}(\mathcal{T}'))$-measurable. ■

So $f : X \mapsto X'$ continuous implies $f^{-1}(A') \in \mathscr{B}(\mathcal{T})$ for every Borel set $A' \in \mathscr{B}(\mathcal{T}')$. In Example 10.5.1 we will see the same does not happen for Lebesgue measurable sets, because the continuous pre-image of a Lebesgue measurable set is not Lebesgue measurable, in general.

Moreover, given topological spaces $(X, \mathcal{T})$, $(X', \mathcal{T}')$, $(X'', \mathcal{T}'')$, if $f : X \mapsto X'$ and $g : X' \mapsto X''$ are continuous and in addition f is Borel or Lebesgue measurable and g is Borel measurable, then the composite $g \circ f : X \mapsto X''$ is of the same type as f.

10.2 $\bar{\mathbb{R}}$-valued Measurable Functions

We turn to maps $f : X \mapsto \bar{\mathbb{R}}$ defined on a measurable space $(X, \mathcal{S})$, where X is non-empty and $\mathcal{S}$ the σ-algebra of subsets of X. In analogy to Definition 10.1.1 we can introduce measurable functions with values in the extended reals.

Definition 10.2.1 (*Measurable functions with values in* $\bar{\mathbb{R}}$) Given a measurable space $(X, \mathcal{S})$, a function $f : X \mapsto \bar{\mathbb{R}}$ is $\mathcal{S}$-measurable or simply measurable if it is $(\mathcal{S}, \mathscr{B}(\bar{\mathbb{R}}))$-measurable.

We indicate with $\mathscr{M}(X, \mathcal{S})$ the collection of $\mathcal{S}$-measurable functions, with $\mathscr{M}^+(X, \mathcal{S})$ that of non-negative $\mathcal{S}$-measurable maps. ◇

The terminology leads to a result summarising the various equivalent ways to describe $\bar{\mathbb{R}}$-valued measurable maps.

Proposition 10.2.1 (Equivalent ways to express measurability of maps) *Let* $(X, \mathcal{S})$ *be a measurable space,* $f : X \mapsto \bar{\mathbb{R}}$ *a map. The following facts are equivalent:*

1. f *is* $\mathcal{S}$*-measurable;*
2. $f^{-1}\big([\alpha, \infty]\big) = \{x \in X : f(x) \in [\alpha, \infty]\} \in \mathcal{S}, \ \forall \alpha \in \mathbb{R}$*;*
3. $f^{-1}\big((\alpha, \infty]\big) = \{x \in X : f(x) \in (\alpha, \infty]\} \in \mathcal{S}, \ \forall \alpha \in \mathbb{R}$*;*
4. $f^{-1}\big([-\infty, \alpha]\big) = \{x \in X : f(x) \in [-\infty, \alpha]\} \in \mathcal{S}, \ \forall \alpha \in \mathbb{R}$*;*
5. $f^{-1}\big([-\infty, \alpha)\big) = \{x \in X : f(x) \in [-\infty, \alpha)\} \in \mathcal{S}, \ \forall \alpha \in \mathbb{R}$*;*
6. $f^{-1}\big(\{+\infty\}\big) = \Big\{x \in X : f(x) \in \bigcap_{n=1}^{\infty}(n, \infty]\Big\} \in \mathcal{S}$*;*
7. $f^{-1}\big(\{-\infty\}\big) = \Big\{x \in X : f(x) \in \bigcap_{n=1}^{\infty}[-\infty, -n)\Big\} \in \mathcal{S}$.

Proof To show *1.* $\Leftrightarrow$ *2.* observe that

$$\mathscr{B}(\bar{\mathbb{R}}) = \big\{[\alpha, \infty] \subseteq \bar{\mathbb{R}} : f^{-1}\big([\alpha, \infty]\big) \in \mathcal{S}, \forall \alpha \in \mathbb{R}\big\},$$

so

$$\forall \alpha \in \mathbb{R}, \exists [\alpha, \infty] \in \mathscr{B}(\bar{\mathbb{R}}) \text{ such that the set } f^{-1}\big([\alpha, \infty]\big) \in \mathcal{S},$$

is equivalent to

$$\forall A' \in \mathcal{S}'(\mathcal{A}), \text{ the set } f^{-1}\big(A'\big) \in \mathcal{S},$$

i.e. the Definition 10.1.1. The equivalences *1.* $\Leftrightarrow$ *3.*, *1.* $\Leftrightarrow$ *4.*, *1.* $\Leftrightarrow$ *5.* are completely similar.

So let us assume f is measurable (as of Definition 10.1.1, or in the sense of *2.*, *3.*, *4.* or *5.*). Since $f^{-1}\big((\alpha, \infty]\big)$ is measurable, setting $n = \alpha$ gives

$$\bigcap_{n=1}^{\infty}\Big(f^{-1}\big((n, +\infty]\big)\Big) = f^{-1}\Big(\bigcap_{n=1}^{\infty}(n, +\infty]\Big) = f^{-1}\big(\{+\infty\}\big) \in \mathcal{S}$$

is measurable. And since $f^{-1}\big([-\infty, \alpha)\big)$ is measurable, using $-n = \alpha$ we find

$$\bigcap_{n=1}^{\infty}\Big(f^{-1}\big([-\infty, -n)\big)\Big) = f^{-1}\Big(\bigcap_{n=1}^{\infty}[-\infty, -n)\Big) = f^{-1}\big(\{-\infty\}\big) \in \mathcal{S}$$

is measurable, too. ■

Remark 10.2.1 1. Based on the above proposition we can also say $f : X \mapsto \bar{\mathbb{R}}$ is $\mathcal{S}$-measurable iff $f^{-1}\big((\alpha, \beta)\big) \in \mathcal{S}$ for every $\alpha, \beta \in \mathbb{R}$, $\alpha \leq \beta$. In fact

$$f^{-1}\big((\alpha, \beta)\big) = f^{-1}\big([-\infty, \beta)\big) \cap f^{-1}\big((\alpha, \infty]\big) \in \mathcal{S},$$

and similarly,

$$f^{-1}\big([\alpha, \beta)\big) = f^{-1}\big([-\infty, \beta)\big) \cap f^{-1}\big([\alpha, \infty]\big) \in \mathcal{S};$$

$$f^{-1}\big((\alpha, \beta]\big) = f^{-1}\big([-\infty, \beta]\big) \cap f^{-1}\big((\alpha, \infty]\big) \in \mathcal{S};$$

$$f^{-1}\big([\alpha, \beta]\big) = f^{-1}\big([-\infty, \beta]\big) \cap f^{-1}\big([\alpha, \infty]\big) \in \mathcal{S}.$$

In conclusion, $f : X \mapsto \bar{\mathbb{R}}$ is $\mathcal{S}$-measurable iff $f^{-1}(I) \in \mathcal{S}$ for every interval $I \subseteq \bar{\mathbb{R}}$. But open sets are countable unions of open pairwise-disjoint intervals, so $f : X \mapsto \bar{\mathbb{R}}$ is $\mathcal{S}$-measurable iff $f^{-1}(G) \in \mathcal{S}$ for every open $G \in \mathscr{B}(\bar{\mathbb{R}})$: namely $G = \bigsqcup_{i=1}^{\infty} I_i$ and

$$f^{-1}(G) = f^{-1}\left(\bigsqcup_{i=1}^{\infty} I_i\right) = \bigsqcup_{i=1}^{\infty} \big(f^{-1}(I_i)\big) \in \mathcal{S}.$$

Take $X = \mathbb{R}$ in Proposition 10.2.1: then $f : \mathbb{R} \mapsto \bar{\mathbb{R}}$ is Lebesgue or Borel measurable iff

a. $f^{-1}(I) \in \mathscr{L}(\mathbb{R})$, or
b. $f^{-1}(I) \in \mathscr{B}(\mathbb{R})$.

respectively, for every interval $I \subseteq \bar{\mathbb{R}}$.

2. For any $A \in \mathcal{S}$ the indicator function $\varphi_A(x)$ is $\mathcal{S}$-measurable. As $\varphi_A = 1$ for every $x \in A$ and $\varphi_A = 0$ for every $x \notin A$, then for all $\beta, \alpha \in \mathbb{R}$

$$\varphi_A^{-1}\big([\beta, \alpha)\big) = \begin{cases} X \text{ if } \beta \leq 0 \text{ and } \alpha > 1 \\ A \text{ if } \beta > 0 \text{ and } \alpha > 1 \\ A^c \text{ if } \beta \leq 0 \text{ and } \alpha \leq 1 \\ \emptyset \text{ if } \alpha \leq 0. \end{cases}$$

This is measurable since $X, A, A^c, \emptyset \in \mathcal{S}$.

3. Constant maps are $\mathcal{S}$-measurable, as continuous. Let $f : X \mapsto \bar{\mathbb{R}}$ be the constant map $f(x) = k$. Then

$$f^{-1}\big((\alpha, \infty]\big) = \big\{x \in X : f(x) > \alpha\big\} = \begin{cases} X \text{ if } \alpha < k \\ \emptyset \text{ if } \alpha \geq k, \end{cases}$$

and $\mathcal{S}$-measurability follows.

4. If $f : X \mapsto \bar{\mathbb{R}}$ is continuous everywhere but for an enclosable set, it is $\mathcal{S}$-measurable. Let $E \in \mathcal{S}$ be an enclosable set on which f is not continuous, and recall enclosable implies $\mathcal{S}$-measurable, of zero measure. Given an interval $I \subseteq \bar{\mathbb{R}}$,

$$\begin{aligned} f^{-1}(I) &= \big\{x \in X : f(x) \in I\big\} \\ &= \big(\big\{x \in X : f(x) \in I\big\} \cap E\big) \cup \big(\big\{x \in X : f(x) \in I\big\} \cap E^c\big). \end{aligned}$$

So $\big(\big\{x \in X : f(x) \in I\big\} \cap E\big)$ is, as subset of E, $\mathcal{S}$-measurable of zero measure. But on E^c f is continuous, so $\big(\big\{x \in X : f(x) \in I\big\} \cap E^c\big)$ is $\mathcal{S}$-measurable. Hence $f^{-1}(I) = \big\{x \in X : f(x) \in I\big\}$ is $\mathcal{S}$-measurable.

5. A monotone map $f : \mathbb{R} \mapsto \mathbb{R}$ (e.g., increasing) preserves the linear order, so for any

$$I \in \big\{[\alpha, \infty) : \alpha \in \mathbb{R}\big\}$$

$f^{-1}(I)$ is $\emptyset$ or an interval of $\mathcal{I}$, hence in class $\mathscr{B}(\mathbb{R})$. ♣

Proposition 10.2.2 (Comparing measurable functions) *Let $f, g : X \mapsto \bar{\mathbb{R}}$ be in $\mathscr{M}(X, \mathcal{S})$. The following sets are $\mathcal{S}$-measurable*

1. $\big\{x \in X : f(x) < g(x)\big\}$;
2. $\big\{x \in X : f(x) > g(x)\big\}$;
3. $\big\{x \in X : f(x) = g(x)\big\}$;
4. $\big\{x \in X : f(x) \geq g(x)\big\}$;
5. $\big\{x \in X : f(x) \leq g(x)\big\}$.

Proof Let us commence with *1.*. For every $x \in X$ there is a $q \in \mathbb{Q}$ such that $f(x) < q < g(x)$, so

$$\begin{aligned} \big\{x \in X : f(x) < g(x)\big\} &= \bigcup_{q \in \mathbb{Q}} \big\{x \in X : f(x) < q < g(x)\big\} \\ &= \bigcup_{q \in \mathbb{Q}} \Big(\big\{x \in X : f(x) < q\big\} \cap \big\{x \in X : q < g(x)\big\}\Big) \\ &= \bigcup_{q \in \mathbb{Q}} \big(f^{-1}([-\infty, q)) \cap g^{-1}((q, \infty])\big). \end{aligned}$$

Clearly $f^{-1}([-\infty, q))$ and $g^{-1}((q, \infty])$ are $\mathcal{S}$-measurable, and so is $\big\{x \in X : f(x) < g(x)\big\}$. From this, also

$$\big\{x \in X : f(x) < g(x)\big\}^c = \big\{x \in X : f(x) \geq g(x)\big\}$$

is $\mathcal{S}$-measurable. And similarly for

$$\begin{aligned}&\big\{x \in X : g(x) < f(x)\big\} \in \mathcal{S}\\ \Leftrightarrow &\big\{x \in X : g(x) < f(x)\big\}^c = \big\{x \in X : g(x) \geq f(x)\big\} \in \mathcal{S}\end{aligned}$$

and

$$\big\{x \in X : f(x) = g(x)\big\} = \big\{x \in X : f(x) \geq g(x)\big\} \cap \big\{x \in X : f(x) \leq g(x)\big\} \in \mathcal{S}.$$

■

Now we shall examine if and when measurability is preserved by operations between measurable functions.

Theorem 10.2.1 (Elementary operations between measurable functions) *Let* $f, g : (X, \mathcal{S}) \mapsto \bar{\mathbb{R}}$ *be in* $\mathscr{M}(X, \mathcal{S})$ *and* $c \in \mathbb{R}$. *Define*

(a) $0 \cdot (\pm\infty) = 0$, *by convention;*
(b) $f(x) + g(x) = 0$ *and* $f(x) \cdot g(x) = 0$ $\forall x \in E_1 \cup E_2$, *where*

$$E_1 = \{x \in X : f(x) = -\infty\} \cap \{x \in X : g(x) = +\infty\}$$

$$E_2 = \{x \in X : f(x) = +\infty\} \cap \{x \in X : g(x) = -\infty\}$$

are measurable (cf. 6.–7. in Proposition 10.2.1).

Then the following belong to $\mathscr{M}(X, \mathcal{S})$*:*

1. $c \cdot f(x) = (c \cdot f)(x)$;
2. $f(x) + g(x) = (f + g)(x)$;
3. $f(x) \cdot f(x) = f^2(x)$;
4. $f(x) \cdot g(x) = (f \cdot g)(x)$;
5. $f(x)/g(x)$ *if* $g(x) \neq 0, \forall x \in X$;
6. $|f(x)|$.

Proof *1.* For $c = 0$ we have

$$(c \cdot f)^{-1}((\alpha, \infty]) = \{x \in X : \underbrace{c \cdot f(x)}_{=0} > \alpha\} = \begin{cases} X & \text{if } \alpha < 0 \\ \emptyset \text{ if } \alpha \geq 0, \end{cases}$$

whence $0 \cdot f(x)$ is $\mathcal{S}$-measurable. For $c \neq 0$,

$$(c \cdot f)^{-1}((\alpha, \infty]) = \big\{x \in X : c \cdot f(x) > \alpha\big\} = \begin{cases} \big\{x \in X : f(x) > \alpha/c\big\} \text{ if } c > 0 \\ \big\{x \in X : f(x) < \alpha/c\big\} \text{ if } c < 0. \end{cases}$$

Now f measurable implies these sets are measurable, so $c \cdot f(x) \in \mathscr{M}(X, \mathcal{S})$.

2. Without loss of generality take $x \in X$ such that $f(x) + g(x) > \alpha$, i.e. $f(x) > \alpha - g(x)$. Then there is a rational number q such that $f(x) > q > \alpha - g(x)$, so $f(x) > q$ and $g(x) > \alpha - q$. Immediately

$$\begin{aligned}(f+g)^{-1}((\alpha,\infty]) &= \big\{x \in X : f(x) + g(x) > \alpha\big\} \\ &= \bigcup_{q\in\mathbb{Q}} \Big[\big\{x \in X : f(x) > q\big\} \cap \big\{x \in X : \alpha - q < g(x)\big\}\Big].\end{aligned}$$

By assumption $f(x), g(x)$ are $\mathcal{S}$-measurable so the last term is a countable union in $\mathcal{S}$, making $f(x) + g(x)$ $\mathcal{S}$-measurable.

3.

$$\begin{aligned}(f^2)^{-1}((\alpha,\infty]) &= \big\{x \in X : f^2(x) > \alpha\big\} \\ &= \begin{cases} X \text{ if } \alpha < 0 \\ \big\{x \in X : f(x) > \sqrt{\alpha}\big\} \cup \big\{x \in X : f(x) < -\sqrt{\alpha}\big\} & \text{if } \alpha \geq 0. \end{cases}\end{aligned}$$

In either case $f^2(x)$ is $\mathcal{S}$-measurable.

4. Since

$$\begin{aligned}f(x)\cdot g(x) &= \frac{f^2(x) + g^2(x) - f^2(x) - g^2(x) + 2f(x)g(x) + 2f(x)g(x)}{4} \\ &= \frac{\big(f(x)+g(x)\big)^2 - \big(f(x)-g(x)\big)^2}{4},\end{aligned}$$

by parts *2.–3.* the product map is $\mathcal{S}$-measurable.

5. If we show $1/g$ is $\mathcal{S}$-measurable, the claim follows from *4.* Clearly $1/g$ is defined on $A = X \setminus \{x \in X : g(x) = 0\}$. Note $\{x \in X : g(x) = 0\} \in \mathcal{S}$, because in this case g is constant. To show $1/g$ is $\mathcal{S}$-measurable we prove that for every $\alpha \in \mathbb{R}$

$$\left(\frac{1}{g(x)}\right)^{-1}\big((\alpha,\infty]\big) = \left\{x \in A : \frac{1}{g(x)} > \alpha\right\} \in \mathcal{S}.$$

Suppose $\alpha = 0$, so $1/g(x) > 0$. This means $g(x) > 0$ and $g(x) \neq \infty$, and then

$$\left\{x \in A : \frac{1}{g(x)} > 0\right\} = \big\{x \in A : g(x) > 0\big\} \setminus \big\{x \in A : g(x) = \infty\big\} \in \mathcal{S}.$$

If $\alpha > 0$, because $1/g(x) > \alpha$, then $g(x) > 0$ and $g(x) < 1/\alpha$, so

$$\left\{x \in A : \frac{1}{g(x)} > \alpha\right\} = \left\{x \in A : 0 < g(x) < \frac{1}{\alpha}\right\} \in \mathcal{S}.$$

Finally if $\alpha < 0$, as $1/g(x) > \alpha$, then we can have $1/g(x) > \alpha$ and $g(x) > 0$ and/or $1/g(x) > \alpha$ and $g(x) < 0$. Consequently

$$\left\{x \in A : \frac{1}{g(x)} > \alpha\right\} = \left\{x \in A : \frac{1}{g(x)} > \alpha \text{ and } g(x) > 0\right\} \cup \left\{x \in A : \frac{1}{g(x)} > \alpha \text{ and } g(x) < 0\right\}.$$

But $\alpha < 0$ so regarding the first set

$$\left\{x \in A : \frac{1}{g(x)} > \alpha \text{ and } g(x) > 0\right\} = \left\{x \in A : g(x) > 0\right\} \in \mathcal{S},$$

while for the second set

$$\begin{aligned}\left\{x \in A : \frac{1}{g(x)} > \alpha \text{ and } g(x) < 0\right\} &= \left\{x \in A : g(x) < \frac{1}{\alpha} \text{ and } g(x) < 0\right\} \\ &= \left\{x \in A : g(x) < \frac{1}{\alpha}\right\} \in \mathcal{S}.\end{aligned}$$

Eventually, $\left\{x \in A : \frac{1}{g(x)} > \alpha\right\}$ is $\mathcal{S}$-measurable.

6. When $\alpha < 0$, then $|f(x)| > \alpha$, so

$$(|f|)^{-1}((\alpha, \infty]) = \left\{x \in X : |f(x)| > \alpha\right\} = X.$$

When $\alpha \geq 0$,

$$\begin{aligned}(|f|)^{-1}((\alpha, \infty]) &= \left\{x \in X : |f(x)| > \alpha\right\} \\ &= \left\{x \in X : f(x) > \alpha\right\} \cup \left\{x \in X : f(x) < -\alpha\right\}.\end{aligned}$$

In either case $(|f|)^{-1}((\alpha, \infty]) \in \mathcal{S}$.

■

Definition 10.2.2 (*Positive and negative part of a map*) We call positive part and negative part of $f : X \mapsto \bar{\mathbb{R}}$ the two non-negative maps

$$f^+(x) = \sup\{f(x), 0\} \quad = \begin{cases} f(x) \text{ if } f(x) \geq 0 \\ 0 \text{ if } f(x) < 0 \end{cases}$$

$$f^-(x) = \sup\{-f(x), 0\} = \begin{cases} -f(x) \text{ if } f(x) < 0 \\ 0 \text{ if } f(x) \geq 0. \end{cases}$$

for every $x \in X$. ◇

It is well known any map can be expresses using these parts

$$f(x) = f^+(x) - f^-(x) |f(x)| = f^+(x) + f^-(x),$$

combining which we obtain:

$$f^+(x) = \frac{1}{2}\big(|f(x)| + f(x)\big) f^-(x) = \frac{1}{2}\big(|f(x)| - f(x)\big).$$

Remark 10.2.2 The statements in Theorem 10.2.1 cannot be reversed. About *1.*, if $(c \cdot f)(x)$ is measurable then f is without doubt measurable when $c \neq 0$, but may be not measurable for $c = 0$.

Regarding *2.*, $(f + g)(x) = 0$ might be measurable and f, g not: take f not measurable and $g = -f$.

3.: let $V \subset [0, 1]$ be the Vitali set and $f : [0, 1] \mapsto \mathbb{R}$

$$f(x) = \begin{cases} 1 + x \text{ if } x \in V \\ -(1 + x) \text{ if } x \in [0, 1] \setminus V. \end{cases} \tag{10.2.1}$$

It is not measurable, for

$$f^{-1}\big((0, \infty)\big) = \{x \in [0, 1] : f(x) > 0\} = V,$$

while $f^2(x) = (1 + x)^2$ is continuous, hence Borel measurable by Proposition 10.1.1.

In item *4.* we can take $g = 0$ and f not measurable, while in *5.* we choose $f = 0$ and $1/g$ not measurable.

For *6.* take f as in (10.2.1), from which $f^{-1}((0, \infty)) = \{x \in [0, 1] : f(x) > 0\} = V$ is not measurable, while

$$|f| = f^+(x) + f^-(x) = 1 + x, \quad \forall x \in [0, 1] = V \cup \big([0, 1] \setminus V\big)$$

is continuous and, by Proposition 10.1.1, Borel measurable. ♣

Now we prove a result that fully describes measurable functions in terms of positive and negative parts.

Corollary 10.2.1 (Characterisation of measurable functions in terms of positive and negative parts) $f : X \mapsto \bar{\mathbb{R}} \in \mathscr{M}(X, \mathcal{S})$ *if and only if* $f^+(x)$, $f^-(x) \in \mathscr{M}^+(X, \mathcal{S})$.

Proof Suppose f measurable. Since

$$f^+(x) = \frac{1}{2}\big(|f(x)| + f(x)\big) \text{ and } f^-(x) = \frac{1}{2}\big(|f(x)| - f(x)\big),$$

by *1.*, *6.* in Theorem 10.2.1 both f^+, f^- are measurable.

Conversely if f^+ and f^- are measurable, by *1.–2.* in Theorem 10.2.1 f is measurable. ■

Theorem 10.2.2 (Sequences of measurable functions: measurability of sup and inf) *For any* $\bar{\mathbb{R}}$*-valued sequence* $\{f_n\}_{n \geq 1} \subseteq \mathscr{M}(X, \mathcal{S})$

$$\sup\{f_n\} \in \mathscr{M}(X,\mathcal{S}) \quad \textit{and} \ \inf\{f_n\} \in \mathscr{M}(X,\mathcal{S}).$$

Proof Fix $\alpha \in \mathbb{R}$. Since

$$\begin{aligned}\sup_{n\geq 1}\left\{f_n^{-1}\big((\alpha,\infty]\big)\right\} &= \left\{x \in X : \sup_{n\geq 1}\{f_n(x) > \alpha\}\right\} \\ &= \bigcup_{n=1}^{\infty}\left\{x \in X : f_n(x) > \alpha\right\},\end{aligned}$$

and $f_n \in \mathscr{M}(X,\mathcal{S})$ for every $n \in \mathbb{N}$, then $\left\{x \in X : f_n(x) > \alpha\right\} \in \mathcal{S}$. But $\mathcal{S}$ is closed under countable unions, so

$$\sup_{n\geq 1} f_n(x) \in \mathscr{M}(X,\mathcal{S}).$$

Moreover

$$\inf_{n\geq 1} f_n(x) = -\sup_{n\geq 1}\big(-f_n(x)\big),$$

and similarly $\inf_{n\geq 1} f_n(x) \in \mathscr{M}(X,\mathcal{S})$.

Clearly $\inf_{n\geq 1} f_n(x)$ and $\sup_{n\geq 1} f_n(x)$ always exist in $\bar{\mathbb{R}}$. ■

Directly from the previous theorem,

Corollary 10.2.2 (Measurability of $\min(f(x), g(x))$ and $\max(f(x), g(x))$) *If $f, g : (X,\mathcal{S}) \mapsto \bar{\mathbb{R}}$ belong in $\mathscr{M}(X,\mathcal{S})$, then*

$$f(x) \vee g(x) = \max\left\{f(x), g(x)\right\} \quad \textit{and} \ f(x) \wedge g(x) = \min\left\{f(x), g(x)\right\}$$

are $\mathcal{S}$-measurable. ■

To clarify, we plot in Fig. 10.1 the functions $\min\{x, x^2\}$ and $\max\{x, x^2\}$ and the corresponding pre-images of $f(x) \geq \alpha'$. We can see $x \vee x^2 = \max\{x, x^2\}$ is simply mapping any x in its domain to the largest of x, x^2, whereas $x \wedge x^2 = \min\{x, x^2\}$ returns the smallest of the two. The set $\left\{x \in X : x \vee x^2 \geq \alpha\right\}$, in red, consists of

$$\left\{x \in X : x \vee x^2 \geq \alpha\right\} = \{x \in X : x \geq \alpha\} \cup \{x \in X : x^2 \geq \alpha\} \in \mathcal{S};$$

the set $\left\{x \in X : x \wedge x^2 \geq \alpha\right\}$, in green, equals

$$\left\{x \in X : x \wedge x^2 \geq \alpha\right\} = \{x \in X : x \geq \alpha\} \cap \{x \in X : x^2 \geq \alpha\} \in \mathcal{S}.$$

Remark 10.2.3 Let $\{f_n\}_{n\geq 1}$ be a sequence of functions from X to $\bar{\mathbb{R}}$. For every $x \in X$

$$\liminf_{n\to\infty} f_n(x) = \limsup_{n\to\infty} f_n(x)$$

iff $\lim_{n\to\infty} f_n(x)$ exists in $\bar{\mathbb{R}}$. The sequence **converges** (pointwise, uniformly etc.) to some $f \in \mathbb{R}$ if the limit (exists and) is finite, i.e. if $\lim_{n\to\infty} f_n(x) \in \mathbb{R}$. Note further that for monotone sequences (increasing or decreasing)

$$\liminf_{n\to\infty} f_n(x) = \limsup_{n\to\infty} f_n(x).$$

Hence $\lim_{n\to\infty} f_n(x) \in \bar{\mathbb{R}}$ always exists. ♣

Theorem 10.2.3 (Sequences of measurable functions: measurability of lim inf *and* lim sup) *Let* $\{f_n\}_{n\geq 1} \subseteq \mathscr{M}(X, \mathcal{S})$ *with values in* $\bar{\mathbb{R}}$. *Then*

$$\liminf_{n\to\infty}\{f_n\} \in \mathscr{M}(X, \mathcal{S}) \quad \textit{and} \quad \limsup_{n\to\infty}\{f_n\} \in \mathscr{M}(X, \mathcal{S}).$$

Proof It is enough to observe

$$\liminf_{n\to\infty} f_n(x) = \sup_{n\geq 1}\left(\inf_{k\geq n} f_k(x)\right) \quad \text{and} \quad \limsup_{n\to\infty} f_n(x) = \inf_{n\geq 1}\left(\sup_{k\geq n} f_k(x)\right),$$

and the claim follows from Theorem 10.2.2.

Here, too, $\liminf_{n\to\infty} f_n(x)$ and $\limsup_{n\to\infty} f_n(x)$ exist in $\bar{\mathbb{R}}$. ■

Theorem 10.2.4 (Sequences of measurable functions: measurability of pointwise limits) *If* $\{f_n(x)\}_{n\geq 1}$ *is a sequence of maps from* X *to* $\bar{\mathbb{R}}$ *in* $\mathscr{M}(X, \mathcal{S})$ *with pointwise limit* $f : X \mapsto \mathbb{R}$, *then* $f \in \mathscr{M}(X, \mathcal{S})$.

Proof Observe that, for every $x \in X$,

$$f(x) = \lim_{n\to\infty} f_n(x) = \liminf_{n\to\infty} f_n(x).$$

■

Theorem 10.2.5 (Monotone sequences of measurable functions: measurability of limits) *If* $\{f_n(x)\}_{n\geq 1}$ *is a monotone sequence of measurable maps from* X *to* $\bar{\mathbb{R}}$:

1. $\lim_{n\to\infty} f_n \in \mathscr{M}(X, \mathcal{S})$;
2. $A = \left\{x \in X : \lim_{n\to\infty} f_n(x) \in \bar{\mathbb{R}}\right\}$ *is measurable.*

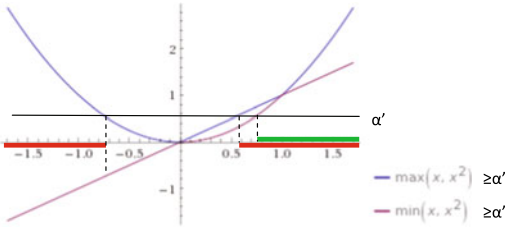


Fig. 10.1 Graphs of $y = \min(x, x^2)$ and $y = \max(x, x^2)$

Proof 1. As $\{f_n(x)\}_{n\geq 1}$ is monotone, $\forall x \in X$

$$\liminf_{n\to\infty} f_n(x) = \lim_{n\to\infty} f_n(x) = \limsup_{n\to\infty} f_n(x);$$

by Theorem 10.2.3 $\lim_{n\to\infty} f_n(x)$ is measurable.

2. Regarding the latter fact,

$$A = \left\{x \in X : \liminf_{n\to\infty} f_n(x) = \limsup_{n\to\infty} f_n(x)\right\}.$$

As both $\liminf_{n\to\infty} f_n(x)$ and $\limsup_{n\to\infty} f_n(x)$ are measurable on X, by part *3.* in Proposition 10.2.2

$$A = \left\{x \in X : \lim_{n\to\infty} f_n(x) \in \bar{\mathbb{R}}\right\}$$

is measurable. ■

Measurability is inherited also in presence of properties valid almost everywhere on X (such as equality a.e. of maps, convergence a.e. of sequences, and so on).

Theorem 10.2.6 (Equality m-almost everywhere of measurable functions) *Let $(X, \mathcal{S}, m)$ be a complete measure space. If $f : X \mapsto \bar{\mathbb{R}} \in \mathscr{M}(X, \mathcal{S})$, and $g : X \mapsto \bar{\mathbb{R}}$ is m-almost everywhere equal to f on X, then $g \in \mathscr{M}(X, \mathcal{S})$.*

Proof $N = \{x \in X : f(x) \neq g(x)\}$ belongs in $\mathcal{N}_m$. Given $\alpha \in \mathbb{R}$

$$\begin{aligned}\{x \in X : g(x) > \alpha\} &= \big(\{x \in X : g(x) > \alpha\} \cap N\big) \cup \big(\{x \in X : g(x) > \alpha\} \cap N^c\big) \\ &= \big(\{x \in X : g(x) > \alpha\} \cap N\big) \cup \big(\{x \in X : f(x) > \alpha\} \cap N^c\big).\end{aligned}$$

Clearly $\big(\{x \in X : f(x) > \alpha\} \cap N^c\big) \in \mathcal{S}$, and by completeness

$$\big(\{x \in X : g(x) > \alpha\} \cap N\big) \in \mathcal{T}_m \subseteq \mathcal{S}.$$

■

Consider a measure space $(X, \mathcal{S}, m)$ and its completion $(X, \bar{\mathcal{S}}, \bar{m})$. The next proposition examines a notion tackling the relationship between $\bar{\mathcal{S}}$- and $\mathcal{S}$-measurability.

Proposition 10.2.3 (Equality m-a.e. on X of $\bar{\mathcal{S}}$- and $\mathcal{S}$-measurable maps) *Let $(X, \mathcal{S}, m)$ be a measure space with completion $(X, \bar{\mathcal{S}}, \bar{m})$, $\bar{f} : X \mapsto \bar{\mathbb{R}}$ an $\bar{\mathcal{S}}$-measurable map. There exists an $\mathcal{S}$-measurable map $f : X \mapsto \bar{\mathbb{R}}$ such that*

$$\bar{f}(x) = f(x)$$

m-almost everywhere on X.

Proof Write $\mathbb{Q} = \{q_n : n \in \mathbb{N}\}$ and define for every $n \in \mathbb{N}$

$$E_n = \{x \in X : \bar{f}(x) < q_n\}.$$

As $\bar{f} \in \mathscr{M}(X, \bar{\mathcal{S}})$, E_n is $\bar{\mathcal{S}}$-measurable. This means $E_n = F_n \cup T_n$, where $F_n \in \mathcal{S}$, $T_n \in \mathcal{T}_m$ is negligible and $T_n \subseteq N_n$. Put $T = \bigcup_{n=1}^{\infty} T_n$, still negligible. Pick $N \in \mathcal{N}_m$ such that $T \subseteq N$ and define $G = N^c$ and $f = \bar{f} \cdot \varphi_G$. Clearly $f = \bar{f}$ $\bar{m}$-a.e., and for every $n \in \mathbb{N}$

$$\begin{aligned}\left\{x \in X : f(x) < q_n\right\} &= \left\{x \in X : \bar{f} \cdot \varphi_G(x) < q_n\right\} \\ &= E_n \cap G = (F_n \cap G) \cup (T_n \cap G) \\ &= (F_n \cap G) \in \mathcal{S},\end{aligned}$$

so $f \in \mathscr{M}(X, \mathcal{S})$. ■

Proposition 10.2.4 (Measurability of Dirichlet-like maps) *Take $f, g : X \mapsto \bar{\mathbb{R}}$ measurable, and measurable sets E_1, E_2 partitioning X. Then the map $h : X \mapsto \bar{\mathbb{R}}$,*

$$h(x) = \begin{cases} f(x) & \text{if } x \in E_1 \\ g(x) & \text{if } x \in E_2 \end{cases}$$

is measurable.

Proof As E_1, E_2 are measurable, their characteristic functions φ_{E_1}, φ_{E_2} are measurable as well. By Theorem 10.2.1 h is Borel measurable, because for every $x \in X$

$$h(x) = f(x) \cdot \varphi_{E_1}(x) + g(x) \cdot \varphi_{E_2}(x).$$

Among the maps like h are Dirichlet-like functions. ■

We come to investigate simple measurable functions, which represent a sort of core for all measurable functions, and are made of linear combinations of characteristic functions. Using these we will be able to approximate more complicated maps.

Definition 10.2.3 ($\bar{\mathbb{R}}$*-valued simple functions*) Consider subsets $A_1, \ldots, A_n$ of X such that $A_i \cap A_j = \emptyset$ and $X = \bigcup_{i=1}^{n} A_i$. A simple function is a map $s : X \mapsto \bar{\mathbb{R}}$

$$s(x) = \sum_{i=1}^{n} a_i \varphi_{A_i}(x), \quad x \in X,$$

where $\{a_1, \ldots, a_n\} = \mathscr{A} \subseteq \bar{\mathbb{R}}$. ◇

A simple function is a (finite) linear combination of characteristic functions of certain sets $A_i \subseteq X$, with $X = \bigsqcup_{i=1}^{n} A_i$, that takes values $a_i \in \bar{\mathbb{R}}$ for every $x \in A_i$. Hence $A_i = \{x \in X : s(x) = a_i\}$.

Remark 10.2.4 (*Simple functions.*) The most trivial simple function is the constant map

$$s(x) = k\varphi_X(x).$$

The representation of a simple function is not unique. Let $\bigcup_{i=1}^{n} A_i = X = \bigcup_{j=1}^{m} B_j$, with $A_i \cap A_k = \emptyset$ and $B_j \cap B_h = \emptyset$ for $i \neq k$ and $j \neq h$, and set

$$s_1(x) = \sum_{i=1}^{n} a_i \varphi_{A_i}(x) \text{ and } s_2(x) = \sum_{j=1}^{m} b_j \varphi_{B_j}(x),$$

such that $a_i = b_j$ if $x \in (A_i \cap B_j)$. Then $\forall x \in X$

$$\begin{aligned} s_1(x) = \sum_{i=1}^{n} a_i \varphi_{A_i}(x) &= \sum_{i=1}^{n} \sum_{j=1}^{m} a_i \varphi_{(A_i \cap B_j)}(x) \\ &= \sum_{i=1}^{n} \sum_{j=1}^{m} b_j \varphi_{(A_i \cap B_j)}(x) = \sum_{j=1}^{m} b_j \varphi_{B_j}(x) = s_2(x). \end{aligned}$$

It follows $s_1(x)$ is essentially the same as $s_2(x)$, so any simple function admits several representations, including:

$$s(x) = \sum_{i=1}^{n} \sum_{j=1}^{m} a_i \varphi_{A_i \cap B_j}(x).$$

♣

The next fact proves that elementary operations preserve simplicity.

Proposition 10.2.5 (Operations between $\bar{\mathbb{R}}$-valued simple functions) *Let s, s_1, s_2 be simple functions. For any $\alpha, \beta \in \mathbb{R}$ the following are simple functions:*

1. $(\alpha s_1 + \beta s_2)$;
2. $(s_1 \cdot s_2)$
3. $|s|$;
4. $(s_1 \vee s_2) = \max(s_1, s_2)$ *and* $(s_1 \wedge s_2) = \min(s_1, s_2)$;
5. $s \cdot \varphi_E$ *for every* $E \subseteq X$.

Proof The proof is just a matter of computing:

1.

$$\begin{aligned}\alpha s_1(x) + \beta s_2(x) &= \alpha \sum_{i=1}^{n} a_i \varphi_{A_i}(x) + \beta \sum_{j=1}^{m} b_j \varphi_{B_j}(x) \\ &= \alpha \sum_{i=1}^{n} \sum_{j=1}^{m} a_i \varphi_{A_i \cap B_j}(x) + \beta \sum_{j=1}^{m} \sum_{i=1}^{n} b_j \varphi_{B_j \cap A_i}(x) \\ &= \sum_{i=1}^{n} \sum_{j=1}^{m} \big(\alpha a_i + \beta b_j\big) \varphi_{A_i \cap B_j}(x) = (\alpha s_1 + \beta s_2)(x).\end{aligned}$$

2.

$$\begin{aligned}s_1(x) \cdot s_2(x) &= \sum_{i=1}^{n} a_i \varphi_{A_i}(x) \cdot \sum_{j=1}^{m} b_j \varphi_{B_j}(x) \\ &= \sum_{i=1}^{n} \sum_{j=1}^{m} a_i \varphi_{A_i \cap B_j}(x) \cdot \sum_{i=1}^{n} \sum_{j=1}^{m} b_j \varphi_{A_i \cap B_j}(x) \\ &= \sum_{i=1}^{n} \sum_{j=1}^{m} a_i b_j \varphi_{A_i \cap B_j}(x) = (s_1 \cdot s_2)(x).\end{aligned}$$

3.

$$|s(x)| = \sum_{i=1}^{n} |a_i| \varphi_{A_i}(x) = s^+(x) + s^-(x).$$

4.

$$\begin{aligned}s_1(x) \vee s_2(x) &= \sum_{i=1}^{n} a_i \varphi_{A_i}(x) \vee \sum_{j=1}^{m} b_j \varphi_{B_j}(x) \\ &= \sum_{i=1}^{n} \sum_{j=1}^{m} a_i \varphi_{A_i \cap B_j}(x) \vee \sum_{i=1}^{n} \sum_{j=1}^{m} b_j \varphi_{A_i \cap B_j}(x) \\ &= \sum_{i=1}^{n} \sum_{j=1}^{m} (a_i \vee b_j) \varphi_{A_i \cap B_j}(x) = \big(s_1 \vee s_2\big)(x).\end{aligned}$$

The second expression is proved similarly.

5.

$$s\varphi_E = \sum_{i=1}^{n} a_i \varphi_{A_i} \varphi_E = \sum_{i=1}^{n} a_i \varphi_{A_i \cap E}.$$

This last argument also shows that restrictions of simple functions are simple. ■

Proposition 10.2.6 (Measurability of simple functions) *A simple function s is $\mathcal{S}$-measurable if $A_i \in \mathcal{S}$ for every i.*

Proof Under the given assumption,

$$s^{-1}(I) = \bigcup_{i:a_i \in I} A_i \in \mathcal{S}.$$

■

On a measurable space $(X, \mathcal{S})$ we indicate the class of $\mathcal{S}$-measurable simple functions by $\mathscr{M}_0(X, \mathcal{S})$, and by $\mathscr{M}_0^+(X, \mathcal{S})$ the subclass of non-negative ones. Obviously s^+ and s^- belong to $\mathscr{M}_0^+(X, \mathcal{S})$.

10.3 Pointwise Convergence A.e. and Almost Uniform Convergence

In this section we study the concepts of pointwise convergence almost everywhere and almost uniform convergence.

Relying on Definitions 4.2.1, 4.2.2 we can address pointwise convergence m-a.e. straightaway, and deal with almost uniform convergence later, **provided the measure spaces are complete**.

Definition 10.3.1 (*Pointwise convergence m-almost everywhere*) A sequence of $\bar{\mathbb{R}}$-valued maps $\{f_n\}_{n\geq 1} \subseteq \mathscr{M}(X, \mathcal{S})$ defined on a complete measure space $(X, \mathcal{S}, m)$ converges pointwise m-a.e. to $f : X \mapsto \mathbb{R}$ whenever

$$\left\{x \in X : \lim_{n\to\infty} f_n(x) = f(x)\right\}^c \in \mathcal{N}_m.$$

◇

Theorem 10.3.1 (Pointwise convergence m-a.e. and measurability of limit) *Let $(X, \mathcal{S}, m)$ be a complete measure space. A sequence $\{f_n\}_{n\geq 1} \subseteq \mathscr{M}(X, \mathcal{S})$ of $\bar{\mathbb{R}}$-valued maps that converges pointwise m-a.e. admits limit $f : X \mapsto \mathbb{R} \in \mathscr{M}(X, \mathcal{S})$.*

Proof The set

$$N = \left\{x \in X : \nexists \lim_{n\to\infty} f_n(x)\right\}$$

belongs to $\mathcal{N}_m$. Define $f : X \mapsto \mathbb{R}$ as follows: for every $x \in X$

$$f(x) = \begin{cases} \lim_{n\to\infty} f_n(x) & \text{if } x \in (X \setminus N) \\ 0 \text{ if } x \in N. \end{cases}$$

This is the well-defined map to which $\{f_n\}_{n\geq 1}$ converges m-almost everywhere.

We claim $f \in \mathscr{M}(X,\mathcal{S})$. Recall $(X,\mathcal{S},m)$ is complete, so $N=\left\{x \in X : \nexists \lim_{n\to\infty} f_n(x)\right\}$ is $\mathcal{S}$-measurable. Given an interval $I \subseteq \bar{\mathbb{R}}$,

$$\begin{aligned} f^{-1}(I) &= \{x \in X : f(x) \in I\} \\ &= \big(\{x \in X : f(x) \in I\} \cap N\big) \cup \big(\{x \in X : f(x) \in I\} \cap N^c\big). \end{aligned}$$

As $\big(\{x \in X : f(x) \in I\} \cap N\big) \subseteq N$, it is also $\mathcal{S}$-measurable. Moreover, since $f = \lim_{n\to\infty} f_n$ on N^c, by *2.* in Theorem 10.2.5 we have $\big(\{x \in X : \lim_{n\to\infty} f_n(x) \in I\} \cap N^c\big) \in \mathcal{S}$. Therefore $f^{-1}(I) = \{x \in X : f(x) \in I\} \in \mathcal{S}$. ■

Remark 10.3.1 The limit of an m-a.e. convergent sequence is unique as a function defined m-a.e.. That is to say:

$$\lim_{n\to\infty} f_n = f_1 \ m\text{-a.e. on } X \quad \text{and} \quad \lim_{n\to\infty} f_n = f_2 \ m\text{-a.e. on } X$$

imply $f_1 = f_2$ m-a.e. on X. ♣

Here is a useful lemma for the sequel.

Lemma 10.3.1 (Sequences of measurable functions: measurability of convergence set and of its complement) *Let $(X,\mathcal{S},m)$ be a complete measure space, $\{f_n\}_{n\geq 1} \subseteq \mathscr{M}(X,\mathcal{S})$ a sequence of $\bar{\mathbb{R}}$-valued maps converging m-a.e. to the $\mathcal{S}$-measurable map $f : X \mapsto \mathbb{R}$. Then $\left\{x \in X : \lim_{n\to\infty} f_n = f\right\}$ and $\left\{x \in X : \lim_{n\to\infty} f_n = f\right\}^c$ are $\mathcal{S}$-measurable.*

Proof By definition, given $x \in X$ the sequence $\{f_n(x)\}_{n\geq 1}$ tends to $f(x)$ iff, for every $\epsilon = 1/k > 0$, there is an $n \in \mathbb{N}$ such that

$$|f_i(x) - f(x)| < \frac{1}{k}$$

for every $i \geq n$. Then

$$\begin{aligned} \left\{x \in X : \lim_{n\to\infty} f_n = f\right\} &= \bigcap_{k=1}^{\infty} \lim_{n\to\infty} \left\{x \in X : |f_n - f| < 1/k\right\} \\ &= \bigcap_{k=1}^{\infty} \bigcup_{n=1}^{\infty} \bigcap_{i=n}^{\infty} \left\{x \in X : |f_i - f| < 1/k\right\} \in \mathcal{S}. \end{aligned}$$

Consequently the complement, too, is in $\mathcal{S}$. ■

Proposition 10.3.1 (Sequences of measurable functions: characterisation of pointwise convergence m-a.e.) *Given a complete measure space $(X,\mathcal{S},m)$, a sequence of $\bar{\mathbb{R}}$-valued maps $\{f_n\}_{n\geq 1} \subseteq \mathscr{M}(X,\mathcal{S})$ and an $\mathcal{S}$-measurable function $f : X \mapsto \mathbb{R}$,*

$$\lim_{n\to\infty} f_n(x) = f(x), \ \ m\text{-a.e. on } X$$
$$\Leftrightarrow m\left(\limsup_{n\to\infty}\left\{x\in X : |f_n - f| \geq \frac{1}{k}\right\}\right) = 0, \ \ \forall k \in \mathbb{N}.$$

Proof Obviously $\lim_{n\to\infty} f_n(x) = f(x)$ m-a.e. iff

$$m\big(\{x \in X : \lim_{n\to\infty} f_n(x) = f(x)\}^c\big) = 0.$$

The latter, by the second relation in Lemma 10.3.1, reads

$$m\left(\bigcup_{k=1}^{\infty}\bigcap_{n=1}^{\infty}\bigcup_{i=n}^{\infty}\{x \in X : |f_i - f| \geq 1/k\}\right) = 0.$$

As m is monotone, for every $k \in \mathbb{N}$

$$m\left(\bigcap_{n=1}^{\infty}\bigcup_{i=n}^{\infty}\{x \in X : |f_i - f| \geq 1/k\}\right) \leq m\left(\bigcup_{k=1}^{\infty}\bigcap_{n=1}^{\infty}\bigcup_{i=n}^{\infty}\{x \in X : |f_i - f| \geq 1/k\}\right) = 0$$
$$\Leftrightarrow m\left(\limsup_{n\to\infty}\{x \in X : |f_n - f| \geq 1/k\}\right) = 0, \ \ \forall k \in \mathbb{N}.$$

■

In a measure space $(X, \mathcal{S}, m)$, the set $\limsup_{n\to\infty} A_n$, where $\{A_n\}_{n\geq 1} \subseteq \mathcal{S}$, consists of the points in X at which infinitely many $A_n \in \mathcal{S}$ meet. Therefore it is only natural to ask how small must the sets A_n be to ensure $m\big(\limsup_{n\to\infty} A_n\big) = 0$. The Borel–Cantelli lemma provides a sufficient condition.

Theorem 10.3.2 (Borel–Cantelli lemma) *Let $(X, \mathcal{S}, m)$ be a measure space, $\{A_n\}_{n\geq 1}$ a sequence in $\mathcal{S}$. If $\sum_{n=1}^{\infty} m(A_n) < \infty$ then $m\big(\limsup_{n\to\infty} A_n\big) = 0$.*

Proof As $\limsup_{n\to\infty} A_n = \bigcap_{n=1}^{\infty}\bigcup_{i=n}^{\infty} A_i$, we can set $B_n = \bigcup_{i=n}^{\infty} A_i$ for every $n \in \mathbb{N}$, making $\{B_n\}_{n\geq 1}$ a decreasing sequence in $\mathcal{S}$ such that

$$\limsup_{n\to\infty} A_n = \lim_{n\to\infty} B_n.$$

But $B_1 = \bigcup_{i=1}^{\infty} A_i$, so

$$m(B_1) = m\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} m(A_i) < \infty.$$

This, together with the fact $\{B_n\}_{n\geq 1}$ decreases, allows to invoke Theorem 8.2.4, from which

$$m\big(\limsup_{n\to\infty} A_n\big) = m\big(\lim_{n\to\infty} B_n\big) = \lim_{n\to\infty} m(B_n)$$
$$= \lim_{n\to\infty} m\left(\bigcup_{i=n}^{\infty} A_i\right) \le \lim_{n\to\infty} \sum_{i=n}^{\infty} m(A_i).$$

By hypothesis $\sum_{n=1}^{\infty} m(A_n) < \infty$, so $\lim_{n\to\infty} \sum_{i=n}^{\infty} m(A_i) = 0$, and then

$$m\big(\limsup_{n\to\infty} A_n\big) = 0.$$

■

Theorem 10.3.3 (Sequences of measurable functions: sufficient condition for pointwise convergence m-a.e.) *Let $(X, \mathcal{S}, m)$ be a complete measure space, $\{f_n\}_{n\ge1}$ a sequence of $\bar{\mathbb{R}}$-valued, $\mathcal{S}$-measurable functions and $f : X \mapsto \mathbb{R}$ an $\mathcal{S}$-measurable map. If there exists a positive sequence $\{\epsilon_n\}_{n\ge1}$ such that*

1. $\lim_{n\to\infty} \epsilon_n = 0$, *and*
2. $\sum_{n=1}^{\infty} m\big(\{x \in X : |f_n(x) - f(x)| \ge \epsilon_n\}\big) < \infty$,

then $\lim_{n\to\infty} f_n(x) = f(x)$, m-almost everywhere.

Proof Set $A_n = \{x \in X : |f_n(x) - f(x)| \ge \epsilon_n\}$ for every $n \in \mathbb{N}$. By *2.* we have $\sum_{n=1}^{\infty} m(A_n) < \infty$, and the Borel–Cantelli lemma guarantees

$$m\left(\limsup_{n\to\infty} A_n\right) = m\left(\limsup_{n\to\infty} \{x \in X : |f_n(x) - f(x)| \ge \epsilon_n\}\right) = 0. \quad (10.3.1)$$

Given $k \in \mathbb{N}$, assumption *1.* tells there is an $\bar{n} \in \mathbb{N}$ such that $\epsilon_n < 1/k$ for every $n \ge \bar{n}$, so

$$\limsup_{n\to\infty} \{x \in X : |f_n - f| \ge \epsilon_n\} \supseteq \limsup_{n\to\infty} \left\{x \in X : |f_n - f| \ge \frac{1}{k}\right\}$$
$$\Leftrightarrow m\left(\limsup_{n\to\infty} \{x \in X : |f_n - f| \ge \epsilon_n\}\right) \ge m\left(\limsup_{n\to\infty} \left\{x \in X : |f_n - f| \ge \frac{1}{k}\right\}\right).$$

Then, using (10.3.1), for every $k \in \mathbb{N}$ we have

$$m\left(\limsup_{n\to\infty} \left\{x \in X : |f_n - f| \ge \frac{1}{k}\right\}\right) = 0.$$

Now Proposition 10.3.1 allows to conclude. ■

Theorem 10.3.4 (Series of measurable functions: sufficient condition for absolute convergence m-a.e.) *Let $(X, \mathcal{S}, m)$ be a complete measure space, $\{f_n\}_{n\ge1}$ a sequence of real-valued, $\mathcal{S}$-measurable functions. If there exists a positive sequence $\{\epsilon_n\}_{n\ge1}$ such that*

1. $\sum_{n=1}^{\infty} \epsilon_n < \infty$,
2. $\sum_{n=1}^{\infty} m\big(\{x \in X : |f_n(x)| \geq \epsilon_n\}\big) < \infty$,

then the series $\sum_{n=1}^{\infty} f_n$ *converges absolutely m-almost everywhere.*

Proof Set $A_n = \{x \in X : |f_n(x)| \geq \epsilon_n\}$ for every $n \in \mathbb{N}$, so by *2.* $\sum_{n=1}^{\infty} m(A_n) < \infty$. Using the Borel–Cantelli lemma,

$$m\left(\limsup_{n\to\infty} A_n\right) = 0. \tag{10.3.2}$$

Let $N = \limsup_{n\to\infty} A_n$ so $N \subseteq X$ and $m(N) = 0$, i.e. N is a null set in $(X, \mathcal{S}, m)$. Note that for every $x \in X$, $x \in N$ iff $x \in A_n$ for infinitely many $n \in \mathbb{N}$, so

$$\begin{aligned} x \in X \setminus N &\Leftrightarrow x \in A_n \text{ for at most finitely many } n \in \mathbb{N} \\ &\Leftrightarrow \exists n_0 \in \mathbb{N} : x \notin A_n, \forall n > n_0 \\ &\Leftrightarrow \exists n_0 \in \mathbb{N} : |f_n(x)| < \epsilon_n, \forall n > n_0. \end{aligned}$$

Then for $x \in X \setminus N$

$$\sum_{n=1}^{\infty} |f_n(x)| = \sum_{n=1}^{n_0} |f_n(x)| + \sum_{n=n_0+1}^{\infty} |f_n(x)| < \sum_{n=1}^{n_0} |f_n(x)| + \sum_{n=n_0+1}^{\infty} \epsilon_n.$$

Since f_n is real-valued for every $n \in \mathbb{N}$, $\sum_{n=1}^{n_0} |f_n(x)| < \infty$. And because of *1.*, similarly, $\sum_{n=n_0+1}^{\infty} \epsilon_n \leq \sum_{n=1}^{\infty} \epsilon_n < \infty$. Hence for every $x \in X \setminus N$

$$\sum_{n=1}^{\infty} |f_n(x)| < \infty,$$

i.e. $\sum_{n=1}^{\infty} f_n(x)$ converges absolutely m-a.e. Notice that it converges also pointwise m-a.e. ■

Let us introduce uniform convergence up to arbitrarily small sets, which we shall call almost uniform convergence (a.u.).

Definition 10.3.2 (*Almost uniform convergence*) On a complete measure space $(X, \mathcal{S}, m)$ a sequence of $\bar{\mathbb{R}}$-valued, $\mathcal{S}$-measurable maps $\{f_n\}_{n\geq 1}$ converges m-almost uniformly on X to an $\mathcal{S}$-measurable map $f : X \mapsto \mathbb{R}$ if, for every $\delta > 0$, there exists a set $E \in \mathcal{S}$ with $m(E) < \delta$ such $\{f_n\}_{n\geq 1}$ converges **uniformly** to f on $(X \setminus E)$. ◇

The measure of E is as small as we like, but not zero. Hence a.u. convergence is uniform convergence up to a set of arbitrarily small, but positive, measure.

The Severini–Egorov theorem explains that pointwise convergence m-a.e. implies m-almost uniform convergence on X provided $m(X) < \infty$. First, though, we need a preliminary fact.

Proposition 10.3.2 (Relevant property of convergence *m-a.e.*) *On a complete measure space* $(X, \mathcal{S}, m)$ *with* $m(X) < \infty$, *let* $\{f_n\}_{n\geq 1}$ *be a sequence of* $\bar{\mathbb{R}}$*-valued,* $\mathcal{S}$*-measurable maps that pointwise converge m-a.e. to an* $\mathcal{S}$*-measurable map* $f : X \mapsto \mathbb{R}$. *Given any* $\epsilon > 0$ *and* $\delta > 0$ *there exist a measurable subset* $E_\epsilon \subseteq X$ *with* $m(E_\epsilon) \leq \delta$, *and an integer* n_ϵ *such that*

$$\sup_{x\in(X\setminus E_\epsilon)} |f_n(x) - f(x)| \leq \epsilon$$

for every $n \geq n_\epsilon$.

Proof Consider

$$N = \left\{x \in X : f(x) \neq \lim_{n\to\infty} f_n(x)\right\} \in \mathcal{N}_m$$

and $A = (X \setminus N)$, so $m(A) = m(X)$. Given $n \in \mathbb{N}$ we introduce

$$E_n = \left\{x \in A : |f_n(x) - f(x)| \geq \epsilon\right\}.$$

The measurability of $f_n(x)$ and $f(x)$ on A implies the measurability of E_n. So let us put, for every $k \in \mathbb{N}$,

$$A_k = \bigcup_{n=k}^{\infty} E_n = \bigcup_{n=k}^{\infty} \left\{x \in A : |f_n(x) - f(x)| \geq \epsilon\right\}.$$

This, too, is measurable as countable union of measurable sets. Clearly $\{A_k\}_{k\geq 1}$ is decreasing, with $m(A_1) \leq m(A) < \infty$, so by the measure's continuity from above

$$m\left(\bigcap_{k=1}^{\infty} A_k\right) = \lim_{k\to\infty} m(A_k).$$

As $f_n(x)$ converges to $f(x)$ pointwise for every $x \in A$, from a certain point onwards, i.e. eventually, $|f_n(x) - f(x)| < \epsilon$. Hence x eventually does not belong to A_{k_ϵ}, say for $n \geq k_\epsilon$. But then the intersection of the A_k is empty and the above limit is zero. Given then any $\delta > 0$, there is a k_ϵ such that $m(A_{k_\epsilon}) \leq \delta$.

Note A_{k_ϵ} depends on ϵ, so what we are talking about is not the uniform convergence of $f_n(x)$ to $f(x)$ on $(X \setminus A_{k_\epsilon})$ (uniform convergence up to an arbitrarily small set). That settled, let us write $E_\epsilon = A_{k_\epsilon} \cup N$. As $A_{k_\epsilon} \subseteq A$ and $(A \cap N) = \emptyset$, the space's completeness implies $m(E_\epsilon) = m(A_{k_\epsilon}) \leq \delta$. If $x \notin E_\epsilon$, then $x \notin E_n$ for every $n \geq k_\epsilon$, and so $x \in (X \setminus E_\epsilon)$ and

$$|f_n(x) - f(x)| < \epsilon,$$

whence

$$\sup_{x\in(X\setminus E_\epsilon)} |f_n(x) - f(x)| \le \epsilon.$$

■

Applying this result repeatedly we can attain almost uniform convergence starting from pointwise convergence m-almost everywhere.

Theorem 10.3.5 (Severini–Egorov theorem, or Littlewood's second principle—pointwise convergence m-a.e. and m-a.u. convergence) *Let $(X, \mathcal{S}, m)$ be a complete measure space with $m(X) < \infty$, and $\{f_n\}_{n\ge1}$ a sequence of $\bar{\mathbb{R}}$-valued, $\mathcal{S}$-measurable maps pointwise converging m-a.e. to an $\mathcal{S}$-measurable map $f(x) : X \mapsto \mathbb{R}$. Then the convergence is uniform on X up to a set of arbitrarily small measure.*

Proof Suppose $\{f_n(x)\}_{n\ge1}$ converges m-a.e. By the previous result, given $\epsilon_n = 1/n$ and $\delta_n = \delta/2^n$ there exists $E_n = A_{\delta_n}$, with $m(E_n) < \delta_n$, such that

$$\sup_{x\in(X\setminus E_n)} |f_n(x) - f(x)| \le \frac{1}{n}.$$

Consider $A_\delta = \bigcup_{n=1}^{\infty} E_n$. By subadditivity

$$m(A_\delta) = m\left(\bigcup_{n=1}^{\infty} E_n\right) \le \sum_{n=1}^{\infty} m(E_n) < \sum_{n=1}^{\infty} \frac{\delta}{2^n} = \delta;$$

furthermore, as $(X \setminus A_\delta) \subseteq (X \setminus E_n)$ for every $n \in \mathbb{N}$,

$$\sup_{x\in(X\setminus A_\delta)} |f_n(x) - f(x)| \le \sup_{x\in(X\setminus E_n)} |f_n(x) - f(x)| \le \frac{1}{n}.$$

The claim now follows. ■

The requirement $m(X) < \infty$ is essential in the proof of the Severini–Egorov theorem. Take in fact $X = \mathbb{R}$ and $f_n(x) = \varphi_{(-n,n)}(x), n \in \mathbb{N}$. The sequence $\{f_n\}_{n\ge1}$ converges everywhere on $\mathbb{R}$ to $f(x) = 1$, but the convergence is not uniform outside any set of finite measure: the set where $f_n(x)$ is different from $f(x)$ has infinite measure, for every $n \in \mathbb{N}$.

The English mathematician J. E. Littlewood condensed the gist of the Severini–Egorov theorem as follows[2]: "a convergent sequence of functions on a finite-measure space is almost uniformly convergent".

[2] Also known as Littlewood's second principle.

10.4 Approximating Measurable Functions

Simple functions play a part of the utmost importance in the theory of Lebesgue integration. A simple function takes the codomain of a real measurable function f, divides it in subsets and labels each with a specific value chosen within it. It then associates each of these values to the pre-image of the corresponding subset. This pre-image is typically a complicated subset, with points scattered all over the domain. We may think of the Dirichlet function. Viewed as a simple function, it assigns the value 1 to a certain codomain subset coming from the domain points $[0, 1] \cap \mathbb{Q}$, and assigns value 0 to the other subset, image of $[0, 1] \setminus \mathbb{Q}$. In general terms, the subsets A_i appearing in the simple function $s(x) = \sum_{i=1}^{n} a_i \varphi_{A_i}(x)$ are not expected to have the elementary structure of an interval, but a much more complex composition spread over the domain, depending on choice of subsets in the codomain of f.

Simple functions include the more modest step functions, which are central in the theory of Riemann integration.

Based on these considerations and in view of a satisfactory discussion on the theory of Lebesgue integral, we believe it useful to define measurable functions also in terms of sequences of simple measurable functions.

Definition 10.4.1 (Functions in $\mathscr{M}(X, \mathcal{S})$) A map $f : X \mapsto \bar{\mathbb{R}}$ belongs to $\mathscr{M}(X, \mathcal{S})$ if there is a sequence of simple functions $\{s_n\}_{n\geq 1} \subseteq \mathscr{M}_0(X, \mathcal{S})$ such that

$$f(x) = \lim_{n\to\infty} s_n(x)$$

for every $x \in X$. ◇

This entitles us to characterise $\mathcal{S}$-measurable maps by means of simple functions.

Theorem 10.4.1 (Characterisation of measurable functions via sequences of simple measurable functions) *Given a measurable space $(X, \mathcal{S})$, a map $f : X \mapsto \bar{\mathbb{R}}$ is $\mathcal{S}$-measurable if and only if there exists a sequence $\{s_n\}_{n\geq 1}$ of simple $\mathcal{S}$-measurable functions such that $\lim_{n\to\infty} s_n(x) = f(x)$ for every $x \in X$.*

Moreover,

1. *if f is bounded the sequence converges uniformly;*
2. *if $f \geq 0$, then $\{s_n\}_{n\geq 1}$ is increasing, and for every $n \in \mathbb{N}$*

$$0 \leq s_n \leq f.$$

Proof Necessary implication: if $f \in \mathscr{M}(X, \mathcal{S})$, by Definition 10.4.1 there is a sequence of simple maps $\{s_n\}_{n\geq 1} \subseteq \mathscr{M}_0(X, \mathcal{S})$ with

$$f(x) = \lim_{n\to\infty} s_n(x)$$

for every $x \in X$.

Sufficient implication: without loss of generality suppose f is non-negative, i.e. $f : X \mapsto [0, \infty]$. This is possible because $s_n(x) = s_n^+(x) - s_n^-(x)$ is the ($\mathcal{S}$-measurable, simple) difference of $\mathcal{S}$-measurable simple functions with values in $[0, \infty]$, and we may apply this theorem to $\{s_n(x)^+\}_{n\geq 1}$ and $\{s_n(x)^-\}_{n\geq 1}$ separately.

Hence let us have $f \geq 0$, meaning the range of f is a subset of $[0, \infty]$. First split $[0, \infty]$ into $[0, n) \cup [n, \infty]$, then divide $[0, n)$, of length n, in $n \cdot 2^n$ parts of length $\frac{1}{2^n}$ each:

$$[0, n) \cup [n, \infty] = \bigcup_{k=1}^{n\cdot 2^n} \left[\frac{k-1}{2^n}, \frac{k}{2^n}\right) \cup [n, \infty].$$

Given $x \in X$ we can have either

$$f(x) \in \bigcup_{k=1}^{n\cdot 2^n} \left[\frac{k-1}{2^n}, \frac{k}{2^n}\right) \text{ or } f(x) \in [n, \infty].$$

For every $n \in \mathbb{N}$ we define the simple function

$$s_n(x) = \begin{cases} \dfrac{k-1}{2^n} & \text{if } x \in f^{-1}\left(\left[\dfrac{k-1}{2^n}, \dfrac{k}{2^n}\right)\right) \\ n & \text{if } x \in f^{-1}\big([n, \infty]\big) \end{cases},$$

and thus

$$s_n(x) = \sum_{k=1}^{n\cdot 2^n} \frac{k-1}{2^n} \varphi_{\left\{f^{-1}\left(\left[\frac{k-1}{2^n}, \frac{k}{2^n}\right)\right)\right\}}(x) + n \cdot \varphi_{\left\{f^{-1}([n,\infty])\right\}}(x).$$

Clearly $s_n(x)$ is non-negative, for its values are n or $\frac{k-1}{2^n}$ for $k = 1, \ldots, n \cdot 2^n$; hence $0 \leq s_n$ for every $n \in \mathbb{N}$. It is a linear combination of characteristic functions, so $s_n(x) = \frac{k-1}{2^n}$ if $f(x) \in \left[\frac{k-1}{2^n}, \frac{k}{2^n}\right)$ and $s_n(x) = n$ if $f(x) \in \big[n, \infty\big]$. Note, moreover, that given $a \in [0, \infty)$, for all values $\frac{k-1}{2^n} < a$, the union of the corresponding sets

$$s_n^{-1}\big([0, a)\big) = \bigcup_k f^{-1}\left(\left[\frac{k-1}{2^n}, \frac{k}{2^n}\right)\right)$$

is measurable since s_n is measurable for every n by hypothesis.

Take now an arbitrary x in X. For every $n \in \mathbb{N}$, $f(x) \in [n, \infty]$ implies $s_n(x) = n$, so $f(x) \to \infty$ and $s_n(x) \to \infty$.

We claim $s_n(x)$ converges to $f(x)$ pointwise. Fix $n \in \mathbb{N}$ and suppose $f(x) \in \left[\frac{k-1}{2^n}, \frac{k}{2^n}\right)$ for some $k = 1, \ldots, n \cdot 2^n$. Then $s_n(x) = \frac{k-1}{2^n}$, and so for $n \geq n_0$

$$\big|f(x) - s_n(x)\big| < \frac{1}{2^{n_0}}.$$

Therefore for every n

$$\left|f(x) - s_n(x)\right| < \frac{1}{2^n},$$

and overall $s_n(x) \to f(x)$.

The sequence $\{s_n(x)\}_{n\geq 1}$ also converges uniformly to f if the latter is non-negative and bounded, for in this case there is an n_0 such that $f(x) \in [0, n_0)$. Hence $\forall x \in X, n \geq n_0$

$$\left|f(x) - s_n(x)\right| < \frac{1}{2^{n_0}}.$$

Now we prove $\{s_n(x)\}_{n\geq 1}$ is increasing, i.e. $s_n(x) \leq s_{n+1}(x)$ for every $n \in \mathbb{N}$, $x \in X$. In analogy to what we did before, split $[0, \infty]$ into $[0, n+1)$ and $[n+1, \infty]$. Then divide $[0, n+1)$ in $(n+1) \cdot 2^{n+1}$ equal bits, all of length $\frac{1}{2^{n+1}}$.

At step $n+1$ a subinterval $\left[\frac{k_0-1}{2^n}, \frac{k_0}{2^n}\right)$ (length $\frac{1}{2^n}$) is split into

$$\left[\frac{2(k_0-1)}{2^{n+1}}, \frac{2k_0-1}{2^{n+1}}\right) \quad \text{and} \left[\frac{2k_0-1}{2^{n+1}}, \frac{2k_0}{2^{n+1}}\right),$$

each $\frac{1}{2^{n+1}}$ wide. In general, step $n+1$ adds to the nth subdivision of $[0, n)$ further points. The ensuing subdivision is finer (it includes the preceding one): for every $k = 1, \ldots, (n+1) \cdot 2^{n+1}$

$$\frac{k-1}{2^n} = \frac{2(k-1)}{2^{n+1}} \quad \text{and} \frac{k}{2^n} = \frac{2k}{2^{n+1}},$$

and each interval from step n gains the intermediate point $\frac{2k-1}{2^{n+1}}$.

Summing up, at step n we have

$$f(x) \in \left[\frac{k_0-1}{2^n}, \frac{k_0}{2^n}\right) \Rightarrow s_n(x) = \frac{k_0-1}{2^n}$$

and at step $n+1$

$$f(x) \in \left[\frac{2(k_0-1)}{2^{n+1}}, \frac{2k_0-1}{2^{n+1}}\right) \text{ or } f(x) \in \left[\frac{2k_0-1}{2^{n+1}}, \frac{2k_0}{2^{n+1}}\right).$$

Hence

$$f(x) \in \left[\frac{2(k_0-1)}{2^{n+1}}, \frac{2k_0-1}{2^{n+1}}\right) \Rightarrow s_{n+1}(x) = \frac{2(k_0-1)}{2^{n+1}} = \frac{k_0-1}{2^n} = s_n(x);$$

and

$$f(x) \in \left[\frac{2k_0-1}{2^{n+1}}, \frac{2k_0}{2^{n+1}}\right) \Rightarrow s_{n+1}(x) = \frac{2k_0-1}{2^{n+1}} > \frac{k_0-1}{2^n} = s_n(x).$$

Recalling k_0 is arbitrary, for every $x \in f^{-1}([0, n+1))$ and $n \in \mathbb{N}$ we have $s_n(x) \leq s_{n+1}(x)$ and $s_n(x) \leq f(x)$.

Similarly, at step n: $f(x) \in [n, \infty] \Rightarrow s_n(x) = n$, and at step $n+1$:

$$f(x) \in [n, n+1) \text{ or } [n+1, \infty].$$

Therefore

$$f(x) \in [n, n+1) \Rightarrow s_{n+1}(x) = n = s_n(x)$$

and

$$f(x) \in [n+1, \infty] \Rightarrow s_{n+1}(x) = n+1 > n = s_n(x).$$

In all cases, then, for every $x \in f^{-1}([0, \infty))$ and every $n \in \mathbb{N}$ we obtain $s_n(x) \leq s_{n+1}(x)$ and $s_n(x) \leq f(x)$.

■

Our next theorem shows instead how a measurable function can be approximated by continuous maps up to a set of arbitrarily small measure.

Theorem 10.4.2 (Lusin–Littlewood's third principle—approximation of measurable functions by continuous maps) *Consider a complete measure space $(X, \mathscr{L}, m)$, where $X \subseteq \mathbb{R}$ is an L-measurable set such that $m(X) < \infty$, $\mathscr{L}$ is the Lebesgue σ-algebra on $\mathbb{R}$ and m is a regular measure on X. If $f : X \mapsto \bar{\mathbb{R}}$ is an m-a.e. bounded, Lebesgue measurable map,*

$$m\big(\{x \in X : f(x) = \pm\infty\}\big) = 0,$$

then for every $\epsilon > 0$:

1. *there exists a compact set $K \subseteq X$ with $m(X \setminus K) < \epsilon$;*
2. *the restriction of f to K is continuous, i.e.: f is continuous on X up to a set of arbitrarily small measure.*

Proof To start with, take as $f(x)$ the indicator function $\varphi_E(x)$, $E \in \mathscr{L}$.

By Proposition 9.4.1, given $\epsilon > 0$ there exists a compact subset $K' \subseteq E$ such that

$$m(E \setminus K') < \epsilon/2.$$

As $X \setminus E$ is measurable, there exists a compact subset $K'' \subseteq (X \setminus E)$ such that

$$m((X \setminus E) \setminus K'') < \epsilon/2.$$

Setting $K = K' \cup K''$ we observe

$$\begin{aligned}(X \setminus E) \setminus K &= [(X \setminus E) \setminus (K' \cup K'')] = \underbrace{[(X \setminus E) \setminus K']}_{=(X \setminus E)} \cap [(X \setminus E) \setminus K''] \\ &= [(X \setminus E) \setminus K'']\end{aligned}$$

and

$$(E \setminus K) = [(E \setminus (K' \cup K'')] = [(E \setminus K') \cap \underbrace{(E \setminus K'')}_{=E}] = (E \setminus K'),$$

and so

$$\begin{aligned}m(X \setminus K) &= m(((X \setminus E) \cup E)) \setminus K) \\ &= m(((X \setminus E) \setminus K) \cup (E \setminus K)) \\ &= m((X \setminus E) \setminus K) + m(E \setminus K) \\ &= m((X \setminus E) \setminus K'') + m(E \setminus K') < \epsilon.\end{aligned}$$

Consider the restriction

$$\varphi_{X|K}(x) = \begin{cases} 1 \text{ if } x \in K \\ 0 \text{ if } x \notin K. \end{cases}$$

As $K' \cap K'' = \emptyset$ and φ_X is constant on K, then φ_X is continuous on K. Since ϵ is arbitrary, this means $\varphi_{X|K}(x)$ is continuous on X up to a set of arbitrarily small measure, proving the claim for characteristic functions.

Now suppose $f(x)$ is the simple function $s : X \mapsto [0, \infty]$,

$$s(x) = \sum_{i=1}^{n} a_i \varphi_{E_i}(x), \quad x \in X.$$

Given $\epsilon > 0$ we can apply the previous part with ϵ/n to every characteristic function in the linear combination $s(x)$.

Pass to the case where $f(x)$ is non-negative and bounded: $0 \le f(x) \le M$. Fix $n \in \mathbb{N}$ and divide the codomain's interval $[0, M)$ in n equal parts. This gives points $y_i = i \cdot (M/n)$, $i = 0, \ldots, n$. For $i = 0, \ldots, n-1$ define sets

$$E_i = \left\{x \in X : y_i \le f(x) < y_{i+1}\right\}.$$

As $f(x)$ is Lebesgue measurable, the E_i are measurable, and satisfy $E_i \cap E_j = \emptyset$, $i \ne j$, plus $\bigcup_{i=0}^{n-1} E_i = X$. We use them to build the non-negative simple function

$$s_n(x) = \sum_{i=0}^{n-1} y_i \varphi_{E_i}(x).$$

For $\epsilon > 0$ and every $n \in \mathbb{N}$, from what we proved above there exists a compact set $K_n \subseteq X$ such that

$$m\big(X \setminus K_n\big) \leq \frac{\epsilon}{2^n}$$

and $s_n(x)$ is continuous on K_n. As $n \in \mathbb{N}$ varies we obtain thus an increasing sequence of compact sets K_n such that

$$K = \bigcap_{n=1}^{\infty} K_n \subseteq X.$$

This K is compact, as countable intersection of compact sets, and further,

$$\begin{aligned} m\big(X \setminus K\big) &= m\left(X \cap \left(\bigcap_{n=1}^{\infty} K_n\right)^c\right) = m\left(X \cap \left(\bigcup_{n=1}^{\infty} K_n^c\right)\right) \\ &= m\left(\bigcup_{n=1}^{\infty} \big(X \setminus K_n\big)\right) \leq \sum_{n=1}^{\infty} \frac{\epsilon}{2^n} = \epsilon. \end{aligned}$$

To prove statement *2.* take $x \in X$. Obviously $x \in E_i$ for some i, so

$$\begin{aligned} |f(x) - s_n(x)| &= |f(x) - y_i \varphi_{E_i}(x)| = |f(x) - y_i| \leq y_{i+1} - y_i \\ &= (i+1) \cdot \frac{M}{n} - i \cdot \frac{M}{n} = \frac{M}{n} \end{aligned}$$

and then

$$\sup_{x \in X} \big|f(x) - s_n(x)\big| \leq \frac{M}{n}.$$

Hence $\{s_n(x)\}_{n \geq 1}$ converges uniformly to $f(x)$ on K, and each $s_n(x)$ is continuous on K, so $f(x)$ too, because the uniform limit of continuous maps is continuous.

Now suppose $f(x) \geq 0$. As $f(x)$ is bounded m-a.e.,

$$m(E) = m\big(\big\{x \in X : f(x) = \infty\big\}\big) = 0.$$

For any $n \in \mathbb{N}$ define

$$E_n = \big\{x \in X : f(x) \geq n\big\}.$$

The sequence is decreasing and such that $m(E_1) \leq m(X) < \infty$ and $E = \bigcap_{n=1}^{\infty} E_n$. The measure's continuity from above forces

$$m(E) = \lim_{n \to \infty} m(E_n) = 0;$$

Therefore for $\epsilon > 0$ there exists an $\bar{n}$ such that

$$m(E_{\bar{n}}) = m\big(\{x \in X : f(x) \geq \bar{n}\}\big) \leq \epsilon/2,$$

and this is the case for every $n \geq \bar{n}$. Put $X' = (X \setminus E_{\bar{n}})$ (measurable) and note $0 \leq f(x) < \bar{n}$. By the previous part, for every $\epsilon > 0$ there exists an K contained in X' such that $m(X' \setminus K) \leq \epsilon/2$ and $f(x)$ is continuous on K. But

$$m(X \setminus K) = m(E_{\bar{n}}) + m(X' \setminus K) \leq \epsilon,$$

so the claim for $f \geq 0$ is settled.

At last, if $f(x)$ is fully arbitrary, the general statement descends from the fact that $f(x) = f^+(x) - f^-(x)$. ■

The English mathematician J. E. Littlewood abridged Lusin's theorem[3] like this: "measurable functions are approximately continuous".

Remark 10.4.1 A Lebesgue measurable function may be discontinuous everywhere, as is the Dirichlet function. So it is interesting that the Dirichlet function satisfies Lusin's theorem on [0, 1]. To see it, take a listing $\{q_i\}_{i \geq 1}$ of the rationals in [0, 1]. Fix $\epsilon > 0$ and associate to q_i the interval

$$Q_i = \left(q_i - \frac{\epsilon}{2^i}, q_i + \frac{\epsilon}{2^i}\right).$$

The set $\bigcup_{i=1}^{\infty} Q_i$ contains all the rationals in [0, 1], it is open, and $\sum_{i=1}^{\infty} \ell(Q_i) < \epsilon$. Then

$$F = [0, 1] \setminus \left(\bigcup_{i=1}^{\infty} Q_i\right)$$

satisfies $\ell\big((0, 1) \setminus F\big) < \epsilon$, and the restriction of the Dirichlet function $f_{|F}$ is identically zero because there are no rationals in F. Therefore the restriction is uniformly continuous on F. ♣

Eventually we come to the so-called Littlewood first principle, whereby a finite-measure subset of $\mathbb{R}$ is approximately a finite union of pairwise-disjoint intervals. This fact follows directly from Proposition 9.3.5.

[3] Also known as Littlewood's third principle.

Proposition 10.4.1 (Littlewood's first principle) *Let $E \subseteq \mathbb{R}$ be a set in $\mathscr{L}$ with measure $m(E) < \infty$. For any $\epsilon > 0$ there exists a finite union of pairwise-disjoint intervals $I_1, \ldots, I_n \in \mathcal{A}(\mathcal{I})$, with $F = \bigsqcup_{i=1}^{n} I_i$, such that*

$$m(E \Delta F) < \epsilon.$$

■

10.5 The Vitali–Cantor Map

We already know the Borel σ-algebra $\mathscr{B}(\mathbb{R})$ is contained in the Lebesgue σ-algebra $\mathscr{L}(\mathbb{R})$, and we have also seen that the axiom of choice brings into being subsets that cannot be Lebesgue measurable.

Here we shall explain how to produce a Lebesgue measurable set that is not Borel measurable. For this purpose we use the Cantor set to construct a function which, among others, was discussed and popularized by Vitali (1905). For this reason we shall call the resulting function the Vitali–Cantor map.

Example 10.5.1 (*The Vitali–Cantor map*) Consider the Cantor set of Example 2.3.3. There we saw that the residual sets produced whilst deleting are:

$$C_1 = \left[0, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right] = [a_{1,1}, b_{1,1}] \cup [a_{1,2}, b_{1,2}]$$

$$C_2 = \left[0, \frac{1}{9}\right] \cup \left[\frac{2}{9}, \frac{3}{9}\right] \cup \left[\frac{6}{9}, \frac{7}{9}\right] \cup \left[\frac{8}{9}, 1\right] = [a_{2,1}, b_{2,1}] \cup [a_{2,2}, b_{2,2}] \cup [a_{2,3}, b_{2,3}] \cup [a_{2,2^2}, b_{2,2^2}]$$

$$\begin{aligned} C_3 = &\left[0, \frac{1}{27}\right] \cup \left[\frac{2}{27}, \frac{3}{27}\right] \cup \left[\frac{6}{27}, \frac{7}{27}\right] \cup \left[\frac{8}{27}, \frac{9}{27}\right] \cup \left[\frac{18}{27}, \frac{19}{27}\right] \cup \left[\frac{20}{27}, \frac{21}{27}\right] \cup \left[\frac{24}{27}, \frac{25}{27}\right] \\ &\cup \left[\frac{26}{27}, 1\right] = [a_{3,1}, b_{3,1}] \cup [a_{3,2}, b_{3,2}] \cup [a_{3,3}, b_{3,3}] \cup [a_{3,4}, b_{3,4}] \cup [a_{3,5}, b_{3,5}] \cup [a_{3,6}, b_{3,6}] \\ &\cup [a_{3,7}, b_{3,7}] \cup [a_{3,2^3}, b_{3,2^3}] \end{aligned}$$

$$\vdots$$

$$C_n = [a_{n,1}, b_{n,1}] \cup [a_{n,2}, b_{n,2}] \cup \cdots \cup [a_{n,k}, b_{n,k}] \cup \cdots \cup [a_{n,2^n}, b_{n,2^n}]$$

$$\vdots$$

At step n one deletes $2^n - 1$ middle intervals, leaving 2^n closed disjoint intervals of length $1/3^n$.

For every $n \in \mathbb{N}$ consider $f_n : [0, 1] \mapsto \mathbb{R}$,

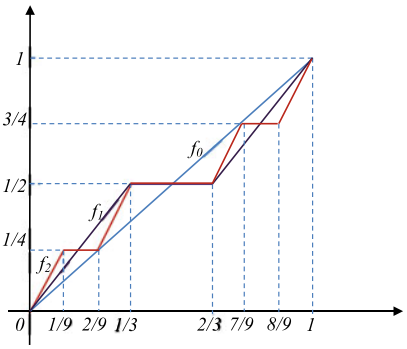


Fig. 10.2 Graph of the Vitali–Cantor map

$$\begin{cases} \dfrac{k-1}{2^n} \le f_n(x) \le \dfrac{k}{2^n} \quad \forall x \in [a_{n,k}, b_{n,k}],\ k = 1, \dots, 2^n \\ f_n(x) = \dfrac{k}{2^n} \forall x \in (b_{n,k}, a_{n,k+1}) \subseteq \big([0,1] \setminus [a_{n,k}, b_{n,k}]\big),\ k = 1, \dots, 2^n - 1. \end{cases}$$

This $f_n(x)$ is

- linearly increasing, ranging between values $\frac{k-1}{2^n}$ and $\frac{k}{2^n}$ inside $[a_{n,k}, b_{n,k}]$, for $k = 1, 2, \dots, 2^n$;
- constant, equal to $\frac{k}{2^n}$, on deleted intervals $(b_{n,k}, a_{n,k+1})$, for $k = 1, \dots, 2^n - 1$.

The graph of $f_n(x)$ at step n is a polygonal path made of as many segments as the overall number $(2^n - 1) + 2^n$ of (deleted and) remaining intervals joining $(a_{n,k}, (k-1)/2^n)$ to $(b_{n,k}, k/2^n)$. Of these intervals, $2^n - 1$ are horizontal and 2^n have slope $(3/2)^n$. Every (upper-)slanted segment is followed by a horizontal one, then another slanted one and so on. This makes $f_n(x)$ increasing.

Moreover, $f_n(x)$ is onto: this happens everywhere and for instance where f_n is constant, for on $x \in (b_{n,k}, a_{n,k+1})$ it has the value $f_n(x) = k/2^n$.

Let us compare the various maps $f_n(x)$, i.e. the paths generated at the different steps. For every $n > 1$ the graph of $f_n(x)$ is obtained from that of $f_{n-1}(x)$ by not touching the $2^{n-1} - 1$ horizontal segments of step $n - 1$, but replacing the 2^{n-1} slanted bits with three segments: a slanted one, then a horizontal one in the middle, then another slanted one (Fig. 10.2).

In general the open intervals along the x-axis corresponding to horizontal segments are pairwise disjoint, because any horizontal segment lies between two slanted bits and remains unchanged during the iterations. This means any two intervals $(b_{n,k}, a_{n,k+1})$ and $(b_{m,k'}, a_{m,k'+1}) \subseteq \big([0,1] \setminus C\big)$, for $n, m \ge 1$, $k = 1, \dots, 2^n - 1$, and $k' = 1, \dots, 2^m - 1$, are disjoint for $k \ne k'$.

We would like to know whether $\{f_n(x)\}_{n \ge 0}$ converges to some limit $f(x)$, and how. Let us fix step n. On intervals where f_{n-1} is constant, f_n has that same value, so $|f_n(x) - f_{n-1}(x)| = 0$. On intervals $[a_{n-1,k}, b_{n-1,k}]$, $k = 1, \dots, 2^{n-1}$,

$$|f_n(x) - f_{n-1}(x)| < |f_{n-1}(a_{n-1,k}) - f_{n-1}(b_{n-1,k})| = \left| \frac{k}{2^{n-1}} - \frac{k-1}{2^{n-1}} \right| = \frac{1}{2^{n-1}},$$

which is, for $n > 1$, the largest increment of $f_{n-1}(x)$ on each $[a_{n-1,k}, b_{n-1,k}]$, $k = 1, \ldots, 2^{n-1}$. When $n = 1$ the biggest increment of f_0 is naturally 1. Fix $m > 1$ and two positive integers $i, j > m$. The maximum increment on each of the $2^i, 2^j$ slanted segments of $f_i(x)$, $f_j(x)$ respectively equals $1/2^i$, $1/2^j$, both less than $1/2^m$. Suppose $i > j$ without loss of generality. Then

$$1/2^m > 1/2^j > 1/2^i$$

and the absolute maximum difference between $f_i(x)$ and $f_j(x)$ is less than $\frac{1}{2^j}$ for every $x \in [a_{j,k}, b_{j,k}]$, $k = 1, \ldots, 2^j$:

$$|f_i(x) - f_j(x)| < \frac{1}{2^j} < \frac{1}{2^m}.$$

This holds for every interval $[a_{j,k}, b_{j,k}]$, $k = 1, \ldots, 2^j$ and hence for every interval on which f_j is constant. In general, then, for every $x \in [0, 1]$ the absolute maximum difference between $f_i(x)$, $f_j(x)$ satisfies

$$|f_i(x) - f_j(x)| < \frac{1}{2^m}.$$

Given any $x \in [0, 1]$ and $\epsilon > 0$, there must exist an m, with $1/2^m < \epsilon$, such that

$$|f_i(x) - f_j(x)| < \epsilon$$

for all i and all $j > m$. Hence the sequence $\{f_n(x)\}_{n \geq 0}$ is Cauchy and, as such, it converges pointwise on $\mathbb{R}$ to some $f : [0, 1] \mapsto [0, 1]$.

The sequence converges also uniformly to f on $[0, 1]$: given $\epsilon > 0$ there exists an m such that, for $j > m$ with $\frac{1}{2^j} < \epsilon$, for every $i \in \mathbb{N}$

$$|f_{j+i}(x) - f_j(x)| < \epsilon, \forall x \in [0, 1].$$

That is to say, for $j > m$ and every $x \in [0, 1]$

$$\lim_{i \to \infty} |f_{j+i}(x) - f_j(x)| < \epsilon,$$

so for every $j > m$ and $x \in [0, 1]$

$$|f(x) - f_j(x)| < \epsilon.$$

Note that each f_n is continuous on $[0, 1]$, so Theorem 4.2.1 forces f to be continuous on $[0, 1]$.

Furthermore f is non decreasing. Taking $x < y$ in $[0, 1]$,

- $x, y \in C \Rightarrow f(x) < f(y)$;
- if $x, y \in [0, 1] \setminus C$, suppose for some $n \in \mathbb{N}$ and $k = 1, \dots, 2^n - 1$ we have $x, y \in (b_{n,k}, a_{n,k+1}) \subseteq [0, 1] \setminus C$, then $f(x) = f(y)$, otherwise $f(x) < f(y)$;
- if $x \in C \wedge y \in \big([0, 1] \setminus C\big)$ or vice versa, then $f(x) \leq f(y)$.

Hence $f(x) \leq f(y)$ whenever $x < y \in [0, 1]$.

What is more, $f(x)$ has zero derivative on $[0, 1]$ except for the zero-measure set C. Take in fact $x \in \big([0, 1] \setminus C\big)$. For some n we have $x \in \big([0, 1] \setminus C_n\big)$. Call $(b_{n,k}, a_{n,k+1})$ the interval in $\big([0, 1] \setminus C_n\big)$ containing x. As f_n is constant on $(b_{n,k}, a_{n,k+1})$ and $f_m(x) = f_n(x)$ on $(b_{n,k}, a_{n,k+1})$ for every $m \geq n$, passing to the limit as $m \to \infty$ gives $f_n(x) = f_{n+1}(x) = \cdots = f(x) = \frac{k}{2^n}$ on $(b_{n,k}, a_{n,k+1})$. Our $x \in \big([0, 1] \setminus C\big)$ is arbitrary so we may set $f(x) = k$ for every $x \in \big([0, 1] \setminus C\big)$, and then $f'(x) = 0$ on $\big([0, 1] \setminus C\big)$. But $\ell(C) = 0$, so $f'(x)$ is zero ℓ-a.e. on $[0, 1]$. We call $f : [0, 1] \mapsto [0, 1]$ the Vitali–Cantor map.

At this juncture consider $g : [0, 1] \mapsto [0, 2]$, $g(x) = f(x) + x$, where $f(x)$ is the Vitali–Cantor map. It is continuous, as sum of continuous maps, and strictly increasing, since $g'(x) = 1$ ℓ-a.e. (in fact $f'(x) = 0$ ℓ-a.e.), hence bijective. The inverse $g^{-1} = h$ is continuous: if $U \subseteq [0, 1]$ is open, then $[0, 1] \setminus U$ is closed and bounded, i.e. compact. But g is continuous so $g([0, 1] \setminus U)$ is compact. As

$$g([0, 1] \setminus U) = h^{-1}([0, 1] \setminus U) = h^{-1}([0, 1]) \setminus h^{-1}(U) = [0, 2] \setminus h^{-1}(U),$$

$[0, 2] \setminus h^{-1}(U)$ is compact. So $h^{-1}(U) \subseteq [0, 2]$ is open, making h continuous. All in all g is a homeomorphism.

We claim $g(C)$ has measure 1. Now, g maps the deleted intervals in $[0, 1]$ (those removed when constructing C) to intervals of the same length in $[0, 2]$. This is because $f(a) = f(b)$ for every removed interval (a, b) belonging to the set $[0, 1] \setminus C$, so

$$\ell(g(a), g(b)) = g(b) - g(a) = f(b) + b - f(a) - a = b - a \tag{10.5.1}$$

and the latter implies

$$\ell\big(g([0, 1] \setminus C)\big) = \ell\big([0, 1] \setminus C\big) = 1.$$

As $[0, 2] = g(C) \bigsqcup g([0, 1] \setminus C)$,

$$\ell\big([0, 2]\big) = \ell\big(g(C)\big) + \ell\big(g([0, 1] \setminus C)\big) = \ell\big(g(C)\big) + 1 = 2,$$

whence $\ell\big(g(C)\big) = 1$.

Since $\ell\big(g(C)\big) > 0$, Remark 9.5.2 tells us there exists a non-measurable set $V \subset g(C)$. Then take $E = h(V) = g^{-1}(V)$. Clearly $E \subset C$ has measure $\ell(C) = 0$. But $\big(\mathbb{R}, \mathscr{L}(\mathbb{R}), \ell\big)$ is complete, making E negligible, i.e. $\ell(E) = 0$.

The map h, inverse of a homeomorphism, is continuous and hence Borel. Therefore for every Borel set B the pre-image $h^{-1}(B)$ is measurable. But we know $h^{-1}(E) = V$ is not measurable, implying E cannot be Borel, albeit measurable with measure zero. The conclusion is $\mathscr{B}(\mathbb{R}) \subset \mathscr{L}(\mathbb{R})$, and the pre-image under a measurable map of a non-Borel measurable set may not be measurable. ★

10.6 Rudiments of the Theory of Random Variables

A random quantity, or random variable, is a function of the outcome of an experiment. The outcome is uncertain before the experiment takes place or before certain informations on the phenomenon under exam become available. A random variable is typically denoted with a capital letter $X, Y, \ldots$, and the outcome of a test, called elementary event, is indicated with $\omega \in \Omega$, where Ω is the sample space.

Definition 10.6.1 (*Random variables*) Let $(\Omega, \mathcal{S}(\Omega))$ and $(\mathbb{R}, \mathscr{B}(\mathbb{R}))$ be measurable spaces. A function $X : \Omega \mapsto \mathbb{R}$ is called random variable (RV) if

$$E = \{\omega \in \Omega : X(\omega) \in B\} = X^{-1}(B) \in \mathcal{S}(\Omega)$$

for every $B \in \mathscr{B}(\mathbb{R})$. A RV is therefore an $\mathcal{S}$-measurable map defined on the measurable space $(\Omega, \mathcal{S}(\Omega))$ with values in Borel sets $\mathscr{B}(\mathbb{R})$. ◇

The main objective of the study of RVs is to compute the probability the variable X assumes values in some $B \in \mathscr{B}(\mathbb{R})$. As the probabilities with which the possible outcomes, or events, manifest themselves are by definition distributed on the sample space Ω (the so-called probability distribution), to determine the probability that a variable X takes value in B we must consider the probability

$$P(X \in B) = P\big(\{\omega \in \Omega : X(\omega) \in B\}\big) = P(X^{-1}(B)). \tag{10.6.1}$$

From formula (10.6.1) the pre-image of $B \in \mathscr{B}(\mathbb{R})$ under X belongs to the event σ-algebra $\mathcal{S}(\Omega)$, which is assumed complete, and so X should be measurable and the measure of $E = \{\omega \in \Omega : X(\omega) \in B\}$ equals $P(X^{-1}(B))$.

Definition 10.6.2 (*Probability distribution functions*) Let $X : \Omega \mapsto \mathbb{R}$ be a RV defined on the probability space $(\Omega, \mathcal{S}(\Omega), P)$. The probability distribution (function) of X, denoted P_X, is the completely additive set function defined by

$$P(E) = P_X(B) = P(X^{-1}(B)), \tag{10.6.2}$$

for every $E = \{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{S}(\Omega)$, where $B \in \mathscr{B}(\mathbb{R})$. ◇

Proposition 10.6.1 (Functions of random variables) *If X is a RV and $f : \mathbb{R} \mapsto \mathbb{R}$ a Borel measurable map, then $Y = f(X)$ is a RV.*

Proof The pre-image $f^{-1}(B)$ of any $B \in \mathscr{B}(\mathbb{R})$, as f is Borel measurable, is a Borel set, so

$$\begin{aligned} Y^{-1}(B) &= \{\omega \in \Omega : f(X(\omega)) \in B\} \\ &= \{\omega \in \Omega : X(\omega) \in f^{-1}(B)\} \\ &= X^{-1}\big(f^{-1}(B)\big) \in \mathcal{S}(\Omega). \end{aligned}$$

Hence Y is a RV. ■

From the above proposition, given a RV X and $\alpha \in \mathbb{R}$,

$$X + \alpha,\ \ \alpha X, |X|,\ \ X^2, 1/X,\ \text{for } X \neq 0,$$

are all RVs.

Proposition 10.6.2 (Sum, difference, product and quotient of RVs) *If X and Y are RVs then so are*

$$Z = X + Y,\ \ Z = X - Y, Z = XY,\ \ Z = X/Y,\ \textit{for } Y \neq 0.$$

Proof To show $Z = X + Y$ is a RV we must prove $\{\omega \in \Omega : Z(\omega) \leq x\} \in \mathcal{S}(\Omega)$ for every $x \in \mathbb{R}$.

Take any $x \in \mathbb{R}$, so $Z(\omega) \leq x$ iff there is a rational q such that $X(\omega) \leq q$ and $Y(\omega) \leq x - q$. Hence

$$\begin{aligned} \{\omega \in \Omega : Z(\omega) \leq x\} &= \bigcup_{q \in \mathbb{Q}} \{\omega \in \Omega : X(\omega) \leq q, Y(\omega) \leq x - q\} \\ &= \bigcup_{q \in \mathbb{Q}} \big(\{\omega \in \Omega : X(\omega) \leq q\} \cap \{\omega \in \Omega : Y(\omega) \leq x - q\}\big). \end{aligned}$$

For every $q \in \mathbb{Q}$

$$\{\omega \in \Omega : X(\omega) \leq q\} \cap \{\omega \in \Omega : Y(\omega) \leq x - q\} \in \mathcal{S}(\Omega)$$

since X, Y are RVs. Hence $\{\omega \in \Omega : Z(\omega) \leq x\} \in \mathcal{S}(\Omega)$.

Now, since $X - Y = X + (-Y)$, by Proposition 10.6.1 $X - Y$ is a RV. Furthermore,

$$XY = \frac{1}{4}\big[(X + Y)^2 - (X - Y)^2\big]$$

shows XY is a RV. Finally, Proposition 10.6.1 implies $1/Y$ is still a RV, and so X/Y is a RV. ■

Sometimes we do know the probability distribution directly (for instance when we toss a coin), while in other cases we have to resort to Eq. (10.6.2). There are various

ways to identify the probability distribution of a RV X. The first is to determine the corresponding cumulative distribution function $F_X(x)$. This is defined, for every $x \in \mathbb{R}$, as

$$F_X(x) = P(X \leq x). \tag{10.6.3}$$

Cumulative distribution functions allow to detect the probability distribution P_X of the sample space Ω and have the advantage of replacing functions defined on sets with functions defined on points. The latter are associated to the Lebesgue–Stieltjes measure of Sect. 9.6 in case

$$\lim_{x\to\infty} \big[F(x) - F(-x)\big]) = \tilde{m}(\mathbb{R}) = 1.$$

Definition 10.6.3 (*Cumulative distribution functions*) We call (cumulative) distribution function of a RV X the map $F_X : \mathbb{R} \mapsto [0, 1]$ given by

$$F_X(x) = P(X \leq x), \quad x \in \mathbb{R}.$$

$\diamond$

In the next proposition we prove the properties of cumulative distribution functions.

Proposition 10.6.3 (Properties of distribution functions) *The distribution function of a RV X is an increasing, right-continuous map such that $P(a < X \leq b) = F_X(b) - F_X(a)$ for every $a < b \in \mathbb{R}$, and*

$$\lim_{x\to-\infty} F_X(x) = 0 \ \textit{and} \ \lim_{x\to\infty} F_X(x) = 1.$$

Proof Let us start by observing that distributions functions (Sect. 9.6) satisfy

$$P(a < X \leq b) = m_{F_X}((a, b]) = F_X(b) - F_X(a),$$

for every $a, b \in \mathbb{R}$, $a < b$. In fact, putting $A = \{\omega \in \Omega : X(\omega) \leq a\}$ and $B = \{\omega \in \Omega : X(\omega) \leq b\}$, we have $A \subseteq B$ and

$$\{\omega \in \Omega : a < X(\omega) \leq b\} = B \cap A^c = B \setminus A.$$

As $P(\Omega) = 1$, the distribution function satisfies

$$P(a < X \leq b) = P(B \setminus A) = P(B) - P(A) = F_X(b) - F_X(a).$$

If $a < b$, moreover, then $A_a = \{\omega \in \Omega : X(\omega) \leq a\}$ is contained in $B_b = \{\omega \in \Omega : X(\omega) \leq b\}$, and hence

$$F_X(a) = P(A_a) = P(X \leq a) \leq P(X \leq b) = P(B_b) = F_X(b),$$

making the distribution function F_X increasing.

As P_X is a measure on the σ-algebra $\mathcal{S}(\Omega)$, we claim F_X is right-continuous. Take $x \in \mathbb{R}$ and $\{x_i\}_{i\geq 1}$ a decreasing sequence tending to x. Define events

$$A = \{\omega \in \Omega : X(\omega) \leq x_1\}, \quad B_1 = \{\omega \in \Omega : X(\omega) \leq x\} \text{ and}$$
$$B_i = \{\omega \in \Omega : x_{i+1} < X(\omega) \leq x_i\}, \ \forall i \geq 1,$$

so $B_i \cap B_j = \emptyset$ for every $i \neq j$, and $A = \bigcup_{i=1}^{\infty} B_i$. As the B_i are independent events we may set

$$P(A) = P\left(\bigcup_{i=1}^{\infty} B_i\right) = \sum_{i=1}^{\infty} P(B_i), \tag{10.6.4}$$

so

$$P(A) = F_X(x_1),\ P(B_1) = F_X(x),\ P(B_i) = F_X(x_i) - F_X(x_{i+1}), \forall i \geq 1.$$

Hence (10.6.4) reads

$$\begin{aligned} F_X(x_1) =& F_X(x) + [F_X(x_1) - F_X(x_2)] + [F_X(x_2) - F_X(x_3)] + \cdots + \\ & [F_X(x_i) - F_X(x_{i+1})] + \cdots, \end{aligned}$$

and then $F_X(x) = \lim_{i\to\infty} F_X(x_i)$, i.e. $F_X(x) = F_X(x^+)$.

Finally, we prove $\lim_{x\to-\infty} F_X(x) = 0$ and $\lim_{x\to+\infty} F_X(x) = 1$.

Regarding the first, let $\{x_i\}_{i\geq 1}$ be a sequence decreasing to $-\infty$. Define events

$$A = \{\omega \in \Omega : X(\omega) \leq x_1\}, \ B_i = \{\omega \in \Omega : x_{i+1} < X(\omega) \leq x_i\}, \ \forall i \geq 1,$$

so $B_i \cap B_j = \emptyset$, $i \neq j$, and $A = \bigcup_{i=1}^{\infty} B_i$. The independence of the B_i forces

$$P(A) = P\left(\bigcup_{i=1}^{\infty} B_i\right) = \sum_{i=1}^{\infty} P(B_i), \tag{10.6.5}$$

and

$$P(A) = F_X(x_1),\ P(B_i) = F_X(x_i) - F_X(x_{i+1}), \forall i \geq 1,$$

so (10.6.5) becomes

$$\begin{aligned} F_X(x_1) =& [F_X(x_1) - F_X(x_2)] + [F_X(x_2) - F_X(x_3)] + \cdots + \\ & [F_X(x_i) - F_X(x_{i+1})] + \cdots \end{aligned}$$

Therefore $F_X(x_1) = F_X(x_1) - \lim_{i\to\infty} F_X(x_i)$. But $\{x_i\}$ tends to $-\infty$, so necessarily $\lim_{i\to\infty} F_X(x_i) = F_X(-\infty) = 0$.

As for the second limit, consider $\{x_i\}_{i\geq 1}$ increasing to ∞. Define

$$\Omega = \{\omega \in \Omega : X(\omega) < \infty\}, \quad B_1 = \{\omega \in \Omega : X(\omega) \leq x_1\},$$
$$\text{and} \quad B_i = \{\omega \in \Omega : x_i < X(\omega) \leq x_{i+1}\}, \ \forall i \geq 1,$$

so $B_i \cap B_j = \emptyset$, $i \neq j$, and $\Omega = \bigcup_{i=1}^{\infty} B_i$. Once more, the B_i are independent so

$$P(\Omega) = P\left(\bigcup_{i=1}^{\infty} B_i\right) = \sum_{i=1}^{\infty} P(B_i), \tag{10.6.6}$$

and

$$P(\Omega) = 1, \ P(B_1) = F_X(x_1), \ P(B_i) = F_X(x_{i+1}) - F_X(x_i), \forall i \geq 1,$$

and (10.6.6) reads

$$\begin{aligned} 1 =& F_X(x_1) + [F_X(x_2) - F_X(x_1)] + [F_X(x_3) - F_X(x_2)] + \cdots + \\ & [F_X(x_{i+1}) - F_X(x_i)] + \cdots, \end{aligned}$$

whence $1 = \lim_{i\to\infty} F_X(x_i)$. As $\{x_i\}$ tends to ∞,

$$\lim_{i\to\infty} F_X(x_i) = F_X(\infty) = 1.$$

■

Proposition 10.6.4 (Uniform continuity of continuous distribution functions) *A continuous distribution function on* $\mathbb{R}$ *is uniformly continuous.*

Proof For every $x \in \mathbb{R}$ we have $0 \leq F_X(x) \leq 1$. Given $\epsilon > 0$ we can find real numbers $a < b$ such that

$$0 \leq F_X(a) < \frac{\epsilon}{2} \ \Rightarrow \forall x < a, \ |F_X(x) - F_X(a)| < \frac{\epsilon}{2}$$

and

$$F_X(b) < 1 - \frac{\epsilon}{2} \ \Rightarrow \forall x > b, \ |F_X(x) - F_X(b)| < \frac{\epsilon}{2}.$$

Let us examine the situation in $[a, b]$, where F_X is continuous and hence uniformly continuous. Given $x_1, x_2 \in [a, b]$ there exists $\eta(\epsilon) > 0$ such that for every $x_1, x_2 \in [a, b]$ with $|x_1 - x_2| < \eta(\epsilon)$,

$$|F_X(x_1) - F_X(x_2)| < \frac{\epsilon}{2}.$$

Choose $\delta = \min(\eta(\epsilon), b - a)$ and x_1, x_2 so that $|x_1 - x_2| < \delta$: if $x_1, x_2 \in [a, b]$ then

$$|F_X(x_1) - F_X(x_2)| < \frac{\epsilon}{2},$$

while if $x_1 < a \leq x_2$

$$\begin{aligned} |F_X(x_1) - F_X(x_2)| &= |F_X(x_1) - F_X(a) + F_X(a) - F_X(x_2)| \\ &\leq |F_X(x_1) - F_X(a)| + |F_X(a) - F_X(x_2)| < \epsilon. \end{aligned}$$

Similarly for $x_1 \leq b < x_2$. In all cases $|F_X(x_1) - F_X(x_2)| < \epsilon$ provided $|x_1 - x_2| < \delta$. ■

We shall prove the probability of the elementary event $P(X = x)$ equals the difference between the one-sided limits of F_X at x.

Proposition 10.6.5 (Probability of elementary events) *If X is a RV and F_X the corresponding distribution function, then*

$$P(X = x) = F_X(x) - F_X(x^-) \geq 0$$

for every $x \in \mathbb{R}$.

Proof Pick an increasing sequence $\{x_i\}_{i \geq 1}$ converging to x and define events

$$\begin{aligned} &A = \{\omega \in \Omega : X(\omega) \leq x\},\ A_1 = \{\omega \in \Omega : X(\omega) \leq x_1\}, \\ &B_i = \{\omega \in \Omega : x_i < X(\omega) \leq x_{i+1}\}, \forall i \geq 1,\ C = \{\omega \in \Omega : X(\omega) = x\}. \end{aligned}$$

Note A_1, B_i, C are mutually independent for any $i \geq 1$, so

$$A = C \cup A_1 \cup \left(\bigcup_{i=1}^{\infty} B_i \right).$$

Define

$$P(A) = P(C) + P(A_1) + \sum_{i=1}^{\infty} P(B_i),$$

and observe $P(A) = F_X(x)$, $P(A_1) = F_X(x_1)$ and $P(B_i) = F_X(x_{i+1}) - F_X(x_i)$ for every $i \geq 1$, whence the previous relation becomes

$$F_X(x) = P(C) + F_X(x_1) + \sum_{i=1}^{\infty} [F_X(x_{i+1}) - F_X(x_i)] = P(C) + \lim_{i \to \infty} F_X(x_i).$$

As $\lim_{i\to\infty} F_X(x_i) = F_X(x^-)$ we see that

$$P(C) = F_X(x) - F_X(x^-).$$

Consequently $P(X = x) > 0$ iff F_x has a jump discontinuity at x. ■

Remark 10.6.1 (*Elementary relationships between probability distributions and distribution functions*) Since a distribution function is right-continuous, the discontinuities of F_X are of first kind and at most countable. By Proposition 10.6.5, moreover, if x is a discontinuity for F_X then $P(X = x) > 0$.

At last, calling F_X the distribution function of the RV X, for any $a < b$ we have

$$\begin{aligned}
P(a < X \le b) &= F_X(b) - F_X(a)\\
P(a < X < b) &= P(a < X \le b) - P(X = b)\\
&= F_X(b) - F_X(a) - F_X(b) + F_X(b^-) = F_X(b^-) - F_X(a)\\
P(a \le X < b) &= P(a < X < b) + P(X = a)\\
&= F_X(b^-) - F_X(a) + F_X(a) - F_X(a^-) = F_X(b^-) - F_X(a^-)\\
P(a \le X \le b) &= P(a \le X < b) + P(X = b)\\
&= F_X(b^-) - F_X(a^-) + F_X(b) - F_X(b^-) = F_X(b) - F_X(a^-).
\end{aligned}$$

♣

Part V
Theory of integration

Introduction

To determine the area of the region below the graph of a map, what one usually does is divide the base, i.e. the interval of integration, in a suitable number of subintervals

$$a = a_0 < a_1 < a_2 < \ldots < a_n = b,$$

and for each (a_{i-1}, a_i) choose one of the infinitely many values $f(x)$ attains on it.

Then compute

$$\begin{aligned} S &= f(x_1)(a_1 - a_0) + f(x_2)(a_2 - a_1) + \cdots + f(x_n)(a_n - a_{n-1}) \\ &= f(x_1)\ell(I_1) + f(x_2)\ell(I_2) + \cdots + f(x_n)\ell(I_n) \\ &= \sum_{i=1}^{n} f(x_i)\ell(I_i). \end{aligned}$$

The roundoff errors made (both up and down) in doing so tend to become smaller, until they vanish, as the subintervals shrink to 0. This is the typical logic of a definition of integral such as the one adopted by Cauchy, which was justified under the assumption the map f to be integrated is continuous. In this case, in fact, to nearby x-values there correspond nearby images $f(x)$. Choosing any one of the infinite values of f over a subinterval would work, because the range of f does not present irregular peaks, like jumps or discontinuities.

As he formulated the notion of integral, Riemann wanted to set up a general theory to include totally discontinuous maps as well. Although his aim was different from Cauchy's, however, he adopted a similar construction.

In practice he divided the integration interval of f in a finite number of subintervals, and assumed that as these tended to zero f behaved, if not exactly as a continuous map, at least with a certain degree of regularity, i.e. with oscillations bounded by some threshold. In this way he managed to capture the integrability of only certain discontinuous maps, certainly not the majority.

Henri Lebesgue, in the 1926 paper "*Sur le développment de la notion d'intégrale*", astonished by the fact that Riemann's method would work for some discontinuous functions, wrote that when that happened it was by sheer accident. The reason for the Riemann integral's failure to generalise as originally intended, as Lebesgue stressed in the aforementioned article, lies in insisting the oscillations of f are confined within certain limits. Put differently, the problem is applying Cauchy's integrability criterion for continuous maps – or a petty modification of it – to discontinuous functions.

Lebesgue therefore scrapped Riemann's method altogether, considering it inadequate in view of generalisations, and adopted instead Cauchy's guideline that close values should be gathered and multiplied by the measure of the corresponding set along the x-axis. Given the discontinuities of f, that set might not be one of the predetermined intervals, but might on the contrary be scattered over the entire domain of integration, to form a complicated linear set of points. Hence the need of a theory of measure capable establishing the 'size' of arbitrary linear subsets of the real line.

The only way to group the values $f(x)$ by proximity was to divide into a finite number of subintervals not the interval of integration, but the interval $[c, d]$ of range values between the infimum and supremum of f:

$$c = c_0 < c_1 < \ldots < c_n = d.$$

Thus over each subinterval the values $f(x)$ are certainly near one another, and one can make them as near as one wants by refining the partition, i.e. increasing the number of subintervals. Moreover, for every $i = 1, 2, \ldots$, nearby values $f(x) \in [c_{i-1}, c_i)$ will have pre-image

$$A_i = \{x \in [a, b] : c_{i-1} \leq f(x) < c_i\}$$

on the x-axis. The latter might not have a naive measure, like Riemann's intervals. On the contrary, in view of the discontinuities of f the structure could turn out to be extremely intricate. Thanks to a suitable theory of measure we will be able to find the measure ℓ^* of any set A_i, and then compute the two sums

$$S = \sum_{i=1}^{n} c_i \ell^*(A_i) \quad \text{and} \quad s = \sum_{i=1}^{n} c_{i-1} \ell^*(A_i).$$

The former is larger, the latter smaller, than the area of the region underneath f. As subintervals tend to zero S and s tend to the respective infimum S_* and supremum s^*. Lebesgue proved that for bounded measurable functions $S_* = s^*$, and this common number is precisely the integral of $f(x)$ over $A = \bigcup_{i=1}^{\infty} A_i$, denoted

$$\int_A f(x)\ell(dx).$$

Needless to say, if A is the entire interval $[a, b]$, then the notation used is slightly different from the Riemann integral's, i.e. $\int_{[a,b]} f(x)\ell(dx)$ as opposed to $\int_a^b f(x)dx$.

In this way the subintervals I_i appearing in the Riemann's definition, whose measure is elementary, are replaced by sets A_i, which may be rather complicated sets.

In the aforementioned survey paper Lebesgue explains the novelties of his integral with an effective analogy, which we transcribe faithfully because of its intelligibility and straightforwardness:

"*[…] with Riemann's procedure, one attempted to sum the indivisibles by taking them in the order in which they are given by the variation of x, one would operate like a merchant with no method who counted coins and bills in the order they came into his hands; whereas we operate like the methodical tradesman who says:*

I have $m(E_1)$ 1*-crown coins with total value* $1 \cdot m(E_1)$,

I have $m(E_2)$ 2*-crown coins with total value* $2 \cdot m(E_2)$,

I have $m(E_3)$ 5*-crown coins with total value* $5 \cdot m(E_3)$

etc, so altogether I have

$S = 1 \cdot m(E_1) + 2 \cdot m(E_2) + 5 \cdot m(E_3) + \ldots$

The two processes will lead, certainly, the merchant to the same result because, irrespective of how rich he is, there is only a finite number of bills to count; but, for us, who have to sum infinitely many indivisibles, the difference of the two ways is crucial".

Integration theory therefore requires we keep conceptually separated the values of f from what we call integration sets A_i or I_i, which for the Lebesgue integral are pre-images under f. The structure of the A_i, which are spread across the x-axis in ways which can be more or less complex, is determined by range values of f close to one another. Roughly, we may say a map f is Lebesgue integrable if every set that f projects on the horizontal axis is measurable.

Within this framework the study we have made of so-called measurable functions is fully justified.

Chapter 11
The Lebesgue Integral



11.1 The Integral of Non-negative, Simple Measurable Maps

Writing a map $f : X \mapsto \bar{\mathbb{R}}$ using its positive and negative parts f^+, f^- has a number of advantages. The first is one can then define integrals just for non-negative functions. The second is that is allows to view integration as a linear operator, i.e. $\int f\, dm = \int f^+\, dm - \int f^-\, dm$.

Throughout this chapter the space $(X, \mathcal{S}, m)$ will be complete and with σ-finite measure, unless said otherwise.

Let us remind $\mathscr{M}_0^+(X, \mathcal{S})$ is the collection of non-negative $\mathcal{S}$-measurable simple maps $s : X \mapsto [0, \infty]$,

$$s(x) = \sum_{i=1}^{n} a_i \varphi_{A_i}(x), \quad x \in X.$$

The $a_i \in [0, \infty]$ are distinct, and $A_i \in \mathcal{S}$, $A_i \cap A_j = \emptyset, \forall i \neq j$ and $\bigcup_{i=1}^n A_i = X$.

Given simple $\mathcal{S}$-measurable maps s, s_1, s_2 with values in $[0, \infty]$, and a constant $\alpha \geq 0$, we know the following are simple and $\mathcal{S}$-measurable (and $[0, \infty]$-valued): (αs), $(s_1 + s_2)$, $(s_1 \cdot s_2)$, $(s_1 \vee s_2)$, $(s_1 \wedge s_2)$, $(s\varphi_E)$ for any $E \in \mathcal{S}$ etc.

We shall define the integral for such class of functions. Keeping in mind the integral of a characteristic function of an $\mathcal{S}$-measurable set $A \subseteq X$ is defined by

$$\int_X \varphi_A\, dm = 1 \cdot m(A),$$

we may define integrals of simple non-negative measurable maps in the following way.

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S. Gentili, *Measure, Integration and a Primer on Probability Theory*, UNITEXT 125, https://doi.org/10.1007/978-3-030-54940-4_11

Definition 11.1.1 (*Integral of non-negative simple maps*) On the measure space $(X, \mathcal{S}, m)$ consider $s = \sum_{i=1}^{n} a_i \varphi_{A_i} \in \mathscr{M}_0^+(X, \mathcal{S})$. We call Lebesgue integral of s with respect to measure m (or simply, in m) the expression

$$\int_X s \, dm = \sum_{i=1}^{n} a_i m(A_i). \qquad \diamond$$

When $a_i = 0$ and $m(A_i) = \infty$ the convention is to set $0 \cdot (\infty) = 0$, so that the integral of 0 over a set of infinite measure vanishes. The integral might be ∞ in case $a_i \in (0, \infty]$ and $m(A_i) = \infty$ for some $i \geq 1$.

This notion does not produce expressions like $\pm\infty \cdot \mp\infty$,[1] and is well defined because it does not depend on the particular representation of s. Suppose in fact $s : X \mapsto [0, \infty]$ simple can be written

$$s(x) = \sum_{i=1}^{n} a_i \varphi_{A_i} = \sum_{j=1}^{m} b_j \varphi_{B_j},$$

where $A_i, B_j \in \mathcal{S}, i = 1, \ldots, n, \ j = 1, \ldots, m$, are pairwise disjoint and $\bigcup_{i=1}^{n} A_i = X = \bigcup_{j=1}^{m} B_j$. The claim is that

$$\int_X s \, dm = \sum_{i=1}^{n} a_i m(A_i) = \sum_{j=1}^{m} b_j m(B_j).$$

But by Remark 10.2.4

$$\begin{aligned} \int_X s \, dm &= \sum_{i=1}^{n} a_i m(A_i) = \sum_{i=1}^{n} a_i m\Big(\bigsqcup_{j=1}^{m} (A_i \cap B_j) \Big) = \sum_{i=1}^{n} a_i \Big(\sum_{j=1}^{m} m(A_i \cap B_j) \Big) \\ &= \sum_{j=1}^{m} b_j \Big(\sum_{i=1}^{n} m(A_i \cap B_j) \Big) = \sum_{j=1}^{m} b_j m(B_j). \end{aligned}$$

Definition 11.1.2 (*Integral of non-negative simple maps on measurable sets*) Given a measure space $(X, \mathcal{S}, m)$, the Lebesgue integral with respect to m of the simple map $s = \sum_{i=1}^{n} a_i \varphi_{A_i} \in \mathscr{M}_0^+(X, \mathcal{S})$ on $E \in \mathcal{S}$ is the quantity

$$\int_E s \, dm = \int_X s \varphi_E \, dm = \sum_{i=1}^{n} a_i m(A_i \cap E). \qquad \diamond$$

[1] As $s \in \mathscr{M}_0^+$ is $[0, \infty]$-valued the worst that can happen is $\int_X s \, dm = \sum_{i=1}^{n} a_i m(A_i) = \infty \cdot \infty$.

Recalling Definition 9.3.1 we immediately see

$$\int_E s\, dm = \sum_{i=1}^{n} a_i m_{|(A_i \cap E)} = \int_X s\, dm_{|E}. \tag{11.1.1}$$

Let us examine the properties of the integral of simple maps, from which the general theory of Lebesgue integration can be developed.

Proposition 11.1.1 (Properties of the integral of non-negative simple maps) *For any* $s, s_1, s_2 \in \mathscr{M}_0^+(X, \mathcal{S})$ *and* $\alpha \geq 0$*:*

1. $\int_X s\, dm \in [0, \infty]$*;*
2. *(homogeneity)* $\int_X \alpha s\, dm = \alpha \int_X s\, dm$
3. *(finite additivity)* $\int_X (s_1 + s_2)\, dm = \int_X s_1\, dm + \int_X s_2\, dm$*;*
4. *(monotonicity)* $\int_X s_1\, dm \leq \int_X s_2\, dm$*, if* $s_1 \leq s_2$ *in* X*;*
5. *(monotonicity w.r.t. integration domain)* $\int_E s\, dm \leq \int_X s\, dm, \quad \forall E \in \mathcal{S}$*;*
6. $\int_X (s_1 \wedge s_2)\, dm \leq \int_X s_i\, dm \leq \int_X (s_1 \vee s_2)\, dm, \ i = 1, 2.$

Proof 1. Regarding 1. it suffices to note the integral is a non-negative extended real number, as sum of non-negative terms in $[0, \infty]$.

2.
$$\int_X \alpha s\, dm = \sum_{i=1}^{n} \alpha a_i m(A_i) = \alpha \sum_{i=1}^{n} a_i m(A_i) = \alpha \int_X s\, dm.$$

3. Writing
$$(s_1 + s_2) = \sum_{i=1}^{n} \sum_{j=1}^{m} (a_i + b_j) \varphi_{A_i \cap B_j},$$
we deduce
$$\begin{aligned} \int_X (s_1 + s_2)\, dm &= \sum_{i=1}^{n} \sum_{j=1}^{m} (a_i + b_j) m(A_i \cap B_j) \\ &= \sum_{i=1}^{n} a_i \sum_{j=1}^{m} m(A_i \cap B_j) + \sum_{j=1}^{m} b_j \sum_{i=1}^{n} m(A_i \cap B_j) \\ &= \sum_{i=1}^{n} a_i m(A_i) + \sum_{j=1}^{m} b_j m(B_j) \end{aligned}$$

$$= \int_X s_1 \, dm + \int_X s_2 \, dm.$$

Hence integration is a linear process.

4. Suppose $s_1 \leq s_2$, i.e. $a_i \leq b_j, \ \forall x \in (A_i \cap B_j), i = 1, \ldots, n, j = 1, \ldots, m$. Then

$$\sum_{i=1}^{n} a_i \varphi_{A_i} = \sum_{i=1}^{n} \sum_{j=1}^{m} a_i \varphi_{A_i \cap B_j} \leq \sum_{i=1}^{n} \sum_{j=1}^{m} b_j \varphi_{A_i \cap B_j} = \sum_{j=1}^{m} b_j \varphi_{B_j}$$

$$\Leftrightarrow \int_X s_1 \, dm = \sum_{i=1}^{n} \sum_{j=1}^{m} a_i m(A_i \cap B_j) \leq \sum_{i=1}^{n} \sum_{j=1}^{m} b_j m(A_i \cap B_j) = \int_X s_2 \, dm.$$

5. For every $E \in \mathcal{S}$ we have $\varphi_E \leq \varphi_X$, and so $s\varphi_E \leq s$. Definition 11.1.2 and the previous item give

$$\int_E s \, dm = \int_X s\varphi_E dm \leq \int_X s \, dm.$$

6. As $\min\{a_i, b_j\} \leq \max\{a_i, b_j\}, \forall x \in (A_i \cap B_j)$, then

$$\begin{aligned} s_1 \wedge s_2 &= \sum_{i=1}^{n} \sum_{j=1}^{m} \min\{a_i, b_j\} \varphi_{(A_i \cap B_j)} \\ &\leq s_i, \ i = 1, 2 \\ &\leq \sum_{i=1}^{n} \sum_{j=1}^{m} \max\{a_i, b_j\} \varphi_{(A_i \cap B_j)} = s_1 \vee s_2. \end{aligned}$$

Using item *4.* ends the proof.

■

Proposition 11.1.2 (Measure with density $s \in \mathscr{M}_0^+(X, \mathcal{S})$ with respect to a reference measure m) *Consider a measure space $(X, \mathcal{S}, m)$ and $s \in \mathscr{M}_0^+(X, \mathcal{S})$. The function $\mu : \mathcal{S} \mapsto [0, \infty]$ defined as*

$$\mu(E) = \int_E s \, dm, \quad \forall E \in \mathcal{S}$$

is a completely additive measure, said to have density s with respect to m.

Proof We have to prove μ is a measure. First, $\emptyset \in \mathcal{S}$, so

$$\mu(\emptyset) = \int_{\emptyset} s\, dm = \sum_{i=1}^{n} a_i m(A_i \cap \emptyset) = 0.$$

Take a pairwise-disjoint sequence $\{E_k\}_{k\geq 1} \subseteq \mathcal{S}$ such that $E = \bigsqcup_{k=1}^{\infty} E_k \in \mathcal{S}$, and we must show

$$\mu(E) = \sum_{k=1}^{\infty} \mu(E_k).$$

Pick $s = \sum_{i=1}^{n} a_i \varphi_{A_i}$. Since positive-term series can be rearranged, we have

$$\begin{aligned}\mu(E) = \int_E s\, dm &= \sum_{i=1}^{n} a_i m(A_i \cap E) = \sum_{i=1}^{n} a_i \sum_{k=1}^{\infty} m(A_i \cap E_k) \\ &= \sum_{k=1}^{\infty} \sum_{i=1}^{n} a_i m(A_i \cap E_k) = \sum_{k=1}^{\infty} \int_{E_k} s\, dm = \sum_{k=1}^{\infty} \mu(E_k).\end{aligned}$$

■

Proposition 11.1.3 (Integral of simple maps—remarkable inequality) *Given a measure space $(X, \mathcal{S}, m)$, a simple map s and increasing sequence $\{s_n\}_{n\geq 1}$ of simple maps in $\mathscr{M}_0^+(X, \mathcal{S})$ such that $s \leq \lim_{n\to\infty} s_n$, then*

$$\int_X s\, dm \leq \lim_{n\to\infty} \int_X s_n dm.$$

Proof Let $\epsilon \in (0, 1)$ and define for any $n \in \mathbb{N}$

$$E_n = \big\{x \in X : (1-\epsilon)s(x) \leq s_n(x)\big\}.$$

We claim $\{E_n\}_{n\geq 1} \subseteq \mathcal{S}$: given $n \in \mathbb{N}$, since $s_n, s \in \mathscr{M}_0^+(X, \mathcal{S})$ then $(1-\epsilon)s \in \mathscr{M}_0^+(X, \mathcal{S})$, so Proposition 10.2.2 guarantees $E_n \in \mathcal{S}$.

Furthermore the sequence $\{E_n\}_{n\geq 1}$ is increasing: given $n \in \mathbb{N}$ and $x \in E_n$,

$$(1-\epsilon)s(x) \leq s_n(x) \leq s_{n+1}(x),$$

i.e. $x \in E_{n+1}$. At last, $X = \bigcup_{n=1}^{\infty} E_n$, because $E_n \subseteq X$ for every $n \in \mathbb{N}$ and so

$$\bigcup_{n=1}^{\infty} E_n \subseteq X.$$

Conversely, take $x \in X$ and set $t(x) = \lim_{n\to\infty} s_n(x)$. By hypothesis $s(x) \leq t(x)$. If $t(x) = \infty$, for n large enough $s_n(x) \geq (1-\epsilon)s(x)$, whilst if $t(x) < \infty$ a large enough n still implies $s_n(x) > (1-\epsilon)t(x) > (1-\epsilon)s(x)$. Anyhow there exists an $n \in \mathbb{N}$ such that $x \in E_n$, whence $X \subseteq \bigcup_{n=1}^{\infty} E_n$.

Take the measure $\mu : \mathcal{S} \mapsto [0, \infty]$ with density s with respect to m and obtain, from Theorem 8.2.3,

$$\mu(X) = \int_X s \, dm = \lim_{n\to\infty} \int_{E_n} s \, dm = \lim_{n\to\infty} \mu(E_n).$$

Since, for every $x \in X$ and every $n \in \mathbb{N}$, $(1-\epsilon)s(x) \leq s_n(x)$, the integral's monotonicity implies

$$\begin{aligned}
&(1-\epsilon) \int_{E_n} s \, dm \leq \int_{E_n} s_n \, dm \leq \int_X s_n \, dm \\
\Rightarrow &(1-\epsilon) \lim_{n\to\infty} \int_{E_n} s \, dm \leq \lim_{n\to\infty} \int_X s_n \, dm \\
\Leftrightarrow &(1-\epsilon) \int_X s \, dm \leq \lim_{n\to\infty} \int_X s_n \, dm;
\end{aligned}$$

To finish, remember ϵ is arbitrary.

■

11.2 The Integral of Non-negative Maps

In view of the comments in the introductions to the present chapter, Sect. 10.4, and the previous section's results, it is only natural to define the Lebesgue integral for non-negative measurable functions in terms of non-negative simple maps.

Definition 11.2.1 (*Integral of maps in* $\mathscr{M}^+(X, \mathcal{S})$) Given a measure space $(X, \mathcal{S}, m)$, define $S_f = \left\{ s \in \mathscr{M}_0^+(X, \mathcal{S}) : 0 \leq s \leq f \right\}$. The integral of $f \in \mathscr{M}^+(X, \mathcal{S})$ with respect to m is

$$\int_X f \, dm = \sup_{s \in S_f} \int_X s \, dm. \qquad \diamond$$

If $f \in \mathscr{M}^+(X, \mathcal{S})$ and $E \in \mathcal{S}$ then $f \cdot \varphi_E \in \mathscr{M}^+(X, \mathcal{S})$, as product of measurable functions. Therefore we may define integrals over sets $E \in \mathcal{S}$.

Definition 11.2.2 (*Integral of maps in* $\mathscr{M}^+(X, \mathcal{S})$ *over* $\mathcal{S}$*-sets*) Given a measure space $(X, \mathcal{S}, m)$ consider $f \in \mathscr{M}^+(X, \mathcal{S})$ and $E \in \mathcal{S}$. We call integral of f with respect to m over the domain $E \subseteq X$ the quantity

$$\int_E f \, dm = \int_X f \cdot \varphi_E \, dm. \qquad \diamond$$

Remark 11.2.1 By (11.1.1) immediately

$$\int_E f\ dm = \sup_{s\in S_f} \int_E s\ dm = \sup_{s\in S_f} \int_X s\ dm_{|E} = \int_X f\ dm_{|E}. \tag{11.2.1}$$

♣

Beside $(f\varphi_E)$, for any $E \in \mathcal{S}$, also measurable and $[0,\infty]$-valued are (αf), $(f+g)$, $(f\cdot g)$, $(f\vee g)$, $(f\wedge g)$ etc. whenever we take $\alpha \geq 0$ and f, g measurable and $[0,\infty]$-valued, by Theorem 10.2.1.

Next up is a result expressing the integral of a non-negative measurable function as a limit of integrals.

Proposition 11.2.1 (Integral of a map in $\mathscr{M}^+(X,\mathcal{S})$ as limit integral of simple maps in $\mathscr{M}_0^+(X,\mathcal{S})$) *Given a measure space $(X,\mathcal{S},m)$, consider $f \in \mathscr{M}^+(X,\mathcal{S})$ and an increasing sequence $\{s_n\}_{n\geq 1} \subseteq \mathscr{M}_0^+(X,\mathcal{S})$ such that $f = \lim_{n\to\infty} s_n$. Then*

$$\int_X f\ dm = \lim_{n\to\infty} \int_X s_n\ dm.$$

Proof When $\int_X f\ dm < \infty$, let us take any $\epsilon > 0$. As $\lim_{n\to\infty} s_n = f$, there exists $s \in S_f$ such that $s \leq \lim_{n\to\infty} s_n$ and by Proposition 11.1.3

$$\int_X s\ dm \leq \lim_{n\to\infty} \int_X s_n dm < \sup_{s\in S_f} \int_X s\ dm + \epsilon = \int_X f\ dm + \epsilon.$$

This is true for all $s \in S_f$ and the arbitrary of ϵ gives the claim. In a similar way one proves the case $\int_X f\ dm = \infty$. ■

This proposition says, in other words, that the integral of a limit function in $\mathscr{M}^+(X,\mathcal{S})$ equals the limit of the integral of the (increasing) sequence (of non-negative simple measurable maps).

Using what we have leant thus far we shall prove the basic properties of integrals.

Proposition 11.2.2 (Properties of the integral of functions in $\mathscr{M}^+(X,\mathcal{S})$) *Given a measure space $(X,\mathcal{S},m)$, $f, g \in \mathscr{M}^+(X,\mathcal{S})$ and $\alpha \geq 0$:*

1. $\int_X f\ dm \in [0,\infty]$;
2. *(homogeneity)* $\int_X \alpha f\ dm = \alpha \int_X f\ dm$
3. *(finite additivity)* $\int_X \big(f+g\big) dm = \int_X f\ dm + \int_X g\ dm$;

4. *(monotonicity) if $f \le g$ on X,*

$$\int_X f\ dm \le \int_X g\ dm;$$

5. *(monotonicity w.r.t. integration domain) for any $E \in \mathcal{S}$*

$$\int_E f\ dm \le \int_X f\ dm.$$

Proof 1. This descends directly from Definition 11.2.1.

2.
$$\int_X \alpha f\ dm = \sup_{s\in S_f} \int_X \alpha s\ dm = \alpha \sup_{s\in S_f} \int_X s\ dm = \alpha \int_X f\ dm.$$

3. Since $(f+g) \in \mathscr{M}^+(X,\mathcal{S})$, pick an increasing sequence $\{s_n + t_n\}_{n\ge 1} \subseteq \mathscr{M}_0^+(X,\mathcal{S})$ such that $(f+g) = \lim_{n\to\infty}(s_n + t_n)$. By Propositions 11.2.1 and 11.1.1 (n.*3.*) we have

$$\begin{aligned}\int_X (f+g)dm &= \lim_{n\to\infty} \int_X (s_n + t_n)dm \\ &= \lim_{n\to\infty} \left(\int_X s_n dm + \int_X t_n dm \right) \\ &= \lim_{n\to\infty} \int_X s_n dm + \lim_{n\to\infty} \int_X t_n dm \\ &= \int_X f\ dm + \int_X g\ dm.\end{aligned}$$

4. Since $f \le g$, then $\sup_{s\in S_f} s \le \sup_{t\in S_g} t$, and Proposition 11.1.1, *4.*, implies

$$\int_X f\ dm = \sup_{s\in S_f} \int_X s\ dm \le \sup_{t\in S_g} \int_X t\ dm = \int_X g\ dm.$$

5. Clearly $\varphi_E \le \varphi_X$, so $f\varphi_E \le f\varphi_X$, and the previous item gives

$$\int_E f\ dm \le \int_X f\ dm.$$

■

Proposition 11.2.3 (Chebyshev inequality) *Given a measure space $(X,\mathcal{S},m)$, for any $f \in \mathscr{M}^+(X,\mathcal{S})$ and $\alpha > 0$*

$$m\big(\{x \in X : f(x) \ge \alpha\}\big) \le \frac{1}{\alpha} \int_X f(x)m(dx).$$

Proof Observe first that the set

$$A_\alpha = \{x \in X : f(x) \geq \alpha\}$$

is measurable since $f(x)$ is measurable on X. Moreover, $f(x)$ is non-negative, so by the integral's monotonicity

$$\int_X f\, dm \geq \int_{A_\alpha} f\, dm \geq \int_{A_\alpha} \alpha\, dm = \alpha \cdot m(A_\alpha)$$
$$\Leftrightarrow\ m\big(\{x \in X : f(x) \geq \alpha\}\big) \leq \frac{1}{\alpha} \int_{A_\alpha} f\, dm \leq \frac{1}{\alpha} \int_X f\, dm.$$

■

Now we shall establish an important theorem for passing limits inside integrals. It was first proved in 1906 by Beppo Levi (1875–1961) and known in the literature as *"monotone convergence theorem"*.

Theorem 11.2.1 (Beppo Levi's monotone convergence theorem) *Let* $\{f_i\}_{i\geq 1} \subseteq \mathscr{M}^+(X, \mathcal{S})$ *be an increasing sequence, and set*

$$\lim_{i\to\infty} f_i = f \quad \text{on } X.$$

Then $f \in \mathscr{M}^+(X, \mathcal{S})$ *and*

$$\lim_{i\to\infty} \int_X f_i\, dm = \int_X f\, dm.$$

Proof As $\{f_i\}_{i\geq 1} \subseteq \mathscr{M}^+(X, \mathcal{S})$ is an increasing sequence of measurable, non-negative functions, by Theorem 10.2.5 f is the same.

Let us prove the second assertion. Since $f_i \leq f_{i+1} \leq f$ on X, item *4.* in Proposition 11.2.2 tells

$$\int_X f_i\, dm \leq \int_X f_{i+1}\, dm \leq \int_X f\, dm$$

for every $i \in \mathbb{N}$, i.e. $\left\{\int_X f_i\, dm\right\}_{i\geq 1}$ is an increasing sequence of numbers, so on X

$$\lim_{i\to\infty} \int_X f_i\, dm \leq \int_X f\, dm.$$

Hence we just need the opposite inequality.

Pick $\alpha \in (0, 1)$ and $s(x) = \sum_{j=1}^n a_j \varphi_{E_j} \in \mathscr{M}_0^+(X, \mathcal{S})$, with $\bigsqcup_{j=1}^n E_j = X$ and $0 \leq s(x) \leq f(x)$. Define

$$A_i = \big\{x \in X : f_i(x) \geq \alpha \cdot s(x)\big\}.$$

By *4.* in Proposition 10.2.2 all sets A_i are $\mathcal{S}$-measurable, and form an increasing sequence: given any $i \in \mathbb{N}$ and $x \in A_i$, we have

$$\alpha \cdot s(x) \leq f_i(x) \leq f_{i+1}(x)$$

and so $x \in A_{i+1}$. Hence $A_i \subseteq A_{i+1}$.

We claim $\bigcup_{i=1}^{\infty} A_i = X$. Take $x \in X$ so that $\alpha \cdot s(x) < s(x) \leq f(x)$. There exists $i_0 \geq 1$ such that $\alpha \cdot s(x) \leq f_{i_0}(x)$, and then $x \in A_{i_0}$. Hence $X \subseteq \bigcup_{i=1}^{\infty} A_i$. Since $A_i \subseteq X, \forall i$, in the end $\bigcup_{i=1}^{\infty} A_i = X$.

Moreover $\alpha \cdot s(x) \leq f_i(x)$ and $f_i(x) \cdot \varphi_{A_i}(x) \leq f_i(x)$, so *4.* in Proposition 11.2.2 gives

$$\alpha \int_{A_i} s \, dm \leq \int_{A_i} f_i \, dm \leq \int_X f_i \, dm.$$

Let $\mu : \mathcal{S} \mapsto [0, \infty]$ denote the measure with density s with respect to m. Since $\{A_i\}_{i \geq 1}$ is increasing and $X = \bigcup_{i=1}^{\infty} A_i$, Theorem 8.2.3 tells

$$\begin{aligned} \alpha \lim_{i \to \infty} \int_{A_i} s \, dm &= \alpha \lim_{i \to \infty} \mu(A_i) = \alpha \mu(X) \\ &= \alpha \int_X s \, dm \\ &\leq \lim_{i \to \infty} \int_X f_i \, dm. \end{aligned}$$

This inequality holds when $\alpha \to 1^-$, so

$$\int_X s \, dm \leq \lim_{i \to \infty} \int_X f_i \, dm.$$

But $s \in \mathscr{M}_0^+(X, \mathcal{S})$ is arbitrary, so the above applies to

$$\sup_{s \in S_f} \int_X s \, dm$$

as well. Eventually, recalling Definition 11.2.1,

$$\int_X f \, dm \leq \lim_{i \to \infty} \int_X f_i \, dm$$

concluding the proof. ∎

Remark 11.2.2 In the monotone convergence theorem we cannot replace the increasing sequence with a decreasing one.

Consider the measure space $(\mathbb{R}, \mathscr{L}, \ell)$ and the decreasing sequence $\{f_n(x)\}_{n\geq 1}$,

$$f_n(x) = \begin{cases} \dfrac{1}{n} & \text{if } x \geq n \\ 0 & \text{if } x < n. \end{cases}$$

Clearly $f_n \in \mathscr{M}^+(\mathbb{R}, \mathscr{L})$ for every n, and for any $x \in \mathbb{R}$

$$\lim_{n\to\infty} f_n(x) = f(x) = 0,$$

hence

$$\int_{\mathbb{R}} f \, d\ell = 0.$$

At the same time

$$\int_{\mathbb{R}} f_n \, d\ell = 0 \cdot \ell((-\infty, n)) + \frac{1}{n} \cdot \ell([n, \infty)) = \infty$$

and

$$\lim_{n\to\infty} \int_{\mathbb{R}} f_n \, d\ell = \infty,$$

and therefore

$$\lim_{n\to\infty} \int_{\mathbb{R}} f_n \, d\ell \neq \int_{\mathbb{R}} f \, d\ell. \qquad \clubsuit$$

Proposition 11.2.4 (Consequences of Beppo Levi's theorem) *The following are corollaries to Beppo Levi's theorem:*

1. *if $\{A_i\}_{i\geq 1}$ are pairwise-disjoint $\mathcal{S}$-measurable sets such that $X = \bigcup_{i=1}^{\infty} A_i$, and $f : X \mapsto \bar{\mathbb{R}}$ belongs in $\mathscr{M}^+(X, \mathcal{S})$,*

$$\int_X f \, dm = \sum_{i=1}^{\infty} \int_{A_i} f \, dm;$$

2. *if $\{f_i\}_{i\geq 1} \subseteq \mathscr{M}^+(X, \mathcal{S})$ then*

$$\int_X \sum_{i=1}^{\infty} f_i \, dm = \sum_{i=1}^{\infty} \int_X f_i \, dm.$$

Proof 1. The sequence $\{g_k\}_{k\geq 1}$, $g_k = \sum_{i=1}^k f\varphi_{A_i} \in \mathscr{M}^+(X,\mathcal{S})$ for every $k \in \mathbb{N}$, is increasing, so by Theorem 10.2.5 $g = f\varphi_X \in \mathscr{M}^+(X,\mathcal{S})$. The integral's additivity and monotone convergence imply

$$\begin{aligned}\int_X f\,dm &= \lim_{k\to\infty}\int_X g_k\,dm = \lim_{k\to\infty}\int_X \sum_{i=1}^k f\varphi_{A_i}\,dm\\ &= \lim_{k\to\infty}\sum_{i=1}^k \int_X f\varphi_{A_i}\,dm = \sum_{i=1}^\infty \int_{A_i} f\,dm.\end{aligned}$$

2. Write $g_k = \sum_{i=1}^k f_i$. Every f_i is non-negative, so $\{g_k\}_{k\geq 1}$ is an increasing sequence of non-negative measurable maps. Again, additivity and monotone convergence give

$$\begin{aligned}\int_X \sum_{i=1}^\infty f_i\,dm &= \int_X \lim_{k\to\infty}\sum_{i=1}^k f_i\,dm = \lim_{k\to\infty}\int_X \sum_{i=1}^k f_i\,dm\\ &= \lim_{k\to\infty}\sum_{i=1}^k \int_X f_i\,dm = \sum_{i=1}^\infty \int_X f_i\,dm.\end{aligned}$$

■

Let us extend Proposition 11.1.2 to non-negative measurable functions.

Proposition 11.2.5 (Measures with density $f \in \mathscr{M}^+(X,\mathcal{S})$ with respect to m) *Given a measure space $(X,\mathcal{S},m)$ and a map $f \in \mathscr{M}^+(X,\mathcal{S})$, the function $\mu : \mathcal{S} \mapsto [0,\infty]$,*

$$\mu(E) = \int_E f\,dm, \quad \forall E \in \mathcal{S}$$

is a completely additive measure. We say μ has density f with respect to m.

Proof First, the set function $\mu : \mathcal{S} \mapsto [0,\infty]$ is a measure on $\mathcal{S}$. As $\emptyset \in \mathcal{S}$ we know

$$\mu(\emptyset) = \int_\emptyset f\,dm = \sup_{s\in S_f}\int_\emptyset s\,dm = 0.$$

Take a sequence $\{E_i\}_{i\geq 1}$ of pairwise-disjoint sets in $\mathcal{S}$ such that $E = \bigsqcup_{i=1}^\infty E_i$, and let us show

$$\mu(E) = \sum_{i=1}^\infty \mu(E_i),$$

i.e.

$$\int_E f\ dm = \sum_{i=1}^{\infty} \int_{E_i} f\ dm.$$

Clearly $\varphi_E = \sum_{i=1}^{\infty} \varphi_{E_i}$. By part *1.* in Proposition 11.2.4

$$\begin{aligned}\int_E f\ dm &= \int_X f\varphi_E\ dm = \int_X f\Big(\sum_{i=1}^{\infty} \varphi_{E_i}\Big)dm = \int_X \Big(\sum_{i=1}^{\infty} f\varphi_{E_i}\Big)dm \\ &= \sum_{i=1}^{\infty} \int_X f\varphi_{E_i} dm = \sum_{i=1}^{\infty} \int_{E_i} f\ dm.\end{aligned}$$

■

From that, we can express the integral of any (non-negative) measurable map in the following terms.

Proposition 11.2.6 (Integrals of measurable maps with respect to density measures) *Given the measure space $(X, \mathcal{S}, m)$, let $f \in \mathscr{M}^+(X, \mathcal{S})$ be the density of $\mu : \mathcal{S} \mapsto [0, \infty]$ with respect to m. For every $g \in \mathscr{M}^+(X, \mathcal{S})$*

$$\int_X g\, d\mu = \int_X gf\ dm.$$

Proof For $s = \sum_{i=1}^{n} a_i \varphi_{A_i}$

$$\begin{aligned}\int_X s\ d\mu &= \sum_{i=1}^{n} a_i \mu(A_i) = \sum_{i=1}^{n} a_i \int_{A_i} f\ dm \\ &= \sum_{i=1}^{n} \int_X a_i f\varphi_{A_i}\ dm = \int_X \Big(\sum_{i=1}^{n} a_i \varphi_{A_i}\Big) f\ dm \\ &= \int_X sf\ dm.\end{aligned}$$

If $g \in \mathscr{M}^+(X, \mathcal{S})$, let $\{s_n\}_{n\geq 1}$ be an increasing sequence of simple, non-negative functions such that $\lim_{n\to\infty} s_n = g$. Beppo Levi's theorem and the above imply

$$\int_X g\, d\mu = \lim_{n\to\infty} \int_X s_n\ d\mu = \lim_{n\to\infty} \int_X s_n f\ dm = \int_X \lim_{n\to\infty} s_n f\ dm = \int_X gf\ dm.$$

■

The following proposition is of interest because it highlights the semi-continuity of the Lebesgue integral of sequences of non-negative measurable maps.

Theorem 11.2.2 (Fatou's lemma) *Given the measure space* $(X, \mathcal{S}, m)$ *and a sequence* $\{f_i\}_{i\geq 1} \subseteq \mathscr{M}^+(X, \mathcal{S})$,

$$\int_X \liminf_{i\to\infty} f_i \, dm \leq \liminf_{i\to\infty} \int_X f_i \, dm.$$

Proof Recall the definition of limit inferior

$$\liminf_{i\to\infty} f_i = \sup_{i\geq 1} \left\{ \inf_{k\geq i} f_k \right\}.$$

By Theorem 10.2.3, $\liminf_{i\to\infty} f_i \in \mathscr{M}^+(X, \mathcal{S})$. Since $\left\{ \inf_{k\geq i} f_k \right\}_{i\geq 1}$ is an increasing sequence in $\mathscr{M}^+(X, \mathcal{S})$ such that

$$\lim_{i\to\infty} \left(\inf_{k\geq i} f_k \right) = \liminf_{i\to\infty} f_i,$$

applying the monotone convergence theorem gives

$$\int_X \liminf_{i\to\infty} f_i \, dm = \lim_{i\to\infty} \int_X \left(\inf_{k\geq i} f_k \right) dm. \tag{11.2.2}$$

Moreover $\left(\inf_{k\geq i} f_k \right) \leq f_i$ for every $i \in \mathbb{N}$, and monotonicity implies

$$\int_X \left(\inf_{k\geq i} f_k \right) dm \leq \int_X f_i \, dm \tag{11.2.3}$$

for every $i \in \mathbb{N}$. Consider the sequence of integrals $\left\{ \int_X \left(\inf_{k\geq i} f_k \right) dm \right\}_{i\geq 1}$: it is increasing and with values in $[0, \infty]$. Hence $\lim_{i\to\infty} \int_X \left(\inf_{k\geq i} f_k \right) dm$ exists, belongs to $[0, \infty]$, and actually equals $\liminf_{i\to\infty} \int_X \left(\inf_{k\geq i} f_k \right) dm$. By (11.2.3) we deduce

$$\lim_{i\to\infty} \int_X \left(\inf_{k\geq i} f_k \right) dm = \liminf_{i\to\infty} \int_X \left(\inf_{k\geq i} f_k \right) dm \leq \liminf_{i\to\infty} \int_X f_i \, dm. \tag{11.2.4}$$

The required inequality descends from (11.2.2), (11.2.4).

■

Remark 11.2.3 *1.* First of all Fatou's lemma is applicable to sequences in $\mathscr{M}^+(X, \mathcal{S})$ that are not necessarily decreasing or increasing: it suffices they are measurable and non-negative.

2. Here is an example fulfilling Fatou's strict inequality: the sequence with general term $f_n(x) = \varphi_{[n,n+1]}(x) \in \mathscr{M}^+(X, \mathcal{S})$ has values in $[0, \infty)$ and

$$f(x) = \lim_{n\to\infty} \varphi_{[n,n+1]}(x) = \varphi_{\emptyset}(x) = 0, \forall x \in [0,\infty).$$

Moreover, $\forall n \in \mathbb{N}$

$$\int_{[0,\infty)} f_n \, dm = \int_{[0,\infty)} \varphi_{[n,n+1]} \, dm = m([n, n+1] = 1$$
$$\Rightarrow \liminf_{n\to\infty} \int_{[0,\infty)} f_n \, dm = 1 > \int_{[0,\infty)} f \, dm = 0.$$

3. If a sequence is made of non-positive measurable functions, applying Fatou's lemma to $\{-f_n\}_{n\geq 1}$ gives

$$\int_X \left(\liminf_{n\to\infty} (-f_n) \right) dm \leq \liminf_{n\to\infty} \int_X (-f_n) \, dm$$
$$\Leftrightarrow \int_X \left(\limsup_{n\to\infty} f_n \right) dm \geq \limsup_{n\to\infty} \int_X f_n \, dm. \tag{11.2.5}$$

4. If we remove the hypothesis of non-negativity or non-positivity, Fatou's lemma becomes false, in general.
Consider the real-valued sequence defined by $f_n(x) = (-1)^n \varphi_{[n,n+1]}(x)$.
Clearly

$$f(x) = \lim_{n\to\infty} f_n(x) = (-1)^n \varphi_{\emptyset}(x) = 0, \forall x \in [0,\infty),$$

and $\forall n \in \mathbb{N}$

$$\int_{[0,\infty)} f_n \, dm = \int_{[0,\infty)} (-1)^n \varphi_{[n,n+1]} \, dm = (-1)^n m([n, n+1] = (-1)^n$$
$$\Rightarrow \liminf_{n\to\infty} \int_{[0,\infty)} f_n \, dm = -1 < \int_{[0,\infty)} f \, dm = 0.$$

5. If there exists a measurable map h such that

$$f_n(x) \geq h(x) \quad \text{or} \quad f_n(x) \leq h(x), \quad \forall x \in X, \quad \forall n \in \mathbb{N},$$

then the measurable sequence $\{f_n\}_{n\geq 1}$ satisfies Fatou's lemma or inequality (11.2.5), regardless of the sign of the f_n. It is enough to consider $\{g_n\}_{n\geq 1}$, where $g_n = f_n - h$, and apply Fatou's lemma or (11.2.5). ♣

Here come two more useful properties.

Theorem 11.2.3 (Vanishing of the integral) *A function $f \in \mathscr{M}^+(X, \mathcal{S})$ has vanishing integral,*

$$\int_X f\, dm = 0,$$

if and only if $f = 0$ *m-almost everywhere.*

Proof Necessary implication: suppose

$$\int_X f\, dm = 0$$

and define for every $n \in \mathbb{N}$

$$E_n = \left\{ x \in X : f(x) \geq \frac{1}{n} \right\}.$$

Chebyshev's inequality implies

$$m(E_n) = m\left(\left\{ x \in X : f(x) \geq \frac{1}{n} \right\} \right) \leq n \int_X f\, dm = 0,$$

so $m(E_n) = 0, \forall n \in \mathbb{N}$. Since

$$E = \left\{ x \in X : f(x) > 0 \right\} = \bigcup_{n=1}^{\infty} E_n, \text{ and } E_1 \subseteq E_2 \subseteq \ldots,$$

the complete subadditivity of m ensures that

$$m(E) = m\big(\{x \in X : f(x) > 0\}\big) = m\left(\bigcup_{n=1}^{\infty} E_n \right) \leq \sum_{n=1}^{\infty} m(E_n) = 0.$$

Therefore the measure of $(X \backslash E)$, where $f(x) = 0$, equals

$$m(X) - m\left(\bigcup_{n=1}^{\infty} E_n \right) = m(X),$$

i.e. $f(x) = 0$ up to a zero-measure set $E \subseteq X$.

Sufficient implication: suppose $f(x)$ is non-negative, $\mathcal{S}$-measurable, and null m-a.e., so

$$m(E) = m\big(\{x \in X : f(x) > 0\}\big) = 0.$$

Pick the increasing sequence $\{s_n(x)\}_{n \geq 1}$ of non-negative simple measurable functions $s_n(x) = n \cdot \varphi_E(x)$.

Clearly

$$f(x) \le \liminf_{n\to\infty} s_n(x). \tag{11.2.6}$$

Moreover, for every $n \in \mathbb{N}$

$$\int_X s_n \, dm = n \int_X \varphi_E \, dm = n \cdot m(E) = 0,$$

and then

$$\liminf_{n\to\infty} \int_X s_n \, dm = 0.$$

Applying Fatou's lemma to inequality (11.2.6) we obtain

$$0 \le \int_X f \, dm \le \int_X \liminf_{n\to\infty} s_n \, dm \le \liminf_{n\to\infty} \int_X s_n \, dm = 0. \qquad \blacksquare$$

Theorem 11.2.4 (Finite integral implies bounded integrand) *Any $f \in \mathscr{M}^+(X, \mathcal{S})$ with*

$$\int_X f \, dm < \infty$$

is bounded m-almost everywhere.

Conversely, if $m(X) < \infty$ and f is bounded m-a.e., then

$$\int_X f \, dm < \infty.$$

Proof Define, for $n \in \mathbb{N}$,

$$E_n = \{x \in X : f(x) \ge n\}.$$

Then $m\big(\{x \in X : f(x) = \infty\}\big) = m\big(\bigcap_{n=1}^{\infty} E_n\big)$, and by Chebyshev's inequality

$$m(E_n) \le \frac{1}{n} \int_X f \, dm \Rightarrow m(E_1) \le \int_X f \, dm < \infty.$$

As $\{E_n\}_{n\ge1}$ is decreasing and $m(E_1) < \infty$, Theorem 8.2.4 forces

$$m\left(\bigcap_{n=1}^{\infty} E_n\right) = \lim_{n\to\infty} m(E_n) \le \lim_{n\to\infty} \frac{1}{n} \int_X f \, dm = 0.$$

Therefore

$$m\big(\{x \in X : f(x) = \infty\}\big) = m\left(\bigcap_{n=1}^{\infty} E_n\right) = 0.$$

On the other hand if $f(x)$ is non-negative, $\mathcal{S}$-measurable, such that $f(x) \leq M$ m-a.e. for some $M > 0$, then

$$m(E) = m\big(\{x \in X : f(x) > M\}\big) = 0.$$

The increasing sequence $\{s_n(x)\}_{n\geq 1}$ of non-negative simple measurable maps $s_n(x) = M \cdot \varphi_{X\setminus E}(x) + n \cdot \varphi_E(x)$ is such that

$$f(x) \leq \liminf_{n\to\infty} s_n(x). \tag{11.2.7}$$

For every $n \in \mathbb{N}$, furthermore,

$$\int_X s_n \, dm = M \int_X \varphi_{X\setminus E} \, dm + n \int_X \varphi_E \, dm = M \cdot m(X),$$

but $m(X) < \infty$, so

$$\liminf_{n\to\infty} \int_X s_n \, dm = M \cdot m(X) < \infty.$$

Using inequality (11.2.7) and Fatou's lemma,

$$0 \leq \int_X f \, dm \leq \int_X \liminf_{n\to\infty} s_n \, dm \leq \liminf_{n\to\infty} \int_X s_n \, dm < \infty. \qquad \blacksquare$$

Theorem 11.2.5 (Comparing integrals of maps in $\mathscr{M}^+(X, \mathcal{S})$) *Consider $f, g \in \mathscr{M}^+(X, \mathcal{S})$ and suppose $f \leq g$, or $f = g$, or $f \geq g$, m-a.e. Correspondingly, then,*

$$\int_X f \, dm \leq \int_X g \, dm; \qquad \int_X f \, dm = \int_X g \, dm; \qquad \int_X f \, dm \geq \int_X g \, dm.$$

Proof Define

$$A = \{x \in X : f(x) > g(x)\},$$

so $m(A) = 0$. Moreover

$$f = f\varphi_{X\setminus A} + f\varphi_A \quad \text{and} \quad g = g\varphi_{X\setminus A} + g\varphi_A,$$

where $f\varphi_A = 0 = g\varphi_A$ m-a.e. By Theorem 11.2.3

$$\int_X f\, dm = \int_X f\varphi_{X\setminus A} dm \quad \text{and} \quad \int_X g\, dm = \int_X g\varphi_{X\setminus A} dm.$$

As $f\varphi_{X\setminus A} \leq g\varphi_{X\setminus A}$ for every $x \in X$, by monotonicity

$$\int_X f\, dm \leq \int_X g\, dm.$$

The cases $f = g$ and $f \geq g$ are completely analogous.

■

11.3 The Integral of Arbitrary Maps

Now we pass to arbitrary functions. **From here on, unless otherwise noted, we shall consider maps f defined on a complete measurable space $(X, \mathcal{S})$ with σ-finite measure** m with values in $\bar{\mathbb{R}}$.

Recall f can be written using its positive and negative parts f^+, f^-, which are non-negative. Furthermore, f^+, f^- are $\mathcal{S}$-measurable iff f is $\mathcal{S}$-measurable (Corollary 10.2.1).

The difference $(f^+ - f^-)$, where $f^+, f^- \in \mathscr{M}^+(X, \mathcal{S})$, is in $\mathscr{M}(X, \mathcal{S})$. The integrals $\int f^+\, dm$ and $\int f^-\, dm$ are well defined and by linearity we can define $\int f\, dm = \int f^+\, dm - \int f^-\, dm$. Note, thought, that this might produce the indeterminacy $(\infty - \infty)$ in case

$$\int f^+\, dm = \int f^-\, dm = \infty.$$

In defining integrable or summable functions we must then exclude those situations.

Definition 11.3.1 (*Integrable and summable functions*) If $f : X \mapsto \bar{\mathbb{R}} \in \mathscr{M}(X, \mathcal{S})$ then:

1. f is integrable with respect to m (in m, for short) if the following integrals exist and at least one of them is finite

$$\int_X f^+\, dm, \quad \int_X f^-\, dm.$$

If so, the Lebesgue integral of f on X is

$$\int_X f\, dm = \int_X f^+\, dm - \int_X f^-\, dm \in [-\infty, \infty];$$

2. f is **summable** (or **Lebesgue integrable**) in m if both integrals

$$\int_X f^+ \, dm \quad \text{and} \quad \int_X f^- \, dm$$

exist and are finite, in which case the Lebesgue integral of f on X is

$$\int_X f \, dm = \int_X f^+ \, dm - \int_X f^- \, dm \in (-\infty, \infty). \qquad \diamond$$

We shall denote by $L_1(X, \mathcal{S}, m)$ the space of m-summable functions. For simplicity's sake we might write $L_1(X)$, $L_1(m)$ or just L_1 whenever no confusion can arise. The first immediate observation is that $f \in L_1 \Rightarrow f \in \mathscr{M}(X, \mathcal{S})$.

Proposition 11.3.1 (Properties of L_1 functions) *For any $f, g : X \mapsto \bar{\mathbb{R}}$ in $\mathscr{M}(X, \mathcal{S})$*

1. $f \in L_1 \Leftrightarrow f^+, f^- \in L_1 \Leftrightarrow |f| \in L_1$;
2. $f \in L_1 \Rightarrow |f| < \infty$ *m-a.e.*;
3. $|f| \leq g$ *m-a.e. and* $g \in L_1 \Rightarrow f \in L_1$.
4. $m(X) < \infty$ *and* $|f| \leq M$ *m-a.e. for some* $M \Rightarrow f \in L_1$;
5. $f \in L_1 \Rightarrow f\varphi_E \in L_1$ *for every* $E \in \mathcal{S}$, *where*

$$\int_E f \, dm = \int_X f\varphi_E \, dm.$$

Proof 1. The equivalence $f \in L_1 \Leftrightarrow f^+, f^- \in L_1$ descends directly from the definition of summability.
Let us show $f^+, f^- \in L_1 \Rightarrow |f| \in L_1$. Since $|f| = (f^+ + f^-)$,

$$\int_X |f| \, dm = \int_X f^+ \, dm + \int_X f^- \, dm \in [0, \infty).$$

Now to $|f| \in L_1 \Rightarrow f^+, f^- \in L_1$. As $f^+, f^- \leq |f|$,

$$\int_X f^+ \, dm \leq \int_X |f| \, dm \in [0, \infty) \quad \text{and} \quad \int_X f^- \, dm \leq \int_X |f| \, dm \in [0, \infty).$$

2. Because

$$\int_X f \, dm = \int_X f^+ \, dm - \int_X f^- \, dm \in (-\infty, \infty),$$

Theorem 11.2.4 gives

$$f^+ \in [0, \infty) \quad \text{and} \quad f^- \in [0, \infty)$$

m-almost everywhere, and hence $|f| = (f^+ + f^-) \in [0, \infty)$ m-a.e.

3. Since

$$|f| = (f^+ + f^-) \le g \ m\text{-a.e. on } X \Leftrightarrow f^+ \le g \ \text{ and } \ f^- \le g \ m\text{-a.e. on } X,$$

by Theorem 11.2.5

$$\int_X f^+ \, dm \le \int_X g \, dm \in [0, \infty) \quad \text{and} \quad \int_X f^- \, dm \le \int_X g \, dm \in [0, \infty),$$

and then $\int_X f \, dm \in (-\infty, \infty)$.
4. From $|f| = f^+ + f^-$ we know $f^+ \le M$ and $f^- \le M$ m-a.e. Theorem 11.2.4 says $f^+, f^- \in L_1$, so $f = \left(f^+ - f^-\right) \in L_1$.
5. As $f \in L_1$, and

$$\begin{aligned} \int_X f \cdot \varphi_E \, dm &= \int_X f^+ \cdot \varphi_E \, dm - \int_X f^- \cdot \varphi_E \, dm \\ &= \int_E f^+ \, dm - \int_E f^- \, dm \\ &= \int_E f \, dm \\ &\le \int_X f \, dm \in (-\infty, \infty); \end{aligned}$$

we have $f\varphi_E \in L_1$. Recalling Remark 11.2.1 we obtain

$$\begin{aligned} \int_E f \, dm &= \int_E f^+ \, dm - \int_E f^- \, dm \\ &= \int_X f^+ \, dm_{|E} - \int_X f^- \, dm_{|E} = \int_X f \, dm_{|E}. \end{aligned} \tag{11.3.1}$$

■

Let us list integral properties of summable functions.

Proposition 11.3.2 (Properties of the integral of L_1 functions) *For any f, $g \in \mathscr{M}(X, \mathcal{S})$ and $\alpha \in \mathbb{R}$,*

1. if $f \in L_1$, then

$$\left| \int_X f \, dm \right| \le \int_X |f| \, dm;$$

2. *if* $f = g$ m*-a.e. and* $f \in L_1$*, then* $g \in L_1$ *and*

$$\int_X f \, dm = \int_X g \, dm;$$

3. *if* $f \in L_1$*, then* $\alpha \cdot f \in L_1$ *and*

$$\int_X \alpha \cdot f \, dm = \alpha \int_X f \, dm; \tag{11.3.2}$$

4. *if* $f, g \in L_1$*, then* $\big(f + g\big) \in L_1$ *and*

$$\int_X \big(f + g\big) \, dm = \int_X f \, dm + \int_X g \, dm.$$

Proof 1. Since $f \in L_1 \Leftrightarrow |f| \in L_1$, then

$$\begin{aligned} \left| \int_X f \, dm \right| &= \left| \int_X f^+ \, dm - \int_X f^- \, dm \right| \\ &\leq \int_X f^+ \, dm + \int_X f^- \, dm \\ &= \int_X |f| \, dm. \end{aligned}$$

2. As $f = g$ m-a.e., then m-almost everywhere we have

$$f^+ = g^+ \quad \text{and} \quad f^- = g^- .$$

Moreover $f \in L_1$, so $f^+, f^- \in L_1$ and by Theorem 11.2.5

$$\int_X f^+ \, dm = \int_X g^+ \, dm \in [0, \infty) \quad \text{and} \quad \int_X f^- \, dm = \int_X g^- \, dm \in [0, \infty).$$

Hence $g = (g^+ - g^-) \in L_1$ and

$$\int_X f \, dm = \int_X g \, dm.$$

3. For $\alpha = 0$ obviously $\alpha f \in L_1$, and (11.3.2) holds. For $\alpha > 0$ we have $(\alpha f)^+ = \alpha f^+$, $(\alpha f)^- = \alpha f^-$ and so

$$\begin{aligned}\int_X \alpha f \, dm &= \int_X \alpha f^+ \, dm - \int_X \alpha f^- \, dm \\ &= \alpha\left(\int_X f^+ \, dm - \int_X f^- \, dm\right) \\ &= \alpha \int_X f \, dm \in (-\infty, \infty).\end{aligned}$$

The case $\alpha < 0$ is completely similar.

4. As $f, g \in L_1$, by part *1.* in Proposition 11.3.1 $|f|, |g| \in L_1$. But

$$|f + g| \leq |f| + |g|,$$

so by linearity and monotonicity

$$\int_X |f + g| \, dm \leq \int_X |f| \, dm + \int_X |g| \, dm \in [0, \infty).$$

That same part *1.* gives $(f + g) \in L_1$. Now, $f, g \in L_1$ implies $f^+, f^-, g^+, g^- \in L_1$, and the latter are all m-almost everywhere finite, non-negative and measurable by part *2.* in Proposition 11.3.1. Hence Proposition 11.2.2, *3.*, ensures

$$\begin{aligned}&\int_X (f+g)^+ dm + \int_X f^- dm + \int_X g^- dm = \int_X f^+ dm + \int_X g^+ dm + \int_X (f+g)^- dm \\ \Rightarrow \quad &\int_X (f+g) \, dm = \int_X f^+ \, dm - \int_X f^- \, dm \\ &\qquad + \int_X g^+ \, dm - \int_X g^- \, dm \\ &\qquad = \int_X f \, dm + \int_X g \, dm.\end{aligned}$$

■

Remark 11.3.1 In contrast to what happened in $\mathscr{M}^+(X, \mathcal{S})$ (Theorem 11.2.3), if $f \in L_1$ is such that

$$\int_X f \, dm = 0,$$

it may not vanish m-almost everywhere. Since

$$\int_X f \, dm = \int_X f^+ \, dm - \int_X f^- \, dm,$$

the integral on X can vanish when

$$\int_X f^+ \, dm = 0 = \int_X f^- \, dm$$

i.e.

$$\int_X |f|\, dm = 0. \tag{11.3.3}$$

In this case $f^+ = 0 = f^-$ m-a.e. $\Rightarrow f = 0$ m-a.e. But

$$\int_X f^+\, dm = \int_X f^-\, dm \in (0, \infty)$$

does not necessarily imply $f = 0$ m-a.e.

The only case, apart from that of (11.3.3), in which zero integral implies the map is m-a.e. zero, is encapsulated in the next result.

It is useful to remember that for L_1 functions the implication $f = 0$ m-a.e. on $X \Rightarrow \int_X f\, dm = 0$ remains valid. ♣

Proposition 11.3.3 (Vanishing integral of L_1 functions) *Suppose $f \in L_1$ is such that*

$$\int_E f\, dm = 0$$

for every $E \in \mathcal{S}$. Then $f = 0$ m-almost everywhere.

Proof Write

$$X^+ = \{x \in X : f(x) \geq 0\}.$$

For any $E \in \mathcal{S}$

$$\int_E f^+\, dm = \int_{E \cap X^+} f\, dm = 0,$$

so Theorem 11.2.3 gives $f^+ = 0$ m-a.e. Similarly, using

$$X^- = \{x \in X : f(x) < 0\}$$

one shows $f^- = 0$ m-a.e. Now consider

$$A = \{x \in X : f^+(x) > 0\} \quad \text{and} \quad B = \{x \in X : -f^-(x) < 0\}.$$

Clearly $m(A) = m(B) = 0$, and

$$\{x \in X : f(x) \neq 0\} = A \cup B \quad \text{and} \quad A \cap B = \emptyset,$$

so $m(\{x \in X : f(x) \neq 0\}) = 0$, i.e. $f = 0$ m-a.e. ■

The Lebesgue integral is finitely and countably additive in the integration domain, in the following sense.

Proposition 11.3.4 (Finite and complete additivity of the integral with respect to the integration domain) *If* $f \in L_1$

1. *for any* $A, B \in \mathcal{S}$ *such that* $A \cap B = \emptyset$,

$$\int_{A\cup B} f\, dm = \int_A f\, dm + \int_B f\, dm;$$

2. *if* $\{E_i\}_{i\geq 1} \subseteq \mathcal{S}$ *is a pairwise-disjoint sequence, then the integrals over* E_i *form an absolutely convergent series:*

$$\sum_{i=1}^{\infty} \left| \int_{E_i} f\, dm \right| < \infty.$$

Setting $E = \bigcup_{i=1}^{\infty} E_i \subseteq X$, *the integral of* f *is completely additive in the integration domain:*

$$\sum_{i=1}^{\infty} \int_{E_i} f\, dm = \int_E f\, dm.$$

Proof 1. As A, B are disjoint,

$$\begin{aligned}\int_{A\cup B} f\, dm &= \int_X f\big(\varphi_A + \varphi_B\big)\, dm = \int_X \big(f\varphi_A + f\varphi_B\big)\, dm \\ &= \int_X f\varphi_A\, dm + \int_X f\varphi_B\, dm = \int_A f\, dm + \int_B f\, dm.\end{aligned}$$

2. Item *1.* in Proposition 11.3.2 says that for every $i \geq 1$

$$\left| \int_{E_i} f\, dm \right| \leq \int_{E_i} |f|\, dm,$$

and $|f| \in \mathscr{M}^+(X, \mathcal{S})$. Hence the set function $\mu : \mathcal{S} \mapsto [0, \infty]$,

$$\mu(E) = \int_E |f|\, dm, \quad \forall E \in \mathcal{S}$$

is a measure with density $|f|$ with respect to m, which is completely additive (Proposition 11.2.5). Therefore

$$\sum_{i=1}^{\infty} \mu(E_i) = \sum_{i=1}^{\infty} \int_{E_i} |f|\, dm = \int_E |f|\, dm = \mu(E) < \infty.$$

But since

$$\sum_{i=1}^{\infty}\left|\int_{E_i} f\,dm\right| \le \sum_{i=1}^{\infty}\int_{E_i}|f|\,dm = \int_E |f|\,dm < \infty,$$

the series converges absolutely. Consequently it converges, and then the sequence

$$S_n = \sum_{i=1}^{n}\int_{E_i} f\,dm$$

converges, meaning

$$\int_E f\,dm = \lim_{n\to\infty}\left(\sum_{i=1}^{n}\int_{E_i} f\,dm\right) = \sum_{i=1}^{\infty}\int_{E_i} f\,dm. \qquad \blacksquare$$

Here is a result used often, that allows to swap limits and integrals of L_1 functions.

Theorem 11.3.1 (Lebesgue's dominated convergence theorem - 1908) *If a sequence* $\{f_i\}_{i\ge 1} \subseteq \mathscr{M}(X,\mathcal{S})$ *converges to* $f = \lim_{i\to\infty} f_i$ *m-almost everywhere, and there exists a map* $g \in L_1$ *such that*

$$|f_i| \le g \ \ m\text{-a.e.} \quad \textit{(so-called Lebesgue condition)}$$

for every $i \in \mathbb{N}$*, then:*

1. $f, f_i \in L_1, \forall i \in \mathbb{N}$;
2. $\int_X f\,dm = \lim_{i\to\infty}\int_X f_i\,dm$;
3. $\lim_{i\to\infty}\int_X |f_i - f|\,dm = 0$.

Proof Without loss of generality we may assume that everywhere on X

$$\lim_{i\to\infty} f_i = f \quad \text{and} \quad |f_i| \le g \quad \text{for every } i \in \mathbb{N}.$$

By Theorem 10.2.4 $f \in \mathscr{M}(X,\mathcal{S})$, and since $\forall i \in \mathbb{N}\ |f_i| \le g$, then for the limit f we also have $|f| \le g$. As $g \in L_1$, by *3.* in Proposition 11.3.1 every f_i and f are L_1. This ends claim *1.*.

For number *2.* pick non-negative measurable functions $\{g + f_i\}_{i\ge 1}$. Using Fatou's lemma and the integral's linearity we have

$$\begin{aligned}\int_X g\,dm + \int_X f\,dm &= \int_X (g+f)\,dm \le \liminf_{i\to\infty} \int_X (g+f_i)\,dm \\ &= \liminf_{i\to\infty} \left(\int_X g\,dm + \int_X f_i\,dm \right) \\ &= \int_X g\,dm + \liminf_{i\to\infty} \int_X f_i\,dm,\end{aligned}$$

and so

$$\int_X f\,dm \le \liminf_{i\to\infty} \int_X f_i\,dm. \tag{11.3.4}$$

Now take the non-negative, measurable maps $\{g - f_i\}_{i\ge 1}$. Using again Fatou's lemma we obtain

$$\begin{aligned}\int_X g\,dm - \int_X f\,dm &= \int_X (g-f)\,dm \le \liminf_{i\to\infty} \int_X (g-f_i)\,dm \\ &= \liminf_{i\to\infty} \left(\int_X g\,dm - \int_X f_i\,dm \right) \\ &= \int_X g\,dm - \limsup_{i\to\infty} \int_X f_i\,dm,\end{aligned}$$

thus

$$\int_X f\,dm \ge \limsup_{i\to\infty} \int_X f_i\,dm. \tag{11.3.5}$$

Then (11.3.4) and (11.3.5) give

$$\int_X f\,dm = \lim_{i\to\infty} \int_X f_i\,dm.$$

Now *3.*. For every $i \in \mathbb{N}$

$$|f_i - f| \le 2g,$$

so the sequence $\{|f_i - f|\}_{i\ge 1}$, that converges to 0, is dominated by $2g$. By dominated convergence we obtain

$$\lim_{i\to\infty} \int_X |f_i - f|\,dm = \int_X \lim_{i\to\infty} |f_i - f|\,dm = 0.$$

■

Remark 11.3.2 In the above proof we simplified a little when we assumed $\{f_i\}_{i\geq 1}$ converged everywhere on X, and said all inequalities $|f_i| \leq g$ held at all points of X. We shall explain why this does not affect the general statement's proof.

Call E the subset of X where convergence fails and some $|f_i| \leq g$ does not hold. As $m(E) = 0$, we may redefine f_i and f to be zero on E. Then $f_i\varphi_{(X\setminus E)}$, $f\varphi_{(X\setminus E)} \in \mathscr{M}(X, \mathcal{S})$ for all $i \in \mathbb{N}$, and $f_i\varphi_{(X\setminus E)} = f_i$ and $f\varphi_{(X\setminus E)} = f$ m-a.e. Theorem 10.2.6 implies f_i, $f \in \mathscr{M}(X, \mathcal{S})$. Similarly, $f_i\varphi_{(X\setminus E)}$, $f\varphi_{(X\setminus E)} \in L_1$, and $f_i\varphi_{(X\setminus E)} = f_i$ and $f\varphi_{(X\setminus E)} = f$ m-a.e. Part *2.* in Proposition 11.3.2 guarantees f_i, $f \in L_1$ and

$$\int_X f_i\varphi_{(X\setminus E)}\, dm = \int_{(X\setminus E)} f_i\, dm = \int_X f_i\, dm, \forall i \in \mathbb{N} \tag{11.3.6}$$

$$\int_X f\varphi_{(X\setminus E)}\, dm = \int_{(X\setminus E)} f\, dm = \int_X f\, dm, \tag{11.3.7}$$

given that

$$\int_E f\, dm = \int_E f_i\, dm = 0,\ \forall i \in \mathbb{N}.$$

Therefore the new statement of the theorem becomes

$$\int_{X\setminus E} f\, dm = \lim_{i\to\infty} \int_{X\setminus E} f_i\, dm,$$

which is indeed the same as the original because of (11.3.6), (11.3.7).

♣

Just as with monotone convergence, Lebesgue's theorem has two corollaries: dominated convergence for series and bounded convergence in spaces with finite measure.

Corollary 11.3.1 (Dominated convergence theorem for series) *If $\{f_i\}_{i\geq 1} \subseteq L_1$ is such that $\sum_{i=1}^{\infty} \int_X |f_i|\, dm < \infty$, then $f = \sum_{i=1}^{\infty} f_i$ is well defined m-a.e., $f \in L_1$ and*

$$\sum_{i=1}^{\infty} \int_X f_i\, dm = \int_X f\, dm.$$

Proof Clearly $\{|f_i|\}_{i\geq 1} \subseteq \mathscr{M}^+(X, \mathcal{S})$, so *2.* in Proposition 11.2.4 implies

$$\int_X \left(\sum_{i=1}^{\infty} |f_i|\right) dm = \sum_{i=1}^{\infty} \int_X |f_i|\, dm < \infty,$$

whence $\sum_{i=1}^{\infty} |f_i| \in L_1$. Theorem 11.2.4 says $\sum_{i=1}^{\infty} |f_i|$ is bounded m-a.e., hence $\sum_{i=1}^{\infty} f_i$ converges absolutely and, in particular, it converges pointwise m-a.e. to f.

Furthermore

$$\int_X \left(\sum_{i=1}^{\infty} f_i \right) dm \leq \int_X \left(\sum_{i=1}^{\infty} |f_i| \right) dm < \infty,$$

so $f = \left(\sum_{i=1}^{\infty} f_i \right) = \lim_{n\to\infty} \left(\sum_{i=1}^{n} f_i \right) \in L_1$. Since $\{f_i\}_{i\geq 1} \subseteq L_1$, by item *4.* in Proposition 11.3.2 $\left(\sum_{i=1}^{n} f_i \right) \in L_1$ for every $n \in \mathbb{N}$. Moreover,

$$\left| \sum_{i=1}^{n} f_i \right| \leq \sum_{i=1}^{\infty} |f_i| \ \forall n \in \mathbb{N} \ m\text{-a.e.} \quad \text{and} \quad |f| = \left| \sum_{i=1}^{\infty} f_i \right| \leq \sum_{i=1}^{\infty} |f_i| \ m\text{-a.e.}\,.$$

By dominated convergence

$$\begin{aligned} \sum_{i=1}^{\infty} \int_X f_i \, dm &= \lim_{n\to\infty} \sum_{i=1}^{n} \int_X f_i \, dm = \lim_{n\to\infty} \int_X \sum_{i=1}^{n} f_i \, dm \\ &= \int_X \lim_{n\to\infty} \sum_{i=1}^{n} f_i \, dm = \int_X \sum_{i=1}^{\infty} f_i \, dm = \int_X f \, dm. \end{aligned}$$

■

Corollary 11.3.2 (Bounded convergence) *Suppose $(X, \mathcal{S}, m)$ is a finite-measure space $(m(X) < \infty)$ and take $\{f_n\}_{n\geq 1} \subseteq \mathscr{M}(X, \mathcal{S})$. Suppose there exists an $M \in (0, \infty)$ such that $|f_n| \leq M$ m-a.e. for every $n \in \mathbb{N}$ and $\lim_{n\to\infty} f_n = f$ m-almost everywhere. Then f and every f_n belong to L_1, and*

$$\int_X f \, dm = \lim_{n\to\infty} \int_X f_n \, dm.$$

Proof Here $g = M$, so

$$\int_X g \, dm = \int_X M \, dm = Mm(X) < \infty,$$

and $g \in L_1$. By dominated convergence

$$\int_X f \, dm = \lim_{n\to\infty} \int_X f_n \, dm.$$

■

In 1907 Giuseppe Vitali had written a paper[2] proving when limits and integrals can be swapped.

Before we prove Vitali's theorem we need to introduce certain notions.

Remark 11.3.3 Let us point out that in contrast to non-negative measurable functions, when we deal with L_1 maps the set function

[2] *Sull'integrazione per serie*, Rend. Circ. Mat. Palermo 23 (1907), 137–155.

$$\mu(E) = \int_E f \, dm \in (-\infty, \infty), \quad E \in \mathcal{S}$$

is not a measure. But if $f \in \mathscr{M}(X, \mathcal{S})$ then $\bar{\mu} : \mathcal{S} \mapsto [0, \infty]$,

$$\bar{\mu}(E) = \int_E |f| \, dm \in [0, \infty], \quad E \in \mathcal{S}$$

is a measure. It is the measure with density $|f|$ with respect to m. Clearly if $f \in L_1$, then $\bar{\mu}$ is **finite**. ♣

Now we want to examine the relationships among several measures on the same σ-algebra.

Definition 11.3.2 (*Absolutely continuous measures*) Let $(X, \mathcal{S})$ be a measurable space and m, μ two measures defined on the σ-algebra $\mathcal{S}$. One says μ is absolutely continuous with respect to m, written $\mu \ll m$, if $m(E) = 0 \Rightarrow \mu(E) = 0$ for every $E \in \mathcal{S}$. ◇

In practice a measure μ is absolutely continuous with respect to another measure m if among its zero-measure sets are all those of m.

The prototypical example of finite and absolutely continuous measure with respect to a given m is the integral of a non-negative summable function, which is absolutely continuous.

Proposition 11.3.5 (Equivalent criteria for summability) *If $f \in L_1$, the following conditions are equivalent:*

1. *the measure $\mu : \mathcal{S} \mapsto [0, \infty)$,*

$$\mu(E) = \int_E |f| \, dm, \quad E \in \mathcal{S}$$

   is absolutely continuous with respect to m: $\forall E \in \mathcal{S}$, $m(E) = 0 \Rightarrow \mu(E) = 0$;
2. *for each $\epsilon > 0$ there exists $\delta > 0$ such that*

$$\mu(E) = \int_E |f| \, dm < \epsilon$$

   for any $E \in \mathcal{S}$ with $m(E) < \delta$.

Proof *1.*: given $\mu \ll m$, suppose by contradiction there was an $\epsilon > 0$ and an $F_n \in \mathcal{S}$, for every $n \in \mathbb{N}$, such that $m(F_n) < 1/2^n$ and

$$\mu(F_n) = \int_{F_n} |f| \, dm \geq \epsilon.$$

The sets $E_n = \bigcup_{i=n}^{\infty} F_i$ form a decreasing sequence in $\mathcal{S}$. Calling

$$E = \bigcap_{n=1}^{\infty} E_n \quad \text{so} \quad E \subseteq E_n, \forall n \in \mathbb{N},$$

by complete subadditivity

$$m(E_n) \leq \sum_{i=n}^{\infty} m(F_i) < \sum_{i=n}^{\infty} \frac{1}{2^i},$$

and hence $\lim_{n\to\infty} m(E_n) = 0$. As $E \subseteq E_n$, we have $m(E) \leq m(E_n)$, $\forall n \in \mathbb{N}$, and so

$$m(E) = 0. \tag{11.3.8}$$

Moreover, integration over $E_n \supseteq F_n$ is monotone, for every $n \in \mathbb{N}$, so

$$\mu(E_n) = \int_{E_n} |f| \, dm \geq \mu(F_n) = \int_{F_n} |f| \, dm \geq \epsilon.$$

Now we use that $f \in L_1$ to have

$$\mu(X) = \int_X |f| \, dm < \infty,$$

so Theorem 8.2.4 applied to the decreasing $\{E_n\}_{n\geq 1}$ gives

$$\mu(E) = \int_E |f| \, dm = \lim_{n\to\infty} \int_{E_n} |f| \, dm \geq \epsilon.$$

This and Eq. (11.3.8) would imply μ is not absolutely continuous with respect to m, contradicting the hypothesis.

2. is immediate, because it actually means $\lim_{m(E)\to 0} \mu(E) = 0$, implying $m(E) = 0 \Rightarrow \mu(E) = 0$ for any $E \in \mathcal{S}$. ■

Example 11.3.1 (*A non-absolutely continuous measure with respect to the Lebesgue measure*) Let us present a measure that is not absolutely continuous measure with respect to the Lebesgue measure ℓ.

Consider the measurable space $(\mathbb{R}, \mathscr{L})$ and let ℓ be the Lebesgue measure on $\mathbb{R}$ and $\delta : \mathscr{L} \mapsto [0, \infty]$ the set function

$$\delta_0(E) = \begin{cases} 1 & \text{if } 0 \in E \\ 0 & \text{if } 0 \notin E. \end{cases}$$

for any $E \in \mathscr{L}$. The claim is δ_0 is a measure. First, $\emptyset \subseteq \mathbb{R}$ and $0 \notin \emptyset$, so $\delta_0(\emptyset) = 0$. For a sequence $\{E_n\}_{n\geq 1}$ of pairwise-disjoint sets in $\mathscr{L}$ we have two possibilities:

- 0 does not belong to $\bigcup_{n=1}^{\infty} E_n$, hence to none of the E_n, and then

$$\delta_0\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \delta_0(E_n) = 0;$$

- 0 belongs to $\bigcup_{n=1}^{\infty} E_n$, i.e. to one E_n only, and so

$$\delta_0\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \delta_0(E_n) = 1.$$

The measure δ_0 concentrates the whole unit mass at the point 0. As $\delta_0(\{0\}) = 1$, it cannot be absolutely continuous with respect to ℓ: given any non-negative, measurable f, $\int_{\{0\}} f \, d\ell = 0 \neq \delta_0(\{0\}) = 1$. ★

Definition 11.3.3 (*Uniformly summable maps*) Given a measure space $(X, \mathcal{S}, m)$, a family $\mathscr{F}$ of summable maps $f : X \mapsto \bar{\mathbb{R}}$ is uniformly summable on X if for every $\epsilon > 0$ there exists $\delta > 0$ such that, for every $E \in \mathcal{S}$ with $m(E) < \delta$,

$$\int_E |f| \, dm < \epsilon \quad \forall f \in \mathscr{F}. \qquad \diamond$$

Proposition 11.3.6 (Sequences of uniformly summable maps) *Let $(X, \mathcal{S}, m)$ be a finite-measure space ($m(X) < \infty$) and $\{f_n : X \mapsto \bar{\mathbb{R}}\}_{n\geq 1}$ a sequence of uniformly summable maps that converge pointwise m-a.e.. Then the limit $f : X \mapsto \mathbb{R}$ is summable on X.*

Proof We need $m(X) < \infty$ to invoke Proposition 9.3.5, so that X can be approximated by a finite disjoint union $E = \bigsqcup_{i=1}^{k} E_i$ of measurable sets, up to a set of arbitrarily small measure. Choosing $\epsilon = 1$, by uniform summability the pairwise-disjoint sets E_i and the leftover $(X \Delta E)$ can be selected so that all of them have measure less than δ. Hence for every $n \in \mathbb{N}$ we have

$$\int_X |f_n| \, dm \leq \sum_{i=1}^{k} \int_{E_i} |f_n| \, dm + \int_{(X\Delta E)} |f_n| \, dm < k \cdot \epsilon + \epsilon = (k+1),$$

and applying Fatou's lemma

$$\int_X |f| \, dm = \int_X \liminf_{n\to\infty} |f_n| \, dm \leq \liminf_{n\to\infty} \int_X |f_n| \, dm < (k+1),$$

whence f is summable on X.

■

At this point we are ready to prove the result guaranteeing when, for summable functions, limit and integral can be swapped.

Theorem 11.3.2 (Vitali's convergence theorem - 1907) *Let $(X, \mathcal{S}, m)$ be a finite-measure space and $\{f_n : X \mapsto \bar{\mathbb{R}}\}_{n\geq 1}$ a sequence of uniformly summable maps that converges to $f : X \mapsto \mathbb{R}$ pointwise m-a.e. Then f is summable on X and*

$$\int_X \lim_{n\to\infty} f_n \, dm = \lim_{n\to\infty} \int_X f_n \, dm.$$

Proof By Proposition 11.3.6 f is summable on X, and further m-a.e. bounded because of Theorem 11.2.4. Part *1.* in Proposition 11.3.2 implies

$$\begin{aligned}\left| \int_X f_n \, dm - \int_X f \, dm \right| &= \left| \int_X (f_n - f) \, dm \right| \\ &\leq \int_X |f_n - f| \, dm\end{aligned}$$

for every $n \in \mathbb{N}$. Then the uniform summability of $\{f_n\}_{n\geq 1}$ guarantees that given $\epsilon > 0$ there is a $\delta > 0$ such that

$$\begin{aligned}\int_X |f_n - f| \, dm &= \int_{X\setminus E} |f_n - f| \, dm + \int_E |f_n - f| \, dm \\ &\leq \int_{X\setminus E} |f_n - f| \, dm + \qquad (11.3.9) \\ &\quad \underbrace{\int_E |f_n| \, dm}_{<\epsilon/3} + \int_E |f| \, dm \qquad (11.3.10)\end{aligned}$$

for every $E \in \mathcal{S}, m(E) < \delta$, and every $n \in \mathbb{N}$. Applying the Fatou lemma to the last integral in (11.3.10) we obtain

$$\int_E |f| \, dm = \int_E \liminf_{n\to\infty} |f_n| \, dm \leq \liminf_{n\to\infty} \int_E |f_n| \, dm < \frac{\epsilon}{3}.$$

Regarding the integral of (11.3.9): since $m(X) < \infty$ the Severini-Egorov theorem applied to E ($m(E) < \delta$) says $\{f_n\}_{n\geq 1}$ converges uniformly on $(X \setminus E)$, for every $n \in \mathbb{N}$. Take $k \in \mathbb{N}$ such that for every $x \in (X \setminus E)$ and every $n \geq k$ we have

$$|f_n - f| < \frac{\epsilon}{3m(X)}.$$

Then

$$\int_{X\setminus E} |f_n - f|\, dm < \frac{\epsilon}{3} \cdot \frac{m(X \setminus E)}{m(X)}.$$

Eventually, for any $n \geq k$ we have

$$\begin{aligned}\left| \int_X f_n\, dm - \int_X f\, dm \right| &\leq \int_{X\setminus E} |f_n - f|\, dm + \int_E |f_n|\, dm + \int_E |f|\, dm \\ &< \frac{\epsilon}{3} \cdot \frac{m(X \setminus E)}{m(X)} + \frac{\epsilon}{3} + \frac{\epsilon}{3} < \epsilon,\end{aligned}$$

and the claim follows. ■

Remark 11.3.4 Let us examine the relationship between Lebesgue's dominated convergence theorem and Vitali's convergence theorem in the case $m(X) < \infty$.

Suppose the sequence $\{f_n\}_{n\geq 1}$ of $\mathcal{S}$-measurable maps converges pointwise m-a.e. to a measurable map f, and consider $g \in L_1$ such that $|f_n| \leq g$ m-almost everywhere, for every $n \in \mathbb{N}$.

We claim the f_n are uniformly summable on X. By Proposition 11.3.5, as $g \in L_1$, for every $\epsilon > 0$ there is a $\delta > 0$ such that

$$\int_E |f_n|\, dm \leq \int_E g\, dm < \epsilon$$

for every $E \in \mathcal{S}$, $m(E) < \delta$, and $n \in \mathbb{N}$. Immediately, then, the f_n are uniformly summable and converge to the pointwise limit f m-a.e. Under these conditions Vitali's theorem holds, since we are assuming $m(X) < \infty$. ♣

Definition 11.3.4 (*Tight families of functions*) Given a measure space $(X, \mathcal{S}, m)$,[3] a family $\mathscr{F}$ of summable functions $f : X \mapsto \bar{\mathbb{R}}$ is tight on X if for any $\epsilon > 0$ there exists a set $K \in \mathcal{S}$, with $m(K) < \infty$, such that

$$\int_{X\setminus K} |f|\, dm < \epsilon \quad \forall f \in \mathscr{F}.$$ ◇

Proposition 11.3.7 (Uniformly summable and tight sequences of functions) *Let $(X, \mathcal{S}, m)$ be a measure space*[4] *and $\{f_n : X \mapsto \bar{\mathbb{R}}\}_{n\geq 1}$ a uniformly summable and tight sequence of maps. If $\{f_n\}$ converges pointwise to $f : X \mapsto \mathbb{R}$ m-a.e., then f is summable on X.*

Proof Fix $\epsilon > 0$, than by assumption there is a set $K \in \mathcal{S}$, with $m(K) < \infty$, such that

[3] Here X is allowed to have infinite measure.

[4] Here X is allowed to have infinite measure.

$$\int_{X\setminus K} |f_n| dm < \epsilon \tag{11.3.11}$$

for every $n \in \mathbb{N}$. Then take $\epsilon = 1$. As $m(K) < \infty$ and the $\{f_n\}_{n\geq 1}$ are uniformly summable, we apply Proposition 11.3.6 to the finite-measure set K, so

$$\int_K |f| dm < (k+1).$$

Using Fatou's lemma on (11.3.11) produces

$$\int_{X\setminus K} |f|\, dm = \int_{X\setminus K} \liminf_{n\to\infty} |f_n|\, dm \leq \liminf_{n\to\infty} \int_{X\setminus K} |f_n|\, dm < 1.$$

Putting these together gives

$$\int_X |f|\, dm = \int_{(X\setminus K)} |f|\, dm + \int_K |f|\, dm < 1 + k + 1 = (k+2),$$

and the claim is proved. ■

With the help of Proposition 11.3.7 Vitali's theorem extends to spaces of infinite measure provided we assume the sequence is tight.

Theorem 11.3.3 (Extension of Vitali's convergence theorem - 1907) *Let $(X, \mathcal{S}, m)$ be a measure space and $\{f_n : X \mapsto \bar{\mathbb{R}}\}_{n\geq 1}$ a tight sequence of uniformly summable maps. If $\{f_n\}$ converges pointwise to $f : X \mapsto \mathbb{R}$ m-a.e., then f is summable on X and*

$$\int_X \lim_{n\to\infty} f_n\, dm = \lim_{n\to\infty} \int_X f_n\, dm.$$

Proof By Proposition 11.3.7 $f \in L_1$, and by Theorem 11.2.4 f is bounded m-a.e. Part *1.* in Proposition 11.3.2 says that for every $n \in \mathbb{N}$

$$\left| \int_X f_n\, dm - \int_X f\, dm \right| = \left| \int_X (f_n - f)\, dm \right| \leq \int_X |f_n - f|\, dm.$$

As the sequence is tight, given $\epsilon > 0$ there is $K \in \mathcal{S}$ with $m(K) < \infty$, such that

$$\int_X |f_n - f|\, dm = \int_{X\setminus K} |f_n - f|\, dm + \int_K |f_n - f|\, dm \tag{11.3.12}$$

for every $n \in \mathbb{N}$. Above, the first integral on the right is smaller than $\epsilon/4$. Moreover, since $m(K) < \infty$ and $\{f_n\}_{n\geq 1}$ is uniformly summable, we can apply Vitali's theorem (the original version, Theorem 11.3.2) to the second integral, obtaining for every $n \geq k$

$$\begin{aligned}\int_K |f_n - f|\, dm &\leq \int_{K\setminus E} |f_n - f|\, dm + \int_E |f_n|\, dm + \int_E |f|\, dm \\ &< \frac{\epsilon}{4} \cdot \frac{m(K \setminus E)}{m(K)} + \frac{\epsilon}{4} + \frac{\epsilon}{4} < \frac{3}{4}\,\epsilon.\end{aligned}$$

Putting all this together,

$$\begin{aligned}\left|\int_X f_n\, dm - \int_X f\, dm\right| &\leq \int_{X\setminus K} |f_n - f|\, dm + \int_K |f_n - f|\, dm \\ &< \frac{\epsilon}{4} + \frac{\epsilon}{4} \cdot \frac{m(K \setminus E)}{m(K)} + \frac{\epsilon}{4} + \frac{\epsilon}{4} < \epsilon.\end{aligned}$$

■

Remark 11.3.5 • Removing tightness in the case $m(X) = \infty$ invalidates the extended version of Vitali's convergence theorem. Consider the sequence of general term

$$f_n = \varphi_{[n,n+1]},$$

so clearly

$$\lim_{n\to\infty} f_n = f = 0.$$

The f_n are uniformly summable, because for every $E \in \mathcal{S}(\mathbb{R}^+)$, $m(E) < \delta$, we have

$$\int_E f_n\, dm = \int_E \varphi_{[n,n+1]}\, dm = m\big([n, n+1] \cap E\big) \leq m(E),$$

for every $n \in \mathbb{N}$. Yet the theorem's conclusion is false, since for every $n \in \mathbb{N}$

$$\begin{aligned}&\int_{\mathbb{R}^+} f_n\, dm = \int_{\mathbb{R}^+} \varphi_{[n,n+1]}\, dm = m\big([n, n+1] \cap \mathbb{R}^+\big) = 1 \\ \Rightarrow &\lim_{n\to\infty} \int_{\mathbb{R}^+} f_n\, dm = 1;\end{aligned}$$

whilst

$$\int_{\mathbb{R}^+} \lim_{n\to\infty} f_n \, dm = \int_{\mathbb{R}^+} f \, dm = 0.$$

- Let us show that the dominated convergence theorem is a consequence of Vitali's convergence result also in case $m(X) = \infty$. Let $\{f_n\}_{n\geq 1}$ be a sequence of measurable functions converging pointwise m-a.e. to a measurable map f, and suppose $g \in L_1$ obeys $|f_n| \leq g$ m-a.e., for every $n \in \mathbb{N}$.
  The sequence is uniformly summable: $g \in L_1$ implies, by Proposition 11.3.5, that for every $\epsilon > 0$ there is a $\delta > 0$ such that, for every $E \in \mathcal{S}$, $m(E) < \delta$,

$$\int_E |f_n| \, dm \leq \int_E g \, dm < \epsilon$$

  for every $n \in \mathbb{N}$. It is also tight, because for every $\epsilon > 0$ exists a set $K \in \mathcal{S}$, with $m(K) < \infty$ such that

$$\int_{X\setminus K} |f_n| \, dm \leq \int_{X\setminus K} g \, dm < \epsilon,$$

  for every $n \in \mathbb{N}$. Since $\{f_n\}_{n\geq 1}$ converges pointwise m-a.e. to f, the extended Vitali theorem holds. ♣

Remark 11.3.6 We remind that the proofs in this chapter hold under the premise the measure space $(X, \mathcal{S}, m)$ is complete and has σ-finite measure, except when we said otherwise. We have seen, with Proposition 9.6.1, that also the Lebesgue–Stieltjes space $\big(\mathbb{R}, \mathcal{S}(\mathbb{R}), m_F\big)$ is complete and with σ-finite measure. In particular, the Lebesgue–Stieltjes measure of $\big([0, 1], \mathcal{S}([0, 1]), m_F\big)$ is totally finite.

Let us call, for $f \in L_1$, Lebesgue–Stieltjes integral the expression

$$\int_X f \, dF,$$

where $X \in \mathcal{S}(\mathbb{R})$ and $dm_F = dF$. From this we deduce a useful observation, namely that the Lebesgue–Stieltjes integral enjoys all the nice properties of the Lebesgue integral if the measure space is complete and the measure is σ-finite. Under these conditions, therefore, the Lebesgue–Stieltjes integral satisfies Beppo Levi's theorem, Fatou's lemma and Lebesgue's dominated convergence theorem, among others. ♣

Chapter 12
Comparing Notions of Integral



12.1 Comparing the Riemann and Lebesgue Integrals

This section is devoted to showing the Lebesgue integral is an extension of the Riemann integral.

Fix $X = \mathbb{R}$, $\mathcal{S} = \mathscr{L}$ (the σ-algebra of Lebesgue measurable sets) and $m = \ell$ (the Lebesgue measure on $\mathbb{R}$). We indicate by $L_1[a, b]$ and $\mathcal{R}[a, b]$ the classes of Lebesgue summable and Riemann integrable bounded maps on the compact domain $[a, b]$ respectively.

The first thing to notice is there exist bounded functions $f : [a, b] \mapsto \mathbb{R}$ that are not Riemann integrable but are Lebesgue summable. That is because the bounded map $f : [a, b] \mapsto \mathbb{R}$,

$$f(x) = \begin{cases} 1 & \text{if } x \in [a, b] \cap \mathbb{Q} \\ 2 & \text{if } x \in [a, b] \setminus \mathbb{Q} \end{cases}$$

is Dirichlet-like, and hence not Riemann integrable. But it is summable, with Lebesgue integral equal to $2(b - a)$.

Hence when we compare the two notions of integral it suffices to show that any Riemann integrable bounded map $f : [a, b] \mapsto \mathbb{R}$ is Lebesgue summable, and the integrals coincide, i.e.

$$\int_a^b f(x)dx = \int_{[a,b]} f(x)\ell(dx)$$

(the integral on the left is *à la* Riemann).

To prove the aforementioned inclusion we must explain which quantities are at stake. For that, given $I = [a, b]$ we define

- $P_n = \{a = x_0 < x_1 < \cdots < x_{j_n} = b\}$ the nth partition of I in j_n parts, and $\mathscr{P}$ the family of all possible partitions of I;

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- $I_{k_n} = [x_{k_n-1}, x_{k_n}]$ the k_nth interval of P_n, $k_n = 1, \dots, j_n$;
- $|P_n| = \max_{k_n=1,\dots,j_n} m_{pj}(I_{k_n})$ the norm, or mesh, of P_n;
- $m_{I_{k_n}} = \inf_{x\in I_{k_n}} f(x)$, $M_{I_{k_n}} = \sup_{x\in I_{k_n}} f(x)$, for a bounded $f : [a, b] \mapsto \mathbb{R}$;
- $s(f, P_n) = \sum_{k_n=1}^{j_n} m_{I_{k_n}} m_{pj}(I_{k_n})$ lower sum (over P_n);
- $S(f, P_n) = \sum_{k_n=1}^{j_n} M_{I_{k_n}} m_{pj}(I_{k_n})$ upper sum (over P_n);
- $s(f) = \sup_{P_n\in\mathscr{P}} s(f, P_n)$ lower (Riemann) integral of $f : [a, b] \mapsto \mathbb{R}$;
- $S(f) = \inf_{P_n\in\mathscr{P}} S(f, P_n)$ upper (Riemann) integral of $f : [a, b] \mapsto \mathbb{R}$;
- $s(f) = \displaystyle\int_a^b f(x)dx = S(f)$ Riemann integral on I of the bounded map f, in which case f is said to be Riemann integrable;
- $\mathcal{R}(I)$ the collection of Riemann integrable, bounded maps on I.

So now we have to prove $\mathcal{R}(I) \subseteq L_1(I)$.

Theorem 12.1.1 (Summability of Riemann integrable maps) *If* $f : [a, b] \mapsto \mathbb{R}$ *belongs to* $\mathcal{R}([a, b])$ *then* $f \in L_1([a, b])$ *and*

$$\int_{[a,b]} f(x)\, \ell(dx) = \int_a^b f(x)dx.$$

Proof Consider an increasing sequence $\{P_n\}_{n\geq 1}$ of partitions of $[a, b]$ such that

$$\lim_{n\to\infty} |P_n| = \lim_{n\to\infty} \left(\max_{k_n=1,\dots,j_n} m_{pj}(I_{k_n}) \right) = 0.$$

For every $n \in \mathbb{N}$, moreover, set

$$P_n = \left\{ a = x_0 < x_1 < x_2 < \cdots < x_{j_n-1} < x_{j_n} = b \right\}$$

and

$$I_{k_n} = [x_{k_n-1}, x_{k_n}],\ M_{I_{k_n}} = \sup_{x\in I_{k_n}} f(x),\ m_{I_{k_n}} = \inf_{x\in I_{k_n}} f(x) \text{ for } k_n = 1, \dots, j_n.$$

Now, for any $n \in \mathbb{N}$ we can define the following step functions with respect to partition P_n of $[a, b]$

$$s_n(x) = \sum_{k_n=1}^{j_n} m_{I_{k_n}} \varphi_{I_{k_n}}(x) \quad \text{and} \quad t_n(x) = \sum_{k_n=1}^{j_n} M_{I_{k_n}} \varphi_{I_{k_n}}(x).$$

The former is a lower bound for $f(x)$, the latter an upper bound. Hence $\{s_n\}_{n\geq 1}$ is increasing, $\{t_n\}_{n\geq 1}$ decreasing, and for every $n \in \mathbb{N}$

$$s_n \leq f \leq t_n.$$

We also have, for every $n \in \mathbb{N}$,

$$\int_a^b s_n(x)dx = \sum_{k_n=1}^{j_n} m_{I_{k_n}} m_{pj}(I_{k_n}) = s(f, P_n)$$
$$\int_a^b t_n(x)dx = \sum_{k_n=1}^{j_n} M_{I_{k_n}} m_{pj}(I_{k_n}) = S(f, P_n),$$

and then

$$\lim_{n\to\infty} \int_a^b s_n(x)dx = \lim_{n\to\infty} s(f, P_n) = s(f)$$
$$\lim_{n\to\infty} \int_a^b t_n(x)dx = \lim_{n\to\infty} S(f, P_n) = S(f).$$

But $f \in \mathcal{R}(I)$, so

$$\lim_{n\to\infty} \int_a^b s_n(x)dx = \int_a^b f(x)dx = \lim_{n\to\infty} \int_a^b t_n(x)dx. \tag{12.1.1}$$

Observe that $m_{pj}(I_{k_n}) = \ell(I_{k_n})$ for every $k_n = 1, \ldots, j_n, n \in \mathbb{N}$. Besides, $\{s_n\}, \{t_n\} \subseteq \mathscr{M}_0([a, b], \mathscr{L})$ are bounded in $[a, b]$. Therefore the Riemann integrals on the far right and far left of (12.1.1) coincide with the Lebesgue integrals. Calling

$$\lim_{n\to\infty} s_n = s, \quad \lim_{n\to\infty} t_n = t,$$

by Corollary 11.3.2 $s, s_n, t, t_n \in L_1\big([a, b], \mathscr{L}\big)$ for every $n \in \mathbb{N}$, and

$$\lim_{n\to\infty} \int_a^b s_n(x)dx = \int_{[a,b]} s(x)\ \ell(dx), \quad \lim_{n\to\infty} \int_a^b t_n(x)dx = \int_{[a,b]} t(x)\ \ell(dx).$$

Formula (12.1.1) implies

$$\int_{[a,b]} s(x)\ \ell(dx) = \int_a^b f(x)dx = \int_{[a,b]} t(x)\ \ell(dx). \tag{12.1.2}$$

Since $(t - s) \in \mathscr{M}\big([a, b], \mathscr{L}\big)$ and

$$\int_{[a,b]} (t - s)(x)\ \ell(dx) = 0,$$

Theorem 11.2.3 guarantees $t = s$ ℓ-a.e. on $[a, b]$, and consequently $s = f = t$ ℓ-a.e. on $[a, b]$. By Theorem 10.2.6 $f \in \mathscr{M}\big([a, b], \mathscr{L}\big)$, so part *2.* of Proposition 11.3.2 gives $f \in L_1\big([a, b], \mathscr{L}\big)$ and

$$\int_{[a,b]} s(x)\ \ell(dx) = \int_{[a,b]} f(x)\ \ell(dx) = \int_{[a,b]} t(x)\ \ell(dx). \tag{12.1.3}$$

Eventually, (12.1.2) and (12.1.3) allow to conclude. ■

Remark 12.1.1 Using Theorem 10.2.6, item *2.* in Proposition 11.3.2 and Theorem 12.1.1 we draw the following conclusion:

"Let $g : [a, b] \mapsto \mathbb{R}$ be Riemann integrable and bounded, and suppose $f : [a, b] \mapsto \mathbb{R}$ coincides with g ℓ-almost everywhere on $[a, b]$. Then $f(x)$ is Lebesgue summable, and

$$\int_a^b g(x)dx = \int_{[a,b]} g(x)\ell(dx) = \int_{[a,b]} f(x)\ell(dx)."$$

Here is an example for this situation: take $f : [0, 1] \mapsto \mathbb{R}$ to be

$$f(x) = \begin{cases} x^2 & \text{if } x \in [0, 1] \setminus \mathbb{Q} \\ \dfrac{1}{x} & \text{if } x \in \big([0, 1] \cap \mathbb{Q}\big) \setminus \{0\} \\ 1 & \text{if } x = 0 \end{cases}.$$

The function $g : [0, 1] \mapsto \mathbb{R}$, $g(x) = x^2$ is Riemann integrable and equals f ℓ-almost everywhere on $[0, 1]$. Hence f is Lebesgue summable and

$$\int_{[0,1]} f(x)\ \ell(dx) = \int_0^1 g(x)dx = \frac{1}{3}.$$ ♣

12.2 Completeness of $L_1([a, b])$

We have seen that if f, g are summable and α a real number, then $f + g$ and αf are summable (Proposition 11.3.2, *3.-4.*). This makes $L_1\big([a, b], \mathscr{L}, \ell\big)$ a real vector space, on which we set

$$\|f\|_1 = \int_{[a,b]} |f| d\ell \qquad \forall f \in L_1([a, b]).$$

This is a semi-norm in that it fulfils the following two properties on a norm (out of three): homogeneity

$$\|\alpha f\|_1 = \alpha \|f\|_1 \quad \forall f \in L_1([a,b]),\ \alpha \in \mathbb{R}$$

and the triangle inequality

$$\|f+g\|_1 \leq \|f\|_1 + \|g\|_1, \quad \forall f, g \in L_1([a,b]).$$

It is not definite in general, though (see Theorem 11.2.3).

To speak about a norm we need to identify ℓ-almost everywhere equal maps, i.e. put on $L_1([a,b])$ the equivalence relation $f \sim g$ iff $f = g$ ℓ-a.e. on $[a,b]$. The ensuing quotient space of $L_1([a,b])$ will become a normed vector space. Rephrasing, we may write

$$f \sim g \Leftrightarrow \int_{[a,b]} |f-g| d\ell = 0.$$

The relation $\sim$ is reflexive, symmetric and transitive and hence an equivalence relation. $L_1([a,b])$ is thus partitioned in equivalence classes, i.e. pairwise-disjoint sets of functions such that for every f, g in one coset $\int_{[a,b]} |f-g| d\ell = 0$. Let us write $[f]$ for the equivalence class of f and consider the quotient $L_1([a,b])/\sim$, whose elements are the equivalence classes.

It is a vector space, for

$$\begin{aligned} [f] + [g] &= [f+g] \\ \alpha[f] &= [\alpha f]. \end{aligned}$$

The coset $[0]$ consists of the zero map and functions whose absolute value has vanishing integral. This equivalence class is the neutral element of the vector space

$$[f] + [0] = [f].$$

On the vector space $L_1([a,b])$ we define the integral norm

$$0 \leq \|[f]\|_1 = \|f\|_1 = \int_{[a,b]} |f| d\ell < \infty, \forall f \in L_1([a,b])$$

and

$$\|[f]\|_1 = \|f\|_1 = \int_{[a,b]} |f| d\ell = 0 \Leftrightarrow f = 0 \ \ \ell\text{-a.e. } on [a,b].$$

Furthermore,

$$\begin{aligned}\|\alpha f\|_1 &= \int_{[a,b]} |\alpha f| d\ell = |\alpha| \|f\|_1, \forall \alpha \in \mathbb{R} \text{ and } f \in L_1\big([a,b]\big)\\ \|f+g\|_1 &= \int_{[a,b]} |f+g| d\ell\\ &\leq \int_{[a,b]} |f| d\ell + \int_{[a,b]} |g| d\ell\\ &= \|f\|_1 + \|g\|_1, \quad \forall f \text{ and } g \in L_1\big([a,b]\big).\end{aligned}$$

All this is coherent since if we take another representative g of $[f]$, by the triangle inequality

$$\|g\|_1 \leq \|g-f\|_1 + \|f\|_1 \quad \text{and} \quad \|f\|_1 \leq \|f-g\|_1 + \|g\|_1;$$

as $f \sim g$, we have $d(g, f) = \|g - f\|_1 = 0$ and then $\|f\|_1 = \|g\|_1$. In fact the integral norm on $L_1([a, b])$ stays the same if we modify functions on zero-measure sets. The quotient $L_1([a, b])/\sim$ with integral norm $\|\cdot\|_1$ is a normed vector space and we indicate it by

$$\big(L_1([a,b]), \|\cdot\|_1\big).$$

Now we show $\big(\mathcal{R}([a,b]), \|\cdot\|_1\big)$ is incomplete, and $\big(L_1([a,b]), \|\cdot\|_1\big)$ is actually its completion. For this we use an example generalising the Cantor set.

Example 12.2.1 (*Generalised Cantor set*) Consider $[0, 1]$ and fix a number $\alpha \in (0, 1/3)$. Step 1: delete the middle open interval of length α

$$G_1^\alpha = \left(\frac{1-\alpha}{2}; \frac{1+\alpha}{2}\right)$$

and call C_1^α what remains

$$C_1^\alpha = \left[0; \frac{1-\alpha}{2}\right] \cup \left[\frac{1+\alpha}{2}, 1\right], \quad \text{where } m(C_1^\alpha) = 1-\alpha.$$

Step 2: from each of the two intervals of C_1^α remove the middle open interval of length α^2, and name the remaining set C_2^α, which now has measure $1 - \alpha - 2\alpha^2$.

Iterating as above, at step n the set C_n^α has measure

$$1 - [\alpha + 2\alpha^2 + 2^2\alpha^3 + \cdots + 2^{n-1}\alpha^n] = 1 - \alpha \sum_{i=0}^{n-1} (2\alpha)^i.$$

The generalised Cantor set is the intersection $C^\alpha = \bigcap_{n=1}^\infty C_n^\alpha$ of a decreasing sequence, and C_1^α has finite measure. Since the Lebesgue measure is continuous from above,

$$m(C^\alpha) = \lim_{n\to\infty} m(C_n^\alpha) = \lim_{n\to\infty}\left[1 - \alpha\sum_{i=0}^{n-1}(2\alpha)^i\right]$$
$$= 1 - \alpha\sum_{i=0}^{\infty}(2\alpha)^i = 1 - \frac{\alpha}{1-2\alpha} = \frac{1-3\alpha}{1-2\alpha}. \tag{12.2.1}$$

For $\alpha = 1/3$ we recover the 'classical' Cantor set, which has measure zero. If we choose $\alpha \in \big(0, 1/3\big)$ the generalised Cantor set has measure $m(C^\alpha) > 0$. ★

Proposition 12.2.1 (Incompleteness of $\big(\mathcal{R}([a,b]), \|\cdot\|_1\big)$) *The vector space of bounded, Riemann integrable maps on an interval $[a,b]$, equipped with norm*

$$\|f(x)\|_1 = \int_{[a,b]} |f| d\ell,$$

is not complete.

Proof Consider $f : [0,1] \mapsto \mathbb{R}$ defined as

$$f = \varphi_{C^\alpha}.$$

It is the characteristic function of the generalised Cantor set, and as such it is discontinuous at the frontier of the set. Since its range does not contain intervals of any (positive) length, C^α has no interior point. All points in C^α lie on the frontier, so C^α is perfect for it coincides with its derived set. Therefore the characteristic function φ_{C^α} is discontinuous everywhere on C^α, which has positive measure. It follows that the discontinuity set is not enclosable, i.e. not PJ measurable, making f not Riemann integrable. On the other hand, φ_{C^α} is Lebesgue integrable and its integral equals the Lebesgue measure of C^α.

Different story if we consider the sets $C_n^\alpha, n \geq 1$. For any such consider $f_n : [0,1] \mapsto \mathbb{R}$, $f_n(x) = \varphi_{C_n^\alpha}(x)$. Each f_n is Riemann integrable, because C_n^α is the finite union of closed intervals, and the frontier consists of finitely many points. Hence f_n is discontinuous at a finite number of points and therefore Riemann integrable. The elements of $\{\varphi_{C_n^\alpha}\}_{n\geq 1}$ all belong to $\mathcal{R}\big([0,1]\big)$ ($\Rightarrow$ Riemann integrable), while

$$\lim_{n\to\infty} f_n = \varphi_{C^\alpha}$$

is only Lebesgue integrable. Moreover, for every $n \geq 1$ $|f_n| \leq 1$, and the constant map is integrable on $[0,1]$. By dominated convergence we have

$$\lim_{n\to\infty} \|\varphi_{C_n^\alpha} - \varphi_{C^\alpha}\|_1 = \lim_{n\to\infty} \int_{[a,b]} |\varphi_{C_n^\alpha} - \varphi_{C^\alpha}| d\ell = 0.$$

This means that for any $\epsilon > 0$ there exists an n_ϵ such that

$$\|\varphi_{C_p^\alpha} - \varphi_{C_q^\alpha}\|_1 < \epsilon$$

for every $p, q \geq n_\epsilon$. Hence $\{\varphi_{C_n^\alpha}\}_{n\geq 1}$ is a Cauchy sequence in $\mathcal{R}([a,b])$, but it does not converge in the space. It converges in $L_1([a,b])$, though, and so $\mathcal{R}([a,b])$ is not complete. ■

Next we show $\|\cdot\|_1$ makes $L_1([a,b])$ complete.

Theorem 12.2.1 (Riesz–Fischer theorem—completeness of $\big(L_1([a,b]), \|\cdot\|_1\big)$) *The normed vector space $\big(L_1([a,b]), \|\cdot\|_1\big)$, defined identifying ℓ-almost everywhere equal functions, is complete.*

Proof In order to show $(L_1([a,b])$ is complete we must prove any Cauchy sequence $\{f_n\}_{n\geq 1} \subseteq L_1([a,b])$ converges to some map $f \in L_1([a,b])$ ℓ-almost everywhere on $[a,b]$.

It suffices to show that from every $\{f_n\}_{n\geq 1} \subseteq L_1([a,b])$ that converges pointwise ℓ-a.e. we can extract a subsequence,[1] similarly convergent ℓ-a.e. on $[a,b]$, to some f in $L_1([a,b])$. So suppose $\{f_n\}_{n\geq 1}$ is Cauchy, so for every $\epsilon > 0$ there is an $n(\epsilon)$ such that for every n and $m \geq n(\epsilon)$ we have

$$\|f_n - f_m\|_1 < \epsilon.$$

Pick $\epsilon = 1/2$, so we have $n\big(\frac{1}{2}\big)$ such that for every $n_1, n_2 \geq n\big(\frac{1}{2}\big)$,

$$\|f_{n_2} - f_{n_1}\|_1 < \frac{1}{2}.$$

Similarly, for $\epsilon = 1/2^2$ there exists $n\big(\frac{1}{2^2}\big)$ such that for every $n_2, n_3 \geq n\big(\frac{1}{2^2}\big)$,

$$\|f_{n_3} - f_{n_2}\|_1 < \frac{1}{2^2}.$$

By induction, given $\epsilon = (1/2^k)$ there is $n\big(\frac{1}{2^k}\big)$ such that for every $n_k, n_{k+1} \geq n\big(\frac{1}{2^k}\big)$,

$$\|f_{n_{k+1}} - f_{n_k}\|_1 < \frac{1}{2^k}.$$

We end up with a subsequence $\{f_{n_k}\}_{k\geq 1}$ such that

$$\|f_{n_1}\|_1 + \sum_{k=1}^{\infty} \|f_{n_{k+1}} - f_{n_k}\|_1 < \|f_{n_1}\|_1 + \sum_{k=1}^{\infty} \frac{1}{2^k} < \infty.$$

By Corollary 11.3.1 this subsequence converges pointwise ℓ-a.e. on $[a,b]$ to some $f(x) \in L_1([a,b])$, i.e.

[1] It is a well-known result that *"any subsequence of a convergent sequence converges to the same limit."*.

$$f = f_{n_1} + \sum_{k=1}^{\infty} \left(f_{n_{k+1}} - f_{n_k} \right) \tag{12.2.2}$$

ℓ-a.e. on $[a,b]$, and

$$\int_{[a,b]} f d\ell = \int_{[a,b]} f_{n_1} d\ell + \sum_{k=1}^{\infty} \int_{[a,b]} (f_{n_k+1} - f_{n_k}) d\ell < \infty. \qquad \blacksquare$$

Exactly as with $\mathbb{R}$ and $\mathbb{Q}$, we will see in a short while that $L_1([a,b])$ is the completion of $\mathcal{R}([a,b])$.

For this it is enough to prove that the space of continuous maps $C([a,b]) \subseteq \mathcal{R}([a,b])$ on $[a,b]$ is a dense subspace in $L_1([a,b])$. All in all we have inclusions

$$C([a,b]) \subseteq \mathcal{R}([a,b]) \subseteq L_1([a,b]) \quad \text{and} \quad \overline{C([a,b])} = L_1([a,b]).$$

To say $C([a,b])$ is dense in $L_1([a,b])$ amounts to assert that every $f \in L_1([a,b])$ can be approximated (as well as we want) by elements of $C([a,b])$: if $f \in L_1([a,b])$ then for every $\epsilon > 0$ there exists $g \in C([a,b])$ such that $\|f - g\|_1 < \epsilon$.

Yet $L_1([a,b])$ is not just important as the completion of $\mathcal{R}([a,b])$, but also because it contains dense subspaces: compactly supported continuous maps and compactly supported smooth maps.

The density of the latter spaces allows to approximate square-integrable functions $f \in L_2(\mathbb{R})$ using sequences in the two spaces. This fact, as we shall see in Volume Two, is paramount to determine Fourier transforms.

Theorem 12.2.2 (Density of $C([a,b])$ *in* $L_1([a,b])$) *The space* $C([a,b])$ *is a dense subspace in* $L_1([a,b])$.

Proof Observe first that by the Weierstraß theorem any $g \in C([a,b])$ is bounded, and being continuous it is Riemann integrable, so $g \in L_1([a,b])$. Consequently $C([a,b]) \subseteq L_1([a,b])$.

There remains to show $C([a,b])$ is dense. Since any $f \in L_1([a,b])$ can be written as $f = f^+ - f^-$, we may restrict to the case $f \geq 0$. So take $f \in L_1([a,b])$, $f \geq 0$. Given $\epsilon > 0$ we must exhibit $g \in C([a,b])$ such that $\|f - g\|_1 < \epsilon$. As $f \in L_1([a,b])$, by Proposition 11.2.1 there is a sequence of non-negative, simple measurable maps $\{s_n\}_{n \geq 1}$ such that

$$\lim_{n \to \infty} \int_{[a,b]} s_n d\ell = \int_{[a,b]} f d\ell.$$

But $s_n \in L_1([a,b])$, so for any $\epsilon > 0$ there is an n' such that for every $n \geq n'$

$$\|s_n - f\|_1 < \epsilon.$$

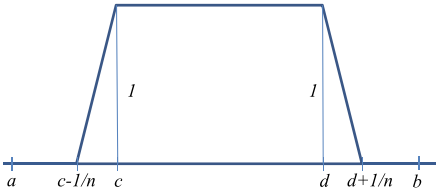


Fig. 12.1 Example of continuous extension

Therefore the collection of non-negative simple measurable maps is dense in $L_1([a,b])$. Now it is enough to show the latter can be approximated by $C([a,b])$. We can simplify even further, since non-negative simple measurable maps are linear combinations of indicator functions of ℓ-measurable sets. This means we can as well reduce to proving the latter can be approximated through $C([a,b])$.

So let $f = \varphi_A$ with $A \in \mathscr{L}$ and $A \subseteq [a,b]$. Since φ_A is Lebesgue measurable and bounded, by Lusin's theorem (Theorem 10.4.2) for any $\epsilon > 0$ there exists a closed set $F \subseteq A$ such that $\ell(A \setminus F) < \epsilon$ and φ_A is continuous on F, whence

$$\|\varphi_A - \varphi_F\|_1 = \ell(A \setminus F) < \epsilon.$$

Urysohn's lemma (Example B.0.1) warrants the existence of a continuous extension g to $\mathbb{R}$ such that $g = \varphi_F$ on F.

To complete the proof, we only have to show that the function g on $[a,b]$ is such that $\|\varphi_F - g\|_1 < \epsilon$. Without loss of generality we can take $F = [c,d]$, where $a < c < d < b$. Therefore we consider the following map (Fig. 12.1)

$$g = \begin{cases} 1 & \text{if } x \in [c,d] \\ \text{linear, increasing from 0 to 1} & \text{if } x \in \left(c - \dfrac{1}{n}, c\right) \\ \text{linear, decreasing from 1 to 0} & \text{if } x \in \left(d, d + \dfrac{1}{n}\right) \\ 0 & \text{if } x \notin \left(c - \dfrac{1}{n}, d + \dfrac{1}{n}\right). \end{cases}$$

Clearly g is continuous on $\mathbb{R}$, $\left\|\varphi_F - g\right\|_1 = 1/n$, and so $\left\|\varphi_F - g\right\|_1 < \epsilon$ for large enough n and any given $\epsilon > 0$. ∎

From Theorems 12.2.1, 12.2.2 we infer $L_1([a,b])$ is the completion of $\mathcal{R}([a,b])$.

Now we study another dense subspace of $L_1(\mathbb{R})$, the space of smooth functions[2] on $\mathbb{R}$, denoted $C^\infty(\mathbb{R})$.

If a map is differentiable at a point it is also continuous at that point, so one says a function is (of class, or in) $C^1(\mathbb{R})$ if its first derivative is continuous on $\mathbb{R}$. By induction, a function is (in) $C^n(\mathbb{R})$ if all derivatives from the 1st to the nth are

[2]Instances of smooth functions on $\mathbb{R}^n$ (in every variable) are exponentials, sine, cosine, any composite thereof, and trivially also $x^3, x^4, \ldots$ etc.

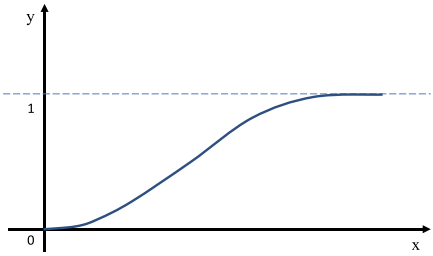


Fig. 12.2 Graph of $f(x) = e^{-\frac{1}{x}}$ on $x > 0$ and $f(x) = 0$ on $x \leq 0$

continuous on $\mathbb{R}$. For uniformity one also sets $C^0(\mathbb{R}) = C(\mathbb{R})$. From this we recover the definition of smooth map.

Definition 12.2.1 (*Smooth functions*) A function is called smooth, or differentiable infinitely many times, if its derivatives of any order are continuous. The space of smooth functions on $\mathbb{R}$ is denoted by $C^\infty(\mathbb{R})$. ◇

We then have an obvious chain of inclusions

$$C^0(\mathbb{R}) \supseteq C^1(\mathbb{R}) \supseteq \cdots \supseteq C^n(\mathbb{R}) \supseteq \cdots,$$

whence

$$C^\infty(\mathbb{R}) = \bigcap_{n=0}^{\infty} C^n(\mathbb{R}).$$

We saw in Theorem 12.2.2 that a map $f \in L_1([a,b])$, so ultimately any $\varphi_A \in L_1([a,b])$, can be approximated by a continuous map g in $C([a,b])$ with norm $\|\cdot\|_1$. Now we are interested in knowing whether, given $\varphi_A \in L_1([a,b])$, there might be a smooth function ϕ_n on $[a,b]$ that approximates φ_A. Put equivalently, we wish to know whether $C^\infty([a,b])$ is dense in $L_1([a,b])$.

To do that we need to construct an explicit function of the space $C^\infty(\mathbb{R})$.

Example 12.2.2 (*Construction of bump functions*) Start from the map (Fig. 12.2)

$$f(x) = \begin{cases} e^{-\frac{1}{x}} & \text{if } x > 0 \\ 0 & \text{if } x \leq 0. \end{cases}$$

We claim it belongs to $C^\infty(\mathbb{R})$. First, when $x > 0$

$$
\begin{aligned}
f'(x) &= \frac{1}{x^2}\, e^{-1/x} \\
f''(x) &= \left(\frac{1}{x^4} - \frac{2}{x^3}\right) e^{-1/x} \\
f'''(x) &= \left(\frac{1}{x^6} - \frac{6}{x^5} + \frac{6}{x^4}\right) e^{-1/x} \\
\vdots \quad &= \qquad\qquad \vdots \\
f^n(x) &= P_{2n}\left(\frac{1}{x}\right) e^{-1/x} \\
\vdots \quad &= \qquad\qquad \vdots
\end{aligned}
$$

where $P_{2n}(1/x)$ is a polynomial of degree $2n$. Second, $f^n(0) = 0$ for every $n \in \mathbb{N}$. This is true for $n = 0$, so assume it true $n - 1$, i.e. $f^{n-1}(0) = 0$. Then

$$
\begin{aligned}
f^n(0) &= \lim_{x\to 0^+} \left(\frac{f^{n-1}(x) - f^{n-1}(0)}{x}\right) = \lim_{x\to 0^+} \left[\frac{P_{2n-1}(1/x)\, e^{-1/x}}{x}\right] \\
&= \lim_{t\to\infty} \left[t\, P_{2n-1}(t)\, e^{-t}\right],
\end{aligned}
$$

where $t = 1/x$. As $t \to \infty$ the map e^t grows faster than t^{2n} for every $n \geq 0$, implying $f^n(0) = 0$. With this we conclude that $f \in C^\infty(\mathbb{R})$.

Using the above map we can then introduce

$$
F(x) = \frac{f(x)}{f(x) + f(1-x)},
$$

where $f(1-x)$ vanishes for $x \geq 1$, so that $F(x) = f(x)/f(x) = 1$ when $x \geq 1$. Let us show the denominator of F is always non-zero. For $x > 0$ we have $f(x) > 0$, and $f(1-x) \geq 0$ always, so

$$
f(x) + f(1-x)) \geq f(x) > 0.
$$

For $x \leq 0$ we have $f(x) = 0$ and $(1-x) \geq 1 \Rightarrow f(1-x) > 0$, so

$$
f(x) + f(1-x) \geq f(1-x) > 0.
$$

So $f(x) + f(1-x)$ is never zero and F, being the quotient of smooth maps with non-vanishing denominator, is smooth as well. Moreover, when $x \leq 0$, $f(x) = 0$ and $f(1-x) > 0$ imply $F(x) = 0$; when $x \geq 1$, then $f(1-x) = 0$ and $f(x) > 0$, so $F(x) = f(x)/f(x) = 1$.

We also claim $F'(x) > 0$ on $(0, 1)$. Since $f'(x) = (1/x^2)e^{-1/x}$ for $x > 0$ and $f'(1-x) = -[1/(1-x)^2]e^{-1/(1-x)}$, if we take $x \in (0, 1)$ then

$$F'(x) = \frac{f'(x)[f(x)+f(1-x)] - [f'(x)+f'(1-x)]f(x)}{[f(x)+f(1-x)]^2}$$
$$= \frac{f'(x)f(1-x) - f'(1-x)f(x)}{[f(x)+f(1-x)]^2}$$
$$= \frac{f(x)f(1-x)}{[f(x)+f(1-x)]^2}\left(\frac{1}{x^2} + \frac{1}{(1-x)^2}\right) > 0.$$

Therefore F is strictly increasing, and so

$$F(x) = \frac{f(x)}{f(x)+f(1-x)} = \begin{cases} 0 & \text{if } x \le 0 \\ \dfrac{1}{1+\frac{e^{1/x}}{e^{1/(1-x)}}} & \text{if } 0 < x < 1 \\ 1 & \text{if } x \ge 1 \end{cases}.$$

The graph of F looks like the map in Fig. 12.3.

Arguing in the same way starting from the C^∞ function

$$g(x) = \begin{cases} e^{\frac{1}{x}} & \text{if } x < 0 \\ 0 & \text{if } x \ge 0 \end{cases}$$

we arrive at the symmetric map to F, i.e.

$$G(x) = \frac{g(x)}{g(x)+g(-1-x)} = \begin{cases} 0 & \text{if } x \ge 0 \\ \dfrac{1}{1+\frac{e^{-1/x}}{e^{-1/(-1-x)}}} & \text{if } -1 < x < 0 \\ 1 & \text{if } x \le -1 \end{cases}.$$

Here, $g(-1-x)$ vanishes when $x \le -1$ and $G(x) = g(x)/g(x) = 1$ for $x \le -1$. The graph of $G(x)$ is represented in Fig. 12.4.

Now shift $F(x)$ leftwards by 2 units, and $G(x)$ rightwards by 2 units. The required bump function is defined piecewise, as follows

$$\phi(x) = \begin{cases} 0 & \text{if } x \le -2 \\ F(x+2) & \text{if } -2 < x < -1 \\ 1 & \text{if } -1 \le x \le 1 \\ G(2-x) & \text{if } 1 < x < 2 \\ 0 & \text{if } x \ge 2. \end{cases} \tag{12.2.3}$$

This is undoubtedly in $C^\infty(\mathbb{R})$, and furthermore:

- $\phi(x) = 0$ if $x \in (-\infty, -2] \cup [2, \infty)$;
- $\phi(x) = 1$ if $x \in [-1, 1]$;

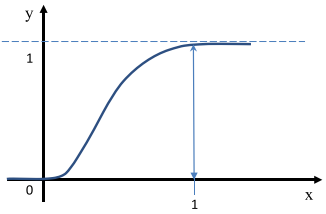


Fig. 12.3 Graph of $F(x) = \frac{f(x)}{f(x)+f(1-x)}$

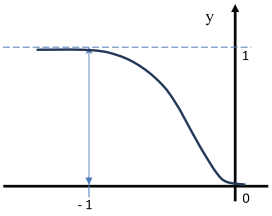


Fig. 12.4 Graph of $G(x) = \frac{g(x)}{g(x)+g(-1-x)}$

- $0 < \phi(x) < 1$ for $x \in (-2,-1)$, i.e. $\phi(x)$ is strictly increasing on $(-2,-1)$; $1 < \phi(x) < 0$ for $x \in (1,2)$, i.e. $\phi(x)$ is strictly decreasing on $(1,2)$ (Figs. 12.3 and 12.4). ★

Theorem 12.2.3 (Density of $C^\infty([a,b])$ in $L_1([a,b])$) *The space $C^\infty([a,b])$ is a dense subspace of $L_1([a,b])$.*

Proof Consider the map in Fig. 12.5, defined for any $n \in \mathbb{N}$,

$$\phi_n(x) = \begin{cases} 0 & \text{if } x \notin \left(c-\frac{2}{n}, d+\frac{2}{n}\right) \\ F\left(x-\left(c-\frac{2}{n}\right)\right) & \text{if } x \in \left(c-\frac{2}{n}, c-\frac{1}{n}\right) \\ 1 & \text{if } x \in \left[c-\frac{1}{n}, d+\frac{1}{n}\right] \\ G\left(\left(d+\frac{2}{n}\right)-x\right) & \text{if } x \in \left(d+\frac{1}{n}, d+\frac{2}{n}\right). \end{cases}$$

Each ϕ_n restricted to $[a,b]$ is of type (12.2.3), hence smooth, and moreover

$$\begin{aligned} \|\varphi_{[c,d]} - \phi_n\|_1 &= d + \frac{1}{n} - c + \frac{1}{n} - d + c + \int_{\left(c-\frac{2}{n}, c-\frac{1}{n}\right)} \phi_n \, d\ell + \int_{\left(d+\frac{1}{n}, d+\frac{2}{n}\right)} \phi_n \, d\ell \\ &< \frac{2}{n} + \frac{2}{n} = \frac{4}{n}. \end{aligned}$$

Given $\epsilon > 0$ there is an $n' \in \mathbb{N}$ such that for every $n \geq n'$

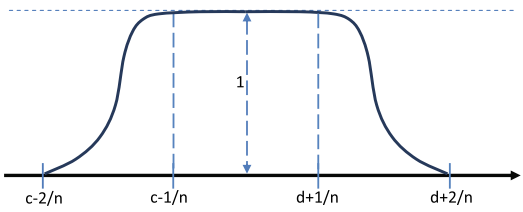


Fig. 12.5 Graph of bump function ϕ_n

$$\|\varphi_{[c,d]} - \phi_n\|_1 < \epsilon,$$

and the claim is proved. ■

Definition 12.2.2 (*Support of a map and space of continuous maps with compact support*) The support of a map $f : \mathbb{R} \mapsto \mathbb{R}$ is the set

$$\text{supp}(f) = \overline{\{x : f(x) \neq 0\}}.$$

One calls

$$C_c(\mathbb{R}) = \{f : \mathbb{R} \mapsto \mathbb{R} : f \in C(\mathbb{R}) \text{ and } \text{supp}(f) \text{ compact}\}.$$

the space of continuous maps with compact support. ◇

The set $\text{supp}(f)$ is the smallest subset of $\mathbb{R}$ outside which f vanishes.

Theorem 12.2.4 (Density of $C_c(\mathbb{R})$ in $L_1(\mathbb{R})$) $C_c(\mathbb{R})$ *is a dense subspace of* $L_1(\mathbb{R})$.

Proof A map $f \in L_1(\mathbb{R})$ is summable and hence

$$\|f\|_1 = \int_{\mathbb{R}} |f| \, d\ell < \infty.$$

Given $\epsilon > 0$, there exists an $n \in \mathbb{N}$ such that

$$\|f - f\varphi_{[-n,n]}\|_1 < \epsilon.$$

This means any $f \in L_1(\mathbb{R})$ can be approximated arbitrarily well by maps in $f \in L_1([a,b])$, i.e. summable functions with compact support. By Theorem 12.2.2 any $f \in L_1([a,b])$ can be approximated by a map of class $C([a,b]) \subseteq C_c(\mathbb{R})$. Therefore if $f \in L_1(\mathbb{R})$, given any $\epsilon > 0$ there is a continuous map g on $\mathbb{R}$ vanishing outside of some compact set, such that

$$\|f - g\|_1 < \epsilon$$

everywhere on $\mathbb{R}$. ■

Also compactly supported smooth functions are dense in $L_1(\mathbb{R})$. We indicate the class of such functions with

$$C_c^\infty(\mathbb{R}) = \big\{ f \in C^\infty(\mathbb{R}) : \operatorname{supp}(f) \text{ is compact} \big\}.$$

Theorem 12.2.5 (Density of $C_c^\infty(\mathbb{R})$ in $L_1(\mathbb{R})$) *The space $C_c^\infty(\mathbb{R})$ is a dense subspace in $L_1(\mathbb{R})$.*

Proof In analogy to Theorem 12.2.4, if $f \in L_1(\mathbb{R})$ then for $\epsilon > 0$ there is an $n \in \mathbb{N}$ satisfying

$$\| f\varphi_{[-n,n]} - f \|_1 < \epsilon.$$

Hence $f \in L_1(\mathbb{R})$ can be approximated as well as we please with maps $f \in L_1([a,b])$, i.e. summable and with compact support.

By Theorem 12.2.3 any $f \in L_1([a,b])$ can be similarly approximated by some map of class $C^\infty([a,b]) \subseteq C_c^\infty(\mathbb{R})$. Overall, if $f \in L_1(\mathbb{R})$, given $\epsilon > 0$ there exists ϕ_ϵ smooth on $\mathbb{R}$ that vanishes outside some compact set and such that $\| f - \phi_\epsilon \|_1 < \epsilon$ in $\mathbb{R}$. ■

12.3 Improper Integrals over Bounded Domains and the Lebesgue Integral

In Sect. 12.1 we established the Lebesgue integral is an extension of Riemann's integral. Here we want to see if, and under which conditions, the same is true for Riemann's so-called 'improper' integrals.

For starters we observe there exist summable functions that are not such in the improper sense. One example is the map $f : [0,1] \mapsto \mathbb{R}$ studied in Remark 12.1.1, which is unbounded in proximity of 0. The improper integral diverges, since f is not integrable in the conventional sense (as a Riemann integral) over any interval $[x,1]$, $0 < x < 1$.

That said, we will see shortly that certain non-Lebesgue summable functions still have improper integrals.

Definition 12.3.1 (*Improper integrals over bounded domains I*) Let $I \subseteq \mathbb{R}$ be an interval such that $\inf I = \alpha$, $\sup I = \beta \in \mathbb{R}$, and $f : I \mapsto \mathbb{R}$ a map defined on I except possibly for the limit point α, or β, or both. Suppose f is unbounded only near the limit point(s) of I. We say f admits improper integral over I if, for any subinterval $[a,b] \subseteq I$, $f \in \mathcal{R}([a,b])$ and the limit

$$\lim_{\substack{a \to \inf I \\ b \to \sup I}} \int_a^b f(x)\, dx = \int_\alpha^\beta f(x)\, dx$$

exists and is finite. The latter is called improper integral of f over I.

We indicate with $\mathscr{G}(I)$ the collection of 'improperly integrable' maps over I (the $\mathscr{G}$ is to convey the idea of 'generalised' Riemann integrability). $\diamond$

If by $\mathscr{G}(I)$ we intend maps $f : I \mapsto \mathbb{R}$ that are for instance unbounded only near β (and Riemann integrable over any $[\alpha, x]$, $x < \beta$), then we can show they are measurable.

Proposition 12.3.1 (Measurability of maps in $\mathscr{G}(I)$) *Consider an interval $I \subseteq \mathbb{R}$ and a map $f : I \mapsto \mathbb{R}$. Then*

$$f \in \mathscr{G}(I) \;\Rightarrow\; f \in \mathscr{M}(I, \mathscr{L}(I)).$$

Proof Let $\{[\alpha, b_n]\}_{n\geq 1}$ be an increasing sequence of closed, bounded intervals such that

$$\bigcup_{n=1}^{\infty} [\alpha, b_n] = I.$$

If E denotes the discontinuity set of f,

$$E \setminus \{\beta\} = \bigcup_{n=1}^{\infty} \big(E \cap [\alpha, b_n]\big).$$

As f is Riemann integrable on any $[\alpha, b_n]$, then $\ell\big(E \cap [\alpha, b_n]\big) = 0$ (it is enclosable) for every $n \geq 1$, and so $\ell(E) = 0$. But then f is measurable because of *4.*, Remark 10.2.1. The same applies in the case $I = (\alpha, \beta]$. ■

The above suggests it makes sense to study the Lebesgue integrability of the aforementioned type of functions.

Theorem 12.3.1 (Cauchy criterion for integrating unbounded functions) *An unbounded map f admits (improper) integral on $I \subseteq \mathbb{R}$ if and only if for any $\epsilon > 0$ there exist a point $a \in I$ (and/or $b \in I$) such that for every $x_1 < x_2 \in (\alpha, a]$ (and/or $[b, \beta)$)*

$$\left| \int_{x_1}^{x_2} f(t)\, dt \right| < \epsilon,$$

where $\alpha = \inf I$, $\beta = \sup I$.

Proof The existence of the improper integral implies the existence in $\mathbb{R}$ of the limit

$$\lim_{\substack{a\to\alpha^+ \\ b\to\beta^-}} \int_a^b f(x)\, dx = \int_\alpha^\beta f(x)\, dx.$$

Therefore $\forall\ x_1, x_2$, with $\alpha < x_1 < x_2 \leq a$ (and/or $b \leq x_1 < x_2 < \beta$) Cauchy's usual convergence criterion applies. ■

Here are sufficient conditions for improper integrals to exist.

Theorem 12.3.2 (Integrability of unbounded dominated maps) *Consider real maps f, g defined on $I \subseteq \mathbb{R}$, with $g \geq 0$ and $g \in \mathscr{G}(I)$. If $|f| \leq g$ then $f \in \mathscr{G}(I)$.*

Proof As g has improper integral, by Theorem 12.3.1 there exists $b \in I$ such that for every $x_1 < x_2 \in [b, \beta)$

$$\left| \int_{x_1}^{x_2} g(t)\ dt \right| < \epsilon,$$

for any $\epsilon > 0$ (we keep the notation where $\beta = \sup I$). Since $|f| \leq g$, then for every $x_1 < x_2 \in [b, \beta)$

$$\left| \int_{x_1}^{x_2} f(t)dt \right| \leq \int_{x_1}^{x_2} |f(t)|dt \leq \int_{x_1}^{x_2} g(t)\ dt < \epsilon. \tag{12.3.1}$$

Similar story for the infimum α of I. The sufficient implication in Cauchy's criterion then gives $f \in \mathscr{G}(I)$. ■

Next is a property relating improper integrals to Lebesgue integrals.

Theorem 12.3.3 (Improper integral of $|f|$ on I) *Given $f : I \subseteq \mathbb{R} \mapsto \mathbb{R}$, if $|f| \in \mathscr{G}(I)$ then $f \in \mathscr{G}(I)$.*

Proof Without loss of generality we may assume $f : I \mapsto \mathbb{R}$ is unbounded near the limit point $\sup I = \beta$ of I.

Take $g = |f|$ in the previous theorem and remember that for $\alpha \leq x < \beta$

$$\left| \int_{\alpha}^{x} f(t)\ dt \right| \leq \int_{\alpha}^{x} |f(t)|\ dt.$$

By this and (12.3.1),

$$\left| \int_{\alpha}^{\beta} f(x)\ dx \right| \leq \int_{\alpha}^{\beta} |f(x)|\ dx.$$

■

We shall prove that only those maps $f \in \mathscr{G}(I)$ for which $|f| \in \mathscr{G}(I)$ are Lebesgue integrable. Equivalently, any improperly integrable map whose absolute value is not improperly integrable cannot be Lebesgue integrable.

Theorem 12.3.4 (Relationship between $\mathscr{G}(I)$ and $L_1(I)$) *Consider $I \subseteq \mathbb{R}$ and a real function f. Then $f \in L_1(I)$ only if $|f| \in \mathscr{G}(I)$.*

Proof Since $|f|_{[a_n,b_n]} \in \mathcal{R}([a_n, b_n])$ for every n,

$$\int_{a_n}^{b_n} |f(x)|dx = \int_{[a_n,b_n]} |f|\, d\ell.$$

By monotone convergence it follows that

$$\begin{aligned}\int_I |f|\, d\ell &= \lim_{n\to\infty} \int_I |f|\varphi_{[a_n,b_n]}\, d\ell = \lim_{n\to\infty} \int_{[a_n,b_n]} |f|\, d\ell \\ &= \lim_{n\to\infty} \int_{a_n}^{b_n} |f(x)|dx = \int_\alpha^\beta |f(x)|dx.\end{aligned}$$

By assumption $\int_\alpha^\beta |f(x)|dx < \infty$, so $\int_I |f|\, d\ell < \infty$ hence $f \in L_1(I)$.

In case $|f| \notin \mathscr{G}(I)$, without loss of generality we may suppose $|f|$ is unbounded at the limit point $\sup I = \beta \in \mathbb{R}$. For any $k > 0$ there exists then $b_n \in [\alpha, \beta)$ such that

$$\int_\alpha^{b_n} |f(x)|dx > k,$$

so for every $n \in \mathbb{N}$

$$\int_{[\alpha,b_n]} |f(x)|d\ell = \int_\alpha^{b_n} |f(x)|dx > k.$$

Hence as $n \to \infty$

$$\int_I |f(x)|d\ell = \int_\alpha^\beta |f(x)|dx \geq \int_\alpha^{b_n} |f(x)|dx > k.$$

But $k > 0$ is arbitrary, so the claim follows. ■

Now we describe an unbounded integrable function whose absolute value is not integrable.

Example 12.3.1 (*A non-summable map* $f \in \mathscr{G}(I)$) Consider $f : (0, 1/\pi] \mapsto \mathbb{R}$,

$$f(x) = \frac{1}{x} \sin \frac{1}{x}, \quad x \in (0, 1/\pi].$$

We shall first prove it has (improper) integral over $[0, 1/\pi]$. As the unique discontinuity is 0, we just need to prove

$$\lim_{x\to 0^+} \int_x^{1/\pi} \frac{1}{t} \sin \frac{1}{t} dt$$

exists and is finite. Multiply and divide by t to obtain

$$\int_x^{1/\pi} \frac{1}{t} \sin \frac{1}{t} dt = \int_x^{1/\pi} t \frac{1}{t^2} \sin \frac{1}{t} dt. \tag{12.3.2}$$

Now rename $f(t) = t$ and $g'(t) = \frac{1}{t^2} \sin \frac{1}{t} \Rightarrow g(t) = \cos \frac{1}{t}$ in (12.3.2), so integrating by parts gives

$$\int_x^{1/\pi} \frac{1}{t} \sin \frac{1}{t} dt = \left[t \cos \frac{1}{t} \right]_x^{1/\pi} - \int_x^{1/\pi} \cos \frac{1}{t} dt.$$

Hence

$$\lim_{x \to 0^+} \left[t \cos \frac{1}{t} \right]_x^{1/\pi} = \lim_{x \to 0^+} \left(\frac{1}{\pi} \cos \pi - x \cos \frac{1}{x} \right) = -\frac{1}{\pi}.$$

Moreover, $\forall x \in [0, 1/\pi]$

$$\left| \cos \frac{1}{x} \right| \leq 1,$$

and Theorem 12.3.2 tells $\cos \frac{1}{x}$ can be integrated over $[0, 1/\pi]$, i.e.

$$\lim_{x \to 0^+} \int_x^{1/\pi} \cos \frac{1}{t} \, dt$$

(exists and) is finite, as claimed.

Now we show this map is not Lebesgue integrable on $[0, 1/\pi]$. For that it suffices to see that on $[0, 1/\pi]$ the integral of

$$|f(x)| = \left| \frac{1}{x} \sin \frac{1}{x} \right|$$

diverges, namely that

$$\lim_{p \to \infty} \int_{1/p\pi}^{1/\pi} \frac{1}{x} \left| \sin \frac{1}{x} \right| dx = \infty. \tag{12.3.3}$$

First of all, the improper integral can be written as an alternated series

$$\mathscr{G} \int_0^{1/\pi} |f(x)| \, dx = \lim_{p \to \infty} \int_{1/p\pi}^{1/\pi} \frac{1}{x} \left| \sin \frac{1}{x} \right| dx = \lim_{p \to \infty} \left[\sum_{n=1}^{p-1} \int_{1/(n+1)\pi}^{1/n\pi} \frac{1}{x} \left| \sin \frac{1}{x} \right| dx \right].$$

Substituting $1/x = t$ we obtain

$$\begin{aligned}\mathscr{G}\int_0^{1/\pi} |f(x)|\,dx &= \lim_{p\to\infty}\left[\sum_{n=1}^{p-1}\int_{1/(n+1)\pi}^{1/n\pi}\frac{1}{x}\left|\sin\frac{1}{x}\right|dx\right]\\ &= \lim_{p\to\infty}\left[\sum_{n=1}^{p-1}\int_{(n+1)\pi}^{n\pi}-\frac{1}{t^2}\cdot t|\sin t|dt\right]\\ &= \lim_{p\to\infty}\left[\sum_{n=1}^{p-1}\int_{n\pi}^{(n+1)\pi}\frac{|\sin t|}{t}dt\right].\end{aligned}$$

As $1/t > 1/(n+1)\pi$, we have

$$\begin{aligned}\mathscr{G}\int_0^{1/\pi} |f(x)|\,dx &= \lim_{p\to\infty}\left[\sum_{n=1}^{p-1}\int_{n\pi}^{(n+1)\pi}\frac{|\sin t|}{t}dt\right]\\ &> \lim_{p\to\infty}\left[\sum_{n=1}^{p-1}\frac{1}{(n+1)\pi}\int_{n\pi}^{(n+1)\pi}|\sin t|dt\right]\\ &= \lim_{p\to\infty}\left[\sum_{n=1}^{p-1}\frac{1}{(n+1)\pi}\left|\cos(t)\right|_{n\pi}^{(n+1)\pi}\right]\\ &= \lim_{p\to\infty}\left[\sum_{n=1}^{p-1}\frac{1}{(n+1)\pi}(1+1)\right]\\ &= \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{1}{(n+1)}.\end{aligned}$$

But $\sum_{n=1}^{\infty} 1/(n+1)$ is harmonic, hence divergent. Summing up, the (improper) integral of $|f|$ over $[0, 1/\pi]$ diverges.

This example shows explicitly that even when an unbounded f is integrable, $|f|$ might not be; and in case $|f| \notin \mathscr{G}(I)$, then by Theorem 12.3.4 $f \notin L_1(I)$. ★

Consequence of Theorem 10.2.6, Proposition 11.3.2 and Theorem 12.3.4 is the following fact.

Proposition 12.3.2 (Almost everywhere equality to maps $|f| \in \mathscr{G}(I)$) *Let $f, g : I \mapsto \mathbb{R}$ be such that $f = g$ ℓ-almost everywhere on I. If $|f| \in \mathscr{G}(I)$ then $g \in L_1(I)$, and furthermore*

$$\int_I g\,d\ell = \mathscr{G}\int_\alpha^\beta f(x)\,dx.$$

■

12.4 Improper Integrals over Unbounded Domains and the Lebesgue Integral

Definition 12.4.1 (*Improper integrals over unbounded real intervals*) Let $I^* \subseteq \mathbb{R}$ denote an unbounded interval, where either: $\inf I^* = -\infty$ and $\sup I^* \in [a, \infty)$, or $\sup I^* = \infty$ and $\inf I^* \in (-\infty, b]$, or $\sup I^* = \infty$ and $\inf I^* = -\infty$. Suppose $f \in \mathcal{R}([a, b])$ or $f \in \mathcal{G}([a, b])$ for every subinterval $[a, b] \subseteq I^*$, and that the limit

$$\lim_{\substack{a \to \inf I^* \\ b \to \sup I^*}} \int_a^b f(x)\, dx = \int_{\inf I^*}^{\sup I^*} f(x)\, dx$$

exists finite. Then we call the above expression (improper) integral of f over I^*. As before, $\mathcal{G}(I^*)$ will indicate the collection of such functions f. ◇

To compare the Lebesgue integral with the (improper) integral over an unbounded domain we can argue as in Sect. 12.3.

In particular, there exist Lebesgue integrable functions that have no improper integral in the above sense. One such is $f : [a, \infty) \mapsto \mathbb{R}$,

$$f(x) = \begin{cases} 0 & \text{if } x \in \big([a, \infty) \setminus \mathbb{Q}\big) \\ x & \text{if } x \in \big([a, \infty) \cap \mathbb{Q}\big) \end{cases},$$

whose Lebesgue integral is zero.

If by $\mathcal{G}(I^*)$ we understand functions $f : [a, \infty) \mapsto \mathbb{R}$ that are Riemann integrable on every domain $[a, x]$, $x > a$ (in the conventional sense or the improper one), it is easy to prove that any $f \in \mathcal{G}(I^*)$ is measurable.

Proposition 12.4.1 (Measurability of functions in $\mathcal{G}(I^*)$) *If $f : I^* \subseteq \mathbb{R} \mapsto \mathbb{R}$ is in $\mathcal{G}(I^*)$ then $f \in \mathcal{M}(I^*, \mathcal{L}(I^*))$.* ■

Mimicking the previous section's proofs, from Definition 12.4.1 we can easily deduce the following facts.

Theorem 12.4.1 (Cauchy criterion for integrating over unbounded intervals) *A map f admits (improper) integral on an unbounded interval $I^* \subseteq \mathbb{R}$ if and only if for any $\epsilon > 0$ there exists a point a (in case $\inf I^* = -\infty$), and/or $b \in \mathbb{R}$ (if $\sup I^* = \infty$), such that for every $x_1 < x_2 \in (-\infty, a]$ (or $[b, \infty)$)*

$$\left| \int_{x_1}^{x_2} f(t)\, dt \right| < \epsilon.$$

■

We also have sufficient conditions for the existence of integrals on I^*.

Theorem 12.4.2 (Integrability of dominated maps on unbounded intervals) *Consider real maps f, g defined on $I^* \subseteq \mathbb{R}$, with $g \geq 0$ and $g \in \mathcal{G}(I^*)$. If $|f| \leq g$ then $f \in \mathcal{G}(I^*)$.* ■

Theorem 12.4.3 (Improper integral of $|f|$ on I^*) *Given $f: I^* \subseteq \mathbb{R} \mapsto \mathbb{R}$, if $|f| \in \mathscr{G}(I^*)$ then $f \in \mathscr{G}(I^*)$.* ■

Regarding Lebesgue integration on unbounded domains:

Theorem 12.4.4 (Relationship between $\mathscr{G}(I^*)$ and $L_1(I^*)$) *Consider $I^* \subseteq \mathbb{R}$ and a real map f. Then $f \in L_1(I^*)$ only if $|f| \in \mathscr{G}(I^*)$.* ■

In complete analogy to $\mathscr{G}(I)$, any integrable f over an unbounded interval I^* might not be summable if $|f|$ cannot be integrated over I^*. Here is an explicit example.

Example 12.4.1 (*A non-summable map $f \in \mathscr{G}(I^*)$*) Consider $f : [\pi, \infty) \mapsto \mathbb{R}$,

$$f(x) = \frac{\sin x}{x}, \quad x \geq \pi. \tag{12.4.1}$$

To prove it is integrable over $[\pi, \infty)$ it it enough to show

$$\lim_{n\to\infty} \int_{\pi}^{n\pi} \frac{\sin x}{x} dx$$

exists in $\mathbb{R}$, whilst to prove $|f(x)| = |\sin x/x|$ is not integrable over $[\pi, \infty)$ we just compute

$$\lim_{n\to\infty} \int_{\pi}^{n\pi} \frac{1}{x} |\sin(x)| dx = \infty.$$

Putting $1/x = t$ we obtain

$$\int_{\pi}^{n\pi} \frac{\sin(x)}{x} dx = \int_{1/\pi}^{1/n\pi} t\left(-\frac{1}{t^2}\right) \sin\left(\frac{1}{t}\right) dt = \int_{1/n\pi}^{1/\pi} \frac{1}{t} \sin\left(\frac{1}{t}\right) dt;$$

$$\int_{\pi}^{n\pi} \frac{1}{x} |\sin(x)| dx = \int_{1/n\pi}^{1/\pi} \frac{1}{t} \left|\sin\left(\frac{1}{t}\right)\right| dt.$$

As $n \to \infty$ we have (cf. Example 12.3.1)

$$\lim_{n\to\infty} \int_{\pi}^{n\pi} \frac{\sin(x)}{x} dx = \lim_{n\to\infty} \int_{1/n\pi}^{1/\pi} \frac{1}{t} \sin\left(\frac{1}{t}\right) dt \in \mathbb{R};$$

$$\lim_{n\to\infty} \int_{\pi}^{n\pi} \frac{1}{x} |\sin(x)| dx = \lim_{n\to\infty} \int_{1/n\pi}^{1/\pi} \frac{1}{t} \left|\sin\left(\frac{1}{t}\right)\right| dt = \infty.$$

The punchline is that $f(x)$ is integrable over $[\pi, \infty)$ whereas $|f(x)|$ is not. Hence $f(x)$ is not Lebesgue integrable on $[\pi, \infty)$. ★

To finish, one last result similar to what we saw at the end of the previous section.

Proposition 12.4.2 (Almost everywhere equality to maps $|f| \in \mathscr{G}(I^*)$) *Let $f, g : I^* \mapsto \mathbb{R}$ be such that $f = g$ ℓ-almost everywhere on I^*. If $|f| \in \mathscr{G}(I^*)$ then $g \in L_1(I^*)$, and furthermore*

$$\int_{I^*} g \, d\ell = \mathscr{G} \int_{\inf I^*}^{\sup I^*} f(x) \, dx.$$

■

Part VI
Fundamental theorems of integral calculus on $\mathbb{R}$

Introduction

The Riemann integral can be considered an evolution of Cauchy's integral, in that certain functions that were not integrable according to Cauchy became integrable in Riemann's theory. At the same time, alas, in the new framework integration is no longer the inverse operation to differentiation. As a matter of fact, there exist Riemann integrable functions f whose integral map $F(x) = \int_a^x f(t)\,dt$ is not differentiable (not even on one side only) at certain points. Thus the fundamental theorem of calculus, in the version for continuous maps proved by Cauchy, loses its status of calculus' highest pinnacle and becomes a mere special case of a much bigger picture.

Realising this fact led mathematicians to view differentiation and integration with a more critical eye, and prompted them to explore the depths of the relationship between continuity, differentiability and integrability.

Hermann Hankel was one among many academics trying to unravel the connection differentiability–continuity. In 1871 Hankel used Riemann's integral function[1] $R(t)$ to define the map

$$G(x) = k + \int_a^x R(t)dt,$$

which is continuous everywhere, but not differentiable at infinitely many points. The example raised much incredulity among the scientific community, for it went against the conventional wisdom that continuity implied differentiability but at a finite number of points.

Many textbooks of the time, to prove that a continuous map $f(x)$ on some interval had to be differentiable, used geometrically-flavoured arguments involving tangents to the curve representing $f(x)$, without fathoming that there do exist continuous maps not allowing for a pictorial description of that kind.

It was Karl Weierstraß who cleared away all doubts when he constructed a function of one real variable that is everywhere continuous but never differentiable.

[1] Here $R(t)$ will denote the function Riemann employed to show his notion of integral was sophisticated enough to allow the integration of maps with infinitely many discontinuities.

Liberating the concept of derivative from continuity allowed to study differentiation independently, and thus extend its scope to discontinuous functions as well. For this reason Ulisse Dini, to make derivatives and integrals both rigorous and more general, and encompass maps with dense discontinuity sets (which he called "*discontinuous dot-functions*"[2]) introduced in the early 1880s the so-called Dini derivatives.

These numbers were created to extend the notion of derivative to pathological functions. Explicitly, the (upper and lower) right Dini derivatives of F at a point $x \in [a, b)$ are

$$D^+F(x) = \limsup_{h\to 0^+} \frac{F(x+h) - F(x)}{h} \quad \text{and} \quad D_+F(x) = \liminf_{h\to 0^+} \frac{F(x+h) - F(x)}{h};$$

the (upper and lower) left Dini derivatives at $x \in (a, b]$ are

$$D^-F(x) = \limsup_{h\to 0^+} \frac{F(x) - F(x-h)}{h} \quad \text{and} \quad D_-F(x) = \liminf_{h\to 0^+} \frac{F(x) - F(x-h)}{h}.$$

Note that $D^+F(x) \geq D_+F(x)$ and $D^-F(x) \geq D_-F(x)$. When $D^+F(x) = D_+F(x) = D^-F(x) = D_-F(x) \in \mathbb{R}$ the map F is differentiable at x and its derivative is $f(x)$, precisely the common value of the four Dini derivatives.

Among other things, these new concepts allowed Dini to state the fundamental theorem of calculus for the Riemann integral in the following form: if F is continuous and one of Dini's derivatives DF is Riemann integrable, then the integral map $F(x) = \int_a^x DF(t)d(t)$ has all four Dini derivatives bounded and integrable, and moreover $\int_a^b DF(x)dx = F(b) - F(a)$. Dini, moreover, explicitly emphasised the necessity of assuming the integrability of DF, and conjectured the existence non-constant, continuous primitives F that become constant over a suitable subinterval of any given interval. This would imply DF is not integrable, because we would not be able to enclose its discontinuities using a finite number of arbitrarily small intervals. By the way, the surmise was proved in 1881 by Vito Volterra, a student of Dini at Scuola Normale Superiore in Pisa, who gave an explicit example (see p.xxx ff.).

The existence of continuous maps with no derivative, and the ensuing demise of the fundamental theorem of Cauchy's integral calculus, persuaded many mathematicians, most notably Lebesgue, to investigate the relationship between integrals and primitives. The first result of Lebesgue's theory is that

"If $f \in L_1[a, b]$ the integral map

$$F(x) = \int_{[a,x]} f(t)\ell(dt)$$

is continuous and has derivative $F'(x) = f(x)$ a.e. on $[a, b]$."

[2] '*funzioni punteggiate discontinue*' in Italian.

Lebesgue also proved, conversely,

"If $F : [a, b] \mapsto \mathbb{R}$ has bounded derivative f, then f is Lebesgue integrable and the equality

$$F(x) - F(a) = \int_{[a,x]} f(t)\ell(dt)$$

holds everywhere on $[a, b]$."

These results entitled him to declare in 1902 that the problem of integral calculus, conjectured by Dini and proved by Volterra within Riemann's set-up, had been completely solved in the case of bounded derivatives.

However, things get more complicated if the derivative f is not bounded, for then f might not be integrable. Lebesgue tried to pin down the features a primitive F should have in order for f to be well defined almost everywhere on $[a, b]$ and integrable on the entire interval. Restricting to the case where one Dini derivative of F is almost everywhere finite, Lebesgue showed that F must have bounded variation. It is at this point that the notion of bounded variation, for which we will provide a brief account, entered the stage.

The idea was elaborated by Jordan for other reasons, namely in relationship to conditions granting that a function $F(x)$ has a Fourier series converging to $F(x)$. Jordan proved a sufficient condition for convergence in terms of a function with bounded variation.

To define bounded variation he took a map $F : [a, b] \mapsto \mathbb{R}$ and fixed a finite partition $P = \{a = x_0, x_1, \ldots, x_{n-1}, x_n = b\}$ of $[a, b]$. He then called variation of $F(x)$ over $[a, b]$ with respect to $P \in \mathscr{P}$ the quantity

$$V_a^b(P, F) = \sum_{i=1}^{n} |F(x_i) - F(x_{i-1})|.$$

The difference $|F(x_i) - F(x_{i-1})|$, $i = 1, \ldots, n$, represents the largest variation of F over $[x_{i-1}, x_i]$. If $\sum_{i=1}^{n} |F(x_i) - F(x_{i-1})| = k$, the set of values $\{k\}_{P\in\mathscr{P}}$ can, as P varies, have either finite or infinite least upper bound. Let us call total variation of F on $[a, b]$ the supremum. When that is finite we say F has bounded variation on $[a, b]$:

$$V_a^b(F) = \sup_{P\in\mathscr{P}} \left\{ V_a^b(P, F) : P \text{ partition of } [a, b] \right\} \leq \infty.$$

A function with bounded variation is bounded, whereas there exist bounded maps whose variation is not bounded. To see this just take the Dirichlet function $f : [0, 1] \mapsto \mathbb{R}$: albeit bounded, since $f(x) \in \{0, 1\}$, it has oscillations of width 1 for every $x \in \mathbb{Q} \cap [0, 1]$. Therefore functions with bounded variation form a proper

subfamily of bounded maps. Moreover, there exist discontinuous maps with bounded variation, such as discontinuous monotone functions, and there also are continuous maps with unbounded variation. But going back to the issue, Jordan proved that a function with bounded variation on $[a, b]$ can be written as difference of two increasing functions (*Cours d'analyse de l'École Polytechnique*, first edition in three volumes published in 1882-1887).

On the matter Lebesgue observed that if $f \in L_1$ is positive, the integral map

$$F(x) = \int_{[a,x]} f(t)\ell(dt), \quad x \in [a, b]$$

is increasing. Moreover, since a summable function f is the difference of positive summable functions f^+, f^-, any integral map is the difference of increasing functions

$$F(x) = F^+(x) - F^-(x).$$

Based on this, in 1904 Lebesgue established the following:

if $F : [a, b] \mapsto \mathbb{R}$ has bounded variation, then it is differentiable a.e. on $[a, b]$, and f is summable on $[a, b]$. He stopped short of saying

$$|F(x) - F(a)| = \int_{[a,x]} |f(t)|\ell(dt)$$

for every $x \in [a, b]$, because in that case the difference of the two sides would be a monotone map G with bounded variation and zero derivative ℓ-almost everywhere on $[a, x]$.

The point is that Lebesgue viewed $F(x) = \int_{[a,x]} f\, d\ell$ (clearly for $f \geq 0$) not as a function of x, but rather as being defined on an interval, or more generally as a set function $\mu_F : \mathcal{S}(X) \mapsto [0, \infty]$, where

$$\mu_F(E) = \int_E f\, d\ell, \quad E \in \mathcal{S}(X)$$

designates a measure on the measurable space $(X, \mathcal{S}(X))$. In practice μ_F is a weighted measure (weighted by the non-negative measurable map f) determined by the formula

$$d\mu_F = f d\ell.$$

Now the question is: is there a way to characterise, among all set functions $\mu_F([a, b])$, indefinite integrals that satisfy

$$F(x) - F(a) = \int_{[a,x]} f(t)\ell(dt)$$

for every $x \in [a, b]$? The answer is yes, there is indeed, and it relies on two measure-theoretical notions that are central to the whole of real analysis: complete additivity and absolute continuity with respect to a given measure. The punchline is this: among functions with bounded variation, the only ones satisfying

$$F(x) - F(a) = \int_{[a,x]} f(t)\ell(dt)$$

must be both

• completely additive – in the words of De la Vallée Poussin:

"Given $E \subseteq X$ measurable, a set function $\mu_F(E)$ is completely additive if, irrespective of how one partitions E in a finite/countable number of measurable subsets E_i,

$$\mu_F(E) = \mu_F\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mu_F(E_i)''$$

• absolutely continuous, as defined by the mathematician Giuseppe Vitali thus:

"Given $E \subseteq X$ measurable, a set function $\mu_F(E)$ is absolutely continuous with respect to measure ℓ if $\mu_F(E)$ tends to 0 as $\ell(E)$ tends to 0."

Lebesgue went on to prove that

"If a set function with bounded variation $\mu_F([a, b])$ is absolutely continuous with respect ℓ and completely additive, then it is differentiable a.e. on $[a, b]$ ($F' = f$ a.e. on $[a, b]$), its derivative f is integrable on $[a, b]$, and for every $x \in [a, b]$

$$F(x) - F(a) = \int_{[a,x]} f(t)\ell(dt).''$$

Hence, F is the indefinite integral of f, and, in turn, f is the derivative of F a.e. on $[a, b]$. Where the derivative f is not defined it can take any value, since the set on which f does not exist has zero measure. The main tool in the proof of this important result is Vitali's covering theorem.

Once we have found a set function with bounded variation $\mu_F(E)$ satisfying the above conditions, we would like to know how to identify the f with indefinite integral $\mu_F(E)$. The procedure for finding f starting from $\mu_F(E)$ generalises the method used to differentiate. Suppose we wish to find f at a point $x_0 \in E$. We compute μ_F on a ball $B(x_0, r)$ centred at x_0, the measure $\ell(B(x_0, r))$ and then the ratio

$$\frac{\mu_F(B(x_0, r))}{\ell(B(x_0, r))} = \frac{\int_{B(x_0,r)} f d\ell}{\ell(B(x_0, r))}.$$

From the complete additivity and absolute continuity of μ_F with respect to ℓ, by letting $\ell(B(x_0, r)) \to 0$ the ratio converges to the value of the non-negative, measurable map f at x_0, where f is precisely the derivative at x_0 of $\mu_F(E)$:

$$f(x_0) = \lim_{r \to 0} \frac{\mu_F(B(x_0, r))}{\ell(B(x_0, r))} = \frac{d\mu_F}{d\ell}(x_0).$$

Clearly, in case the limit does not exist one has to resort to the Dini derivatives, which generalise the derivative. This is not the right place to discuss the existence of the derivative f of μ_F, which requires tools – such as the Radon-Nikodym theorem – that will be dealt with appropriately in Volume Two.

Despite the enormous advancements made in the pursuit of primitives, it must be said that the Lebesgue integral does not settle the matter once and for all. There are situations in which f is neither Riemann integrable nor Lebesgue integrable, and yet $\mu_F(E)$ exists and has f as derivative ℓ-almost everywhere on E. To find these primitives we need a novel idea, and a broader concept of integral. The new integral was fleshed out by Denjoy and Perron, who independently developed the same theory, starting from separate premises and different viewpoints. The technique devised by Denjoy, called *complete totalisation*, is constructive: it starts from the Lebesgue integral, and defines a hierarchy (finite or transfinite) of operations meant to determine explicitly the primitive of any derivative. Complete totalisation is the inverse process to differentiation in the aforementioned generalised sense. Denjoy's approach combines Lebesgue's analysis with the transfinite methods used by Baire, and as it happens, it is completely analogous to the method of Baire's 1899 PhD thesis.

Chapter 13
Functions with Bounded Variation and Absolutely Continuous Functions



13.1 Functions with Bounded Variation

Definition 13.1.1 (*Functions with bounded variation*) Given $f : [a, b] \mapsto \mathbb{R}$ and a partition

$$P = \{a = x_0 < x_1 < \cdots < x_n = b\}$$

of $[a, b]$, the variation of f over $[a, b]$, with respect to $P \in \mathscr{P}$, is the number

$$V_a^b(P, f) = \sum_{i=1}^{n} |f(x_i) - f(x_{i-1})|.$$

We call total variation of f over $[a, b]$ the supremum of all variations of f over the possible partitions P of $[a, b]$:

$$V_a^b(f) = \sup_{P \in \mathscr{P}} \left\{ V_a^b(P, f) : P \text{ partition of } [a, b] \right\} \leq \infty.$$

We say f has bounded variation over $[a, b]$ whenever its total variation over $[a, b]$ is finite:

$$V_a^b(f) < \infty.$$

The total variation of $f : \mathbb{R} \mapsto \mathbb{R}$ over $\mathbb{R}$ is the quantity

$$V(f) = \sup_{-\infty < a < b < \infty} \left\{ V_a^b(f) : a < b, \ a \text{ and } b \in \mathbb{R} \right\} \leq \infty,$$

and f is said to have bounded variation if the above is a number:

$$V(f) < \infty. \qquad \diamond$$

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Remark 13.1.1

1. If f has bounded variation on $\mathbb{R}$, then $V_a^b(f) < \infty$ for every $[a, b] \subseteq \mathbb{R}$. The opposite is false. For instance $f(x) = \sin(x)$ has bounded variation over any $[a, b] \subseteq \mathbb{R}$, but not over the entire real line. Similarly, if f has bounded variation on $[a, b]$, it has bounded variation on any subinterval $[c, d] \subseteq [a, b]$.
2. A map f is constant iff it has bounded variation and its total variation is 0.
3. Monotone maps $f : [a, b] \mapsto \mathbb{R}$ have bounded variation:

$$V_a^b(f) = |f(b) - f(a)|.$$

   The Vitali–Cantor map f, for example, has $V_0^1 f(x) = 1$.
4. A monotone function $f : \mathbb{R} \mapsto \mathbb{R}$ has bounded variation iff it is bounded, in which case

$$V(f) = |f(\infty) - f(-\infty)| < \infty.$$

5. Any function with bounded variation, $f : [a, b] \mapsto \mathbb{R}$ or $f : \mathbb{R} \mapsto \mathbb{R}$, is bounded. In the former case, $\forall x \in [a, b]$

$$\begin{aligned} |f(x)| - |f(a)| &\leq |f(x) - f(a)| \\ &\leq |f(x) - f(a)| + |f(b) - f(x)| \\ &\leq \sum_{i=1}^{n} |f(x_i) - f(x_{i-1})| = V_a^b(P, f) \\ &\leq \sup_{P \in \mathscr{P}} \left\{ V_a^b(P, f) \right\} = V_a^b(f) < \infty; \end{aligned}$$

   hence $|f(x)| \leq |f(a)| + V_a^b(f)$. Similarly in the latter case: $\forall x \in \mathbb{R}$

$$\begin{aligned} |f(x)| - |f(-\infty)| &\leq |f(x) - f(-\infty)| \\ &\leq |f(x) - f(-\infty)| + |f(\infty) - f(x)| \\ &\leq \sup_{-\infty < a < b < \infty} \left\{ V_a^b(f) \right\} = V(f) < \infty. \end{aligned}$$

   and so $|f(x)| \leq |f(-\infty)| + V(f)$.
6. Lipschitz functions $f : [a, b] \mapsto \mathbb{R}$ have bounded variation: since for any $x', x'' \in [a, b]$ there exists k such that

$$|f(x') - f(x'')| \leq k|x' - x''|,$$

   it follows

$$\sum_{i=1}^{n} |f(x_i) - f(x_{i-1})| \le k(b-a), \quad \forall P \in \mathscr{P}$$
$$\Rightarrow \sup_{P\in\mathscr{P}} \left\{V_a^b(P, f)\right\} \le k(b-a)$$
$$\Rightarrow V_a^b(f) \le k(b-a) < \infty.$$

7. Functions with bounded derivative on $[a, b]$ have bounded variation: by the mean-value theorem for any $x < z < y$ in $[a, b]$

$$f(x) - f(y) = f'(z)(x - y),$$

so f is Lipschitz.
Example: $f : [0, 1] \mapsto \mathbb{R}$ given by

$$f(x) = \begin{cases} x^2 \sin \dfrac{1}{x} & \text{if } x \in (0, 1] \\ 0 & \text{if } x = 0. \end{cases}$$

The derivative

$$f'(x) = 2x \cdot \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right)$$

at 0^+ equals

$$f'(0^+) = \lim_{x\to 0^+} \frac{f(0+x) - f(0)}{x - 0} = \lim_{x\to 0^+} \frac{f(x) - 0}{x}$$
$$= \lim_{x\to 0^+} \frac{x^2 \cdot \sin\left(\frac{1}{x}\right)}{x} = \lim_{x\to 0^+} x \sin\left(\frac{1}{x}\right) = 0,$$

and for all $x \in [0, 1]$

$$|f'(x)| = \left|2x \cdot \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right)\right| \le \left|2x \cdot \sin\left(\frac{1}{x}\right)\right| + \left|\cos\left(\frac{1}{x}\right)\right| < 3.$$

We claim the total variation on $[0, 1]$ is finite, despite the graph curve oscillates infinitely many times over the unit interval. Fix $n \in \mathbb{N}$ and consider the partition

$$P_n = \{0\} \cup \left\{\frac{1}{(n-k)+1}\right\}_{k=1}^{n},$$

i.e.

$$x_0 = 0; x_1 = \frac{1}{n}; x_2 = \frac{1}{n-1}; \ldots; x_{k-1} = \frac{1}{n-k+2}; x_k = \frac{1}{n-k+1} \ldots; x_n = 1.$$

For any $x \in [0, 1]$ we have

$$f(x_k) = x_k^2 \sin\left(1/x_k\right) \leq \begin{cases} x_k^2(-1)^{n-k} = x_k^2 & \text{if } (n-k) \text{ is even} \\ x_k^2(-1)^{n-k} = -x_k^2 & \text{if } (n-k) \text{ is odd.} \end{cases}$$

Therefore

$$\begin{aligned} \left|f(x_k) - f(x_{k-1})\right| &= \left|x_k^2 \cdot \sin\left(\frac{1}{x_k}\right) - x_{k-1}^2 \cdot \sin\left(\frac{1}{x_{k-1}}\right)\right| \\ &\leq \left|x_k^2(-1)^{n-k} - x_{k-1}^2(-1)^{n-k+1}\right| \\ &= \left|(-1)^{n-k}(x_k^2 + x_{k-1}^2)\right| \\ &\leq 2x_k^2 = 2\left[\frac{1}{n-k+1}\right]^2, \end{aligned}$$

so the total variation of f over $[0, 1]$ with respect to P_n equals

$$\sum_{k=1}^{n} \left|f(x_k) - f(x_{k-1})\right| \leq 2\sum_{k=1}^{n}\left[\frac{1}{n-k+1}\right]^2 = 2\left(1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{n^2}\right).$$

The sum within brackets is the partial sum of a series that converges to $\pi^2/6$, so $V_0^1(f) \leq \pi^2/3$. In a similar fashion one can prove

$$f(x) = \begin{cases} x^2 \cos\dfrac{1}{x} & \text{if } x \in (0, 1] \\ 0 & \text{if } x = 0 \end{cases}$$

has bounded variation.

8. Continuous maps do not necessarily have bounded variation.
The map

$$f(x) = \begin{cases} x \sin\dfrac{1}{x} & \text{if } x \in (0, 2/\pi] \\ 0 & \text{if } x = 0; \end{cases}$$

is continuous on $[0, 2/\pi]$, but has unbounded derivative

$$f'(x) = \sin\left(\frac{1}{x}\right) - \frac{1}{x} \cdot \cos\left(\frac{1}{x}\right).$$

The claim is that its total variation is infinite. Fix $n \in \mathbb{N}$ and the following partition of $[0, 2/\pi]$:

$$P_n = \{0\} \cup \left\{ \frac{1}{(n-k+1/2)\pi} \right\}_{k=1}^{n}.$$

From

$$\begin{aligned} |f(x_k) - f(x_{k-1})| &= \left| x_k \cdot \sin\left(\frac{1}{x_k}\right) - x_{k-1} \cdot \sin\left(\frac{1}{x_{k-1}}\right) \right| \\ &= \left| x_k(-1)^{n-k} - x_{k-1}(-1)^{n-k+1} \right| \\ &= \left| (-1)^{n-k}(x_k + x_{k-1}) \right| \geq x_k > \frac{1}{\pi}\left(\frac{1}{n-k+1}\right) \end{aligned}$$

the total variation of f on $[0, 2/\pi]$ (with respect to P_n) is

$$\sum_{k=1}^{n} |f(x_k) - f(x_{k-1})| \geq \frac{1}{\pi} \sum_{k=1}^{n} \frac{1}{\left[(n-k)+1\right]} = \frac{1}{\pi}\left(1 + \frac{1}{2} + \cdots + \frac{1}{n}\right).$$

The term in brackets is the partial sum of the divergent harmonic series, so

$$V_0^{2/\pi}(f) = \sup_{P_n \in \mathscr{P}} \left\{ V_0^{2/\pi}(P_n, f) : P_n \text{ partition of } [0, 2/\pi] \right\} = \infty.$$

A similar argument applies to

$$f(x) = \begin{cases} x \cos \dfrac{1}{x} & \text{if } x \in (0, 1/\pi] \\ 0 & \text{if } x = 0. \end{cases}$$

Given $n \in \mathbb{N}$ and

$$P_n = \{0\} \cup \left\{ \frac{1}{(n-k+1)\pi} \right\}_{k=1}^{n},$$

we have

$$\begin{aligned} |f(x_k) - f(x_{k-1})| &= \left| x_k \cdot \cos\left(\frac{1}{x_k}\right) - x_{k-1} \cdot \cos\left(\frac{1}{x_{k-1}}\right) \right| \\ &= \left| (-1)^{n-k}(x_k + x_{k-1}) \right| \\ &\geq x_k = \frac{1}{(n-k+1)\pi}. \end{aligned}$$

Hence the total variation of f over $[0, 1/\pi]$ (with respect to P_n) is

$$\sum_{k=1}^{n} |f(x_k) - f(x_{k-1})| \geq \frac{1}{\pi} \sum_{k=1}^{n} \frac{1}{(n-k+1)} = \frac{1}{\pi}\left(1 + \frac{1}{2} + \cdots + \frac{1}{n}\right).$$

This is still controlled by the harmonic series, whence $V_0^{1/\pi}(f) = \infty$.

9. The characteristic function of the rational numbers in $[0, 1]$,

$$f(x) = \begin{cases} 1 & \text{if } x \in [0,1] \cap \mathbb{Q} \\ 0 & \text{if } x \in [0,1] \setminus \mathbb{Q}. \end{cases}$$

does not have bounded variation. Take $n \in \mathbb{N}$ even, divide $[0, 1]$ so that x_i is rational if i even and irrational if i odd, with $i = 0, \ldots, n$. For this partition

$$V_0^1(P, f) = \sum_{i=1}^{n} |f(x_i) - f(x_{i-1})| = n.$$

But $\big(\mathbb{Q} \cap [0, 1]\big)$ is countable, so

$$\sup_{P \in \mathscr{P}} \{V_0^1(P, f)\} = \infty. \qquad \clubsuit$$

Theorem 13.1.1 (Additivity of bounded total variation) *If $f : [a, b] \mapsto \mathbb{R}$ has bounded variation over both $[a, c]$ and $[c, b]$, for some $a < c < b$, then it has bounded variation over $[a, b]$, and moreover*

$$V_a^b(f(x)) = V_a^c(f(x)) + V_c^b(f(x)).$$

Proof Consider a partition $P \in \mathscr{P}$ of $[a, b]$

$$P = \{a = x_0 < x_1 < \cdots < x_{m-1} < x_m < \cdots < x_n = b\}.$$

If c coincides with one of the partition points, e.g. $c = x_m$, then

$$\begin{aligned} \sum_{i=1}^{n} |f(x_i) - f(x_{i-1})| &= \sum_{i=1}^{m} |f(x_i) - f(x_{i-1})| + \sum_{i=m+1}^{n} |f(x_i) - f(x_{i-1})| \\ &\leq V_a^c(f) + V_c^b(f). \end{aligned}$$

If c is an extra point, instead,

$$\sum_{i=1}^{n} |f(x_i) - f(x_{i-1})| \leq \sum_{i=1}^{n+1} |f(x_i) - f(x_{i-1})|.$$

In general then

$$\sum_{i=1}^{n} |f(x_i) - f(x_{i-1})| \leq V_a^c(f) + V_c^b(f).$$

This holds for any P, so

$$V_a^b(f) \leq V_a^c(f) + V_c^b(f).$$

Let us prove the opposite inequality. Take $\epsilon > 0$, so that $V_a^c(f) - \epsilon/2$ and $V_c^b(f) - \epsilon/2$ are less than the supremum. Then there exist partitions P_1 of $[a, c]$, P_2 of $[c, b]$ such that

$$V_a^c(P_1, f) > V_a^c(f) - \epsilon/2 \quad \text{and} \quad V_c^b(P_2, f) > V_c^b(f) - \epsilon/2.$$

But $P = P_1 \cup P_2$ is a partition of $[a, b]$, for which

$$V_a^b(P, f) = V_a^c(P_1, f) + V_c^b(P_2, f) > V_a^c(f) + V_c^b(f) - \epsilon;$$

since the above holds for any $P \in \mathscr{P}$, we conclude

$$V_a^b(f) \geq V_a^c(f) + V_c^b(f) - \epsilon$$

and the claim follows. ■

Proposition 13.1.1 (Operations between maps with bounded variation) *If $f, g : [a, b] \mapsto \mathbb{R}$ have bounded variation and k is a real number, then the following have bounded variation:*
1. $|f|$; 2. $k \cdot f$; 3. $f + g$; 4. $f \cdot g$; 5. f/g, if $g \geq m > 0$ on $[a, b]$.

Proof 1. By assumption, given $n \in \mathbb{N}$ and $P_n \in \mathscr{P}$ we have[1]

$$\sum_{k=1}^{n} \big||f(x_k)| - |f(x_{k-1})|\big| \leq \sum_{k=1}^{n} \big|f(x_k) - f(x_{k-1})\big| < \infty$$
$$\Rightarrow \qquad V_a^b(|f|) \leq V_a^b(f) < \infty.$$

---

[1]

$$\big||a| - |b|\big| \leq |a - b| \quad \Leftrightarrow \quad \big||a| - |b|\big|^2 \leq |a - b|^2$$
$$\Leftrightarrow \quad a^2 - 2|ab| + b^2 \leq a^2 - 2ab + b^2 \quad \Leftrightarrow \quad |ab| \geq ab$$

and the latter is true.

2. For every $P_n \in \mathscr{P}$ of $[a, b]$

$$\begin{aligned} V_a^b(P_n, k \cdot f) &= \sum_{i=1}^{n} |k \cdot f(x_i) - k \cdot f(x_{i-1})| = |k| \cdot \sum_{i=1}^{n} |f(x_i) - f(x_{i-1})| \\ &= |k| \cdot V_a^b(P_n, f) \\ \Rightarrow \quad V_a^b(k \cdot f) &= |k| \cdot V_a^b(f) < \infty. \end{aligned}$$

3. For every partition $P_n \in \mathscr{P}$ of $[a, b]$

$$\begin{aligned} V_a^b(P_n, f + g) &= \sum_{i=1}^{n} \big|f(x_i) - f(x_{i-1}) + g(x_i) - g(x_{i-1})\big| \\ &\le \sum_{i=1}^{n} \big|f(x_i) - f(x_{i-1})\big| + \sum_{i=1}^{n} \big|g(x_i) - g(x_{i-1})\big| \\ &= V_a^b(P_n, f) + V_a^b(P_n, g) \\ \Rightarrow \quad V_a^b(f + g) &\le V_a^b(f) + V_a^b(g) < \infty. \end{aligned}$$

4. As f, g are bounded by some $K > 0$, for all $x \in [a, b]$

$$|f(x)| \le K \quad \text{and} \quad |g(x)| \le K.$$

Choose any partition $P_n \in \mathscr{P}$ of $[a, b]$. Then

$$\begin{aligned} V_a^b\big(P_n, (f \cdot g)\big) &= \sum_{i=1}^{n} |f(x_i)g(x_i) - f(x_{i-1})g(x_{i-1})| \\ &= \sum_{i=1}^{n} |f(x_i)g(x_i) - f(x_i)g(x_{i-1}) + f(x_i)g(x_{i-1}) - f(x_{i-1})g(x_{i-1})| \\ &= \sum_{i=1}^{n} |f(x_i)[g(x_i) - g(x_{i-1})] + g(x_{i-1})[f(x_i) - f(x_{i-1})]| \\ &\le \sum_{i=1}^{n} \big[|f(x_i)| \cdot |g(x_i) - g(x_{i-1})| + |g(x_{i-1})| \cdot |f(x_i) - f(x_{i-1})|\big] \\ &\le \sum_{i=1}^{n} \big[K \cdot |g(x_i) - g(x_{i-1})| + K \cdot |f(x_i) - f(x_{i-1})|\big] \\ &= K \sum_{i=1}^{n} \big[|g(x_i) - g(x_{i-1})| + |f(x_i) - f(x_{i-1})|\big] \\ &= K\big[V_a^b(P_n, f) + V_a^b(P_n, g)\big]; \\ \Rightarrow \quad V_a^b\big(f \cdot g\big) &\le K\big[V_a^b(f) + V_a^b(g)\big] < \infty. \end{aligned}$$

5. Based on the previous result we may just show $1/g$ has bounded variation. Take $P_n \in \mathscr{P}$ of $[a, b]$, so

$$V_a^b\left(P_n, \frac{1}{g}\right) = \sum_{i=1}^{n} \left| \frac{1}{g(x_i)} - \frac{1}{g(x_{i-1})} \right| = \sum_{i=1}^{n} \left| \frac{g(x_i) - g(x_{i-1})}{g(x_i) \cdot g(x_{i-1})} \right|$$

$$\leq \sum_{i=1}^{n} \left| \frac{g(x_i) - g(x_{i-1})}{m^2} \right| \leq \frac{1}{m^2} \sum_{i=1}^{n} |g(x_i) - g(x_{i-1})| = \frac{1}{m^2} V_a^b(P_n, g);$$

$$\Rightarrow \quad V_a^b\left(\frac{1}{g}\right) \leq \frac{1}{m^2} V_a^b(g) < \infty. \qquad \blacksquare$$

13.2 Positive and Negative Total Variations

Let us introduce positive and negative total variations. Consider a partition $P \in \mathscr{P}$ of $[a, b]$,

$$P = \{x_0 = a, x_1, \ldots, x_{i-1}, x_i, \ldots, x_{n-1}, x_n = b\}.$$

The variation of f over $[x_{i-1}, x_i]$ is positive if $[f(x_i) - f(x_{i-1})] \geq 0$, in which case we write $[f(x_i) - f(x_{i-1})]^+$. It is negative if $[f(x_i) - f(x_{i-1})] < 0$, in which case $[f(x_i) - f(x_{i-1})]^- = -[f(x_i) - f(x_{i-1})] > 0$. Let us define

$$P_+ = \{k = 0, 1, \ldots, n : f(x_k) - f(x_{k-1}) \geq 0\}$$
$$P_- = \{k = 0, 1, \ldots, n : f(x_k) - f(x_{k-1}) < 0\}$$

so that $P = P_+ \cup P_-$. We also indicate the positive and negative variations of f over $[x_{k-1}, x_k]$ respectively by

$$V_{x_{i-1}}^{x_i}(P_+, f) = \begin{cases} [f(x_i) - f(x_{i-1})]^+ & \text{if} \quad [f(x_i) - f(x_{i-1})] \geq 0 \\ 0 & \text{if} \quad [f(x_i) - f(x_{i-1})] < 0 \end{cases}$$

and

$$V_{x_{i-1}}^{x_i}(P_-, f) = \begin{cases} [f(x_i) - f(x_{i-1})]^- & \text{if} \quad [f(x_i) - f(x_{i-1})] < 0 \\ 0 & \text{if} \quad [f(x_i) - f(x_{i-1})] \geq 0. \end{cases}$$

These simple relations lead to the definition of positive and negative total variations of a function on $[a, b]$.

Definition 13.2.1 (*Positive and negative total variations*) Consider a function $f : [a, b] \mapsto \mathbb{R}$ and a partition $P \in \mathscr{P}$ of $[a, b]$:

$$P = \{a = x_0 < x_1 < \cdots < x_{i-1} < x_i < \cdots < x_{n-1} < x_n = b\}.$$

The positive variation of f over $[a, b]$ with respect to $P \in \mathscr{P}$ is

$$V_a^b(P_+, f) = \sum_{k \in P_+} [f(x_k) - f(x_{k-1})]^+;$$

the negative variation of f over $[a, b]$ with respect to $P \in \mathscr{P}$ is

$$V_a^b(P_-, f) = \sum_{k \in P_-} [f(x_k) - f(x_{k-1})]^-.$$

Similarly, the total positive variation and total negative variation of f over $[a, b]$ are the quantities

$$VP_a^b(f) = \sup_{P_+ \in \mathscr{P}} \left\{ V_a^b(P_+, f) \right\} \quad \text{and} \quad VN_a^b(f) = \sup_{P_- \in \mathscr{P}} \left\{ V_a^b(P_-, f) \right\}. \qquad \diamond$$

The two total signed variations of a map with bounded variation satisfy a number of properties that we now examine.

Proposition 13.2.1 (Relationships among f, $V_a^b(f)$, $VP_a^b(f)$ and $VN_a^b(f)$) *Let $f : [a, b] \mapsto \mathbb{R}$ be a map with bounded variation on $[a, b]$. Then*

1. $V_a^b(f) = VP_a^b(f) + VN_a^b(f)$;
2. $VP_a^b(f) - VN_a^b(f) = f(b) - f(a)$;
3. $VP_a^b(f) = \frac{1}{2} V_a^b(f) + \frac{f(b)-f(a)}{2}$;
4. $VN_a^b(f) = \frac{1}{2} V_a^b(f) - \frac{f(b)-f(a)}{2}$.

Proof

1. Let $P \in \mathscr{P}$ be a partition of $[a, b]$. Then

$$\begin{aligned} V_a^b(P_+, f) + V_a^b(P_-, f) &= \sum_{k \in P_+} [f(x_k) - f(x_{k-1})]^+ + \sum_{k \in P_-} [f(x_k) - f(x_{k-1})]^- \\ &= \sum_{k \in P_+} |f(x_k) - f(x_{k-1})| + \sum_{k \in P_-} |f(x_k) - f(x_{k-1})| \\ &= \sum_{k \in P} |f(x_k) - f(x_{k-1})| = V_a^b(P, f). \end{aligned}$$

   As $P \in \mathscr{P}$ is arbitrary, the claim follows.

2.

$$\begin{aligned} V_a^b(P_+, f) - V_a^b(P_-, f) &= \sum_{k \in P_+} \left[f(x_k) - f(x_{k-1}) \right]^+ - \sum_{k \in P_-} \left[f(x_k) - f(x_{k-1}) \right]^- \\ &= \sum_{k \in P_+} \left[f(x_k) - f(x_{k-1}) \right] + \sum_{k \in P_-} \left[f(x_k) - f(x_{k-1}) \right] \\ &= \sum_{k \in P} \left[f(x_k) - f(x_{k-1}) \right] = f(b) - f(a). \end{aligned}$$

As $P \in \mathscr{P}$ is arbitrary, the claim follows.

3. Regarding statements *3* and *4.*, the result is immediate once we solve the system

$$\begin{cases} V_a^b(f) = VP_a^b(f) + VN_a^b(f) \\ VP_a^b(f) - VN_a^b(f) = f(b) - f(a) \end{cases}.$$

■

13.3 Total Variations as Functions

The total variations seen previously can be interpreted as functions of the variations themselves.

Definition 13.3.1 (*Total variations as functions*) Given $f : [a, b] \mapsto \mathbb{R}$ we call:

1. $v : [a, b] \mapsto \bar{\mathbb{R}}$, $v(x) = V_a^x(f)$, total variation function of f;
2. $v_+ : [a, b] \mapsto [0, \infty]$, $v_+(x) = VP_a^x(f)$, positive total variation function of f;
3. $v_- : [a, b] \mapsto [0, \infty]$, $v_-(x) = VN_a^x(f)$, negative total variation function of f.

◇

These functions enjoy various properties, described below.

Proposition 13.3.1 (Properties of the total variation function) *Let $f : [a, b] \mapsto \mathbb{R}$ have bounded variation. Then v is*

1. *increasing;*
2. *right-continuous or left-continuous at any point where f is right- or left-continuous.*

Proof

1. Take $x < y$ in $[a, b]$:

$$\begin{aligned} & v(y) - v(x) = V_a^y(f) - V_a^x(f) = V_x^y(f) \geq 0 \\ \Leftrightarrow \quad & V_a^x(f) \leq V_a^y(f) \\ \Leftrightarrow \quad & v(x) \leq v(y). \end{aligned}$$

2. To prove right-continuity pick $x_0 \in [a, b)$ and suppose f is continuous at x_0^+:

$$\lim_{x \to x_0^+} f(x) = f(x_0).$$

Given $\epsilon > 0$, the total variation $V_{x_0}^b(f) - \epsilon/2$ is smaller than the supremum, so there is a partition

$$P = \{x_0 = t_0 < t_1 < \cdots < t_n = b\}$$

of $[x_0, b]$ such that

$$V_{x_0}^b(P, f) > V_{x_0}^b(f) - \frac{\epsilon}{2}.$$

Now pick $x_1 \in (x_0, t_1)$ so that

$$|f(x_1) - f(x_0)| < \frac{\epsilon}{2}.$$

Clearly $P \subseteq \big(P \cup \{x_1\}\big) = P_1$, and P_1 is finer, so

$$V_{x_0}^b(P_1, f) \geq V_{x_0}^b(P, f) > V_{x_0}^b(f) - \frac{\epsilon}{2}.$$

Moreover $P_2 = P_1 \setminus \{x_0\}$ partitions $[x_1, b] \subseteq [x_0, b]$, and so

$$V_{x_1}^b(P_2, f) = V_{x_0}^b(P_1, f) - |f(x_1) - f(x_0)| > V_{x_0}^b(P_1, f) - \frac{\epsilon}{2}.$$

Hence

$$\begin{aligned} V_{x_1}^b(f) &\geq V_{x_1}^b(P_2, f) = V_{x_0}^b(P_1, f) - |f(x_1) - f(x_0)| \\ &\geq V_{x_0}^b(P, f) - |f(x_1) - f(x_0)| > V_{x_0}^b(f) - \epsilon. \end{aligned}$$

But the total variation is additive (Theorem 13.1.1),

$$\begin{aligned} V_a^{x_1}(f) &= V_a^b(f) - V_{x_1}^b(f) < V_a^b(f) - V_{x_0}^b(f) + \epsilon \\ &= V_a^{x_0}(f) + \epsilon. \end{aligned}$$

We conclude

$$v(x_1) < v(x_0) + \epsilon.$$

As the total variation is increasing, then for every $x \in (x_0, x_1)$

$$v(x_0) \leq v(x) \leq v(x_1) < v(x_0) + \epsilon,$$

and hence

$$\lim_{x \to x_0^+} v(x) = v(x_0).$$

The proof of left-continuity is essentially the same. ■

Proposition 13.3.2 (Relationships among f, v, v_+, v_-) *For any $f : [a, b] \mapsto \mathbb{R}$ with bounded variation:*

1. $v(x) = v_+(x) + v_-(x)$;
2. $f(x) = v_+(x) - v_-(x) + f(a)$;

3. $v_+(x) = \frac{1}{2}\left(v(x) + f(x)\right) - \frac{f(a)}{2}$;
4. $v_-(x) = \frac{1}{2}\left(v(x) - f(x)\right) + \frac{f(a)}{2}$.

Proof Immediate from Proposition 13.2.1. ■

Proposition 13.3.3 (Properties of signed total variation functions) *Let $f : [a, b] \mapsto \mathbb{R}$ have bounded variation. Then the functions v_+, v_- are*

1. *increasing;*
2. *right- or left-continuous at any point where f is right- or left-continuous.*

Proof Immediate from the previous two results. ■

Theorem 13.3.1 (Jordan) *A map $f : [a, b] \mapsto \mathbb{R}$ has bounded variation over $[a, b]$ if and only if it is the difference of two increasing maps on $[a, b]$.*

Proof The sufficient part descends directly from properties *2.*, *3.* of Proposition 13.1.1 (subtraction preserves bounded variations) and Remark 13.1.1, 3 (monotonicity on $[a, b]$ implies bounded variation).

Necessary implication: suppose f has bounded variation. By Proposition 13.3.2, *3.-4.* we deduce

$$\begin{aligned} v(x) &= 2v_+(x) + f(a) - f(x) \\ v(x) &= 2v_-(x) + f(x) - f(a), \end{aligned}$$

subtracting which produces

$$\begin{aligned} v(x) - v(x) &= 2v_+(x) - 2v_-(x) + f(a) - f(x) - f(x) + f(a) \\ 0 &= v_+(x) - v_-(x) + f(a) - f(x), \end{aligned}$$

hence $f(x) = f(a) + v_+(x) - v_-(x)$. Part *1.* in Proposition 13.3.3 says the total signed variation functions are increasing. The claim then follows from the observation that $f(a) + v_+$ is increasing on $[a, b]$, as well. ■

13.4 Differentiability of Monotone Functions

The last theorem said a map with bounded variation can be written as difference of two increasing maps. If we prove that monotonicity forces differentiability almost everywhere, and that the derivative is summable, by the linearity of differentiation and integration we obtain a similar result for maps with bounded variation. Precisely:

"*A map $f : [a, b] \mapsto \mathbb{R}$ with bounded variation is differentiable almost everywhere on $[a, b]$, and the derivative f' is summable.*"

Initially we shall prove this for monotone functions. But to do it we need a few preliminaries, including establishing the cardinality of the discontinuity set of a monotone function.

Proposition 13.4.1 (Discontinuities of monotone functions) *A monotone function $f : [a, b] \mapsto \mathbb{R}$ has at most countably many jump discontinuities.*

Proof Let f be monotone, say increasing, without loss of generality.

The discontinuities of a monotone map are of first kind, and we recall that the jump of f at $x \in [a, b]$ is

$$\omega(f, x) = \lim_{\delta \to 0} \omega\big(f, U_\delta(x) \cap [a, b]\big).$$

As f is increasing, $\omega(f, x) \geq 0$ for every $x \in [a, b]$. If f is discontinuous at x then $\omega(f, x) > 0$. We seek the cardinality of the set

$$D(f) = \{x \in [a, b] : \omega(f, x) > 0\}.$$

By monotonicity the sum of all jumps cannot exceed $f(b) - f(a)$, so there exist at most n_1 points

$$a \leq x_1 < x_2 < \cdots < x_{n_1} \leq b$$

such that

$$\omega(f, x_i)_{n_1} > \frac{1}{2}, \quad i = 1, \ldots, n_1$$

and

$$\sum_{i=1}^{n_1} \omega(f, x_i)_{n_1} \leq f(b) - f(a).$$

Hence the cardinality of $D(f)_1 = \{x \in [a, b] : \omega(f, x) > 1/2\}$ is finite, and equal to n_1. Moving on, there exist n_2 points

$$a \leq x_1 < x_2 < \cdots < x_{n_2} \leq b$$

where f jumps by $\omega(f, x_i) > 1/2^2$ and whose sum cannot exceed $\big[f(b) - f(a) - \sum_{i=1}^{n_1} \omega(f, x_i)_{n_1}\big]$. Therefore

$$\frac{1}{2^2} < \omega(f, x_i)_{n_2} \leq \frac{1}{2}, \quad i = 1, \ldots, n_2, \quad \text{and} \quad \sum_{i=1}^{n_2} \omega(f, x_i)_{n_2} \leq \left[f(b) - f(a) - \sum_{i=1}^{n_1} \omega(f, x_i)_{n_1} \right].$$

Clearly

$$D(f)_2 = \left\{ x \in [a,b] : \frac{1}{2^2} < \omega(f,x) \le \frac{1}{2} \right\}$$

is finite, of cardinality n_2. After the kth iteration we obtain a finite set

$$D(f)_k = \left\{ x \in [a,b] : \frac{1}{2^k} < \omega(f,x) \le \frac{1}{2^{k-1}} \right\}$$

with n_k elements. But

$$D(f) = \left\{ x \in [a,b] : \omega(f,x) > 0 \right\} = \bigcup_{k \in \mathbb{N}} D(f)_k$$

is a countable union of finite sets, so it cannot be more than countable. ■

The proof is completely similar in case f is decreasing, or even if defined on $\mathbb{R}$. In the latter situation it suffices to impose the hypothesis on every interval $[-n, n]$, $n \in \mathbb{N}$.

Definition 13.4.1 (*Vitali coverings*) A family of intervals $\mathscr{I}$ is a Vitali covering of $E \subseteq \mathbb{R}$ if for every $\epsilon > 0$ and $x \in E$ there exists an $I \in \mathscr{I}$ such that $x \in I$ and $\ell(I) < \epsilon$. ◇

The next is an important lemma due to Giuseppe Vitali.

Lemma 13.4.1 (Vitali covering lemma) *Let $E \subseteq \mathbb{R}$ have finite outer measure and suppose $\mathscr{I}$ is a Vitali covering of E. For any $\epsilon > 0$ there exists a finite subset $\{I_1, \ldots, I_n\}$ of $\mathscr{I}$ of pairwise-disjoint intervals such that*

$$\ell^*\left(E \setminus \bigcup_{i=1}^{n} I_i \right) < \epsilon$$

(ℓ^* is the outer measure on $\mathbb{R}$).

Proof Let G be an open set such that $E \subseteq G$ and $\ell^*(G) < \infty$.

The family $\tilde{\mathscr{I}} = \{I \in \mathscr{I} : I \subseteq G\}$ is still, by Definition 13.4.1, a Vitali covering of E, so without loss of generality we can assume that all intervals in $\mathscr{I}$ are contained in the open set G, and replace $\mathscr{I}$ with the subfamily $\tilde{\mathscr{I}}$. As a consequence we can put

$$k_0 = \sup \left\{ \ell(I) : I \in \mathscr{I} \right\} < \infty$$

and choose I_1 so that

$$\frac{k_0}{2} < \ell(I_1) \quad \text{i.e.} \quad k_0 < 2\ell(I_1).$$

If $E \subseteq I_1$, the proof ends. If not, there exists an $x_1 \in \big(E \setminus I_1\big)$. As $\mathscr{I}$ is a Vitali covering, at least one $I \in \mathscr{I}$ contains x_1 and is disjoint from I_1. Hence

$$\mathcal{E}_1 = \big\{I \in \mathscr{I} : I \cap I_1 = \emptyset \text{ and } x_1 \in I\}$$

is not empty, and we set

$$k_1 = \sup\big\{\ell(I) : I \in \mathscr{I}, I \cap I_1 = \emptyset \text{ and } x_1 \in I\} > 0.$$

Now pick $I_2 \in \mathcal{E}_1$ so that

$$\frac{k_1}{2} < \ell(I_2) \quad \text{i.e.} \quad k_1 < 2\ell(I_2).$$

If $E \subseteq (I_1 \cup I_2)$ we are done, if not there exists $x_2 \in E \setminus (I_1 \cup I_2)$ etc. At step n we will have pairwise-disjoint intervals $I_1, I_2, \ldots, I_n$ of decreasing length not larger than each bound $k_0, k_1, \ldots, k_{n-1}$.

The proposition is proved if we show $E \subseteq \big(\bigcup_{i=1}^n I_i\big)$. If that is not the case, there exist $x_n \in \big(E \setminus \bigcup_{i=1}^n I_i\big)$ and $I \in \mathscr{I}$, containing x_n and disjoint from the other I_i, such that the family

$$\mathcal{E}_n = \left\{I \in \mathscr{I} : I \cap I_i = \emptyset, \text{ for } i = 1, \ldots, n \text{ and } x_n \in I\right\},$$

is not empty. Set

$$k_n = \sup\{\ell(I) : I \in \mathscr{I}, I \cap I_i = \emptyset, \text{ for } i = 1, \ldots, n \text{ and } x_n \in I\} > 0.$$

Induction generates a sequence of pairwise-disjoint intervals $I_1, I_2, \ldots, I_n, \ldots$ of $\mathscr{I}$ such that, for every $n \geq 0$,

$$\frac{k_n}{2} < \ell(I_{n+1}) \quad \text{i.e.} \quad k_n < 2\ell(I_{n+1}).$$

The I_n are all subsets of G and disjoint, so

$$\ell^*\left(\bigcup_{n=1}^{\infty} I_n\right) = \sum_{n=1}^{\infty} \ell(I_n) \leq \ell^*(G) < \infty$$

and $\ell(I_n) \to 0$ as $n \to \infty$. Hence for any $\epsilon > 0$ there exists an $n_\epsilon \in \mathbb{N}$ such that

$$\sum_{n=n_\epsilon+1}^{\infty} \ell(I_n) < \frac{\epsilon}{5}.$$

Consider $F = \left(E \setminus \bigcup_{i=1}^{n_\epsilon} I_i\right)$. The proposition is proved if $\ell^*(F) < \epsilon$, so let us argue as follows. Take $x \in F$ so $x \notin \left(\bigcup_{i=1}^{n_\epsilon} I_i\right)$. As said earlier, we choose $I \in \mathscr{I}$ so that $I \cap I_i = \emptyset$, $i = 1, \ldots, n_\epsilon$, and $x \in I$. Because $\ell(I_n) \to 0$ as $n \to \infty$, and $k_n < 2\ell(I_{n+1})$, there always exists an $m > n_\epsilon$ such that $k_m < \ell(I)$, in which case $I \cap I_m \neq \emptyset$. Call m_0 the smallest integer such that $I \cap I_{m_0} \neq \emptyset$, where clearly $m_0 > n_\epsilon$. As $I \cap I_{m_0} \neq \emptyset$ and $x \in I$, the distance between x and the midpoint x_{m_0} of I_{m_0} is at most $\ell(I) + \ell(I_{m_0})/2$. Furthermore,

$$\ell(I) + \frac{\ell(I_{m_0})}{2} < k_{m_0-1} + \frac{\ell(I_{m_0})}{2} < 2\ell(I_{m_0}) + \frac{\ell(I_{m_0})}{2} = 5\frac{\ell(I_{m_0})}{2}.$$

Take $\ell(J_{m_0}) = 5\ell(I_{m_0})$, where J_{m_0} is the closed interval centred at x_{m_0} such that $x \in J_{m_0}$. Then we have proved that for every $x \notin \bigcup_{i=1}^{n_\epsilon} I_i$ there exists an $n > n_\epsilon$ such that $x \in J_n$ and $\ell(J_n) = 5\ell(I_n)$. Therefore $F \subseteq \bigcup_{n=n_\epsilon+1}^{\infty} J_n$ and so

$$\ell^*(F) = \ell^*\left(E \setminus \bigcup_{i=1}^{n_\epsilon} I_i\right) \leq \sum_{n=n_\epsilon+1}^{\infty} \ell(J_n) = 5 \sum_{n=n_\epsilon+1}^{\infty} \ell(I_n) < 5 \cdot \frac{\epsilon}{5} = \epsilon. \qquad \blacksquare$$

Theorem 13.4.1 (Lebesgue—Differentiability a.e of monotone maps) *If $f : [a, b] \mapsto \mathbb{R}$ is monotone, then f' exists and is finite ℓ-almost everywhere on $[a, b]$, it is everywhere summable, and furthermore*

$$\int_{[a,b]} |f'(x)|\ell(dx) \leq |f(b) - f(a)|.$$

Proof Without loss of generality we can take f increasing, so if $f'(x)$ exists at some point $x \in (a, b)$, then $f'(x) \geq 0$.

To prove that the limit

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

exists and is finite ℓ-almost everywhere we shall use the Dini derivatives. Because of the monotonicity, f is differentiable at x iff

$$0 \leq D_- f(x) = D^- f(x) = D_+ f(x) = D^+ f(x) < \infty,$$

and we want this to hold ℓ-a.e. on $[a, b]$.

Since in general

$$D_- f(x) \leq D^- f(x) \quad \text{and} \quad D_+ f(x) \leq D^+ f(x),$$

we must have

$$D_-f(x) > D^-f(x) \quad \text{and} \quad D_+f(x) > D^+f(x)$$

ℓ-a.e. on $[a, b]$. This can be obtained by showing that the set of points in $[a, b]$ where any two of the four derivatives differ is null. For example, putting $E = \{x \in (a, b) : D^+f(x) > D_-f(x)\}$, it suffices to show $\ell(E) = 0$. The other null sets are obtained similarly, by choosing two out of four derivatives.

So consider

$$E = \{x \in (a, b) : D^+f(x) > D_-f(x)\}.$$

As f is increasing, $D \geq 0$ on (a, b), so for rational numbers $p > q > 0$ the set E can be written as

$$E = \bigcup_{p,q\in\mathbb{Q}\cap(0,\infty)} \left\{x \in (a, b) : D^+f(x) > p > q > D_-f(x)\right\} = \bigcup_{p,q\in\mathbb{Q}\cap(0,\infty)} E_{p,q}.$$

Write $r = \ell^*(E_{p,q})$. As $r < (b - a),\ \forall p, q \in \mathbb{Q} \cap (0, \infty)$, if we fix any $\epsilon > 0$ there exists an open set G such that $E_{p,q} \subseteq G$ and

$$\ell^*(G) < \ell^*(E_{p,q}) + \epsilon = r + \epsilon.$$

But $D_-f(x) < q$ for every $x \in E_{p,q}$, and since $x \in G$ we can choose an arbitrarily small real number $h > 0$ such that

$$[x - h, x] \subseteq G \quad \text{and} \quad \frac{f(x) - f(x - h)}{h} < q \Leftrightarrow f(x) - f(x - h) < qh.$$

The closed intervals

$$\left\{[x - h, x] : x \in E_{p,q}, h > 0 \ \text{ and } \ f(x) - f(x - h) < qh\right\}$$

form a Vitali covering of $E_{p,q}$. Therefore by Lemma 13.4.1 there is a finite number n of pairwise-disjoint intervals $I_i = [x_i - h_i, x_i]$, $i = 1, \ldots, n$, such that

$$f(x_i) - f(x_i - h_i) < qh_i$$

for every i and

$$\ell^*\left(E_{p,q} \setminus \bigcup_{i=1}^{n} I_i\right) < \epsilon.$$

Summing over i

$$\sum_{i=1}^{n}\big[f(x_i)-f(x_i-h_i)\big]<q\sum_{i=1}^{n}h_i=q\sum_{i=1}^{n}[x_i+h_i-x_i]\le q\cdot\ell^*(G)<q\big(r+\epsilon\big). \tag{13.4.1}$$

Set $A=\big[E_{p,q}\cap\big(\bigcup_{i=1}^{n}(x_i-h_i,x_i)\big)\big]=\big(E_{p,q}\cap\big(\bigcup_{i=1}^{n}\overset{\circ}{I}_i\big)\big)$, and observe

$$r=\ell^*\big(E_{p,q}\big)=\ell^*(A)+\underbrace{\ell^*\left(E_{p,q}\setminus\bigcup_{i=1}^{n}I_i\right)}_{<\epsilon}<\ell^*(A)+\epsilon\quad\Leftrightarrow\quad\ell^*(A)>r-\epsilon. \tag{13.4.2}$$

Similarly, $D^+f(y)>p\ \forall\, y\in A$ implies we can pick $k>0$ so small that $[y,y+k]\subseteq I_i$ for some $i\in\{1,\dots,n\}$, whence

$$[y,y+k]\subseteq G\quad\text{and}\quad\frac{f(y+k)-f(y)}{k}>p\Leftrightarrow f(y+k)-f(y)>pk.$$

The family

$$\big\{[y,y+k]:y\in A,k>0\text{ and }f(y+k)-f(y)>pk\big\}$$

is a Vitali covering of A, and Lemma 13.4.1 ensures there are finitely many pairwise-disjoint intervals $J_j=[y_j,y_j+k_j]$, $j=1,\dots,m$ such that

$$f(y_j+k_j)-f(y_j)>pk_j\quad\text{and}\quad\ell^*\left(A\setminus\bigcup_{j=1}^{m}J_j\right)<\epsilon, \tag{13.4.3}$$

so

$$\ell^*\left(A\cap\bigcup_{j=1}^{m}J_j\right)>\ell^*\big(A\big)-\epsilon>r-2\epsilon. \tag{13.4.4}$$

Summing over j

$$\sum_{j=1}^{m}\big(f(y_j+k_j)-f(y_j)\big)>p\sum_{j=1}^{m}k_j=p\sum_{j=1}^{m}[y_j+k_j-y_j]\ge p\cdot\ell^*\left(A\cap\bigcup_{j=1}^{m}J_j\right)>p(r-2\epsilon). \tag{13.4.5}$$

Note that every J_j is contained in some I_i, and since the J_j are pairwise disjoint and f is increasing, for a given i we obtain

$$\sum_{j:J_j\subseteq I_i}\big(f(y_j+k_j)-f(y_j)\big)\le f(x_i)-f(x_i-h_i).$$

Summing over i and considering that every J_j is contained in some I_i we have

$$\begin{aligned}\sum_{i=1}^{n}\big(f(x_i)-f(x_i-h_i)\big)&\ge\sum_{i=1}^{n}\sum_{j:J_j\subseteq I_i}\big(f(y_j+k_j)-f(y_j)\big)\\&=\sum_{j=1}^{m}\big(f(y_j+k_j)-f(y_j)\big).\end{aligned}$$

Considering inequalities (13.4.1) and (13.4.5)

$$p(r-2\epsilon)<q(r+\epsilon)$$

and let ϵ go to zero, to obtain

$$p\cdot r\le q\cdot r.$$

But from the start $p>q$, so $r=\ell^*(E_{p,q})=0$. This concludes the proof that the limit exists almost everywhere. There remains to show f' is finite ℓ-almost everywhere.

Let us extend f to $(b,b+1]$ by setting $f(x)=f(b)$ when $x\ge b$, and consider the sequence

$$g_n(x)=\frac{f\big(x+\frac{1}{n}\big)-f(x)}{\frac{1}{n}},\quad\forall x\in[a,b].$$

As f is monotone, it is certainly Riemann integrable, in particular measurable. Every map g_n is Riemann integrable, too, hence measurable. The relation

$$\lim_{n\to\infty}g_n(x)=f'(x)$$

holds ℓ-almost everywhere on $[a,b]$, and proves f' is measurable. Furthermore, $\{g_n\}_{n\ge1}\subseteq\mathscr{M}^+\big([a,b],\mathscr{L}\big)$ since f is increasing. Recalling we have set $f(x)=f(b)$ when $x\ge b$, and since $f(x)\ge f(a)$ in the integral $\displaystyle\int_{[a,a+\frac{1}{n}]}f(x)\ell(dx)$, we apply Fatou's lemma and deduce

$$\begin{aligned}\int_{[a,b]} f'(x)\,\ell(dx) &\le \liminf_{n\to\infty} \int_{[a,b]} g_n(x)\ell(dx)\\ &= \liminf_{n\to\infty}\left[n\int_{[a,b]} f\left(x+\frac{1}{n}\right)\ell(dx) - n\int_{[a,b]} f(x)\ell(dx)\right]\\ &= \liminf_{n\to\infty}\left[n\int_{\left[a+\frac{1}{n},b+\frac{1}{n}\right]} f(x)\ell(dx) - n\int_{[a,b]} f(x)\ell(dx)\right]\\ &= \liminf_{n\to\infty}\left[n\int_{\left[b,b+\frac{1}{n}\right]} f(x)\ell(dx) - n\int_{[a,a+\frac{1}{n}]} f(x)\ell(dx)\right]\\ &\le \liminf_{n\to\infty}\left[n\cdot\frac{f(b)}{n} - n\cdot\frac{f(a)}{n}\right] = f(b)-f(a).\end{aligned}$$

We have thus proved f' is summable on $[a,b]$, whence it exists, finite, ℓ-a.e. on $[a,b]$. ■

13.5 Differentiability of Functions with Bounded Variation

An immediate consequence of Theorems 13.3.1, 13.4.1 is the following.

Theorem 13.5.1 (Differentiability and summability of maps with bounded variation) *If $f:[a,b]\mapsto\mathbb{R}$ has bounded variation on $[a,b]$, then f' exists, finite, ℓ-almost everywhere on $[a,b]$. Additionally, f' is summable on $[a,b]$ and*

$$\int_{[a,b]} |f'(x)|\ell(dx) \le |f(b)-f(a)|.$$ ■

With the help of Theorems 13.4.1 and 13.3.1 we can prove the next result on bounded variations.

Theorem 13.5.2 *If $f:[a,b]\mapsto\mathbb{R}$ has bounded variation on $[a,b]$ then*

$$\int_{[a,b]} |f'(x)|\ell(dx) \le V_a^b(f).$$

Proof By *2.*, Proposition 13.3.2, we can write

$$f(x) = v_+(x) - v_-(x) + f(a);$$

where v_+, v_- are increasing on $[a,b]$, hence ℓ-a.e. differentiable by Theorem 13.4.1. Therefore

$$f'(x) = v'_+(x) - v'_-(x)$$

holds ℓ-a.e. on $[a, b]$. The same happens for the total variation function v, namely

$$v'(x) = v'_+(x) + v'_-(x) \quad \ell\text{-a.e. on } [a, b].$$

Hence

$$|f'(x)| \leq |v'_+(x)| + |v'_-(x)| = v'_+(x) + v'_-(x) = v'(x).$$

All of the above are summable as well on $[a, b]$, so

$$\begin{aligned}\int_{[a,b]} |f'(x)|\ell(dx) &\leq \int_{[a,b]} v'(x)\ell(dx) \\ &\leq v(b) - v(a) = V_a^b(f).\end{aligned}$$

■

Bounded step functions, or the Vitali–Cantor map, are instances of functions with bounded variation where the total-variation inequality is strict.

Earlier we saw that when the total variation is bounded, the map is the difference of monotone functions, and as such differentiable almost everywhere. Yet integrating the derivative will not necessarily return the initial function, because the difference between the initial map and the final outcome is a function whose derivative is zero almost everywhere. This difference map is called singular, and might not be constant. As a matter of fact it can assume any value in a given interval. Lebesgue proved that a function with bounded variation can be decomposed in the sum of an absolutely continuous map plus a singular map.

Absolutely continuous functions are precisely the maps with bounded variation that coincide with the integral of their derivative. The next section is devoted to this class of maps.

13.6 Absolutely Continuous Functions

We set forth to study maps verifying

$$f(x) - f(a) = \int_{[a,x]} f'(t)\ell(dt).$$

They are called absolutely continuous, a notion introduced by Giuseppe Vitali.

Definition 13.6.1 (*Absolute continuity*) A function $f : [a, b] \mapsto \mathbb{R}$ is called absolutely continuous on $[a, b] \subseteq \mathbb{R}$ if the following occurs. Given $\epsilon > 0$ there exists $\delta > 0$ such that

$$\sum_{i=1}^{n} |f(b_i) - f(a_i)| < \epsilon$$

for any choice of n subintervals

$$[a_1, b_1], \ldots, [a_n, b_n] \subseteq [a, b]$$

such that $\sum_{i=1}^{n}(b_i - a_i) < \delta$ and whose possible pairwise intersections have empty interiors. ◇

Remark 13.6.1

1. The definition is rather close to that of continuity, and actually it generalises uniform continuity. In fact, take f absolutely continuous on $[a, b]$, so for $\epsilon > 0$ there exists $\delta > 0$ such that

$$|f(b_1) - f(a_1)| < \epsilon$$

for every $[a_1, b_1] \subset [a, b]$ with $(b_1 - a_1) < \delta$. But this is uniform continuity. Therefore absolutely continuous $\Rightarrow$ uniformly continuous $\Rightarrow$ continuous, but not vice versa. Uniform continuity is much weaker a notion than absolute continuity. The Vitali–Cantor map is uniformly but not absolutely continuous, as we shall see.
2. Also absolutely continuous are Lipschitz functions: for these

$$|f(x') - f(x'')| \leq k|x' - x''|$$

for any $x', x'' \in [a, b]$ and some $k > 0$. Then, corresponding to each $\epsilon > 0$ we may take $\delta = \epsilon/k$. In that way

$$\sum_{i=1}^{n} |f(b_i) - f(a_i)| \leq k \sum_{i=1}^{n} |b_i - a_i| < k \cdot \delta = \epsilon$$

for any choice of subintervals $[a_1, b_1], \ldots, [a_n, b_n]$ with $\sum_{i=1}^{n}(b_i - a_i) < \delta$ and no interior points in the possible pairwise intersections.
Any affine map $f(x) = ax + b$ is Lipschitz, with Lipschitz constant $k = |a|$, so affine maps are absolutely continuous. In fact

$$\sum_{i=1}^{n} |ab_i + b - aa_i - b| = |a| \sum_{i=1}^{n} |b_i - a_i| < |a| \cdot \delta = |a| \frac{\epsilon}{|a|} = \epsilon. \qquad \clubsuit$$

Proposition 13.6.1 (Absolute continuity of integral functions) *If $f : [a, b] \mapsto \mathbb{R}$ is summable, the integral map $F : [a, b] \mapsto \mathbb{R}$,*

$$F(x) = \int_{[a,x]} f(t)\ell(dt), \quad x \in [a, b]$$

is absolutely continuous.

Proof Take $f \in L_1$. By *2.*, Proposition 11.3.5, picking $\epsilon > 0$ gives $\delta > 0$ such that

$$\int_E |f|\, d\ell < \epsilon$$

for every $E \in \mathscr{L}(\mathbb{R})$ with $\ell(E) < \delta$. Take $E = \bigcup_{i=1}^n [a_i, b_i]$, where the intervals' possible pairwise intersections have empty interiors, so that

$$\ell(E) = \sum_{i=1}^n (b_i - a_i) < \delta.$$

Since F is the integral map of f,

$$\begin{aligned}\sum_{i=1}^n |F(b_i) - F(a_i)| &= \sum_{i=1}^n \left| \int_{[a,b_i]} f\, d\ell - \int_{[a,a_i)} f\, d\ell \right| = \sum_{i=1}^n \left| \int_{[a_i,b_i]} f\, d\ell \right| \\ &\le \sum_{i=1}^n \int_{[a_i,b_i]} |f|\, d\ell = \int_E |f|\, d\ell < \epsilon. \end{aligned}$$

■

Proposition 13.6.2 (Operations on absolutely continuous maps) *If $f, g : [a, b] \mapsto \mathbb{R}$ are absolutely continuous and $k \in \mathbb{R}$, the following are absolutely continuous:*

1. $|f(x)|$; *2.* $f(x) + k \cdot g(x)$; *3.* $f(x) \cdot g(x)$; *4.* $f(x)/g(x)$, *if* $g(x) \ge m > 0$ $\forall x \in [a, b]$.

Proof

1. From Proposition 13.1.1, *1*,

$$\sum_{i=1}^n ||f(b_i)| - |f(a_i)|| \le \sum_{i=1}^n |f(b_i) - f(a_i)|,$$

   and the absolute continuity of f allows to conclude.
2. For $k = 0$ we obviously have $f(x) + kg(x) = f(x)$, which by assumption is absolutely continuous. So given $\epsilon > 0$ there is $\delta_1 > 0$ such that, for any choice of subintervals $[a_1, b_1], \ldots, [a_{n_1}, b_{n_1}]$ with $\sum_{i=1}^{n_1} (b_i - a_i) < \delta_1$ and whose possible pairwise intersections have empty interiors,

$$\sum_{i=1}^{n_1} |f(b_i) - f(a_i)| < \frac{\epsilon}{2}.$$

   Corresponding to the same ϵ there is also a $\delta_2 > 0$ such that, for any subintervals $[a_1, b_1], \ldots, [a_{n_2}, b_{n_2}]$ whose possible pairwise intersections have empty interiors and satisfying $\sum_{i=1}^{n_2} (b_i - a_i) < \delta_2$,

$$\sum_{i=1}^{n_2} |g(b_i) - g(a_i)| < \frac{\epsilon}{2|k|}.$$

Take $\delta = \min\{\delta_1, \delta_2\}$. Again, for any subintervals as above with $\sum_{i=1}^{n}(b_i - a_i) < \delta$ we have

$$\begin{aligned}\sum_{i=1}^{n} |f(b_i) + k \cdot g(b_i) - f(a_i) - k \cdot g(a_i)| &= \sum_{i=1}^{n} |[f(b_i) - f(a_i)] + k \cdot [g(b_i) - g(a_i)]| \\ &\le \sum_{i=1}^{n} |f(b_i) - f(a_i)| + |k| \cdot |g(b_i) - g(a_i)| \\ &< \frac{\epsilon}{2} + |k| \frac{\epsilon}{2|k|} = \epsilon.\end{aligned}$$

3. f, g are absolutely continuous hence continuous, and the Weierstraß theorem says they have an absolute maximum on $[a, b]$:

$$|f(x)| \le M_1 \quad \text{and} \quad |g(x)| \le M_2$$

for some $M_1, M_2 > 0$ and all $x \in [a, b]$. Given $\epsilon > 0$ there exist δ_1 and δ_2 such that

$$\sum_{i=1}^{n_1} |f(b_i) - f(a_i)| < \frac{\epsilon}{2M_2} \quad \text{and} \quad \sum_{i=1}^{n_2} |g(b_i) - g(a_i)| < \frac{\epsilon}{2M_1},$$

in accordance with absolute continuity. Let $\delta = \min\{\delta_1, \delta_2\}$, so for any subintervals $[a_1, b_1], \ldots, [a_n, b_n]$ whose possible pairwise intersections have empty interiors, and $\sum_{i=1}^{n}(b_i - a_i) < \delta$, we have

$$\begin{aligned}\sum_{i=1}^{n} |f(b_i)g(b_i) - f(a_i)g(a_i)| &= \sum_{i=1}^{n} |f(b_i)g(b_i) - f(a_i)g(b_i) + f(a_i)g(b_i) - f(a_i)g(a_i)| \\ &= \sum_{i=1}^{n} |f(a_i)[g(b_i) - g(a_i)] + g(b_i)[f(b_i) - f(a_i)]| \\ &\le \sum_{i=1}^{n} \big[|f(a_i)| \cdot |g(b_i) - g(a_i)| + |g(b_i)| \cdot |f(b_i) - f(a_i)|\big] \\ &\le M_1 \sum_{i=1}^{n} |g(b_i) - g(a_i)| + M_2 \sum_{i=1}^{n} |f(b_i) - f(a_i)| \\ &< M_1 \frac{\epsilon}{2M_1} + M_2 \frac{\epsilon}{2M_2} = \epsilon.\end{aligned}$$

4. Based on the previous fact, it suffices to show $1/g$ is absolutely continuous. Fix $\epsilon > 0$, so by absolute continuity there is a $\delta > 0$ such that

$$\sum_{i=1}^{n} |g(a_i) - g(b_i)| < m^2 \cdot \epsilon.$$

Hence

$$\sum_{i=1}^{n} \left| \frac{1}{g(b_i)} - \frac{1}{g(a_i)} \right| = \sum_{i=1}^{n} \left| \frac{g(a_i) - g(b_i)}{g(b_i) \cdot g(a_i)} \right| \leq \sum_{i=1}^{n} \left| \frac{g(a_i) - g(b_i)}{m^2} \right|$$
$$\leq \frac{1}{m^2} \sum_{i=1}^{n} |g(a_i) - g(b_i)| < \frac{1}{m^2} \cdot m^2 \cdot \epsilon = \epsilon. \qquad \blacksquare$$

Theorem 13.6.1 (Bounded variation of absolutely continuous maps) *An absolutely continuous map $f : [a, b] \mapsto \mathbb{R}$ has bounded variation on its domain.*

Proof For any $\epsilon > 0$ there is a $\delta > 0$ satisfying the condition of absolute continuity. Consider a partition $P = \{a = x_0 < \cdots < x_n = b\}$ with $(x_i - x_{i-1}) < \delta$. Given any partition $P_t = \{x_{i-1} = t_0 < \cdots < t_m = x_i\}$ of $[x_{i-1}, x_i]$, by absolute continuity

$$V_{x_{i-1}}^{x_i}(f) = \sup_{P_t \in \mathscr{P}} \left\{ \sum_{j=1}^{m} |f(t_j) - f(t_{j-1})| < \epsilon : P_t = \{x_{i-1} = t_0 < \cdots < t_m = x_i\} \right\} \leq \epsilon.$$

Hence f has bounded variation on every $[x_{i-1}, x_i]$, $i = 1, \ldots, n$, and Theorem 13.1.1 forces bounded variation on $[a, b]$. $\blacksquare$

Remark 13.6.2 The continuous functions of Remark 13.1.1, nr. *8.* are not absolutely continuous because their variations are not bounded: given $\epsilon > 0$ we can choose suitable subintervals $[a_i, b_i]$, with $\sum_{i=1}^{n}(b_i - a_i) < \delta$ as small as we like, over which the variation $\sum_{i=1}^{n} |f(b_i) - f(a_i)|$ exceeds ϵ. This is always possible since the total variation of f is infinite.

Observe moreover that there exist continuous maps with bounded variation that are not absolutely continuous, one being the Vitali–Cantor map (Example 10.5.1). To see this, note that on $[0, 1]$ the map increases by $1/2^n$ on each interval $C_{n,k}$, $k = 1, 2, \ldots, 2^n$. Supposing $C_{n,k} = [a_k^n, b_k^n]$, we have

$$\sum_{k=1}^{2^n} |f(b_k^n) - f(a_k^n)| = \sum_{k=1}^{2^n} \frac{1}{2^n} = 2^n \cdot \frac{1}{2^n} = 1, \tag{13.6.1}$$

whereas

$$\sum_{k=1}^{2^n} |b_k^n - a_k^n| = \left(\frac{2}{3}\right)^n.$$

But n can be chosen so that $(2/3)^n$ is smaller than any $\delta > 0$. In view of (13.6.1), then, f cannot be absolutely continuous.

The conclusion is that the converse of Theorem 13.6.1 is false. Having bounded variation on some set is not sufficient for a map to be absolutely continuous on that set. ♣

Chapter 14
Fundamental Theorems of Calculus for the Lebesgue Integral



14.1 The 1st and 2nd Fundamental Theorems of Integral Calculus

In this section we shall examine the differentiability properties of absolutely continuous maps. The first property is a straightforward consequence of Theorems 13.5.1 and 13.6.1.

Theorem 14.1.1 (Differentiability and summability of absolutely continuous maps) *If $f : [a, b] \mapsto \mathbb{R}$ is absolutely continuous, then f' exists, finite, ℓ-almost everywhere on $[a, b]$, and it is summable on $[a, b]$.* ■

Before we get to the first fundamental theorems of calculus we state a result that is an immediate consequence of Proposition 11.3.3.

Lemma 14.1.1 (Vanishing condition for integrals of summable maps) *If $f : [a, b] \mapsto \mathbb{R}$ is summable and*

$$\int_{[a,x]} f(t)\ell(dt) = 0 \quad \textit{for every } x \in [a, b],$$

then $f = 0$ ℓ-almost everywhere. ■

Theorem 14.1.2 (First fundamental theorem of integral calculus) *Let $f : [a, b] \mapsto \mathbb{R}$ be summable on $[a, b]$, and call*

$$F(x) = \int_{[a,x]} f(t)\ell(dt), \quad x \in [a, b]$$

its integral map. Then F' exists and is finite ℓ-almost everywhere. Moreover, $F' = f$ ℓ-almost everywhere.

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Proof Notice $F'(x)$ exists, finite, ℓ-a.e. on $[a,b]$ because F is absolutely continuous (Proposition 13.6.1 and Theorem 14.1.1).

So there remains to prove the relation $F' = f$, ℓ-almost everywhere on $[a,b]$. To start with, we shall assume f is bounded, i.e. $|f(x)| \leq M,\ \forall x \in [a,b], M > 0$. For every $n \in \mathbb{N}$, consider $f_n : [a,b] \mapsto \mathbb{R}$, such that for all $x \in [a,b]$

$$f_n(x) = \frac{\left[F\left(x + \frac{1}{n}\right) - F(x)\right]}{1/n},$$

where we have set $F(x) = F(b)$ on $(b, b+1]$.

Since F is the integral map, for every $n \in \mathbb{N}$

$$\begin{aligned} |f_n(x)| &= n\left|\int_{\left[a,x+\frac{1}{n}\right]} f(t)\ell(dt) - \int_{[a,x]} f(t)\ell(dt)\right| \\ &= n\left|\int_{\left[x,x+\frac{1}{n}\right]} f(t)\ell(dt)\right| \leq n\int_{\left[x,x+\frac{1}{n}\right]} |f(t)|\ell(dt) \leq M, \end{aligned}$$

whence $|f_n(x)| \leq M$ on $[a,b]$. Taking $h = 1/n$, and using the absolute continuity of F, Theorem 14.1.1 ensures that the derivative

$$F'(x) = \lim_{h \to 0^+} \frac{F(x+h) - F(x)}{h} = \lim_{n \to \infty} f_n(x)$$

exists and is finite ℓ-almost everywhere. But $f_n(x)$ is measurable as continuous, so also $F'(x)$ is measurable (Theorem 10.3.1). Now take a point $y \in [a,b]$ and apply the bounded convergence theorem to obtain

$$\int_{[a,y]} F'(x)\ell(dx) = \lim_{n \to \infty} \int_{[a,y]} f_n(x)\ell(dx).$$

F is absolutely continuous, hence continuous, and so $F \in \mathcal{R}([a,b])$. Similarly $\{f_n\}_{n \geq 1} \subseteq \mathcal{R}([a,b])$, so for every $n \in \mathbb{N}$

$$\int_{[a,y]} f_n(x)\ell(dx) = \int_a^y f_n(x)dx.$$

Consequently

$$\begin{aligned} \int_{[a,y]} F'(x)\ell(dx) &= \lim_{n \to \infty} \int_{[a,y]} f_n(x)\ell(dx) = \lim_{n \to \infty} \int_a^y f_n(x)dx \\ &= \lim_{n \to \infty} \int_a^y n\left[F\left(x + \frac{1}{n}\right) - F(x)\right]dx \\ &= \lim_{n \to \infty} \left[n\int_{a+1/n}^{y+1/n} F(x)dx - n\int_a^y F(x)dx\right] \end{aligned}$$

$$= \lim_{n\to\infty} \left[n \int_y^{y+1/n} F(x)dx - n \int_a^{a+1/n} F(x)dx \right].$$

The continuity of F says that for $\epsilon > 0$ there exists $\delta > 0$ such that, for every $x \in [a, b]$ with $|x - y| < \delta$,

$$\big|F(x) - F(y)\big| < \epsilon.$$

So taking $h \in (0, \delta)$ and remembering

$$\frac{1}{h} \int_y^{y+h} F(y)dx = \frac{1}{h} F(y) \cdot h = F(y) \quad \forall y \in [a, b]$$

we deduce

$$\left| \frac{1}{h} \int_y^{y+h} F(x)dx - F(y) \right| \le \frac{1}{h} \int_y^{y+h} \big|F(x) - F(y)\big| dx < \frac{1}{h} \int_y^{y+h} \epsilon \, dx = \epsilon,$$

and then

$$\lim_{n\to\infty} n \int_y^{y+1/n} F(x)dx = \lim_{h\to 0^+} \frac{1}{h} \int_y^{y+h} F(x)dx = F(y).$$

In a similar way

$$\lim_{n\to\infty} n \int_a^{a+1/n} F(x)dx = \lim_{h\to 0^+} \frac{1}{h} \int_a^{a+h} F(x)dx = F(a).$$

All in all,

$$\int_{[a,y]} F'(x)\ell(dx) = F(y) - F(a) = \int_{[a,y]} f(x)\ell(dx). \tag{14.1.1}$$

So, for any $y \in [a, b]$ and any $[a, y] \subseteq [a, b]$

$$\int_{[a,y]} [F'(x) - f(x)]\ell(dx) = 0.$$

By Lemma 14.1.1

$$F'(x) = f(x) \quad \ell\text{-a.e. on } [a, b]. \tag{14.1.2}$$

Now we remove the boundedness of f, and note it is enough to prove the statement when $f \ge 0$, due to the decomposition $f = f^+ - f^-$.

We extend f to the value 0 where it is not defined on $[a, b]$. Then non-negativity $f \geq 0$ implies that

$$F(x) = \int_{[a,x]} f(t)\ell(dt)$$

is increasing on $[a, b]$, and Theorem 13.4.1 warrants finite derivative F' ℓ-almost everywhere on $[a, b]$. Moreover,

$$\int_{[a,x]} F'(t)\ell(dt) \leq F(x) - F(a) \tag{14.1.3}$$

for every $x \in [a, b]$. Take $n \in \mathbb{N}$ and call $f_n : [a, b] \mapsto \mathbb{R}$ the map

$$f_n(x) = \begin{cases} f(x) & \text{if } 0 \leq f(x) \leq n \\ n & \text{if } f(x) > n. \end{cases}$$

Let

$$F_n(x) = \int_{[a,x]} f_n(t)\ell(dt)$$

be the integral map of f_n. Since each f_n is summable by assumption, and bounded by construction, from the first part of this proof we know

$$F_n'(x) = \frac{d}{\ell(dx)} \int_{[a,x]} f_n(t)\ell(dt) = f_n(x) \quad \ell\text{-a.e. on } [a, b].$$

The maps $g_n(x) = f(x) - f_n(x)$ are non-negative, and

$$G_n(x) = \int_{[a,x]} [f(t) - f_n(t)]\ell(dt)$$

is increasing, so by Theorem 13.4.1 $G_n(x)$ is ℓ-almost everywhere differentiable, and $G_n'(x) \geq 0$ ℓ-almost everywhere. Since, for every x,

$$F(x) = G_n(x) + F_n(x) = \int_{[a,x]} [f(t) - f_n(t)]\ell(dt) + \int_{[a,x]} f_n(t)\ell(dt),$$

we have

$$F'(x) = G_n'(x) + F_n'(x) = G_n'(x) + \frac{d}{\ell(dx)} \int_{[a,x]} f_n(t)\ell(dt) \geq f_n(x)$$

ℓ-almost everywhere. This holds for any $n \in \mathbb{N}$, whence

$$F'(x) \geq f(x) \quad \ell\text{-a.e. on } [a,b].$$

By (14.1.1)

$$\int_{[a,x]} F'(t)\, \ell(dt) \geq \int_{[a,x]} f(t)\, \ell(dt) = F(x) - F(a).$$

Combining the above with (14.1.3),

$$\int_{[a,x]} F'(t)\, \ell(dt) = F(x) - F(a) = \int_{[a,x]} f(t)\, \ell(dt)$$

for every $x \in [a,b]$. Therefore

$$\int_{[a,x]} [F'(t) - f(t)] \ell(dt) = 0 \ \ \forall x \in [a,b].$$

But $F'(x) - f(x) = 0$ ℓ-a.e., so finally

$$F'(x) = f(x) \quad \ell\text{-almost everywhere on } [a,b]. \qquad \blacksquare$$

For the second fundamental theorem we shall need a lemma, itself an application of Vitali's covering theorem.

Lemma 14.1.2 (Absolutely continuous maps with almost everywhere-vanishing derivative) *If $f : [a,b] \mapsto \mathbb{R}$ is absolutely continuous and $f'(x) = 0$ ℓ-almost everywhere, then f is constant on $[a,b]$.*

Proof Selecting $c \in (a,b]$ arbitrarily, we have to prove $f(c) = f(a)$. Let E denote the set of critical points in (a,c):

$$E = \{x \in (a,c) : f'(x) = 0\}.$$

As the derivative is ℓ-almost everywhere zero, $\ell(E) = (c-a)$.

Moreover, for every $x \in E$

$$f'(x) = \lim_{h \to 0} \frac{|f(x+h) - f(x)|}{h} = 0.$$

Fix $\epsilon > 0$ and let $\delta > 0$ be the corresponding number from the definition of absolute continuity. Given $\alpha > 0$ and any $x \in E$ we can choose $h \in (0, \alpha)$ so that

$$[x, x+h] \subseteq (a,c), \quad \text{and} \quad \frac{|f(x+h) - f(x)|}{h} \leq \epsilon \Leftrightarrow |f(x+h) - f(x)| < h \cdot \epsilon.$$

The family of closed intervals

$$\big\{[x, x+h] \subseteq (a, c) : x \in E, h \in (0, \alpha) \quad \text{and} \quad |f(x+h) - f(x)| < h \cdot \epsilon\big\}$$

is a Vitali covering for E. By Lemma 13.4.1 in that family we have a finite number of pairwise-disjoint intervals, say $I_i = [x_i, x_i + h_i]$, $i = 1, \ldots, n$, such that

$$\bigcup_{i=1}^{n} [x_i, x_i + h_i] \subseteq (a, c), \ |f(x_i + h_i) - f(x_i)| < \epsilon \cdot h_i \ \ \forall i = 1, \ldots, n \quad (14.1.4)$$

and

$$\ell\left(E \setminus \bigcup_{i=1}^{n} [x_i, x_i + h_i]\right) < \delta.$$

Let us order these intervals so that $x_i < x_{i+1}$ and set $y_i = x_i + h_i$, so

$$y_0 = a < x_1 < y_1 < x_2 < y_2 < \cdots < x_n < y_n < c = x_{n+1}, \quad (14.1.5)$$

and hence

$$\begin{aligned} \sum_{i=0}^{n} (x_{i+1} - y_i) &= (c - a) - \sum_{i=1}^{n} [y_i - x_i] \\ &= \ell\left(E \setminus \bigcup_{i=1}^{n} [x_i, x_i + h_i]\right) < \delta. \end{aligned} \quad (14.1.6)$$

Due to (14.1.5) we have

$$\begin{aligned} |f(c) - f(a)| &= |f(x_{n+1}) - f(y_0)| \\ &= \big|[f(x_1) - f(y_0)] + [f(y_1) - f(x_1)] + [f(x_2) - f(y_1)] + \\ &\quad [f(y_2) - f(x_2)] + \cdots + [f(x_{n+1}) - f(y_n)]\big| \\ &= \big|\big([f(x_1) - f(y_0)] + [f(x_2) - f(y_1)] + \cdots + [f(x_{n+1}) - f(y_n)]\big) + \\ &\quad \big([f(y_1) - f(x_1)] + [f(y_2) - f(x_2)] + \cdots + [f(y_n) - f(x_n)]\big)\big| \\ &= \left|\sum_{i=0}^{n} [f(x_{i+1}) - f(y_i)] + \sum_{i=1}^{n} [f(y_i) - f(x_i)]\right| \\ &\leq \left|\sum_{i=0}^{n} [f(x_{i+1}) - f(y_i)]\right| + \left|\sum_{i=1}^{n} [f(y_i) - f(x_i)]\right|. \end{aligned} \quad (14.1.7)$$

Now absolute continuity implies

$$\sum_{i=1}^{n} |f(x_{i+1}) - f(y_i)| < \epsilon.$$

In view of the inequality in (14.1.4), relation (14.1.7) reads

$$|f(c) - f(a)| < \epsilon + \epsilon \sum_{i=1}^{n} h_i < \epsilon + \epsilon(c-a).$$

By varying $\epsilon > 0$ we obtain $f(c) = f(a)$ and finally the claim, since $c \in (a, b]$ was completely arbitrary. ■

Theorem 14.1.3 (Second fundamental theorem of integral calculus) *If $F : [a, b] \mapsto \mathbb{R}$ is absolutely continuous on $[a, b]$ then: F' exists and is finite ℓ-almost everywhere on $[a, b]$, $F' \in L_1([a, b])$, and*

$$F(y) - F(x) = \int_{[x,y]} F'(t)\ell(dt) \tag{14.1.8}$$

for every $x < y$ in $[a, b]$.

Proof By Theorem 13.6.1 we know F has bounded variation, so Theorem 13.5.2 implies $F' \in L_1([a, b])$, and finally by Theorem 14.1.1 F' exists finite ℓ-almost everywhere.

There remains to show (14.1.8). Since constants are absolutely continuous maps we can set

$$G(x) = \int_{[a,x]} F'(t)\ell(dt) + F(a), \quad x \in [a, b].$$

By Proposition 13.6.1 and *2.*, Proposition 13.6.2, the function $G(x)$ is absolutely continuous, so the first fundamental theorem of calculus (Theorem 14.1.2) says $G'(x) = F'(x)$ ℓ-almost everywhere. Hence $H = G - F$ is absolutely continuous (as difference of absolutely continuous maps), $H'(x) = F'(x) - G'(x) = 0$ ℓ-a.e., and

$$H(a) = G(a) - F(a) = \int_{\{a\}} F'(t)\ell(dt) + F(a) - F(a) = 0.$$

Lemma 14.1.2 implies H is constant on $[a, b]$, so $H(x) = H(a) = 0$. Then $G(x) = F(x)$ for every $x \in [a, b]$. Given $x < y$ in $[a, b]$,

$$\begin{aligned} G(x) = F(x), \quad \text{where} \quad F(x) &= \int_{[a,x]} F'(t)\ell(dt) + F(a), \\ G(y) = F(y), \quad \text{where} \quad F(y) &= \int_{[a,y]} F'(t)\ell(dt) + F(a), \end{aligned} \tag{14.1.9}$$

from which

$$F(y) - F(x) = \int_{[a,y]} F'(t)\ell(dt) + F(a) - \int_{[a,x]} F'(t)\ell(dt) - F(a)$$
$$= \int_{[x,y]} F'(t)\ell(dt). \qquad \blacksquare$$

Using the fundamental theorems of calculus we can prove the following.

Proposition 14.1.1 (Absolute continuity and bounded variation in integral calculus) *For any map $f : [a, b] \mapsto \mathbb{R}$ the following are equivalent:*

1. *f is absolutely continuous on $[a, b]$;*
2. *f has bounded variation on $[a, b]$ and*

$$\int_{[a,b]} |f'(x)|\ell(dx) = V_a^b(f(x)).$$

Proof (*1.* $\Rightarrow$*2.*) Take

$$P = \{a = x_0 < x_1 < \cdots < x_{i-1} < x_i < \cdots < x_{n-1} < x_n = b\},$$

so by Theorem 14.1.3

$$V_a^b(P, f) = \sum_{i=1}^{n} |f(x_i) - f(x_{i-1})| = \sum_{i=1}^{n} \left| \int_{[x_{i-1},x_i]} f'(t)\ell(dt) \right|$$
$$\leq \sum_{i=1}^{n} \int_{[x_{i-1},x_i]} |f'(t)|\ell(dt) = \int_{[a,b]} |f'(x)|\ell(dx).$$

Varying the partition we deduce

$$V_a^b(f) \leq \int_{[a,b]} |f'(x)|\ell(dx).$$

Moreover, by Theorem 13.5.2

$$\int_{[a,b]} |f'(x)|\ell(dx) \leq V_a^b(f),$$

and so

$$\int_{[a,b]} |f'(x)|\ell(dx) = V_a^b(f).$$

(*2.* $\Rightarrow$*1.*) Bounded variation means that for any $x \in [a, b]$

$$V_a^x(f) \geq \int_{[a,x]} |f'(t)|\ell(dt) = \int_{[a,b]} |f'(t)|\ell(dt) - \int_{[x,b]} |f'(t)|\ell(dt)$$
$$= V_a^b(f) - \int_{[x,b]} |f'(t)|\ell(dt) = V_a^b(f) - V_x^b(f) = V_a^x(f);$$

i.e.

$$V_a^x(f) = \int_{[a,x]} |f'(t)|\ell(dt), \ \forall x \in [a,b].$$

Again, the hypothesis implies $\int_{[a,x]} |f'(t)|\ell(dt) < \infty$ for every $x \in [a,b]$, i.e. $|f'| \in L_1[a,b]$. By part *1.* of Proposition 11.3.1 $f' \in L_1[a,b]$, and by Proposition 13.6.1 the integral function

$$f(x) = \int_{[a,x]} f'(x)\ell(dx)$$

is absolutely continuous on $[a,b]$. ■

Let us apply the above result to monotone functions.

Corollary 14.1.1 (Absolute continuity and monotonicity in integral calculus) *For any map $f : [a,b] \mapsto \mathbb{R}$ the following are equivalent:*

1. *f is absolutely continuous on $[a,b]$;*
2. *f is monotone on $[a,b]$ and*

$$\int_{[a,b]} |f'(x)|\ell(dx) = |f(b) - f(a)|.$$ ■

It was mentioned earlier that the indefinite integral of the derivative of a function with bounded variation will not, in general, give back the initial function. The resulting map and the function with bounded variation differ by a map whose derivative vanishes ℓ-almost everywhere. The difference is a singular map, which we now define.

Definition 14.1.1 (*Singular maps*) A function $f : [a,b] \mapsto \mathbb{R}$ is called singular on $[a,b]$ if $f' = 0$ on $[a,b]$ ℓ-almost everywhere. ◇

Now we come to Lebesgue's decomposition theorem for maps with bounded variation, which clarifies that only the absolutely continuous part of a map equals the indefinite integral of its derivative.

Theorem 14.1.4 (Lebesgue decomposition theorem) *Let $F : [a,b] \mapsto \mathbb{R}$ have bounded variation. There exist an absolutely continuous map G and a singular map H, both with bounded variation, such that $F = G + H$. The decomposition is unique up to constants.*

Proof A map F with bounded variation is the difference of increasing maps F_1, F_2, so we can restrict the argument to F increasing.

Then by Theorem 13.4.1 F' exists and is finite a.e., it is non-negative and summable on $[a, b]$. Therefore the integral map

$$G(x) = \int_{[a,x]} F'(t)\ell(dt)$$

is increasing on $[a, b]$ and real-valued. By Proposition 13.6.1 G is absolutely continuous, so Theorem 14.1.2 says $G' = F'$ ℓ-a.e. on $[a, b]$.

Define $H : [a, b] \mapsto \mathbb{R}$, $H(x) = F(x) - G(x)$, so $H'(x) = F'(x) - G'(x) = 0$ ℓ-almost everywhere. Then H is singular. Furthermore, it is increasing, because $x_1 < x_2$ implies, by Theorem 13.4.1,

$$\int_{[x_1,x_2]} F'(t)\ \ell(dt) \leq F(x_2) - F(x_1). \tag{14.1.10}$$

As $F' = G' + H' = G'$ ℓ-almost everywhere, and since G is absolutely continuous, the second fundamental theorem of calculus (Theorem 14.1.3) tells

$$\int_{[x_1,x_2]} F'(t)\ \ell(dt) = \int_{[x_1,x_2]} G'(t)\ell(dt) = G(x_2) - G(x_1).$$

And comparing the above with (14.1.10) gives

$$G(x_2) - G(x_1) \leq F(x_2) - F(x_1) \Leftrightarrow (F - G)(x_1) \leq (F - G)(x_2) \Leftrightarrow H(x_1) \leq H(x_2),$$

which ends the proof that H is increasing. Altogether $F = G + H$ is the required decomposition.

Regarding uniqueness: suppose $F = G_1 + H_1 = G_2 + H_2$ with G_1, G_2 increasing and absolutely continuous, H_1, H_2 increasing and singular. The map $F_0 = G_1 - G_2 = H_2 - H_1$ is absolutely continuous ($G_1 - G_2$ is) and singular ($H_2 - H_1$ is), so it is constant on $[a, b]$, which is another way of saying $G_1 - G_2$ and $H_2 - H_1$ are constant, i.e. the decomposition is unique up to some number. ■

14.2 Integration by Parts and by Substitution

One application of the fundamental theorems of calculus to the Lebesgue integral is the formula customarily referred to as 'integration by parts'.

Theorem 14.2.1 (Integration by parts) *If $F, G : [a, b] \mapsto \mathbb{R}$ are absolutely continuous, then $F \cdot g$, $f \cdot G$ are summable on $[a, b]$, and for any $x \in [a, b]$*

$$\int_{[a,b]} F(x) \cdot g(x)\ell(dx) + \int_{[a,b]} f(x) \cdot G(x)\ell(dx) = F(b) \cdot G(b) - F(a) \cdot G(a).$$

Proof From 3., Proposition 13.6.2 we know $(F \cdot G)$ is absolutely continuous, and $(F \cdot G)$ is differentiable ℓ-almost everywhere. By the second fundamental theorem of calculus, for any $x \in [a, b]$

$$\int_{[a,x]} [F(t) \cdot G(t)]'\ell(dt) = [F(x) \cdot G(x)] - [F(a) \cdot G(a)].$$

By the first fundamental theorem, then, $G' = g$ and $F' = f$ ℓ-almost everywhere, so

$$[F(x) \cdot G(x)]' = F(x) \cdot g(x) + f(x) \cdot G(x) \quad \ell\text{-a.e. on } [a, b].$$

Integrating each side we obtain

$$\int_{[a,x]} [F(t) \cdot G(t)]'\ell(dt) = \int_{[a,x]} [F(t) \cdot g(t) + f(t) \cdot G(t)]\ell(dt)$$

and so

$$\int_{[a,x]} [F(t) \cdot g(t)]\ell(dt) + \int_{[a,x]} [f(t) \cdot G(t)]\ell(dt) = [F(x) \cdot G(x)] - [F(a) \cdot G(a)]$$

for any $x \in [a, b]$, whence the claim. ■

In Riemann's theory integration by substitution is a well-known technique, and goes something like this. Consider $f : [a, b] \mapsto \mathbb{R}$ continuous, and $g : [\alpha, \beta] \mapsto \mathbb{R}$ increasing, with derivative g' continuous on $[\alpha, \beta]$, and with $g(\alpha) = a$, $g(\beta) = b$. If we write $x = g(y)$ for $x \in [a, b]$, then formally $dx/dy = g'(y)$ i.e. $dx = g'(y) \cdot dy$. Substituting all this in $\int_a^b f(x)dx$ produces

$$\int_a^b f(x)dx = \int_\alpha^\beta f(g(y))g'(y)dy.$$

Note that the continuity of g' on $[\alpha, \beta]$ implies g is absolutely continuous on $[a, b]$.

The corresponding theorem for the Lebesgue integral, which we will see shortly, is another application of the fundamental theorems of calculus. Lebesgue's version of the substitution formula does not require g' to be continuous, in stark contrast to Riemann's theorem.

Lemma 14.2.1 (to integration by substitution) *Given g absolutely continuous, strictly increasing and onto on $[\alpha, \beta]$ such that $g([\alpha, \beta]) = [a, b]$, define $A = \{y \in [\alpha, \beta] : g'(y) > 0\}$. Then:*

1. if $G \subseteq [a, b]$ is a G_δ set, then $\ell(G) = \int_{g^{-1}(G)} g'(y)\ell(dy)$;

2. *if $E \subseteq [a,b]$ is L-measurable with $\ell(E) = 0$, then $\left(g^{-1}(E) \cap A\right)$ is L-measurable, of measure zero;*
3. *if $E \subseteq [a,b]$ is L-measurable then $\left(g^{-1}(E) \cap A\right)$ is L-measurable,- $\varphi_{g^{-1}(E)} g'$ is summable on $[\alpha, \beta]$, and*

$$\ell(E) = \int_{[\alpha,\beta]} \varphi_{g^{-1}(E)}(y) g'(y) \ell(dy). \tag{14.2.1}$$

Proof First, note $g' \geq 0$, and also that g' exists, finite, a.e. on $[\alpha, \beta]$ and is summable by Theorem 13.4.1. Hence it is measurable on $[\alpha, \beta]$, which amounts to saying that $A = (g')^{-1}\big((0, \infty]\big)$ is measurable.

To prove 1., the **first step** is assuming G is an interval $(a_0, b_0) \subseteq [a, b]$. As g is continuous then $g^{-1}\big((a_0, b_0)\big)$ is open and $g^{-1}(\{a_0\})$ and $g^{-1}(\{b_0\})$ are closed, and all contained in $[\alpha, \beta]$. Define $\alpha_0 = \min g^{-1}(\{a_0\})$ and $\beta_0 = \max g^{-1}(\{b_0\})$. By the continuity and monotonicity of g we have $g(\alpha_0) = a_0$, $g(\beta_0) = b_0$ and $g^{-1}\big((a_0, b_0)\big) = (\alpha_0, \beta_0)$. At this point we apply Theorem 14.1.3,

$$\begin{aligned} \ell((a_0, b_0)) = b_0 - a_0 = g(\beta_0) - g(\alpha_0) &= \int_{[\alpha_0,\beta_0]} g'(y) \ell(dy) \\ &= \int_{g^{-1}\left((a_0,b_0)\right)} g'(y) \ell(dy). \end{aligned}$$

The **second step** is to suppose G is any open subset of $[a, b]$, i.e. $G = \bigcup_{i=1}^{\infty} (a_i, b_i)$ where $\{(a_i, b_i)\}_{i \geq 1}$ are open pairwise-disjoint subintervals of $[a, b]$. Then

$$\begin{aligned} \ell\left(\bigcup_{i=1}^{\infty} (a_i, b_i)\right) = \sum_{i=1}^{\infty} \ell((a_i, b_i)) &= \sum_{i=1}^{\infty} \int_{g^{-1}\left((a_i,b_i)\right)} g'(y) \ell(dy) \\ &= \int_{\bigcup_{i=1}^{\infty} g^{-1}\left((a_i,b_i)\right)} g'(y) \ell(dy) \\ &= \int_{g^{-1}(G)} g'(y) \ell(dy). \end{aligned} \tag{14.2.2}$$

In **step three** we take G to be a G_δ set, $G = \bigcap_{i=1}^{\infty} G_i$, G_i open in $[a, b]$, $i = 1, 2, \ldots$ Clearly for some $k \in \mathbb{N}$ there is a $G_k \in \{G_i\}_{i \geq 1}$ such that $G \subseteq G_k$. Without loss of generality suppose $G_k \subseteq (a - 1, b + 1)$, for otherwise we could take $G_k \cap (a - 1, b + 1) \subseteq (a - 1, b + 1)$. Now we extend g to $[a - 1, b + 1]$, assuming the extension absolutely continuous. Let g_1 be given by

$$g_1(y) = \begin{cases} g(y) & \forall y \in [\alpha, \beta] \\ y + a - \alpha & \forall y \in [\alpha - 1, \alpha) \\ y + b - \beta & \forall y \in (\beta, \beta + 1] \end{cases}.$$

The above is absolutely continuous and strictly increasing on $[\alpha-1,\beta+1]$, and moreover $g_1(\alpha-1)=(a-1)$, $g_1(\beta+1)=(b+1)$ and $g_1([\alpha-1,\beta+1])=[a-1,b+1]$. For every $n\in\mathbb{N}$ define $B_n=\bigcap_{i=1}^n G_i$. Clearly $\{B_n\}_{n\geq 1}$ is a decreasing sequence of open sets contained in $(a-1,b+1)$, with $\bigcap_{n=1}^\infty B_n=G$. As $\ell((a-1,b+1))<\infty$, then $\ell(G)=\lim_{n\to\infty}\ell(B_n)$. Applying Eq. (14.2.2) to g_1 (absolutely continuous on $[\alpha-1,\beta+1]$) gives

$$\ell(B_n)=\int_{g_1^{-1}(B_n)} g_1'(y)\ell(dy)$$

for every $n\in\mathbb{N}$. But $\{B_n\}_{n\geq 1}$ is decreasing, and so is $\{g_1^{-1}(B_n)\}_{n\geq 1}$. Hence

$$\lim_{n\to\infty} g_1^{-1}(B_n)=\bigcap_{n=1}^\infty g_1^{-1}(B_n)=g_1^{-1}\left(\bigcap_{n=1}^\infty B_n\right)=g_1^{-1}(G).$$

Let us apply the result to the characteristic function, so that for any $y\in[\alpha-1,\beta+1]$

$$\lim_{n\to\infty}\varphi_{g_1^{-1}(B_n)}(y)=\varphi_{g_1^{-1}(G)}(y).$$

Since $0\leq\varphi_{g_1^{-1}(B_n)}(y)g_1'(y)\leq g_1'(y)$ and $g_1'(y)$ is summable on $[\alpha-1,\beta+1]$, Lebesgue's dominated convergence theorem gives

$$\begin{aligned}\ell(G)=\lim_{n\to\infty}\ell(B_n)&=\lim_{n\to\infty}\int_{g_1^{-1}(B_n)} g_1'(y)\ell(dy)\\&=\lim_{n\to\infty}\int_{[\alpha-1,\beta+1]}\varphi_{g_1^{-1}(B_n)}g_1'(y)\ell(dy)\\&=\int_{[\alpha-1,\beta+1]}\varphi_{g_1^{-1}(G)}g_1'(y)\ell(dy)\\&=\int_{g_1^{-1}(G)} g_1'(y)\ell(dy)=\int_{g^{-1}(G)} g'(y)\ell(dy).\end{aligned}\tag{14.2.3}$$

This ends the proof of *1*.

Part 2. Take $E\subseteq[a,b]$ L-measurable, of zero measure.

By Theorem 9.4.3 there exists $G\in G_\delta$, $E\subseteq G$, such that $\ell(G\setminus E)=0$. Without loss of generality we assume $G\subseteq(a-1,b+1)$. As $\ell(G\setminus E)=0$, then $\ell(G)=\ell(E)+\ell(G\setminus E)=0$. By part *1*.

$$\ell(G)=\int_{g_1^{-1}(G)} g_1'(y)\ell(dy)=0.$$

Define

$$A'=\left\{y\in[\alpha-1,\beta+1]:g_1'(y)>0\right\}.$$

In analogy to A, here $A' = (g_1')^{-1}\big((0,\infty]\big)$ is measurable, and by Theorem 13.4.1 the derivative $g_1' \geq 0$ exists finite a.e. on $[\alpha-1, \beta+1]$, and is summable. As $g_1' \geq 0$ on $[\alpha-1, \beta+1]$ and $g_1^{-1}(G) \cap A' \subseteq g_1^{-1}(G)$, the integral's monotonicity in the integration domain implies

$$\int_{g_1^{-1}(G)\cap A'} g_1'(y)\ell(dy) \leq \int_{g_1^{-1}(G)} g_1'(y)\ell(dy) = 0.$$

Since $g_1' > 0$ on $g_1^{-1}(G) \cap A'$, then $\ell\big(g_1^{-1}(G) \cap A'\big) = 0$. On the other hand $E \subseteq G$, and so $g^{-1}(E) \cap A \subseteq g_1^{-1}(G) \cap A'$, whence $g^{-1}(E) \cap A$ is a subset in the zero-measure set $g_1^{-1}(G) \cap A'$ in the complete measure space $\big([\alpha-1, \beta+1], \mathscr{L}, \ell\big)$. Therefore $g^{-1}(E) \cap A$ is L-measurable of zero measure.

Part 3. Take $E \subseteq [a, b]$ L-measurable with $\ell(E) \in (0, \infty)$. As in part *2.*, there is a $G \in G_\delta$, $E \subseteq G \subseteq (a-1, b+1)$, such that $\ell(G \setminus E) = 0$, so $\ell(G) = \ell(E)$ up to the zero-measure set $(G \setminus E)$. Applying then part *1.* we obtain

$$\ell(E) = \ell(G) = \int_{g_1^{-1}(G)} g_1'(y)\ell(dy). \tag{14.2.4}$$

Clearly $\big(g_1^{-1}(G)\big) \cap A' \subseteq g_1^{-1}(G)$. But $g_1'(y) \geq 0$ a.e. on $[\alpha-1, \beta+1]$, and $g_1'(y) > 0$ on $\big(g_1^{-1}(G)\big) \cap A'$, therefore $g_1'(y) = 0$ on $\big(g_1^{-1}(G)\big) \setminus A'$. Consequently

$$\int_{\left(g_1^{-1}(G)\right)\cap A'} g_1'(y)\ell(dy) = \int_{g_1^{-1}(G)} g_1'(y)\ell(dy).$$

The above, together with Eq. (14.2.4), produces

$$\ell(E) = \ell(G) = \int_{g_1^{-1}(G)} g_1'(y)\ell(dy) = \int_{g_1^{-1}(G)\cap A'} g_1'(y)\ell(dy). \tag{14.2.5}$$

As $(G \setminus E) \subseteq [a-1, b+1]$, $(G \setminus E)$ is L-measurable of zero measure, and then part *2.* tells $g_1^{-1}(G \setminus E) \cap A'$ is L-measurable, and $\ell\big(g_1^{-1}(G \setminus E) \cap A'\big) = 0$. Moreover $E = G \setminus (G \setminus E)$, so $g_1^{-1}(E) = g_1^{-1}(G) \setminus g_1^{-1}(G \setminus E)$ and

$$g_1^{-1}(E) \cap A' = [g_1^{-1}(G) \cap A'] \setminus [g_1^{-1}(G \setminus E) \cap A']. \tag{14.2.6}$$

As G is $\mathscr{B}$-measurable, the measurability of g_1 forces $g_1^{-1}(G)$ to be L-measurable.[1] Hence $g_1^{-1}(G) \cap A'$ is L-measurable, and so in Eq. (14.2.6) the set $g_1^{-1}(E) \cap A'$ is L-measurable. As, moreover, $\ell\big(g_1^{-1}(G \setminus E) \cap A'\big) = 0$, then

$$\ell(g_1^{-1}(E) \cap A') = \ell(g_1^{-1}(G) \cap A').$$

[1] The L-measurability of g implies $g^{-1}(E) \in \mathscr{L}(\mathbb{R})$ for every $E \in \mathscr{B}(\mathbb{R})$. Conversely, $E \in \mathscr{L}(\mathbb{R})$ could also satisfy $g^{-1}(E) \notin \mathscr{L}(\mathbb{R})$. In this respect see the discussion at the end of Example 10.5.1.

Because $g_1^{-1}(E) \cap A' \subseteq g_1^{-1}(G) \cap A'$ and $g_1' \geq 0$, the sets' equal measures imply

$$\int_{g_1^{-1}(E)\cap A'} g_1'(y)\ell(dy) = \int_{g_1^{-1}(G)\cap A'} g_1'(y)\ell(dy).$$

Inserting this result in (14.2.4) produces

$$\begin{aligned}\ell(E) &= \int_{g_1^{-1}(E)\cap A'} g_1'(y)\ell(dy) = \int_{g^{-1}(E)\cap A} g'(y)\ell(dy)\\ &= \int_{[\alpha,\beta]} \varphi_{g^{-1}(E)}(y)\varphi_A(y)g'(y)\ell(dy) = \int_{[\alpha,\beta]} \varphi_{g^{-1}(E)}(y)g'(y)\ell(dy),\end{aligned}$$

i.e. Eq. (14.2.1). ■

Theorem 14.2.2 (Integration by substitution) *Let $f : [a, b] \mapsto \mathbb{R}$ be measurable and g absolutely continuous, strictly increasing, and with $g(\alpha) = a$, $g(\beta) = b$. Then the map $f(g)g'$ is summable on $[\alpha, \beta]$ and*

$$\int_{[a,b]} f(x)\ell(dx) = \int_{[\alpha,\beta]} f\big(g(y)\big)g'(y)\ell(dy).$$

Proof We begin by supposing $f(x) = \varphi_E(x)$, where E is an L-measurable subset in $[a, b]$. Part *3.* in Lemma 14.2.1 gives

$$\begin{aligned}\int_{[a,b]} f(x)\ell(dx) &= \int_{[a,b]} \varphi_E(x)\ell(dx) = \ell(E) = \int_{[\alpha,\beta]} \varphi_{g^{-1}(E)}(y)g'(y)\ell(dy)\\ &= \int_{[\alpha,\beta]} \varphi_E\big(g(y)\big)g'(y)\ell(dy) = \int_{[\alpha,\beta]} f\big(g(y)\big)g'(y)\ell(dy).\end{aligned}$$

By linearity, this equality is valid when f is simple and non-negative, since any such is a linear combination of indicator functions.

Now assume $f : [a, b] \mapsto \mathbb{R}$ non-negative and measurable. Theorem 10.4.1 guarantees there is an increasing sequence of non-negative, simple measurable maps $\{s_n(x)\}_{n\geq 1}$ with limit f on $[a, b]$. As g is absolutely continuous and strictly increasing, $g' \geq 0$ a.e. on $[\alpha, \beta]$. We prolong it as $g' = 0$ on the zero-measure set, so that $g' \geq 0$ on the entire $[\alpha, \beta]$ and thus

$$\lim_{n\to\infty} \big[s_n(g(y))\big]g'(y) = f(g(y))g'(y).$$

Beppo Levi's monotone convergence theorem yields

$$\lim_{n\to\infty} \int_{[a,b]} s_n(x)\ell(dx) = \int_{[a,b]} f(x)\ell(dx)$$

and so

$$\lim_{n\to\infty}\int_{[\alpha,\beta]}\big[s_n(g(y))\big]g'(y)\ell(dy)=\int_{[\alpha,\beta]}f(g(y))g'(y)\ell(dy).$$

As $s_n(x)$ is a non-negative simple map, from what we have proved for simple functions we obtain

$$\int_{[a,b]}s_n(x)\ell(dx)=\int_{[\alpha,\beta]}\big[s_n(g(y))\big]g'(y)\ell(dy)$$

for any $n\in\mathbb{N}$. Therefore

$$\int_{[a,b]}f(x)\ell(dx)=\int_{[\alpha,\beta]}f(g(y))g'(y)\ell(dy).$$

The theorem is then proved for f non-negative and measurable, and hence in general because $f=f^+-f^-$. ■

14.3 Foundations of the Theory of Discrete and Continuous Random Variables

Random variables (RV) are typically sorted in two large classes: discrete random variables and continuous random variables.

We have seen the function F_X has a jump at x iff $P(X(\omega)=x)>0$ (Proposition 10.6.5). Since a monotone function $F:\mathbb{R}\mapsto\mathbb{R}$ is only allowed a countable (at most) number of jump discontinuities, for every random variable X the set

$$\big\{a\in\mathbb{R}:P(X(\omega)=a)>0\big\}$$

is at most countable. This observation leads to the following notion.

Definition 14.3.1 (*Discrete and continuous random variables*) Relatively to a **partition** of the sample space Ω, a random variable X is called:

1. discrete, if it has finite or countable (discrete) range $\{x_i\}_{i\geq 1}\subseteq\mathbb{R}$ such that

$$P(X(\omega)=x_i)=P(A_i)=p_i\geq 0\ \text{ and }\ P\left(\bigcup_i A_i\right)=\sum_i P(A_i)=\sum_i p_i=1$$

   for every $x_i\in\{x_i\}_{i\geq 1}$, where $A_i=\big\{\omega\in\Omega:X(\omega)=x_i\big\}$;
2. continuous, if its range is a (bounded or unbounded) interval $I\subseteq\mathbb{R}$ such that

$$P(X(\omega)=x)=0,\ \ P(X(\omega)\leq x)\geq 0\ \text{ and }\ P(-\infty<X(\omega)<\infty)=1$$

for every $x \in I \subseteq \mathbb{R}$. ◇

From the definition of discrete RV we deduce:

- $A_i = \{\omega \in \Omega : X(\omega) = x_i\} \in \mathcal{S}(\Omega),\ \forall i = 1, 2, \ldots;$
- $A_i \cap A_j = \emptyset$ for every $i \neq j$;
- $P\left(\bigcup_i A_i\right) = \sum_i P(A_i) = \sum_i p_i = 1$, where $\{p_i\}_{i\geq 1}$ is the probability distribution of the discrete random variable X.

For a discrete RV to be well defined we must necessarily have $p_i = P(A_i) \geq 0$ for every i and $\sum_i p_i = 1$.

A discrete RV is determined once we know its range values $x_1, x_2, \ldots$ and the latter's probabilities of occurrence $p_1 = P(A_1)$, $p_2 = P(A_2), \ldots$ A discrete RV may then be represented in the following form

$$X = \begin{pmatrix} x_1, x_2, \ldots \\ p_1, p_2, \ldots \end{pmatrix}.$$

It may happen that a discrete random variable X takes not points x_i, but intervals $\{I_i\}_{i\geq 1}$ (at most countably many) as values, on which the probability distribution concentrates a fraction of the unit mass, assigning to each a value p_i.

This occurs when, for practical purposes, one discretises a RV whose range is a 'continuum' of real numbers (read: an uncountable set). Think for example of the weight of a randomly chosen person in a given population. The possible values of such RV form a set of positive real numbers. The working practice is to decide a priori to round off weights in kilos to the first decimal, so that the RV becomes discrete because it takes values in a finite number of range intervals, between a minimum weight and a maximum weight. To each interval, e.g. (75.500, 75.599 kg), we assign the probability p_i that the weight falls in it. In the specific case at hand the RV is made discrete (called 'discretisation' of a continuous RV) by allowing only finitely many possible values, as for honest discrete RVs. Said discretisation is also employed to compute the relative frequency of a continuous RV.

We know that the collection of probability values p_i forms the probability distribution of a discrete random variable X. We have also already mentioned the cumulative distribution function of such X is the tool that detects the probability distribution on the sample space Ω of the RV. In this case it is called discrete distribution function and is defined as follows.

Definition 14.3.2 (*Discrete distribution functions*) The step function $F_X : \mathbb{R} \mapsto [0, 1]$ defined by

$$P(X \leq x) = F_X(x) = \sum_{x_i \leq x} p_i \geq 0 \quad \text{and} \quad P(-\infty < X < \infty) = \sum_i p_i = 1$$

is the (cumulative) distribution function of the discrete RV $X = \begin{pmatrix} x_1, x_2, \ldots \\ p_1, p_2, \ldots \end{pmatrix}$. ◇

We might want to do the same for continuous random variables and define probability distributions P_X through the corresponding cumulative distribution functions $F_X(x)$. Here, though, we cannot use the procedure seen for discrete RVs, because there is no real value of a continuous X at which F_X has probability bigger than 0.

One possibility to solve the problem, as seen, is to discretise a continuous RV. Besides that, there are two situations regarding the identification of P_X.

1. The probability distribution P_X does not concentrate fractions of the unit mass in any point, but it spreads them over a subset of $\mathbb{R}$ of positive measure. The cumulative distribution function F_X is absolutely continuous, and there exists ℓ-a.e. on $\mathbb{R}$ a finite function f_X such that

$$F_X(x) = \int_{(-\infty,x]} f_X(t)\ell(dt).$$

   This f_X is non-negative (since F_X is increasing), summable, and $F'_X = f_X$, ℓ-a.e. on $\mathbb{R}$. It is called probability density function (or just density) because if the derivative f_X exists at x, the mass contained in $(x + \Delta x)$ is precisely $f(x)\Delta x$ up to infinitesimals of order higher than one in Δx:

$$\begin{aligned} f(x) = F'_X(x) &= \lim_{\ell(\Delta x)\to 0} \frac{F_X(x+\Delta x) - F_X(x)}{\Delta x} \\ &= \lim_{\ell(\Delta x)\to 0} \frac{P(x < X \le x + \Delta x)}{\Delta x} \\ \Leftrightarrow\ P(x < X \le x + \Delta x) &= f(x)\Delta x + O(\Delta(x)). \end{aligned}$$

   $O(\Delta(x))$ gathers terms that are infinitesimal as $\ell(\Delta x) \to 0$;
2. The probability distribution P_X does not concentrate parts of the unit mass in any point, but scatters them over a zero-measure set. This is the situation in which F_X is a singular function for which $F'_X = 0$ ℓ-a.e. on $\mathbb{R}$.
   This is obviously a limiting situation with a purely theoretical interest, and hence not so relevant in concrete cases.

Based on these circumstances we may say a cumulative distribution function F_X of a RV is, in general, a convex linear combination

$$F_X(x) = a_1 F_X^1(x) + a_2 F_X^2(x) + a_3 F_X^3(x),$$

where $a_i \ge 0$, $\sum_i a_i = 1$, and F_X^i are distribution functions of three types: discrete, absolutely continuous, and singular.

Notice that X is a continuous RV iff F_X is continuous on $\mathbb{R}$, whilst it is a discrete RV iff the sum of the jumps of the corresponding distributions F_X equals 1. Moreover, if F_X has jumps the RV X is not necessarily a discrete one. To see this we can take $\mathbb{Q}$ as set where X has jumps. In this case $\sum_i p_i = 0$.

Definition 14.3.3 (*Distribution function of absolutely continuous RVs*) Given a random variable X, suppose there exists, ℓ-almost everywhere on $\mathbb{R}$, a non-negative summable map f_X such that

$$P(X \leq x) = F_X(x) = \int_{(-\infty,x]} f_X(t)\ \ell(dt). \tag{14.3.1}$$

Then F_X is called the (cumulative) distribution function of the absolutely continuous random variable X. The map f_X is the probability density function of X, and the differential form $f_X(x)\ell(dx)$ is called elementary probability of X. ◇

Equation (14.3.1) has two consequences—by the way satisfied by discrete RVs as well—that ensure an absolutely continuous RV X is well defined:

$$f_X \geq 0 \quad \text{and} \quad \lim_{x\to\infty} F_X(x) = \int_{\mathbb{R}} f_X(t)\ell(dt) = 1. \tag{14.3.2}$$

Any non-negative map f whose integral on $\mathbb{R}$ equals $\alpha \in (0, \infty)$ can be the density of a continuous RV X, because f/α fulfils (14.3.2).

Proposition 14.3.1 (Probability density of functions of RVs) *Let X be an absolutely continuous RV with density f_X, and $\phi : \mathbb{R} \mapsto \mathbb{R}$ a strictly monotone, differentiable function. Then $Y = \phi(X)$ is a random variable with density function*

$$f_Y(y) = F'_Y(y) = f_X\big(\phi^{-1}(\{y\})\big) \cdot \left|\frac{d\phi^{-1}(\{y\})}{dy}\right|.$$

Proof That Y is a RV descends directly from Proposition 10.6.1.

Assume ϕ is strictly increasing, so ϕ^{-1} is well defined: for every $y \in Y$ there exists a unique x such that $\phi(x) = y$, i.e. $x = \phi^{-1}(\{y\})$. Then

$$F_Y(y) = P(Y \leq y) = P(\phi(X) \leq y) = P(X \leq \phi^{-1}(\{y\})) = F_X(\phi^{-1}(\{y\})) = F_X(x).$$

But F_X is absolutely continuous, so

$$f_Y(y) = \frac{dF_X(x)}{dy} = F'_X(x)\frac{dx}{dy} = f_X\big(\phi^{-1}(\{y\})\big)\frac{d\phi^{-1}(\{y\})}{dy}.$$

If ϕ is strictly decreasing $d\phi^{-1}(\{y\})/dy$ is negative, and the same argument returns

$$f_Y(y) = -f_X\big(\phi^{-1}(\{y\})\big)\frac{d\phi^{-1}(\{y\})}{dy}.$$

Remembering that $f_Y \geq 0$, the claim follows. ■

To determine the probability distribution of a RV X, apart from finding the distribution function F_X it is often useful to compute certain specific quantities, like the expectation, the variance and other characteristic indicators extracted from the RV's so-called moments. This is necessary, for instance, when we want to compare different experiments, or when we seek a global picture of our study based on the characteristic quantities of the distribution P_X.

For example, suppose we want to compute the expected value of a discrete RV. Then we use the formula

$$E(X) = \sum_{i \in I} p_i x_i, \tag{14.3.3}$$

provided the series $\sum_{i\in I} |x_i| < \infty$, i.e. $\sum_{i\in I} x_i$ is absolutely convergent.

Similarly, if we wish to find the expectation of an absolutely continuous RV X we have to use, in analogy to the discrete situation, the following expression

$$E(X) = \int_{\mathbb{R}} x f(x)\, \ell(dx), \tag{14.3.4}$$

provided the integral exists and is finite.

As proven in general in the next proposition, formulas (14.3.3), (14.3.4) (and others, too) can be recovered from a single theoretical framework, consisting of Propositions 9.6.1, 9.6.2, 10.6.3 and Remark 11.3.6.

Proposition 14.3.2 (Integrals of measurable functions of RV) *Let $(\Omega, \mathcal{S}(\Omega), P)$ be a probability space and $X : \Omega \mapsto \mathbb{R}$ a RV with distribution function F_X. Suppose $g : \mathbb{R} \mapsto \mathbb{R}$ is a measurable map such that $g(X(\omega))$ is summable with respect to P. Then*

$$\int_\Omega g(X(\omega)) dP(\omega) = \int_{\mathbb{R}} g(x) dF_X(x), \tag{14.3.5}$$

where the left side is a Lebesgue integral, the right side is Lebesgue–Stieltjes.

Proof If g is the characteristic function φ_B, it suffices to check (14.3.5) for $B = (a, b] \in \mathscr{B}(\mathbb{R})$, so

$$\varphi_B(X(\omega)) = \begin{cases} 1 & \text{if } x \in (a, b] \\ 0 & \text{if } x \notin (a, b] \end{cases}$$

i.e.

$$\int_\Omega \varphi_B\big(X(\omega)\big) dP(\omega) = P\big(\{\omega \in \Omega : X(\omega) \in B\}\big) = \int_{\mathbb{R}} \varphi_B(x) dF_X(x)$$
$$= \int_{(a,b]} dF_X(x) = F_X(b) - F_X(a).$$

By linearity the above result extends to non-negative simple measurable functions.

Now suppose $g : \mathbb{R} \mapsto [0, \infty)$ is measurable, so by Theorem 10.4.1 there is an increasing sequence $\{s_n(x)\}_{n\geq 1}$ of simple measurable functions with limit $g(x)$. Moreover, the increasing sequence $\{s_n(X(\omega))\}_{n\geq 1}$ of non-negative simple RVs converges to the RV $g(X(\omega))$, measurable with respect to P. Then we have

$$\int_\Omega g(X(\omega))dP(\omega) = \lim_{n\to\infty} \int_\Omega s_n(X(\omega))dP(\omega) = \lim_{n\to\infty} \int_{\mathbb{R}} s_n(x)dF_X(x)$$
$$= \int_{\mathbb{R}} g(x)dF_X(x),$$

as required. The general statement follows immediately from the summability of g and $g = g^+ - g^-$. ■

This proposition allows to define, whenever existing and finite, not only the expectation, but any indicator characterising the RV. If we specify the type of function g we can compute the so-called moments of any random variable X, which are then used to find the other significant indicators.

The moments that are useful to determine all the indicators characterising a RV are the first four. From these we may compute the expected value of X, the variance, the skewness and the kurtosis, which measure of the shape of X.

Definition 14.3.4 (*Moments of a random variable*) Let X be a RV with distribution function F_X. **Assuming the following exist and are finite**, we call:

1. nth (raw) moment of X the mean value of the nth power of X:

$$m_n = E(X^n) = \int_{\mathbb{R}} x^n dF_X(x) = \begin{cases} \int_{\mathbb{R}} x^n f(x)\ell(dx) & \text{if } X \text{ is absolutely continuous} \\ \sum_{i=1}^{\infty} x_i^n p_i & \text{if } X \text{ is discrete.} \end{cases}$$

   m_1 is the mean value of X, often simply written m, while $m_0 = 1$;
2. nth absolute moment of X the mean value of the nth power of $|X|$:

$$|m|_n = E(|X|^n) = \int_{\mathbb{R}} |x|^n dF_X(x) = \begin{cases} \int_{\mathbb{R}} |x|^n f(x)\ell(dx) & \text{if } X \text{ is absolutely continuous} \\ \sum_{i=1}^{\infty} |x_i|^n p_i & \text{if } X \text{ is discrete;} \end{cases}$$

3. nth central moment of X the mean value of the nth power of $(X - E(X))$:

$$\bar{m}_n = E((X - E(X))^n) = \int_{\mathbb{R}} (x - m)^n dF_X(x) = \begin{cases} \int_{\mathbb{R}} (x - m)^n f(x)\ell(dx) & \text{if } X \text{ is a.c.} \\ \sum_{i=1}^{\infty} (x_i - m)^n p_i & \text{if } X \text{ is discrete.} \end{cases}$$

Note that $E(X - E(X)) = 0$ (zero expectation of the first central moment), because $E(X)$ is constant so $E(X) - E(E(X)) = 0$;

4. variance of X the mean value of the square of $(X - E(X))$:

$$\bar{m}_2 = \mathrm{Var}(X) = \sigma^2(X) = \int_{\mathbb{R}} (x-m)^2 dF_X(x) = \begin{cases} \displaystyle\int_{\mathbb{R}} (x-m)^2 f(x)\ell(dx) & \text{if } X \text{ is a.c.} \\ \displaystyle\sum_{i=1}^{\infty} (x_i - m)^2 p_i & \text{if } X \text{ is discrete.} \end{cases}$$

Finally, the standard deviation of X is

$$\sqrt{\mathrm{Var}(X)} = \sigma(X). \qquad \diamond$$

Let us make a few comments. If the variance of a RV is zero, the RV assumes only one value with probability 1, namely the expected one. Such a RV is called degenerate.

Even-order absolute moments coincide with the corresponding ordinary moments. Lastly, $E(|X|^n) < \infty \Rightarrow |E(X^n)| < \infty$ for every $n \in \mathbb{N}$. Hence if $|m|_n$ exists and is finite then m_n exists and is finite. As prescribed by the next proposition, furthermore, all moments m_k, $|m|_k$ of smaller order exist and are finite.

Proposition 14.3.3 (Existence of moments of lower order than a given moment) *If the nth absolute moment of a RV (exists and) is finite, then all kth raw and absolute moments (exist and) are finite, for any $k < n$.*

Proof If

$$\int_{\mathbb{R}} |x|^n dF_X(x) < \infty$$

then m_n exists and is finite, and so

$$\int_{(-\infty,-1)} |x|^n dF_X(x) + \int_{(1,\infty)} |x|^n dF_X(x) < \infty.$$

But $|x|^k < |x|^n$ whenever $x < -1$ or $x > 1$ and $0 < k < n$, so

$$\int_{(-\infty,-1)} |x|^k dF_X(x) + \int_{(1,\infty)} |x|^k dF_X(x) < \int_{(-\infty,-1)} |x|^n dF_X(x) + \int_{(1,\infty)} |x|^n dF_X(x) < \infty,$$

and then

$$\int_{\mathbb{R}} |x|^k dF_X(x) < \infty$$

for $0 < k < n$. ■

Proposition 14.3.4 (Vanishing of mass fraction at infinity) *Let F_X be the distribution function of a random variable X whose expectation $E(X)$ is finite. Then*

$$\lim_{x\to\infty} x\big(1 - F_X(x)\big) = 0 \quad \textit{and} \quad \lim_{x\to-\infty} x\ F_X(x) = 0$$

Proof As $E(X)$ is finite

$$\int_{\mathbb{R}} |x| dF_X(x) < \infty,$$

so for $x > 0$ we have

$$\int_{(x,\infty)} t\ dF_X(t) \geq x \int_{(x,\infty)} dF_X(t) = x\big(1 - F_X(x)\big) \geq 0.$$

Hence

$$0 \leq \lim_{x\to\infty} x\big(1 - F(x)\big) \leq \lim_{x\to\infty} \int_{(x,\infty)} t\ dF_X(t) = 0$$

and so $\lim_{x\to\infty} x\big(1 - F(x)\big) = 0$. For $x < 0$, we similarly have

$$\int_{(-\infty,x]} t\ dF_X(t) \leq x \int_{(-\infty,x]} dF_X(t) = x\ F_X(x) \leq 0,$$

so

$$0 \geq \lim_{x\to-\infty} x\ F_X(x) \geq \lim_{x\to-\infty} \int_{(-\infty,x]} t\ dF_X(t) = 0$$

and then $\lim_{x\to-\infty} x\ F_X(x) = 0$. ■

Proposition 14.3.5 (Schwarz inequality) *If the random variables X, Y admit moments of order 2, the random variable XY has first moment, and*

$$E(|XY|) \leq \sqrt{E(X)^2 E(Y)^2}.$$

Proof Define the random variable $Z = (X - \alpha Y)^2$, with $\alpha \in \mathbb{R}$, so that

$$E(Z) = E(X)^2 - 2\alpha E(XY) + \alpha^2 E(Y)^2.$$

As $Z \geq 0$, then $E(Z) \geq 0$ for every $\alpha \in \mathbb{R}$, and then

$$E(X)^2 - 2\alpha E(XY) + \alpha^2 E(Y)^2 \geq 0.$$

Choose

$$\alpha = \frac{E(XY)}{E(Y)^2},$$

whence

$$\begin{aligned} 0 &\le E(X)^2 - 2\alpha E(XY) + \alpha^2 E(Y)^2 \\ &= E(X)^2 - 2\frac{E(XY)}{E(Y)^2}E(XY) + \left(\frac{E(XY)}{E(Y)^2}\right)^2 E(Y)^2 \\ &= E(X)^2 - \frac{[E(XY)]^2}{E(Y)^2} \\ \Leftrightarrow\ [E(XY)]^2 &\le E(X)^2 E(Y)^2 \\ \Leftrightarrow\ E(|XY|) &\le \sqrt{E(X)^2 E(Y)^2}. \end{aligned}$$

■

Proposition 14.3.6 (Markov inequality) *If a random variable X with expectation $m = E(X)$ only assumes positive values, $P(X > 0) = 1$, then for every $\alpha > 0$*

$$P(X > \alpha \cdot E(X)) \le \frac{1}{\alpha}. \tag{14.3.6}$$

Proof If $0 < \alpha \le 1$ then $1/\alpha \ge 1$ and the inequality is trivial. Suppose $\alpha > 1$. As X is positive,

$$m = \int_{(0,\infty)} x\, dF(x) > 0.$$

Then

$$\begin{aligned} m &= \int_{(0,\alpha m)} x\, dF(x) + \int_{(\alpha m,\infty)} x\, dF(x) \ge \int_{(\alpha m,\infty)} x\, dF(x) \\ &\ge \alpha m \int_{(\alpha m,\infty)} dF(x) = \alpha m\big(1 - F(\alpha m)\big), \end{aligned}$$

and so

$$1 - P(X \le \alpha \cdot m) \le \frac{1}{\alpha}.$$

■

Next up is the Bienaymé–Chebyshev inequality, according to which given an expectation m, a standard deviation σ and a number $\alpha > 0$, the probability a RV lands between $m \pm \alpha\sigma$ is bigger than $1 - 1/\alpha^2$. Hence the larger the interval (read: σ) the more certain the event.

Proposition 14.3.7 (Bienaymé–Chebyshev inequality) *Suppose a random variable X has finite second moment, so in particular $E(X)$ and $E(X-m)^2=\sigma^2$ are well defined. Then*

$$P\big(|X-m|>\alpha\,\sigma\big)\le\frac{1}{\alpha^2}.$$

Proof Just replace X in Markov's inequality with $|X-m|^2$ and $\alpha>0$ with α^2:

$$P\big(|X-m|^2>\alpha^2E\big(|X-m|^2\big)\big)\le\frac{1}{\alpha^2}\Leftrightarrow P\big(|X-m|>\alpha\sigma\big)\le\frac{1}{\alpha^2}.$$ ■

What this means is that for any RV X with $E(X)=m$ and $\mathrm{Var}(x)=\sigma^2<\infty$, and any $\alpha>0$,

$$P\big(|X-m|<\alpha\sigma\big)\ge 1-\frac{1}{\alpha^2}.$$

The Bienaymé–Chebyshev inequality holds for any $\alpha>0$, but only the case $\alpha>1$ is useful. When $\alpha=1$ the inequality is trivial

$$P\big(|X-m|<\alpha\sigma\big)\ge 0,$$

and for $\alpha<1$ it states the even more obvious fact that probabilities are bigger than negative numbers. If we take $\alpha\sigma=k>0$, the inequality reads

$$P\big(|X-m|<k\big)\ge 1-\frac{\sigma^2}{k^2}.$$

To finish we shall discuss a few properties of discrete RVs.
First of all, the expectation $E(\cdot)$ is a linear operator:

$$E(\alpha X+\beta Y)=\alpha E(X)+\beta E(Y),\quad \text{for any } \alpha,\beta\in\mathbb{R}.$$

Proposition 14.3.8 (Linearity of a discrete RV's expected value) *Given random variables X, Y and $\alpha,\beta\in\mathbb{R}$, suppose $E(X)$, $E(Y)$ exist finite. Then so does*

$$E(\alpha X+\beta Y)=\alpha E(X)+\beta E(Y).$$

Proof Let us call the events of X, Y respectively $A_i=\{\omega\in\Omega: X(\omega)=x_i\}$, $i\in I$ and $B_j=\{\omega\in\Omega: Y(\omega)=y_j\}$, $j\in J$, where

$$\bigcup_{i\in I}A_i=\Omega=\bigcup_{j\in J}B_j.$$

On Ω we can define events

$$C_{i,j} = \{\omega \in \Omega : X(\omega) = x_i, Y(\omega) = y_j\}$$

where

$$\sum_{i\in I} P(C_{i,j}) = P(B_j) \quad \text{and} \quad \sum_{j\in J} P(C_{i,j}) = P(A_i).$$

Then

$$\begin{aligned}
E(\alpha X + \beta Y) &= \sum_{i\in I, j\in J} (\alpha x_i + \beta y_j) P(C_{i,j}) \\
&= \sum_{i\in I, j\in J} \alpha x_i P(C_{i,j}) + \sum_{i\in I, j\in J} \beta y_j P(C_{i,j}) \\
&= \alpha \left[\sum_{i\in I} x_i \left(\sum_{j\in J} P(C_{i,j}) \right) \right] + \beta \left[\sum_{j\in J} y_j \left(\sum_{i\in I} P(C_{i,j}) \right) \right] \\
&= \alpha \left(\sum_{i\in I} x_i P(A_i) \right) + \beta \left(\sum_{j\in J} y_j P(B_j) \right) = \alpha \left(\sum_{i\in I} x_i p_i \right) + \beta \left(\sum_{j\in J} y_j p_j \right) \\
&= \alpha E(X) + \beta E(Y). \qquad \blacksquare
\end{aligned}$$

What we have proved above can naturally be extended by induction to a finite, even countable, number of RVs, but in the latter case the series of expected values $\sum_{i=1}^{\infty} E(|X_i|)$ should converge.

Proposition 14.3.9 (Variance in terms of moments) *For any discrete RV X with finite second moment,*

$$Var(X) = E(X^2) - [E(X)]^2.$$

Proof

$$\begin{aligned}
\text{Var}(X) &= E\big(X - E(X)\big)^2 = E\big(X^2 - 2XE(X) + [E(X)]^2\big) \\
&= E(X^2) - 2E(XE(X)) + E([E(X)]^2) \\
&= \sum_{i\in I} P(X = x_i) x_i^2 - 2 \sum_{i\in I} P(X = x_i) \left(x_i \cdot \sum_{i\in I} P(X = x_i) x_i \right) + \\
&\quad \sum_{i\in I} P(X = x_i) \left(\left[\sum_{i\in I} P(X = x_i) x_i \right]^2 \right) \\
&= \sum_{i\in I} P(X = x_i) x_i^2 - \left(\left[\sum_{i\in I} P(X = x_i) x_i \right]^2 \right)
\end{aligned}$$

$$= E(X^2) - [E(X)]^2. \qquad \blacksquare$$

We recall that two events $A, B \in \Omega$ are said independent if the occurrence of A does not affect the probability of occurrence of B. Equivalently, since $P(A) > 0$,

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(A)P(B)}{P(A)} = P(B). \tag{14.3.7}$$

The definition requires $P(A) > 0$, which is why in general the notion of independent events relies on formula

$$P(A \cap B) = P(A)P(B),$$

valid for $P(A) \geq 0$, instead of Eq. (14.3.7).

Definition 14.3.5 (*Independent discrete random variables*) Two discrete RVs X, Y are called independent if, for every $A_i \in \mathcal{S}(\Omega_X)$, $B_j \in \mathcal{S}(\Omega_Y)$, $i \in I,\ j \in J$,

$$P(A_i \cap B_j) = P(A_i)P(B_j). \qquad \diamond$$

Proposition 14.3.10 (Expected value of product of independent discrete RVs) *If X, Y are independent discrete RVs with finite expectations $E(X)$, $E(Y)$, then $E(XY)$ exists, is finite, and equals*

$$E(XY) = E(X)E(Y).$$

Proof In analogy to Proposition 14.3.8,

$$\begin{aligned} E(XY) &= \sum_{i\in I, j\in J} (x_i y_j) P(C_{i,j}) \\ &= \left[\sum_{i\in I, j\in J} x_i P(C_{i,j})\right]\left[\sum_{i\in I, j\in J} y_j P(C_{i,j})\right] \\ &= \left(\sum_{i\in I} x_i P(A_i)\right)\left(\sum_{j\in J} y_j P(B_j)\right) \\ &= E(X)E(Y). \end{aligned}$$

$\blacksquare$

Definition 14.3.6 (*Conditional expectation of discrete RVs*) Given a discrete RV X and an event B with $P(B) > 0$, the conditional expectation of X with respect to B is

$$E(X|B) = \sum_{i\in I} x_i P(A_i|B) = \sum_{i\in I} x_i P\big((X = x_i)|B\big), \tag{14.3.8}$$

assuming the series $\sum_{i\in I} \big|x_i P\big((X = x_i)|B\big)\big|$ converges. ◇

Proposition 14.3.11 *Let X be a discrete RV with finite expectation $E(X)$ and $\{B_i\}_{i\in I}$ a sequence of pairwise-disjoint events such that $\bigcup_{i\in I} B_i = \Omega$ and $P(B_i) > 0$. Then*

$$E(X) = \sum_{i\in I} P(B_i) E(X|B_i).$$

Proof Take any $A_k \in \mathcal{S}(\Omega)$. Then $A_k = A_k \cap \Omega = A_k \cap \big(\bigcup_{i=1}^{\infty} B_i\big) = \bigcup_{i=1}^{\infty}(A_k \cap B_i)$, and also

$$(A_k \cap B_i) \cap (A_k \cap B_j) = A_k \cap (B_i \cap B_j) = \emptyset.$$

Hence $P(A_k) = \sum_{i=1}^{\infty} P(A_k \cap B_i)$.

Substituting $P(A_k \cap B_i) = P(B_i) P(A_k|B_i)$ above gives

$$\begin{aligned} P(A_k) &= \sum_{i\in I} P(B_i) P(A_k|B_i) \qquad (14.3.9)\\ \Leftrightarrow P(X = x_k) &= \sum_{i\in I} P(B_i) P((X = x_k)|B_i). \end{aligned}$$

By (14.3.8), then,

$$\begin{aligned} E(X) &= \sum_{k\in K} x_k P(X = x_k) = \sum_{k\in K} x_k \left[\sum_{i\in I} P(B_i) P((X = x_k)|B_i)\right] \\ &= \sum_{i\in I} P(B_i) \left[\sum_{k\in K} x_k P((X = x_k)|B_i)\right] = \sum_{i\in I} P(B_i) E(X|B_i). \end{aligned}$$ ■

Definition 14.3.7 (*Almost constant RVs*) A random variable X is almost constant if the event $\{\omega \in \Omega : X(\omega) = K \in \mathbb{R}\}$ is almost sure, i.e. its probability is 1 up to zero-measure sets:

$$P(X = K) = 1.$$ ◇

Proposition 14.3.12 (Variance vanishing) *The variance of a discrete RV X is zero if and only if X is almost constant.*

Proof If X is almost constant it assumes values $\{x_i \neq K\}_{i\in I}$ with probability $P(X \neq K) = 0$ on event

$$B = \{\omega \in \Omega : X(\omega) \neq K\},$$

while it assumes value K with probability $P(X = K) = 1$ on

$$A = \{\omega \in \Omega : X(\omega) = K\},$$

and $A \cap B = \emptyset$. Therefore

$$\begin{aligned} & E(X) = \sum_{i\in I} P(X = x_i)x_i + P(X = K)K = P(X \neq K)\sum_{i\in I} x_i + K = K, \\ \Leftrightarrow \quad & E(X^2) = \sum_{i\in I} P(X = x_i)x_i^2 + P(X = K)K^2 = P(X \neq K)\sum_{i\in I} x_i^2 + K^2 = K^2, \end{aligned}$$

and then

$$\begin{aligned} \mathrm{Var}(X) &= E\big(X - E(X)\big)^2 = E(X - K)^2 \\ &= E(X^2 - 2KX + K^2) = E(X^2) - 2KE(X) + K^2 = E(X^2) - K^2 = 0. \end{aligned}$$

Conversely, suppose

$$\mathrm{Var}(X) = E\big(X - E(X)\big)^2 = \sum_{i\in I} \big[x_i - E(X)\big]^2 P(X = x_i) = 0.$$

By definition $P(X = x_i) \geq 0$ for every $i \in I$, and hence

$$\begin{cases} x_k - E(X) = 0 \Rightarrow P(X = x_k) > 0 & , \forall k \in I_1 \\ x_j - E(X) \neq 0 \Rightarrow P(X = x_j) = 0 & , \forall j \in I_2 \end{cases},$$

where $I_1 \cap I_2 = \emptyset$ and $I_1 \cup I_2 = I$. Consequently $x_k = E(X) = K$ has $P(X = x_k) > 0$ for every $k \in I_1$, while $x_j \neq K$ has $P(X = x_j) = 0$ for every $j \in I_2$. In the end

$$P(X = K) = \sum_{k\in I_1} P(X = x_k) = 1 - \sum_{j\in I_2} P(X = x_j) = 1 - P(X \neq K) = 1. \qquad \blacksquare$$

14.4 A Miscellany of Discrete and Continuous Distributions

Example 14.4.1 (*Binomial random variable*) Suppose we have an urn containing the same number of white and red balls. Five balls are drawn at random, one at a time, and after each extraction we put the draw back in the urn. We want to calculate the probability of picking two whites and three reds, in any order.

First, we find the probability of extracting a white ball:

$$P\big([W][\ ][\ ][\ ][\ ]\big) = \frac{1}{2}.$$

The probability the first ball is white and the second is also white equals

$$P\big([W][W][\ ][\ ][\ ]\big) = P\big([W][\ ][\ ][\ ][\ ]\big) \times P\big([\ ][W][\ ][\ ][\ ]\big) = \left(\frac{1}{2}\right)^2.$$

The probability of drawing three whites and two reds, in this order, is

$$P\big([W][W][W][R][R]\big) = \left(\frac{1}{2}\right)^3 \times \left(\frac{1}{2}\right)^2.$$

But extracting first three whites and then two reds is not the only way to obtain the required unordered outcome. The possible arrangements

$[R][R][W][W][W]$	$[W][R][W][R][W]$
$[R][W][R][W][W]$	$[W][R][W][W][R]$
$[R][W][W][R][W]$	$[W][W][R][R][W]$
$[R][W][W][W][R]$	$[W][W][R][W][R]$
$[W][R][R][W][W]$	$[W][W][W][R][R]$

correspond, as a matter of fact, to the number of **permutations with repetition** of $k \leq n$ elements of one type and $n-k$ of another type[2]:

$$\binom{5}{3} = \frac{5!}{3!2!} = \frac{5!}{3!(5-3)!} = 10.$$

Clearly each outcome has the same probability, e.g.

$$P\big([R][W][W][R][W]\big) = \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2 = P\big([R][R][W][W][W]\big).$$

and so

[2] If k of the n elements are labelled W and $n-k$ are labelled R, the general number of permutations with repetitions

$$P_n^r = \frac{n!}{k_1! \cdot k_2! \cdot k_3! \cdots k_i!}$$

reduces to the number of combinations of k objects out of n

$$P_{k,n-k} = \frac{n!}{k!(n-k)!}.$$

$$P\big([R][W][W][R][W] \text{ or } [R][R][W][W][W]\big) = \left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^2.$$

Eventually

$$P\big([R][R][W][W][W] \text{ or } \ldots \text{ or } [W][W][W][R][R]\big) = \underbrace{\left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^2 + \cdots + \left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^2}_{10 \text{ terms}}$$
$$= \binom{5}{3}\left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^2.$$

In general, then, if we draw n balls and reintroduce the extracted one every time, the probability (also known as the mass) of getting $0 \le k \le n$ white balls equals

$$p_k = P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

We call binomial distribution, or binomial random variable, of order n and parameter p the RV

$$\mathscr{B}(n, p) = \begin{pmatrix} 0 & , & 1 & , & 2 & , \ldots , & n \\ (1-p)^n, & \binom{n}{1} p\,(1-p)^{n-1}, & \binom{n}{2} p^2\,(1-p)^{n-2}, & \ldots , & p^n \end{pmatrix}.$$

The sum of all masses p_k equals

$$\sum_{k=0}^{n} p_k = \underbrace{\sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k}}_{\text{Newton's binomial formula}} = [p + (1-p)]^n = 1.$$

Set $q = 1 - p$ to compute the expectation

$$E(X) = \sum_{k=0}^{n} k p_k = \sum_{k=1}^{n} k \binom{n}{k} p^k q^{n-k} \tag{14.4.1}$$

of the binomial distribution. Differentiating in t the identity

$$(pt + q)^n = \sum_{k=1}^{n} \binom{n}{k} p^k t^k q^{n-k}$$

gives

$$np(pt + q)^{n-1} = \sum_{k=1}^{n} k \binom{n}{k} p^k t^{k-1} q^{n-k}.$$

As $p+q=1$, when $t=1$ we find

$$np = \sum_{k=1}^{n} k \binom{n}{k} p^k q^{n-k},$$

so from (14.4.1) we deduce $E(X) = np$.

Let us compute the binomial variance. Since

$$\mathrm{Var}(X) = E(X^2) - E(X)^2 = E(X(X-1)) + E(X) - E(X)^2$$

the only unknown term is $E(X(X-1))$. We relabel terms by $i = k-2$ and $m = n-2$, so $k = 2, \ldots, n \Rightarrow i = 0, \ldots, n-2 = m$. Then

$$\begin{aligned}
E(X(X-1)) &= \sum_{k=0}^{n} k(k-1)\binom{n}{k} p^k q^{n-k} = \sum_{k=0}^{n} k(k-1)\frac{n!}{k!(n-k)!} p^k q^{n-k} \\
&= \sum_{k=2}^{n} \frac{n!}{(k-2)!(n-k)!} p^k q^{n-k} = n(n-1)p^2 \sum_{k=2}^{n} \frac{(n-2)!}{(k-2)!(n-k)!} p^{k-2} q^{n-k} \\
&= n(n-1)p^2 \sum_{i=0}^{m} \frac{m!}{i!(m-i)!} p^i q^{m-i} = n(n-1)p^2 \sum_{i=0}^{m} \binom{m}{i} p^i q^{m-i} \\
&= n(n-1)p^2 (p+q)^m = n(n-1)p^2,
\end{aligned}$$

and finally

$$\begin{aligned}
\mathrm{Var}(X) &= E(X(X-1)) + E(X) - E(X)^2 \\
&= n(n-1)p^2 + np - n^2p^2 = n^2p^2 - np^2 + np - n^2p^2 \\
&= np - np^2 = np(1-p) = npq.
\end{aligned}$$

★

Example 14.4.2 (*Poisson random variable*) In contrast to many discrete random variables (e.g. the hypergeometric RV, or the negative binomial RV) the Poisson process is not generated by a sequence of Bernoulli trials, but concerns a number of events over a certain time interval in a spacial, social, or other, setting. The events are completely independent in the sense that they occur with constant rate, and same-setting event sequences over non-overlapping time intervals do not interact.

Consider then a time interval $(0, t+\Delta t]$ and suppose we want to register how many events of a certain type occur in a given spatial region. We speak of Poisson distributions when the following key conditions are met:

- the probability that an event occurs in $(0, t+\Delta t]$ is independent of the instant t (stationary Poisson distribution), nor depends on earlier history, i.e. what happened before or at time t;
- between t and $t+\Delta t$ only one event of the type considered can occur, with probability proportional to Δt by a factor λ. The probability further events occur in

that interval is infinitesimal of order bigger than Δt, so that the probability of observing one or more events at time $t + \Delta t$ equals

$$p_{k\geq 1}(t + \Delta t) = \lambda \Delta t + o(\Delta t);$$

- no event of the type examined has occurred at the initial time $t = 0$.

Due to this, we assume no event took place in the system at time 0, and that

$p_{k\geq 1}(t + \Delta t) = \lambda \Delta t + o(\Delta t)$	probability that at least one event occurs in the period $(t, t + \Delta t]$
$p_{k>1}(t + \Delta t) = o(\Delta t)$	probability that $k \geq 2$ events occur in $(t, t + \Delta t]$
$p_0(t + \Delta t) = 1 - \lambda \Delta t - o'(\Delta t)$	probability that no event occurs in $(t, t + \Delta t]$,

where $o'(\Delta t)$ gathers all infinitesimals. We seek the probability that within $(t, t + \Delta t]$ exactly n system events occur, i.e.

$$p_n(t + \Delta t).$$

For this let us indicate by S_n, S_{n-1}, S_{n-k} the system's state by the time n, $n - 1$, $n - k$ events have already occurred, respectively. In the following table infinitesimals $o(\Delta t)$ are added, rather than subtracted, since they are irrelevant, in practice:

state at t	events at $t + \Delta t$	state at $t + \Delta t$	probability
S_n	0	S_n	$p_n(t)[1 - \lambda \Delta t + o'(\Delta t)]$
S_{n-1}	1	S_n	$p_{n-1}(t)[\lambda \Delta t + o(\Delta t)]$
S_{n-k}	k	S_n	$p_{n-k}(t)[o(\Delta t)]$

The probability the system at time $t + \Delta t$ is in state S_n (n events have been witnessed) is

$$p_n(t + \Delta t) = p_n(t)[1 - \lambda \Delta t] + p_{n-1}(t)[\lambda \Delta t] + o^*(\Delta t),$$

where $o^*(\Delta t)$ collects all infinitesimal, and hence

$$\frac{p_n(t + \Delta t) - p_n(t)}{\Delta t} = -\lambda p_n(t) + \lambda p_{n-1}(t) + \frac{o^*(\Delta t)}{\Delta t}.$$

Since $o^*(\Delta t)$ is a higher-order infinitesimal than Δt, as the latter tends to 0 the ratio $o^*(\Delta t)/\Delta t$ goes to zero, too. Hence the above right-hand side converges, and

$$\frac{dp_n(t)}{dt} = -\lambda p_n(t) + \lambda p_{n-1}(t). \tag{14.4.2}$$

Thus we have tied $p_n(t)$ to its derivative and the preceding term in the sequence. This is a recurrence differential equation.

To solve it we first determine a few values of $p_n(t)$ using the assumptions. In particular

- $p_0(0) = 1$, saying it is certain that no events occurred at time 0;
- $p_0(t) = 0$, meaning it is impossible no event occurred at time t;
- $p_{-n}(\cdot) = 0$, expressing the fact we admit only positive integers n.

We may solve (14.4.2) for $p_0(t)$:

$$\frac{dp_0(t)}{dt} = -\lambda p_0(t) \Leftrightarrow \frac{1}{p_0(t)}\frac{dp_0(t)}{dt} = -\lambda,$$

integrating which

$$\begin{aligned}\int \frac{1}{p_0(t)}\frac{dp_0(t)}{dt}\cdot dt &= \int -\lambda dt \\ &\Leftrightarrow \ln p_0(t) = -\lambda t + A_0.\end{aligned}$$

A_0 is to all effects the initial condition of the ODE, and actually $A_0 = 0$ because nothing happened at the very beginning of the process, so

$$\ln(p_0(t)) = (\ln p_0(t))\ln e = \ln\left(e^{\ln p_0(t)}\right) = \ln\left(e^{-\lambda t}\right) \Leftrightarrow p_0(t) = e^{-\lambda t}.$$

The probability of witnessing an event at time t according to Eq. (14.4.2) is

$$\begin{aligned}\frac{dp_1(t)}{dt} &= \lambda p_0(t) - \lambda p_1(t) \\ \Leftrightarrow \frac{dp_1(t)}{dt} &+ \lambda p_1(t) = \lambda p_0(t) \\ \Leftrightarrow \frac{dp_1(t)}{dt} &+ \lambda p_1(t) = \lambda e^{-\lambda t}.\end{aligned}$$

Multiplying both sides by $e^{\lambda t}$ gives

$$\frac{dp_1(t)}{dt}e^{\lambda t} + \lambda e^{\lambda t} p_1(t) = \lambda.$$

Note $\frac{dp_1(t)}{dt}e^{\lambda t} + \lambda e^{\lambda t} p_1(t)$ is the derivative of $p_1(t)\cdot e^{\lambda t}$, so integrating produces

$$\begin{aligned}\int \left(\frac{dp_1(t)}{dt}e^{\lambda t} + \lambda e^{\lambda t} p_1(t)\right)dt &= \int \lambda dt \\ &\Leftrightarrow p_1(t)e^{\lambda t} = \lambda t + A_1,\end{aligned}$$

and the initial condition $p_1(0) = 0$ forces $A_1 = 0$, and then $p_1(t)e^{\lambda t} = \lambda t$. Multiplying by $e^{-\lambda t}$,

$$p_1(t) = \frac{\lambda t}{1!}\, e^{-\lambda t}.$$

We may assume by induction that the probability of occurrence of $n-1$ events at time t equals

$$p_{n-1}(t) = \frac{\lambda^{n-1} t^{n-1}}{(n-1)!}\, e^{-\lambda t}.$$

Then, by (14.4.2), the probability of having n events at time t is

$$\begin{aligned} \frac{dp_n(t)}{dt} &= \lambda p_{n-1}(t) - \lambda p_n(t) \\ \Leftrightarrow \frac{dp_n(t)}{dt} + \lambda p_n(t) &= \lambda p_{n-1}(t) \\ \Leftrightarrow \frac{dp_n(t)}{dt} + \lambda p_n(t) &= \frac{\lambda^{n-1} t^{n-1}}{(n-1)!}\, \lambda e^{-\lambda t}. \end{aligned}$$

Multiplying by $e^{\lambda t}$,

$$\frac{dp_n(t)}{dt} e^{\lambda t} + \lambda e^{\lambda t} p_n(t) = \frac{\lambda^n t^{n-1}}{(n-1)!}.$$

As noticed earlier, the left side is the derivative of $e^{\lambda t} p_n(t)$, and integrating we obtain

$$\begin{aligned} \int \left(\frac{dp_n(t)}{dt} e^{\lambda t} + \lambda e^{\lambda t} p_n(t) \right) dt &= \int \frac{\lambda^n t^{n-1}}{(n-1)!} dt. \\ \Leftrightarrow\ p_n(t) e^{\lambda t} &= \frac{(\lambda t)^n}{n(n-1)!} + A_n. \end{aligned}$$

Again, $p_n(0) = 0$ forces $A_n = 0$, so multiplying by $e^{-\lambda t}$ eventually gives

$$p_n(t) = \frac{(\lambda t)^n e^{-\lambda t}}{n!},$$

the so-called Poisson distribution, or Poisson random variable. In general this distribution is defined at every t, so we set, for $k = 1, \ldots, n, \ldots$

$$p_k = \frac{\lambda^k e^{-\lambda}}{k!} = \begin{pmatrix} 0 & , & 1 & , & 2 & , \ldots , & n & , \ldots \\ e^{-\lambda} & , & \frac{\lambda}{1!} e^{-\lambda} & , & \frac{\lambda^2}{2!} e^{-\lambda} & , \ldots , & \frac{\lambda^n}{n!} e^{-\lambda} & , \ldots \end{pmatrix}.$$

The Poisson random variable is well defined, since $P(X = k) \geq 0$ for any k and

$$\sum_{k=0}^{\infty} P(X=k) = \sum_{k=0}^{\infty} \frac{\lambda^k e^{-\lambda}}{k!} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{-\lambda} e^{\lambda} = 1.$$

Let us find its expectation and variance:

$$E(X) = \sum_{k=0}^{\infty} k \frac{\lambda^k e^{-\lambda}}{k!} = \lambda \left[\sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} \right] e^{-\lambda} = \lambda e^{\lambda} e^{-\lambda} = \lambda;$$

and from

$$\begin{aligned} E(X^2) &= \sum_{k=0}^{\infty} k^2 \frac{\lambda^k e^{-\lambda}}{k!} = \sum_{k=0}^{\infty} (k^2 - k + k) \frac{\lambda^k e^{-\lambda}}{k!} \\ &= \sum_{k=0}^{\infty} k(k-1) \frac{\lambda^k e^{-\lambda}}{k!} + \sum_{k=0}^{\infty} k \frac{\lambda^k e^{-\lambda}}{k!} \\ &= \lambda^2 \left[\sum_{k=2}^{\infty} \frac{\lambda^{k-2}}{(k-2)!} \right] e^{-\lambda} + E(X) = \lambda^2 e^{-\lambda} e^{\lambda} + \lambda = \lambda^2 + \lambda, \end{aligned}$$

$$\operatorname{Var}(X) = \lambda^2 + \lambda - \lambda^2 = \lambda.$$

The parameter λ, initially taken as proportionality factor of the event's occurrence during a Δt lapse, is therefore both the expected value and the variance of our event at time t.

Next we want to explain how the binomial random variable under certain conditions has an asymptotic behaviour that turns it into a Poisson distribution. Suppose $np = \lambda$, in the sense that as $n \to \infty$ the constant p tends to 0 and therefore np becomes asymptotic to a constant λ. Let us examine the expansion

$$\begin{aligned} \lim_{\substack{n\to\infty \\ p\to 0}} \binom{n}{k} p^k (1-p)^{n-k} &= \lim_{\substack{n\to\infty \\ p\to 0}} \frac{n!}{k!(n-k)!} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} \\ &= \frac{\lambda^k}{k!} \lim_{\substack{n\to\infty \\ p\to 0}} \frac{n!}{(n-k)!} \left(\frac{1}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-k} \end{aligned}$$

term by term, starting from $\left(1 - \frac{\lambda}{n}\right)^{-k}$. Since $\lim\limits_{\substack{n\to\infty \\ p\to 0}} (np) = \lambda$ and hence the double limit boils down to $\lim_{n\to\infty}$, for any $k \in \mathbb{N}$ we have

$$\lim_{\substack{n\to\infty \\ p\to 0}} \left(1 - \frac{\lambda}{n}\right)^{-k} = (1)^{-k} = 1.$$

Regarding $\lim\limits_{\substack{n\to\infty\\ p\to 0}} \left(1-\dfrac{\lambda}{n}\right)^n$: because $\left(1-\frac{\lambda}{n}\right)^n > 0$ one has

$$\begin{aligned} &\ln\left(1-\frac{\lambda}{n}\right)^n \ln e = \ln\left(e^{\ln\left(1-\frac{\lambda}{n}\right)^n}\right) \\ \Leftrightarrow &\left(1-\frac{\lambda}{n}\right)^n = e^{\ln\left(1-\frac{\lambda}{n}\right)^n}, \end{aligned}$$

and we consider the limit of the exponent on the right:

$$\lim_{n\to\infty} \ln\left(1-\frac{\lambda}{n}\right)^n = \lim_{n\to\infty} n\ln\left(\frac{n-\lambda}{n}\right) = \lim_{n\to\infty}\left[\ln\left(\frac{n-\lambda}{n}\right)\Big/\frac{1}{n}\right].$$

As $n\to\infty$ the indeterminacy $0/0$ is sorted by de l'Hôpital's rule. For the numerator

$$\frac{d}{dn}\left(\ln\left(\frac{n-\lambda}{n}\right)\right) = \frac{1}{\frac{n-\lambda}{n}}\cdot\frac{d}{dn}\left(\frac{n-\lambda}{n}\right) = \frac{n}{n-\lambda}\cdot\frac{\lambda}{n^2} = \frac{\lambda}{n(n-\lambda)},$$

while the denominator's derivative equals $-1/n^2$. Therefore

$$\lim_{n\to\infty} \ln\left(1-\frac{\lambda}{n}\right)^n = \lim_{n\to\infty}\left(\frac{\frac{\lambda}{n(n-\lambda)}}{-\frac{1}{n^2}}\right) = \lim_{n\to\infty}\left(-\frac{n\lambda}{n-\lambda}\right) = -\lambda$$

and hence

$$\lim_{n\to\infty}\left(1-\frac{\lambda}{n}\right)^n = \lim_{n\to\infty} e^{\ln\left(1-\frac{\lambda}{n}\right)^n} = e^{-\lambda}.$$

Finally,

$$\begin{aligned} \lim_{\substack{n\to\infty\\ p\to 0}} \frac{n!}{(n-k)!}\left(\frac{1}{n}\right)^k &= \lim_{n\to\infty}\frac{n!}{(n-k)!n^k} \\ &= \lim_{n\to\infty}\frac{n(n-1)\cdots(n-k+1)(n-k)(n-k-1)\cdots 1}{[(n-k)(n-k-1)\cdots 1]n^k} \\ &= \lim_{n\to\infty}\frac{n(n-1)\cdots(n-k+1)}{nn\cdots n} \\ &= \lim_{n\to\infty} 1\left(1-\frac{1}{n}\right)\ldots\left(1-\frac{k-1}{n}\right) = 1. \end{aligned}$$

Putting everything back together,

$$\lim_{\substack{n\to\infty\\ p\to 0}} \binom{n}{k} p^k (1-p)^{n-k} = \frac{\lambda^k}{k!} e^{-\lambda}.$$

We have thus proved that as the number n of trials increases and the probability p of event E lowers in such a way that np is a constant λ, a binomial random variable tends to a Poisson random variable.

The fact that a binomial distribution converges to a Poisson distribution whenever the probability of witnessing an event is close to zero induced some scholars to call the Poisson random variable, somehow misleadingly, "random variable of rare events". The Poisson random variable is well defined even when events are not rare at all ($p \nrightarrow 0$). We have proved that if $n \to \infty$ and $p \to 0$ but $np = \lambda$, then $\mathscr{B}(n, p) \to \frac{\lambda^k}{k!} e^{-\lambda}$. This is not the case when $p \nrightarrow 0$, nor when $n \to \infty$, because then $p > (\lambda/n)$ and $(1-p) < (1-\lambda/n)$. ★

Now we want to examine a few classical continuous distributions.

Example 14.4.3 (*Uniform random variable*) A RV on $[a, b]$ with probability density

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } x \in [a,b] \\ 0 & \text{if } x \notin [a,b]. \end{cases}$$

is called uniform, and denoted $\mathscr{U}_{[a,b]}$ (Fig. 14.1).

To start with, a uniform RV satisfies conditions (14.3.2). Let us find its cumulative distribution function. If $x < a$, $f(t) = 0$ for $t \le x$, and so $F(x) = 0$. If $a \le x \le b$, we have $f(t) = 1/(b-a)$ and then (Fig. 14.1)

$$F(x) = \int_{(-\infty,a)} f(t)dt + \int_{[a,x]} f(t)dt = \frac{1}{b-a} \int_{[a,x]} dt = \frac{x-a}{b-a}.$$

If $x > b$,

$$F(x) = \int_{(-\infty,a)} f(t)dt + \int_{[a,b]} f(t)dt + \int_{(b,x]} f(t)dt = \frac{1}{b-a} \int_{[a,b]} dt = 1.$$

Altogether

$$F(x) = \begin{cases} 0 & \text{if } x < a \\ \frac{x-a}{b-a} & \text{if } a \le x \le b \\ 1 & \text{if } x > b. \end{cases}$$

Out of the distribution function we recover the mean and variance:

$$E(X) = \int_{(-\infty,a)} x dF(x) + \int_{[a,b]} x dF(x) + \int_{(b,\infty)} x dF(x) = \frac{1}{b-a} \int_{[a,b]} x dx = \frac{a+b}{2}.$$

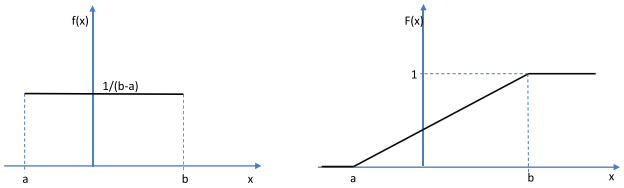


Fig. 14.1 Density and distribution functions of a uniform RV $\mathscr{U}_{a,b}$

$$E(X^2) = \int_{(-\infty,a)} x^2 dF(x) + \int_{[a,b]} x^2 dF(x) + \int_{(b,\infty)} x^2 dF(x)$$
$$= \frac{1}{b-a} \int_{[a,b]} x^2 dx = \frac{a^2 + ab + b^2}{3},$$

so

$$\mathrm{Var}(X) = E(X^2) - (E(X))^2 = \frac{(b-a)^2}{12}.$$

In particular, the distribution function of a uniform RV over [0, 1] equals

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } 0 \le x \le 1 \\ 1 & \text{if } x > 1. \end{cases}$$

Taking linear combinations of independent uniform RVs generates random variables with density varying according to the number of RVs involved. In particular, for $n = 2$ we obtain a triangular RV; for higher n, putting $S_n = \sum_{i=1}^{n} \mathscr{U}^i_{[0,1]}$, the density

$$\frac{S_n - E(S_n)}{\sqrt{\mathrm{Var}(S_n)}}$$

quickly tends to a bell-shaped curve. ★

Notwithstanding its simplicity, the uniform RV is very useful towards applications, because any RV can be transformed in a uniform one and vice versa. The random-number algorithms sometimes employed to generate uniform random values are not typically object-oriented, and pass all randomness tests. Hence these generators are suitable for random variables of other kinds as well. Sequences of random numbers have several applications in the fields of experimental simulations, numerical analysis involving Monte Carlo methods (evaluating integrals and solving systems of differential equations) and polling models.

Example 14.4.4 (*Negative-exponential random variable*) The negative-exponential RV is deduced from memoryless probabilistic models. These are used to compute the lifetime of devices either not subject to decay, or subject to negligible deterioration.

In general a RV X is said to be without memory, or memoryless, if the probability the phenomenon survives to time $t_0 + t$ (or the device does not break down before that instant), assuming it survives until t_0, is equal to the probability the phenomenon survives during the entire lapse $[0, t]$:

$$P(X > t_0 + t | X > t_0) = P(X > t) \Leftrightarrow \frac{P(X > t_0 + t)}{P(X > t_0)} = P(X > t).$$

This relation formalises the lack of memory.

If $F_X(t) = P(X \leq t)$ denotes the cumulative distribution function of X, the survival function $S : [0, \infty) \mapsto [0, 1]$

$$S(t) = 1 - F_X(t) = P(X > t), \quad t \geq 0$$

encodes memorylessness, because

$$\frac{S(t_0 + t)}{S(t_0)} = S(t) \quad \text{for } t_0, t > 0.$$

The map S is differentiable on $(0, \infty)$, it has right derivative at 0 and solves system

$$\begin{cases} S(t_0 + t) = S(t_0) \cdot S(t) \\ S(0) = 1 - F_X(0) = 1, \ S(\infty) = 1 - F_X(\infty) = 0. \end{cases}$$

When $t > 0$

$$\begin{aligned} S'(t) &= \lim_{t_1 \to 0^+} \frac{S(t + t_1) - S(t)}{t_1} = \lim_{t_1 \to 0^+} \frac{S(t) \cdot S(t_1) - S(t)}{t_1} \\ &= S(t) \cdot \lim_{t_1 \to 0^+} \frac{S(t_1) - 1}{t_1} = S(t) \cdot S'(0^+). \end{aligned}$$

As $F_X(t)$ is increasing, $1 - F_X(t) = S(t)$ is positive and decreasing, i.e. $S'(t) < 0$ for every $t \in [0, \infty)$. From

$$\frac{S'(t)}{S(t)} = S'(0^+) = -\lambda \leq 0$$

we find

$$\big(\ln S(t)\big)' = \frac{S'(t)}{S(t)} = -\lambda \Leftrightarrow \ln S(t) = -\lambda t + a \Leftrightarrow S(t) = e^{-\lambda t + a}.$$

Now $S(0) = 1 \Rightarrow S(0) = e^a = 1 \Rightarrow a = 0 \Rightarrow S(t) = e^{-\lambda t}$, so $F_X(t) = 1 - e^{-\lambda t}$. In addition, S vanishes for $t < 0$. The random variable X then has exponential distribution function with rate parameter $\lambda > 0$. The choice $\lambda = 0$ is excluded since

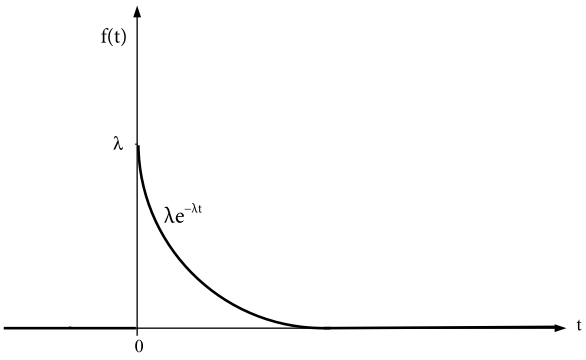


Fig. 14.2 Density function of a negative-exponential RV

$S(t) = 1$ is not the survival function of any RV (S must decrease from $S(0) = 1$ to $S(\infty) = 0$).

In summary, for $\lambda > 0$

$$F_X(t) = \begin{cases} 1 - e^{-\lambda t} & \text{if } t \in [0, \infty) \\ 0 & \text{if } t \in (-\infty, 0). \end{cases} \quad \text{and} \quad f_X(t) = \begin{cases} \lambda e^{-\lambda t} & \text{if } t \in [0, \infty) \\ 0 & \text{if } t \in (-\infty, 0). \end{cases}$$

The density function of a negative-exponential RV looks like in Fig. 14.2. The mean and variance are found integrating by parts:

$$\begin{aligned} E(X) &= \int_{[0,\infty)} t\lambda e^{-\lambda t} dt = -\frac{1}{\lambda} e^{-\lambda t} \lambda t - \int_{[0,\infty)} \lambda \left[-\frac{1}{\lambda} e^{-\lambda t} \right] dt \\ &= -te^{-\lambda t} + \int_{[0,\infty)} e^{-\lambda t} dt = \left| -te^{-\lambda t} - \frac{1}{\lambda} e^{-\lambda t} \right|_0^\infty = \frac{1}{\lambda} \end{aligned}$$

$$\begin{aligned} \mathrm{Var}(X) &= E(X^2) - (E(X))^2 = \left[\int_{[0,\infty)} t^2 \lambda e^{-\lambda t} dt \right] - \frac{1}{\lambda^2} \\ &= -\lambda t^2 \frac{1}{\lambda} e^{-\lambda t} - \int_{[0,\infty)} 2\lambda t \left(-\frac{1}{\lambda} e^{-\lambda t} \right) dt - \frac{1}{\lambda^2} \\ &= \left[-t^2 e^{-\lambda t} + \frac{2}{\lambda} \int_{[0,\infty)} t\lambda e^{-\lambda t} dt \right] - \frac{1}{\lambda^2} \\ &= \left| -t^2 e^{-\lambda t} \right|_0^\infty + \frac{2}{\lambda} \cdot \frac{1}{\lambda} - \frac{1}{\lambda^2} = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}. \end{aligned}$$

★

Example 14.4.5 (*Normal random variable*) The normal, or Gaussian, random variable is supported on the entire real axis. Its density function

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\frac{(x-m)^2}{\sigma^2}}. \tag{14.4.3}$$

has the graph as in Fig. 14.3. We claim

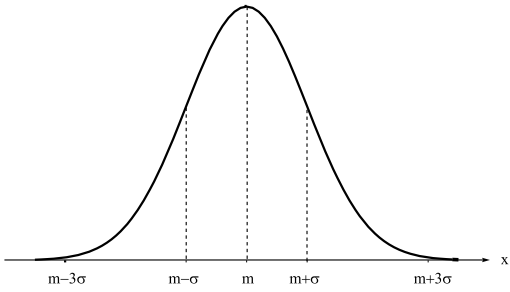


Fig. 14.3 Density function of a normal RV

$$\int_{\mathbb{R}} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\frac{(x-m)^2}{\sigma^2}} dx = 1. \tag{14.4.4}$$

To see this set $y = \frac{x-m}{\sigma}$, so $dx = \sigma dy$ and

$$A = \int_{\mathbb{R}} \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy.$$

Moreover,

$$A^2 = \frac{1}{2\pi} \iint\limits_{\mathbb{R}^2} e^{-\frac{y^2+z^2}{2}} dydz.$$

Let C_r indicate the disc centred at the origin with radius $0 < r < \infty$:

$$C_r = \left\{(y, z) \in \mathbb{R}^2 : \frac{y^2 + z^2}{2} \leq r^2\right\}.$$

Then

$$A^2 = \frac{1}{2\pi} \iint\limits_{\mathbb{R}^2} e^{-\frac{y^2+z^2}{2}} dydz = \frac{1}{2\pi} \lim_{r\to\infty} \iint\limits_{C_r} e^{-\frac{y^2+z^2}{2}} dydz.$$

Passing to polar coordinates (ρ, θ), since $\rho = \sqrt{y^2 + z^2}$ we can write

$$A^2 = \frac{1}{2\pi} \lim_{r\to\infty} \int_0^r \rho e^{-\frac{\rho^2}{2}} d\rho \int_0^{2\pi} d\theta.$$

Substituting

$$\frac{d\left(e^{-\frac{\rho^2}{2}}\right)}{d\rho} = -\rho e^{-\frac{\rho^2}{2}} \Leftrightarrow -\frac{d\left(e^{-\frac{\rho^2}{2}}\right)}{d\rho} = \rho e^{-\frac{\rho^2}{2}}$$

in the integral we conclude

$$A^2 = \frac{1}{2\pi}\lim_{r\to\infty}\int_0^r -\frac{d\left(e^{-\frac{\rho^2}{2}}\right)}{d\rho}\,d\rho\int_0^{2\pi} d\theta = -\frac{1}{2\pi}\left[e^{-\frac{\rho^2}{2}}\right]_0^{\infty}\cdot\left[\theta\right]_0^{2\pi}$$
$$= -\frac{1}{2\pi}\left[0-1\right]\cdot\left[2\pi\right] = \frac{1}{2\pi}\cdot 2\pi = 1,$$

i.e.

$$\iint\limits_{\mathbb{R}^2} e^{-\frac{y^2+z^2}{2}}dydz = 2\pi.$$

Furthermore,

$$\iint\limits_{\mathbb{R}^2} e^{-\frac{x^2+y^2}{2}}dxdy = \int_{\mathbb{R}} e^{-\frac{x^2}{2}}dx\int_{\mathbb{R}} e^{-\frac{y^2}{2}}dy$$
$$= \int_{\mathbb{R}} e^{-\frac{x^2}{2}}dx\int_{\mathbb{R}} e^{-\frac{x^2}{2}}dx \quad \text{(relabelling } y \text{ as } x\text{)}$$
$$= \left[\int_{\mathbb{R}} e^{-\frac{x^2}{2}}dx\right]^2 = 2\pi$$
$$\Leftrightarrow \int_{\mathbb{R}} e^{-\frac{x^2}{2}}dx = \sqrt{2\pi} \Leftrightarrow \int_{\mathbb{R}} e^{-x^2}dx = \sqrt{\pi}.$$

With the substitution $x = \sigma y + m$, $dx = \sigma dy$, the expectation of X is

$$E(X) = \int_{\mathbb{R}} x\cdot\frac{1}{\sigma\sqrt{2\pi}}\,e^{-\frac{1}{2}\frac{(x-m)^2}{\sigma^2}}dx = \int_{\mathbb{R}}(\sigma y+m)\cdot\frac{1}{\sqrt{2\pi}}\,e^{-\frac{y^2}{2}}dy$$
$$= \frac{\sigma}{\sqrt{2\pi}}\int_{\mathbb{R}} y\cdot\,e^{-\frac{y^2}{2}}dy + m\cdot\frac{1}{\sqrt{2\pi}}\underbrace{\int_{\mathbb{R}} e^{-\frac{y^2}{2}}dy}_{=\sqrt{2\pi}}.$$

Substituting

$$\frac{d\left(e^{-\frac{y^2}{2}}\right)}{dy} = -ye^{-\frac{y^2}{2}}, \text{ or } -\frac{d\left(e^{-\frac{y^2}{2}}\right)}{dy} = ye^{-\frac{y^2}{2}},$$

gives

$$E(x) = \frac{\sigma}{\sqrt{2\pi}}\int_{\mathbb{R}} -\frac{d\left(e^{-\frac{y^2}{2}}\right)}{dy}dy + m = \frac{\sigma}{\sqrt{2\pi}}\Big| - e^{-\frac{y^2}{2}}\Big|_{-\infty}^{\infty} + m = m.$$

Choosing, instead,

$$\frac{(x-m)^2}{2\sigma^2} = t^2, \text{ so } x = \sigma\sqrt{2}t + m \text{ and } dx = \sigma\sqrt{2}dt,$$

the variance equals

$$\begin{aligned}
\mathrm{Var}(x) &= \frac{1}{\sigma\sqrt{2\pi}} \int_{\mathbb{R}} x^2 e^{-\frac{1}{2}\frac{(x-m)^2}{\sigma^2}} dx - (E(X))^2 = \frac{\sigma\sqrt{2}}{\sigma\sqrt{2\pi}} \int_{\mathbb{R}} (\sigma\sqrt{2}t + m)^2 e^{-t^2} dt - m^2 \\
&= \frac{1}{\sqrt{\pi}} \left(2\sigma^2 \int_{\mathbb{R}} t^2 e^{-t^2} dt + 2\sigma\sqrt{2}m \underbrace{\int_{\mathbb{R}} t e^{-t^2} dt}_{=0} + m^2 \underbrace{\int_{\mathbb{R}} e^{-t^2} dt}_{=\sqrt{\pi}} \right) - m^2; \\
&= \frac{2\sigma^2}{\sqrt{\pi}} \int_{\mathbb{R}} t^2 e^{-t^2} dt.
\end{aligned}$$

Since

$$\frac{d(e^{-t^2})}{dt} = -2te^{-t^2}, \text{ so } -\frac{t}{2} \cdot \frac{d(e^{-t^2})}{dt} = t^2 e^{-t^2},$$

replacing in the expression and integrating by parts we obtain

$$\begin{aligned}
\mathrm{Var}(x) &= \frac{2\sigma^2}{\sqrt{\pi}} \int_{\mathbb{R}} t^2 \, e^{-t^2} dt = \frac{2\sigma^2}{\sqrt{\pi}} \int_{\mathbb{R}} -\frac{t}{2} \cdot \frac{d(e^{-t^2})}{dt} dt \\
&= \frac{2\sigma^2}{\sqrt{\pi}} \left(\left| -\frac{t}{2} \cdot e^{-t^2} \right|_{-\infty}^{\infty} + \frac{1}{2} \int_R e^{-t^2} dt \right) \\
&= \frac{2\sigma^2}{\sqrt{\pi}} \cdot \frac{1}{2} \int_{\mathbb{R}} e^{-t^2} dt = \frac{\sigma^2 2\sqrt{\pi}}{2\sqrt{\pi}} = \sigma^2.
\end{aligned}$$

The RV X with density (14.4.3) is completely characterised by the two parameters m, σ. One writes

$$X \sim N(m, \sigma^2),$$

where $\sim$ should be understood as "has the same distribution as", to mean such an X is "normally distributed".

To finish, a few observations concerning Gaussian random variables.

- The density function is symmetric about the straight line $x = m$, so the skewness vanishes. At $x = m$ the function has an absolute maximum.
- The x-axis is a horizontal asymptote for the density function.
- The points $x_1 = m - \sigma, x_2 = m + \sigma$ are the density's inflections.
- Almost the entire unit mass is contained in $[m - 3\sigma, m + 3\sigma]$:

$$P(m - 3\sigma \leq X \leq m + 3\sigma) = 0.99.$$

- The expected value m of the normal RV determines where the density is placed in the plane. Given two density functions with $m_1 < m_2$, the graph with mean m_1 somehow sits to the right of the one with mean m_2.
- The variance σ^2 of the normal RV determines the density's shape. For smaller values of σ the density function is more concentrated around the expected value m, and hence its peak is higher than a density with larger σ.
- The so-called standard normal RV, obtained setting $z = (x - m)/\sigma$, has probability density

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}.$$

It enjoys all the properties of normal RVs but has mean 0 and variance 1. It is customary to call it

$$Z \sim N(0, 1).$$

To transform a normal RV X into a standard normal RV Z it suffices to put

$$F_Z(z) = F_X\left(\frac{x - m}{\sigma}\right).$$

This formula allows to compute the cumulative distribution function of $N(m, \sigma)$ by knowing that of $N(0, 1)$.

By the aforementioned symmetry, moreover, the table of values of the standard normal distribution function only contains positive arguments. ★

Appendix A
Compact and Totally Bounded Metric Spaces

Definition A.0.1 (*Diameter of a set*) The diameter of a subset E in a metric space (X, d) is the quantity

$$\operatorname{diam}(E) = \sup_{x,y \in E} d(x, y).$$

$\diamond$

Definition A.0.2 (*Bounded and totally bounded sets*)A subset E in a metric space (X, d) is said to be

1. bounded, if there exists a number M such that $E \subseteq B(x, M)$ for every $x \in E$, where $B(x, M)$ is the ball centred at x of radius M. Equivalently, $d(x_1, x_2) \leq M$ for every $x_1, x_2 \in E$ and some $M \in \mathbb{R}$;
2. totally bounded, if for any $\epsilon > 0$ there exists a finite subset $\{x_1, \ldots, x_{n_\epsilon}\}$ of X such that the (open or closed) balls $B(x_i, \epsilon)$, $i = 1, \ldots, n_\epsilon$ cover E, i.e. $E = \bigcup_{i=1}^{n_\epsilon} B(x_i, \epsilon)$.

$\diamond$

Clearly a set is bounded if its diameter is finite.

We will see later that a bounded set may not be totally bounded. Now we prove the converse is always true.

Proposition A.0.1 (Total boundedness implies boundedness) *In a metric space* (X, d) *any totally bounded set* E *is bounded.*

Proof Total boundedness means we may choose $\epsilon = 1$ and find $A = \{x_1, \ldots, x_n\}$ in X such that for every $x \in E$

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S. Gentili, *Measure, Integration and a Primer on Probability Theory*, UNITEXT 125, https://doi.org/10.1007/978-3-030-54940-4

$$\inf_{1\le i\le n} d(x_i, x) \le 1.$$

Set $M = \max_{1\le i\le n} d(x_1, x_i)$. We know there is an index $i \in \{1, \ldots, n\}$ such that $d(x_i, x) \le 1$, so the triangle inequality implies

$$d(x_1, x) \le d(x_1, x_i) + d(x_i, x) \le M + 1.$$

Hence $E \subseteq B(x_1, M + 1)$, and E is bounded.

■

Example A.0.1 (*Bounded versus totally bounded sets*) The notions of boundedness and total boundedness differ, because the former is weaker. The discrete distance of distinct points is always 1, and equals 0 for coinciding points.

- The space $X = \{a, b, c\}$ with discrete metric is surely bounded. It is totally bounded, too, since $\{a\}, \{b\}, \{c\}$ have zero diameter (less than any $\epsilon > 0$), there is a finite number of them, and their union is X.
- The space $[0, 1]$ equipped with the discrete distance, rather than the usual Euclidean metric, is bounded: its diameter is 1, the distance of any pair of distinct points. But it is not totally bounded. The subsets of diameter less than 1 are the singletons $\{x\}$, of which there is no finite number covering the whole space.
- $\mathbb{N}$ with the discrete metric is bounded (distinct natural numbers are all 1 apart), but not totally bounded. We cannot cover $\mathbb{N}$ by finitely many balls of radius $\epsilon < 1$ (a ball consists of a single point, its centre, but $\mathbb{N}$ is infinite). A Cauchy sequence of natural numbers is eventually constant, hence it converges. $\mathbb{N}$ is therefore a complete metric space.

★

Proposition A.0.2 (Separability of totally bounded metric spaces) *Totally bounded metric spaces (X, d) are separable.*

Proof As X is totally bounded, for every $n \in \mathbb{N}$ there exists $A_n = \{x_1, \ldots, x_n\} \subseteq X$, finite, such that

$$X = \bigcup_{i=1}^{n} B\left(x_i, \frac{1}{n}\right).$$

The union $D = \bigcup_{n=1}^{\infty} A_n$ is countable (as countable union of finite sets).

We claim that $D = X$. Let $x \in X$ be an arbitrary point. For any $n \in \mathbb{N}$, there exists some $x_{i_n} \in A_n$ such that $x \in B\big(x_{i_n}, 1/n\big)$. This gives a sequence $\{x_{i_n}\}_{n\ge1} \subseteq D$

with $d(x, x_{i_n}) < 1/n$. Therefore $\lim x_i = x$. This shows that every $x \in X$ lies in the closure of D and so $\bar{D} = X$.

■

Intuitively a separable space is one where any element is close to one from some (dense) countable subset. In a compact space, instead, any element is close to another taken from a finite number of subsets, so compact spaces can be considered kind of "pseudo-finite". In analogy to the notion of compactness in $\mathbb{R}$ we have the following definition for generic metric spaces.

Definition A.0.3 (*Compactness*) A subset E in a metric space (X, d) is said to be compact if any open cover of E admits a finite subcover (so-called Heine–Pincherle–Borel property).

◇

The next results generalises the Bolzano–Weierstraß theorem to compact metric spaces.

Theorem A.0.1 (Bolzano–Weierstraß theorem for metric spaces) *If (X, d) is a compact metric space, any infinite subset $E \subseteq X$ has a limit point.*

Proof Let (X, d) be compact and $E \subseteq X$ infinite. Assume, by contradiction, E has no limit points. Then for every $x \in X$ there is an open ball $B(x, r_x)$ such that $B(x, r_x) \cap E \subseteq \{x\}$. The family $\{B(x, r_x)\}_{x \in X}$ is an open cover of X, so there exists a finite subcover $B(x_i, r_{x_i})$, $i = 1, \ldots, n$. But then

$$E = E \cap X = E \cap \left(\bigcup_{i=1}^{n} B(x_i, r_{x_i}) \right) \subseteq \{x_1, \ldots, x_n\},$$

i.e. E is finite. The contradiction proves the claim.

■

Definition A.0.4 (*Sequentially compact metric spaces*) A metric space (X, d) is sequentially compact if every sequence in X admits a convergent subsequence.

◇

We will see straightaway that sequential compactness implies total boundedness.

Proposition A.0.3 (Sequential compactness versus total boundedness) *A sequentially compact metric space (X, d) is totally bounded.*

Proof Suppose (X, d) is not totally bounded. Then there is an ϵ_0 such that X cannot be covered by balls centred at the points of any finite set $A_0 \subseteq X$. Taking $x_1 \in X$, then, we can find $x_2 \in X$ with $d(x_1, x_2) > \epsilon_0$, for otherwise $X = B(x_1, \epsilon_0)$ and $A_0 = \{x_1\}$.

Next, take $x_1, x_2 \in X$, and there is an $x_3 \in X$ with $d(x_3, x_1) > \epsilon_0$ and $d(x_3, x_2) > \epsilon_0$, because otherwise $X = B(x_1, \epsilon_0) \cup B(x_2, \epsilon_0)$ and $A_0 = \{x_1, x_2\}$.

Inductively, at step $k + 1$ we have $\{x_1, \ldots, x_k\} \subseteq X$ and there is an $x_{k+1} \in X$ with $d(x_{k+1}, x_i) > \epsilon_0$ for every $i \in \{1, \ldots, k\}$, because if not, once again,

$$X = \bigcup_{i=1}^{k} B(x_i, \epsilon_0)$$

and $A_0 = \{x_1, \ldots, x_k\}$.

Overall we have a whole sequence $\{x_k\}_{k\geq 1}$ such that $d(x_k, x_j) > \epsilon_0$, $k \neq j$. It must diverge, hence it certainly cannot have a convergent subsequence.

But then X would not be sequentially compact, contradicting the hypothesis.

■

Definition A.0.5 (*Countably compact metric spaces*) A metric space (X, d) is countably compact if any countable open cover of X has a finite subcover.

◇

Proposition A.0.4 (Sequentially compact metric spaces and countable compactness) *A sequentially compact metric space (X, d) is countably compact.*

Proof Let $\{G_n\}_{n\geq 1}$ be a countable open cover of X

$$X = \bigcup_{n=1}^{\infty} G_n.$$

Suppose X not countably compact, so $\{G_n\}_{n\geq 1}$ does not have a finite subcover, meaning for every $m \geq 1$

$$\bigcup_{n=1}^{m} G_n \subseteq X.$$

The proof is by induction. Take $x_{m_1} \in X$ such that $x_{m_1} \notin \bigcup_{n=1}^{m} G_n$, and since $\{G_n\}_{n\geq 1}$ covers X, there is an m_1 such that $x_{m_1} \in G_{m_1}$. Choose $x_{m_2} \in X$ such that $x_{m_2} \notin \bigcup_{n=1}^{m_1} G_n$. Then there is an m_2 such that $x_{m_2} \in G_{m_2}$. Then pick $x_{m_3} \in X$ such that $x_{m_3} \notin \bigcup_{n=1}^{m_2} G_n$, so $x_{m_3} \in G_{m_3}$ for some index m_3.

At step k

$$x_{m_k} \in G_{m_k} \quad \text{and} \quad x_{m_k} \notin \bigcup_{n=1}^{m_{k-1}} G_n.$$

We end up with a sequence $\{x_{m_k}\}_{k\geq 1}$ in the sequentially compact space X, so there is a subsequence $\{x_{k_n}\}_{n\geq 1}$ with some limit point $x \in X$. But $\{G_n\}_{n\geq 1}$ is a cover, so x belongs in some G_{n_0}. Furthermore,

$$\lim_{n\to\infty} x_{k_n} = x$$

forces $x_{k_n} \in G_{n_0}$ for all $n \geq n_0$. But because of how the sequence was constructed, there exists an $m_{k_n} > n_0$ such that $x_{k_n} \in G_{m_{k_n}}$, and $x_{k_n} \notin G_{n_0} \subseteq \bigcup_{n=1}^{m_{k_n}-1} G_n$. Hence no subsequence of $\{x_{m_k}\}_{k\geq 1}$ can converge to x. But this means X is not sequentially compact which is a contradiction. Therefore X is countably compact.

■

In metric spaces compactness can be characterised by means of sequences. Namely,

Theorem A.0.2 (Characterisation of compact metric spaces in terms of sequential compactness) *A metric space (X, d) is compact if and only if it is sequentially compact.*

Proof ($\Rightarrow$) Let (X, d) be compact, and let us prove it is sequentially compact.

Given a compact subset $E \subseteq X$, suppose there is a sequence $\{x_n\}_{n\geq 1}$ none of whose subsequences converge in E.

Any such must have infinite range, otherwise it would become eventually constant and hence converge in E. Nor can it accumulate at a point in E, otherwise this point would be the limit of a subsequence.

Take $x \in E$. This is not a limit point for $\{x_n\}_{n\geq 1}$, so there is an $\epsilon_x > 0$ such that the ball $B(x, \epsilon_x)$ contains at most one term of the sequence.

Clearly $\left\{B(x, \epsilon_x)\right\}_{x\in E}$ is an open cover of E, and by compactness we extract a finite subcover such that

$$E \subseteq \bigcup_{i=1}^{n} B(x_i, \epsilon_{x_i}).$$

But $\{x_n\}_{n\geq 1} \subseteq E$, and since in $\bigcup_{i=1}^{n} B(x_i, \epsilon_{x_i})$ there are at most n distinct points, we deduce the sequence can only attain finitely many values. Hence it must have a convergent subsequence, contradicting the hypothesis.

($\Leftarrow$) Consider a set of indices I and a family of open sets $\mathcal{G} = \{G_i\}_{i\in I}$ with

$$X = \bigcup_{i\in I} G_i.$$

As X is sequentially compact, it is totally bounded (Proposition A.0.3), and by Proposition A.0.2 also separable.

Call $D = \{x_1, \ldots, x_n, \ldots\} \subseteq X$ the countable dense subset. Using this we construct the countable collection of balls

$$\mathcal{B} = \big\{B(x_n, q) : x_n \in D, q \in \mathbb{Q}\big\}.$$

Take the subcollection

$$\mathcal{B}_{\mathcal{G}} = \big\{B(x_n, q) \in \mathcal{B} : \exists G_i \in \mathcal{G} \text{ such that } B(x_n, q) \subseteq G_i\big\}.$$

Clearly the latter is at most countable. At this point it is enough to show $\mathcal{B}_{\mathcal{G}}$ is an open cover of X. If so, in fact, by Proposition A.0.4 and Definition A.0.5 we can extract from it a finite subcover $\big\{B(x_1, q_1), \ldots, B(x_k, q_k)\big\}$. Each $B(x_j, q_j)$, $j = 1, \ldots, k$, is contained in some G_i of the original cover $\mathcal{G}$. Renaming these G_j gives a finite subcollection $\{G_j : 1 \leq j \leq k\}$ of $\mathcal{G}$ such that

$$X = \bigcup_{j=1}^{k} G_j,$$

whence compactness follows.

So take an x in X. As $\mathcal{G}$ covers X, x belongs to some G_i. The latter is open, so for $\epsilon > 0$ there is an open ball $B(x, \epsilon) \subseteq G_i$.

But D is dense in X, so there exists $x_n \in D$ such that $d(x, x_n) < \epsilon/2$, i.e. $x \in B(x_n, \epsilon/2)$. Moreover, for every $y \in B(x_n, \epsilon/2)$ we have $d(y, x_n) < \epsilon/2$, so

$$d(y, x) \leq d(y, x_n) + d(x, x_n) < \epsilon$$

from the triangle inequality. Hence

$$x \in B(x_n, \epsilon/2) \subseteq B(x, \epsilon) \subseteq G_i.$$

Pick $q \in \mathbb{Q}$ such that $\epsilon/2 < q < \epsilon$, so that

$$x \in B(x_n, q) \subseteq B(x, \epsilon) \subseteq G_i.$$

As $B(x_n, q)$ has rational radius and centre $x_n \in D$, it belongs to the family $\mathcal{B}$. But $B(x_n, q)$ is a subset of G_i, so it also belong to $\mathcal{B}_{\mathcal{G}}$. In conclusion, any $x \in X$ belongs to a ball of $\mathcal{B}_{\mathcal{G}}$, making the latter an open countable cover of X.

■

There is a second description of compactness in metric spaces in terms of completeness and total boundedness.

Theorem A.0.3 (Characterisation of compact metric spaces in terms of completeness and total boundedness) *A metric space (X, d) is compact if and only if it is complete and totally bounded.*

Proof If X is compact, it is sequentially compact by what we said previously. Then X is complete and by Proposition A.0.3 it is totally bounded as well.

Sufficient implication: suppose X complete, totally bounded and not compact, by contradiction. Hence it has an open cover $\left\{B(x_\alpha, r_\alpha)\right\}_{\alpha\in\mathcal{I}}$ without finite subcovers.

Total boundedness guarantees that for $\epsilon = 1$ there exists a finite set $\{x_{1_1}, \ldots, x_{n_1}\}$ such that the (open or closed) balls $B(x_{i_1}, 1)$, $i = 1, \ldots, n$, cover X. Call x_1 the element in $\{x_{1_1}, \ldots, x_{n_1}\}$ with the property that no finite subcollection of $\left\{B(x_\alpha, r_\alpha)\right\}_{\alpha\in\mathcal{I}}$ is a cover of $B(x_1, 1)$. Such an element must exist, because if the balls of radius 1 centred at $x_{1_1}, \ldots, x_{n_1}$ possessed a finite subcover extracted from $\left\{B(x_\alpha, r_\alpha)\right\}_{\alpha\in\mathcal{I}}$ then X would be compact.

Similarly, choosing $\epsilon = 1/2$ we let x_2 be the element among $x_{1_{1/2}}, \ldots, x_{n_{1/2}}$ for which $B(x_2, 1/2) \cap B(x_1, 1) \neq \emptyset$ and $B(x_2, 1/2)$ is not covered by any finite subfamily of the open cover of X. This x_2 exists, too: if all balls of radius $1/2$ centred at $x_{1_{1/2}}, \ldots, x_{n_{1/2}}$ and intersecting $B(x_1, 1)$ had a finite subcover of the open cover, then $B(x_1, 1)$ would admit a finite subcover of the original family.

By induction, for $\epsilon = 1/2^m$ we call x_m the element among $x_{1_{1/2^m}}, \ldots, x_{n_{1/2^m}}$ such that $B(x_m, 1/2^m) \cap B(x_{m-1}, 1/2^{m-1}) \neq \emptyset$ and with the property that no finite subfamily of the open cover of X is a cover of $B(x_m, 1/2^m)$.

Take $x \in B(x_m, 1/2^m) \cap B(x_{m-1}, 1/2^{m-1})$, so the triangle inequality forces

$$d(x_{m-1}, x_m) \leq d(x_{m-1}, x) + d(x_m, x) \leq \frac{1}{2^{m-1}} + \frac{1}{2^m} \leq \frac{1}{2^{m-2}},$$

and for $k < m$

$$\begin{aligned} d(x_k, x_m) &\leq d(x_k, x_{k+1}) + d(x_{k+1}, x_{k+2}) + \ldots + d(x_{m-1}, x_m) \\ &\leq \frac{1}{2^{k-1}} + \frac{1}{2^k} + \frac{1}{2^{k+1}} + \ldots + \frac{1}{2^{m-2}} \leq \frac{1}{2^{k-2}}. \end{aligned}$$

Hence $\{x_i\}_{i\geq 1}$ is a Cauchy sequence in the complete space X, so it converges to a point $x \in X$.

Call $B(x_{\alpha_0}, r_{\alpha_0})$ the ball of the open cover $\left\{B(x_\alpha, r_\alpha)\right\}_{\alpha\in\mathcal{I}}$ that contains x. There exists an $\epsilon > 0$ such that $B(x, \epsilon) \subseteq B(x_{\alpha_0}, r_{\alpha_0})$. By construction, moreover, there is an m such that $1/2^m < \epsilon/2$ and $d(x_m, x) < \epsilon/2$, so finally

$$B(x_m, 1/2^m) \subseteq B(x, \epsilon) \subseteq B(x_{\alpha_0}, r_{\alpha_0}).$$

But then one ball from the open cover is enough to cover $B(x_m, 1/2^m)$, contradicting the assumption.

■

Theorems A.0.2 and A.0.3 imply that a metric space is sequentially compact iff it is complete and totally bounded.

Proposition A.0.5 (Compactness in complete metric spaces) *In a complete metric space (X, d) a subset F is compact if and only if it is closed and totally bounded.*

Proof The $\Rightarrow$ part is self-evident: F is compact (also sequentially) and it contains all of its limit points.

Conversely, take a sequence $\{x_n\}_{n\geq 1}$ in F. By assumption F is totally bounded, so for every $\epsilon = 1/m$, $m \geq 1$, F can be covered by finitely many balls $B(x_n, 1/m) = \{y \in X : d(x_n, y) \leq 1/m\}$, $n = 1, \ldots, k$.

Choosing $\epsilon = 1$, let $\{x_{1_1}, \ldots, x_{k_1}\}$ denote the subset of X such that the closed balls $B(x_{i_1}, 1)$ of radius 1 finitely cover F. Among these there is at least one that contains infinitely many terms of $\{x_n\}_{n\geq 1} \subseteq F$, call it $B_1 = B(x_{n_1}, 1)$. Then define $N_1 = \{n : x_n \in B_1\}$, an infinite set.

Pick $n_1 \in N_1$ and $\epsilon = 1/2$. Let $\{x_{1_2}, \ldots, x_{k_2}\}$ be the finite subset of X for which the closed balls $B(x_{i_2}, 1/2)$ cover F. One of these, say $B_2 = B(x_{n_2}, 1/2)$, contains infinitely many terms of $\{x_n\}_{n\geq 1} \subseteq F$. Define $N_2 = \{n > n_1 : x_n \in B_1 \cap B_2\}$ and so forth.

Eventually we obtain a subsequence $\{x_{n_i}\}_{i\geq 1}$ of $\{x_n\}_{n\geq 1}$ such that, for every $n_j \geq n_i$, the terms x_{n_j} belong to $B_i = B(x_{n_i}, 1/i)$, and therefore $\{x_{n_i}\}_{i\geq 1}$ is Cauchy in $B_i \subseteq F$.

By completeness $\{x_n\}_{n\geq 1}$ converges in X, and as a matter of fact in F, since F is closed.

Thus we proved $\{x_n\}_{n\geq 1} \subseteq F$ has a convergent subsequence, and F is compact.

■

The Heine–Pincherle–Borel theorem descends in a straightforward manner from the the above result.

Theorem A.0.4 (Heine–Pincherle–Borel theorem for complete metric spaces) *Let (X, d) be a complete metric space and $F \subseteq X$ a closed and totally bounded subset. Any cover of open balls $\{B(x_\alpha, r_\alpha)\}_{\alpha\in\mathcal{I}}$ of F admits a finite subcover.*

■

At last, here is the version of Cantor's intersection theorem for complete metric spaces.

Theorem A.0.5 (Cantor's intersection theorem in complete metric spaces) *In a complete metric space (X, d) the intersection of a decreasing sequence of non-empty, closed and totally bounded subsets is non-empty.*

■

Theorem A.0.6 (Cardinality of perfect sets) *In a complete metric space (X, d) any non-empty, perfect and totally bounded set $P \subseteq X$ has the cardinality of the continuum.*

Proof Let P be a perfect, non-empty and totally bounded set in X. As any point in P is a limit point of P, the set must be infinite. Suppose P is countable. Take one point x_1 and

$$B(x_1, r_1) = \big\{x \in X : d(x, x_1) < r_1\big\}.$$

As x_1 is a limit point of P, $B(x_1, r_1)$ contains infinitely many points of P different from x_1, so $K_1 = \bar{B}(x_1, r_1) \cap P \neq \emptyset$, where $\bar{B}(x_1, r_1) = \big\{x \in X : d(x, x_1) \leq r_1\big\}$. Among those pick $x_2 \neq x_1$ and let $r_2 > 0$ be such that

$$r_2 < \min\big\{d(x_1, x_2), r_1 - d(x_1, x_2)\big\}.$$

Call

$$B(x_2, r_2) = \big\{x \in X : d(x, x_2) < r_2\big\},$$

so $\bar{B}(x_2, r_2) \subseteq \bar{B}(x_1, r_1)$. Since $d(x_1, x_2) > r_2$, then $x_1 \notin \bar{B}(x_2, r_2)$. Furthermore, x_2 is a limit point of P, so $B(x_2, r_2)$ contains infinitely many points of P other than x_2, and $K_2 = \bar{B}(x_2, r_2) \cap P \neq \emptyset$. Among them pick $x_3 \neq x_2$, and so forth.

Suppose, by induction, to have built $B(x_n, r_n)$ such that $\bar{B}(x_n, r_n) \cap P \neq \emptyset$ contains infinitely many points of P different from x_n. Choose $x_{n+1} \neq x_n$ and take

$$r_{n+1} < \min\big\{d(x_n, x_{n+1}), r_n - d(x_n, x_{n+1})\big\}.$$

Then

$$B(x_{n+1}, r_{n+1}) = \big\{x \in X : d(x, x_{n+1}) < r_{n+1}\big\}$$

and

1. $\bar{B}(x_{n+1}, r_{n+1}) \subseteq \bar{B}(x_n, r_n)$;
2. $x_n \notin \bar{B}(x_{n+1}, r_{n+1})$, because $d(x_n, x_{n+1}) > r_{n+1}$;
3. $K_{n+1} = \bar{B}(x_{n+1}, r_{n+1}) \cap P \neq \emptyset$.

The set in 3. satisfies the induction hypothesis, so we can proceed with the construction.

Since $x_n \notin K_{n+1}$, at the end of the process none of the points of P belongs in $\bigcap_{n=1}^{\infty} K_n$. Moreover $K_n \subseteq P$ for every n, so $\bigcap_{n=1}^{\infty} K_n = \emptyset$. But $\{K_n\}$ is a decreasing sequence of non-empty compact sets (Proposition 2.1.4) and with empty intersection. This violates Theorem A.0.5, and the claim follows.

■

Appendix B
Urysohn's Lemma and Tietze's Theorem

In a metric space (X, d) any two non-empty, closed and disjoint sets $A, B \subseteq X$ can be separated by open disjoint sets $U, V \subseteq X$. This is called normality property.

Proving that metric spaces are normal is rather easy. Take in fact $a \in A$, $b \in B$ and $r_a = (1/3) \cdot d(a, B) > 0$, $r_b = (1/3) \cdot d(b, A) > 0$, where $d(a, B) = \inf \{d(a, b) : b \in B\}$ and $d(b, A) = \inf \{d(b, a) : a \in A\}$. Then

$$U = \bigcup_{a \in A} B(a, r_a) \quad \text{and} \quad V = \bigcup_{b \in B} B(b, r_b)$$

are open, and disjoint since if $y \in B(a, r_a) \cap B(b, r_b)$ (assuming $r_a \geq r_b$ without loss of generality), the triangle inequality would give

$$d(a, b) \leq d(a, y) + d(y, b) \leq r_a + r_b \leq 2r_a,$$

whilst by construction $d(a, b) \geq 3r_a$.

We set out to prove a general fact concerning normal topological spaces—hence valid for metric spaces—called Urysohn lemma. The objective is to build a continuous function f on a normal space X mapping non-empty closed subsets A, B to $0, 1$ respectively. To do this we need a very large family of open sets containing A that do not meet B, and then decide which value f takes of a given $x \in X$ by looking at which open sets it belongs.

Theorem B.0.1 (Urysohn's lemma[1]) *Let X be a normal space and A, B non-empty, closed, disjoint subsets. Call $\mathcal{U}$ the family of neighbourhoods of A that do not intersect*

[1] Pavel Samuilovich Urysohn (1898–1924) was born in Odessa. He studied under D.F. Egorov and N.N. Lusin in Moscow, where he was awarded a Ph.D. in 1921. Urysohn was one of the most promising Soviet mathematicians of his generation at the time of his death, at the age of 25, in a tragic accident while swimming off the coast of Brittany. Urysohn's lemma, albeit trivial in metric spaces, allows to generalise Tietze's extension theorem to normal spaces.

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B, and D the set of rationals of the form $m/2^n$, $m, n \in \mathbb{N}$ *and* $0 \leq m/2^n \leq 1$. *Then there exists a continuous map* $f : X \mapsto [0, 1]$ *such that* $f(A) = 0$, $f(B) = 1$, *and* $0 < f(x) < 1$ *if* $x \in X \setminus (A \cup B)$.

Proof The set

$$D = \left\{ \frac{m}{2^n} : m, n \in \mathbb{N} \text{ and } 0 \leq \frac{m}{2^n} \leq 1 \right\} = \left\{ 0, 1, \frac{1}{2}, \frac{1}{4}, \frac{3}{4}, \frac{1}{8}, \frac{3}{8}, \frac{5}{8}, \frac{7}{8}, \dots \right\},$$

consists of the dyadic numbers in [0, 1]. For each $q \in D$ take a neighbourhood $U_q \subseteq X$. Initially, choose $U_1 = X$. As X is normal, there are open disjoint neighbourhoods U_A, U_B such that $U_A \cap U_B = \emptyset$, so $\bar{U}_A \cap B = \emptyset$ and $\bar{U}_A \subseteq (X \setminus B) \subseteq U_1 = X$.

Put $U_0 = U_A$, and by normality we have

$$\bar{U}_0 \subseteq U_{1/2} \subseteq \bar{U}_{1/2} \subseteq (X \setminus B) \subseteq U_1 = X.$$

Iterating in this way we can find $U_{1/4}$ such that

$$\bar{U}_0 \subseteq U_{1/4} \subseteq \bar{U}_{1/4} \subseteq U_{1/2}.$$

Analogously, we can find $U_{3/4}$ satisfying a similar relationship with $U_{1/2}$ and $(X \setminus B)$, and so on. By induction we end up with

1. $A \subseteq U_q$ for every $q \in D$;
2. $B \subseteq U_1$ and $B \cap U_q = \emptyset$ for every $q < 1$;
3. $\bar{U}_p \subseteq U_q$ for every $p < q$ in D.

Now define $f : X \mapsto [0, 1]$ by

$$f(x) = \inf \left\{ t \in D : x \in U_t \right\}.$$

This map equals 0 on A (condition 1. above) and 1 on B (condition 2.).

We need to show f is continuous, i.e. that the pre-image of any open subset in [0, 1] is open in X.

Suppose $V = (a, b)$ intersects [0, 1]. Given $x \in f^{-1}(V)$, clearly $f(x) \in (a, b)$, so there exist rational numbers p, q such that

$$a < p < f(x) < q < b.$$

As $p < f(x)$ and $f(x) = \inf \left\{ t \in D : x \in U_t \right\}$, there exists some p' such that $p < p' < f(x)$. But $p' < f(x)$ implies $x \notin U_{p'}$, and since $\bar{U}_p \subseteq U_{p'}$ then $x \notin \bar{U}_p$.

On the other hand, $f(x) < q$ and $f(x) = \inf \left\{ t \in D : x \in U_t \right\}$ force $x \in U_q$. Combining these facts leads to $x \in U_q \setminus \bar{U}_p = U$, which is open in X.

We claim $f(U) \subseteq (a, b)$.

Pick $y \in U$, so by construction $y \in U_q \subseteq \bar{U}_q$ and by definition $f(y) \leq q < b$. On the other hand by construction $y \notin \bar{U}_p \supseteq U_p$, and by definition $f(y) \geq p > a$. Hence $f(y) \in [p, q] \subseteq (a, b)$.

Clearly $f : X \mapsto [0, 1]$ is 0 on A and 1 on B, and it is continuous on X, so for $x \in X \setminus (A \cup B)$ it must assume all values in $(0, 1)$.

■

Example B.0.1 (*Example of Urysohn function on a metric space*) Take A, B nonempty, closed and disjoint in a metric space (X, d), and let us construct $f : X \mapsto [0, 1]$ such that $f(x) = 0$ for every $x \in A$, $f(x) = 1$ for every $x \in B$ and $0 < f(x) < 1$ for every $x \in X \setminus (A \cup B)$.

Define continuous maps (see Lemma 5.1.2)

$$d(x, A) = \inf \left\{ d(x, a) : a \in A \right\} \quad \text{and} \quad d(x, B) = \inf \left\{ d(x, b) : b \in B \right\}.$$

Clearly

$$d(x, A) = \begin{cases} = 0 & \text{if } x \in A \\ > 0 & \text{if } x \in A^c \end{cases} \quad \text{and} \quad d(x, B) = \begin{cases} = 0 & \text{if } x \in B \\ > 0 & \text{if } x \in B^c. \end{cases}$$

Since $d(x, A) + d(x, B)$ is always positive and continuous, we introduce the continuous map

$$f(x) = \frac{d(x, A)}{d(x, A) + d(x, B)}.$$

In practice

$$f(x) \begin{cases} = 0 & \text{if } x \in A \\ = 1 & \text{if } x \in B \\ \in (0, 1) & \text{if } x \in X \setminus (A \cup B) \end{cases},$$

furnishing the required function.

★

Theorem B.0.2 (Tietze's extension theorem[2]) *Let X be a normal topological space and E a closed subset. Then*

[2]Heinrich Tietze (1880–1964) was born in Schleinz, today's Austria. He studied in Vienna, where he obtained his Habilitation in 1908. From 1910 until 1919 he was professor in Brno, during which time he proved the extension theorem for metric spaces. He spent the rest of his career teaching at universities in Erlangen and Munich.

1. a continuous map $f : E \mapsto [c, d]$ extends to a continuous map $g : X \mapsto [c, d]$;
2. a continuous map $f : E \mapsto \mathbb{R}$ extends to a continuous map $g : X \mapsto \mathbb{R}$.

'Extends' means $f(x) = g(x)$ for every $x \in E$.

Proof We start by *1*. Replacing f with $(f - c)/(d - c)$ allows us to prove the statement with respect to the standard interval [0, 1].

Pick $\alpha \in (0, 1]$ and a continuous mapping $f : E \mapsto [0, \alpha]$. Divide $[0, \alpha]$ in three subintervals of length $(1/3)\alpha$:

$$I_1 = \left[0, \frac{1}{3}\alpha\right], \quad I_2 = \left[\frac{1}{3}\alpha, \frac{2}{3}\alpha\right] \quad I_3 = \left[\frac{2}{3}\alpha, \alpha\right].$$

Now set

$$A = \left\{x \in E : f(x) \leq \frac{1}{3}\alpha\right\} \quad \text{and} \quad B = \left\{x \in E : f(x) \geq \frac{2}{3}\alpha\right\}.$$

These are disjoint, and because $f \in C^0(E)$ (the class of continuous maps on E) they are closed in E, hence in X.

By Urysohn's lemma there exists a continuous function $\hat{g} : X \mapsto [0, 1]$ mapping A to 0 and B to 1. Then $g_1(x) = (\alpha/3)\hat{g}(x)$, $x \in X$, is continuous and

$$\begin{aligned} &g_1(x) = 0, && \forall x \in A; \\ &g_1(x) = \tfrac{\alpha}{3}, && \forall x \in B; \\ &g_1(x) \in \left(0, \tfrac{\alpha}{3}\right), && \forall x \in X \setminus (A \cup B). \end{aligned}$$

Furthermore, for $x \in E$

$$0 \leq f(x) - g_1(x) \leq \alpha - \frac{1}{3}\alpha = \frac{2}{3}\alpha.$$

When $\alpha = 1$ we have $g_1 : E \mapsto [0, 1/3]$. Set

$$(f - g_1) = f_1 : E \mapsto \left[0, \frac{2}{3}\right].$$

Repeating the previous step for $\alpha = 2/3$ gives the three subintervals

$$I_1 = \left[0, \frac{1}{3} \cdot \frac{2}{3}\right], \quad I_2 = \left[\frac{1}{3} \cdot \frac{2}{3}, \left(\frac{2}{3}\right)^2\right], \quad I_3 = \left[\left(\frac{2}{3}\right)^2, \frac{2}{3}\right],$$

and we redefine

$$A = \left\{ x \in E : f_1(x) \leq \frac{1}{3} \cdot \frac{2}{3} \right\} \quad \text{and} \quad B = \left\{ x \in E : f_1(x) \geq \left(\frac{2}{3}\right)^2 \right\}.$$

still closed and disjoint in X. The Urysohn lemma gives a new continuous map $\hat{g} : X \mapsto [0, 1]$ sending A to 0 and B to 1. The continuous map $g_2(x) = (1/3) \cdot (2/3)\hat{g}(x)$ satisfies

$$\begin{aligned} &g_2(x) = 0, && \forall x \in A; \\ &g_2(x) = \left(\tfrac{1}{3} \cdot \tfrac{2}{3}\right), && \forall x \in B; \\ &g_2(x) \in \left(0, \left(\tfrac{1}{3} \cdot \tfrac{2}{3}\right)\right), && \forall x \in X \setminus (A \cup B); \end{aligned}$$

and for every $x \in E$

$$0 \leq [f(x) - (g_1(x) + g_2(x))] \leq \left(\frac{2}{3}\right)^2.$$

Next, consider

$$[f - (g_1 + g_2)] = f_2 : E \mapsto \left[0, \left(\frac{2}{3}\right)^2\right].$$

By induction, at the nth iteration we have a continuous map g_n such that

$$\begin{aligned} &g_n(x) = 0, && \forall x \in A; \\ &g_n(x) = \tfrac{1}{3}\left(\tfrac{2}{3}\right)^{n-1}, && \forall x \in B; \\ &g_n(x) \in \left(0, \tfrac{1}{3} \cdot \left(\tfrac{2}{3}\right)^{n-1}\right), && \forall x \in X \setminus (A \cup B); \end{aligned}$$

and

$$0 \leq f(x) - \sum_{i=1}^{n} g_i(x) \leq \left(\frac{2}{3}\right)^n$$

for every $x \in E$. Taking the limit as $n \to \infty$ we finally obtain

$$g(x) = \sum_{i=1}^{\infty} g_i(x), \quad x \in E.$$

The above turns into the required extension $g : X \mapsto [0, 1]$, provided we show it is continuous, i.e. $\sum_{i=1}^{\infty} g_i(x)$ converges uniformly on X, and that $g(x) = f(x)$ for every $x \in E$.

Define $S_n(x) = g_1(x) + g_2(x) + \ldots + g_n(x)$. For any $n \in \mathbb{N}$ the partial sum S_n is continuous, and whenever $m > n$

$$\begin{aligned} |S_m(x) - S_n(x)| &= \left| \sum_{k=n+1}^{m} g_k(x) \right| \leq \sum_{k=n+1}^{m} |g_k(x)| \\ &\leq \frac{1}{3} \sum_{k=n+1}^{m} \left(\frac{2}{3}\right)^{k-1} = \frac{1}{3} \cdot \frac{\left(\frac{2}{3}\right)^n - \left(\frac{2}{3}\right)^m}{1 - \frac{2}{3}} \\ &= \left(\frac{2}{3}\right)^n - \left(\frac{2}{3}\right)^m \leq \left(\frac{2}{3}\right)^n \end{aligned}$$

for every $x \in X$. Hence $\{S_n\}_{n\geq 1}$ is a Cauchy sequence. Taking the limit as $m \to \infty$, we know

$$\lim_{m\to\infty} S_m(x) = g(x)$$

so the above inequality reads

$$|g(x) - S_n(x)| \leq \left(\frac{2}{3}\right)^n, \quad x \in X.$$

Therefore $\{S_n\}_{n\geq 1}$ converges uniformly on X to $g(x)$, whence the continuity of g follows.

Furthermore $g(x)$ is bounded, since

$$0 \leq g(x) = \sum_{i=1}^{\infty} g_i(x) \leq \frac{1}{3} \sum_{i=1}^{\infty} \left(\frac{2}{3}\right)^{i-1} = \frac{1}{3} \cdot \frac{1}{1 - \frac{2}{3}} = 1.$$

At last, given $x' \in E$, by construction

$$f(x') - S_n(x') = f(x') - \left[g_1(x') + \ldots + g_n(x')\right] \leq \left(\frac{2}{3}\right)^n.$$

As $n \to \infty$ we obtain $f(x') - g(x') = 0$, so $f(x) = g(x)$ for every $x \in E$.

Proof of *2*. Now we have a continuous extension $g : X \mapsto [0, 1]$ of $f : E \mapsto [0, 1]$. Since $\mathbb{R}$ is homeomorphic to $(0, 1)$, it will suffice to exhibit a continuous extension ϕ of f with image $\phi(X) \subseteq (0, 1)$. Notice that by replacing f with $(2f - 1)$ and g with $(2g - 1)$ we can prove *2.* for the interval $[-1, 1]$. So consider

$$F = g^{-1}(\{-1, 1\}) \subseteq X.$$

Clearly F is closed, and disjoint from E because $g(E) \subseteq (-1, 1)$. Urysohn's lemma guarantees there exists a continuous function $h : X \mapsto [0, 1]$ such that $h(F) = \{0\}$ and $h(E) = \{1\}$.

The continuous map $\phi = h \cdot g$ extends f, since for every $x \in E$

$$\phi(x) = h(x) \cdot g(x) = g(x) = f(x);$$

its image lies in $(-1, 1)$, because for $x \in F$

$$\phi(x) = g(x) \cdot h(x) = 0;$$

and when $x \notin F$

$$|\phi(x)| = |g(x) \cdot h(x)| \leq |g(x)| < 1.$$

■

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List of Symbols

$A \subset B$	A is a proper subset of B, 19
$A \subseteq B$	A is a subset of B, 4
$\mathcal{A}(\mathcal{C})$	Algebra generated by a family $\mathcal{C}$, 124
$\mathcal{A}^+$	Algebra generated by pairwise-disjoint sets of a semi-algebra, 125
$\mathscr{B}(n, p)$	Binomial random variable of order n and parameter p, 413
$\mathcal{B}(\mathbb{R})$	Borel σ-algebra over $\mathbb{R}$, 142
$B(A)$	Boundary of a set A, 22
C	Cantor set, 40
$\mathbf{c}$	Cardinality of the continuum, 27
$\aleph_0$	Cardinality of the set $\mathbb{N}$, 27
φ_A	Characteristic function of a set A, 67
$\mathcal{I}$	Class of all real intervals, 122
Π^0_1	Class of closed subsets of $\mathbb{R}$, 56
Π^0_α	Class of countable intersections of sets in $\bigcup_{\beta<\alpha} \Sigma^0_\beta$, 56
Π^0_β	Class of countable intersections of sets in $\Sigma^0_{\beta-1}$, 56
Σ^0_α	Class of countable unions of sets in $\bigcup_{\beta<\alpha} \Pi^0_\beta$, 58
Σ^0_β	Class of countable unions of sets in $\Pi^0_{\beta-1}$, 56
$\mathscr{M}_0^+(X, \mathcal{S})$	Class of non-negative $\mathcal{S}$-measurable simple maps, 256
Σ^0_1	Class of open subsets of $\mathbb{R}$, 56
$\mathcal{A}_{pj}(\mathcal{I})$	Class of Peano-Jordan measurable subsets of $\mathbb{R}$, 172
$\tilde{\mathcal{I}}$	Class of real right-closed intervals, 122
$\mathscr{M}_0(X, \mathcal{S})$	Class of $\mathcal{S}$-measurable simple maps, 256
$\mathcal{N}$	Class of subsets of X with zero outer measure, 204
$\bar{B}(x, r)$	Closed ball, 30
$\bar{A}$	Closure of a set A, 4

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$\mathscr{G}(I)$	Collection of improperly integrable maps on a bounded interval I, 339
$\mathscr{G}(I^*)$	Collection of improperly integrable maps on unbounded interval I, 344
$\mathcal{R}(I)$	Collection of Riemann integrable, bounded maps, 324
A^c	Complement set to A, 22
$C(f)$	Continuity set of f, 64
$F_{\sigma,\delta}$	Countable intersections of F_σ sets, 57
G_δ	Countable intersections of open sets of $\mathbb{R}$, 55
F_σ	Countable unions of closed sets of $\mathbb{R}$, 57
$G_{\delta,\sigma}$	Countable unions of G_δ sets, 55
$F_X(x) = \sum_{x_i<x} p_i$	Cumulative distribution function of a discrete RV X, 399
$F_X(x) = P(X \le x)$	Cumulative distribution function of a RV X, 276
$F_X(x) = \int_{(-\infty,x]} f_X(t)\, \ell(dt)$	Cumulative distribution function of an absolutely continuous RV X, 401
$f_X(x)$	Density function of an absolutely continuous RV X, 401
A'	Derived set of A, 13
$\operatorname{diam}(E)$	Diameter of a set E, 429
$D^+F(x)$, $D_+F(x)$, $D^-F(x)$, $D_-F(x)$	Dini derivatives, 349
$D(f)$	Discontinuity set of f, 64
$X = \binom{x_1,x_2,\ldots}{p_1,p_2,\ldots}$	Discrete RV X, 399
$f_X(x)\ell(dx)$	Elementary probability of a continuous random variable X, 401
$\emptyset$	Empty set, 3
$\diamond$	End of a definition, 4
$\blacksquare$	End of a proof, 4
$\clubsuit$	End of a remark, 12
$\bigstar$	End of an example, 16
$\{\omega \in \Omega : X(\omega) \in B\}$	Events corresponding to a set B, 274
$f : X \mapsto Y$	f is a map on X with values in Y, 29
Δ^0_α	Family of ambiguous sets of class α, 75
$\mathscr{B}_\alpha$	Family of Baire functions of class α, 90
$\mathscr{B}_\beta$	Family of Baire functions of class β, 90
$\mathscr{B}_\Omega$	Family of Baire functions of class Ω, 90
$\mathscr{B}_0$	Family of Baire functions of class 0, 90
Δ^0_1	Family of clopen subsets of $\mathbb{R}$ (ambiguous of class 1), 74
$\mathcal{T}_m$	Family of negligible sets w.r.t. m, 213
$\mathscr{M}^+(X,\mathcal{S})$	Family of non-negative $\mathcal{S}$-measurable functions, 243
$\mathcal{N}_m$	Family of sets with zero measure w.r.t. m, 213
$\mathscr{M}(X,\mathcal{S})$	Family of $\mathcal{S}$-measurable functions, 243
$\forall$	For all, 5

∂A	Frontier of a set A, 7
inf	Greatest lower bound or infimum, 108
$\sim$	has the same distribution as, 426
$\Leftrightarrow$	If and only if, 5
$\Rightarrow$	Implies, 4
$F(x) = \int_{[a,x]} f(t)\ell(dt)$	Integral function, 349
$\int_X f\ dm$	Integral over X of map $\in \mathscr{M}(X, \mathcal{S})$ with respect to m, 290
$\int_X s\ dm$	Integral over X of a simple map $\in \mathscr{M}_0^+(X, \mathcal{S})$ with respect to m, 286
$\mathring{A}$	Interior of a set A, 5
$\bigcap, \cap$	Intersection, 5
$\mathbb{R} \setminus \mathbb{Q}$	Irrational numbers, 25
sup	Least upper bound or supremum, 63
$\int_{[a,b]} f(x)\ell(dx)$	Lebesgue integral of f on $[a, b]$, 323
ℓ^*	Lebesgue outer measure, 218
$\mathscr{L}$	Lebesgue σ-algebra of measurable subsets of X, 202
$\mathscr{L}(\mathbb{R})$	Lebesgue σ-algebra on $\mathbb{R}$, 218
m_F	Lebesgue-Stieltjes measure generated by F, 234
$s(f)$	Lower Riemann integral of a bounded map $f : [a, b] \mapsto \mathbb{R}$, 324
$s(f, P_n)$	Lower sum over partition P_n, 324
$E(X) = \sum_{i \in I} p_i x_i$	Mean value of a discrete RV X, 402
$E(X) = \int_{\mathbb{R}} xf(x)\ell(dx)$	Mean value of an absolutely continuous RV X, 402
$(X, \mathcal{S})$	Measurable space, 213
$(X, \mathcal{S}, m)$	Measure space, 213
(X, d)	Metric space, 29
$\mathscr{M}$	Monotone class, 134
$\mathscr{M}(\mathcal{C})$	Monotone class generated by $\mathcal{C}$, 134
$\int_{\mathbb{R}} \mid x \mid^n f(x)\ell dx$	nth absolute moment of an absolutely continuous RV, 403
$\sum_{i=1}^{\infty} \mid x_i \mid^n p_i$	nth absolute moment of a discrete RV, 403
$\int_{\mathbb{R}} (x - m)^n f(x)\ell dx$	nth central moment of an absolutely continuous RV, 403
$\sum_{i=1}^{\infty} (x_i - m)^n p_i$	nth central moment of a discrete RV, 403
$\int_{\mathbb{R}} x^n f(x)\ell dx$	nth moment of an absolutely continuous RV, 403
$\sum_{i=1}^{\infty} x_i^n p_i$	nth moment of a discrete RV, 403
f^-	Negative part of a map, 248
$v_-(x) = VN_a^x(f)$	Negative total variation function of f, 365
$V_a^b(P_-, f)$	Negative variation of f over $[a, b]$ for partition P, 364
U_x	Neighbourhood of a point x, 4
$N(m, \sigma^2)$	Normal RV, 426
Ω	First uncountable ordinal, 55

$B(x, r)$	Open ball, 30
$\omega(f, x)$	Oscillation of f at x, 63
$\omega(f, E)$	Oscillation of f over E, 63
$m^*(A)$	Outer measure of a set A, 197
$\bigsqcup$	Pairwise-disjoint union, 59
$m_{pj}(A)$	Peano-Jordan measure of a set A, 172
$p_k = \frac{\lambda^k e^{-\lambda}}{k!}$	Poisson RV, 417
f^+	Positive part of a map, 248
$\mathbb{Q}^+$	Positive rationals, 25
$v_+(x) = V P_a^x(f)$	Positive total variation function of f, 365
$V_a^b(P_+, f)$	Positive variation of f over $[a, b]$ for partition P, 364
$\mathcal{P}(X)$	Power set of X: collection of subsets of X, 123
$f^{-1}(E)$	Pre-image $\{x \in X : f(x) \in E\}$, 74
$\{p_i\}_{i\geq 1}$	Probability distribution of a discrete RV, 399
P_X	Probability distribution of a RV X, 274
$P(\Omega)$	Probability of certain event, 276
$P(X = x)$	Probability of elementary event, 279
$P(A)$	Probability of event A, 276
p_i	Probability of value x_i for a discrete RV, 399
$(\Omega, \mathcal{S}(\Omega), P)$	Probability space, 274
$P(X \in B)$	Probability the variable X takes values in B, 274
$X : \Omega \mapsto \mathbb{R}$	Random variable (RV), 274
$\mathbb{Q}$	Rational numbers, 25
ℓ	Restriction of measure ℓ^* to $\mathscr{L}(\mathbb{R})$ or $\mathscr{B}(\mathbb{R})$, 222
$\int_a^b f(x)dx$	Riemann integral of f on $[a, b]$, 323
Ω	Sample space, 274
$\{A_n\}_{n\geq 1}$	Sequence of sets with nth term A_n, 49
m_F	Set function induced by map F, 234
$A \setminus B$	Set of elements of A that do not belong to B, 3
$\bar{\mathbb{R}}$	Set of extended reals, 11
$\mathbb{Z}$	Set of integers, 28
$\mathbb{N}$	Set of natural numbers, 19
$\mathbb{R}$	Set of real numbers, 3
$\mathcal{S}(\Omega)$	σ-algebra of events, 274
$\mathcal{S}(\mathcal{A})$	σ-algebra generated by an algebra, 138
$\mathcal{S}(\mathcal{C})$	σ-algebra generated by $\mathcal{C}$, 131
$s(x)$	Simple functions, 106
$\mathcal{R}([a, b])$	Space of maps in $\mathcal{R}(I)$, 328
$C_c^\infty(\mathbb{R})$	Space of compactly supported smooth maps on $\mathbb{R}$, 338
$C([a, b])$	Space of continuous maps on $[a, b]$, 331
$C_c(\mathbb{R})$	Space of continuous maps with compact support on $\mathbb{R}$, 337

$L_1(X, \mathcal{S}, m)$ *or* L_1	Space of m-summable functions, 304
$C^\infty([a, b])$	Space of smooth functions on $[a, b]$, 336
$C^\infty(\mathbb{R})$	Space of smooth functions on $\mathbb{R}$, 333
$L_1([a, b])$	Space of summable functions on $[a, b]$, 326
$\sqrt{Var(X)}$ or $\sigma(X)$	Standard deviation of a RV X, 403
$N(0, 1)$	Standard normal RV, 427
$\mathcal{C} \cap Y$	Subsets of Y contained in $\mathcal{C}$, 145
$\text{supp}(f)$	Support of a map, 337
$(X, \mathcal{T})$	Topological space, 3
$\mathcal{T}$	Topology, 3
$VN_a^b(f)$	Total negative variation of f on $[a, b]$, 364
$VP_a^b(f)$	Total positive variation of f on $[a, b]$, 364
$v(x) = V_a^x(f)$	Total variation function of f, 365
$V_a^b(f)$	Total variation of f over $[a, b]$, 355
$V(f)$	Total variation of f over $\mathbb{R}$, 355
$\mathcal{U}_{[a,b]}$	Uniform RV, 420
$\bigcup, \cup$	Union, 5
$S(f)$	Upper Riemann integral of a bounded map $f : [a, b] \mapsto \mathbb{R}$, 324
$S(f, P_n)$	Upper sum over partition P_n, 324
$Var(X) = \sum_{i=1}^{\infty}(x_i - m)^2 p_i$	Variance of a discrete RV X, 403
$Var(X)$ *or* $\sigma^2(X)$	Variance of a RV X, 403
$Var(X) = \int_{\mathbb{R}}(x - m)^2 f(x)\ell dx$	Variance of an absolutely continuous RV X, 403
$V_a^b(P, f)$	Variation of f over $[a, b]$ with respect to partition P, 355
$x \in A$	x belongs to the set A, 4
$x \notin A$	x does not belong to the set A, 8

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